

# WEST CENTRAL RAILWAY

## CONTRACT AGREEMENT OF WORKS

CONTRACT AGREEMENT NO.: WCR/WCRC/Civil/2024/0055

DATE: AS MENTIONED BELOW SIGNATURE

ARTICLES OF AGREEMENT made between the President of India acting through the Railway Administration hereafter called the "Railway" of the one part and **SURENDRA SINGH KALRA-BILASPUR, FLAT NO. A 507 HARSH ASHIYANA GREEN PARK COLONY JARHABHATA ROAD, BILASPUR, Chhattisgarh-495001 India** herein after called the "Contractor" of other part.

WHEREAS the Contractor has agreed with the Railway for performance of the works "**Execution of balance works in newly constructed staff quarters at Rewa in connection with the Rewa-Sidhi new BG rail line project.**" set forth in the Schedule hereto annexed upon the Standard General Conditions of Contract, corrected upto latest correction slips and the Specifications of West Central Railway corrected upto latest correction slips and the Schedule of Rates of West Central Railway, corrected upto latest correction slips and the Special Conditions and Special Specifications, if any and in conformity with the drawings hereinto annexed AND WHEREAS the performance of the said works is an act in which the public are interested.

NOW THIS INDENTURE WITNESSETH that in consideration to the payments to be made by the Railways, the Contractor will duly perform the said works in the said schedule set forth and shall execute the same with great promptness, care and accuracy in a workman like manner to the satisfaction of the Railway and will complete the same in accordance with the said specification and said drawings and said conditions of the contract on or before **17/04/2025** and will maintain the said works for a period of 6.0 Calendar months from the certified date of their completion and will observe, fulfill and keep all the conditions therein mentioned (which shall be deemed and taken to be part of this contract, as if the same have been fully set forth herein), AND the Railway, both hereby agree that if the Contractor shall duly perform the said works in the manner aforesaid and observe and keep the said terms and conditions, the Railway will pay or cause to be paid to the Contractor for the said works on the final completion thereof the amount due in respect thereof at the rates specified in the Schedule hereto annexed.

Contractor: (Signature)

Name: SURENDRA SINGH KALRA-BILASPUR

Address: FLAT NO. A 507 HARSH  
ASHIYANA GREEN PARK COLONY  
JARHABHATA ROAD, BILASPUR,  
Chhattisgarh-495001 India,

Railway: Dy.CE/C-II/REWA  
(For President of India)

**VETTED**

## AGREEMENT DETAILS

|                                 |                                  |                                |                                       |                               |                |
|---------------------------------|----------------------------------|--------------------------------|---------------------------------------|-------------------------------|----------------|
| <b>Railway Name:</b>            | West Central Railway             | <b>Unit Name:</b>              | WCR<br>/CONSTRUC<br>TION,<br>JABALPUR | <b>Department<br/>Name:</b>   | Civil          |
| <b>Tender Number:</b>           | DYCEC1REWA-07-<br>2024           | <b>Tender Type:</b>            | Open                                  |                               |                |
| <b>Nature of<br/>Contract:</b>  | Works                            | <b>Date of<br/>Completion:</b> | 17/04/2025                            | <b>LOA Number:</b>            | 00912200115255 |
| <b>LOA Date:</b>                | 17/10/2024                       | <b>LOA Amount<br/>(Rs.):</b>   | 25988562                              | <b>PG Amount<br/>(Rs.):</b>   | 1299428        |
| <b>Contractor's<br/>Name:</b>   | SURENDRA SINGH KALRA-BILASPUR    |                                |                                       | <b>Contractor's<br/>Code:</b> | 91220          |
| <b>Bidding Style:</b>           | Single Rate for Each<br>Schedule | <b>Bidding Unit:</b>           |                                       | <b>Bid Rate:</b>              |                |
| <b>Tender Amount<br/>(Rs.):</b> | 28013419                         | <b>Rebate :</b>                | 0.0                                   | <b>Bid Amount:</b>            | 25988562       |

## SCHEDULE DETAILS

| SN                                                     | Item Type                                                                                                                                                                                                                                                                                                                                                                                     | Item Code | Item Qty | Qty Unit      | Unit Rate | Basic Value | Escl. (%)                | Advt.Value (Rs) | Bid Rate/Unit Rate |         | Bid Amount (Rs) | Sch. Bid Total |
|--------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|----------|---------------|-----------|-------------|--------------------------|-----------------|--------------------|---------|-----------------|----------------|
| Sch. Code & Desc. : A Earthwork and concrete work etc. |                                                                                                                                                                                                                                                                                                                                                                                               |           |          | Sch. Type:SOR |           |             | SOR Dir.: CP WD DSR 2021 | 25089061        | 8.08               | % Below | 23061865        | 23061865       |
| Chapter Name: EARTH WORK                               |                                                                                                                                                                                                                                                                                                                                                                                               |           |          |               |           |             |                          |                 |                    |         |                 |                |
| 1                                                      | Indi vidu al                                                                                                                                                                                                                                                                                                                                                                                  | 2.8.1     | 1150     | cum           | 286.85    | 329878      | 10.91 Below w            | 293888          |                    |         |                 |                |
|                                                        | Description: Earth work in excavation by mechanical means (Hydraulic excavator) / manual means in foundation trenches or drains (not exceeding 1.5 m in width or 10 sqm on plan), including dressing of sides and ramming of bottoms, lift upto 1.5 m, including getting out the excavated soil and disposal of surplus excavated soil as directed, within a lead of 50 m. All kinds of soil. |           |          |               |           |             |                          |                 |                    |         |                 |                |
| 2                                                      | Indi vidu al                                                                                                                                                                                                                                                                                                                                                                                  | 2.25      | 225      | cum           | 253.95    | 57139       | 10.91 Below w            | 50905           |                    |         |                 |                |
|                                                        | Description: Filling available excavated earth (excluding rock) in trenches, plinth, sides of foundations etc. in layers not exceeding 20cm in depth, consolidating each deposited layer by ramming and watering, lead up to 50 m and lift upto 1.5 m.                                                                                                                                        |           |          |               |           |             |                          |                 |                    |         |                 |                |
| 3                                                      | Indi vidu al                                                                                                                                                                                                                                                                                                                                                                                  | 2.25A     | 450      | cum           | 368.65    | 165893      | 10.91 Below w            | 147794          |                    |         |                 |                |
|                                                        | Description: Excavating, supplying and filling of local earth (including royalty) by mechanical transport upto a lead of 5km also including ramming and watering of the earth in layers not exceeding 20 cm in trenches, plinth, sides of foundation etc. complete.                                                                                                                           |           |          |               |           |             |                          |                 |                    |         |                 |                |
| 4                                                      | Indi vidu al                                                                                                                                                                                                                                                                                                                                                                                  | 2.26.1    | 410.5    | cum           | 104.5     | 42897       | 10.91 Below w            | 38217           |                    |         |                 |                |
|                                                        | Description: Extra for every additional lift of 1.5 m or part thereof in excavation / banking excavated or stacked materials. All kinds of soil                                                                                                                                                                                                                                               |           |          |               |           |             |                          |                 |                    |         |                 |                |
| 5                                                      | Indi vidu al                                                                                                                                                                                                                                                                                                                                                                                  | 2.27      | 40.5     | cum           | 2161.2    | 87529       | 10.91 Below w            | 77979           |                    |         |                 |                |
|                                                        | Description: Supplying and filling in plinth with sand under floors, including watering, ramming, consolidating and dressing complete.                                                                                                                                                                                                                                                        |           |          |               |           |             |                          |                 |                    |         |                 |                |

| SN                                 | Item Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Item Code    | Item Qty | Qty Unit | Unit Rate   | Basic Value | Escl. (%)          | Advt.Value (Rs) | Bid Rate/Unit Rate |  | Bid Amount (Rs) | Sch. Bid Total |
|------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|----------|----------|-------------|-------------|--------------------|-----------------|--------------------|--|-----------------|----------------|
| 6                                  | Individual                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 2.28.1       | 60       | Sqm      | 28.15       | 1689        | 10.9<br>1<br>Below | 1505            |                    |  |                 |                |
|                                    | Description: Surface dressing of the ground including removing vegetation and inequalities not exceeding 15 cm deep and disposal of rubbish, lead up to 50 m and lift up to 1.5 m. All kinds of soil                                                                                                                                                                                                                                                                                                                                                                 |              |          |          |             |             |                    |                 |                    |  |                 |                |
| 7                                  | Individual                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 2.34.1       | 190      | Litre    | 200.9       | 38171       | 10.9<br>1<br>Below | 34007           |                    |  |                 |                |
|                                    | Description: Supplying chemical emulsion in sealed containers including delivery as specified. Chlorpyriphos emulsifiable concentrate of 20%                                                                                                                                                                                                                                                                                                                                                                                                                         |              |          |          |             |             |                    |                 |                    |  |                 |                |
| 8                                  | Individual                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 2.35.1<br>.1 | 190      | Metre    | 32.3        | 6137        | 10.9<br>1<br>Below | 5467            |                    |  |                 |                |
|                                    | Description: Along external wall where the apron is not provided using chemical emulsion @ 7.5 litres / sqm of the vertical surface of the substructure to a depth of 300mm including excavation channel along the wall & rodding etc. complete: With Chlorpyriphos E.C. 20% with 1% concentration                                                                                                                                                                                                                                                                   |              |          |          |             |             |                    |                 |                    |  |                 |                |
| <b>Chapter Name: CONCRETE WORK</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |              |          |          |             |             |                    |                 |                    |  |                 |                |
| 9                                  | Individual                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 4.1.3        | 90       | cum      | 7365.<br>15 | 662864      | 10.9<br>1<br>Below | 590545          |                    |  |                 |                |
|                                    | Description: Providing and laying in position cement concrete of specified grade excluding the cost of centering and shuttering - All work up to plinth level : 1:2:4 (1 cement : 2 coarse sand (zone-III) derived from natural sources : 4 graded stone aggregate 20 mm nominal size derived from natural sources)                                                                                                                                                                                                                                                  |              |          |          |             |             |                    |                 |                    |  |                 |                |
| 10                                 | Individual                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 4.1.5        | 75       | cum      | 6833.<br>4  | 512505      | 10.9<br>1<br>Below | 456591          |                    |  |                 |                |
|                                    | Description: Providing and laying in position cement concrete of specified grade excluding the cost of centering and shuttering - All work up to plinth level : 1:3:6 (1 Cement : 3 coarse sand (zone-III) derived from natural sources : 6 graded stone aggregate 20 mm nominal size derived from natural sources)                                                                                                                                                                                                                                                  |              |          |          |             |             |                    |                 |                    |  |                 |                |
| 11                                 | Individual                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 4.1.6        | 80       | cum      | 6670.<br>25 | 533620      | 10.9<br>1<br>Below | 475402          |                    |  |                 |                |
|                                    | Description: Providing and laying in position cement concrete of specified grade excluding the cost of centering and shuttering - All work up to plinth level : 1:3:6 (1 Cement : 3 coarse sand (zone-III) derived from natural sources: 6 graded stone aggregate 40 mm nominal size derived from natural sources)                                                                                                                                                                                                                                                   |              |          |          |             |             |                    |                 |                    |  |                 |                |
| 12                                 | Individual                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 4.2.3        | 250      | cum      | 9375.<br>2  | 2343800     | 10.9<br>1<br>Below | 2088091         |                    |  |                 |                |
|                                    | Description: Providing and laying cement concrete in retaining walls, return walls, walls (any thickness) including attached pilasters, columns, piers, abutments, pillars, posts, struts, buttresses, string or lacing courses, parapets, coping, bed blocks, anchor blocks, plain window sills, fillets, sunken floor etc., up to floor five level, excluding the cost of centering, shuttering and finishing: 1:2:4 (1 Cement : 2 coarse sand (zone-III) derived from natural sources : 4 graded stone aggregate 20 mm nominal size derived from natural sources) |              |          |          |             |             |                    |                 |                    |  |                 |                |
| 13                                 | Individual                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 4.3.1        | 200      | Sqm      | 307.9<br>5  | 61590       | 10.9<br>1<br>Below | 54871           |                    |  |                 |                |
|                                    | Description: Centering and shuttering including strutting, propping etc. and removal of form work for : Foundations, footings, bases for columns                                                                                                                                                                                                                                                                                                                                                                                                                     |              |          |          |             |             |                    |                 |                    |  |                 |                |
| 14                                 | Individual                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 4.3.2        | 1800     | Sqm      | 669.5<br>5  | 1205190     | 10.9<br>1<br>Below | 1073704         |                    |  |                 |                |
|                                    | Description: Centering and shuttering including strutting, propping etc. and removal of form work for : Retaining walls, return walls, walls (any thickness) including attached pilasters, buttresses, plinth and string courses fillets, kerbs and steps etc.                                                                                                                                                                                                                                                                                                       |              |          |          |             |             |                    |                 |                    |  |                 |                |

| SN                                              | Item Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Item Code | Item Qty | Qty Unit | Unit Rate | Basic Value | Escl. (%)          | Advt.Value (Rs) | Bid Rate/Unit Rate |  | Bid Amount (Rs) | Sch. Bid Total |
|-------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|----------|----------|-----------|-------------|--------------------|-----------------|--------------------|--|-----------------|----------------|
| 15                                              | Individual                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 4.4.1     | 6        | cum      | 7783.65   | 46702       | 10.9<br>1<br>Below | 41607           |                    |  |                 |                |
|                                                 | Description: Providing and laying cement concrete in kerbs, steps and the like at or near ground level excluding the cost of centering, shuttering and finishing. 1:1.5:3 (1 Cement: 1.5 coarse sand(zone-III) derived from natural sources : 3 graded stone aggregate 20 mm nominal size derived from natural sources)                                                                                                                                                                                   |           |          |          |           |             |                    |                 |                    |  |                 |                |
| 16                                              | Individual                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 4.5.1     | 6        | cum      | 9277.75   | 55667       | 10.9<br>1<br>Below | 49593           |                    |  |                 |                |
|                                                 | Description: Providing and fixing up to floor five level precast cement concrete string or lacing courses, copings, bed plates, anchor blocks, plain window sills, shelves, louvers, steps, stair cases, etc., including hoisting and setting in position with cement mortar 1:3 (1 Cement : 3 coarse sand), cost of required Centering complete. 1:1.5:3 (1 cement : 1.5 coarse sand(zone-III) derived from natural sources: 3 graded stone aggregate 20mm nominal size derived from natural sources)    |           |          |          |           |             |                    |                 |                    |  |                 |                |
| 17                                              | Individual                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 4.11      | 7.5      | Sqm      | 452.25    | 3392        | 10.9<br>1<br>Below | 3022            |                    |  |                 |                |
|                                                 | Description: Providing and laying damp-proof course 50mm thick with cement concrete 1:2:4 (1 cement : 2 coarse sand(zone-III) derived from natural sources : 4 graded stone aggregate 20mm nominal size derived from natural sources).                                                                                                                                                                                                                                                                    |           |          |          |           |             |                    |                 |                    |  |                 |                |
| <b>Chapter Name: REINFORCED CEMENT CONCRETE</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |           |          |          |           |             |                    |                 |                    |  |                 |                |
| 18                                              | Individual                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 5.3       | 14       | cum      | 10719.3   | 150070      | 10.9<br>1<br>Below | 133698          |                    |  |                 |                |
|                                                 | Description: Reinforced cement concrete work in beams, suspended floors, roofs having slope up to 15 degree landings, balconies, shelves, chajjas, lintels, bands, plain window sills, staircases and spiral stair cases above plinth level up to floor five level, excluding the cost of centering, shuttering, finishing and reinforcement with 1:1.5:3 (1 cement : 1.5 coarse sand(zone-III) derived from natural sources : 3 graded stone aggregate 20 mm nominal size derived from natural sources). |           |          |          |           |             |                    |                 |                    |  |                 |                |
| 19                                              | Individual                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 5.5       | 4        | cum      | 11291.2   | 45165       | 10.9<br>1<br>Below | 40237           |                    |  |                 |                |
|                                                 | Description: Reinforced cement concrete work in arches, archribs, domes, vaults, shells, folded plate and roofs having slope more than 15 degree up to floor five level, excluding the cost of centering, shuttering, finishing and reinforcement with 1:1.5:3 (1 cement : 1.5 coarse sand (zone-III) derived from natural sources: 3 graded stone aggregate 20 mm nominal size derived from natural sources).                                                                                            |           |          |          |           |             |                    |                 |                    |  |                 |                |
| 20                                              | Individual                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 5.33.1.1  | 165      | cum      | 8683.8    | 1432827     | 10.9<br>1<br>Below | 1276506         |                    |  |                 |                |
|                                                 | Description: All works upto plinth level                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |           |          |          |           |             |                    |                 |                    |  |                 |                |
|                                                 | Concrete of M25 grade with minimum cement content of 330 kg /cum                                                                                                                                                                                                                                                                                                                                                                                                                                          |           |          |          |           |             |                    |                 |                    |  |                 |                |
| 21                                              | Individual                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 5.33.2.1  | 150      | cum      | 10306.2   | 1545930     | 10.9<br>1<br>Below | 1377269         |                    |  |                 |                |
|                                                 | Description: All works above plinth level upto floor V level                                                                                                                                                                                                                                                                                                                                                                                                                                              |           |          |          |           |             |                    |                 |                    |  |                 |                |
|                                                 | Concrete of M25 grade with minimum cement content of 330 kg /cum                                                                                                                                                                                                                                                                                                                                                                                                                                          |           |          |          |           |             |                    |                 |                    |  |                 |                |
| 22                                              | Individual                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 5.9.1     | 245      | Sqm      | 307.95    | 75448       | 10.9<br>1<br>Below | 67216           |                    |  |                 |                |
|                                                 | Description: Centering and shuttering including strutting, propping etc. and removal of form for Foundations, footings, bases of columns, etc. for mass concrete                                                                                                                                                                                                                                                                                                                                          |           |          |          |           |             |                    |                 |                    |  |                 |                |

| SN                                 | Item Type                                                                                                                                                                                                                                                                       | Item Code | Item Qty | Qty Unit | Unit Rate | Basic Value | Escl. (%)   | Advt.Value (Rs) | Bid Rate/Unit Rate |  | Bid Amount (Rs) | Sch. Bid Total |
|------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|----------|----------|-----------|-------------|-------------|-----------------|--------------------|--|-----------------|----------------|
| 23                                 | Individual                                                                                                                                                                                                                                                                      | 5.9.3     | 356      | Sqm      | 766.55    | 272892      | 10.91 Below | 243119          |                    |  |                 |                |
|                                    | Description: Centering and shuttering including strutting, propping etc. and removal of form for Suspended floors, roofs, landings, balconies and access platform                                                                                                               |           |          |          |           |             |             |                 |                    |  |                 |                |
| 24                                 | Individual                                                                                                                                                                                                                                                                      | 5.9.5     | 627      | Sqm      | 608.35    | 381435      | 10.91 Below | 339821          |                    |  |                 |                |
|                                    | Description: Centering and shuttering including strutting, propping etc. and removal of form for Lintels, beams, plinth beams, girders, bressumers and cantilevers                                                                                                              |           |          |          |           |             |             |                 |                    |  |                 |                |
| 25                                 | Individual                                                                                                                                                                                                                                                                      | 5.9.15    | 110      | Sqm      | 307.95    | 33875       | 10.91 Below | 30179           |                    |  |                 |                |
|                                    | Description: Centering and shuttering including strutting, propping etc. and removal of form for Small lintels not exceeding 1.5 m clear span, moulding as in cornices, window sills, string courses, bands, copings, bed plates, anchor blocks and the like                    |           |          |          |           |             |             |                 |                    |  |                 |                |
| 26                                 | Individual                                                                                                                                                                                                                                                                      | 5.9.16.1  | 150      | Metre    | 181.9     | 27285       | 10.91 Below | 24308           |                    |  |                 |                |
|                                    | Description: Edges of slabs and breaks in floors and walls Under 20 cm wide                                                                                                                                                                                                     |           |          |          |           |             |             |                 |                    |  |                 |                |
| Chapter Name: MASONRY WORK         |                                                                                                                                                                                                                                                                                 |           |          |          |           |             |             |                 |                    |  |                 |                |
| 27                                 | Individual                                                                                                                                                                                                                                                                      | 6.1.1     | 55       | cum      | 6882      | 378510      | 10.91 Below | 337215          |                    |  |                 |                |
|                                    | Description: Brick work with common burnt clay F.P.S. (non modular) bricks of class designation 7.5 in foundation and plinth in: Cement mortar 1:4 (1 cement : 4 coarse sand)                                                                                                   |           |          |          |           |             |             |                 |                    |  |                 |                |
| 28                                 | Individual                                                                                                                                                                                                                                                                      | 6.13.2    | 15       | Sqm      | 1018.05   | 15271       | 10.91 Below | 13605           |                    |  |                 |                |
|                                    | Description: Half brick masonry with common burnt clay F.P.S. (non modular) bricks of class designation 7.5 in superstructure above plinth level up to floor V level. Cement mortar 1:4 (1 cement :4 coarse sand)                                                               |           |          |          |           |             |             |                 |                    |  |                 |                |
| 29                                 | Individual                                                                                                                                                                                                                                                                      | 6.15      | 11       | Sqm      | 86.45     | 951         | 10.91 Below | 847             |                    |  |                 |                |
|                                    | Description: Extra for providing and placing in position 2 Nos 6mm dia. M.S. bars at every third course of half brick masonry.                                                                                                                                                  |           |          |          |           |             |             |                 |                    |  |                 |                |
| 30                                 | Individual                                                                                                                                                                                                                                                                      | 6.3.1     | 269      | cum      | 8030.6    | 2160231     | 10.91 Below | 1924550         |                    |  |                 |                |
|                                    | Description: Brick work with common burnt clay machine moulded perforated bricks of class designation 12.5 conforming to IS: 2222 in superstructure above plinth level up to floor five level in cement mortar 1:6 (1 cement : 6 coarse sand) : With F.P.S.(non modular) bricks |           |          |          |           |             |             |                 |                    |  |                 |                |
| Chapter Name: CLADDING WORK        |                                                                                                                                                                                                                                                                                 |           |          |          |           |             |             |                 |                    |  |                 |                |
| 31                                 | Individual                                                                                                                                                                                                                                                                      | 8.3.2     | 12.5     | Metre    | 418.85    | 5236        | 10.91 Below | 4664            |                    |  |                 |                |
|                                    | Description: Providing edge moulding to 18 mm thick marble stone counters, Vanities etc., including machine polishing to edge to give high gloss finish etc. complete as per design approved by Engineer-in-Charge. Granite work                                                |           |          |          |           |             |             |                 |                    |  |                 |                |
| Chapter Name: WOOD AND P.V.C. WORK |                                                                                                                                                                                                                                                                                 |           |          |          |           |             |             |                 |                    |  |                 |                |
| 32                                 | Individual                                                                                                                                                                                                                                                                      | 9.7.7.2   | 157      | Sqm      | 2231.2    | 350298      | 10.91 Below | 312081          |                    |  |                 |                |

[illegible]

| SN                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Item Type  | Item Code | Item Qty | Qty Unit | Unit Rate | Basic Value | Escl. (%)   | Advt.Value (Rs) | Bid Rate/Unit Rate |  | Bid Amount (Rs) | Sch. Bid Total |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----------|----------|----------|-----------|-------------|-------------|-----------------|--------------------|--|-----------------|----------------|
| 42                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Individual | 9.66.1    | 224      | Each     | 35.2      | 7885        | 10.91 Below | 7025            |                    |  |                 |                |
| Description: Providing and fixing ISI marked oxidised M.S. handles conforming to IS:4992 with necessary screws etc. complete : (copper oxidized as per IS: 1378) 125 mm                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |            |           |          |          |           |             |             |                 |                    |  |                 |                |
| 43                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Individual | 9.96.2    | 25       | Each     | 234.9     | 5873        | 10.91 Below | 5232            |                    |  |                 |                |
| Description: Providing and fixing aluminium sliding door bolts, ISI marked anodised (anodic coating not less than grade AC 10 as per IS : 1868), transparent or dyed to required colour or shade, with nuts and screws etc. complete : 250x16 mm                                                                                                                                                                                                                                                                                                                                                                                                                 |            |           |          |          |           |             |             |                 |                    |  |                 |                |
| Chapter Name: STEEL WORK                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |            |           |          |          |           |             |             |                 |                    |  |                 |                |
| 44                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Individual | 10.1      | 170      | Kg       | 93.05     | 15819       | 10.91 Below | 14093           |                    |  |                 |                |
| Description: Structural steel work in single section, fixed with or without connecting plate, including cutting, hoisting, fixing in position and applying a priming coat of approved steel primer all complete.                                                                                                                                                                                                                                                                                                                                                                                                                                                 |            |           |          |          |           |             |             |                 |                    |  |                 |                |
| 45                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Individual | 10.6.1    | 36       | Sqm      | 3008.8    | 108317      | 10.91 Below | 96499           |                    |  |                 |                |
| Description: Supplying and fixing rolling shutters of approved make, made of required size M.S. laths, interlocked together through their entire length and jointed together at the end by end locks, mounted on specially designed pipe shaft with brackets, side guides and arrangements for inside and outside locking with push and pull operation complete, including the cost of providing and fixing necessary 27.5 cm long wire springs manufactured from high tensile steel wire of adequate strength conforming to IS: 4454 - part 1 and M.S. top cover of required thickness for rolling shutters. 80x1.25 mm M.S. laths with 1.25 mm thick top cover |            |           |          |          |           |             |             |                 |                    |  |                 |                |
| 46                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Individual | 10.8.1    | 4        | Sqm      | 1108.7    | 4435        | 10.91 Below | 3951            |                    |  |                 |                |
| Description: Extra for providing mechanical device chain and crank operation for operating rolling shutters. Exceeding 10.00 sqm and upto 16.80 sqm in the area                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |            |           |          |          |           |             |             |                 |                    |  |                 |                |
| 47                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Individual | 10.14.2.1 | 112      | Metre    | 471.25    | 52780       | 10.91 Below | 47022           |                    |  |                 |                |
| Description: Profile C Fixing with adjustable lugs with split end tail to each jamb                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |            |           |          |          |           |             |             |                 |                    |  |                 |                |
| 48                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Individual | 10.14.3.1 | 8        | Metre    | 511.4     | 4091        | 10.91 Below | 3645            |                    |  |                 |                |
| Description: Profile E Fixing with adjustable lugs with split end tail to each jamb                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |            |           |          |          |           |             |             |                 |                    |  |                 |                |
| 49                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Individual | 10.16.1   | 500      | Kg       | 154.9     | 77450       | 10.91 Below | 69000           |                    |  |                 |                |
| Description: Steel work in built up tubular (round, square or rectangular hollow tubes etc.) trusses etc., including cutting, hoisting, fixing in position and applying a priming coat of approved steel primer, including welding and bolted with special shaped washers etc. complete. Hot finished welded type tubes                                                                                                                                                                                                                                                                                                                                          |            |           |          |          |           |             |             |                 |                    |  |                 |                |
| 50                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Individual | 10.17     | 26       | Each     | 185.95    | 4835        | 10.91 Below | 4307            |                    |  |                 |                |
| Description: Providing and fixing M.S. fan clamp type I or II of 16 mm dia M.S. bar, bent to shape with hooked ends in R.C.C. slabs or beams during laying, including painting the exposed portion of loop, all as per standard design complete.                                                                                                                                                                                                                                                                                                                                                                                                                 |            |           |          |          |           |             |             |                 |                    |  |                 |                |
| 51                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Individual | 10.25.2   | 2200     | Kg       | 142.3     | 313060      | 10.91 Below | 278905          |                    |  |                 |                |

| SN                      | Item Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Item Code | Item Qty | Qty Unit | Unit Rate | Basic Value | Escl. (%)   | Advt.Value (Rs) | Bid Rate/Unit Rate | Bid Amount (Rs) | Sch. Bid Total |
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|                         | Description: Steel work welded in built up sections/ framed work, including cutting, hoisting, fixing in position and applying a priming coat of approved steel primer using structural steel etc. as required. In gratings, frames, guard bar, ladder, railings, brackets, gates and similar works                                                                                                                                                                                                                                                                                                                                    |           |          |          |           |             |             |                 |                    |                 |                |
| 52                      | Individual                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 10.26.1   | 145      | Kg       | 157.15    | 22787       | 10.91 Below | 20301           |                    |                 |                |
|                         | Description: Providing and fixing hand rail of approved size by welding etc. to steel ladder railing, balcony railing, staircase railing and similar works, including applying priming coat of approved steel primer. M.S. tube                                                                                                                                                                                                                                                                                                                                                                                                        |           |          |          |           |             |             |                 |                    |                 |                |
| Chapter Name: FLOORING  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |           |          |          |           |             |             |                 |                    |                 |                |
| 53                      | Individual                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 11.41.2   | 608      | Sqm      | 1416.65   | 861323      | 10.91 Below | 767353          |                    |                 |                |
|                         | Description: Providing and laying vitrified floor tiles in different sizes (thickness to be specified by the manufacturer) with water absorption less than 0.08% and conforming to IS: 15622, of approved make, in all colours and shades, laid on 20mm thick cement mortar 1:4 (1 cement : 4 coarse sand), jointing with grey cement slurry @ 3.3 kg/ sqm including grouting the joints with white cement and matching pigments etc., complete. Size of Tile 600x600 mm                                                                                                                                                               |           |          |          |           |             |             |                 |                    |                 |                |
| 54                      | Individual                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 11.56.1   | 12       | Sqm      | 3908.8    | 46906       | 10.91 Below | 41788           |                    |                 |                |
|                         | Description: Providing and laying Polished Granite stone flooring in required design and patterns, in linear as well as curvilinear portions of the building all complete as per the architectural drawings with 18 mm thick stone slab over 20 mm (average) thick base of cement mortar 1:4 (1 cement : 4 coarse sand) laid and jointed with cement slurry and pointing with white cement slurry admixed with pigment of matching shade including rubbing , curing and polishing etc. all complete as specified and as directed by the Engineer-in-Charge. Polished Granite stone slab colour of Black, Cherry/Ruby Red or equivalent |           |          |          |           |             |             |                 |                    |                 |                |
| Chapter Name: FINISHING |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |           |          |          |           |             |             |                 |                    |                 |                |
| 55                      | Individual                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 13.1.1    | 1355     | Sqm      | 294.85    | 399522      | 10.91 Below | 355934          |                    |                 |                |
|                         | Description: 12 mm cement plaster of mix : 1:4 (1 cement: 4 fine sand)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |           |          |          |           |             |             |                 |                    |                 |                |
| 56                      | Individual                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 13.2.1    | 1370     | Sqm      | 339.7     | 465389      | 10.91 Below | 414615          |                    |                 |                |
|                         | Description: 15 mm cement plaster on the rough side of single or half brick wall of mix : 1:4 (1 cement: 4 fine sand)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |           |          |          |           |             |             |                 |                    |                 |                |
| 57                      | Individual                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 13.3.1    | 100      | Sqm      | 402.15    | 40215       | 10.91 Below | 35828           |                    |                 |                |
|                         | Description: 20 mm cement plaster of mix : 1:4 (1 cement: 4 fine sand)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |           |          |          |           |             |             |                 |                    |                 |                |
| 58                      | Individual                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 13.26     | 7585     | Sqm      | 214.3     | 1625466     | 10.91 Below | 1448127         |                    |                 |                |
|                         | Description: Providing and applying plaster of paris putty of 2 mm thickness over plastered surface to prepare the surface even and smooth complete.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |           |          |          |           |             |             |                 |                    |                 |                |
| 59                      | Individual                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 13.39.1   | 8660     | Sqm      | 44.25     | 383205      | 10.91 Below | 341397          |                    |                 |                |
|                         | Description: New work (two or more coats) with a base coat of white washing with lime                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |           |          |          |           |             |             |                 |                    |                 |                |
| 60                      | Individual                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 13.41.1   | 7585     | Sqm      | 162.55    | 1232942     | 10.91 Below | 1098428         |                    |                 |                |
|                         | Description: New work (two or more coats) over and including water thinnable priming coat with cement primer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |           |          |          |           |             |             |                 |                    |                 |                |



| SN                             | Item Type                                                                                                                                                                                                                                                                                                                                                                                                            | Item Code | Item Qty | Qty Unit | Unit Rate | Basic Value | Escl. (%)          | Advt.Value (Rs) | Bid Rate/Unit Rate |  | Bid Amount (Rs) | Sch. Bid Total |
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| 61                             | Individual                                                                                                                                                                                                                                                                                                                                                                                                           | 13.48.2   | 700      | Sqm      | 144.9     | 101430      | 10.9<br>1<br>Below | 90364           |                    |  |                 |                |
|                                | Description: Finishing with Deluxe Multi surface paint system for interiors and exteriors using Primer as per manufacturers specifications : Painting wood work with Deluxe Multi Surface Paint of required shade. Two or more coat applied @ 0.90 ltr/10 sqm over an under coat of primer applied @ 0.75 ltr/10 sqm of approved brand and manufacture                                                               |           |          |          |           |             |                    |                 |                    |  |                 |                |
| 62                             | Individual                                                                                                                                                                                                                                                                                                                                                                                                           | 13.50.1   | 600      | Sqm      | 61.45     | 36870       | 10.9<br>1<br>Below | 32847           |                    |  |                 |                |
|                                | Description: Applying priming coat: With ready mixed pink or Grey primer of approved brand and manufacture on wood work (hard and soft wood)                                                                                                                                                                                                                                                                         |           |          |          |           |             |                    |                 |                    |  |                 |                |
| 63                             | Individual                                                                                                                                                                                                                                                                                                                                                                                                           | 13.61.1   | 300      | Sqm      | 131.45    | 39435       | 10.9<br>1<br>Below | 35133           |                    |  |                 |                |
|                                | Description: Two or more coats on new work                                                                                                                                                                                                                                                                                                                                                                           |           |          |          |           |             |                    |                 |                    |  |                 |                |
| <b>Chapter Name: ROAD WORK</b> |                                                                                                                                                                                                                                                                                                                                                                                                                      |           |          |          |           |             |                    |                 |                    |  |                 |                |
| 64                             | Individual                                                                                                                                                                                                                                                                                                                                                                                                           | 16.1      | 825      | Sqm      | 180.5     | 148913      | 10.9<br>1<br>Below | 132666          |                    |  |                 |                |
|                                | Description: Preparation and consolidation of sub grade with power road roller of 8 to 12 tonne capacity after excavating earth to an average of 22.5 cm depth, dressing to camber and consolidating with road roller including making good the undulations etc. and re-rolling the sub grade and disposal of surplus earthwith lead upto 50 metres.                                                                 |           |          |          |           |             |                    |                 |                    |  |                 |                |
| 65                             | Individual                                                                                                                                                                                                                                                                                                                                                                                                           | 16.3.2    | 85       | cum      | 1624.5    | 138083      | 10.9<br>1<br>Below | 123018          |                    |  |                 |                |
|                                | Description: Supplying and stacking at site. 63 mm to 45 mm size stone aggregate                                                                                                                                                                                                                                                                                                                                     |           |          |          |           |             |                    |                 |                    |  |                 |                |
| 66                             | Individual                                                                                                                                                                                                                                                                                                                                                                                                           | 16.3.3    | 65       | cum      | 1837.25   | 119421      | 10.9<br>1<br>Below | 106392          |                    |  |                 |                |
|                                | Description: Supplying and stacking at site. 53 mm to 22.4 mm size stone aggregate                                                                                                                                                                                                                                                                                                                                   |           |          |          |           |             |                    |                 |                    |  |                 |                |
| 67                             | Individual                                                                                                                                                                                                                                                                                                                                                                                                           | 16.3.10   | 475      | cum      | 888.3     | 421943      | 10.9<br>1<br>Below | 375909          |                    |  |                 |                |
|                                | Description: Supplying and stacking at site. Moorum                                                                                                                                                                                                                                                                                                                                                                  |           |          |          |           |             |                    |                 |                    |  |                 |                |
| 68                             | Individual                                                                                                                                                                                                                                                                                                                                                                                                           | 16.4      | 150      | cum      | 865.8     | 129870      | 10.9<br>1<br>Below | 115701          |                    |  |                 |                |
|                                | Description: Laying, spreading and compacting stone aggregate of specified sizes to WBM specifications in uniform thickness, hand picking, rolling with 3 wheeled road/vibratory roller 8-10 tonne capacity in stages to proper grade and camber, applying and brooming requisite type of screening / binding material to fill up interstices of coarse aggregate, watering and compacting to the required density . |           |          |          |           |             |                    |                 |                    |  |                 |                |
| 69                             | Individual                                                                                                                                                                                                                                                                                                                                                                                                           | 16.6.1    | 825      | Sqm      | 21.8      | 17985       | 10.9<br>1<br>Below | 16023           |                    |  |                 |                |
|                                | Description: Supplying, stacking and Spreading 6 mm thick red bajri, watering and rolling complete including preparation of the surface and rolling. With road roller/ hand roller                                                                                                                                                                                                                                   |           |          |          |           |             |                    |                 |                    |  |                 |                |
| 70                             | Individual                                                                                                                                                                                                                                                                                                                                                                                                           | 16.42     | 85       | cum      | 7399.2    | 628932      | 10.9<br>1<br>Below | 560316          |                    |  |                 |                |
|                                | Description: Cement concrete 1:2:4 (1 cement : 2 coarse sand : 4 graded stone aggregate 40 mm nominal size) in pavements, laid to required slope and camber in panels as required including consolidation finishing and tamping complete.                                                                                                                                                                            |           |          |          |           |             |                    |                 |                    |  |                 |                |

| SN                                   | Item Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Item Code | Item Qty | Qty Unit | Unit Rate | Basic Value | Escl. (%)   | Advt.Value (Rs) | Bid Rate/Unit Rate |  | Bid Amount (Rs) | Sch. Bid Total |
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| 71                                   | Individual                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 16.43.1   | 165      | cum      | 9492.9    | 1566329     | 10.91 Below | 1395442         |                    |  |                 |                |
|                                      | Description: Providing and laying design mix cement concrete of M-30 grade, in roads/ taxi tracks/ runways, using cement content as per design mix, using coarse sand and graded stone aggregate of 40 mm nominal size in appropriate proportions as per approved & specified design criteria, providing dowel bars with sleeve/ tie bars wherever required, laying at site, spreading and compacting mechanically by using needle and surface vibrators, levelling to required slope/ camber, finishing with required texture, including steel form work with sturdy M.S. channel sections, curing, making provision for contraction/ expansion, construction & longitudinal joints (10 mm wide x 50 mm deep) by groove cutting machine, providing and filling joints with approved joint filler and sealants, complete all as per direction of Engineer-in-charge (Item of joint fillers, sealants, dowel bars with sleeve/ tie bars to be paid separately). Cement concrete prepared with batch mixing machine |           |          |          |           |             |             |                 |                    |  |                 |                |
| 72                                   | Individual                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 16.69     | 15       | cum      | 8613.55   | 129203      | 10.91 Below | 115107          |                    |  |                 |                |
|                                      | Description: Providing and laying at or near ground level factory made kerb stone of M-25 grade cement concrete in position to the required line, level and curvature, jointed with cement mortar 1:3 (1 cement: 3 coarse sand), including making joints with or without grooves (thickness of joints except at sharp curve shall not to more than 5mm), including making drainage opening wherever required complete etc. as per direction of Engineer-in-charge (length of finished kerb edging shall be measured for payment). (Precast C.C. kerb stone shall be approved by Engineer-in-charge).                                                                                                                                                                                                                                                                                                                                                                                                              |           |          |          |           |             |             |                 |                    |  |                 |                |
| 73                                   | Individual                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 16.91.1   | 450      | Sqm      | 932.35    | 419558      | 10.91 Below | 373784          |                    |  |                 |                |
|                                      | Description: Providing and laying factory made chamfered edge Cement Concrete paver blocks in footpath, parks, lawns, drive ways or light traffic parking etc, of required strength, thickness & size/ shape, made by table vibratory method using PU mould, laid in required colour & pattern over 50mm thick compacted bed of sand, compacting and proper embedding/laying of inter locking paver blocks into the sand bedding layer through vibratory compaction by using plate vibrator, filling the joints with sand and cutting of paver blocks as per required size and pattern, finishing and sweeping extra sand. complete all as per direction of Engineer-in-Charge. 60mm thick cement concrete paver block of M-35 grade with approved colour, design & pattern.                                                                                                                                                                                                                                      |           |          |          |           |             |             |                 |                    |  |                 |                |
| Chapter Name: SANITARY INSTALLATIONS |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |           |          |          |           |             |             |                 |                    |  |                 |                |
| 74                                   | Individual                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 17.2.1    | 31       | Each     | 5540.55   | 171757      | 10.91 Below | 153018          |                    |  |                 |                |
|                                      | Description: Providing and fixing white vitreous china pedestal type water closet (European type W.C. pan) with seat and lid, 10 litre low level white P.V.C. flushing cistern, including flush pipe, with manually controlled device (handle lever), conforming to IS : 7231, with all fittings and fixtures complete, including cutting and making good the walls and floors wherever required : W.C. pan with ISI marked white solid plastic seat and lid                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |           |          |          |           |             |             |                 |                    |  |                 |                |
| 75                                   | Individual                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 17.7.2    | 5        | Each     | 1913.75   | 9569        | 10.91 Below | 8525            |                    |  |                 |                |
|                                      | Description: Providing and fixing wash basin with C.I. brackets, 15 mm C.P. brass pillar taps, 32 mm C.P. brass waste of standard pattern, including painting of fittings and brackets, cutting and making good the walls wherever require: White Vitreous China Wash basin size 630x450 mm with a single 15 mm C.P. brass pillar tap                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |           |          |          |           |             |             |                 |                    |  |                 |                |
| 76                                   | Individual                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 17.7.4    | 29       | Each     | 1679.6    | 48708       | 10.91 Below | 43394           |                    |  |                 |                |
|                                      | Description: Providing and fixing wash basin with C.I. brackets, 15 mm C.P. brass pillar taps, 32 mm C.P. brass waste of standard pattern, including painting of fittings and brackets, cutting and making good the walls wherever require: White Vitreous China Flat back wash basin size 550x 400 mm with single 15 mm C.P. brass pillar tap                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |           |          |          |           |             |             |                 |                    |  |                 |                |
| 77                                   | Individual                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 17.7B     | 10       | Each     | 1404      | 14040       | 10.91 Below | 12508           |                    |  |                 |                |

| SN                         | Item Type                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Item Code     | Item Qty | Qty Unit | Unit Rate   | Basic Value | Escl. (%)              | Advt.Value (Rs) | Bid Rate/Unit Rate | Bid Amount (Rs) | Sch. Bid Total |
|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|----------|----------|-------------|-------------|------------------------|-----------------|--------------------|-----------------|----------------|
|                            | Description: Providing and fixing wash basin with C.I. brackets, 15 mm PTMT pillar cock, 32 mm PTMT waste coupling of standard pattern, including painting of fittings and brackets, cutting and making good the walls wherever required. White Vitreous China Flat back wash basin size 550x400 mm with single 15 mm PTMT pillar cock.                                                                                                                       |               |          |          |             |             |                        |                 |                    |                 |                |
| 78                         | Indi<br>vidu<br>al                                                                                                                                                                                                                                                                                                                                                                                                                                            | 17.10.<br>1.1 | 28       | Each     | 6008.<br>45 | 168237      | 10.9<br>1<br>Belo<br>w | 149882          |                    |                 |                |
|                            | Description: Kitchen sink with drain board 510x1040 mm bowl depth 250 mm                                                                                                                                                                                                                                                                                                                                                                                      |               |          |          |             |             |                        |                 |                    |                 |                |
| 79                         | Indi<br>vidu<br>al                                                                                                                                                                                                                                                                                                                                                                                                                                            | 17.10.<br>2.2 | 32       | Each     | 3337.<br>85 | 106811      | 10.9<br>1<br>Belo<br>w | 95158           |                    |                 |                |
|                            | Description: Kitchen sink without drain board 610x460 mm bowl depth 200 mm                                                                                                                                                                                                                                                                                                                                                                                    |               |          |          |             |             |                        |                 |                    |                 |                |
| 80                         | Indi<br>vidu<br>al                                                                                                                                                                                                                                                                                                                                                                                                                                            | 17.28.<br>2.1 | 8        | Each     | 104.3<br>5  | 835         | 10.9<br>1<br>Belo<br>w | 744             |                    |                 |                |
|                            | Description: Flexible pipe 32 mm dia                                                                                                                                                                                                                                                                                                                                                                                                                          |               |          |          |             |             |                        |                 |                    |                 |                |
| 81                         | Indi<br>vidu<br>al                                                                                                                                                                                                                                                                                                                                                                                                                                            | 17.71         | 6        | Each     | 146.3       | 878         | 10.9<br>1<br>Belo<br>w | 782             |                    |                 |                |
|                            | Description: Providing and fixing PTMT liquid soap container 109 mm wide, 125 mm high and 112 mm distance from wall of standard shape with bracket of the same materials with snap fittings of approved quality and colour, weighing not less than 105 gms.                                                                                                                                                                                                   |               |          |          |             |             |                        |                 |                    |                 |                |
| 82                         | Indi<br>vidu<br>al                                                                                                                                                                                                                                                                                                                                                                                                                                            | 17.73.<br>2   | 6        | Each     | 600.3<br>5  | 3602        | 10.9<br>1<br>Belo<br>w | 3209            |                    |                 |                |
|                            | Description: Providing and fixing PTMT towel rail complete with brackets fixed to wooden cleats with CP brass screws with concealed fittings arrangement of approved quality and colour. 600 mm long towel rail with total length of 645 mm, width 78 mm and effective height of 88 mm, weighing not less than 190 gms.                                                                                                                                       |               |          |          |             |             |                        |                 |                    |                 |                |
| Chapter Name: WATER SUPPLY |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |               |          |          |             |             |                        |                 |                    |                 |                |
| 83                         | Indi<br>vidu<br>al                                                                                                                                                                                                                                                                                                                                                                                                                                            | 18.7.1        | 25       | Metre    | 255.9       | 6398        | 10.9<br>1<br>Belo<br>w | 5700            |                    |                 |                |
|                            | Description: Providing and fixing Chlorinated Polyvinyl Chloride (CPVC) pipes, having thermal stability for hot & cold water supply, including all CPVC plain & brass threaded fittings, including fixing the pipe with clamps at 1.00 m spacing. This includes jointing of pipes & fittings with one step CPVC solvent cement and testing of joints complete as per direction of Engineer in Charge.-Internal work - Exposed on wall 15 mm nominal dia Pipes |               |          |          |             |             |                        |                 |                    |                 |                |
| 84                         | Indi<br>vidu<br>al                                                                                                                                                                                                                                                                                                                                                                                                                                            | 18.7.2        | 25       | Metre    | 325.1       | 8128        | 10.9<br>1<br>Belo<br>w | 7241            |                    |                 |                |
|                            | Description: Providing and fixing Chlorinated Polyvinyl Chloride (CPVC) pipes, having thermal stability for hot & cold water supply, including all CPVC plain & brass threaded fittings, including fixing the pipe with clamps at 1.00 m spacing. This includes jointing of pipes & fittings with one step CPVC solvent cement and testing of joints complete as per direction of Engineer in Charge.-Internal work - Exposed on wall 20 mm nominal dia Pipes |               |          |          |             |             |                        |                 |                    |                 |                |
| 85                         | Indi<br>vidu<br>al                                                                                                                                                                                                                                                                                                                                                                                                                                            | 18.7.3        | 25       | Metre    | 408.5<br>5  | 10214       | 10.9<br>1<br>Belo<br>w | 9099            |                    |                 |                |
|                            | Description: Providing and fixing Chlorinated Polyvinyl Chloride (CPVC) pipes, having thermal stability for hot & cold water supply, including all CPVC plain & brass threaded fittings, including fixing the pipe with clamps at 1.00 m spacing. This includes jointing of pipes & fittings with one step CPVC solvent cement and testing of joints complete as per direction of Engineer in Charge.-Internal work - Exposed on wall 25 mm nominal dia Pipes |               |          |          |             |             |                        |                 |                    |                 |                |
| 86                         | Indi<br>vidu<br>al                                                                                                                                                                                                                                                                                                                                                                                                                                            | 18.7.5        | 15       | Metre    | 674.3<br>5  | 10115       | 10.9<br>1<br>Belo<br>w | 9012            |                    |                 |                |

| SN | Item Type  | Item Code                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Item Qty | Qty Unit | Unit Rate | Basic Value | Escl. (%)     | Advt.Value (Rs) | Bid Rate/Unit Rate | Bid Amount (Rs) | Sch. Bid Total |
|----|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|----------|-----------|-------------|---------------|-----------------|--------------------|-----------------|----------------|
|    |            | <b>Description:</b> Providing and fixing Chlorinated Polyvinyl Chloride (CPVC) pipes, having thermal stability for hot & cold water supply, including all CPVC plain & brass threaded fittings, including fixing the pipe with clamps at 1.00 m spacing. This includes jointing of pipes & fittings with one step CPVC solvent cement and testing of joints complete as per direction of Engineer in Charge.-Internal work - Exposed on wall<br>40 mm nominal dia Pipes                                                                                             |          |          |           |             |               |                 |                    |                 |                |
| 87 | Individual | 18.7.6                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 15       | Metre    | 927       | 13905       | 10.91 Below w | 12388           |                    |                 |                |
|    |            | <b>Description:</b> Providing and fixing Chlorinated Polyvinyl Chloride (CPVC) pipes, having thermal stability for hot & cold water supply, including all CPVC plain & brass threaded fittings, including fixing the pipe with clamps at 1.00 m spacing. This includes jointing of pipes & fittings with one step CPVC solvent cement and testing of joints complete as per direction of Engineer in Charge.-Internal work - Exposed on wall<br>50 mm nominal dia Pipes                                                                                             |          |          |           |             |               |                 |                    |                 |                |
|    | Individual | 18.8.1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 60       | Metre    | 441.15    | 26469       | 10.91 Below w | 23581           |                    |                 |                |
| 88 |            | <b>Description:</b> Providing and fixing Chlorinated Polyvinyl Chloride (CPVC) pipes, having thermal stability for hot & cold water supply, including all CPVC plain & brass threaded fittings, i/c fixing the pipe with clamps at 1.00 m spacing. This includes jointing of pipes & fittings with one step CPVC solvent cement and the cost of cutting chases and making good the same including testing of joints complete as per direction of Engineer in Charge, Concealed work, including cutting chases and making good the wall etc. 15 mm nominal dia Pipes |          |          |           |             |               |                 |                    |                 |                |
|    | Individual | 18.8.2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 60       | Metre    | 513.75    | 30825       | 10.91 Below w | 27462           |                    |                 |                |
| 89 |            | <b>Description:</b> Providing and fixing Chlorinated Polyvinyl Chloride (CPVC) pipes, having thermal stability for hot & cold water supply, including all CPVC plain & brass threaded fittings, i/c fixing the pipe with clamps at 1.00 m spacing. This includes jointing of pipes & fittings with one step CPVC solvent cement and the cost of cutting chases and making good the same including testing of joints complete as per direction of Engineer in Charge, Concealed work, including cutting chases and making good the wall etc. 20 mm nominal dia Pipes |          |          |           |             |               |                 |                    |                 |                |
|    | Individual | 18.8.3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 60       | Metre    | 626.05    | 37563       | 10.91 Below w | 33465           |                    |                 |                |
| 90 |            | <b>Description:</b> Providing and fixing Chlorinated Polyvinyl Chloride (CPVC) pipes, having thermal stability for hot & cold water supply, including all CPVC plain & brass threaded fittings, i/c fixing the pipe with clamps at 1.00 m spacing. This includes jointing of pipes & fittings with one step CPVC solvent cement and the cost of cutting chases and making good the same including testing of joints complete as per direction of Engineer in Charge, Concealed work, including cutting chases and making good the wall etc. 25 mm nominal dia Pipes |          |          |           |             |               |                 |                    |                 |                |
|    | Individual | 18.13.1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 29       | Each     | 757.1     | 21956       | 10.91 Below w | 19561           |                    |                 |                |
| 91 |            | <b>Description:</b> Making connection of G.I. distribution branch with G.I. main of following sizes by providing and fixing tee, including cutting and threading the pipe etc. complete : 25 to 40 mm nominal bore                                                                                                                                                                                                                                                                                                                                                  |          |          |           |             |               |                 |                    |                 |                |
|    | Individual | 18.13.2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 29       | Each     | 1513.7    | 43897       | 10.91 Below w | 39108           |                    |                 |                |
| 92 |            | <b>Description:</b> Making connection of G.I. distribution branch with G.I. main of following sizes by providing and fixing tee, including cutting and threading the pipe etc. complete : 50 to 80 mm nominal bore                                                                                                                                                                                                                                                                                                                                                  |          |          |           |             |               |                 |                    |                 |                |
|    | Individual | 18.17.1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 29       | Each     | 532.35    | 15438       | 10.91 Below w | 13754           |                    |                 |                |
| 93 |            | <b>Description:</b> Providing and fixing gun metal gate valve with C.I. wheel of approved quality (screwed end) : 25 mm nominal bore                                                                                                                                                                                                                                                                                                                                                                                                                                |          |          |           |             |               |                 |                    |                 |                |
|    | Individual | 18.17.3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 29       | Each     | 707.3     | 20512       | 10.91 Below w | 18274           |                    |                 |                |
| 94 |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |          |          |           |             |               |                 |                    |                 |                |

| SN  | Item Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Item Code | Item Qty | Qty Unit | Unit Rate | Basic Value | Escl. (%)   | Advt.Value (Rs) | Bid Rate/Unit Rate | Bid Amount (Rs) | Sch. Bid Total |
|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|----------|----------|-----------|-------------|-------------|-----------------|--------------------|-----------------|----------------|
|     | Description: Providing and fixing gun metal gate valve with C.I. wheel of approved quality (screwed end) : 40 mm nominal bore                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |           |          |          |           |             |             |                 |                    |                 |                |
| 95  | Individual                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 18.18.3   | 29       | Each     | 399.15    | 11575       | 10.91 Below | 10312           |                    |                 |                |
|     | Description: Providing and fixing ball valve (brass) of approved quality, High or low pressure, with plastic floats complete : 25 mm nominal bore                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |           |          |          |           |             |             |                 |                    |                 |                |
| 96  | Individual                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 18.19.3.1 | 28       | Each     | 815.05    | 22821       | 10.91 Below | 20332           |                    |                 |                |
|     | Description: 40 mm nominal bore Horizontal                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |           |          |          |           |             |             |                 |                    |                 |                |
| 97  | Individual                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 18.21.1.1 | 29       | Each     | 74.8      | 2169        | 10.91 Below | 1933            |                    |                 |                |
|     | Description: 30 cm length 15 mm nominal bore                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |           |          |          |           |             |             |                 |                    |                 |                |
| 98  | Individual                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 18.22.1   | 9        | Each     | 166.35    | 1497        | 10.91 Below | 1334            |                    |                 |                |
|     | Description: Providing and fixing C.P. brass shower rose with 15 or 20 mm inlet : 100 mm diameter                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |           |          |          |           |             |             |                 |                    |                 |                |
|     | Individual                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 18.34.1   | 65       | Each     | 17577.9   | 1142564     | 10.91 Below | 1017910         |                    |                 |                |
| 99  | Description: Constructing masonry Chamber 90x90x100 cm inside, in brick work in cement mortar 1:4 (1 cement : 4 coarse sand) for sluice valve, with C.I. surface box 100 mm top diameter, 160 mm bottom diameter and 180 mm deep (inside) with chained lid and RCC top slab 1:2:4 mix (1 cement : 2 coarse sand : 4 graded stone aggregate 20 mm nominal size ), i/c necessary excavation, foundation concrete 1:5:10 (1 cement : 5 fine sand : 10 graded stone aggregate 40 mm nominal size ) and inside plastering with cement mortar 1:3 (1 cement : 3 coarse sand) 12 mm thick, finished with a floating coat of neat cement complete as per standard design : With common burnt clay F.P.S.(non modular) bricks of class designation 7.5 |           |          |          |           |             |             |                 |                    |                 |                |
| 100 | Individual                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 18.48     | 28000    | Litre    | 9.7       | 271600      | 10.91 Below | 241968          |                    |                 |                |
|     | Description: Providing and placing on terrace (at all floor levels) polyethylene water storage tank, IS : 12701 marked, with cover and suitable locking arrangement and making necessary holes for inlet, outlet and overflow pipes but without fittings and the base support for tank.                                                                                                                                                                                                                                                                                                                                                                                                                                                       |           |          |          |           |             |             |                 |                    |                 |                |
| 101 | Individual                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 18.54.4   | 124      | Each     | 170.05    | 21086       | 10.91 Below | 18786           |                    |                 |                |
|     | Description: Providing and fixing PTMT bib cock of approved quality and colour. 15 mm nominal bore, 90 mm long, weighing not less than 93 gms                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |           |          |          |           |             |             |                 |                    |                 |                |
| 102 | Individual                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 18.55.1   | 76       | Each     | 103.15    | 7839        | 10.91 Below | 6984            |                    |                 |                |
|     | Description: Providing and fixing PTMT stop cock of approved quality and colour. 15 mm nominal bore, 86 mm long, weighing not less than 88 gms                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |           |          |          |           |             |             |                 |                    |                 |                |
| 103 | Individual                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 18.55.3   | 76       | Each     | 183.4     | 13938       | 10.91 Below | 12418           |                    |                 |                |
|     | Description: Providing and fixing PTMT stop cock of approved quality and colour. Concealed stop cock, 15 mm nominal bore, 108 mm long, weighing not less than 108 gms                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |           |          |          |           |             |             |                 |                    |                 |                |
| 104 | Individual                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 18.56.2   | 76       | Each     | 194.15    | 14755       | 10.91 Below | 13146           |                    |                 |                |

| SN                     | Item Type                                                                                                                                                                                                                         | Item Code | Item Qty | Qty Unit | Unit Rate | Basic Value | Escl. (%)   | Advt.Value (Rs) | Bid Rate/Unit Rate | Bid Amount (Rs) | Sch. Bid Total |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|----------|----------|-----------|-------------|-------------|-----------------|--------------------|-----------------|----------------|
|                        | Description: Providing and fixing PTMT pillar cock of approved quality and colour. 15 mm nominal bore, 125 mm long foam flow, weighing not less than 120 gms                                                                      |           |          |          |           |             |             |                 |                    |                 |                |
| 105                    | Individual                                                                                                                                                                                                                        | 18.58.1.2 | 76       | Each     | 45.25     | 3439        | 10.91 Below | 3064            |                    |                 |                |
|                        | Description: Circular type 125 mm nominal dia with 25 mm waste hole                                                                                                                                                               |           |          |          |           |             |             |                 |                    |                 |                |
| 106                    | Individual                                                                                                                                                                                                                        | 18.62.3   | 29       | Each     | 459.4     | 13323       | 10.91 Below | 11869           |                    |                 |                |
|                        | Description: Providing and fixing PTMT Ball cock of approved quality, colour and make complete with Epoxy coated aluminium rod with L.P./ H.P.H.D. plastic ball. 25 mm nominal bore, 152mm long, weighing not less than 440 gms   |           |          |          |           |             |             |                 |                    |                 |                |
| 107                    | Individual                                                                                                                                                                                                                        | 18.63     | 80       | Each     | 136.6     | 10928       | 10.91 Below | 9736            |                    |                 |                |
|                        | Description: Providing and fixing PTMT angle stop cock 15 mm nominal bore, weighing not less than 85 gms                                                                                                                          |           |          |          |           |             |             |                 |                    |                 |                |
| 108                    | Individual                                                                                                                                                                                                                        | 18.64     | 38       | Each     | 106.15    | 4034        | 10.91 Below | 3594            |                    |                 |                |
|                        | Description: Providing and fixing PTMT swivelling shower, 15 mm nominal bore, weighing not less than 40 gms                                                                                                                       |           |          |          |           |             |             |                 |                    |                 |                |
| 109                    | Individual                                                                                                                                                                                                                        | 18.74.1   | 38       | Each     | 85.5      | 3249        | 10.91 Below | 2895            |                    |                 |                |
|                        | Description: Providing and fixing unplasticised P.V.C. connection pipe with PTMT Nuts, collar and bush of approved quality and colour. 15 mm nominal bore with 30cm length                                                        |           |          |          |           |             |             |                 |                    |                 |                |
| 110                    | Individual                                                                                                                                                                                                                        | 18.75.1   | 38       | Each     | 43.05     | 1636        | 10.91 Below | 1457            |                    |                 |                |
|                        | Description: Providing and fixing PTMT extension nipple for water tank pipe, fittings of approved quality and colour. 15 mm nominal bore, weighing not less than 32 gms                                                           |           |          |          |           |             |             |                 |                    |                 |                |
| 111                    | Individual                                                                                                                                                                                                                        | 18.75.3   | 38       | Each     | 100.55    | 3821        | 10.91 Below | 3404            |                    |                 |                |
|                        | Description: Providing and fixing PTMT extension nipple for water tank pipe, fittings of approved quality and colour. 25mm nominal bore, weighing not less than 62 gms                                                            |           |          |          |           |             |             |                 |                    |                 |                |
| Chapter Name: DRAINAGE |                                                                                                                                                                                                                                   |           |          |          |           |             |             |                 |                    |                 |                |
| 112                    | Individual                                                                                                                                                                                                                        | 19.2.1    | 160      | Metre    | 895.5     | 143280      | 10.91 Below | 127648          |                    |                 |                |
|                        | Description: Providing and laying cement concrete 1:5:10 (1 cement : 5 coarse sand : 10 graded stone aggregate 40 mm nominal size) all-round S.W. pipes including bed concrete as per standard design : 100 mm diameter S.W. pipe |           |          |          |           |             |             |                 |                    |                 |                |
| 113                    | Individual                                                                                                                                                                                                                        | 19.2.2    | 450      | Metre    | 1095.15   | 492818      | 10.91 Below | 439051          |                    |                 |                |
|                        | Description: Providing and laying cement concrete 1:5:10 (1 cement : 5 coarse sand : 10 graded stone aggregate 40 mm nominal size) all-round S.W. pipes including bed concrete as per standard design : 150 mm diameter S.W. pipe |           |          |          |           |             |             |                 |                    |                 |                |
| 114                    | Individual                                                                                                                                                                                                                        | 19.4.1.1  | 29       | Each     | 2453.55   | 71153       | 10.91 Below | 63390           |                    |                 |                |

| SN                                                      | Item Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Item Code | Item Qty | Qty Unit      | Unit Rate | Basic Value | Escl. (%)                           | Advt.Value (Rs) | Bid Rate/Unit Rate | Bid Amount (Rs) | Sch. Bid Total |         |
|---------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|----------|---------------|-----------|-------------|-------------------------------------|-----------------|--------------------|-----------------|----------------|---------|
|                                                         | Description: 100x100 mm size P type With common burnt clay F.P.S. (non modular) bricks of class designation 7.5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |           |          |               |           |             |                                     |                 |                    |                 |                |         |
| 115                                                     | Indi vidu al                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 19.7.1.1  | 54       | Each          | 11687.1   | 631103      | 10.91 Below                         | 562250          |                    |                 |                |         |
|                                                         | Description: Inside size 90x80 cm and 45 cm deep including C.I. cover with frame (light duty) 455x610 mm internal dimensions, total weight of cover and frame to be not less than 38 kg (weight of cover 23 kg and weight of frame 15 kg) : With common burnt clay F.P.S. (non modular) bricks of class designation 7.5                                                                                                                                                                                                                                                                                 |           |          |               |           |             |                                     |                 |                    |                 |                |         |
| Chapter Name: ALUMINIUM WORK                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |           |          |               |           |             |                                     |                 |                    |                 |                |         |
| 116                                                     | Indi vidu al                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 21.1.2.1  | 450      | Kg            | 531.8     | 239310      | 10.91 Below                         | 213201          |                    |                 |                |         |
|                                                         | Description: For shutters of doors, windows & ventilators including providing and fixing hinges/ pivots and making provision for fixing of fittings wherever required including the cost of EPDM rubber / neoprene gasket required (Fittings shall be paid for separately) Anodised aluminium (anodised transparent or dyed to required shade according to IS: 1868, Minimum anodic coating of grade AC 15)                                                                                                                                                                                             |           |          |               |           |             |                                     |                 |                    |                 |                |         |
| 117                                                     | Indi vidu al                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 21.3.1    | 230      | Sqm           | 1019.8    | 234554      | 10.91 Below                         | 208964          |                    |                 |                |         |
|                                                         | Description: Providing and fixing glazing in aluminium door, window, ventilator shutters and partitions etc. with EPDM rubber / neoprene gasket etc. complete as per the architectural drawings and the directions of engineer-in-charge . (Cost of aluminium snap beading shall be paid in basic item): With float glass panes of 4.0 mm thickness (weight not less than 10kg/sqm)                                                                                                                                                                                                                     |           |          |               |           |             |                                     |                 |                    |                 |                |         |
| 118                                                     | Indi vidu al                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 21.4.1    | 4        | Each          | 2448.85   | 9795        | 10.91 Below                         | 8727            |                    |                 |                |         |
|                                                         | Description: Providing and fixing double action hydraulic floor spring of approved brand and manufacture conforming to IS : 6315, having brand logo embossed on the body / plate with double spring mechanism and door weight upto 125 kg, for doors, including cost of cutting floors, embedding in floors as required and making good the same matching to the existing floor finishing and cover plates with brass pivot and single piece M.S. sheet outer box with slide plate etc. complete as per the direction of Engineer-in-charge. With stainless steel cover plate minimum 1.25 mm thickness |           |          |               |           |             |                                     |                 |                    |                 |                |         |
| Sch. Code & Desc. : B Supply of Steel                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |           |          | Sch. Type:SOR |           |             | SOR Dir.: WC R-HQ-IR-USS OR 2021 -2 | 2494373         | 0.08               | % Above         | 2496368        | 2496368 |
| Chapter Name: CHAPTER - 2 : Bridge Works - Substructure |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |           |          |               |           |             |                                     |                 |                    |                 |                |         |
| 1                                                       | Indi vidu al                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 025082    | 29640    | Kg            | 118.78    | 3520639     | 29.15 Below                         | 2494373         |                    |                 |                |         |
|                                                         | Description: Supply of steel reinforcement of approved brands/makes for R.C.C. work including straightening, cutting, bending, placing in position and binding all complete. Thermo-Mechanically Treated bars of grade Fe-500D or more of approved brands/makes.                                                                                                                                                                                                                                                                                                                                        |           |          |               |           |             |                                     |                 |                    |                 |                |         |
| Sch. Code & Desc. : C Supply of Cement                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |           |          | Sch. Type:SOR |           |             | SOR Dir.: WC R-HQ-IR-USS OR 2021 -2 | 429984          | 0.08               | % Above         | 430328         | 430328  |
| Chapter Name: CHAPTER - 2 : Bridge Works - Substructure |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |           |          |               |           |             |                                     |                 |                    |                 |                |         |

| SN              | Item Type                                                                                                   | Item Code                     | Item Qty | Qty Unit      | Unit Rate  | Basic Value   | Escl. (%)               | Advt.Value (Rs)  | Bid Rate/Unit Rate |               | Bid Amount (Rs) | Sch. Bid Total |
|-----------------|-------------------------------------------------------------------------------------------------------------|-------------------------------|----------|---------------|------------|---------------|-------------------------|------------------|--------------------|---------------|-----------------|----------------|
| 1               | Individual                                                                                                  | 025072                        | 50       | MT            | 9075.23    | 453762        | 5.24 Below              | 429984           |                    |               |                 |                |
|                 | Description: Supply and using Cement at Worksite Ordinary Portland Cement 53 grade of approved brands/makes |                               |          |               |            |               |                         |                  |                    |               |                 |                |
|                 |                                                                                                             |                               |          | Advt. Total : |            | 28,013,418.64 |                         | Sch. Bid Total : |                    | 25,988,561.97 |                 |                |
| Bidding Style : |                                                                                                             | Single Rate for Each Schedule |          |               | Bid Rate : |               |                         | Bid Amount :     |                    | 25988562      |                 |                |
| Rebate :        |                                                                                                             |                               |          |               | 0.0        |               | Total Agreement Value : |                  |                    | 25988562      |                 |                |



2206900/2024/O/o CAO/C/WCR

WEST CENTRAL RAILWAY

Office of  
Chief Administrative Officer (Const.)  
second floor WCR, HQ  
Jabalpur-(M.P.)-482001

No. EW/HQ/Const./ B.G.

Date 20/11/2024.

Dy.CE/C/REWA

Sub. : Scrutiny of Bank Guarantee Bond No. 0282NDL900034625 Submitted by M/s  
**SURENDRA SINGH KALRA; BILASPUR (C.G.)**  
Ref. : Your letter no/ Computer no - 290739 - Dated 20/11/2024.

Xerox copy of extension deed of Bank Guarantee Bond/Bank Guarantee Bond  
No. 0282NDL900034625 Dated 28.10.24 executed on behalf of M/s **Surendra Singh Kalra**  
forwarded vide your letter of 11.11.24 has been examined. **BLs (C.G.)**

The deed having been drawn on stamp paper worth Rs. 3250/- and signed  
executed by the **ICICI Bank Ltd. Bilaspur** during the ordinary courses of its business  
and considered to be adequate, properly stamped, properly executed and the same may be accepted as  
in order, subject to the following:

1. Department concerned may confirm the genuineness of the above deed by tallying the xerox copy of the deed with the original deed in Railways custody.
2. Call for the confirmation advice from the issuing bank.
3. Check and confirm that the details such as contract agreement no., date etc. have been correctly indicated in the deed.
4. Check and confirm that the amount of the Bank guarantee and its validity as per the terms and conditions of the contract.
5. Guarantee bond hereby vetted subject to verification of terms of tender security deposit from tender committee proceedings by the acceptance authority personally because tender committee proceeding details and acceptance letter have not been sent to this office for verification while forwarding the Bank Guarantee Bond.
6. Payment of legal vetting charges by the firm.

It may be ensured that the Bank is advice in time about recovery, if any against the party within the time limit stipulated in the deed i.e. on or before 31.08.2025.

10.10.24  
20/11/24  
CLA/Const  
WCR, JBP/HQ

| Customer Care Centres | Phone Nos.    |
|-----------------------|---------------|
| Retail Customer care  | 1800 1080     |
| Wealth Management     | 1800 103 8181 |
| Business Banking      | 1860 120 6699 |
| i-Direct              | 1860 123 1122 |





18

**M/s. Surendra Singh Kalra**

Head Office :

A-507, Harsh Ashiyana, Green Park Colony,

Jarhabhata Road, Bilaspur (C.G.) 495001

Mob. : 9425220510

GSTIN : 23AAQFS2993A1Z4

**BRANCH OFFICE :**

A-904, Ojas Imperia

Bandariya Tiraha

Jabalpur (M.P.)

Pin- 482001

Mob. : 7999647629

Email ID : surendra.rlycontractor@gmail.com

Ref.No. ....

Date : 28 OCT 2024

सेवा में

उप मुख्य अभियन्ता

पश्चिम मध्य रेल्वे

रीवा (मध्य प्रदेश)

विषय : नई BG जमा करने बाबत ।

संदर्भ : Acceptance Letter No. DYCE-C-I-JBP-ENGINEERING/DYCEC1REWA-07-2024/00912200115255  
dated 17/10/2024.कार्य का नाम : Execution of balance works in newly constructed staff quarters at Rewa in connection  
with the Rewa-Sidhi new BG rail line project.

महोदय,

उपरोक्त कार्य के अंतर्गत अनुबंध कार्य वैल्यू Rs.2,59,88,561.97 का Bank Guarantee  
Rs.12,99,428.10/- की जो कि ICICI Bank Ltd., व्यापार विहार शाखा, बिलासपुर (छ.ग.) द्वारा  
जारी की गई है ।एमाउंट Rs.12,99,428.10 BG No.0282NDLG00034625 वैलिड upto 31/08/2025. हम इस पत्र के  
साथ सलग्न कर रहे हैं ।अतः आपसे विनम्र निवेदन है कि उपरोक्त कार्य का अनुबंध अतिशीघ्र बनाने के लिये संबंधित अधिकारियों  
को आदेश देने की कृपा करें ताकि हम अनुबंध दस्तखत कर सकें ।

यह पत्र सादर सूचनार्थ एवं आवश्यक कार्यवाही हेतु ।

आशा है आप पूर्ववत् सहयोग करेंगे ।

धन्यवाद

सलग्न:ओरिजिनल BG No. 0282NDLG00034625

Amount Rs.12,99,428.10 /-

प्रतिलिपि:

1) Dy.FA &amp; CAO(Con)

पश्चिम मध्य रेल्वे

जबलपुर (मध्य प्रदेश)

भवदीय

M/s. Surendra Singh Kalra

(मेसेंस सुरेन्द्र सिंह कालरा)

Received

29.10.2024  
SSE/DIC



Date: 28-10-2024  
Ref: 0282NDLG00034625

To,  
DY,FA AND CAO/C/WR/JABALPUR  
JABALPUR

Jabalpur  
MADHYA PRADESH  
INDIA  
482001

Sub: Issuance of Bank Guarantee

Dear Sir/Madam,

Please find enclosed Bank Guarantee issued by ICICI Bank Limited ("ICICI Bank") favoring yourself on behalf of: SURENDRA SINGH KALRA, A 507, HARSH AASHIYANA, GREEN PARK COLONY, ARIYA TIRAHA, BILASPUR, CHHATTISGARH, INDIA, 495001 ("Bank Guarantee") with the tender and claim period as requested by you. For ease of reference the details have been reproduced as below:

| Bank Guarantee No. & Date of Issue | Expiry Date | Claim Expiry Date | Currency | Amount of Bank Guarantee |
|------------------------------------|-------------|-------------------|----------|--------------------------|
| 0282NDLG00034625<br>28-10-2024     | 31-08-2025  | 31-08-2025        | INR      | 12,99,428.10             |

We confirm that the officials who have signed the above Bank Guarantee are authorized to sign such documents on behalf of ICICI Bank. You may verify genuineness of the Bank Guarantee by writing to us at [bgconfirmation@icicibank.com](mailto:bgconfirmation@icicibank.com) for receiving the confirmation over email.

Alternatively, you may also write to the following address to verify the genuineness of the BG:

ICICI Bank Limited, Trade Finance Operations Group,  
ICICI Bank Towers, Survey No.115/27, Tower 3, South Wing, 6th Floor,  
Plot No. 12, Nanakramguda, Serilingampally, Hyderabad - 500032, Telangana

In the event of invocation, we request you to please ensure compliance with the terms and conditions of the Bank Guarantee in order to ensure timely payment. You are requested to ensure special care inter alia with respect to the following in the invocation claim letter -

- Bank Guarantee Number
- Expiry/Claim Expiry date
- Claim Amount
- Designated Bank branch for submission of invocation claim
- Any declaration / certification that may be required as part of the guarantee text.
- Any other requisite document including the original Bank Guarantee.

Please note that ICICI Bank shall not be liable under the Bank Guarantee post expiry of the claim period as requested by you.

Thanking you,

Yours faithfully,

For ICICI Bank Limited  
Authorized Signatory



ICICI Bank Limited  
Upper Ground Floor,  
Aditya Tower,  
Near Shilpa Mandir,  
Gandhi Chowk, Main Road,  
Dayalbandh,  
Bilaspur - 495 001,  
Chhattisgarh, India.

Website [www.icicibank.com](http://www.icicibank.com)  
CIN L65190GJ1994PLC021012

Regd. Office : ICICI Bank Tower, Near Chokli Circle,  
Old Padra Road, Vadodara 390 007,  
India.  
Corp. Office : ICICI Bank Towers, Bandra-Kurla  
Complex, Mumbai 400051, India.

Sr.No. 1610385

**BANK GUARANTEE**  
**ICICI Bank Limited**  
(Incorporated in India)

BG Number: 0282NDLG00034625  
Issuance Date: October 28, 2024



1 Bank Guarantee Band for Performance Bank Guarantee

2 Name of Bank: ICICI Bank Limited

3 President of India

4 Acting Through

5 West Central Railway

6 Beneficiary: Dy.FA& CAO/C/WCR/Jabalpur

7 MADHYA PRADESH, 482001

8 Bank Guarantee Bond No.0282NDLG00034625

9 In consideration of the President of India, Dy.FA & CAO/C/WCR/Jabalpur (hereinafter called  
10 "the Government") having agreed to exempt M/s Surendra Singh Kalra, A-507, Harsh  
11 Ashlyana, Green Park colony, Jarhabhata Road, Bilaspur (C.G.), 495001, (hereinafter called "the  
12 said Contractor(s)" from the demand, under the terms and conditions of Agreement as  
13 Acceptance letter no. DYCE-C-I-JBP-ENGINEERING/DYCECI/REWA-82-2024/00912200115255  
14 dated 17/10/2024 made between Dy. Chief Engineer (Con-I), W.C.Rly, Rewa (M.P.) and M/s  
15 Surendra Singh Kalra, A-507, Harsh Ashlyana, Green Park colony, Jarhabhata  
16 Road, Bilaspur (C.G.), 495001, for Execution of balance works in newly constructed staff  
17 quarters at Rewa in connection with the Rewa-Sidhi new BG rail line project. (hereinafter  
18 called "the said Agreement"), of Security deposit for the due fulfillment by the said  
19 Contractor(s) of the terms and conditions contained in the said Agreement, on production of a  
20 bank Guarantee for Rs.12,99,428.10 (Rupees Twelve Lakh Ninety Nine Thousand Four  
21 Hundred Twenty Eight and Ten Paise only).

22 1. We, ICICI Bank Ltd, Surya Chambers, Plot No A/09, Vyapar Vihar, Bilaspur - 495001 having  
23 its registered office at ICICI Bank Tower, Near Chakli Circle, Old Padra Road, Vadodara,  
24 Gujarat, Pin - 390007 (hereinafter referred to as "the Bank" at the request of M/s Surendra  
25 Singh Kalra (contractor(s) do hereby undertake to pay to the Government an amount not  
26 exceeding Rs.12,99,428.10 (Rupees Twelve Lakh Ninety Nine Thousand Four Hundred Twenty  
27 Eight and Ten Paise only) against any loss or damage caused to or suffered or would be  
28 caused to or suffered by the Government by reason of any breach by the said Contractor(s) of  
29 any of the terms or conditions contained in the said Acceptance/Agreement.

30 2. We ICICI Bank Ltd, Surya Chambers, Plot No A/09, Vyapar Vihar, Bilaspur - 495001 do  
31 hereby undertake to pay the amounts due and payable under this guarantee without any  
32 demur, merely on a demand from the Government stating that the amount claimed is due by



The beneficiary may, in its own interest, verify the genuineness of the bank guarantee by seeking confirmation of its issuance by writing to the email ID [bgconfirmation@icicibank.com](mailto:bgconfirmation@icicibank.com) or to ICICI Bank Limited, Trade Finance Operations Group, ICICI Bank Towers, Survey No.115/27, Tower 3, South Wing, 8th Floor, Plot No. 12, Nanakramguda, Serilingampally, Hyderabad - 500032, Telangana.

Regd. Office: ICICI Bank Ltd., ICICI Bank Tower, Near Chakli Circle, Old Padra Road, Vadodara, Pin code- 390 007, Gujarat  
Phone : +91-265-6722286, CIN L65190GJ1994PLC021012

Sr.No. 1610387

**BANK GUARANTEE**  
**ICICI Bank Limited**  
(Incorporated in India)

BG Number: 0282NDLG00034625  
Issuance Date: October 28, 2024



33 way of loss or damage caused to or would be caused to or suffered by the Government by  
34 reason of breach by the said contractor(s) of any of the terms of conditions contained in the  
35 said Agreement or by reason of the contractor(s) failure to perform the said Agreement. Any  
36 such demand made on the bank shall be conclusive as regards the amount due and payable  
37 by the Bank under this guarantee. However, our liability under this guarantee shall be  
38 restricted to an amount not exceeding Rs.12,99,428.10(Rupees Twelve Lakh Ninety Nine  
39 Thousand Four Hundred Twenty Eight and Ten Paise only)

40 3.We under-take to pay to the Government any money so demanded notwithstanding any  
41 dispute or disputes raised by the contractor(s)/supplier (s) in any suite or proceeding pending  
42 before any court or Tribunal relating thereto our liability under this present being absolute and  
43 unequivocal.  
44 The payment so made by us under this bond shall be a valid discharge of our liability for  
45 payment there under and the contractor(s)/supplier(s) shall have no claim against us for  
46 making such payment.

47 4.We ICICI Bank Ltd, Surya Chambers, Plot NoA/09, Vyapar Vihar, Bilaspur - 495001 further  
48 agreed that the guarantee herein contained shall remain in full force and effect during the  
49 period that would be taken for the performance of the said Agreement and that it shall  
50 continue to be enforceable till all the dues of the Government under or by virtue of the said  
51 Agreement have been fully paid and its claims satisfied or discharged or till Dy. Chief Engineer  
52 (Can-I) ,West Central Railway Rewa(M.P.) (office/Department) Ministry of Railways certifies  
53 that the terms and conditions of the said Agreement have been fully and properly carried out  
54 by the said Contractor(s) and accordingly discharges this guarantee. Unless a demand or  
55 claim under this guarantee is made on us in writing on or before the 31-Aug-2025 we shall be  
56 discharged from all liability under this guarantee thereafter.

57 5.We ICICI Bank Ltd, Surya Chambers, Plot NoA/09, Vyapar Vihar, Bilaspur - 495001 further  
58 agree with the government that the Government shall have the fullest liberty without our  
59 consent and without affecting in any manner our obligations hereunder to vary any of the  
60 terms and conditions of the said Agreement or to extend time of performance by the said  
61 contractor(s) from time to time or to postpone for any time or from time to time any of the  
62 powers exercisable by the Government against the said, Contractor(s) and forbear or enforce  
63 any of the terms and conditions relating to the said agreement and we shall not be relieved  
64 from our liability by reason of any such variation, or extension being granted to the said  
65 Contractor(s) or for any forbearance, act or omission on the part of the Government or any  
66 indulgence by the Government to the said Contractor(s) or any such matter or thing  
67 whatsoever which under the law relating to sureties would, but for this provision, have effect



The beneficiary may, in its own interest, verify the genuineness of the bank guarantee by seeking confirmation of its issuance by writing to the email ID [bgconfirmation@icicibank.com](mailto:bgconfirmation@icicibank.com) or to ICICI Bank Limited, Trade Finance Operations Group, ICICI Bank Towers, Survey No.115/27, Tower 3, South Wing, 6th Floor, Plot No. 12, Nanakramguda, Serilingampally, Hyderabad - 500032, Telangana.

Regd. Office: ICICI Bank Ltd., ICICI Bank Tower, Near Chakli Circle, Old Padra Road, Vadodara, Pin code- 390 007, Gujarat  
Phone : +91-265-8722288, CIN L65190GJ1994PLC021012

Sr.No. 1610388

**BANK GUARANTEE**  
**ICICI Bank Limited**  
(Incorporated in India)

BG Number: 0282NDLG00034625  
Issuance Date: October 28, 2024



68 of so relieving us.

69 6.This guarantee will not be discharged due to the change in the constitution of the bank or  
70 the contractor(s)/Supplier(s).

71 7.We ICICI Bank Ltd, Surya Chambers, Plot NoA/09, Vyapar Vihar, Bilaspur - 495001 lastly  
72 undertake not to revoke this guarantee during its currency except with the previous consent of  
73 the Government in writing.

74 The liability of the Guarantor under this Guarantee shall not exceed INR 1299428.10 ( Twelve  
75 Lakh Ninety Nine Thousand Four Hundred Twenty Eight Rupee and Ten Paise Only) (the  
76 "Guaranteed Amounts").

77 This Guarantee shall be valid up to 31/08/2025 (the "Expiry Date").

78 Notwithstanding anything to the contrary contained herein, no obligation of the Guarantor to  
79 pay any amount under this Guarantee shall arise prior to the fulfillment of the following  
80 conditions precedent:

81 (a) written claim/demand(s) in terms of this Guarantee of an aggregate amount less than or  
82 equal to the Guaranteed Amounts is/are made by the Beneficiary hereunder; and

83 (b) such written claim/demand(s) is/are delivered to the Guarantor on or before the  
84 31/08/2025 at the ICICI Bank branch located at ICICI Bank Ltd, Surya Chambers, Plot NoA/09,  
85 Vyapar Vihar, Bilaspur - 495001.

86 Date:28/10/2024

87 Place:Bilaspur - Vyapar Vihar

88 FOR ICICI BANK LIMITED

89 Authorised Signatories

90 Signature: \_\_\_\_\_

Signature: \_\_\_\_\_

91 Name: Anjali V Singh

Name: \_\_\_\_\_

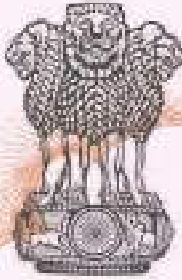
92 Signature Code: 418190

Signature Code: \_\_\_\_\_

The beneficiary may, in its own interest, verify the genuineness of the bank guarantee by seeking confirmation of its issuance by writing to the email ID [bgconfirmation@icicibank.com](mailto:bgconfirmation@icicibank.com) or to ICICI Bank Limited, Trade Finance Operations Group, ICICI Bank Towers, Survey No.115/27, Tower 3, South Wing, 8th Floor, Plot No. 12, Nanakramguda, Serilingampally, Hyderabad - 500032, Telangana.

Regd. Office: ICICI Bank Ltd., ICICI Bank Tower, Near Chakli Circle, Old Padra Road, Vadodara, Pin code- 390 007, Gujarat  
Phone : +91-265-6722286, CIN L65190GJ1994PLC021012





सत्यमेव जयते

INDIA NON JUDICIAL

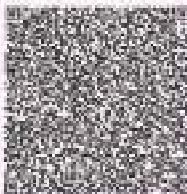
Government of Chhattisgarh

₹3,250

e-Stamp

Certificate No. : IN-CG60127552051236W  
Certificate Issued Date : 28-Oct-2024 03:59 PM  
Account Reference : SHCIL (FI)/ cgshcil01/ BILASPUR1/ CG-BP  
Unique Doc. Reference : SUBIN-CGCGSHCIL0102298141929988W  
Purchased by : BIMAL KUMAR  
Description of Document : Article 5 Agreement or Memorandum of an agreement  
Property Description : AGREEMENT  
Consideration Price (Rs.) : 0  
(Zero)  
First Party : DY FA AND CAO C WCR JABALPUR  
Second Party : ICICI BANK LTD  
Stamp Duty Paid By : ICICI BANK LTD  
Stamp Duty Amount(Rs.) : 3,250  
(Three Thousand Two Hundred And Fifty only)

सत्यमेव जयते



₹3,250



Please write or type below the line

IN-CG60127552051236W

This Stamp Paper is  
an integral part of

SG No.

0182NDL400034C25

Dated

28 Oct 2024



Chattisgarh



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Statutory Alert:

1. The authenticity of this Stamp certificate should be verified at 'www.echidnarec.com' or using e-Stamp Mobile App of Stock Holding.
2. Any discrepancy in the details on this Certificate and as available on the website / Mobile App renders it void.
3. The onus of checking the legitimacy is on the users of the certificate.
4. The Government is not responsible for the consequences of any misuse of this Stamp.



**DYCE-C-I-JBP-ENGINEERING/WEST CENTRAL RLY  
TENDER DOCUMENT**

**Tender No:** DYCEC1REWA-07-2024

**Closing Date/Time:** 12/08/2024 15:00

**Dy.CE/C-1/JBP** acting for and on behalf of The President of India invites E-Tenders against Tender No **DYCEC1REWA-07-2024** Closing Date/Time 12/08/2024 15:00 Hrs. Bidders will be able to submit their original/revised bids upto closing date and time only. Manual offers are not allowed against this tender, and any such manual offer received shall be ignored.

**1. NIT HEADER**

|                                    |                                                                                                                                    |                                                    |                      |
|------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------|----------------------|
| <b>Name of Work</b>                | Execution of balance works in newly constructed staff quarters at Rewa in connection with the Rewa-Sidhi new BG rail line project. |                                                    |                      |
| <b>Bidding type</b>                | Normal Tender                                                                                                                      |                                                    |                      |
| <b>Tender Type</b>                 | Open                                                                                                                               | <b>Bidding System</b>                              | Single Packet System |
| <b>Tender Closing Date Time</b>    | 12/08/2024 15:00                                                                                                                   | <b>Date Time Of Uploading Tender</b>               | 12/07/2024 19:51     |
| <b>Pre-Bid Conference Required</b> | No                                                                                                                                 | <b>Pre-Bid Conference Date Time</b>                | Not Applicable       |
| <b>Advertised Value</b>            | 28013418.64                                                                                                                        | <b>Tendering Section</b>                           | DRAWING              |
| <b>Bidding Style</b>               | Single Rate for Each Schedule                                                                                                      | <b>Bidding Unit</b>                                |                      |
| <b>Earnest Money (Rs.)</b>         | 290100.00                                                                                                                          | <b>Validity of Offer ( Days)</b>                   | 60                   |
| <b>Tender Doc. Cost (Rs.)</b>      | 0.00                                                                                                                               | <b>Period of Completion</b>                        | 6 Months             |
| <b>Contract Type</b>               | Works                                                                                                                              | <b>Contract Category</b>                           | Expenditure          |
| <b>Bidding Start Date</b>          | 29/07/2024                                                                                                                         | <b>Are Joint Venture (JV) firms allowed to bid</b> | No                   |
| <b>Ranking Order For Bids</b>      | Lowest to Highest                                                                                                                  | <b>Expenditure Type</b>                            | Capital (Works)      |

**2. SCHEDULE**

| S.No.                                              | Item Code                                                                                                                                                                                                                                                                                                                                                                                                              | Item Qty | Qty Unit | Unit Rate | Basic Value | Escl.(%)  | Amount      | Bidding Unit        |
|----------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|----------|-----------|-------------|-----------|-------------|---------------------|
| <b>Schedule A-Earthwork and concrete work etc.</b> |                                                                                                                                                                                                                                                                                                                                                                                                                        |          |          |           |             |           | 25089061.37 | Above/<br>Below/Par |
| 1                                                  | 2.8.1                                                                                                                                                                                                                                                                                                                                                                                                                  | 1150.00  | cum      | 286.85    | 329877.50   | (-) 10.91 | 293887.86   |                     |
|                                                    | <b>Description:- EARTH WORK</b> // Earth work in excavation by mechanical means (Hydraulic excavator) / manual means in foundation trenches or drains (not exceeding 1.5 m in width or 10 sqm on plan), including dressing of sides and ramming of bottoms, lift upto 1.5 m, including getting out the excavated soil and disposal of surplus excavated soil as directed, within a lead of 50 m. // All kinds of soil. |          |          |           |             |           |             |                     |
| 2                                                  | 2.25                                                                                                                                                                                                                                                                                                                                                                                                                   | 225.00   | cum      | 253.95    | 57138.75    | (-) 10.91 | 50904.91    |                     |
|                                                    | <b>Description:- EARTH WORK</b> // Filling available excavated earth (excluding rock) in trenches, plinth, sides of foundations etc. in layers not exceeding 20cm in depth, consolidating each deposited layer by ramming and watering, lead up to 50 m and lift upto 1.5 m.                                                                                                                                           |          |          |           |             |           |             |                     |
| 3                                                  | 2.25A                                                                                                                                                                                                                                                                                                                                                                                                                  | 450.00   | cum      | 368.65    | 165892.50   | (-) 10.91 | 147793.63   |                     |
|                                                    | <b>Description:- EARTH WORK</b> // Excavating, supplying and filling of local earth (including royalty) by mechanical transport upto a lead of 5km also including ramming and watering of the earth in layers not exceeding 20 cm in trenches, plinth, sides of foundation etc. complete.                                                                                                                              |          |          |           |             |           |             |                     |
| 4                                                  | 2.26.1                                                                                                                                                                                                                                                                                                                                                                                                                 | 410.50   | cum      | 104.50    | 42897.25    | (-) 10.91 | 38217.16    |                     |
|                                                    | <b>Description:- EARTH WORK</b> // Extra for every additional lift of 1.5 m or part thereof in excavation / banking excavated or stacked materials. // All kinds of soil                                                                                                                                                                                                                                               |          |          |           |             |           |             |                     |
| 5                                                  | 2.27                                                                                                                                                                                                                                                                                                                                                                                                                   | 40.50    | cum      | 2161.20   | 87528.60    | (-) 10.91 | 77979.23    |                     |
|                                                    | <b>Description:- EARTH WORK</b> // Supplying and filling in plinth with sand under floors, including watering, ramming, consolidating and dressing complete.                                                                                                                                                                                                                                                           |          |          |           |             |           |             |                     |
| 6                                                  | 2.28.1                                                                                                                                                                                                                                                                                                                                                                                                                 | 60.00    | Sqm      | 28.15     | 1689.00     | (-) 10.91 | 1504.73     |                     |
|                                                    | <b>Description:- EARTH WORK</b> // Surface dressing of the ground including removing vegetation and in-equalities not exceeding 15 cm deep and disposal of rubbish, lead up to 50 m and lift up to 1.5 m. // All kinds of soil                                                                                                                                                                                         |          |          |           |             |           |             |                     |
| 7                                                  | 2.34.1                                                                                                                                                                                                                                                                                                                                                                                                                 | 190.00   | Litre    | 200.90    | 38171.00    | (-) 10.91 | 34006.54    |                     |
|                                                    | <b>Description:- EARTH WORK</b> // Supplying chemical emulsion in sealed containers including delivery as specified. // Chlorpyriphos emulsifiable concentrate of 20%                                                                                                                                                                                                                                                  |          |          |           |             |           |             |                     |
|                                                    | 2.35.1.1                                                                                                                                                                                                                                                                                                                                                                                                               | 190.00   | Metre    | 32.30     | 6137.00     | (-) 10.91 | 5467.45     |                     |

**DYCE-C-I-JBP-ENGINEERING/WEST CENTRAL RLY  
TENDER DOCUMENT**

**Tender No:** DYCEC1REWA-07-2024

**Closing Date/Time:** 12/08/2024 15:00

|    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |         |     |          |            |           |            |
|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|-----|----------|------------|-----------|------------|
| 8  | <b>Description:- EARTH WORK</b> // Diluting and injecting chemical emulsion for POST-CONSTRUCTIONAL anti-termite treatment (excluding the cost of chemical emulsion) : // Along external wall where the apron is not provided using chemical emulsion @ 7.5 litres / sqm of the vertical surface of the substructure to a depth of 300mm including excavation channel along the wall & rodding etc. complete: // With Chlorpyrifos E.C. 20% with 1% concentration                                                                                                                                                                  |         |     |          |            |           |            |
| 9  | 4.1.3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 90.00   | cum | 7365.15  | 662863.50  | (-) 10.91 | 590545.09  |
|    | <b>Description:- CONCRETE WORK</b> // CEMENT CONCRETE (CAST IN SITU) // Providing and laying in position cement concrete of specified grade excluding the cost of centering and shuttering - All work up to plinth level : // 1:2:4 (1 cement : 2 coarse sand (zone-III) derived from natural sources : 4 graded stone aggregate 20 mm nominal size derived from natural sources)                                                                                                                                                                                                                                                  |         |     |          |            |           |            |
| 10 | 4.1.5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 75.00   | cum | 6833.40  | 512505.00  | (-) 10.91 | 456590.70  |
|    | <b>Description:- CONCRETE WORK</b> // CEMENT CONCRETE (CAST IN SITU) // Providing and laying in position cement concrete of specified grade excluding the cost of centering and shuttering - All work up to plinth level : // 1:3:6 (1 Cement : 3 coarse sand (zone-III) derived from natural sources : 6 graded stone aggregate 20 mm nominal size derived from natural sources)                                                                                                                                                                                                                                                  |         |     |          |            |           |            |
| 11 | 4.1.6                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 80.00   | cum | 6670.25  | 533620.00  | (-) 10.91 | 475402.06  |
|    | <b>Description:- CONCRETE WORK</b> // CEMENT CONCRETE (CAST IN SITU) // Providing and laying in position cement concrete of specified grade excluding the cost of centering and shuttering - All work up to plinth level : // 1:3:6 (1 Cement : 3 coarse sand (zone-III) derived from natural sources: 6 graded stone aggregate 40 mm nominal size derived from natural sources)                                                                                                                                                                                                                                                   |         |     |          |            |           |            |
| 12 | 4.2.3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 250.00  | cum | 9375.20  | 2343800.00 | (-) 10.91 | 2088091.42 |
|    | <b>Description:- CONCRETE WORK</b> // CEMENT CONCRETE (CAST IN SITU) // Providing and laying cement concrete in retaining walls, return walls, walls (any thickness) including attached pilasters, columns, piers, abutments, pillars, posts, struts, buttresses, string or lacing courses, parapets, coping, bed blocks, anchor blocks, plain window sills, fillets, sunken floor etc., up to floor five level, excluding the cost of centering, shuttering and finishing: // 1:2:4 (1 Cement : 2 coarse sand (zone-III) derived from natural sources : 4 graded stone aggregate 20 mm nominal size derived from natural sources) |         |     |          |            |           |            |
| 13 | 4.3.1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 200.00  | Sqm | 307.95   | 61590.00   | (-) 10.91 | 54870.53   |
|    | <b>Description:- CONCRETE WORK</b> // CEMENT CONCRETE (CAST IN SITU) // Centering and shuttering including strutting, propping etc. and removal of form work for : // Foundations, footings, bases for columns                                                                                                                                                                                                                                                                                                                                                                                                                     |         |     |          |            |           |            |
| 14 | 4.3.2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 1800.00 | Sqm | 669.55   | 1205190.00 | (-) 10.91 | 1073703.77 |
|    | <b>Description:- CONCRETE WORK</b> // CEMENT CONCRETE (CAST IN SITU) // Centering and shuttering including strutting, propping etc. and removal of form work for : // Retaining walls, return walls, walls (any thickness) including attached pilasters, buttresses, plinth and string courses fillets, kerbs and steps etc.                                                                                                                                                                                                                                                                                                       |         |     |          |            |           |            |
| 15 | 4.4.1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 6.00    | cum | 7783.65  | 46701.90   | (-) 10.91 | 41606.72   |
|    | <b>Description:- CONCRETE WORK</b> // CEMENT CONCRETE (CAST IN SITU) // Providing and laying cement concrete in kerbs, steps and the like at or near ground level excluding the cost of centering, shuttering and finishing. // 1:1.5:3 (1 Cement: 1.5 coarse sand(zone-III) derived from natural sources : 3 graded stone aggregate 20 mm nominal size derived from natural sources)                                                                                                                                                                                                                                              |         |     |          |            |           |            |
| 16 | 4.5.1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 6.00    | cum | 9277.75  | 55666.50   | (-) 10.91 | 49593.28   |
|    | <b>Description:- CONCRETE WORK</b> // CEMENT CONCRETE PRECAST // Providing and fixing up to floor five level precast cement concrete string or lacing courses, copings, bed plates, anchor blocks, plain window sills, shelves, louvers, steps, stair cases, etc., including hoisting and setting in position with cement mortar 1:3 (1 Cement : 3 coarse sand), cost of required Centering complete. // 1:1.5:3 (1 cement : 1.5 coarse sand(zone-III) derived from natural sources: 3 graded stone aggregate 20mm nominal size derived from natural sources)                                                                      |         |     |          |            |           |            |
| 17 | 4.11                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 7.50    | Sqm | 452.25   | 3391.88    | (-) 10.91 | 3021.83    |
|    | <b>Description:- CONCRETE WORK</b> // DAMP-PROOF COURSE // Providing and laying damp-proof course 50mm thick with cement concrete 1:2:4 (1 cement : 2 coarse sand(zone-III) derived from natural sources : 4 graded stone aggregate 20mm nominal size derived from natural sources).                                                                                                                                                                                                                                                                                                                                               |         |     |          |            |           |            |
| 18 | 5.3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 14.00   | cum | 10719.30 | 150070.20  | (-) 10.91 | 133697.54  |
|    | <b>Description:- REINFORCED CEMENT CONCRETE</b> // CAST IN SITU // Reinforced cement concrete work in beams, suspended floors, roofs having slope up to 15 degree landings, balconies, shelves, chajjas, lintels, bands, plain window sills, staircases and spiral stair cases above plinth level up to floor five level, excluding the cost of centering, shuttering, finishing and reinforcement with 1:1.5:3 (1 cement : 1.5 coarse sand(zone-III) derived from natural sources : 3 graded stone aggregate 20 mm nominal size derived from natural sources).                                                                    |         |     |          |            |           |            |
| 19 | 5.5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 4.00    | cum | 11291.20 | 45164.80   | (-) 10.91 | 40237.32   |
|    | <b>Description:- REINFORCED CEMENT CONCRETE</b> // CAST IN SITU // Reinforced cement concrete work in arches, archribs, domes, vaults, shells, folded plate and roofs having slope more than 15 degree up to floor five level, excluding the cost of centering, shuttering, finishing and reinforcement with 1:1.5:3 (1 cement : 1.5 coarse sand (zone-III) derived from natural sources: 3 graded stone aggregate 20 mm nominal size derived from natural sources).                                                                                                                                                               |         |     |          |            |           |            |

**DYCE-C-I-JBP-ENGINEERING/WEST CENTRAL RLY  
TENDER DOCUMENT**

**Tender No:** DYCEC1REWA-07-2024

**Closing Date/Time:** 12/08/2024 15:00

|    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |        |       |          |            |           |            |  |
|----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|-------|----------|------------|-----------|------------|--|
| 20 | 5.33.1.1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 165.00 | cum   | 8683.80  | 1432827.00 | (-) 10.91 | 1276505.57 |  |
|    | <b>Description:- REINFORCED CEMENT CONCRETE</b> // DESIGN MIX CONCRETE // Providing and laying in position ready mixed or site batched design mix cement concrete for reinforced cement concrete work; using coarse aggregate and fine aggregate derived from natural sources, Portland Pozzolana / Ordinary Portland /Portland Slag cement, admixtures in recommended proportions as per IS: 9103 to accelerate / retard setting of concrete, to improve durability and workability without impairing strength; including pumping of concrete to site of laying, curing, carriage for all leads; but excluding the cost of centering, shuttering, finishing and reinforcement as per direction of the engineer-in-charge; for the following grades of concrete. Note: Extra cement up to 10% of the minimum specified cement content in design mix shall be payable separately. In case the cement content in design mix is more than 1.10 times of the specified minimum cement content, the contractor shall have discretion to either re-design the mix or bear the cost of extra cement. // All works upto plinth level // Concrete of M25 grade with minimum cement content of 330 kg /cum                     |        |       |          |            |           |            |  |
| 21 | 5.33.2.1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 150.00 | cum   | 10306.20 | 1545930.00 | (-) 10.91 | 1377269.04 |  |
|    | <b>Description:- REINFORCED CEMENT CONCRETE</b> // DESIGN MIX CONCRETE // Providing and laying in position ready mixed or site batched design mix cement concrete for reinforced cement concrete work; using coarse aggregate and fine aggregate derived from natural sources, Portland Pozzolana / Ordinary Portland /Portland Slag cement, admixtures in recommended proportions as per IS: 9103 to accelerate / retard setting of concrete, to improve durability and workability without impairing strength; including pumping of concrete to site of laying, curing, carriage for all leads; but excluding the cost of centering, shuttering, finishing and reinforcement as per direction of the engineer-in-charge; for the following grades of concrete. Note: Extra cement up to 10% of the minimum specified cement content in design mix shall be payable separately. In case the cement content in design mix is more than 1.10 times of the specified minimum cement content, the contractor shall have discretion to either re-design the mix or bear the cost of extra cement. // All works above plinth level upto floor V level // Concrete of M25 grade with minimum cement content of 330 kg /cum |        |       |          |            |           |            |  |
| 22 | 5.9.1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 245.00 | Sqm   | 307.95   | 75447.75   | (-) 10.91 | 67216.40   |  |
|    | <b>Description:- REINFORCED CEMENT CONCRETE</b> // FORM WORK // Centering and shuttering including strutting, propping etc. and removal of form for // Foundations, footings, bases of columns, etc. for mass concrete                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |        |       |          |            |           |            |  |
| 23 | 5.9.3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 356.00 | Sqm   | 766.55   | 272891.80  | (-) 10.91 | 243119.30  |  |
|    | <b>Description:- REINFORCED CEMENT CONCRETE</b> // FORM WORK // Centering and shuttering including strutting, propping etc. and removal of form for // Suspended floors, roofs, landings, balconies and access platform                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |        |       |          |            |           |            |  |
| 24 | 5.9.5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 627.00 | Sqm   | 608.35   | 381435.45  | (-) 10.91 | 339820.84  |  |
|    | <b>Description:- REINFORCED CEMENT CONCRETE</b> // FORM WORK // Centering and shuttering including strutting, propping etc. and removal of form for // Lintels, beams, plinth beams, girders, bressumers and cantilevers                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |        |       |          |            |           |            |  |
| 25 | 5.9.15                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 110.00 | Sqm   | 307.95   | 33874.50   | (-) 10.91 | 30178.79   |  |
|    | <b>Description:- REINFORCED CEMENT CONCRETE</b> // FORM WORK // Centering and shuttering including strutting, propping etc. and removal of form for // Small lintels not exceeding 1.5 m clear span, moulding as in cornices, window sills, string courses, bands, copings, bed plates, anchor blocks and the like                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |        |       |          |            |           |            |  |
| 26 | 5.9.16.1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 150.00 | Metre | 181.90   | 27285.00   | (-) 10.91 | 24308.21   |  |
|    | <b>Description:- REINFORCED CEMENT CONCRETE</b> // FORM WORK // Edges of slabs and breaks in floors and walls // Under 20 cm wide                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |        |       |          |            |           |            |  |
| 27 | 6.1.1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 55.00  | cum   | 6882.00  | 378510.00  | (-) 10.91 | 337214.56  |  |
|    | <b>Description:- MASONRY WORK</b> // Brick work with common burnt clay F.P.S. (non modular) bricks of class designation 7.5 in foundation and plinth in: // Cement mortar 1:4 (1 cement : 4 coarse sand)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |        |       |          |            |           |            |  |
| 28 | 6.13.2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 15.00  | Sqm   | 1018.05  | 15270.75   | (-) 10.91 | 13604.71   |  |
|    | <b>Description:- MASONRY WORK</b> // HALF BRICK MASONRY // Half brick masonry with common burnt clay F.P.S. (non modular) bricks of class designation 7.5 in superstructure above plinth level up to floor V level. // Cement mortar 1:4 (1 cement :4 coarse sand)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |        |       |          |            |           |            |  |
| 29 | 6.15                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 11.00  | Sqm   | 86.45    | 950.95     | (-) 10.91 | 847.20     |  |
|    | <b>Description:- MASONRY WORK</b> // HALF BRICK MASONRY // Extra for providing and placing in position 2 Nos 6mm dia. M.S. bars at every third course of half brick masonry.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |        |       |          |            |           |            |  |
| 30 | 6.3.1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 269.00 | cum   | 8030.60  | 2160231.40 | (-) 10.91 | 1924550.15 |  |
|    | <b>Description:- MASONRY WORK</b> // Brick work with common burnt clay machine moulded perforated bricks of class designation 12.5 conforming to IS: 2222 in superstructure above plinth level up to floor five level in cement mortar 1:6 (1 cement : 6 coarse sand) : // With F.P.S.(non modular) bricks                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |        |       |          |            |           |            |  |
|    | 8.3.2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 12.50  | Metre | 418.85   | 5235.63    | (-) 10.91 | 4664.42    |  |
|    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |        |       |          |            |           |            |  |

**DYCE-C-I-JBP-ENGINEERING/WEST CENTRAL RLY  
TENDER DOCUMENT**

**Tender No:** DYCEC1REWA-07-2024

**Closing Date/Time:** 12/08/2024 15:00

|    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |         |      |         |           |           |           |
|----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|------|---------|-----------|-----------|-----------|
| 31 | <b>Description:- CLADDING WORK</b> // Providing and fixing 18 mm thick gang saw cut, mirror polished, premoulded and prepolished, machine cut for kitchen platforms, vanity counters, window sills, facias and similar locations of required size, approved shade, colour and texture laid over 20 mm thick base cement mortar 1:4 (1 cement : 4 coarse sand), joints treated with white cement, mixed with matching pigment, epoxy touch ups, including rubbing, curing, moulding and polishing of edges to give high gloss finish etc. complete at all levels. // Providing edge moulding to 18 mm thick marble stone counters, Vanities etc., including machine polishing to edge to give high gloss finish etc. complete as per design approved by Engineer-in-Charge. // Granite work |         |      |         |           |           |           |
|    | 9.7.7.2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 157.00  | Sqm  | 2231.20 | 350298.40 | (-) 10.91 | 312080.84 |
| 32 | <b>Description:- WOOD AND P.V.C. WORK</b> // Providing and fixing panelled or panelled and glazed shutters for doors, windows and clerestory windows, fixing with butt hinges of required size with necessary screws, excluding panelling which will be paid for separately, all complete as per direction of Engineer-in-charge. (Note:- Butt hinges and necessary screws shall be paid separately) // Float glass panes // 5.0 mm thick glass panes (weight not less than 12.50 kg/sqm).                                                                                                                                                                                                                                                                                                 |         |      |         |           |           |           |
|    | 9.20.1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 79.00   | Sqm  | 3086.10 | 243801.90 | (-) 10.91 | 217203.11 |
| 33 | <b>Description:- WOOD AND P.V.C. WORK</b> // Providing and fixing 25 mm thick shutters for cup board etc. : // Providing and fixing ISI marked flush door shutters conforming to IS : 2202 (Part I) decorative type, core of block board construction with frame of 1st class hard wood and well matched teak 3 ply veneering with vertical grains or cross bands and face veneers on both faces of shutters. // 35 mm thick including ISI marked Stainless Steel butt hinges with necessary screws                                                                                                                                                                                                                                                                                        |         |      |         |           |           |           |
|    | 9.20.3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 130.00  | Sqm  | 2512.35 | 326605.50 | (-) 10.91 | 290972.84 |
| 34 | <b>Description:- WOOD AND P.V.C. WORK</b> // Providing and fixing 25 mm thick shutters for cup board etc. : // Providing and fixing ISI marked flush door shutters conforming to IS : 2202 (Part I) decorative type, core of block board construction with frame of 1st class hard wood and well matched teak 3 ply veneering with vertical grains or cross bands and face veneers on both faces of shutters. // 25 mm thick (for cupboard) including ISI marked nickel plated bright finished M.S. Piano hinges IS : 3818 marked with necessary screws                                                                                                                                                                                                                                    |         |      |         |           |           |           |
|    | 9.21.2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 36.00   | Sqm  | 1881.95 | 67750.20  | (-) 10.91 | 60358.65  |
| 35 | <b>Description:- WOOD AND P.V.C. WORK</b> // Providing and fixing 25 mm thick shutters for cup board etc. : // Providing and fixing ISI marked flush door shutters conforming to IS : 2202 (Part I) non-decorative type, core of block board construction with frame of 1st class hard wood and well matched commercial 3 ply veneering with vertical grains or cross bands and face veneers on both faces of shutters: // 30 mm thick including ISI marked Stainless Steel butt hinges with necessary screws                                                                                                                                                                                                                                                                              |         |      |         |           |           |           |
|    | 9.27.2.1.1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 114.00  | Sqm  | 3863.65 | 440456.10 | (-) 10.91 | 392402.34 |
| 36 | <b>Description:- WOOD AND P.V.C. WORK</b> // Providing and fixing wire gauge shutters using galvanized M.S. wire gauge of average width of aperture 1.4 mm in both directions with wire of dia 0.63 mm, for doors, windows and clerestory windows with hinges and necessary screws : // 30 mm thick shutters // with ISI marked M.S. pressed butt hinges bright finished of required size // Second class teak wood                                                                                                                                                                                                                                                                                                                                                                        |         |      |         |           |           |           |
|    | 9.48.1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 1500.00 | Kg   | 181.00  | 271500.00 | (-) 10.91 | 241879.35 |
| 37 | <b>Description:- WOOD AND P.V.C. WORK</b> // MISCELLANEOUS-WOOD WORK // Providing and fixing M.S. grills of required pattern in frames of windows etc. with M.S. flats, square or round bars etc. including priming coat with approved steel primer all complete. // Fixed to steel windows by welding                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |         |      |         |           |           |           |
|    | 9.62.1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 62.00   | Each | 180.55  | 11194.10  | (-) 10.91 | 9972.82   |
| 38 | <b>Description:- WOOD AND P.V.C. WORK</b> // COPER OXIDISED MILD STEEL FITTINGS (COPPER OXIDISED AS PER IS: 1378) // Providing and fixing ISI marked oxidised M.S. sliding door bolts with nuts and screws etc. complete : (copper oxidized as per IS: 1378) // 300x16 mm                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |         |      |         |           |           |           |
|    | 9.62.2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 288.00  | Each | 165.90  | 47779.20  | (-) 10.91 | 42566.49  |
| 39 | <b>Description:- WOOD AND P.V.C. WORK</b> // COPER OXIDISED MILD STEEL FITTINGS (COPPER OXIDISED AS PER IS: 1378) // Providing and fixing ISI marked oxidised M.S. sliding door bolts with nuts and screws etc. complete : (copper oxidized as per IS: 1378) // 250x16 mm                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |         |      |         |           |           |           |
|    | 9.63.3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 76.00   | Each | 53.05   | 4031.80   | (-) 10.91 | 3591.93   |
| 40 | <b>Description:- WOOD AND P.V.C. WORK</b> // COPER OXIDISED MILD STEEL FITTINGS (COPPER OXIDISED AS PER IS: 1378) // Providing and fixing ISI marked oxidised M.S. tower bolt black finish, (Barrel type) with necessary screws etc. complete : (copper oxidized as per IS: 1378) // 150x10 mm                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |         |      |         |           |           |           |
|    | 9.63.4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 150.00  | Each | 39.10   | 5865.00   | (-) 10.91 | 5225.13   |
| 41 | <b>Description:- WOOD AND P.V.C. WORK</b> // COPER OXIDISED MILD STEEL FITTINGS (COPPER OXIDISED AS PER IS: 1378) // Providing and fixing ISI marked oxidised M.S. tower bolt black finish, (Barrel type) with necessary screws etc. complete : (copper oxidized as per IS: 1378) // 100x10 mm                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |         |      |         |           |           |           |
|    | 9.66.1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 224.00  | Each | 35.20   | 7884.80   | (-) 10.91 | 7024.57   |
| 42 | <b>Description:- WOOD AND P.V.C. WORK</b> // COPER OXIDISED MILD STEEL FITTINGS (COPPER OXIDISED AS PER IS: 1378) // Providing and fixing ISI marked oxidised M.S. handles conforming to IS:4992 with necessary screws etc. complete : (copper oxidized as per IS: 1378) // 125 mm                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |         |      |         |           |           |           |
|    | 9.96.2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 25.00   | Each | 234.90  | 5872.50   | (-) 10.91 | 5231.81   |

**DYCE-C-I-JBP-ENGINEERING/WEST CENTRAL RLY  
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|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|-------|---------|-----------|-----------|-----------|
| 43 | <b>Description:- WOOD AND P.V.C. WORK</b> // ANODISED ALUMINIUM FITTINGS (ALL FITTINGS SHALL BE ISI MARKED) // Providing and fixing aluminium sliding door bolts, ISI marked anodised (anodic coating not less than grade AC 10 as per IS : 1868), transparent or dyed to required colour or shade, with nuts and screws etc. complete : // 250x16 mm                                                                                                                                                                                                                                                                                                                                                                   |         |       |         |           |           |           |
| 44 | 10.1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 170.00  | Kg    | 93.05   | 15818.50  | (-) 10.91 | 14092.70  |
| 44 | <b>Description:- STEEL WORK</b> // Structural steel work in single section, fixed with or without connecting plate, including cutting, hoisting, fixing in position and applying a priming coat of approved steel primer all complete.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |         |       |         |           |           |           |
| 45 | 10.6.1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 36.00   | Sqm   | 3008.80 | 108316.80 | (-) 10.91 | 96499.44  |
| 45 | <b>Description:- STEEL WORK</b> // Supplying and fixing rolling shutters of approved make, made of required size M.S. laths, interlocked together through their entire length and jointed together at the end by end locks, mounted on specially designed pipe shaft with brackets, side guides and arrangements for inside and outside locking with push and pull operation complete, including the cost of providing and fixing necessary 27.5 cm long wire springs manufactured from high tensile steel wire of adequate strength conforming to IS: 4454 - part 1 and M.S. top cover of required thickness for rolling shutters. // 80x1.25 mm M.S. laths with 1.25 mm thick top cover                               |         |       |         |           |           |           |
| 46 | 10.8.1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 4.00    | Sqm   | 1108.70 | 4434.80   | (-) 10.91 | 3950.96   |
| 46 | <b>Description:- STEEL WORK</b> // Extra for providing mechanical device chain and crank operation for operating rolling shutters. // Exceeding 10.00 sqm and upto 16.80 sqm in the area                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |         |       |         |           |           |           |
| 47 | 10.14.2.1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 112.00  | Metre | 471.25  | 52780.00  | (-) 10.91 | 47021.70  |
| 47 | <b>Description:- STEEL WORK</b> // Providing and fixing pressed steel door frames conforming to IS: 4351, manufactured from commercial mild steel sheet of 1.60 mm thickness, including hinges, jamb, lock jamb, bead and if required angle threshold of mild steel angle of section 50x25 mm, or base ties of 1.60 mm, pressed mild steel welded or rigidly fixed together by mechanical means, including M.S. pressed butt hinges 2.5 mm thick with mortar guards, lock strike-plate and shock absorbers as specified and applying a coat of approved steel primer after pre-treatment of the surface as directed by Engineer-in-charge: // Profile C // Fixing with adjustable lugs with split end tail to each jamb |         |       |         |           |           |           |
| 48 | 10.14.3.1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 8.00    | Metre | 511.40  | 4091.20   | (-) 10.91 | 3644.85   |
| 48 | <b>Description:- STEEL WORK</b> // Providing and fixing pressed steel door frames conforming to IS: 4351, manufactured from commercial mild steel sheet of 1.60 mm thickness, including hinges, jamb, lock jamb, bead and if required angle threshold of mild steel angle of section 50x25 mm, or base ties of 1.60 mm, pressed mild steel welded or rigidly fixed together by mechanical means, including M.S. pressed butt hinges 2.5 mm thick with mortar guards, lock strike-plate and shock absorbers as specified and applying a coat of approved steel primer after pre-treatment of the surface as directed by Engineer-in-charge: // Profile E // Fixing with adjustable lugs with split end tail to each jamb |         |       |         |           |           |           |
| 49 | 10.16.1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 500.00  | Kg    | 154.90  | 77450.00  | (-) 10.91 | 69000.21  |
| 49 | <b>Description:- STEEL WORK</b> // Steel work in built up tubular (round, square or rectangular hollow tubes etc.) trusses etc., including cutting, hoisting, fixing in position and applying a priming coat of approved steel primer, including welding and bolted with special shaped washers etc. complete. // Hot finished welded type tubes                                                                                                                                                                                                                                                                                                                                                                        |         |       |         |           |           |           |
| 50 | 10.17                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 26.00   | Each  | 185.95  | 4834.70   | (-) 10.91 | 4307.23   |
| 50 | <b>Description:- STEEL WORK</b> // Providing and fixing M.S. fan clamp type I or II of 16 mm dia M.S. bar, bent to shape with hooked ends in R.C.C. slabs or beams during laying, including painting the exposed portion of loop, all as per standard design complete.                                                                                                                                                                                                                                                                                                                                                                                                                                                  |         |       |         |           |           |           |
| 51 | 10.25.2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 2200.00 | Kg    | 142.30  | 313060.00 | (-) 10.91 | 278905.15 |
| 51 | <b>Description:- STEEL WORK</b> // Steel work welded in built up sections/ framed work, including cutting, hoisting, fixing in position and applying a priming coat of approved steel primer using structural steel etc. as required. // In gratings, frames, guard bar, ladder, railings, brackets, gates and similar works                                                                                                                                                                                                                                                                                                                                                                                            |         |       |         |           |           |           |
| 52 | 10.26.1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 145.00  | Kg    | 157.15  | 22786.75  | (-) 10.91 | 20300.72  |
| 52 | <b>Description:- STEEL WORK</b> // Providing and fixing hand rail of approved size by welding etc. to steel ladder railing, balcony railing, staircase railing and similar works, including applying priming coat of approved steel primer. // M.S. tube                                                                                                                                                                                                                                                                                                                                                                                                                                                                |         |       |         |           |           |           |
| 53 | 11.41.2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 608.00  | Sqm   | 1416.65 | 861323.20 | (-) 10.91 | 767352.84 |
| 53 | <b>Description:- FLOORING</b> // VITRIFIED FLOOR TILES // Providing and laying vitrified floor tiles in different sizes (thickness to be specified by the manufacturer) with water absorption less than 0.08% and conforming to IS: 15622, of approved make, in all colours and shades, laid on 20mm thick cement mortar 1:4 (1 cement : 4 coarse sand), jointing with grey cement slurry @ 3.3 kg/ sqm including grouting the joints with white cement and matching pigments etc., complete. // Size of Tile 600x600 mm                                                                                                                                                                                                |         |       |         |           |           |           |
|    | 11.56.1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 12.00   | Sqm   | 3908.80 | 46905.60  | (-) 10.91 | 41788.20  |

**DYCE-C-I-JBP-ENGINEERING/WEST CENTRAL RLY  
TENDER DOCUMENT**

**Tender No:** DYCEC1REWA-07-2024

**Closing Date/Time:** 12/08/2024 15:00

|    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |         |     |         |            |           |            |
|----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|-----|---------|------------|-----------|------------|
| 54 | <b>Description:- FLOORING</b> // VITRIFIED FLOOR TILES // Providing and laying Polished Granite stone flooring in required design and patterns, in linear as well as curvilinear portions of the building all complete as per the architectural drawings with 18 mm thick stone slab over 20 mm (average) thick base of cement mortar 1:4 (1 cement : 4 coarse sand) laid and jointed with cement slurry and pointing with white cement slurry admixed with pigment of matching shade including rubbing , curing and polishing etc. all complete as specified and as directed by the Engineer-in-Charge. // Polished Granite stone slab colour of Black, Cherry/Ruby Red or equivalent |         |     |         |            |           |            |
|    | 13.1.1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 1355.00 | Sqm | 294.85  | 399521.75  | (-) 10.91 | 355933.93  |
| 55 | <b>Description:- FINISHING</b> // CEMENT PLASTER (IN FINE SAND) // 12 mm cement plaster of mix : // 1:4 (1 cement: 4 fine sand)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |         |     |         |            |           |            |
|    | 13.2.1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 1370.00 | Sqm | 339.70  | 465389.00  | (-) 10.91 | 414615.06  |
| 56 | <b>Description:- FINISHING</b> // CEMENT PLASTER (IN FINE SAND) // 15 mm cement plaster on the rough side of single or half brick wall of mix : // 1:4 (1 cement: 4 fine sand)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |         |     |         |            |           |            |
|    | 13.3.1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 100.00  | Sqm | 402.15  | 40215.00   | (-) 10.91 | 35827.54   |
| 57 | <b>Description:- FINISHING</b> // CEMENT PLASTER (IN FINE SAND) // 20 mm cement plaster of mix : // 1:4 (1 cement: 4 fine sand)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |         |     |         |            |           |            |
|    | 13.26                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 7585.00 | Sqm | 214.30  | 1625465.50 | (-) 10.91 | 1448127.21 |
| 58 | <b>Description:- FINISHING</b> // Extra for plastering: // Providing and applying plaster of paris putty of 2 mm thickness over plastered surface to prepare the surface even and smooth complete.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |         |     |         |            |           |            |
|    | 13.39.1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 8660.00 | Sqm | 44.25   | 383205.00  | (-) 10.91 | 341397.33  |
| 59 | <b>Description:- FINISHING</b> // Colour washing such as green, blue or buff to give an even shade : // New work (two or more coats) with a base coat of white washing with lime                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |         |     |         |            |           |            |
|    | 13.41.1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 7585.00 | Sqm | 162.55  | 1232941.75 | (-) 10.91 | 1098427.81 |
| 60 | <b>Description:- FINISHING</b> // Distemping with oil bound washable distemper of approved brand and manufacture to give an even shade : // New work (two or more coats) over and including water thinnable priming coat with cement primer                                                                                                                                                                                                                                                                                                                                                                                                                                            |         |     |         |            |           |            |
|    | 13.48.2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 700.00  | Sqm | 144.90  | 101430.00  | (-) 10.91 | 90363.99   |
| 61 | <b>Description:- FINISHING</b> // EXTERIOR FINISHING // Finishing with Deluxe Multi surface paint system for interiors and exteriors using Primer as per manufacturers specifications : // Painting wood work with Deluxe Multi Surface Paint of required shade. Two or more coat applied @ 0.90 ltr/10 sqm over an under coat of primer applied @0.75 ltr/10 sqm of approved brand and manufacture                                                                                                                                                                                                                                                                                    |         |     |         |            |           |            |
|    | 13.50.1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 600.00  | Sqm | 61.45   | 36870.00   | (-) 10.91 | 32847.48   |
| 62 | <b>Description:- FINISHING</b> // EXTERIOR FINISHING // Applying priming coat: // With ready mixed pink or Grey primer of approved brand and manufacture on wood work (hard and soft wood)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |         |     |         |            |           |            |
|    | 13.61.1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 300.00  | Sqm | 131.45  | 39435.00   | (-) 10.91 | 35132.64   |
| 63 | <b>Description:- FINISHING</b> // Painting with synthetic enamel paint of approved brand and manufacture to give an even shade : // Two or more coats on new work                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |         |     |         |            |           |            |
|    | 16.1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 825.00  | Sqm | 180.50  | 148912.50  | (-) 10.91 | 132666.15  |
| 64 | <b>Description:- ROAD WORK</b> // ROADS // Preparation and consolidation of sub grade with power road roller of 8 to 12 tonne capacity after excavating earth to an average of 22.5 cm depth, dressing to camber and consolidating with road roller including making good the undulations etc. and re-rolling the sub grade and disposal of surplus earthwith lead upto 50 metres.                                                                                                                                                                                                                                                                                                     |         |     |         |            |           |            |
|    | 16.3.2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 85.00   | cum | 1624.50 | 138082.50  | (-) 10.91 | 123017.70  |
| 65 | <b>Description:- ROAD WORK</b> // ROADS // Supplying and stacking at site. // 63 mm to 45 mm size stone aggregate                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |         |     |         |            |           |            |
|    | 16.3.3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 65.00   | cum | 1837.25 | 119421.25  | (-) 10.91 | 106392.39  |
| 66 | <b>Description:- ROAD WORK</b> // ROADS // Supplying and stacking at site. // 53 mm to 22.4 mm size stone aggregate                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |         |     |         |            |           |            |
|    | 16.3.10                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 475.00  | cum | 888.30  | 421942.50  | (-) 10.91 | 375908.57  |
| 67 | <b>Description:- ROAD WORK</b> // ROADS // Supplying and stacking at site. // Moorum                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |         |     |         |            |           |            |
|    | 16.4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 150.00  | cum | 865.80  | 129870.00  | (-) 10.91 | 115701.18  |
| 68 | <b>Description:- ROAD WORK</b> // ROADS // Laying, spreading and compacting stone aggregate of specified sizes to WBM specifications in uniform thickness, hand picking, rolling with 3 wheeled road/vibratory roller 8-10 tonne capacity in stages to proper grade and camber, applying and brooming requisite type of screening / binding material to fill up interstices of coarse aggregate, watering and compacting to the required density .                                                                                                                                                                                                                                     |         |     |         |            |           |            |
|    | 16.6.1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 825.00  | Sqm | 21.80   | 17985.00   | (-) 10.91 | 16022.84   |
| 69 | <b>Description:- ROAD WORK</b> // ROADS // Supplying, stacking and Spreading 6 mm thick red bajri, watering and rolling complete including preparation of the surface and rolling. // With road roller/ hand roller                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |         |     |         |            |           |            |
|    | 16.42                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 85.00   | cum | 7399.20 | 628932.00  | (-) 10.91 | 560315.52  |

**DYCE-C-I-JBP-ENGINEERING/WEST CENTRAL RLY  
TENDER DOCUMENT**

**Tender No:** DYCEC1REWA-07-2024

**Closing Date/Time:** 12/08/2024 15:00

|    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |        |      |         |            |           |            |
|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|------|---------|------------|-----------|------------|
| 70 | <b>Description:- ROAD WORK // CONCRETE PAVEMENTS //</b> Cement concrete 1:2:4 (1 cement : 2 coarse sand : 4 graded stone aggregate 40 mm nominal size) in pavements, laid to required slope and camber in panels as required including consolidation finishing and tamping complete.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |        |      |         |            |           |            |
| 71 | 16.43.1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 165.00 | cum  | 9492.90 | 1566328.50 | (-) 10.91 | 1395442.06 |
|    | <b>Description:- ROAD WORK // CONCRETE PAVEMENTS //</b> Providing and laying design mix cement concrete of M-30 grade, in roads/ taxi tracks/ runways, using cement content as per design mix, using coarse sand and graded stone aggregate of 40 mm nominal size in appropriate proportions as per approved & specified design criteria, providing dowel bars with sleeve/ tie bars wherever required, laying at site, spreading and compacting mechanically by using needle and surface vibrators, levelling to required slope/ camber, finishing with required texture, including steel form work with sturdy M.S. channel sections, curing, making provision for contraction/ expansion, construction & longitudinal joints (10 mm wide x 50 mm deep) by groove cutting machine, providing and filling joints with approved joint filler and sealants, complete all as per direction of Engineer-in-charge (Item of joint fillers, sealants, dowel bars with sleeve/ tie bars to be paid separately). // Cement concrete prepared with batch mixing machine |        |      |         |            |           |            |
| 72 | 16.69                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 15.00  | cum  | 8613.55 | 129203.25  | (-) 10.91 | 115107.18  |
|    | <b>Description:- ROAD WORK // MISCELLANEOUS //</b> Providing and laying at or near ground level factory made kerb stone of M-25 grade cement concrete in position to the required line, level and curvature, jointed with cement mortar 1:3 (1 cement: 3 coarse sand), including making joints with or without grooves (thickness of joints except at sharp curve shall not be more than 5mm), including making drainage opening wherever required complete etc. as per direction of Engineer-in-charge (length of finished kerb edging shall be measured for payment). (Precast C.C. kerb stone shall be approved by Engineer-in-charge).                                                                                                                                                                                                                                                                                                                                                                                                                      |        |      |         |            |           |            |
| 73 | 16.91.1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 450.00 | Sqm  | 932.35  | 419557.50  | (-) 10.91 | 373783.78  |
|    | <b>Description:- ROAD WORK // MISCELLANEOUS //</b> Providing and laying factory made chamfered edge Cement Concrete paver blocks in footpath, parks, lawns, drive ways or light traffic parking etc, of required strength, thickness & size/ shape, made by table vibratory method using PU mould, laid in required colour & pattern over 50mm thick compacted bed of sand, compacting and proper embedding/laying of inter locking paver blocks into the sand bedding layer through vibratory compaction by using plate vibrator, filling the joints with sand and cutting of paver blocks as per required size and pattern, finishing and sweeping extra sand. complete all as per direction of Engineer-in-Charge. // 60mm thick cement concrete paver block of M-35 grade with approved colour, design & pattern.                                                                                                                                                                                                                                           |        |      |         |            |           |            |
| 74 | 17.2.1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 31.00  | Each | 5540.55 | 171757.05  | (-) 10.91 | 153018.36  |
|    | <b>Description:- SANITARY INSTALLATIONS //</b> Providing and fixing white vitreous china pedestal type water closet (European type W.C. pan) with seat and lid, 10 litre low level white P.V.C. flushing cistern, including flush pipe, with manually controlled device (handle lever), conforming to IS : 7231, with all fittings and fixtures complete, including cutting and making good the walls and floors wherever required : // W.C. pan with ISI marked white solid plastic seat and lid                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |        |      |         |            |           |            |
| 75 | 17.7.2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 5.00   | Each | 1913.75 | 9568.75    | (-) 10.91 | 8524.80    |
|    | <b>Description:- SANITARY INSTALLATIONS //</b> Providing and fixing wash basin with C.I. brackets, 15 mm C.P. brass pillar taps, 32 mm C.P. brass waste of standard pattern, including painting of fittings and brackets, cutting and making good the walls wherever require: // White Vitreous China Wash basin size 630x450 mm with a single 15 mm C.P. brass pillar tap                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |        |      |         |            |           |            |
| 76 | 17.7.4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 29.00  | Each | 1679.60 | 48708.40   | (-) 10.91 | 43394.31   |
|    | <b>Description:- SANITARY INSTALLATIONS //</b> Providing and fixing wash basin with C.I. brackets, 15 mm C.P. brass pillar taps, 32 mm C.P. brass waste of standard pattern, including painting of fittings and brackets, cutting and making good the walls wherever require: // White Vitreous China Flat back wash basin size 550x 400 mm with single 15 mm C.P. brass pillar tap                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |        |      |         |            |           |            |
| 77 | 17.7B                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 10.00  | Each | 1404.00 | 14040.00   | (-) 10.91 | 12508.24   |
|    | <b>Description:- SANITARY INSTALLATIONS //</b> Providing and fixing wash basin with C.I. brackets, 15 mm PTMT pillar cock, 32 mm PTMT waste coupling of standard pattern, including painting of fittings and brackets, cutting and making good the walls wherever required. White Vitreous China Flat back wash basin size 550x400 mm with single 15 mm PTMT pillar cock.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |        |      |         |            |           |            |
| 78 | 17.10.1.1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 28.00  | Each | 6008.45 | 168236.60  | (-) 10.91 | 149881.99  |
|    | <b>Description:- SANITARY INSTALLATIONS //</b> Providing and fixing Stainless Steel A ISI 304 (18/8) kitchen sink as per IS:13983 with C.I. brackets and stainless steel plug 40 mm, including painting of fittings and brackets, cutting and making good the walls wherever required : // Kitchen sink with drain board // 510x1040 mm bowl depth 250 mm                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |        |      |         |            |           |            |
| 79 | 17.10.2.2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 32.00  | Each | 3337.85 | 106811.20  | (-) 10.91 | 95158.10   |
|    | <b>Description:- SANITARY INSTALLATIONS //</b> Providing and fixing Stainless Steel A ISI 304 (18/8) kitchen sink as per IS:13983 with C.I. brackets and stainless steel plug 40 mm, including painting of fittings and brackets, cutting and making good the walls wherever required : // Kitchen sink without drain board // 610x460 mm bowl depth 200 mm                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |        |      |         |            |           |            |
|    | 17.28.2.1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 8.00   | Each | 104.35  | 834.80     | (-) 10.91 | 743.72     |

**DYCE-C-I-JBP-ENGINEERING/WEST CENTRAL RLY  
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|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-------|--------|----------|-----------|----------|
| 80 | <b>Description:- SANITARY INSTALLATIONS</b> // Providing and fixing P.V.C. waste pipe for sink or wash basin including P.V.C. waste fittings complete. // Flexible pipe // 32 mm dia                                                                                                                                                                                                                                                                                                                                                                                                                      |       |       |        |          |           |          |
|    | 17.71                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 6.00  | Each  | 146.30 | 877.80   | (-) 10.91 | 782.03   |
| 81 | <b>Description:- SANITARY INSTALLATIONS</b> // Providing and fixing PTMT liquid soap container 109 mm wide, 125 mm high and 112 mm distance from wall of standard shape with bracket of the same materials with snap fittings of approved quality and colour, weighing not less than 105 gms.                                                                                                                                                                                                                                                                                                             |       |       |        |          |           |          |
|    | 17.73.2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 6.00  | Each  | 600.35 | 3602.10  | (-) 10.91 | 3209.11  |
| 82 | <b>Description:- SANITARY INSTALLATIONS</b> // Providing and fixing PTMT towel rail complete with brackets fixed to wooden cleats with CP brass screws with concealed fittings arrangement of approved quality and colour. // 600 mm long towel rail with total length of 645 mm, width 78 mm and effective height of 88 mm, weighing not less than 190 gms.                                                                                                                                                                                                                                              |       |       |        |          |           |          |
|    | 18.7.1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 25.00 | Metre | 255.90 | 6397.50  | (-) 10.91 | 5699.53  |
| 83 | <b>Description:- WATER SUPPLY</b> // C.P.V.C. PIPES // Providing and fixing Chlorinated Polyvinyl Chloride (CPVC) pipes, having thermal stability for hot & cold water supply, including all CPVC plain & brass threaded fittings, including fixing the pipe with clamps at 1.00 m spacing. This includes jointing of pipes & fittings with one step CPVC solvent cement and testing of joints complete as per direction of Engineer in Charge.-Internal work - Exposed on wall // 15 mm nominal dia Pipes                                                                                                |       |       |        |          |           |          |
|    | 18.7.2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 25.00 | Metre | 325.10 | 8127.50  | (-) 10.91 | 7240.79  |
| 84 | <b>Description:- WATER SUPPLY</b> // C.P.V.C. PIPES // Providing and fixing Chlorinated Polyvinyl Chloride (CPVC) pipes, having thermal stability for hot & cold water supply, including all CPVC plain & brass threaded fittings, including fixing the pipe with clamps at 1.00 m spacing. This includes jointing of pipes & fittings with one step CPVC solvent cement and testing of joints complete as per direction of Engineer in Charge.-Internal work - Exposed on wall // 20 mm nominal dia Pipes                                                                                                |       |       |        |          |           |          |
|    | 18.7.3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 25.00 | Metre | 408.55 | 10213.75 | (-) 10.91 | 9099.43  |
| 85 | <b>Description:- WATER SUPPLY</b> // C.P.V.C. PIPES // Providing and fixing Chlorinated Polyvinyl Chloride (CPVC) pipes, having thermal stability for hot & cold water supply, including all CPVC plain & brass threaded fittings, including fixing the pipe with clamps at 1.00 m spacing. This includes jointing of pipes & fittings with one step CPVC solvent cement and testing of joints complete as per direction of Engineer in Charge.-Internal work - Exposed on wall // 25 mm nominal dia Pipes                                                                                                |       |       |        |          |           |          |
|    | 18.7.5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 15.00 | Metre | 674.35 | 10115.25 | (-) 10.91 | 9011.68  |
| 86 | <b>Description:- WATER SUPPLY</b> // C.P.V.C. PIPES // Providing and fixing Chlorinated Polyvinyl Chloride (CPVC) pipes, having thermal stability for hot & cold water supply, including all CPVC plain & brass threaded fittings, including fixing the pipe with clamps at 1.00 m spacing. This includes jointing of pipes & fittings with one step CPVC solvent cement and testing of joints complete as per direction of Engineer in Charge.-Internal work - Exposed on wall // 40 mm nominal dia Pipes                                                                                                |       |       |        |          |           |          |
|    | 18.7.6                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 15.00 | Metre | 927.00 | 13905.00 | (-) 10.91 | 12387.96 |
| 87 | <b>Description:- WATER SUPPLY</b> // C.P.V.C. PIPES // Providing and fixing Chlorinated Polyvinyl Chloride (CPVC) pipes, having thermal stability for hot & cold water supply, including all CPVC plain & brass threaded fittings, including fixing the pipe with clamps at 1.00 m spacing. This includes jointing of pipes & fittings with one step CPVC solvent cement and testing of joints complete as per direction of Engineer in Charge.-Internal work - Exposed on wall // 50 mm nominal dia Pipes                                                                                                |       |       |        |          |           |          |
|    | 18.8.1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 60.00 | Metre | 441.15 | 26469.00 | (-) 10.91 | 23581.23 |
| 88 | <b>Description:- WATER SUPPLY</b> // C.P.V.C. PIPES // Providing and fixing Chlorinated Polyvinyl Chloride (CPVC) pipes, having thermal stability for hot & cold water supply, including all CPVC plain & brass threaded fittings, i/c fixing the pipe with clamps at 1.00 m spacing. This includes jointing of pipes & fittings with one step CPVC solvent cement and the cost of cutting chases and making good the same including testing of joints complete as per direction of Engineer in Charge, Concealed work, including cutting chases and making good the wall etc. // 15 mm nominal dia Pipes |       |       |        |          |           |          |
|    | 18.8.2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 60.00 | Metre | 513.75 | 30825.00 | (-) 10.91 | 27461.99 |
| 89 | <b>Description:- WATER SUPPLY</b> // C.P.V.C. PIPES // Providing and fixing Chlorinated Polyvinyl Chloride (CPVC) pipes, having thermal stability for hot & cold water supply, including all CPVC plain & brass threaded fittings, i/c fixing the pipe with clamps at 1.00 m spacing. This includes jointing of pipes & fittings with one step CPVC solvent cement and the cost of cutting chases and making good the same including testing of joints complete as per direction of Engineer in Charge, Concealed work, including cutting chases and making good the wall etc. // 20 mm nominal dia Pipes |       |       |        |          |           |          |
|    | 18.8.3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 60.00 | Metre | 626.05 | 37563.00 | (-) 10.91 | 33464.88 |
| 90 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |       |       |        |          |           |          |



**DYCE-C-I-JBP-ENGINEERING/WEST CENTRAL RLY  
TENDER DOCUMENT**

**Tender No:** DYCEC1REWA-07-2024

**Closing Date/Time:** 12/08/2024 15:00

|     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |          |       |          |            |           |            |
|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-------|----------|------------|-----------|------------|
|     | <b>Description:- WATER SUPPLY // C.P.V.C. PIPES // Providing and fixing Chlorinated Polyvinyl Chloride (CPVC) pipes, having thermal stability for hot &amp; cold water supply, including all CPVC plain &amp; brass threaded fittings, i/c fixing the pipe with clamps at 1.00 m spacing. This includes jointing of pipes &amp; fittings with one step CPVC solvent cement and the cost of cutting chases and making good the same including testing of joints complete as per direction of Engineer in Charge, Concealed work, including cutting chases and making good the wall etc. // 25 mm nominal dia Pipes</b>                                                                                                                                                                                                        |          |       |          |            |           |            |
| 91  | 18.13.1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 29.00    | Each  | 757.10   | 21955.90   | (-) 10.91 | 19560.51   |
|     | <b>Description:- WATER SUPPLY // C.P.V.C. PIPES // Making connection of G.I. distribution branch with G.I. main of following sizes by providing and fixing tee, including cutting and threading the pipe etc. complete : // 25 to 40 mm nominal bore</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |          |       |          |            |           |            |
| 92  | 18.13.2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 29.00    | Each  | 1513.70  | 43897.30   | (-) 10.91 | 39108.10   |
|     | <b>Description:- WATER SUPPLY // C.P.V.C. PIPES // Making connection of G.I. distribution branch with G.I. main of following sizes by providing and fixing tee, including cutting and threading the pipe etc. complete : // 50 to 80 mm nominal bore</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |          |       |          |            |           |            |
| 93  | 18.17.1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 29.00    | Each  | 532.35   | 15438.15   | (-) 10.91 | 13753.85   |
|     | <b>Description:- WATER SUPPLY // BRASS FITTINGS // Providing and fixing gun metal gate valve with C.I. wheel of approved quality (screwed end) : // 25 mm nominal bore</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |          |       |          |            |           |            |
| 94  | 18.17.3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 29.00    | Each  | 707.30   | 20511.70   | (-) 10.91 | 18273.87   |
|     | <b>Description:- WATER SUPPLY // BRASS FITTINGS // Providing and fixing gun metal gate valve with C.I. wheel of approved quality (screwed end) : // 40 mm nominal bore</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |          |       |          |            |           |            |
| 95  | 18.18.3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 29.00    | Each  | 399.15   | 11575.35   | (-) 10.91 | 10312.48   |
|     | <b>Description:- WATER SUPPLY // BRASS FITTINGS // Providing and fixing ball valve (brass) of approved quality, High or low pressure, with plastic floats complete : // 25 mm nominal bore</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |       |          |            |           |            |
| 96  | 18.19.3.1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 28.00    | Each  | 815.05   | 22821.40   | (-) 10.91 | 20331.59   |
|     | <b>Description:- WATER SUPPLY // Providing and fixing gun metal non- return valve of approved quality (screwed end) : // 40 mm nominal bore // Horizontal</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |          |       |          |            |           |            |
| 97  | 18.21.1.1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 29.00    | Each  | 74.80    | 2169.20    | (-) 10.91 | 1932.54    |
|     | <b>Description:- WATER SUPPLY // Providing and fixing uplasticised PVC connection pipe with brass unions : // 30 cm length // 15 mm nominal bore</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |          |       |          |            |           |            |
| 98  | 18.22.1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 9.00     | Each  | 166.35   | 1497.15    | (-) 10.91 | 1333.81    |
|     | <b>Description:- WATER SUPPLY // Providing and fixing uplasticised PVC connection pipe with brass unions : // Providing and fixing C.P. brass shower rose with 15 or 20 mm inlet : // 100 mm diameter</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |          |       |          |            |           |            |
| 99  | 18.34.1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 65.00    | Each  | 17577.90 | 1142563.50 | (-) 10.91 | 1017909.82 |
|     | <b>Description:- WATER SUPPLY // C.I. SLUICE VALVES/ FIRE HYDRANTS &amp; FIXTURES // Constructing masonry Chamber 90x90x100 cm inside, in brick work in cement mortar 1:4 (1 cement : 4 coarse sand) for sluice valve, with C.I. surface box 100 mm top diameter, 160 mm bottom diameter and 180 mm deep (inside) with chained lid and RCC top slab 1:2:4 mix (1 cement : 2 coarse sand : 4 graded stone aggregate 20 mm nominal size ), i/c necessary excavation, foundation concrete 1:5:10 (1 cement : 5 fine sand : 10 graded stone aggregate 40 mm nominal size ) and inside plastering with cement mortar 1:3 (1 cement : 3 coarse sand) 12 mm thick, finished with a floating coat of neat cement complete as per standard design : // With common burnt clay F.P.S.(non modular) bricks of class designation 7.5</b> |          |       |          |            |           |            |
| 100 | 18.48                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 28000.00 | Litre | 9.70     | 271600.00  | (-) 10.91 | 241968.44  |
|     | <b>Description:- WATER SUPPLY // C.I. SLUICE VALVES/ FIRE HYDRANTS &amp; FIXTURES // Providing and placing on terrace (at all floor levels) polyethylene water storage tank, IS : 12701 marked, with cover and suitable locking arrangement and making necessary holes for inlet, outlet and overflow pipes but without fittings and the base support for tank.</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                          |          |       |          |            |           |            |
| 101 | 18.54.4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 124.00   | Each  | 170.05   | 21086.20   | (-) 10.91 | 18785.70   |
|     | <b>Description:- WATER SUPPLY // PTMT FITTINGS // Providing and fixing PTMT bib cock of approved quality and colour. // 15 mm nominal bore, 90 mm long, weighing not less than 93 gms</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |          |       |          |            |           |            |
| 102 | 18.55.1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 76.00    | Each  | 103.15   | 7839.40    | (-) 10.91 | 6984.12    |
|     | <b>Description:- WATER SUPPLY // PTMT FITTINGS // Providing and fixing PTMT stop cock of approved quality and colour. // 15 mm nominal bore, 86 mm long, weighing not less than 88 gms</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |          |       |          |            |           |            |
| 103 | 18.55.3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 76.00    | Each  | 183.40   | 13938.40   | (-) 10.91 | 12417.72   |
|     | <b>Description:- WATER SUPPLY // PTMT FITTINGS // Providing and fixing PTMT stop cock of approved quality and colour. // Concealed stop cock, 15 mm nominal bore, 108 mm long, weighing not less than 108 gms</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |          |       |          |            |           |            |
| 104 | 18.56.2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 76.00    | Each  | 194.15   | 14755.40   | (-) 10.91 | 13145.59   |
|     | <b>Description:- WATER SUPPLY // PTMT FITTINGS // Providing and fixing PTMT pillar cock of approved quality and colour. // 15 mm nominal bore, 125 mm long foam flow, weighing not less than 120 gms</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |          |       |          |            |           |            |
|     | 18.58.1.2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 76.00    | Each  | 45.25    | 3439.00    | (-) 10.91 | 3063.81    |

**DYCE-C-I-JBP-ENGINEERING/WEST CENTRAL RLY  
TENDER DOCUMENT**

**Tender No:** DYCEC1REWA-07-2024

**Closing Date/Time:** 12/08/2024 15:00

|     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |        |       |          |           |           |           |
|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|-------|----------|-----------|-----------|-----------|
| 105 | <b>Description:- WATER SUPPLY</b> // Providing and fixing PTMT grating of approved quality and colour. // Circular type // 125 mm nominal dia with 25 mm waste hole                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |        |       |          |           |           |           |
|     | 18.62.3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 29.00  | Each  | 459.40   | 13322.60  | (-) 10.91 | 11869.10  |
| 106 | <b>Description:- WATER SUPPLY</b> // AIR VALVE & WATER METER (BULK TYPE) // Providing and fixing PTMT Ball cock of approved quality, colour and make complete with Epoxy coated aluminium rod with L.P./ H.P.H.D. plastic ball. // 25 mm nominal bore, 152mm long, weighing not less than 440 gms                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |        |       |          |           |           |           |
|     | 18.63                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 80.00  | Each  | 136.60   | 10928.00  | (-) 10.91 | 9735.76   |
| 107 | <b>Description:- WATER SUPPLY</b> // AIR VALVE & WATER METER (BULK TYPE) // Providing and fixing PTMT angle stop cock 15 mm nominal bore, weighing not less than 85 gms                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |        |       |          |           |           |           |
|     | 18.64                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 38.00  | Each  | 106.15   | 4033.70   | (-) 10.91 | 3593.62   |
| 108 | <b>Description:- WATER SUPPLY</b> // AIR VALVE & WATER METER (BULK TYPE) // Providing and fixing PTMT swivelling shower, 15 mm nominal bore, weighing not less than 40 gms                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |        |       |          |           |           |           |
|     | 18.74.1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 38.00  | Each  | 85.50    | 3249.00   | (-) 10.91 | 2894.53   |
| 109 | <b>Description:- WATER SUPPLY</b> // AIR VALVE & WATER METER (BULK TYPE) // Providing and fixing unplasticised P.V.C. connection pipe with PTMT Nuts, collar and bush of approved quality and colour. // 15 mm nominal bore with 30cm length                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |        |       |          |           |           |           |
|     | 18.75.1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 38.00  | Each  | 43.05    | 1635.90   | (-) 10.91 | 1457.42   |
| 110 | <b>Description:- WATER SUPPLY</b> // AIR VALVE & WATER METER (BULK TYPE) // Providing and fixing PTMT extension nipple for water tank pipe, fittings of approved quality and colour. // 15 mm nominal bore, weighing not less than 32 gms                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |        |       |          |           |           |           |
|     | 18.75.3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 38.00  | Each  | 100.55   | 3820.90   | (-) 10.91 | 3404.04   |
| 111 | <b>Description:- WATER SUPPLY</b> // AIR VALVE & WATER METER (BULK TYPE) // Providing and fixing PTMT extension nipple for water tank pipe, fittings of approved quality and colour. // 25mm nominal bore, weighing not less than 62 gms                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |        |       |          |           |           |           |
|     | 19.2.1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 160.00 | Metre | 895.50   | 143280.00 | (-) 10.91 | 127648.15 |
| 112 | <b>Description:- DRAINAGE</b> // STONE WARE PIPES AND FITTINGS // Providing and laying cement concrete 1:5:10 (1 cement : 5 coarse sand : 10 graded stone aggregate 40 mm nominal size) all-round S.W. pipes including bed concrete as per standard design : // 100 mm diameter S.W. pipe                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |        |       |          |           |           |           |
|     | 19.2.2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 450.00 | Metre | 1095.15  | 492817.50 | (-) 10.91 | 439051.11 |
| 113 | <b>Description:- DRAINAGE</b> // STONE WARE PIPES AND FITTINGS // Providing and laying cement concrete 1:5:10 (1 cement : 5 coarse sand : 10 graded stone aggregate 40 mm nominal size) all-round S.W. pipes including bed concrete as per standard design : // 150 mm diameter S.W. pipe                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |        |       |          |           |           |           |
|     | 19.4.1.1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 29.00  | Each  | 2453.55  | 71152.95  | (-) 10.91 | 63390.16  |
| 114 | <b>Description:- DRAINAGE</b> // Providing and fixing square-mouth S.W. gully trap class SP-1 complete with C.I. grating brick masonry chamber with water tight C.I. cover with frame of 300 x300 mm size (inside) the weight of cover to be not less than 4.50 kg and frame to be not less than 2.70 kg as per standard design: // 100x100 mm size P type // With common burnt clay F.P.S. (non modular) bricks of class designation 7.5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |        |       |          |           |           |           |
|     | 19.7.1.1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 54.00  | Each  | 11687.10 | 631103.40 | (-) 10.91 | 562250.02 |
| 115 | <b>Description:- DRAINAGE</b> // Constructing brick masonry manhole in cement mortar 1:4 ( 1 cement : 4 coarse sand ) with R.C.C. top slab with 1:1.5:3 mix (1 cement : 1.5 coarse sand (zone-III) : 3 graded stone aggregate 20 mm nominal size), foundation concrete 1:4:8 mix (1 cement : 4 coarse sand (Zone-III) : 8 graded stone aggregate 40 mm nominal size), inside plastering 12 mm thick with cement mortar 1:3 (1 cement : 3 coarse sand) finished with floating coat of neat cement and making channels in cement concrete 1:2:4 (1 cement : 2 coarse sand : 4 graded stone aggregate 20 mm nominal size) finished with a floating coat of neat cement complete as per standard design : // Inside size 90x80 cm and 45 cm deep including C.I. cover with frame (light duty) 455x610 mm internal dimensions, total weight of cover and frame to be not less than 38 kg (weight of cover 23 kg and weight of frame 15 kg) : // With common burnt clay F.P.S. (non modular) bricks of class designation 7.5                                                                                                                                                                                                                                 |        |       |          |           |           |           |
|     | 21.1.2.1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 450.00 | Kg    | 531.80   | 239310.00 | (-) 10.91 | 213201.28 |
| 116 | <b>Description:- ALUMINIUM WORK</b> // Providing and fixing aluminium work for doors, windows, ventilators and partitions with extruded built up standard tubular sections/ appropriate Z sections and other sections of approved make conforming to IS: 733 and IS: 1285, fixing with dash fasteners of required dia and size, including necessary filling up the gaps at junctions, i.e. at top, bottom and sides with required EPDM rubber/ neoprene gasket etc. Aluminium sections shall be smooth, rust free, straight, mitred and jointed mechanically wherever required including cleat angle, Aluminium snap beading for glazing / paneling, C.P. brass / stainless steel screws, all complete as per architectural drawings and the directions of Engineer-in-charge. (Glazing, paneling and dash fasteners to be paid for separately) : // For shutters of doors, windows & ventilators including providing and fixing hinges/ pivots and making provision for fixing of fittings wherever required including the cost of EPDM rubber / neoprene gasket required (Fittings shall be paid for separately) // Anodised aluminium (anodised transparent or dyed to required shade according to IS: 1868, Minimum anodic coating of grade AC 15) |        |       |          |           |           |           |
|     | 21.3.1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 230.00 | Sqm   | 1019.80  | 234554.00 | (-) 10.91 | 208964.16 |

**DYCE-C-I-JBP-ENGINEERING/WEST CENTRAL RLY  
TENDER DOCUMENT**

**Tender No:** DYCEC1REWA-07-2024

**Closing Date/Time:** 12/08/2024 15:00

| 117                                | <b>Description:- ALUMINIUM WORK</b> // Providing and fixing aluminium work for doors, windows, ventilators and partitions with extruded built up standard tubular sections/ appropriate Z sections and other sections of approved make conforming to IS: 733 and IS: 1285, fixing with dash fasteners of required dia and size, including necessary filling up the gaps at junctions, i.e. at top, bottom and sides with required EPDM rubber/ neoprene gasket etc. Aluminium sections shall be smooth, rust free, straight, mitred and jointed mechanically wherever required including cleat angle, Aluminium snap beading for glazing / paneling, C.P. brass / stainless steel screws, all complete as per architectural drawings and the directions of Engineer-in-charge. (Glazing, paneling and dash fasteners to be paid for separately) : // Providing and fixing glazing in aluminium door, window, ventilator shutters and partitions etc. with EPDM rubber / neoprene gasket etc. complete as per the architectural drawings and the directions of engineer-in-charge . (Cost of aluminium snap beading shall be paid in basic item): // With float glass panes of 4.0 mm thickness (weight not less than 10kg/sqm)                                                                                                                                                                                                                     |          |          |           |             |           |            |                     |
|------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|----------|-----------|-------------|-----------|------------|---------------------|
| 118                                | 21.4.1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 4.00     | Each     | 2448.85   | 9795.40     | (-) 10.91 | 8726.72    |                     |
|                                    | <b>Description:- ALUMINIUM WORK</b> // Providing and fixing aluminium work for doors, windows, ventilators and partitions with extruded built up standard tubular sections/ appropriate Z sections and other sections of approved make conforming to IS: 733 and IS: 1285, fixing with dash fasteners of required dia and size, including necessary filling up the gaps at junctions, i.e. at top, bottom and sides with required EPDM rubber/ neoprene gasket etc. Aluminium sections shall be smooth, rust free, straight, mitred and jointed mechanically wherever required including cleat angle, Aluminium snap beading for glazing / paneling, C.P. brass / stainless steel screws, all complete as per architectural drawings and the directions of Engineer-in-charge. (Glazing, paneling and dash fasteners to be paid for separately) : // Providing and fixing double action hydraulic floor spring of approved brand and manufacture conforming to IS : 6315, having brand logo embossed on the body / plate with double spring mechanism and door weight upto 125 kg, for doors, including cost of cutting floors, embedding in floors as required and making good the same matching to the existing floor finishing and cover plates with brass pivot and single piece M.S. sheet outer box with slide plate etc. complete as per the direction of Engineer-in-charge. // With stainless steel cover plate minimum 1.25 mm thickness |          |          |           |             |           |            |                     |
| S.No.                              | Item Code                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Item Qty | Qty Unit | Unit Rate | Basic Value | Escl.(%)  | Amount     | Bidding Unit        |
| <b>Schedule B-Supply of Steel</b>  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |          |          |           |             |           | 2494372.87 | Above/<br>Below/Par |
| 1                                  | 025082                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 29640.00 | Kg       | 118.78    | 3520639.20  | (-) 29.15 | 2494372.87 |                     |
|                                    | <b>Description:-</b> Supply of steel reinforcement of approved brands/makes for R.C.C. work including straightening, cutting, bending, placing in position and binding all complete. Thermo-Mechanically Treated bars of grade Fe-500D or more of approved brands/makes.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |          |          |           |             |           |            |                     |
| S.No.                              | Item Code                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Item Qty | Qty Unit | Unit Rate | Basic Value | Escl.(%)  | Amount     | Bidding Unit        |
| <b>Schedule C-Supply of Cement</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |          |          |           |             |           | 429984.40  | Above/<br>Below/Par |
| 1                                  | 025072                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 50.00    | MT       | 9075.23   | 453761.50   | (-) 5.24  | 429984.40  |                     |
|                                    | <b>Description:-</b> Supply and using Cement at Worksite Ordinary Portland Cement 53 grade of approved brands/makes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |          |          |           |             |           |            |                     |

**3. ITEM BREAKUP**

|                        |
|------------------------|
| No item break up added |
|------------------------|

**4. ELIGIBILITY CONDITIONS**

**Standard Financial Criteria**

| S.No. | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Confirmation Required | Remarks Allowed | Documents Uploading |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|-----------------|---------------------|
| 1     | The tenderer must have minimum average annual contractual turnover of V/N or 'V' whichever is less; V= Advertised value of the tender in crores of Rupees N= Number of years prescribed for completion of work for which bids have been invited. The average annual contractual turnover shall be calculated as an average of "total contractual payments" in the previous three financial years, as per the audited balance sheet. However, in case balance sheet of the previous year is yet to be prepared/ audited, the audited balance sheet of the fourth previous year shall be considered for calculating average annual contractual turnover. The tenderers shall submit requisite information as per Annexure- XXX, along with copies of Audited Balance Sheets duly certified by the Chartered Accountant/ Certificate from Chartered Accountant duly supported by Audited Balance Sheet. | No                    | No              | Allowed (Mandatory) |

**DYCE-C-I-JBP-ENGINEERING/WEST CENTRAL RLY  
TENDER DOCUMENT**

**Tender No:** DYCEC1REWA-07-2024

**Closing Date/Time:** 12/08/2024 15:00

**Special Financial Criteria**

| S.No. | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Confirmation Required | Remarks Allowed | Documents Uploading |
|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|-----------------|---------------------|
| 1     | Bid Capacity: - For tenders of advertised value more than Rs. 20 Crores, the tenderers who meet the Minimum Eligibility Criteria will be qualified only if their Available Bid Capacity is equal to or more than the Total Bid Value of the Present Tender. Available Bid Capacity shall be calculated as under: Available Bid Capacity = $[A \times N \times 2] - 0.33 \times N \times B$ Where, A=Maximum value of construction works executed and payment received in any one of the previous three financial years or the current financial year (up to the date of inviting of tender) taking into account the completed as well as works in progress. N= Number of years prescribed for completion of work for which bids has been invited. B=Existing commitments and balance amount of ongoing works with the tenderer as per prescribed proforma of Railway for statement of all works in progress and also the works which are awarded to tenderer but yet not started up to the date of inviting of tender. The tenderer(s) must furnish the details of all existing commitments and balance amount of all ongoing works including works which are awarded to tenderer but yet not started and also works for which letter of acceptance received up to the date of inviting of the tender as per Annexure G (in case of no works in hand, a 'NIL' statement should be furnished) and available Bid capacity calculation as per annexure G-1 of Chapter -6 of tender document. These statements should be submitted duly verified by Chartered Accountant. In case, the statement is submitted without verification by Chartered Accountant, the offer shall be considered as incomplete and will be SUMMARILY REJECTED. | No                    | No              | Allowed (Mandatory) |
| 1.1   | For Maximum value of construction works executed and payment received in any one financial year i.e., Value- "A" considered for calculation of available Bid capacity shall be duly certified by CA supported with audited balance sheet and in case balance sheet is yet to be prepared /audited than it shall be supported by Form-16A issued by client (duly certified by CA) and/or payment received under relevant section for contractual payment of Form-26AS generated through TRACES of Income Tax Department (duly certified by CA). Value of "A" not supported by duly certified audited balance sheet/Form- 16A/Form-26AS by Chartered Accountant will not be considered for Bid capacity calculation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | No                    | No              | Allowed (Mandatory) |

**Special Technical Criteria**

| S.No. | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Confirmation Required | Remarks Allowed | Documents Uploading |
|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|-----------------|---------------------|
| 1     | Note for Technical Criteria: Work experience certificate from private individual shall not be considered. However, in addition to work experience certificates issued by any Govt. Organisation, work experience certificate issued by Public listed company having average annual turnover of Rs 500 crore and above in last 3 financial years excluding the current financial year, listed on National Stock Exchange or Bombay Stock Exchange, incorporated/registered at least 5 years prior to the date of closing of tender, shall also be considered provided the work experience certificate has been issued by a person authorized by the Public listed company to issue such certificates. In case tenderer submits work experience certificate issued by public listed company, the tenderer shall also submit along with work experience certificate, the relevant copy of work order, bill of quantities, bill wise details of payment received duly certified by Chartered Accountant, TDS certificates for all payments received and copy of final/last bill paid by company in support of above work experience certificate. For such work experience certificate following details shall also be submitted: - i) Details of Average Annual turnover of the public listed company in last 3 financial years excluding the current financial year, (company should be 500 Cr and above) issued by Chartered Accountant, ii) The copy of the documents regarding listing in the National stock exchange or Bombay stock exchange with details of status of listing as on date of opening of tender, duly self-attested ,iii) The copy of the document of incorporation/ registration of the Public listed company (should be at least 5 years prior to date of closing of tender), duly self-attested. Iv) The copy of document regarding Person Authorized by the Public listed Company to issue such certificate, duly self-attested. | No                    | No              | Allowed (Mandatory) |
| 1.1   | Definition of Similar Nature of Work :- Any civil engineering work                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | No                    | No              | Allowed (Mandatory) |

**Standard Technical Criteria**

| S.No. | Description | Confirmation Required | Remarks Allowed | Documents Uploading |
|-------|-------------|-----------------------|-----------------|---------------------|
|-------|-------------|-----------------------|-----------------|---------------------|

**DYCE-C-I-JBP-ENGINEERING/WEST CENTRAL RLY  
TENDER DOCUMENT**

**Tender No:** DYCEC1REWA-07-2024

**Closing Date/Time:** 12/08/2024 15:00

|   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |    |    |                        |
|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|------------------------|
| 1 | <p>a) The tenderer must have successfully completed or substantially completed any of the following during last 07 (seven) years, ending last day of month previous to the one in which tender is invited (Date of publishing tender notice on IREPS website): Three similar works each costing not less than the amount equal to 30% of advertised value of the tender, OR Two similar works each costing not less than the amount equal to 40% of advertised value of the tender, OR One similar work costing not less than the amount equal to 60% of advertised value of the tender. (b) (i) In case of tenders for composite works (e.g. works involving more than one distinct component, such as Civil Engineering works, S&amp;T works, Electrical works, OHE works etc. and in the case of major bridges - substructure, superstructure etc.), tenderer must have successfully completed or substantially completed any of the following during last 07 (seven) years, ending last day of month previous to the one in which tender is invited*: Three similar works, each costing not less than the amount equal to 30% of advertised value of each component of tender, OR Two similar works, each costing not less than the amount equal to 40 % of advertised value of each component of tender, OR One similar work costing not less than the amount equal to 60% of advertised value of each component of tender. Note for b (i): Separate completed works of minimum required values shall also be considered for fulfilment of technical eligibility criteria for different components. (b) (ii) In such cases, what constitutes a component in a composite work shall be clearly pre- defined with estimated tender cost of it, as part of the tender documents without any ambiguity. (b)(iii) To evaluate the technical eligibility of tenderer, only components of work as stipulated in tender documents for evaluation of technical eligibility, shall be considered. The scope of work covered in other remaining components shall be either executed by tenderer himself if he has work experience as mentioned in clause 7 of the Standard General Conditions of Contract or through subcontractor fulfilling the requirements as per clause 7 of the Standard General Conditions of Contract or jointly i.e., partly himself and remaining through subcontractor, with prior approval of Chief Engineer in writing.</p> | No | No | Allowed<br>(Mandatory) |
|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|------------------------|

**Bidders shall confirm and certify on the behalf of the tenderer including its constituents as under:**

| S.No. | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1     | I/we the tenderer (s) am/are signing this document after carefully reading the contents.                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 2     | I/We the tenderer(s) also accept all the conditions of the tender and have signed all the pages in confirmation thereof.                                                                                                                                                                                                                                                                                                                                                                                         |
| 3     | I/we hereby declare that I/we have downloaded the tender documents from Indian Railway website www.ireps.gov.in . I/we have verified the content of the document from the website and there is no addition, no deletion or no alteration to the content of the tender document. In case of any discrepancy noticed at any stage i.e. evaluation of tenders, execution of work or final payment of the contract, the master copy available with the railway Administration shall be final and binding upon me/us. |
| 4     | I/we declare and certify that I/we have not made any misleading or false representation in the forms, statements and attachments in proof of the qualification requirements.                                                                                                                                                                                                                                                                                                                                     |
| 5     | I/We also understand that my/our offer will be evaluated based on the documents/credentials submitted along with the offer and same shall be binding upon me/us.                                                                                                                                                                                                                                                                                                                                                 |
| 6     | I/We declare that the information and documents submitted along with the tender by me/us are correct and I/we are fully responsible for the correctness of the information and documents, submitted by us.                                                                                                                                                                                                                                                                                                       |
| 7     | I/we certify that I/we the tenderer(s) is/are not blacklisted or debarred by Railways or any other Ministry / Department of Govt. of India from participation in tender on the date of submission of bids, either in individual capacity or as a HUF/ member of the partnership firm/LLP/JV/Society/Trust.                                                                                                                                                                                                       |
| 8     | I/we understand that if the contents of the certificate submitted by us are found to be forged/false at any time during process for evaluation of tenders, it shall lead to forfeiture of the Bid Security and may also lead to any other action provided in the contract including banning of business for a period of upto two year. Further, I/we and all my/our constituents understand that my/our offer shall be summarily rejected.                                                                       |
| 9     | I/we also understand that if the contents of the certificate submitted by us are found to be false/forged at any time after the award of the contract, it will lead to termination of the contract, along with forfeiture of Bid Security/Security Deposit and Performance guarantee and may also lead to any other action provided in the contract including banning of business for a period of upto two year.                                                                                                 |
| 10    | I/We have read the clause regarding restriction on procurement from a bidder of a country which shares a land border with India and certify that I am/We are not from such a country or, if from such a country, have been registered with the competent Authority. I/We hereby certify that I/we fulfil all the requirements in this regard and am/are eligible to be considered (evidence of valid registration by the competent authority is enclosed)                                                        |

**Partnership firm/Joint Venture (JV) / Hindu Undivided Family (HUF) / Limited Liability Partnership (LLP) etc.**

| S.No. | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1     | <p><b>Partnership firm/Joint Venture (JV) / Hindu Undivided Family (HUF) / Limited Liability Partnership (LLP) etc.</b> Please submit a certificate in the prescribed format (please download the format from the link given below). Non submission of the certificate, or submission of certificate either not properly filled in, or in a format other than the prescribed format shall lead to summary rejection of your offer.</p> <p>( <a href="#">Click here</a> to download the Format of Self Certification)</p> |

**DYCE-C-I-JBP-ENGINEERING/WEST CENTRAL RLY  
TENDER DOCUMENT**

**Tender No:** DYCEC1REWA-07-2024

**Closing Date/Time:** 12/08/2024 15:00

**5. COMPLIANCE**

**Commercial-Compliance**

| S.No. | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Confirmation Required | Remarks Allowed | Documents Uploading |
|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|-----------------|---------------------|
| 1     | If Bid Security is submitted through mode of Bank Guarantee, then tenderer must submit the scanned copy of the Bank Guarantee (as per Specimen Format at Annexure-Q of Chapter-6 of Tender Document) submitted towards Bid Security and the original bank guarantee shall be submitted/delivered in-person to the official /office mentioned under item no.-16 of Top sheet-1 (as per para-3 of Instruction to Tenderers (ITT) of Chapter-2 of tender Document) before closing date of submission of bids (i.e. excluding the last date of submission of bids). Non submission of scanned copy of Bank Guarantee with the bid on e-tendering portal (IREPS) and/or non-submission of original Bank Guarantee within above specified period shall lead to summarily rejection of bid. In case of the Bid Security deposited through e- payment gateway, then the same should be confirmed by the tenderer.                                                                                                                                                                                                             | No                    | Yes             | Allowed (Optional)  |
| 2     | Tenderer shall submit certificate as per Annexure- A & A-1 (as required) of Chapter-6 of the Tender document. However, if the bidder on behalf of tenderer including its constituents confirms and certify the provisions of these certificate on IREPS, the uploading of physical signed Annexure- A & A -1 on IREPS portal is not mandatory.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | No                    | Yes             | Allowed (Optional)  |
| 3     | Tenderer shall submit details of works successfully completed or substantially completed during last 07 years ending last day of month previous to the one in which tender has been invited as per the format vide Annexure-F & F-1 (as applicable) of Chapter-6 of the Tender Document along with work completion/experience certificate. Non-compliance with this condition shall lead to summarily rejection of the offer.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | No                    | No              | Allowed (Mandatory) |
| 4     | Tenderer shall submit the details of the constitution of Firm i.e. Proprietary Firm/Partnership Firm/HUF/JV/Company Registered under Companies Act-2013/Limited Liability Partnership/Registered Society & Registered Trust etc. as per Annexure-C of Chapter-6 of the Tender Document. Tenderer shall also submit the mandatory supporting documents such as copies of Partnership Deed, Power of Attorney, Memorandum of Articles, Articles of Association, Authorisation, Certificate of Incorporation, Certificate of Registration, Deed of Formation, MOU of JV, Copy of PAN card etc. as applicable under Para 14, 15, 17 & 18 of Chapter-2 of Tender Document. Important Note: - (a) In case one or more of the members of JV is/are partnership firm(s)/Proprietary Firm/HUF/Companies/LLP Firm(s)/Society(s)/Trust(s), the required documents of each individual member of JV shall be submitted along with the documents to be submitted on behalf of the JV, (b) non-submission of above documents (except PAN Card) or submission of incomplete documents shall lead to summarily rejection of the offer. | No                    | No              | Allowed (Mandatory) |
| 5     | Tenderer shall submit Power of attorney duly notarized/authorization in favour of a single individual as per para-14,15,17 & 18 of tender form (second sheet) of SGCC and para 15 of Chapter-2 of Tender document.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | No                    | Yes             | Allowed (Mandatory) |
| 6     | Tenderer participating as Joint Venture (JV) firm shall submit MOU for Joint Venture (JV) participation as per annexure-L of Chapter-6 of the Tender Document along with other mandatory documents of constituent members as stipulated. Non-submission of above documents or submission of incomplete documents shall lead to summarily rejection of the offer.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | No                    | Yes             | Allowed (Optional)  |
| 7     | Tenderer shall submit list of Plant & Machinery available on hand and proposed to be inducted and hired for the tendered work as per the format vide Annexure-D of Chapter-6 of Tender Document. Non-compliance with this condition is liable to result in the tender being rejected.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | No                    | No              | Allowed (Mandatory) |
| 8     | Tenderer shall submit list of personnel/organization on hand and proposed to be engaged for the tendered work as per the format vide Annexure-E of Chapter-6 of the Tender Document. Non-compliance with this condition is liable to result in the tender being rejected.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | No                    | No              | Allowed (Mandatory) |
| 9     | Tenderer shall submit the information/declaration regarding association of Railway Officers with tenderer(s) as per Annexure-H of Chapter-6 in terms of Clause 16 of Chapter-2 of tender Document. On non-submission of this information, contract is liable to be dealt in accordance with provision of Clause 62 of Standard General Conditions of Contract.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | No                    | No              | Allowed (Mandatory) |



**DYCE-C-I-JBP-ENGINEERING/WEST CENTRAL RLY  
TENDER DOCUMENT**

**Tender No:** DYCEC1REWA-07-2024

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|    |                                                                                                                                                                                                                                                                                                                                                                       |    |     |                     |
|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|-----|---------------------|
| 10 | Tenderer shall submit list of court cases during previous three financial years as per Annexure-I of Chapter-6 of the Tender Document.                                                                                                                                                                                                                                | No | No  | Allowed (Mandatory) |
| 11 | Tenderer shall submit list of arbitration cases during previous three financial years as per Annexure-J of Chapter-6 of the Tender Document.                                                                                                                                                                                                                          | No | No  | Allowed (Mandatory) |
| 12 | Tenderer shall submit the Certificate of familiarization to the work site as per Chapter- 5 of the Tender Document.                                                                                                                                                                                                                                                   | No | No  | Allowed (Mandatory) |
| 13 | Tenderer shall enter the percentage of local content in the material being offered. Please enter 0 for fully imported items, and 100 for fully indigenous items. The definition and calculation of local content shall be in accordance with the Make in India policy as incorporated in the Tender Document.                                                         | No | Yes | Allowed (Optional)  |
| 14 | Tenderer shall submit broad plan of execution of work as per Annexure-K of Chapter-6 of the Tender Document.                                                                                                                                                                                                                                                          | No | No  | Allowed (Optional)  |
| 15 | Tenderer shall submit details of other credentials/facilities as per Annexure-N, Chapter-6 of the Tender Document.                                                                                                                                                                                                                                                    | No | No  | Allowed (Optional)  |
| 16 | Tenderer shall submit bank details i.e. Name of Bank along with Bank Branch Code, Accounts Number as appearing in the cheque book, IFSC Code and PAN Number, duly certified by the authorised official of the bank, to facilitate payment through ECS/NEFT/RTGS as per Annexure-O of Chapter-6 of Tender Document.                                                    | No | No  | Allowed (Mandatory) |
| 17 | For tenders having advertised cost of Rs. 10 lakhs and above:- If tenderer desired to take payment through Letter of Credit (LC) arrangement then tenderer shall exercise the option at the time of bidding itself and shall affirm having read for and agreed to the terms and conditions of LC option. For details refer Clause-7 of Chapter- 4 of Tender Document. | No | No  | Allowed (Optional)  |
| 18 | Please enter the percentage of local content in the material being offered. Please enter 0 for fully imported items, and 100 for fully indigenous items. The definition and calculation of local content shall be in accordance with the Make in India policy as incorporated in the tender conditions.                                                               | No | Yes | Allowed (Optional)  |

**General Instructions**

| S.No. | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Confirmation Required | Remarks Allowed | Documents Uploading |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|-----------------|---------------------|
| 1     | Payment of Bid Security shall be accepted either in cash through e-Payment Gateway or as Bank Guarantee bond from a Scheduled Commercial Bank of India. BG shall be as per Annexure-Q of Chapter-6 of Tender Document. No other mode of payment shall be accepted. Any firm recognized by Department of Industrial Policy and Promotion (DIPP) as 'Startups' shall be exempted from payment of Bid Security. Labour Cooperative Societies shall submit only 50% of the Bid Security.                                                                                                                                                                                                                       | No                    | No              | Not Allowed         |
| 2     | In case the date of closing mentioned in the NIT Header of e- tender document is declared a holiday/bandh/strike etc. on any account, the date of closing tender online will not be changed as the application in the website of IREPS does not permit submission of any offer after closing date and time of the tender. However, opening of tenders online will be on the following working day after the closing date/time of tenders.                                                                                                                                                                                                                                                                  | No                    | No              | Not Allowed         |
| 3     | Rates quoted by the tenderer in the relevant fields of the Financial Bid only will be the ruling terms for deciding inter-se-ranking and any condition having financial repercussions, if quoted, anywhere else shall not be considered for deciding the inter-se-ranking.                                                                                                                                                                                                                                                                                                                                                                                                                                 | No                    | No              | Not Allowed         |
| 4     | Tenderer participating as JV shall refer the guidelines under Clause-17 of Tender Document and relevant clauses of IR-SGCC 2022 regarding participation of Joint Venture Firms in Works Tender and documents to be submitted .                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | No                    | No              | Not Allowed         |
| 5     | (a) In case of any information submitted by tenderer is found to be forged/false at any time during process for evaluation of tenders, it shall lead to forfeiture of the Bid Security and may also lead to any other action provided in the contract including banning of business for a period of upto two year. (b) In case of any information submitted by tenderer is found to be false/forged at any time after the award of the contract, it will lead to termination of the contract, along with forfeiture of Bid Security/Security Deposit and Performance guarantee and may also lead to any other action provided in the contract including banning of business for a period of upto two year. | No                    | No              | Not Allowed         |

**DYCE-C-I-JBP-ENGINEERING/WEST CENTRAL RLY  
TENDER DOCUMENT**

**Tender No:** DYCEC1REWA-07-2024

**Closing Date/Time:** 12/08/2024 15:00

|    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |    |    |             |
|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|-------------|
| 6  | The tender shall be finalised based on submitted documents & no post tender correspondence shall be entertained. However, if any clarification is required by the Railway, the same will be sought from the tenderer which need to be provided by tenderer(s) within stipulated time failing which his/their offer(s) shall be dealt based on available information with Railways interpretation only.                                                                                                                                                                                                                                                                                               | No | No | Not Allowed |
| 7  | While submitting offers the tenderer(s) generally commit mistakes in submitting the mandatory documents like Certificates regarding contracting experience, Audited Balance Sheet duly certified by the Chartered Accountant, Certificate in respect to true and factuality of documents, information in connection with evaluation of bid capacity, Power of Attorney, submission of tenders by authorised signatory, JV related documents etc. As such, the tenderer(s) should go through the tender document very carefully before submitting his/their offer and accordingly upload all the mandatory documents and other relevant documents as per the formats enclosed in the tender document. | No | No | Not Allowed |
| 8  | Tenderers must go through the tender documents to ascertain and get themselves acquainted with general conditions of contract, e-tendering process and two packet bid system (as applicable). The tenderer must also go through the other documents such as special conditions, additional technical conditions, specification, drawings, scope of work etc. attached as documents on the website.                                                                                                                                                                                                                                                                                                   | No | No | Not Allowed |
| 9  | It is the responsibility of the tenderer to keep watch on IREPS website regarding Addendum/Corrigendum etc. (if any) to tender notice up to the date of Bid closing. No separate intimation/ information shall be given in Newspapers or through any other means.                                                                                                                                                                                                                                                                                                                                                                                                                                    | No | No | Not Allowed |
| 10 | Tenderers are advised to visit the site of work to ascertain all aspects of the site conditions viz accessibility, availability of approach road, nature of the soil, availability of material, labour, water etc. that may be encountered in course of execution of work.                                                                                                                                                                                                                                                                                                                                                                                                                           | No | No | Not Allowed |
| 11 | No Advance payment shall be made for mobilization.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | No | No | Not Allowed |
| 12 | No advance payment against Plant and Machinery shall be admissible.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | No | No | Not Allowed |
| 13 | No advance/stage Payment will be payable for supply of Cement and Reinforcement Steel. Cement and Reinforcement Steel shall be paid only after its actual use in work                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | No | No | Not Allowed |
| 14 | The quoted rates shall be inclusive of all taxes, levies etc (including GST and royalty) leviable by State/Central Govt. or by local authorities. The rate shall be inclusive of all sales / turn over tax, Service Tax including all other cess etc., on contract by respective State Govt.                                                                                                                                                                                                                                                                                                                                                                                                         | No | No | Not Allowed |
| 15 | Tenderers are requested to go through the enclosed Additional Special Technical Conditions for various items included in tender schedule/Bid of quantities, etc                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | No | No | Not Allowed |
| 16 | Price Variation Clause (PVC) shall be applicable for Tenders having advertised value above Rs.2 Crores.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | No | No | Not Allowed |
| 17 | ELECTRONIC REVERSE AUCTION (e-RA) will be applicable for tenders having advertised value more than Rs. 50 Cr.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | No | No | Not Allowed |

**Undertakings**

| S.No. | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Confirmation Required | Remarks Allowed | Documents Uploading |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|-----------------|---------------------|
| 1     | I/ We have visited the works site and I / We am / are aware of the site conditions.                                                                                                                                                                                                                                                                                                                                                                                                   | No                    | No              | Not Allowed         |
| 2     | I/We have read the various conditions attached/referred to in this tender document and agree to abide by the said conditions.                                                                                                                                                                                                                                                                                                                                                         | No                    | No              | Not Allowed         |
| 3     | I/We also hereby agree to abide by the Indian Railways Standard General Conditions of Contract-2022 with all Correction Slips up to date of closing of tender and to carry out the work according to the Special/Technical Conditions of Contract and Specifications of Materials and Works as laid down by Railway in the Special Conditions/Specifications, CPWD DSR-2021 Schedule of Rates with all correction slips up to the date of closing of tender for the present contract. | No                    | No              | Not Allowed         |



**DYCE-C-I-JBP-ENGINEERING/WEST CENTRAL RLY  
TENDER DOCUMENT**

**Tender No:** DYCEC1REWA-07-2024

**Closing Date/Time:** 12/08/2024 15:00

|     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |    |    |             |
|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|-------------|
| 4   | I/We hereby confirm that the rates, rebates and/or other financial terms, if any, quoted by us in the relevant fields of the Financial Bid will only be the ruling terms for deciding the inter-se-ranking and any such conditions having financial repercussions, if quoted by us anywhere else including attached documents shall not be considered for deciding inter- se-ranking.                                                                                                                                                                                                     | No | No | Not Allowed |
| 5   | I/We agree to keep our offer open for acceptance for the period of days as mentioned on NIT HEADER from the date of closing of tender and in default thereof, I/We will be liable for forfeiture of my/our Bid Security.                                                                                                                                                                                                                                                                                                                                                                  | No | No | Not Allowed |
| 6   | The amount as stipulated in tender document is herewith forwarded as Bid Security shall stand forfeited without prejudice if :-                                                                                                                                                                                                                                                                                                                                                                                                                                                           | No | No | Not Allowed |
| 6.1 | I/We resile from my/our offer or modify the terms and conditions thereof in a manner not acceptable to the Railways during validity period of the offer as mentioned on NIT HEADER.                                                                                                                                                                                                                                                                                                                                                                                                       | No | No | Not Allowed |
| 6.2 | I/We do not submit a Performance Guarantee in any of the prescribed form within 60 days from the date of issue of letter of acceptance.                                                                                                                                                                                                                                                                                                                                                                                                                                                   | No | No | Not Allowed |
| 6.3 | I/We do not execute the contract within 7 days after receipt of notice by Railway that such documents are ready.                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | No | No | Not Allowed |
| 6.4 | I/We do not commence the work within fifteen days after receipt of Letter of Acceptance to that effect.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | No | No | Not Allowed |
| 7   | Until a formal agreement is prepared and executed, acceptance of this tender shall constitute a binding contract between us subject to modifications, as may be mutually agreed to between us and indicated in the letter of acceptance of my/our offer to this work.                                                                                                                                                                                                                                                                                                                     | No | No | Not Allowed |
| 8   | I/we hereby confirm that I/we have satisfied myself/ourselves by actual inspection of the site and locality of the work that all conditions liable to be encountered during execution of the work are taken into account and that the rates quoted in the tender are adequate and all inclusive.                                                                                                                                                                                                                                                                                          | No | No | Not Allowed |
| 9   | I/ We have read the tender document and NIT conditions about the Mandatory Documents required to be uploaded up to the time and date of Bid closing of Tender, in absence of which, the offer shall be SUMMARILY REJECTED. I/We understand that no Suo moto cognizance of any document available in public domain (i.e., on internet etc.) or in Railways record/office files etc. Will be taken for consideration of the tender, if no such mention is available in tender offer submitted. I / we submitted all the mandatory documents and have no additional document to add further. | No | No | Not Allowed |
| 10  | I/we hereby confirm that for tenders involving supply of track ballast, I/we undertake that we will supply the track ballast at all times conforming to "Specification for Track ballast" (IS/RDSO-GE/0001:2023)" issued by RDSO and amended from time to time. "                                                                                                                                                                                                                                                                                                                         | No | No | Not Allowed |

**6. Documents attached with tender**

| S.No. | Document Name                         | Document Description              |
|-------|---------------------------------------|-----------------------------------|
| 1     | NewTenderdocuments.pdf                | Tender Document                   |
| 2     | Specialconditionandspecifications.pdf | Scope and Technical Specification |
| 3     | GCC-2022-ACS6.pdf                     | GCC 2022 updated upto ACS 6       |
| 4     | TOPSHEET-1.pdf                        | TOP Sheet                         |

This tender complies with Public Procurement Policy (Make in India) Order 2017, dated 15/06/2017, issued by Department of Industrial Promotion and Policy, Ministry of Commerce, circulated vide Railway Board letter no. 2015/RS(G)/779/5 dated 03/08/2017 and 27/12/2017 and amendments/ revisions thereof.

As a Tender Inviting Authority, the undersigned has ensured that the issue of this tender does not violate provisions of GFR regarding procurement through GeM.

**Signed By:** GHAMANDI LAL MEENA

**Designation :** Dy.CE/C-1/JBP



Government eMarketplace  
Indian Railways E-Procurement System  
Works

GHAMANDI LAL MEENA (Dy.CE/C-1/JBP), DYCE-C-I-JBP-ENGINEERING, WEST CENTRAL RLY (IREPS ID- 601)12-Aug-2024 15:52:55 IST

Published Corrigendum

|              |                                                                                                                                                     |          |                                           |                   |                         |
|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|----------|-------------------------------------------|-------------------|-------------------------|
| Tender No.   | DYCEC1REWA-07-2024                                                                                                                                  | Dept/Rly | DYCE-C-I-JBP-ENGINEERING/WEST CENTRAL RLY | Closing Date/Time | 12/08/2024 15:52:55 IST |
| Tender Title | Execution of balance works in newly constructed staff quarte.....  |          |                                           |                   |                         |

No Corrigendum issued against this tender till date

Home

**WEST CENTRAL RLY**

**DYCE-C-I-JBP-  
ENGINEERING  
Railways Officer  
Rest House, Near  
Rewa Railway  
Station  
Rewa, 486450  
Madhya Pradesh ,  
India**

**Letter No: DYCE-C-I-JBP-ENGINEERING /  
DYCEC1REWA-07-2024 / 00912200115255**

**Dated: 17/10/2024**

**M/s SURENDRA SINGH KALRA-BILASPUR**  
FLAT NO. A 507 HARSH ASHIYANA  
GREEN PARK COLONY JARHABHATA ROAD  
BILASPUR- 495001  
Chhattisgarh, India

**Sub: Letter Of Acceptance**

- Ref:**
1. Tender No. DYCEC1REWA-07-2024 closing date 12-08-2024 15:00 for Execution of balance works in newly constructed staff quarters at Rewa in connection with the Rewa-Sidhi new BG rail line project.
  2. Your bid ID **17127455** dated **12/08/2024 14:30**

The Competent Authority has accepted your offered rates in connection with the subject work. The total cost of the work at the accepted rates works out to Rs. 25988561.97 (Rupees Two Crore Fifty-Nine Lakh Eighty-Eight Thousand Five Hundred And Sixty-One Rupees And Ninety-Seven Paise Only)

A sum of Rs.290100 deposited as Earnest Money vide IREPS reference ID PE412323362659 has been retained towards initial Security Money for due and faithful fulfillment of the contract, and the balance Security Money will be recovered from the progressive bills @ 6 % of the bill amount till it reaches 5 % of the contract value in terms of clause 16(1) of Indian Railway Standard GCC April-2022.

You are requested to submit Performance Guarantee in the form as given in Clause 16(4) of Indian Railway Standard GCC April-2022 equivalent to 5 % of the contract value amounting to Rs. 1299428.1 (Rupees Twelve Lakh Ninety-Nine Thousand Four Hundred And Twenty-Eight Rupees And Ten Paise Only) within 21 days from the date of issue of Letter of Acceptance, valid up to stipulated date of completion plus 60 days so that contract agreement can be executed. The Deposit Receipt/ Bank Guarantee etc. should be in favour of DY.FA&CAO/C/WCR/Jabalpur. Extension of time for submission of PG beyond 21(Twenty one) days and up to the date of submission of the PG from the date of issue of LOA may be given by the competent authority who is competent to sign the contract agreement. However, a panel interest of 12% per annum shall be charged for the delay beyond 21 (Twenty one) days, i.e. from 22nd day after the date of issue of LOA. Further, if 60th day happens to be a declared Holiday in the concerned office of the Railway, submission of PG can be

accepted on the next working day. In case failing to submit the requisite PG even after 60 (Sixty) days from the date of issue of LOA, the contract is liable to be terminated. In case contract is terminated railway shall be entitled to forfeit Bid Security and other dues payable to you against this particular contract, subject to maximum of PG amount and you will be debarred from participating in retender for this work.

The entire work shall be completed within 6 month from the date of issue of Letter of Acceptance.

Executive in field: Further instructions for the above noted work shall be given by the AXEN/Con/Rewa, who may be contacted in this regards.

Contract Agreement signing: You are required to execute the contract agreement in compliance to the tender document.

Regarding contract agreement:- The Letter of acceptance (LOA) shall be treated as final and part and parcel of the contract agreement and binding between you and railway administration till such time a final contract agreement is executed.

commencement of the work: work shall be started within 15 days since issue of LOA.

Quarterly Statement of Claims: The Contractor shall prepare and furnish to the Engineer once in every quarter commencing from the month following the month of issue of Letter of Acceptance, an account giving full and detailed particulars of all claims for any additional expenses to which the Contractor may consider himself entitled to and of all extra or additional works ordered by the Engineer which he has executed during the preceding quarter and no claim for payment for such work will be considered which has not been included in such particulars.

shramik kalyan: Contractor is to abide by the provisions of Payment of Wages Act & Minimum Wages Act in terms of clause 54 & 55 of Indian Railways General Conditions of Contract. Corrections if any:: While every care has been taken to fully incorporate the decision of the Accepting Authority regarding acceptance of your offer, the Railway administration reserves the rights to correct the inadvertent mistakes which may have occurred in communication of the accepted rates and other terms and conditions at the time of signing the formal contract agreement between the contractor and the Railway administration

labour licence : You are also required to obtain the necessary license for employing contract labour from ALC (C) Jabalpur under intimation to this office. The certificate in form rule 12(2) of the labour regulations Act is enclosed. Please get yourself registered immediately under the contractual labour (Regulation and Abolition) Act 1970 with the Regional / Assistant Labour Commissioner, Government of India and intimate the date commencement and completion of work to the concerned Regional Labour Commissioner

Site order book: It is mentioned for your information that the site order book and register will be maintained at site by the engineer incharge and instruction communicated regarding the execution of the work shall be duly recorded and daily labour and work done will be reflected in daily register.

NOTE:- while every care has been taken to fully incorporate, the decision of the accepting authority regarding acceptance of your offer, the railway administration reserves the rights to correct the inadvertent mistakes which may be have occurred in communication of the accepted rates and other terms and conditions at the time of signing the formal contract agreement between the contractor and railway administration.

All Other terms and conditions, as stipulated in the tender documents shall be applicable.

**GHAMANDI LAL MEENA**  
**Dy.CE/C-1/JBP**  
**Digitally Signed**  
[View Signature Details](#)

#### Awarded Quantities And Rates

| Item Sno.                                                                                 | Item Desc                                                                                                                                                                                                                                                                                                                                                                        | Item Code | Item Qty | Qty Unit | Unit Rate(Rs) | Escl. (%) | Advt.Value (Rs)    | Bid Rate/ Unit Rate | Bid Amount (Rs) |
|-------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|----------|----------|---------------|-----------|--------------------|---------------------|-----------------|
| <b>Schedule A-Earthwork and concrete work etc. (Item Directory - CPWD DSR 2021-Ver-2)</b> |                                                                                                                                                                                                                                                                                                                                                                                  |           |          |          |               |           | <b>25089061.37</b> | 8.08 % Below        | 23061865.21     |
| 1                                                                                         | Earth work in excavation by mechanical means (Hydraulic excavator) / manual means in foundation trenches or drains (not exceeding 1.5 m in width or 10 sqm on plan), including dressing of sides and ramming of bottoms, lift upto 1.5 m, including getting out the excavated soil and disposal of surplus excavated soil as directed, within a lead of 50 m. All kinds of soil. | 2.8.1     | 1150     | cum      | 286.85        | (-) 10.91 | 293887.86          |                     |                 |
| 2                                                                                         | Filling available excavated earth (excluding rock) in trenches, plinth, sides of foundations etc.                                                                                                                                                                                                                                                                                | 2.25      | 225      | cum      | 253.95        | (-) 10.91 | 50904.91           |                     |                 |

|   |                                                                                                                                                                                                                                                        |        |       |       |         |              |           |  |  |
|---|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|-------|-------|---------|--------------|-----------|--|--|
|   | in layers not exceeding 20cm in depth, consolidating each deposited layer by ramming and watering, lead up to 50 m and lift upto 1.5 m.                                                                                                                |        |       |       |         |              |           |  |  |
| 3 | Excavating, supplying and filling of local earth (including royalty) by mechanical transport upto a lead of 5km also including ramming and watering of the earth in layers not exceeding 20 cm in trenches, plinth, sides of foundation etc. complete. | 2.25A  | 450   | cum   | 368.65  | (-)<br>10.91 | 147793.63 |  |  |
| 4 | Extra for every additional lift of 1.5 m or part thereof in excavation / banking excavated or stacked materials. All kinds of soil                                                                                                                     | 2.26.1 | 410.5 | cum   | 104.50  | (-)<br>10.91 | 38217.16  |  |  |
| 5 | Supplying and filling in plinth with sand under floors, including watering, ramming, consolidating and dressing complete.                                                                                                                              | 2.27   | 40.5  | cum   | 2161.20 | (-)<br>10.91 | 77979.23  |  |  |
| 6 | Surface dressing of the ground including removing vegetation and in-equalities not exceeding 15 cm deep and disposal of rubbish, lead up to 50 m and lift up to 1.5 m. All kinds of soil                                                               | 2.28.1 | 60    | Sqm   | 28.15   | (-)<br>10.91 | 1504.73   |  |  |
| 7 | Supplying chemical emulsion in sealed containers including delivery as specified. Chlorpyriphos emulsifiable concentrate of 20%                                                                                                                        | 2.34.1 | 190   | Litre | 200.90  | (-)<br>10.91 | 34006.54  |  |  |

|    |                                                                                                                                                                                                                                                                                                        |          |     |       |         |              |           |  |  |
|----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-----|-------|---------|--------------|-----------|--|--|
| 8  | Along external wall where the apron is not provided using chemical emulsion @ 7.5 litres / sqm of the vertical surface of the substructure to a depth of 300mm including excavation channel along the wall & rodding etc. complete: With Chlorpyrifos E.C. 20% with 1% concentration                   | 2.35.1.1 | 190 | Metre | 32.30   | (-)<br>10.91 | 5467.45   |  |  |
| 9  | Providing and laying in position cement concrete of specified grade excluding the cost of centering and shuttering - All work up to plinth level : 1:2:4 (1 cement : 2 coarse sand (zone-III) derived from natural sources : 4 graded stone aggregate 20 mm nominal size derived from natural sources) | 4.1.3    | 90  | cum   | 7365.15 | (-)<br>10.91 | 590545.09 |  |  |
| 10 | Providing and laying in position cement concrete of specified grade excluding the cost of centering and shuttering - All work up to plinth level : 1:3:6 (1 Cement : 3 coarse sand (zone-III) derived from natural sources : 6 graded stone aggregate 20 mm nominal size derived from natural sources) | 4.1.5    | 75  | cum   | 6833.40 | (-)<br>10.91 | 456590.70 |  |  |
| 11 | Providing and laying in position cement concrete of specified grade excluding the cost of centering and shuttering - All work up to plinth level : 1:3:6 (1 Cement : 3 coarse sand                                                                                                                     | 4.1.6    | 80  | cum   | 6670.25 | (-)<br>10.91 | 475402.06 |  |  |

|    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |       |      |     |         |              |            |  |  |
|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|------|-----|---------|--------------|------------|--|--|
|    | (zone-III)<br>derived from<br>natural sources:<br>6 graded stone<br>aggregate 40 mm<br>nominal size<br>derived from<br>natural sources)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |       |      |     |         |              |            |  |  |
| 12 | Providing and<br>laying cement<br>concrete in<br>retaining walls,<br>return walls,<br>walls (any<br>thickness)<br>including<br>attached<br>pilasters,<br>columns, piers,<br>abutments,<br>pillars, posts,<br>struts,<br>buttresses, string<br>or lacing courses,<br>parapets, coping,<br>bed blocks,<br>anchor blocks,<br>plain window<br>sills, fillets,<br>sunken floor etc.,<br>up to floor five<br>level, excluding<br>the cost of<br>centering,<br>shuttering and<br>finishing: 1:2:4<br>(1 Cement : 2<br>coarse sand<br>(zone-III)<br>derived from<br>natural sources :<br>4 graded stone<br>aggregate 20 mm<br>nominal size<br>derived from<br>natural sources) | 4.2.3 | 250  | cum | 9375.20 | (-)<br>10.91 | 2088091.42 |  |  |
| 13 | Centering and<br>shuttering<br>including<br>strutting,<br>propping etc. and<br>removal of form<br>work for :<br>Foundations,<br>footings, bases<br>for columns                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 4.3.1 | 200  | Sqm | 307.95  | (-)<br>10.91 | 54870.53   |  |  |
| 14 | Centering and<br>shuttering<br>including<br>strutting,<br>propping etc. and<br>removal of form<br>work for :<br>Retaining walls,<br>return walls,<br>walls (any<br>thickness)<br>including<br>attached                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 4.3.2 | 1800 | Sqm | 669.55  | (-)<br>10.91 | 1073703.77 |  |  |



|    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |       |     |     |         |           |          |  |  |
|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-----|-----|---------|-----------|----------|--|--|
|    | pilasters, buttresses, plinth and string courses fillets, kerbs and steps etc.                                                                                                                                                                                                                                                                                                                                                                                                            |       |     |     |         |           |          |  |  |
| 15 | Providing and laying cement concrete in kerbs, steps and the like at or near ground level excluding the cost of centering, shuttering and finishing. 1:1.5:3 (1 Cement: 1.5 coarse sand(zone-III) derived from natural sources : 3 graded stone aggregate 20 mm nominal size derived from natural sources)                                                                                                                                                                                | 4.4.1 | 6   | cum | 7783.65 | (-) 10.91 | 41606.72 |  |  |
| 16 | Providing and fixing up to floor five level precast cement concrete string or lacing courses, copings, bed plates, anchor blocks, plain window sills, shelves, louvers, steps, stair cases, etc., including hoisting and setting in position with cement mortar 1:3 (1 Cement : 3 coarse sand), cost of required Centering complete. 1:1.5:3 (1 cement : 1.5 coarse sand(zone-III) derived from natural sources: 3 graded stone aggregate 20mm nominal size derived from natural sources) | 4.5.1 | 6   | cum | 9277.75 | (-) 10.91 | 49593.28 |  |  |
| 17 | Providing and laying damp-proof course 50mm thick with cement concrete 1:2:4 (1 cement : 2 coarse sand(zone-III) derived from natural sources : 4 graded stone                                                                                                                                                                                                                                                                                                                            | 4.11  | 7.5 | Sqm | 452.25  | (-) 10.91 | 3021.83  |  |  |

|    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |          |     |     |          |           |            |  |  |
|----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-----|-----|----------|-----------|------------|--|--|
|    | aggregate 20mm nominal size derived from natural sources).                                                                                                                                                                                                                                                                                                                                                                                                                                   |          |     |     |          |           |            |  |  |
| 18 | Reinforced cement concrete work in beams, suspended floors, roofs having slope up to 15 degree landings, balconies, shelves, chajjas, lintels, bands, plain window sills, staircases and spiral stair cases above plinth level up to floor five level, excluding the cost of centering, shuttering, finishing and reinforcement with 1:1.5:3 (1 cement : 1.5 coarse sand(zone-III) derived from natural sources : 3 graded stone aggregate 20 mm nominal size derived from natural sources). | 5.3      | 14  | cum | 10719.30 | (-) 10.91 | 133697.54  |  |  |
| 19 | Reinforced cement concrete work in arches, archribs, domes, vaults, shells, folded plate and roofs having slope more than 15 degree up to floor five level, excluding the cost of centering, shuttering, finishing and reinforcement with 1:1.5:3 (1 cement : 1.5 coarse sand (zone-III) derived from natural sources: 3 graded stone aggregate 20 mm nominal size derived from natural sources).                                                                                            | 5.5      | 4   | cum | 11291.20 | (-) 10.91 | 40237.32   |  |  |
| 20 | All works upto plinth level Concrete of M25 grade with minimum cement content of 330 kg /cum                                                                                                                                                                                                                                                                                                                                                                                                 | 5.33.1.1 | 165 | cum | 8683.80  | (-) 10.91 | 1276505.57 |  |  |

|    |                                                                                                                                                                                                                                                 |          |     |       |          |              |            |  |  |
|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-----|-------|----------|--------------|------------|--|--|
| 21 | All works above plinth level upto floor V level<br>Concrete of M25 grade with minimum cement content of 330 kg /cum                                                                                                                             | 5.33.2.1 | 150 | cum   | 10306.20 | (-)<br>10.91 | 1377269.04 |  |  |
| 22 | Centering and shuttering including strutting, propping etc. and removal of form for Foundations, footings, bases of columns, etc. for mass concrete                                                                                             | 5.9.1    | 245 | Sqm   | 307.95   | (-)<br>10.91 | 67216.40   |  |  |
| 23 | Centering and shuttering including strutting, propping etc. and removal of form for Suspended floors, roofs, landings, balconies and access platform                                                                                            | 5.9.3    | 356 | Sqm   | 766.55   | (-)<br>10.91 | 243119.30  |  |  |
| 24 | Centering and shuttering including strutting, propping etc. and removal of form for Lintels, beams, plinth beams, girders, bressumers and cantilevers                                                                                           | 5.9.5    | 627 | Sqm   | 608.35   | (-)<br>10.91 | 339820.84  |  |  |
| 25 | Centering and shuttering including strutting, propping etc. and removal of form for Small lintels not exceeding 1.5 m clear span, moulding as in cornices, window sills, string courses, bands, copings, bed plates, anchor blocks and the like | 5.9.15   | 110 | Sqm   | 307.95   | (-)<br>10.91 | 30178.79   |  |  |
| 26 | Edges of slabs and breaks in floors and walls Under 20 cm wide                                                                                                                                                                                  | 5.9.16.1 | 150 | Metre | 181.90   | (-)<br>10.91 | 24308.21   |  |  |
| 27 | Brick work with common burnt clay F.P.S. (non modular) bricks of class designation 7.5                                                                                                                                                          | 6.1.1    | 55  | cum   | 6882.00  | (-)<br>10.91 | 337214.56  |  |  |

|    |                                                                                                                                                                                                                                                                    |         |      |       |         |              |            |  |  |
|----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|------|-------|---------|--------------|------------|--|--|
|    | in foundation and plinth in: Cement mortar 1:4 (1 cement : 4 coarse sand)                                                                                                                                                                                          |         |      |       |         |              |            |  |  |
| 28 | Half brick masonry with common burnt clay F.P.S. (non modular) bricks of class designation 7.5 in superstructure above plinth level up to floor V level. Cement mortar 1:4 (1 cement :4 coarse sand)                                                               | 6.13.2  | 15   | Sqm   | 1018.05 | (-)<br>10.91 | 13604.71   |  |  |
| 29 | Extra for providing and placing in position 2 Nos 6mm dia. M.S. bars at every third course of half brick masonry.                                                                                                                                                  | 6.15    | 11   | Sqm   | 86.45   | (-)<br>10.91 | 847.20     |  |  |
| 30 | Brick work with common burnt clay machine moulded perforated bricks of class designation 12.5 conforming to IS: 2222 in superstructure above plinth level up to floor five level in cement mortar 1:6 (1 cement : 6 coarse sand) : With F.P.S.(non modular) bricks | 6.3.1   | 269  | cum   | 8030.60 | (-)<br>10.91 | 1924550.15 |  |  |
| 31 | Providing edge moulding to 18 mm thick marble stone counters, Vanities etc., including machine polishing to edge to give high gloss finish etc. complete as per design approved by Engineer-in-Charge. Granite work                                                | 8.3.2   | 12.5 | Metre | 418.85  | (-)<br>10.91 | 4664.42    |  |  |
| 32 | Float glass panes 5.0 mm thick glass panes (weight not less than 12.50 kg/sqm).                                                                                                                                                                                    | 9.7.7.2 | 157  | Sqm   | 2231.20 | (-)<br>10.91 | 312080.84  |  |  |

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| 33 | Providing and fixing ISI marked flush door shutters conforming to IS : 2202 (Part I) decorative type, core of block board construction with frame of 1st class hard wood and well matched teak 3 ply veneering with vertical grains or cross bands and face veneers on both faces of shutters. 35 mm thick including ISI marked Stainless Steel butt hinges with necessary screws                                                     | 9.20.1 | 79  | Sqm | 3086.10 | (-)<br>10.91 | 217203.11 |  |  |
| 34 | Providing and fixing ISI marked flush door shutters conforming to IS : 2202 (Part I) decorative type, core of block board construction with frame of 1st class hard wood and well matched teak 3 ply veneering with vertical grains or cross bands and face veneers on both faces of shutters. 25 mm thick (for cupboard) including ISI marked nickel plated bright finished M.S. Piano hinges IS : 3818 marked with necessary screws | 9.20.3 | 130 | Sqm | 2512.35 | (-)<br>10.91 | 290972.84 |  |  |
| 35 | Providing and fixing ISI marked flush door shutters conforming to IS : 2202 (Part I) non-decorative type, core of block board construction with frame of 1st class hard wood and well matched commercial 3 ply                                                                                                                                                                                                                        | 9.21.2 | 36  | Sqm | 1881.95 | (-)<br>10.91 | 60358.65  |  |  |

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|    | veneering with vertical grains or cross bands and face veneers on both faces of shutters: 30 mm thick including ISI marked Stainless Steel butt hinges with necessary screws                                                |            |      |      |         |              |           |  |  |
| 36 | with ISI marked M.S. pressed butt hinges bright finished of required size Second class teak wood                                                                                                                            | 9.27.2.1.1 | 114  | Sqm  | 3863.65 | (-)<br>10.91 | 392402.34 |  |  |
| 37 | Providing and fixing M.S. grills of required pattern in frames of windows etc. with M.S. flats, square or round bars etc. including priming coat with approved steel primer all complete. Fixed to steel windows by welding | 9.48.1     | 1500 | Kg   | 181.00  | (-)<br>10.91 | 241879.35 |  |  |
| 38 | Providing and fixing ISI marked oxidised M.S. sliding door bolts with nuts and screws etc. complete : (copper oxidized as per IS: 1378) 300x16 mm                                                                           | 9.62.1     | 62   | Each | 180.55  | (-)<br>10.91 | 9972.82   |  |  |
| 39 | Providing and fixing ISI marked oxidised M.S. sliding door bolts with nuts and screws etc. complete : (copper oxidized as per IS: 1378) 250x16 mm                                                                           | 9.62.2     | 288  | Each | 165.90  | (-)<br>10.91 | 42566.49  |  |  |
| 40 | Providing and fixing ISI marked oxidised M.S. tower bolt black finish, (Barrel type) with necessary screws etc. complete : (copper oxidized as per IS: 1378) 150x10 mm                                                      | 9.63.3     | 76   | Each | 53.05   | (-)<br>10.91 | 3591.93   |  |  |
| 41 | Providing and fixing ISI marked oxidised M.S. tower bolt black                                                                                                                                                              | 9.63.4     | 150  | Each | 39.10   | (-)<br>10.91 | 5225.13   |  |  |

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|    | finish, (Barrel type) with necessary screws etc. complete : (copper oxidized as per IS: 1378) 100x10 mm                                                                                                                                                                                                                    |        |     |      |         |           |          |  |  |
| 42 | Providing and fixing ISI marked oxidised M.S. handles conforming to IS:4992 with necessary screws etc. complete : (copper oxidized as per IS: 1378) 125 mm                                                                                                                                                                 | 9.66.1 | 224 | Each | 35.20   | (-) 10.91 | 7024.57  |  |  |
| 43 | Providing and fixing aluminium sliding door bolts, ISI marked anodised (anodic coating not less than grade AC 10 as per IS : 1868), transparent or dyed to required colour or shade, with nuts and screws etc. complete : 250x16 mm                                                                                        | 9.96.2 | 25  | Each | 234.90  | (-) 10.91 | 5231.81  |  |  |
| 44 | Structural steel work in single section, fixed with or without connecting plate, including cutting, hoisting, fixing in position and applying a priming coat of approved steel primer all complete.                                                                                                                        | 10.1   | 170 | Kg   | 93.05   | (-) 10.91 | 14092.70 |  |  |
| 45 | Supplying and fixing rolling shutters of approved make, made of required size M.S. laths, interlocked together through their entire length and jointed together at the end by end locks, mounted on specially designed pipe shaft with brackets, side guides and arrangements for inside and outside locking with push and | 10.6.1 | 36  | Sqm  | 3008.80 | (-) 10.91 | 96499.44 |  |  |

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|    | pull operation complete, including the cost of providing and fixing necessary 27.5 cm long wire springs manufactured from high tensile steel wire of adequate strength conforming to IS: 4454 - part 1 and M.S. top cover of required thickness for rolling shutters. 80x1.25 mm M.S. laths with 1.25 mm thick top cover |           |     |       |         |              |          |  |  |
| 46 | Extra for providing mechanical device chain and crank operation for operating rolling shutters. Exceeding 10.00 sqm and upto 16.80 sqm in the area                                                                                                                                                                       | 10.8.1    | 4   | Sqm   | 1108.70 | (-)<br>10.91 | 3950.96  |  |  |
| 47 | Profile C Fixing with adjustable lugs with split end tail to each jamb                                                                                                                                                                                                                                                   | 10.14.2.1 | 112 | Metre | 471.25  | (-)<br>10.91 | 47021.70 |  |  |
| 48 | Profile E Fixing with adjustable lugs with split end tail to each jamb                                                                                                                                                                                                                                                   | 10.14.3.1 | 8   | Metre | 511.40  | (-)<br>10.91 | 3644.85  |  |  |
| 49 | Steel work in built up tubular (round, square or rectangular hollow tubes etc.) trusses etc., including cutting, hoisting, fixing in position and applying a priming coat of approved steel primer, including welding and bolted with special shaped washers etc. complete. Hot finished welded type tubes               | 10.16.1   | 500 | Kg    | 154.90  | (-)<br>10.91 | 69000.21 |  |  |
| 50 | Providing and fixing M.S. fan clamp type I or II of 16 mm dia M.S. bar, bent to shape with                                                                                                                                                                                                                               | 10.17     | 26  | Each  | 185.95  | (-)<br>10.91 | 4307.23  |  |  |



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|    | hooked ends in R.C.C. slabs or beams during laying, including painting the exposed portion of loop, all as per standard design complete.                                                                                                                                                                                                                                                                                  |         |      |     |         |              |           |  |  |
| 51 | Steel work welded in built up sections/ framed work, including cutting, hoisting, fixing in position and applying a priming coat of approved steel primer using structural steel etc. as required. In gratings, frames, guard bar, ladder, railings, brackets, gates and similar works                                                                                                                                    | 10.25.2 | 2200 | Kg  | 142.30  | (-)<br>10.91 | 278905.15 |  |  |
| 52 | Providing and fixing hand rail of approved size by welding etc. to steel ladder railing, balcony railing, staircase railing and similar works, including applying priming coat of approved steel primer. M.S. tube                                                                                                                                                                                                        | 10.26.1 | 145  | Kg  | 157.15  | (-)<br>10.91 | 20300.72  |  |  |
| 53 | Providing and laying vitrified floor tiles in different sizes (thickness to be specified by the manufacturer) with water absorption less than 0.08% and conforming to IS: 15622, of approved make, in all colours and shades, laid on 20mm thick cement mortar 1:4 (1 cement : 4 coarse sand), jointing with grey cement slurry @ 3.3 kg/ sqm including grouting the joints with white cement and matching pigments etc., | 11.41.2 | 608  | Sqm | 1416.65 | (-)<br>10.91 | 767352.84 |  |  |

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|    | complete. Size of Tile 600x600 mm                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |         |      |     |         |           |            |  |  |
| 54 | Providing and laying Polished Granite stone flooring in required design and patterns, in linear as well as curvilinear portions of the building all complete as per the architectural drawings with 18 mm thick stone slab over 20 mm (average) thick base of cement mortar 1:4 (1 cement : 4 coarse sand) laid and jointed with cement slurry and pointing with white cement slurry admixed with pigment of matching shade including rubbing , curing and polishing etc. all complete as specified and as directed by the Engineer-in-Charge. Polished Granite stone slab colour of Black, Cherry/Ruby Red or equivalent | 11.56.1 | 12   | Sqm | 3908.80 | (-) 10.91 | 41788.20   |  |  |
| 55 | 12 mm cement plaster of mix : 1:4 (1 cement: 4 fine sand)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 13.1.1  | 1355 | Sqm | 294.85  | (-) 10.91 | 355933.93  |  |  |
| 56 | 15 mm cement plaster on the rough side of single or half brick wall of mix : 1:4 (1 cement: 4 fine sand)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 13.2.1  | 1370 | Sqm | 339.70  | (-) 10.91 | 414615.06  |  |  |
| 57 | 20 mm cement plaster of mix : 1:4 (1 cement: 4 fine sand)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 13.3.1  | 100  | Sqm | 402.15  | (-) 10.91 | 35827.54   |  |  |
| 58 | Providing and applying plaster of paris putty of 2 mm thickness over plastered surface to prepare the surface even and smooth complete.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 13.26   | 7585 | Sqm | 214.30  | (-) 10.91 | 1448127.21 |  |  |

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| 59 | New work (two or more coats) with a base coat of white washing with lime                                                                                                                                                                                                                                                                  | 13.39.1 | 8660 | Sqm | 44.25  | (-)<br>10.91 | 341397.33  |  |  |
| 60 | New work (two or more coats) over and including water thinnable priming coat with cement primer                                                                                                                                                                                                                                           | 13.41.1 | 7585 | Sqm | 162.55 | (-)<br>10.91 | 1098427.81 |  |  |
| 61 | Finishing with Deluxe Multi surface paint system for interiors and exteriors using Primer as per manufacturers specifications : Painting wood work with Deluxe Multi Surface Paint of required shade. Two or more coat applied @ 0.90 ltr/10 sqm over an under coat of primer applied @ 0.75 ltr/10 sqm of approved brand and manufacture | 13.48.2 | 700  | Sqm | 144.90 | (-)<br>10.91 | 90363.99   |  |  |
| 62 | Applying priming coat: With ready mixed pink or Grey primer of approved brand and manufacture on wood work (hard and soft wood)                                                                                                                                                                                                           | 13.50.1 | 600  | Sqm | 61.45  | (-)<br>10.91 | 32847.48   |  |  |
| 63 | Two or more coats on new work                                                                                                                                                                                                                                                                                                             | 13.61.1 | 300  | Sqm | 131.45 | (-)<br>10.91 | 35132.64   |  |  |
| 64 | Preparation and consolidation of sub grade with power road roller of 8 to 12 tonne capacity after excavating earth to an average of 22.5 cm depth, dressing to camber and consolidating with road roller including making good the undulations etc. and re-rolling the sub grade and disposal of surplus earthwith lead upto 50 metres.   | 16.1    | 825  | Sqm | 180.50 | (-)<br>10.91 | 132666.15  |  |  |

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| 65 | Supplying and stacking at site. 63 mm to 45 mm size stone aggregate                                                                                                                                                                                                                                                                                                                                     | 16.3.2  | 85  | cum | 1624.50 | (-) 10.91 | 123017.70 |  |  |
| 66 | Supplying and stacking at site. 53 mm to 22.4 mm size stone aggregate                                                                                                                                                                                                                                                                                                                                   | 16.3.3  | 65  | cum | 1837.25 | (-) 10.91 | 106392.39 |  |  |
| 67 | Supplying and stacking at site. Moorum                                                                                                                                                                                                                                                                                                                                                                  | 16.3.10 | 475 | cum | 888.30  | (-) 10.91 | 375908.57 |  |  |
| 68 | Laying, spreading and compacting stone aggregate of specified sizes to WBM specifications in uniform thickness, hand picking, rolling with 3 wheeled road/vibratory roller 8-10 tonne capacity in stages to proper grade and camber, applying and brooming requisite type of screening / binding material to fill up interstices of coarse aggregate, watering and compacting to the required density . | 16.4    | 150 | cum | 865.80  | (-) 10.91 | 115701.18 |  |  |
| 69 | Supplying, stacking and Spreading 6 mm thick red bajri, watering and rolling complete including preparation of the surface and rolling. With road roller/ hand roller                                                                                                                                                                                                                                   | 16.6.1  | 825 | Sqm | 21.80   | (-) 10.91 | 16022.84  |  |  |
| 70 | Cement concrete 1:2:4 (1 cement : 2 coarse sand : 4 graded stone aggregate 40 mm nominal size) in pavements, laid to required slope and camber in panels as required including consolidation finishing and tamping complete.                                                                                                                                                                            | 16.42   | 85  | cum | 7399.20 | (-) 10.91 | 560315.52 |  |  |

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| 71 | Providing and laying design mix cement concrete of M-30 grade, in roads/ taxi tracks/ runways, using cement content as per design mix, using coarse sand and graded stone aggregate of 40 mm nominal size in appropriate proportions as per approved & specified design criteria, providing dowel bars with sleeve/ tie bars wherever required, laying at site, spreading and compacting mechanically by using needle and surface vibrators, levelling to required slope/ camber, finishing with required texture, including steel form work with sturdy M.S. channel sections, curing, making provision for contraction/ expansion, construction & longitudinal joints (10 mm wide x 50 mm deep) by groove cutting machine, providing and filling joints with approved joint filler and sealants, complete all as per direction of Engineer-in-charge (Item of joint fillers, sealants, dowel bars with sleeve/ tie bars to be paid separately). Cement concrete prepared with batch mixing machine | 16.43.1 | 165 | cum | 9492.90 | (-)<br>10.91 | 1395442.06 |  |  |
| 72 | Providing and laying at or near ground level factory made kerb stone of M-25 grade cement                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 16.69   | 15  | cum | 8613.55 | (-)<br>10.91 | 115107.18  |  |  |

|    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |         |     |     |        |           |           |  |  |
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|    | concrete in position to the required line, level and curvature, jointed with cement mortar 1:3 (1 cement: 3 coarse sand), including making joints with or without grooves (thickness of joints except at sharp curve shall not to more than 5mm), including making drainage opening wherever required complete etc. as per direction of Engineer-in-charge (length of finished kerb edging shall be measured for payment). (Precast C.C. kerb stone shall be approved by Engineer-in-charge).                                            |         |     |     |        |           |           |  |  |
| 73 | Providing and laying factory made chamfered edge Cement Concrete paver blocks in footpath, parks, lawns, drive ways or light traffic parking etc, of required strength, thickness & size/ shape, made by table vibratory method using PU mould, laid in required colour & pattern over 50mm thick compacted bed of sand, compacting and proper embedding/laying of inter locking paver blocks into the sand bedding layer through vibratory compaction by using plate vibrator, filling the joints with sand and cutting of paver blocks | 16.91.1 | 450 | Sqm | 932.35 | (-) 10.91 | 373783.78 |  |  |

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|    | as per required size and pattern, finishing and sweeping extra sand. complete all as per direction of Engineer-in-Charge. 60mm thick cement concrete paver block of M-35 grade with approved colour, design & pattern.                                                                                                                                                                                                                          |        |    |      |         |              |           |  |  |
| 74 | Providing and fixing white vitreous china pedestal type water closet (European type W.C. pan) with seat and lid, 10 litre low level white P.V.C. flushing cistern, including flush pipe, with manually controlled device (handle lever), conforming to IS : 7231, with all fittings and fixtures complete, including cutting and making good the walls and floors wherever required : W.C. pan with ISI marked white solid plastic seat and lid | 17.2.1 | 31 | Each | 5540.55 | (-)<br>10.91 | 153018.36 |  |  |
| 75 | Providing and fixing wash basin with C.I. brackets, 15 mm C.P. brass pillar taps, 32 mm C.P. brass waste of standard pattern, including painting of fittings and brackets, cutting and making good the walls wherever require: White Vitreous China Wash basin size 630x450 mm with a single 15 mm C.P. brass pillar tap                                                                                                                        | 17.7.2 | 5  | Each | 1913.75 | (-)<br>10.91 | 8524.80   |  |  |
| 76 | Providing and fixing wash basin with C.I.                                                                                                                                                                                                                                                                                                                                                                                                       | 17.7.4 | 29 | Each | 1679.60 | (-)<br>10.91 | 43394.31  |  |  |

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|    | brackets, 15 mm C.P. brass pillar taps, 32 mm C.P. brass waste of standard pattern, including painting of fittings and brackets, cutting and making good the walls wherever require: White Vitreous China Flat back wash basin size 550x400 mm with single 15 mm C.P. brass pillar tap                                     |           |    |      |         |           |           |  |  |
| 77 | Providing and fixing wash basin with C.I. brackets, 15 mm PTMT pillar cock, 32 mm PTMT waste coupling of standard pattern, including painting of fittings and brackets, cutting and making good the walls wherever required. White Vitreous China Flat back wash basin size 550x400 mm with single 15 mm PTMT pillar cock. | 17.7B     | 10 | Each | 1404.00 | (-) 10.91 | 12508.24  |  |  |
| 78 | Kitchen sink with drain board 510x1040 mm bowl depth 250 mm                                                                                                                                                                                                                                                                | 17.10.1.1 | 28 | Each | 6008.45 | (-) 10.91 | 149881.99 |  |  |
| 79 | Kitchen sink without drain board 610x460 mm bowl depth 200 mm                                                                                                                                                                                                                                                              | 17.10.2.2 | 32 | Each | 3337.85 | (-) 10.91 | 95158.10  |  |  |
| 80 | Flexible pipe 32 mm dia                                                                                                                                                                                                                                                                                                    | 17.28.2.1 | 8  | Each | 104.35  | (-) 10.91 | 743.72    |  |  |
| 81 | Providing and fixing PTMT liquid soap container 109 mm wide, 125 mm high and 112 mm distance from wall of standard shape with bracket of the same materials with snap fittings of approved quality and colour, weighing not less than 105 gms.                                                                             | 17.71     | 6  | Each | 146.30  | (-) 10.91 | 782.03    |  |  |



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| 82 | Providing and fixing PTMT towel rail complete with brackets fixed to wooden cleats with CP brass screws with concealed fittings arrangement of approved quality and colour. 600 mm long towel rail with total length of 645 mm, width 78 mm and effective height of 88 mm, weighing not less than 190 gms.                                                                                                                                       | 17.73.2 | 6  | Each  | 600.35 | (-)<br>10.91 | 3209.11 |  |  |
| 83 | Providing and fixing Chlorinated Polyvinyl Chloride (CPVC) pipes, having thermal stability for hot & cold water supply, including all CPVC plain & brass threaded fittings, including fixing the pipe with clamps at 1.00 m spacing. This includes jointing of pipes & fittings with one step CPVC solvent cement and testing of joints complete as per direction of Engineer in Charge.-Internal work - Exposed on wall 15 mm nominal dia Pipes | 18.7.1  | 25 | Metre | 255.90 | (-)<br>10.91 | 5699.53 |  |  |
| 84 | Providing and fixing Chlorinated Polyvinyl Chloride (CPVC) pipes, having thermal stability for hot & cold water supply, including all CPVC plain & brass threaded fittings, including fixing the pipe with clamps at 1.00 m spacing. This includes jointing of pipes & fittings with one step CPVC solvent cement and testing of joints complete as per direction of Engineer in                                                                 | 18.7.2  | 25 | Metre | 325.10 | (-)<br>10.91 | 7240.79 |  |  |

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|    | Charge.-Internal work - Exposed on wall 20 mm nominal dia Pipes                                                                                                                                                                                                                                                                                                                                                                                  |        |    |       |        |              |          |  |  |
| 85 | Providing and fixing Chlorinated Polyvinyl Chloride (CPVC) pipes, having thermal stability for hot & cold water supply, including all CPVC plain & brass threaded fittings, including fixing the pipe with clamps at 1.00 m spacing. This includes jointing of pipes & fittings with one step CPVC solvent cement and testing of joints complete as per direction of Engineer in Charge.-Internal work - Exposed on wall 25 mm nominal dia Pipes | 18.7.3 | 25 | Metre | 408.55 | (-)<br>10.91 | 9099.43  |  |  |
| 86 | Providing and fixing Chlorinated Polyvinyl Chloride (CPVC) pipes, having thermal stability for hot & cold water supply, including all CPVC plain & brass threaded fittings, including fixing the pipe with clamps at 1.00 m spacing. This includes jointing of pipes & fittings with one step CPVC solvent cement and testing of joints complete as per direction of Engineer in Charge.-Internal work - Exposed on wall 40 mm nominal dia Pipes | 18.7.5 | 15 | Metre | 674.35 | (-)<br>10.91 | 9011.68  |  |  |
| 87 | Providing and fixing Chlorinated Polyvinyl Chloride (CPVC) pipes, having thermal stability for hot & cold water supply, including all CPVC plain & brass threaded fittings, including                                                                                                                                                                                                                                                            | 18.7.6 | 15 | Metre | 927.00 | (-)<br>10.91 | 12387.96 |  |  |

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|    | fixing the pipe with clamps at 1.00 m spacing. This includes jointing of pipes & fittings with one step CPVC solvent cement and testing of joints complete as per direction of Engineer in Charge.-Internal work - Exposed on wall 50 mm nominal dia Pipes                                                                                                                                                                                                                                                                                      |        |    |       |        |              |          |  |  |
| 88 | Providing and fixing Chlorinated Polyvinyl Chloride (CPVC) pipes, having thermal stability for hot & cold water supply, including all CPVC plain & brass threaded fittings, i/c fixing the pipe with clamps at 1.00 m spacing. This includes jointing of pipes & fittings with one step CPVC solvent cement and the cost of cutting chases and making good the same including testing of joints complete as per direction of Engineer in Charge, Concealed work, including cutting chases and making good the wall etc. 15 mm nominal dia Pipes | 18.8.1 | 60 | Metre | 441.15 | (-)<br>10.91 | 23581.23 |  |  |
| 89 | Providing and fixing Chlorinated Polyvinyl Chloride (CPVC) pipes, having thermal stability for hot & cold water supply, including all CPVC plain & brass threaded fittings, i/c fixing the pipe with clamps at 1.00 m spacing. This includes jointing of pipes & fittings with one step CPVC solvent cement and the                                                                                                                                                                                                                             | 18.8.2 | 60 | Metre | 513.75 | (-)<br>10.91 | 27461.99 |  |  |

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|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|----|-------|---------|--------------|----------|--|--|
|    | cost of cutting chases and making good the same including testing of joints complete as per direction of Engineer in Charge, Concealed work, including cutting chases and making good the wall etc. 20 mm nominal dia Pipes                                                                                                                                                                                                                                                                                                                     |         |    |       |         |              |          |  |  |
| 90 | Providing and fixing Chlorinated Polyvinyl Chloride (CPVC) pipes, having thermal stability for hot & cold water supply, including all CPVC plain & brass threaded fittings, i/c fixing the pipe with clamps at 1.00 m spacing. This includes jointing of pipes & fittings with one step CPVC solvent cement and the cost of cutting chases and making good the same including testing of joints complete as per direction of Engineer in Charge, Concealed work, including cutting chases and making good the wall etc. 25 mm nominal dia Pipes | 18.8.3  | 60 | Metre | 626.05  | (-)<br>10.91 | 33464.88 |  |  |
| 91 | Making connection of G.I. distribution branch with G.I. main of following sizes by providing and fixing tee, including cutting and threading the pipe etc. complete : 25 to 40 mm nominal bore                                                                                                                                                                                                                                                                                                                                                  | 18.13.1 | 29 | Each  | 757.10  | (-)<br>10.91 | 19560.51 |  |  |
| 92 | Making connection of G.I. distribution branch with G.I. main of following sizes by                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 18.13.2 | 29 | Each  | 1513.70 | (-)<br>10.91 | 39108.10 |  |  |

|    |                                                                                                                                                                                                                                                                                                                                                                   |           |    |      |          |           |            |  |  |
|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|----|------|----------|-----------|------------|--|--|
|    | providing and fixing tee, including cutting and threading the pipe etc. complete : 50 to 80 mm nominal bore                                                                                                                                                                                                                                                       |           |    |      |          |           |            |  |  |
| 93 | Providing and fixing gun metal gate valve with C.I. wheel of approved quality (screwed end) : 25 mm nominal bore                                                                                                                                                                                                                                                  | 18.17.1   | 29 | Each | 532.35   | (-) 10.91 | 13753.85   |  |  |
| 94 | Providing and fixing gun metal gate valve with C.I. wheel of approved quality (screwed end) : 40 mm nominal bore                                                                                                                                                                                                                                                  | 18.17.3   | 29 | Each | 707.30   | (-) 10.91 | 18273.87   |  |  |
| 95 | Providing and fixing ball valve (brass) of approved quality, High or low pressure, with plastic floats complete : 25 mm nominal bore                                                                                                                                                                                                                              | 18.18.3   | 29 | Each | 399.15   | (-) 10.91 | 10312.48   |  |  |
| 96 | 40 mm nominal bore Horizontal                                                                                                                                                                                                                                                                                                                                     | 18.19.3.1 | 28 | Each | 815.05   | (-) 10.91 | 20331.59   |  |  |
| 97 | 30 cm length 15 mm nominal bore                                                                                                                                                                                                                                                                                                                                   | 18.21.1.1 | 29 | Each | 74.80    | (-) 10.91 | 1932.54    |  |  |
| 98 | Providing and fixing C.P. brass shower rose with 15 or 20 mm inlet : 100 mm diameter                                                                                                                                                                                                                                                                              | 18.22.1   | 9  | Each | 166.35   | (-) 10.91 | 1333.81    |  |  |
| 99 | Constructing masonry Chamber 90x90x100 cm inside, in brick work in cement mortar 1:4 (1 cement : 4 coarse sand) for sluice valve, with C.I. surface box 100 mm top diameter, 160 mm bottom diameter and 180 mm deep (inside) with chained lid and RCC top slab 1:2:4 mix (1 cement : 2 coarse sand : 4 graded stone aggregate 20 mm nominal size ), i/c necessary | 18.34.1   | 65 | Each | 17577.90 | (-) 10.91 | 1017909.82 |  |  |

|     |                                                                                                                                                                                                                                                                                                                                                                 |         |       |       |        |              |           |  |  |
|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|-------|-------|--------|--------------|-----------|--|--|
|     | excavation, foundation concrete 1:5:10 (1 cement : 5 fine sand : 10 graded stone aggregate 40 mm nominal size ) and inside plastering with cement mortar 1:3 (1 cement : 3 coarse sand) 12 mm thick, finished with a floating coat of neat cement complete as per standard design : With common burnt clay F.P.S. (non modular) bricks of class designation 7.5 |         |       |       |        |              |           |  |  |
| 100 | Providing and placing on terrace (at all floor levels) polyethylene water storage tank, IS : 12701 marked, with cover and suitable locking arrangement and making necessary holes for inlet, outlet and overflow pipes but without fittings and the base support for tank.                                                                                      | 18.48   | 28000 | Litre | 9.70   | (-)<br>10.91 | 241968.44 |  |  |
| 101 | Providing and fixing PTMT bib cock of approved quality and colour. 15 mm nominal bore, 90 mm long, weighing not less than 93 gms                                                                                                                                                                                                                                | 18.54.4 | 124   | Each  | 170.05 | (-)<br>10.91 | 18785.70  |  |  |
| 102 | Providing and fixing PTMT stop cock of approved quality and colour. 15 mm nominal bore, 86 mm long, weighing not less than 88 gms                                                                                                                                                                                                                               | 18.55.1 | 76    | Each  | 103.15 | (-)<br>10.91 | 6984.12   |  |  |
| 103 | Providing and fixing PTMT stop cock of approved quality and colour. Concealed stop cock, 15 mm nominal bore, 108 mm                                                                                                                                                                                                                                             | 18.55.3 | 76    | Each  | 183.40 | (-)<br>10.91 | 12417.72  |  |  |

|     |                                                                                                                                                                                                                    |           |    |      |        |              |          |  |  |
|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|----|------|--------|--------------|----------|--|--|
|     | long, weighing not less than 108 gms                                                                                                                                                                               |           |    |      |        |              |          |  |  |
| 104 | Providing and fixing PTMT pillar cock of approved quality and colour. 15 mm nominal bore, 125 mm long foam flow, weighing not less than 120 gms                                                                    | 18.56.2   | 76 | Each | 194.15 | (-)<br>10.91 | 13145.59 |  |  |
| 105 | Circular type 125 mm nominal dia with 25 mm waste hole                                                                                                                                                             | 18.58.1.2 | 76 | Each | 45.25  | (-)<br>10.91 | 3063.81  |  |  |
| 106 | Providing and fixing PTMT Ball cock of approved quality, colour and make complete with Epoxy coated aluminium rod with L.P./ H.P.H.D. plastic ball. 25 mm nominal bore, 152mm long, weighing not less than 440 gms | 18.62.3   | 29 | Each | 459.40 | (-)<br>10.91 | 11869.10 |  |  |
| 107 | Providing and fixing PTMT angle stop cock 15 mm nominal bore, weighing not less than 85 gms                                                                                                                        | 18.63     | 80 | Each | 136.60 | (-)<br>10.91 | 9735.76  |  |  |
| 108 | Providing and fixing PTMT swivelling shower, 15 mm nominal bore, weighing not less than 40 gms                                                                                                                     | 18.64     | 38 | Each | 106.15 | (-)<br>10.91 | 3593.62  |  |  |
| 109 | Providing and fixing unplasticised P.V.C. connection pipe with PTMT Nuts, collar and bush of approved quality and colour. 15 mm nominal bore with 30cm length                                                      | 18.74.1   | 38 | Each | 85.50  | (-)<br>10.91 | 2894.53  |  |  |
| 110 | Providing and fixing PTMT extension nipple for water tank pipe, fittings of approved quality and colour. 15 mm nominal bore, weighing not less than 32 gms                                                         | 18.75.1   | 38 | Each | 43.05  | (-)<br>10.91 | 1457.42  |  |  |

|     |                                                                                                                                                                                                                                                                                                            |          |     |       |          |              |           |  |  |
|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-----|-------|----------|--------------|-----------|--|--|
| 111 | Providing and fixing PTMT extension nipple for water tank pipe, fittings of approved quality and colour. 25mm nominal bore, weighing not less than 62 gms                                                                                                                                                  | 18.75.3  | 38  | Each  | 100.55   | (-)<br>10.91 | 3404.04   |  |  |
| 112 | Providing and laying cement concrete 1:5:10 (1 cement : 5 coarse sand : 10 graded stone aggregate 40 mm nominal size) all-round S.W. pipes including bed concrete as per standard design : 100 mm diameter S.W. pipe                                                                                       | 19.2.1   | 160 | Metre | 895.50   | (-)<br>10.91 | 127648.15 |  |  |
| 113 | Providing and laying cement concrete 1:5:10 (1 cement : 5 coarse sand : 10 graded stone aggregate 40 mm nominal size) all-round S.W. pipes including bed concrete as per standard design : 150 mm diameter S.W. pipe                                                                                       | 19.2.2   | 450 | Metre | 1095.15  | (-)<br>10.91 | 439051.11 |  |  |
| 114 | 100x100 mm size P type With common burnt clay F.P.S. (non modular) bricks of class designation 7.5                                                                                                                                                                                                         | 19.4.1.1 | 29  | Each  | 2453.55  | (-)<br>10.91 | 63390.16  |  |  |
| 115 | Inside size 90x80 cm and 45 cm deep including C.I. cover with frame (light duty) 455x610 mm internal dimensions, total weight of cover and frame to be not less than 38 kg (weight of cover 23 kg and weight of frame 15 kg) : With common burnt clay F.P.S. (non modular) bricks of class designation 7.5 | 19.7.1.1 | 54  | Each  | 11687.10 | (-)<br>10.91 | 562250.02 |  |  |



|     |                                                                                                                                                                                                                                                                                                                                                                                                |          |     |      |         |              |           |  |  |
|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-----|------|---------|--------------|-----------|--|--|
| 116 | For shutters of doors, windows & ventilators including providing and fixing hinges/ pivots and making provision for fixing of fittings wherever required including the cost of EPDM rubber / neoprene gasket required (Fittings shall be paid for separately) Anodised aluminium (anodised transparent or dyed to required shade according to IS: 1868, Minimum anodic coating of grade AC 15) | 21.1.2.1 | 450 | Kg   | 531.80  | (-)<br>10.91 | 213201.28 |  |  |
| 117 | Providing and fixing glazing in aluminium door, window, ventilator shutters and partitions etc. with EPDM rubber / neoprene gasket etc. complete as per the architectural drawings and the directions of engineer-in-charge . (Cost of aluminium snap beading shall be paid in basic item): With float glass panes of 4.0 mm thickness (weight not less than 10kg/sqm)                         | 21.3.1   | 230 | Sqm  | 1019.80 | (-)<br>10.91 | 208964.16 |  |  |
| 118 | Providing and fixing double action hydraulic floor spring of approved brand and manufacture conforming to IS : 6315, having brand logo embossed on the body / plate with double spring mechanism and door weight upto 125 kg, for doors, including cost of cutting floors,                                                                                                                     | 21.4.1   | 4   | Each | 2448.85 | (-)<br>10.91 | 8726.72   |  |  |

|                                                                                    |                                                                                                                                                                                                                                                                                                                 |        |       |    |         |           |                                  |      |                    |
|------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|-------|----|---------|-----------|----------------------------------|------|--------------------|
|                                                                                    | embedding in floors as required and making good the same matching to the existing floor finishing and cover plates with brass pivot and single piece M.S. sheet outer box with slide plate etc. complete as per the direction of Engineer-in-charge. With stainless steel cover plate minimum 1.25 mm thickness |        |       |    |         |           |                                  |      |                    |
| <b>Schedule Totals</b>                                                             |                                                                                                                                                                                                                                                                                                                 |        |       |    |         |           | <b>23061865.21</b>               |      |                    |
| <b>Schedule B-Supply of Steel (Item Directory - WCR-HQ-IR-USSOR 2021-2-Ver-1)</b>  |                                                                                                                                                                                                                                                                                                                 |        |       |    |         |           | <b>2494372.87</b>                | 0.08 | % Above 2496368.37 |
| 1                                                                                  | Supply of steel reinforcement of approved brands/makes for R.C.C. work including straightening, cutting, bending, placing in position and binding all complete. Thermo-Mechanically Treated bars of grade Fe-500D or more of approved brands/makes.                                                             | 025082 | 29640 | Kg | 118.78  | (-) 29.15 | 2494372.87                       |      |                    |
| <b>Schedule Totals</b>                                                             |                                                                                                                                                                                                                                                                                                                 |        |       |    |         |           | <b>2496368.37</b>                |      |                    |
| <b>Schedule C-Supply of Cement (Item Directory - WCR-HQ-IR-USSOR 2021-2-Ver-1)</b> |                                                                                                                                                                                                                                                                                                                 |        |       |    |         |           | <b>429984.40</b>                 | 0.08 | % Above 430328.39  |
| 1                                                                                  | Supply and using Cement at Worksite Ordinary Portland Cement 53 grade of approved brands/makes                                                                                                                                                                                                                  | 025072 | 50    | MT | 9075.23 | (-) 5.24  | 429984.40                        |      |                    |
| <b>Schedule Totals</b>                                                             |                                                                                                                                                                                                                                                                                                                 |        |       |    |         |           | <b>430328.39</b>                 |      |                    |
| <b>Total Value</b>                                                                 |                                                                                                                                                                                                                                                                                                                 |        |       |    |         |           | <b>28013418.64</b>               |      | <b>25988561.97</b> |
|                                                                                    |                                                                                                                                                                                                                                                                                                                 |        |       |    |         |           | <b>Rebate on Total Value (%)</b> |      | <b>0.00</b>        |
|                                                                                    |                                                                                                                                                                                                                                                                                                                 |        |       |    |         |           | <b>Net Bid Value</b>             |      | <b>25988561.97</b> |

### Item Breakup

No break up item added

## CHAPTER-1

### WEST CENTRAL RAILWAY CONSTRUCTION ORGANISATION

#### TOP SHEET-1

|     |                                                                                                                                               |                                                                                                                                      |
|-----|-----------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| 1.  | Tender Notice No.                                                                                                                             | <b><u>DYCEC1REWA-07-2024</u></b>                                                                                                     |
| 2.  | Bidding System.                                                                                                                               | Single packet system                                                                                                                 |
| 3.  | Pre-bid Conference                                                                                                                            | NA                                                                                                                                   |
| 4.  | Name of Work:                                                                                                                                 | “Execution of balance works in newly constructed staff quarters at Rewa in connection with the Rewa-Sidhi new BG rail line project”. |
| 4-A | Cost of Work (Rs.)                                                                                                                            | 28013418.64                                                                                                                          |
| 4-B | Bid Security (Rs.)                                                                                                                            | 290100.00                                                                                                                            |
| 4-C | Completion Period (In Month)                                                                                                                  | 06 Month                                                                                                                             |
| 5.  | Dy. Chief Engineer (Con) Unit                                                                                                                 | Dy CE/Con/Rewa                                                                                                                       |
| 6.  | Chief Engineer (Con) Unit                                                                                                                     | CE/Con/Rewa                                                                                                                          |
| 7.  | Applicability of Eligibility Criteria                                                                                                         | Applicable as per Chapter-2 of the Tender documents                                                                                  |
| 7-A | <b><i>Similar Nature of Work</i></b>                                                                                                          | <b>“Any civil engineering work.”</b>                                                                                                 |
| 8.  | Applicability of Bid Capacity<br><b><i>(Applicable in tender having advertisedvalue more than Rs. 20 Crores)</i></b>                          | No                                                                                                                                   |
| 9.  | Permissibility of Participation of Joint Venture Firms<br><b><i>(Applicable in tender having advertisedvalue more than Rs. 10 Crores)</i></b> | No                                                                                                                                   |
| 10. | Applicability of Reverse Auction<br><b><i>(Applicable in tender having advertisedvalue more than Rs. 50 Crores.)</i></b>                      | NO                                                                                                                                   |
| 11. | Applicability of Price Variation Clause<br><b><i>(Applicable in tenders having advertisedvalue above Rs. 2 Crores)</i></b>                    | YES                                                                                                                                  |

|     |                                                                                                                                      |                                                                   |
|-----|--------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|
| 12. | Applicability of Advances to the Contractor.<br><i>(May be Applicable in tender having advertised value more than Rs. 50Crores.)</i> | NA                                                                |
| 13. | Measurement of Works by Railway/Measurement of Works by Contractor                                                                   | Measurement of Works by Contractor                                |
| 14. | Estimate No. & Allocation No.                                                                                                        | 88/W-2/NL/C/23 Vol. I Pt.(E-3372794) New Delhi, Dated 27.01.2023. |
| 15. | Location of work site (Dist & State)                                                                                                 | Rewa (Madhya Pradesh)                                             |
| 16. | Address for physical submission of original B.G. towards Bid Security. <i>(Refer Para-(3) of Chapter-2)</i>                          | Dy CE/Con office Near Railway Station, Rewa Madhya Pradesh-486001 |
| 17. | Tender Number                                                                                                                        | <b>DYCEC1REWA-07-2024</b>                                         |
| 18. | Tender Amount                                                                                                                        | 28013418.64                                                       |
| 19. | LOA Amount                                                                                                                           | 25988561.97                                                       |
| 20. | LOA number                                                                                                                           | DYCE-C-I-JBP-ENGINEERING/DYCEC1 REWA -07-2024/00912200115255      |
| 21. | LOA Date                                                                                                                             | 17.10.2024                                                        |
| 22. | Date of Completion                                                                                                                   | 17.04.2025                                                        |

**Note:** In case it happens to be a holiday on the Date of Opening, the Tender will be opened on the next working day at the same time of Opening.

**Not Transferable**

Issued by : - Chief Administrative Officer (Const.) WCR, JBP

Issued to : - M/s Surendra Singh Kalra, A-507, Harsh Ashiyana, Green Park Colony, Jarhabhata Road Bilaspur (C.G.) 495001, India.



**West Central Railway  
(Construction Organization)**

**Tender Document**

**(Based on IR-SGCC -2022 updated up to ACS-06)**

**Office of  
Deputy Chief Engineer (Construction)  
For Chief Administrative Officer Construction)  
West Central Railway, Jabalpur**

**For Chief Administrative Officer (Const)  
West Central Railway, Jabalpur.**

**WEST CENTRAL RAILWAY CONSTRUCTION ORGANISATION**  
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| <b>Note</b>    | <b>1. Additional Special Technical Conditions for items included in Schedule/Bid of Quantities are enclosed separately.</b><br><br><b>2. Copy of latest IR-GCC incorporating all amendment/correction slips issued is enclosed.</b> |                 |

## CHAPTER-1

### WEST CENTRAL RAILWAY CONSTRUCTION ORGANISATION

#### TOP SHEET-1

|     |                                                                                                                                                |                                                                                                                                      |
|-----|------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| 1.  | Tender Notice No.                                                                                                                              | <b><u>DYCEC1REWA-07-2024</u></b>                                                                                                     |
| 2.  | Bidding System.                                                                                                                                | Single Packet System                                                                                                                 |
| 3.  | Pre-bid Conference                                                                                                                             | NA                                                                                                                                   |
| 4.  | Name of Work:                                                                                                                                  | “Execution of balance works in newly constructed staff quarters at Rewa in connection with the Rewa-Sidhi new BG rail line project”. |
| 5.  | Dy. Chief Engineer (Con) Unit                                                                                                                  | Dy CE/Con/Rewa                                                                                                                       |
| 6.  | Chief Engineer (Con) Unit                                                                                                                      | CE/Con/Rewa                                                                                                                          |
| 7.  | Applicability of Eligibility Criteria                                                                                                          | Applicable as per Chapter-2 of the Tender documents                                                                                  |
| 7-A | <b><i>Similar Nature of Work</i></b>                                                                                                           | <b>“Any civil engineering work”</b>                                                                                                  |
| 8.  | Applicability of Bid Capacity<br><b><i>(Applicable in tender having advertised value more than Rs. 20 Crores)</i></b>                          | No                                                                                                                                   |
| 9.  | Permissibility of Participation of Joint Venture Firms<br><b><i>(Applicable in tender having advertised value more than Rs. 10 Crores)</i></b> | No                                                                                                                                   |
| 10. | Applicability of Reverse Auction<br><b><i>(Applicable in tender having advertised value more than Rs. 50 Crores.)</i></b>                      | NO                                                                                                                                   |
| 11. | Applicability of Price Variation Clause<br><b><i>(Applicable in tenders having advertised value above Rs. 2 Crores)</i></b>                    | YES                                                                                                                                  |
| 12. | Applicability of Advances to the Contractor.<br><b><i>(May be Applicable in tender having advertised value more than Rs. 50 Crores.)</i></b>   | NA                                                                                                                                   |
| 13. | Measurement of Works by Railway/Measurement of Works by Contractor                                                                             | Measurement of Works by Contractor                                                                                                   |
| 14. | Estimate No. & Allocation No.                                                                                                                  | 88/W-2/NL/C/23 Vol. I Pt.(E-3372794) New Delhi, Dated 27.01.2023.                                                                    |
| 15. | Location of work site (Dist & State)                                                                                                           | Rewa (Madhya Pradesh)                                                                                                                |
| 16. | Address for physical submission of original B.G. towards Bid Security.<br><b><i>(Refer Para- (3) of Chapter-2)</i></b>                         | Dy CE/Con office Near Railway Station, Rewa Madhya Pradesh-486001                                                                    |

**TOP SHEET-2**  
**CHECK LIST FOR SUBMISSION OF TENDERS**

|     |                                                                                                                                                                                                                                                                                                                                                                                             |
|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1   | All Optional documents to be submitted before signing of Contract Agreement/or at the time of submission of Bid.                                                                                                                                                                                                                                                                            |
| 2   | <b>List of documents which are to be submitted by the tenderer along with their offer failing which the tender shall be summarily rejected.</b>                                                                                                                                                                                                                                             |
| 2.1 | If Bid Security is submitted through mode of Bank Guarantee, then submission of scanned copy of the Bank Guarantee towards Bid Security of the bid on e-tendering portal (IREPS) and original of above Bank Guarantee shall be submitted in the office of as mention under item no.-16 of Top Sheet-1 (as per Para-3 of Instructions to Tenderers (ITT) in Chapter-2)                       |
| 2.2 | Technical Eligibility Criteria as per Para 10.1 & Para 17.15.1 of Chapter-2. Details of works completed/substantially completed during previous 07 years ending last day of month previous to the one in which tender has been invited as shall be submitted as per Annexure-F and F-1 (as applicable), Chapter-6 of the Tender Document along with Work completion/experience certificate. |
| 2.3 | Financial Eligibility Criteria as per Para 10.2 & 17.15.2 of Chapter-2. For Annual contractual turnover, Certificate as per Annexure-B of Chapter-6 of Tender Document along with the copies of Audited Balance Sheet duly certified by the Chartered Accountant/Certificate from Chartered Accountant duly supported by Audited Balance Sheet shall be submitted.                          |
| 2.4 | Details of Bid Capacity & Existing commitment as per Para 10.3 & 17.15.3 shall be submitted as per Annexure-G & G-1 of Chapter-6 of Tender Document.                                                                                                                                                                                                                                        |
| 2.5 | Certificate as per the proforma enclosed vide Annexure-A & A-1 (as required) Chapter-6 of the Tender Document.                                                                                                                                                                                                                                                                              |
| 2.6 | Power of attorney duly notarized and in favour of a single individual should be submitted. As per Para 14, 15, 17 & 18 of tender form (Second sheet) of SGCC and Para 15 of Chapter-2 of the Tender Document.                                                                                                                                                                               |
| 3   | <b>List of important documents which are to be submitted by the tenderer along with their offer.</b>                                                                                                                                                                                                                                                                                        |
| 3.1 | Certificate of familiarization to the work site as per Chapter-5 of the Tender Document.                                                                                                                                                                                                                                                                                                    |
| 3.2 | List of Plant & Machinery available on hand and proposed to be inducted and hired for the tendered work as per the format vide Annexure-D of Chapter-6 of Tender Document.                                                                                                                                                                                                                  |
| 3.3 | List of Personnel/Organization on hand and proposed to be engaged for the tendered work as per the format vide Annexure-E of Chapter-6 of the Tender Document.                                                                                                                                                                                                                              |
| 3.4 | Declaration regarding association of Railway Officer(s) with tenderer(s) as per the format vide Annexure-H of Chapter-6 of the Tender Document.                                                                                                                                                                                                                                             |
| 3.5 | Annexure-C, I, J, K, O, P of Chapter-6 of the Tender Document.                                                                                                                                                                                                                                                                                                                              |

For Chief Administrative Officer (Const)  
West Central Railway, Jabalpur.



## CHAPTER-2

### WEST CENTRAL RAILWAY CONSTRUCTION ORGANISATION

#### Instructions to Tenderers (ITT)

#### TENDERS FOR WORKS

**1. Tender Form:** Tender Forms shall embody the contents of the contract documents either directly or by reference and shall be as per specimen form, Annexure-I. e-Tender Forms shall be issued free of cost to all tenderers.

**2. Omissions & Discrepancies:** Should a tenderer find discrepancies in or omissions from the drawings or any of the Tender Forms or should he be in doubt as to their meaning, he should at once notify the authority inviting tenders. The tender inviting authority may, if deemed necessary, clarify the same to all tenderers. It shall be understood that every endeavor has been made to avoid any error which can materially affect the basis of tender and successful tenderer shall take upon himself and provide for the risk of any error which may subsequently be discovered and shall make no subsequent claim on account thereof.

#### 3. Bid Security:

(1)(a) The tenderer shall be required to submit the Bid Security with the tender for the due performance with the stipulation to keep the offer open till such date as specified in the tender, under the conditions of tender. The Bid Security shall be as under:

| Value of the Work                               | Bid Security                                                                                                                   |
|-------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| For works estimated to cost up to ₹ 1 crore     | 2% of the estimated cost of the work                                                                                           |
| For works estimated to cost more than ₹ 1 crore | ₹ 2 lakh plus ½% (half percent) of the excess of the estimated cost of work beyond ₹ 1 crore subject to a maximum of ₹ 1 crore |

Note:

- (i) The Bid Security shall be rounded to the nearest ₹100. This Bid Security shall be applicable for all modes of tendering.
  - (ii) Any firm recognized by Department of Industrial Policy and Promotion (DIPP) as 'Startups' shall be exempted from payment of Bid Security detailed above.
  - (iii) Labour Cooperative Societies shall submit only 50% of above Bid Security detailed above.
- (b) It shall be understood that the tender documents have been issued to the tenderer and the tenderer is permitted to tender in consideration of stipulation on his part, that after submitting his tender he will not resile from his offer or modify the terms and conditions thereof in a manner not acceptable to the Engineer. Should the tenderer fail to observe or comply with the said stipulation, the aforesaid amount shall be liable to be forfeited to the Railway.
- (c) If his tender is accepted this Bid Security mentioned in sub clause (a) above will be retained as part security for the due and faithful fulfillment of the contract in terms of Clause 16 of the Standard General Conditions of Contract. The Bid Security of other Tenderers shall, save as herein before provided, be returned to them, but the Railway shall not be responsible for any loss or depreciation that may happen thereto while in their possession, nor be liable to pay interest thereon.

- (2) The Bid Security shall be deposited in cash through e-payment gateway or submitted as Bank Guarantee bond from a scheduled commercial bank of India or as mentioned in tender documents. The Bank Guarantee bond shall be as per Annexure-Q of Chapter-6 and shall be valid for a period of 90 days beyond the bid validity period.
- (3) In case, submission of Bid Security in the form of Bank Guarantee, following shall be ensured:
  - i. A scanned copy of the Bank Guarantee shall be uploaded on e-Procurement Portal (IREPS) while applying to the tender.
  - ii. The original Bank Guarantee should be delivered in person to the official nominated as indicated in the tender document before closing date for submission of bids (i.e. excluding the last date of submission of bids).
  - iii. Non submission of scanned copy of Bank Guarantee with the bid on e-tendering portal (IREPS) and/or non submission of original Bank Guarantee within the specified period shall lead to **summary rejection of bid.**
  - iv. The Tender Security shall remain valid for a period of 90 days beyond the validity period for the Tender.
  - v. The details of the BG, physically submitted should match with the details available in the scanned copy and the data entered during bid submission time, failing which the bid will be rejected.
  - vi. The Bank Guarantee shall be placed in an envelope, which shall be sealed. The envelope shall clearly bear the identification “**Bid for the.....Project**” and shall clearly indicate the name and address of the Bidder. In addition, the Bid Due Date should be indicated on the right-hand top corner of the envelope.
  - vii. The envelope shall be addressed to the officer and address as mentioned in the tender document.
  - viii. If the envelope is not sealed and marked as instructed above, the Railway assumes no responsibility for the misplacement or premature opening of the contents of the Bid submitted and consequent losses, if any, suffered by the Bidder.

#### **4. Care in Submission of Tenders:**

- (a)(i) Before submitting a tender, the tenderer will be deemed to have satisfied himself by actual inspection of the site and locality of the works, that all conditions liable to be encountered during the execution of the works are taken into account and that the rates he enters in the tender forms are adequate and all-inclusive to accord with the provisions in Clause-37 of the Standard General Conditions of Contract for the completion of works to the entire satisfaction of the Engineer.
- (a)(ii) Tenderers will examine the various provisions of The Central Goods and Services Tax Act, 2017(CGST)/ Integrated Goods and Services Tax Act, 2017(IGST)/ Union Territory Goods and Services Tax Act, 2017(UTGST)/respective state's State Goods and Services Tax Act (SGST) also, as notified by Central/State Govt. & as amended from time to time and applicable taxes before bidding. Tenderers will ensure that full benefit of Input Tax Credit (ITC) likely to be availed by them is duly considered while quoting rates.
- (a)(iii) The successful tenderer who is liable to be registered under CGST/IGST/UTGST/SGST Act shall submit GSTIN along with other details required under CGST/IGST/UTGST/SGST Act to railway immediately after the award of contract, without which no payment shall be released to the Contractor. The Contractor shall be responsible for deposition of applicable GST to the concerned authority.

- (a)(iv) In case the successful tenderer is not liable to be registered under CGST/IGST/UTGST/ SGST Act, the railway shall deduct the applicable GST from his/their bills under reverse charge mechanism (RCM) and deposit the same to the concerned authority.
- (a) When work is tendered for by a firm or company, the tender shall be signed by the individual legally authorized to enter into commitments on their behalf.
- (b) The Railway will not be bound by any power of attorney granted by the tenderer or by changes in the composition of the firm made subsequent to the execution of the contract. It may, however, recognize such power of attorney and changes after obtaining proper legal advice, the cost of which will be chargeable to the Contractor.

**4.1** The tenderers shall submit a copy of **certificate** stating that all their statements/documents submitted along with bid are true and factual. Standard format of certificate to be submitted by the bidder is enclosed as **Annexure-A of Chapter-6**. In addition to Annexure-A, in case of other than Company/Proprietary firm, **Annexure-A-1** shall also be submitted by each member of a Partnership Firm/ Joint Venture (JV)/ Hindu Undivided Family (HUF)/ Limited Liability Partnership (LLP) etc. as the case may be. Non submission of above certificate by the bidder shall result in **summarily rejection** of his/their bid. It shall be mandatorily incumbent upon the tenderer to identify, state and submit the supporting documents duly **self-attested/digitally signed** by which they/he is qualifying the Qualifying Criteria mentioned in the Tender Document.

## **CONSIDERATION OF TENDERS**

**5. Right of Railway to Deal with Tenders:** The Railway reserves the right of not to invite tenders for any of Railway work or works or to invite open or limited tenders and when tenders are called to accept a tender in whole or in part or reject any tender or all tenders without assigning reasons for any such action. In case if tender is accepted in part by Railway administration, Letter of Acceptance shall be issued as counter offer to the Tenderer, which shall be subject to acceptance by the Tenderer.

**5A. Two Packets System of Tendering:** With a view to assess the tenders technically without being influenced by the financial bids, 'Two Packets System of tendering' shall be adopted wherein tender documents provide for the same.

**5B. Pre-Bid Conference:** Intenders having advertised value more than Rs 50 Crore or as mentioned in the tender document, Railway shall conduct Pre-Bid Conference(s) with the prospective bidders.

**5C. Make in India Policy:** Provisions of Make in India Policy 2017 issued by Govt. of India, as amended from time to time, shall be followed for consideration of tenders.

**5D. Permission to Bid for a bidder from a country which shares Land boundary with India:** Any bidder from the countries sharing a land border with India will be eligible to bid in any procurement of works (including turnkey projects) only if the bidder is registered with the Competent Authority. The Competent Authority for registration will be the Registration Committee constituted by the Department for Promotion of Industry and Internal Trade (DPIIT), Government of India. For interpretation of this para, Department of Expenditure, Ministry of Finance, Government of India letter F.No.6/18/2019-PPD dated 23/07/2020 shall be referred.

**5E. Clarification of Bids:** To assist in the examination, evaluation & comparison and prequalification of the Tender, the Railway may, at its discretion, ask any Bidder for a clarification of its Bid. Any clarification submitted by a Bidder that is not in response to a request by the Railway shall not be entertained or considered. The Railway request for clarification and the response of the bidder in this regard shall be in writing.

However, if a Bidder does not provide clarification of its bid by the date and time communicated in the Railway request for clarification, the bid shall be evaluated as per the documents submitted along with the bid.

**6. Execution of Contract Document:** The Tenderer whose tender is accepted shall be required to appear in person at the office of General Manager/General Manager (Construction), Chief Administrative Officer (Construction), Divisional Railway Manager or concerned Engineer, as the case may be, or if tenderer is a firm or corporation, a duly authorized representative shall appear (there would be no need for appear in person if agreement is signed digitally) and execute the contract agreement within seven days of notice from Railways that the Contract Agreement is ready. The Contract Agreement shall be entered into by Railway only after submission of valid Performance Guarantee by the Contractor. Failure to do so shall constitute a breach of the agreement affected by the acceptance of the tender. In such cases the Railway may determine that such tenderer has abandoned the contract and there upon his tender and acceptance thereof shall be treated as cancelled and the Railway shall be entitled to forfeit the full amount of the Bid Security and other dues payable to the Contractor under this contract. The failed Contractor shall be debarred from participating in the re-tender for that work.

**7. Form of Contract Document:** Every contract shall be complete in respect of the document it shall so constitute. Not less than 2 copies of the contract document shall be signed by the competent authority and the Contractor and one copy given to the Contractor (there would be no need for appear in person if agreement is signed digitally).

(a) For contracts for specific works, the contract document required to be executed by the tenderer whose tender is accepted shall be an agreement as per specimen form **Annexure-2 of Chapter-7**.

**8. Order of Precedence of Documents:** In a contract agreement, in case of any difference, contradiction, discrepancy, with regard to conditions of tender/contract, specifications, drawings, Bill(s) of Quantities etc., forming part of the tender/contract, the following shall be the order of precedence:

- (i) Letter of Award/Acceptance (LOA)
- (ii) Bill(s) of Quantities
- (iii) Special Conditions of Contract
- (iv) Technical Specifications as given in tender documents
- (v) Drawings
- (vi) Indian Railways Standard General Conditions of Contract updated with correction slips issued up to date of inviting tender or as otherwise specified in the tender documents.

- (vii) Indian Railways Unified Standard Specification (IRUSS-2019) updated with correction slips issued up to date of inviting tender or as otherwise specified in the tender documents, if applicable in the contract.
- (viii) CPWD Specifications 2019 Vol I & II updated with correction slips issued up to date of inviting tender or as otherwise specified in the tender documents, if applicable in the contract.
- (ix) Indian Railways Unified Standard Specifications (Works and Material) 2010 updated with correction slips issued up to date of inviting tender or as otherwise specified in the tender documents, if applicable in the contract.
- (x) IR Specifications/Guidelines updated with correction slips issued up to date of inviting tender or as otherwise specified in the tender documents.
- (xi) Relevant B.I.S. Codes updated with correction slips issued up to date of inviting tender or as otherwise specified in the tender documents.

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## ANNEXURE - I

### WEST CENTRAL RAILWAY CONSTRUCTION ORGANISATION

#### TENDER FORM (First Sheet)

Tender No. DYCEC1REWA-07-2024

Name of Work: ***“Execution of balance works in newly constructed staff quarters at Rewa in connection with the Rewa-Sidhi new BG rail line project.”***

To

The President of India

Acting through the Chief Administrative Officer (Con), West Central Railway

1. I/We \_\_\_\_\_ have read the various conditions to tender attached hereto and agree to abide by the said conditions. I/We also agree to keep this tender open for acceptance for a period of **60** days from the date fixed for opening the same and in default thereof, I/We will be liable for forfeiture of my/our “Bid Security”. I/We offer to do the work for West Central Railway, at the rates quoted in the attached bills of quantities and hereby bind myself/ourselves to complete the work in all respects within **06 months** from the date of issue of letter of acceptance of the tender.
2. I/We also hereby agree to abide by the Indian Railways Standard General Conditions of Contract, with all correction slips up-to-date and to carry out the work according to the Special Conditions of Contract and Specifications of materials and works as laid down by Railway in the annexed Special Conditions/Specifications, Standard Schedule of Rates (SSOR) with all correction slips up-to-date for the present contract.
3. A Bid Security of ₹..... has already been deposited online/submitted as Bank Guarantee bond. Full value of the Bid Security shall stand forfeited without prejudice to any other right or remedies in case my/our Tender is accepted and if:
  - (a) I/We do not submit the Performance Guarantee within the time specified in the Tender document;
  - (b) I/We do not execute the contract documents within seven days after receipt of notice issued by the Railway that such documents are ready; and
  - (c) I/We do not commence the work within fifteen days after receipt of orders to that effect.
4. (a) I/We am/are a Startup firm registered by ..... Department of Industrial Policy and Promotion (DIPP) and my registration number is ..... valid Up to ..... (Copy enclosed) and hence exempted from submission of Bid Security.
5. We are a Labour Cooperative Society and our Registration No. is ..... with ..... and hence required to deposit only 50% of Bid Security.
6. Until a formal agreement is prepared and executed, acceptance of this tender shall constitute a binding contract between us subject to modifications, as may be mutually agreed to between us and indicated in the letter of acceptance of my/our offer for this work.

Signature of Witnesses:

(1) \_\_\_\_\_

(2) \_\_\_\_\_

\_\_\_\_\_  
Signature of Tenderer(s)

Date \_\_\_\_\_

\_\_\_\_\_  
Address of the Tenderer(s)

For Chief Administrative Officer (Const)  
West Central Railway, Jabalpur.

**TENDER FORM (Second Sheet)**

**1. Instructions to Tenderers and Conditions of Tender:** The following documents form part of Tender / Contract:

- (a) Tender Forms – First Sheet and Second Sheet
- (b) Special Conditions/Specifications (enclosed)
- (c) Bill(s) of quantities (as per E-Tender Document)
- (d) Standard General Conditions of Contract and Standard Specifications (Works and Materials) of Indian Railways as amended/corrected upto latest correction slips, copies of which can be seen in the office of Dy. Chief Engineer (Con) or obtained from the office of the Chief Engineer (Con), West Central Railway on payment of prescribed charges.
- (e) Standard Schedule of Rates (SSOR) as amended / corrected upto latest correction slips, copies of which can be seen in the office of Dy. Chief Engineer (Con) or obtained from the office of the Chief Engineer (Con), West Central Railway on payment of prescribed charges.
- (f) All general and detailed drawings pertaining to this work which will be issued by the Engineer or his representatives (from time to time) with all changes and modifications.

**2. Drawings for the Work:** The Drawing for the work can be seen in the office of the Dy. Chief Engineer (Con) at any time during the office hours. The drawings are only for the guidance of Tenderer(s). Detailed working drawings (if required) based generally on the drawing mentioned above, will be given by the Engineer or his representative from time to time.

**3.** The Tenderer(s) shall quote his / their rates as a percentage above or below the Standard Schedule of Rates (SSOR) of West Central Railway as applicable to Engineering Department (Construction) except where he/they are required to quote item rates and must tender for all the items shown in the Bill(s) of Quantities attached. The quantities shown in the attached Bill(s) of Quantities are given as a guide and are approximate only and are subject to variation according to the needs of the Railway. The Railway does not guarantee work undereach item of the Bill(s) of Quantities. The tenderer(s) shall quote rates / rebates only at specified place in Tender Form supplied by Railway. *Any revision of rates / rebates submitted (quoted) through a separate letter whether enclosed with the bid (Tender Form) or submitted separately or mentioned elsewhere in the document other than specified place shall be summarily ignored and will not be considered.*

**3.1** In addition to check list given in E-Tender Document under 'Compliance' the tenderer should check once more on important items as under:

- i) Whether the tender offer is accompanied by Power of Attorney and it is accepted by Power of Attorney holder.
- ii) Whether the details submitted in required format duly signed by tenderer and Chartered Accountant as in Annexure-B, G & G1.

iii) Whether all other mandatory documents uploaded **without which tender offer is to be summarily rejected or is liable to be rejected.**

4. Tenders containing erasures and / or alterations of tender documents **are liable to be rejected.** Any correction made by tender(s) in his/their entries must be attested by him / them.

**4.1 The tender shall be finalised based on submitted documents & no post tender correspondence shall be entertained. However, if any clarification is required, the Railway at its discretion may sought such clarification from the Tenderers.**

The Railway's request for clarification and the response of the contractor shall be in writing. No change in substance of the Bid shall be sought, offered or permitted Except required clarification needed by Railway during evaluation. If the Bidder does not provide clarifications of its Bid by the date and time set in the Railway's request for clarification then his/their offer(s) shall be dealt based on available information with Railways interpretation only.

**4.2 While submitting offers the tenderer(s) generally commit mistakes in submitting the mandatory documents like Certificates regarding contracting experience, Audited Balance Sheet duly certified by the Chartered Accountant, Certificate in respect to true and factuality of documents, information in connection with evaluation of bid capacity, Power of Attorney, submission of tenders by authorized signatory, JV related documents etc. As such, the tenderer(s) should go through the tender document very carefully before submitting his/their offer and accordingly upload all the mandatory documents and other relevant documents as per the formats annexed in the tender document.**

**4.3 The works are required to be completed within a period of months as mentioned on the NIT Header of E-Tender document** from the date of issue of acceptance letter.

## **5. Bid Security:**

(a) *Subject to exemptions provided under para 3(1) (a) of Chapter-2 (ITT) of this document,* the tender must be accompanied by a Bid Security as mentioned in tender documents, **failing which the tender shall be summarily rejected.**

(b) The Tenderer(s) shall keep the offer open for a minimum period of **60 days (in case of two packet system of tendering 90 days)**, from the date of closing of the Tender. It is understood that the tender documents have been issued to the Tenderer(s) and the Tenderer(s), is / are permitted to tender in consideration of the stipulation on his / their part that after submitting his / their tender subject to the period being extended further, if required by mutual agreement from time to time, he will not resile from his offer or modify the terms and conditions thereof in a manner not acceptable to the Chief Engineer/Dy. Chief Engineer/Divisional Engineer of West Central Railway (Construction). Should the tenderer fail to observe or comply with the foregoing stipulation, the amount deposited or Bank guarantee bond submitted as Bid Security for the due performance of the above stipulation, shall be forfeited to the Railway.

(c) If his tender is accepted,

(i) the Bid Security mentioned in sub para(a) above deposited in cash through e-payment gateway will be retained as part security for the due and faithful fulfillment of the contract interms of Clause 16 of the Standard General Conditions of Contract;

For Chief Administrative Officer (Const)  
West Central Railway, Jabalpur.



- (ii) The Bid Security mentioned in sub para(a) above submitted as Bank guarantee bond, will be encashed as part security for the due and faithful fulfillment of the contract in terms of Clause 16 of the Standard General Conditions of Contract.

The Bid Security of other Tenderers shall, save as herein before provided, be returned to them, but the Railway shall not be responsible for any loss or depreciation to the Bid Security that may happen thereto while in their possession, nor be liable to pay interest thereon.

- (d) In case Contractor submits the Term Deposit Receipt/Bank Guarantee Bond towards either the Full Security Depositor the Part Security Deposit equal to or more than Bid Security, the Railway shall return the Bid Security so retained as per sub para(c) above, to the Contractor.

**6. Rights of the Railway to deal with Tender:** The authority for the acceptance of the tender will rest with the Railway. It shall not be obligatory on the said authority to accept the lowest tender or any other tender, and tenderer(s) shall neither demand any explanation for the cause of rejection of his/ their tender nor the Railway to assign reasons for declining to consider or reject any particular tender or tenders.

7. If the tenderer(s) deliberately gives / give wrong information in his / their tender or creates / create circumstances for the acceptance of his / their tender, the Railway reserves the right to reject such tender at any stage.

8. *If any partner(s) of a partnership firm expires after the submission of its tender or after the acceptance of its tender, the Railway shall deem such tender as cancelled/contract as terminated under clause 61 of the Standard General Conditions of Contract, unless the firm retains its character as per partnership agreement. If a sole proprietor expires after the submission of tender or after the acceptance of tender, the Railway shall deem such tender as cancelled / contract as terminated under clause 61 of the Standard General Conditions of Contract.*

## **9. Eligibility Criteria:**

### **10.1 Technical Eligibility Criteria:**

(a) The tenderer must have successfully completed or substantially completed any one of the following categories of works during **last 07 (seven) years**, ending last day of month previous to the one in which tender is invited:

Three similar works each costing not less than the amount equal to **30%** of advertised value of the tender,

or

Two similar works each costing not less than the amount equal to **40%** of advertised value of the tender,

or

One similar work each costing not less than the amount equal to **60%** of advertised value of the tender.

(b) (1) **In case of composite works** (e.g. works involving more than one distinct component, such as Civil Engineering works, S&T works, Electrical works, OHE works etc. and in the case of major bridges – substructure, superstructure etc.), tenderer must have successfully completed or substantially completed any one of the following categories of works during last 07 (seven) years, ending last day of month previous to the one in which tender is invited:

Three similar works **each** costing not less than the amount equal to **30%** of advertised value of each component of tender,

or

Two similar works **each** costing not less than the amount equal to **40%** of advertised value of each component of tender,

or

One similar work **each** costing not less than the amount equal to **60%** of advertised value of each component of tender.

**Note for**

b (1) **Separate completed works** of minimum required values shall also be considered for fulfillment of technical eligibility criteria **for different components**.

(b) (2) ***In such cases, what constitutes a component in a composite work shall be clearly pre-defined with estimated tender cost of it, as part of the tender documents without any ambiguity.***

(b) (3) To evaluate the technical eligibility of tenderer, ***only components of work as stipulated in tender documents for evaluation of technical eligibility, shall be considered.*** The scope of work covered in other remaining components shall be either executed by tenderer himself if he has work experience as mentioned in clause-7 of the Standard General Conditions of Contract or through subcontractor fulfilling the requirements as per clause-7 of the Standard General Conditions of Contract or jointly i.e., partly himself and remaining through subcontractor, with prior approval of Chief Engineer in writing.

However, if required in tender documents by way of Special Conditions, a formal agreement duly notarised, legally enforceable in the court of law, shall be executed by the main contractor with the subcontractor for the component(s) of work proposed to be executed by the subcontractor(s), and shall be submitted along with the offer for considering subletting of that scope of work towards fulfilment of technical eligibility. Such subcontractor must fulfill technical eligibility criteria as follows:

The sub-contractor shall have successfully completed at least one work similar to work proposed for subcontract, costing not less than 35% value of work to be subletted, in last 5 years, ending last day of month previous to the one in which tender is invited through a works contract.

**Note: for subletting of work costing up to Rs 50 lakh, no previous work experience of subcontractor shall be asked for by the Railway.**

*In case after award of contract or during execution of work it becomes necessary for contractor to change subcontractor, the same shall be done with subcontractor(s) fulfilling the requirements as per clause-7 of the Standard General Conditions of Contract, with prior approval of Chief Engineer in writing.*

**Note for Item 10.1:**

Work experience certificate from private individual shall not be considered. However, in addition to work experience certificates issued by any Govt. Organisation, work experience certificate issued by Public listed company having average annual turnover of Rs. 500 crore and above in last 3 financial years excluding the current financial year, listed on National Stock Exchange or Bombay Stock Exchange, incorporation/registered at least 5 year prior to the date of closing of tender, shall also be considered provided the work experience certificate has been issued by a person authorized by the Public listed company to issue such certificates.

For Chief Administrative Officer (Const)  
West Central Railway, Jabalpur.

In case tenderer submits work experience certificate issued by public listed company, the tenderer shall also submit along with work experience certificate, the relevant copy of work order, bill of quantities, bill wise details of payment received duly certified by Chartered Accountant. TDS certificates for all payments received and copy of final/last bill paid by company in support of above work experience certificate.

**10.2. Financial Eligibility Criteria:** The tenderer must have minimum average annual contractual turnover of **V/N or 'V' whichever is less;**

where

**V=** Advertised value of the tender in crores of Rupees

**N=** Number of years prescribed for completion of work for which bids have been invited.

The average annual contractual turnover shall be calculated as an average of “total contractual payments” in the previous three financial years, as per the audited balance sheet. However, in case balance sheet of the previous year is yet to be prepared/ audited, the audited balance sheet of the fourth previous year shall be considered for calculating average annual contractual turnover.

The tenderers shall submit requisite information as per **Annexure-B of Chapter-6** along with copies of Audited Balance Sheets duly certified by the Chartered Accountant/Certificate from Chartered Accountant duly supported by Audited Balance Sheet.

**10.3. Bid Capacity:** The tender/technical bid will be evaluated based on bid capacity formula detailed as **Annexure-G & G1 of Chapter-6**.

**10.4** Credentials if submitted in foreign currency shall be converted into Indian currency i.e., Indian Rupee as under:

The conversion rate of US Dollars into Rupees shall be the daily representative exchange rates published by the Reserve Bank of India or entity authorized by RBI to do so for the relevant date or immediately previous date for which rates have been published. Where, relevant date shall be as on the last day of month previous to the one in which tender is invited. In case of any other currency, the same shall first be converted to US Dollars as on the last day of month previous to the one in which tender is invited, and the amount so derived in US Dollars shall be converted into Rupees at the aforesaid rate. The conversion rate of such currencies shall be the daily representative exchange rates published by the International Monetary Fund for the relevant date or immediately previous date for which rates have been published.

***Explanation for clause 10 including clause 10.1 to 10.4 - Eligibility Criteria:***

**1. Substantially Completed Work** means an ongoing work in which payment equal to or more than 90% of the present contract value (**excluding the payment made for adjustment of Price variation (PVC), if any**) has been made to the contractor in that ongoing contract and no proceedings of termination of contract on Contractor's default has been initiated. The credential certificate in this regard should have been issued not prior to 60 days of date of invitation of present tender.

**2.** In case a work is started prior to 07 (seven) years, ending last day of month previous to the one in which tender is invited, but completed in last 07 (seven) years, ending last day of month previous to the one in which tender is invited, the completed work shall be considered for fulfillment of credentials.

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3. If a work is physically completed and completion certificate to this extent is issued by the concerned organization but final bill is pending, such work shall be considered for fulfillment of credentials.
4. ***In case of completed work***, the value of final bill (gross amount) including the PVC amount (if paid) shall be considered as the completion cost of work. In case final bill is pending, only the total gross amount already paid including the PVC amount (if paid) shall be considered as the completion cost of work.  
  
***In case of substantially completed work***, the total gross amount already paid including the PVC amount (if paid), as mentioned in the certificate, shall be considered as the cost of substantially completed work.
5. If a bidder has successfully completed a work as subcontractor and the work experience certificate has been issued for such work to the subcontractor by a Govt. Organization or public listed company as defined in Note for Item 10.1 Para 10 of the Tender Form (Second Sheet), the same shall be considered for the purpose of fulfillment of credentials.
6. In case a work is considered similar in nature for fulfillment of technical credentials, the **overall cost including the PVC amount (if paid) of that completed work or substantially completed work, shall be considered** and no separate evaluation for each component of that work shall be made to decide eligibility.
7. In case of newly formed partnership firm, the credentials of individual partners from previous proprietary firm(s) or dissolved previous partnership firm(s) or split previous partnership firm(s), shall be considered only to the extent of their share in previous entity on the date of dissolution / split and their share in newly formed partnership firm. For example, a partner A had 30% share in previous entity and his share in present partnership firm is 20%. In the present tender under consideration, the credentials of partner A will be considered to the extent of  $0.3 \times 0.2 \times \text{value of the work done in the previous entity}$ . For this purpose, the tenderer shall submit along with his bid all the relevant documents which include copy of previous partnership deed(s), dissolution deed(s) and proof of surrender of PAN No.(s) in case of dissolution of partnership firm(s) etc.
8. In case of existing partnership firm, **if any one or more partners quit the partnership firm**, the credentials of remaining partnership firm shall be re-worked out i.e., the quitting partner(s) shall take away his credentials to the extent of his share on the date of quitting the partnership firm (e.g. in a partnership firm of partners A, B & C having share 30%, 30% & 40% respectively and credentials of Rs 10 crore; in case partner C quits the firm, the credentials of this partnership firm shall remain as Rs 6 crore). For this purpose, the tenderer shall submit along with his bid all the relevant documents which include copy of previous partnership deed(s), dissolution deed(s) and proof of surrender of PAN No.(s) in case of dissolution of partnership firm(s) etc.
9. In case of existing partnership firm if any other partner(s) joins the firm without any modification in the name and PAN/TAN no. of the firm, the credentials of partnership firm shall get enhanced to the extent of credentials of newly added partner(s) on the same principles as mentioned in item 6 above. For this purpose, the tenderer shall submit along with his bid all the relevant documents which include copy of previous partnership deeds, dissolution/splitting deeds and proof of surrender of PAN No.(s) in case of dissolution of partnership firm etc.

10. Any partner in a partnership firm cannot use or claim his credentials in any other firm without leaving the partnership firm i.e., In a partnership firm of A&B partners, A or B partner cannot use credentials of partnership firm of A&B partners in any other partnership firm or propriety firm without leaving partnership firm of A&B partners.
11. In case a partner in a partnership firm is replaced due to succession as per succession law, the proportion of credentials of the previous partner will be passed on to the successor.
12. If the percentage share among partners of a partnership firm is changed, but the partners remain the same, the credentials of the firm before such modification in the share will continue to be considered for the firm as it is without any change in their value. Further, in case a partner of partnership firm retires without taking away any credentials from the firm, the credentials of partnership firm shall remain the same as it is without any change in their value.
13. In a partnership firm "AB" of A&B partners, in case A also works as propriety firm "P" or partner in some other partnership firm "AX", credentials of A in propriety firm "P" or in other partnership firm "AX" earned after the date of becoming a partner of the firm AB shall not be added in partnership firm AB.
14. In case a tenderer is LLP, the credentials of tenderer shall be worked out on above lines similar to a partnership firm.
15. In case company A is merged with company B, then company B would get the credentials of company A also.

#### **11. Tenderer Credentials:**

Documents testifying tenderer previous experience and financial status should be produced along with the tender.

Tenderer(s) who is / are not borne on the approved list of the Contractors of West Central Railway shall submit along with his / their tender:

- (i) Certificates and testimonials regarding contracting experience for the type of job for which tender is invited with list of works carried out in the past.
- (ii) Audited Balance Sheet duly certified by the Chartered Accountant etc regarding contractual payments received in the past.
- (iii) The list of personnel / organization on hand and proposed to be engaged for the tendered work. Similarly list of Plant & Machinery available on hand and proposed to be inducted and hired for the tendered work.
- (iv) A copy of **certificate** stating that they are not liable to be disqualified and all their statements/documents submitted along with bid are true and factual. Standard format of the **certificate** to be submitted by the bidder is enclosed as **Annexure-A of Chapter-6**. In addition to Annexure-A, in case of other than Company/Proprietary firm, **Annexure-A-1** shall also be submitted by each member of a Partnership Firm/ Joint Venture (JV)/ Hindu Undivided Family (HUF)/ Limited Liability Partnership (LLP) etc. as the case may be. Non submission of **certificate** by the bidder shall result in **summarily rejection** of his/their bid. It shall be mandatorily incumbent upon the tenderer to identify state and submit the supporting documents duly self-attested/digitally signed by which they/he are/is qualifying the Qualifying Criteria mentioned in the Tender Document.

(v) The Railway reserves the right to verify all statements, information and documents submitted by the bidder in his tender offer, and the bidder shall, when so required by the Railway, make available all such information, evidence and documents as may be necessary for such verification. Any such verification or lack of such verification, by the Railway shall not relieve the bidder of its obligations or liabilities hereunder nor will it affect any rights of the Railway thereunder.

(vi) (a) In case of any information submitted by tenderer is found to be false, forged or incorrect at any time during process for evaluation of tenders, it shall lead to forfeiture of the tender Bid Security besides banning of business for a period of upto two years.

(b) In case of any information submitted by tenderer is found to be false, forged or incorrect after the award of contract, the contract shall be terminated. Bid Security, Performance Guarantee and Security Deposit available with the railway shall be forfeited. In addition, other dues of the contractor, if any, under this contract shall be forfeited and agency shall be banned for doing business for a period of upto two years.

**12. Non-compliance with any of the conditions set forth therein above is liable to result in the tender being rejected.**

**13. Execution of Contract Documents:** The successful Tenderer(s) shall be required to execute an agreement with the President of India acting through the Chief Engineer (Con)/Dy. Chief Engineer (Con), West Central Railway for carrying out the work according to Standard General Conditions of Contract, Special Conditions / Specifications annexed to the tender and Standard Specifications (Works and Materials) of Railway as amended/corrected upto latest correction slips, mentioned in tender form (First Sheet).

**14. Documents to be submitted along with Tender:**

(i) The tenderer shall clearly specify whether the tender is submitted on his own (Proprietary Firm) or on behalf of a Partnership Firm / company / Joint Venture (JV) / Registered Society / Registered Trust/Hindu Undivided Family (HUF)/Limited Liability Partnership (LLP) etc. The tenderer(s) shall enclose the attested copies of the constitution of their concern, and copy of PAN Card along with their tender. Tender Documents in such cases are to be signed by such persons as may be legally competent to sign them on behalf of the firm, company, association, trust or society, as the case may be.

(ii) Following documents shall be submitted by the tenderer:

**(a) Sole Proprietorship Firm:**

All documents in terms of Para 10 of the Tender Form (Second Sheet) above.

**(b) HUF:**

(i) A copy of notarized affidavit on Stamp Paper declaring that he who is submitting the tender on behalf of HUF is in the position of 'Karta' of Hindu Undivided Family (HUF) and he has the authority, power and consent given by other members to act on behalf of HUF.

(ii) All other documents in terms of Para 10 of the Tender Form (Second Sheet) above.

**(c) Partnership Firm:**

All documents as mentioned in para 18 of the Tender Form (Second Sheet).

**(d) Joint Venture (JV):** All documents as mentioned in para 17 of the Tender Form (Second Sheet).

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**(e) Company registered under Companies Act 2013:**

- (i) The copies of **MOA (Memorandum of Association) / AOA (Articles of Association)** of the company
- (ii) A copy of Certificate of Incorporation
- (iii) A copy of Authorization/Power of Attorney issued by the Company (backed by the resolution of Board of Directors) in favour of the individual to sign the tender on behalf of the company and create liability against the company.
- (iv) All other documents in terms of Para 10 of the Tender Form (Second Sheet) above.

**(f) LLP (Limited Liability Partnership):**

- (i) A copy of LLP Agreement
- (ii) A copy of Certificate of Incorporation
- (iii) A copy of Power of Attorney/Authorization issued by the LLP in favour of the individual to sign the tender on behalf of the LLP and create liability against the LLP.
- (iv) **An undertaking by all partners of the LLP** that they are not blacklisted or debarred by Railways or any other Ministry / Department of the Govt. of India from participation intenders / contracts as on the date of submission of bids, either in their individual capacity or in any firm/LLP or JV in which they were / are partners/members. Concealment / wrong information in regard to above shall make the contract liable for determination under Clause 62 of the Standard General Conditions of Contract.
- (v) All other documents in terms of Para 10 of the Tender Form (Second Sheet) above.

**(g) Registered Society & Registered Trust:**

- (i) A copy of the Certificate of Registration
  - (ii) A copy of Memorandum of Association of Society/Trust Deed
  - (iii) A copy of Power of Attorney in favour of the individual to sign the tender documents and create liability against the Society/Trust.
  - (iv) A copy of Rules & Regulations of the Society
  - (v) All other documents in terms of Para 10 of the Tender Form (Second Sheet) above
- (iii) *If it is NOT mentioned in the submitted tender that tender is being submitted on behalf of a Sole Proprietorship firm / Partnership firm / Joint Venture / Registered Company etc., then the tender shall be treated as having been submitted by the individual who has signed the tender.*
- (iv) After opening of the tender, any document pertaining to the constitution of Sole Proprietorship Firm / Partnership Firm / Registered Company/ Registered Trust / Registered Society / HUF/LLP etc. shall be neither asked nor considered, if submitted. Further, no suo moto cognizance of any document available in public domain (i.e., on internet etc.) or in Railway's record/office files etc. will be taken for consideration of the tender, if no such mention is available in tender offer submitted.
  - (v) A tender from JV / Partnership firm etc. shall be considered only where permissible as per the tender conditions.

- (vi) The Railway will not be bound by any change of power of attorney or in the composition of the firm made subsequent to the submission of tender. Railway may, however, recognize such power of attorney and changes after obtaining proper legal advice, the cost of which will be chargeable to the Contractor.

**15. The tenderer** whether sole proprietor/ a company or a partnership firm / registered society/ registered trust/ HUF/ LLP etc **if they want to act through agent or individual partner(s), should submit along with the tender, a copy of power of attorney duly stamped and authenticated by a Notary Public or by Magistrate in favour of the specific person whether he/they be partner(s) of the firm or any other person specifically authorizing him/them** to submit the tender, and further to deal with the Tender/ Contract up to the stage of signing the agreement **except in case where such specific person is authorized for above purposes through a provision made in the partnership deed / Memorandum of Understanding / Article of Association /Board resolution, failing which tender shall be summarily rejected.**

A separate power of attorney duly stamped and authenticated by a Notary Public or by Magistrate in favour of the specific person whether he/they be partner(s) of the firm or any other person, shall be submitted after award of work, specifically authorizing him/them to deal with all other contractual activities subsequent to signing of agreement, if required.

*Note: A Power of Attorney executed and issued overseas, the document will also have to be legalized by the Indian Embassy and notarized in the jurisdiction where the Power of Attorney is being issued. However, the Power of Attorney provided by Bidders from countries that have signed the Hague Legislation Convention 1961 are not required to be legalized by the Indian Embassy if it carries a conforming Appostille certificate.*

- **For JV participation, all the members** (i.e. Partnership Firm/ Proprietary Firm or HUF/ Company/LLP Firm/Society or Trust) **shall invariably** submit Power of Attorney/Affidavit/Authorization in favour of the individual to sign the MOU/JV Agreement on behalf of the partnership firm and create liability against the firm as per para-17.14.1(iii), 17.14.2(i), 17.14.3(iv), 17.14.4(iv), 17.14.5(iv) of SGCC, otherwise offer will be **summarily rejected**.

**16. Employment/Partnership etc. of Retired Railway Employees (Para-16 of SGCC):**

- (a) Should a tenderer
- i) be a retired Engineer of the gazetted rank or any other gazetted officer working before his retirement, whether in the executive or administrative capacity or whether holding a pensionable post or not, in the Engineering or any other department of any of the railways owned and administered by the President of India for the time being,  
OR
  - ii) being partnership firm / joint venture (JV) / registered society / registered trust etc have as one of its partners/members a retired Engineer of the gazetted rank or any other gazetted officer working before his retirement,  
OR
  - iii) being an incorporated company have any such retired Engineer of the gazetted rank or any other gazetted officer working before his retirement as one of its directors

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in case where such Engineer or officer had not retired from government service at least 1 year prior to the date of submission of the tender

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the tenderer will give full information as to the date of retirement of such Engineer or gazetted officer from the said service and as to whether permission for taking such contract, or if the Contractor be a partnership firm or an incorporated company, to become a partner or director as the case may be, has been obtained by the tenderer or the Engineer or officer, as the case may be from the President of India or any officer, duly authorized by him in this behalf, shall be clearly stated in writing at the time of submitting the tender.

- b) In case, upon successful award of contract, should a tenderer depute for execution of the works under or to deal matters related with this contract, any retired Engineer of gazette rank or retired gazetted officer working before his retirement in the Engineering or any other department of any of the railways owned and administered by the President of India for the time being, and now in his employment, then the tenderer will ensure that retired Engineer or retired gazetted officer had retired from government service at least 1 year prior to the date of his employment with tenderer and in case he had retired from service within a year then he possesses the requisite permission from the President of India or any officer, duly authorized by him in this behalf, to get associated with the tenderer.
- c) Should a tenderer or Contractor being an individual, have member(s) of his family or in the case of partnership firm/ company / joint venture (JV) / registered society / registered trust etc. one or more of his partner(s)/shareholder(s) or member(s) of the family of partner(s)/shareholder(s) having share of more than 1% in the tendering entity employed in gazetted capacity in the Engineering or any other department of the railway, then the tenderer at the time of submission of tender, will inform the authority inviting tenders the details of such persons.

**Note: -If information as required as per 16. (a), (b) & (c) above has not been furnished, contract is liable to be dealt in accordance with provision of clause 62 of Standard General Condition of contract.**

#### **JOINT VENTURE (JV) IN WORKS TENDERS**

**17. Participation of Joint Venture (JV) in Works Tender:** This para shall be applicable for works tenders wherein tender documents provide for the same.

17.1 Separate identity/name shall be given to the Joint Venture.

17.2 Number of members in a JV shall not be more than three, if the work involves only one department (say Civil or S&T or Electrical or Mechanical) and shall not be more than five, if the work involves more than one Department. One of the members of the JV shall be the Lead Member who shall have a majority (at least 51%) share of interest in the JV. The other members shall have a share of not less than 20% each in case of JV with upto three members and not less than 10% each in case of JV with more than three members. In case of JV with foreign member(s), the Lead Member has to be an Indian firm/company with a minimum share of 51%.

17.3 A member of JV shall not be permitted to participate either in individual capacity or as a member of another JV in the same tender.

**17.4** The tender form shall be purchased and submitted only in the name of the JV and not in the name of any constituent member. **The tender form can however be submitted by JV or any of its constituent member or any person authorized by JV through Power of Attorney to submit tender.**

**17.5 Bid Security shall be submitted by JV or authorized person of JV either as:**

- (i) Cash through e-payment gateway or as mentioned in tender document, or
- (ii) Bank Guarantee bond either in the name of JV, or in the name of all members of JV as per MOU irrespective of their share in the JV if the JV has not been constituted legally till the date of submission of tender.

**17.6** A copy of Memorandum of Understanding (MoU) duly executed by the JV members on a stamp paper, shall be submitted by the JV along with the tender. The complete details of the members of the JV, their share and responsibility in the JV etc. particularly with reference to financial, technical and other obligations shall be furnished in the MoU.

**17.7 Once the tender is submitted, the MoU shall not normally be modified / altered / terminated during the validity of the tender.** In case the tenderer fails to observe/comply with this stipulation, the full Bid Security shall be liable to be forfeited.

**17.8** Approval for change of constitution of JV shall be at the sole discretion of the Railway. The constitution of the JV shall not normally be allowed to be modified after submission of the tender bid by the JV, except when modification becomes inevitable due to succession laws etc. provided further that there is no change in qualification of minimum eligibility criteria by JV after change of composition. However, the Lead Member shall continue to be the Lead Member of the JV. Failure to observe this requirement would render the offer **invalid**.

**17.9** Similarly, after the contract is awarded, the constitution of JV shall not be allowed to be altered during the currency of contract except when modification becomes inevitable due to succession laws etc. and minimum eligibility criteria should not get vitiated. Failure to observe this stipulation shall be deemed to be breach of contract with all consequential penal action as per contract conditions.

**17.10** On award of contract to a JV, **a single Performance Guarantee shall be submitted by the JV as per tender conditions.** All the Guarantees like Performance Guarantee, Bank Guarantee for Mobilization Advance, Machinery Advance etc. shall be accepted only in the name of the JV and no splitting of guarantees amongst the members of the JV shall be permitted.

**17.11** On issue of LOA (Letter of Acceptance), the JV entity to whom the work has been awarded, with the same shareholding pattern as was declared in the MOU/JV Agreement submitted along with the tender, shall be got registered before the Registrar of the Companies under 'The Companies Act - 2013' (in case JV entity is to be registered as Company) or before the Registrar/Sub-Registrar under the 'The Indian Partnership Act, 1932' (in case JV entity is to be registered as Partnership Firm) or under 'The LLP Act 2008' (in case JV entity is to be registered as LLP). **A separate PAN shall be obtained for this entity.** The documents pertaining to this entity including its PAN shall be furnished to the Railways before signing the contract agreement for the work. In case the tenderer fails to observe/comply with this stipulation within 60 days of issue of LOA, contract is liable to be terminated. In case contract is terminated railway shall be entitled to forfeit the full amount of the Bid Security and other dues payable to the Contractor under this contract. The entity so registered, in the registered documents, shall have, inter-alia, following Clauses:

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**17.11.1 Joint And Several Liability** - Members of the entity to which the contract is awarded, shall be jointly and severally liable to the Railway for execution of the project in accordance with General and Special Conditions of Contract. The members of the entity shall also be liable jointly and severally for the loss, damages caused to the Railways during the course of execution of the contract or due to non-execution of the contract or part thereof.

**17.11.2 Duration of the Registered Entity** - It shall be valid during the entire currency of the contract including the period of extension, if any and the maintenance period after the work is completed.

**17.11.3 Governing Laws** - The Registered Entity shall in all respect be governed by and interpreted in accordance with Indian Laws.

**17.12 Authorized Member** - Joint Venture members in the JV MoU shall **authorize Lead member on behalf of the Joint Venture** to deal with the Contract, sign the agreement or enter into contract in respect of the said tender, to receive payment, to witness joint measurement of work done, to sign measurement books and similar such action in respect of the said tender/contract. All notices/correspondences with respect to the contract would be sent only to this authorized member of the JV.

**17.13** No member of the Joint Venture shall have the right to assign or transfer the interest right or liability in the contract without the written consent of the other members and that of the Railway in respect of the said tender/contract.

**17.14 Documents to be enclosed by the JV along with the tender:**

**17.14.1 In case one or more of the members of the JV is/are partnership firm(s)**, following documents shall be submitted:

(i) A notarized copy of the Partnership Deed or a copy of the Partnership deed registered with the Registrar.

(ii) A copy of consent of all the partners or individual authorized by partnership firm, to enter into the Joint Venture Agreement on a stamp paper,

(iii) A notarized or registered copy of Power of Attorney (duly registered as per prevailing law) in favour of the individual to sign the MOU/JV Agreement on behalf of the partnership firm and create liability against the firm.

(iv) An undertaking by all partners of the partnership firm that they are not blacklisted or debarred by Railways or any other Ministry / Department of the Govt. of India from participation in tenders / contracts as on the date of submission of bids, either in their individual capacity or in any firm/LLP in which they were / are partners/members. Any Concealment / wrong information in regard to above shall make the bid ineligible or the contract shall be determined under Clause 62 of the Standard General Conditions of Contract.

**17.14.2 In case one or more members is/are Proprietary Firm or HUF**, the following documents shall be enclosed:

(i) A copy of notarized affidavit on Stamp Paper declaring that his Concern is a proprietary Concern and he is sole proprietor of the Concern OR he who is signing the affidavit on behalf of HUF is in the position of 'Karta' of Hindu Undivided Family (HUF) and he has the authority, power and consent given by other members to act on behalf of HUF.

**17.14.3 In case one or more members of the JV is/are companies,** the following documents shall be submitted:

- (i) A copy of resolutions of the Directors of the Company, permitting the company to enter into a JV agreement,
- (ii) The copies of **MOA (Memorandum of Association) / AOA (Articles of Association)** of the company
- (iii) A copy of Certificate of Incorporation
- (iv) A copy of Authorization/copy of Power of Attorney issued by the Company (backed by the resolution of Board of Directors) in favour of the individual to sign the tender, sign MOU/JV Agreement on behalf of the company and create liability against the company.

**17.14.4 In case one or more members of the JV is/are LLP firm/s,** the following documents shall be submitted:

- (i) A copy of LLP Agreement
- (ii) A copy of Certificate of Incorporation of LLP
- (iii) A copy of resolution passed by partners of LLP firm, permitting the Firm to enter into a JV agreement
- (iv) A copy of Authorization /copy of Power of Attorney issued by the LLP firm (backed by resolution passed by the Partners) in favour of the individual, to sign the tender and/or sign the MOU/ JV agreement on behalf of the LLP and create liability against the LLP.
- (v) An undertaking by all partners of the LLP that they are not blacklisted or debarred by Railways or any other Ministry / Department of the Govt. of India from participation in tenders / contracts as on the date of submission of bids, either in their individual capacity or in any firm/LLP or JV in which they were / are partners/members. Any Concealment / wrong information in regard to above shall make the contract liable for determination under Clause 62 of the Standard General Conditions of Contract.

**17.14.5 In case one or more members of the JV is/are Society/s or Trust/s,** the following documents shall be submitted:

- (i) A copy of Certificate of Registration
- (ii) A copy of Memorandum of Association of Society/Trust Deed
- (iii) A copy of Rules & Regulations of the Society
- (iv) A copy of Power of Attorney, in favour of the individual to sign the tender documents and create liability against the Society/Trust.

**17.14.6 All other documents in terms of Para 10 of the Tender Form (Second Sheet) above.**

*17.14.7 A Power of Attorney executed and issued overseas, the document will also have to be legalized by the Indian Embassy and notarized in the jurisdiction where the Power of Attorney is being issued. However, the Power of Attorney provided by Bidders from countries that have signed the Hague Legislation Convention 1961 are not required to be legalized by the Indian Embassy if it carries a conforming Apostille certificate.*

**17.15 Credentials & Qualifying Criteria:** *Technical, financial eligibility and Bid capacity of the JV shall be adjudged based on satisfactory fulfillment of the following criteria:*

**17.15.1 Technical Eligibility Criteria** ('a' or 'b' mentioned hereunder):

**(a) For Works without composite components**

The technical eligibility for the work as per para 10.1 above, shall be satisfied by **either** the '**JV in its own name & style**' or '**Lead member of the JV**'.

Each other (non-lead) member(s) of JV, who is/ are not satisfying the technical eligibility for the work as per para 10.1 above, shall have technical capacity of minimum 10% of the cost of work i.e., each non-lead member of JV member must have satisfactorily completed or substantially completed during the last 07 (seven) years, ending last day of month previous to the one in which tender is invited, one similar single work for a minimum of 10% of advertised value of the tender.

**(b) For works with composite components**

The technical eligibility for major component of work as per para 10.1 above, shall be satisfied by either the '**JV in its own name & style**' or '**Lead member of the JV**' and technical eligibility for other component(s) of work as per para 10.1 above, shall be satisfied by either the '**JV in its own name & style**' or '**any member of the JV**'.

Each other (non-lead) member(s) of JV, who is/ are not satisfying the technical eligibility for any component of the work as per para 10.1 above, shall have technical capacity of minimum 10% of the cost of any component of work mentioned in technical eligibility criteria. i.e., each other (non lead) member of must have satisfactorily completed or substantially completed during the last 07 (seven) years, ending last day of month previous to the one in which tender is invited, one similar single work for a minimum of 10% of cost of any component of work mentioned in technical eligibility criteria.

**Note for Clause 17.15.1:**

- (a) The Major component of the work for this purpose shall be the component of work having highest value. In cases where value of two or more component of work is same, any one work can be classified as Major component of work.*
- (b) Value of a completed work done by a Member in an earlier JV shall be reckoned only to the extent of the concerned member's share in that JV for the purpose of satisfying his/her compliance to the above-mentioned technical eligibility criteria in the tender under consideration.*

**17.15.2 Financial Eligibility Criteria**

The JV shall satisfy the requirement of "Financial Eligibility" mentioned at para 10.2 above. **The "financial capacity" of the lead partner of JV shall not be less than 51% of the financial eligibility criteria** mentioned at para 10.2 above.

*The arithmetic sum of individual "financial capacity" of all the members shall be taken as JV's "financial capacity" to satisfy this requirement.*

Note: Contractual payment received by a Member in an earlier JV shall be reckoned only to the extent of the concerned member's share in that JV for the purpose of satisfying compliance of the above-mentioned financial eligibility criteria in the tender under consideration.

### **17.15.3 Bid Capacity**

The JV shall satisfy the requirement of “Bid Capacity” requirement mentioned at para 10.3 above. The arithmetic sum of individual “Bid capacity” of all the members shall be taken as JV’s “Bid capacity” to satisfy this requirement.

### **18 Participation of Partnership Firms in works tenders:**

**18.2** The Partnership Firms participating in the tender should be legally valid under the provisions of the Indian Partnership Act.

**18.3** The partnership firm should have been in existence or should have been formed prior to submission of tender. Partnership firm should have either been registered with the Registrar or the partnership deed should have been notarized prior to date of tender opening as per the Indian Partnership Act prior to submission of tender.

**18.4** Separate identity / name should be given to the partnership firm. The partnership firm should have PAN / TAN number in its own name and PAN /TAN number in the name of any of the constituent partners shall not be considered. The valid constituents of the firm shall be called partners.

**18.5** *Once the tender has been submitted, the constitution of the firm shall not be allowed to be modified / altered / terminated during the validity of the tender as well as the currency of the contract except when modification becomes inevitable due to succession laws etc., in which case prior permission should be taken from Railway and in any case the minimum eligibility criteria should not get vitiated. The re-constitution of firm in such cases should be followed by a notary certified Supplementary Deed. The approval for change of constitution of the firm, in any case, shall be at the sole discretion of the Railways and the tenderer shall have no claims what-so-ever. Any change in the constitution of Partnership firm after submission of tender shall be with the consent of all partners and with the signatures of all partners as that in the Partnership Deed. Failure to observe this requirement shall render the offer invalid and full Bid Security shall be forfeited.*

**If any Partner/s withdraws from the firm after submission of the tender and before the award of the contract, the offer shall be rejected and Bid Security of the tenderer will be forfeited.** If any new partner joins the firm after submission of tender but prior to award of contract, his / her credentials shall not qualify for consideration towards eligibility criteria either individually or in proportion to his share in the previous firm. In case the tenderer fails to inform Railway beforehand about any such changes / Modification in the constitution which is inevitable due to succession laws etc. and the contract is awarded to such firm, then it will be considered a breach of the contract conditions liable for determination of the contract under Clause 62 of the Standard General Conditions of Contract.

**18.6** A partner of the firm shall not be permitted to participate either in his individual capacity or as a partner of any other firm in the same tender.

**18.7** The tender form shall be submitted only in the name of partnership firm. **The Bid Security shall be submitted by partnership firm through e-payment gateway or as mentioned in tender document. The EMD submitted in the name of any individual partner or in the name of authorized partner (s) shall not be considered.**

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**18.8** On issue of Letter of Acceptance (LOA) to the partnership firm, all the guarantees like Performance Guarantee, Guarantee for various Advances to the Contractor shall be submitted only in the name of the partnership firm and no splitting of guarantees among the partners shall be acceptable.

**18.9** On issue of Letter of Acceptance (LOA), contract agreement with partnership firm shall be executed in the name of the firm only and not in the name of any individual partner.

**18.10 In case the Letter of Acceptance (LOA) is issued to a partnership firm, the following undertakings shall be furnished by all the partners through a notarized affidavit, before signing of contract agreement.**

(a) Joint and several liabilities:

The partners of the firm to which the Letter of Acceptance (LOA) is issued, shall be jointly and severally liable to the Railway for execution of the contract in accordance with General and Special Conditions of the Contract. The partners shall also be liable jointly and severally for the loss, damages caused to the Railway during the course of execution of the contract or due to non-execution of the contract or part thereof.

(b) Duration of the partnership deed and partnership firm agreement:

The partnership deed/partnership firm agreement shall normally not be modified/altered/ terminated during the currency of contract and the maintenance period after the work is completed as contemplated in the conditions of the contract. Any change carried out by partners in the constitution of the firm without permission of Railway, shall constitute a breach of the contract, liable for determination of the contract under Clause 62 of the Standard General Conditions of Contract.

(c) Governing laws: The partnership firm agreement shall in all respect be governed by and interpreted in accordance with the Indian laws.

(d) No partner of the firm shall have the right to assign or transfer the interest right or liability in the contract without the written consent of the other partner/s and that of the Railway.

**18.11 The tenderer shall clearly specify that the tender is submitted on behalf of a partnership firm. The following documents shall be submitted by the partnership firm, with the tender:**

- (i) A notarized copy of partnership Deed or a copy of the Partnership deed registered with the Registrar.
- (ii) A notarized or registered copy of Power of Attorney in favour of the individual to tender for the work, sign the agreement etc. and create liability against the firm.
- (iii) An undertaking by all partners of the partnership firm that they are not blacklisted or debarred by Railways or any other Ministry / Department of the Govt. of India from participation in tenders / contracts as on the date of submission of bids, either in their individual capacity or in any firm in which they were / are partners/members. Any Concealment / wrong information in regard to above shall make the bid ineligible or the contract shall be determined under Clause 62 of the Standard General Conditions of Contract.

(iv) All other documents in terms of **Para 10 of the Tender Form (Second Sheet) above.**

**18.12 Evaluation of eligibility of a partnership firm:**

Technical and financial eligibility of the firm shall be adjudged based on satisfactory fulfillment of the eligibility criteria laid down in **Para 10 of the Tender Form (Second Sheet) above.**

**19 Advances to Contractor –**

If specifically provided in Tender Documents of tender having advertised value more than Rs 50 Crores, Railway shall make payment, as an Interest-bearing advances, as per Contractor's request. These advances shall carry a simple interest **@ RBI Bank Rate + 5% (Ref- Rly Board's Lr. No. 2018/CE-I/CT/1 dtd. 10.03.2022).** The payment and recovery of such advances shall be made as per manners prescribed in Clause 46.4 of the Standard General Conditions of Contract.

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**WEST CENTRAL RAILWAY CONSTRUCTION ORGANISATION  
EXCERPTS OF STANDARD GENERAL CONDITIONS OF CONTRACT**

**GENERAL OBLIGATIONS**

**1. (1) Law Governing the Contract:** The contract shall be governed by the law for the time being in force in the Republic of India. (Para-3.(1) of SGCC-2022)

**1.(2) Compliance to Regulations and Bye-Laws:** The Contractor shall conform to the provision of any statute relating to the works and regulations and bye-laws of any local authority and of any water and lighting companies or undertakings, with whose system the work is proposed to be connected and shall before making any variation from the drawings or the specifications that may be necessitated by so confirming give to the Engineer notice specifying the variation proposed to be made and the reason for making the variation and shall not carry out such variation until he has received instructions from the Engineer in respect thereof. The Contractor shall be bound to give all notices required by statute, regulations or bye-laws as aforesaid and to pay all fees and taxes payable to any authority in respect thereof. (Para-3.(2) of SGCC-2022)

**1.(3) Environmental and Forest clearances:**

The Railway represents and warrants that the environmental and forest clearances pertaining to the work commensurate with the progress of work/agreed programme, will be obtained by Engineer. In the event of any delay in securing respective clearances leading to delay in execution of work, the Contractor shall be entitled to Extension of Time for the period of such delay in accordance with the provisions of Clause-17A(ii) of SGCC-2022. (Para-3.(3) of SGCC-2022)

**2. Communications to be in Writing:** All notices, communications, reference and complaints made by the Railway or the Engineer or the Engineer's Representative or the Contractor inter-se concerning the works shall be in writing or e-mail on registered e-mail IDs and no notice, communication, reference or complaint not in writing or through e-mail, shall be recognized. (Para-4 of SGCC-2022)

**3. Service of Notices on Contractors:** The Contractor shall furnish to the Engineer the name, designation and address of his authorized agent and all complaints, notices, communications and references shall be deemed to have been duly given to the Contractor, if delivered to the Contractor or his authorized agent or left at or posted to the address so given and shall be deemed to have been so given in the case of posting on day on which they would have reached such address in the ordinary course of post/e-mail or on the day on which they were so delivered or left. In the case of contract by partners, any change in the constitution of the firm shall be forthwith notified by the Contractor to the Engineer. (Para-5 of SGCC-2022)

**4. Occupation and Use of Land:** No land belonging to or in the possession of the Railway shall be occupied by the Contractor without the permission of the Railway. The Contractor shall not use, or allow to be used the site for any purposes other than that of executing the works. Whenever non-railway bodies/persons are permitted to use railway premises with competent authority's approval, conservancy charges as applicable from time to time may be levied. (Para-6 of SGCC-2022)

**5. Assignment or Subletting of Contract:** The Contractor shall not assign or sublet the contract or any part thereof or allow any person to become interested therein any manner whatsoever without the special permission in writing of the Chief Engineer, save as provided below. Any breach of this condition shall entitle the Railway to rescind the contract under Clause 62 of Standard General Conditions of Contract and also render the Contractor liable for payment to the Railway in respect of any loss or damage arising or ensuing from such cancellation; provided always that execution of the details of the work by petty Contractor under the direct and personal supervision of the Contractor or his agent shall not be deemed to be sub-letting under this clause.

In case Contractor intends to subcontract part of work, he shall submit a proposal in writing seeking permission of Chief Engineer for the same. While submitting the proposal to railway, Contractor shall ensure the following:

**(a)** (i) Total value of work to be assigned to sub-contractor(s) shall not be more than 50% of total contract value.

(ii) The subcontractor shall have successfully completed at least one work similar to work proposed for subcontract in last 5 years, ending date of submission of proposal by Contractor to Railway, costing not less than 35% value of work to be subletted, through a works contract. For fulfilment of above, Work Experience Certificate issued by a Govt. Department/Organisation shall be considered. Further, Work Experience Certificate issued by a Public listed company shall be considered provided the company is having average annual turnover of Rs 500 crore and above in last 3 financial years excluding the current financial year, listed on National Stock Exchange or Bombay Stock Exchange, registered at least 5 years back from the date of submission of proposal by Contractor to Railway and work experience certificate issued by a person authorised by the Public Listed Company to issue such certificates.

**Note:** for subletting of work costing up to Rs 50 lakh no previous work experience shall be asked for by the Railway.

In case contractor submits subcontractor's work experience certificate issued by public listed company, the contractor shall also submit along with work experience certificate, the relevant copy of work order, bill of quantities, bill wise details of payment received duly certified by Chartered Accountant, TDS certificates for all payments received and copy of final/last bill paid by company in support of above work experience certificate.

(iii) There is no banning with the sub-contractor in force over IR.

**(b)** The Contractor shall provide to the Engineer a copy of the agreement to be entered into by Contractor with subcontractor. No subcontractor shall be permitted without a formal agreement between Contractor and subcontractor. This agreement shall clearly define the scope of work to be carried out by subcontractor and the terms of payment in clear & unambiguous manner.

**(c)** On receipt of approval from Chief Engineer, Contractor shall enter into a formal agreement legally enforceable in Court of Law with subcontractor and submit a copy of the same to the Engineer.

**(d)** The Contractor shall intimate to the Engineer not less than 7 days in advance, the intended date of commencement of subcontractor's work.

(e) Once having entered into above agreement, Contractor shall discontinue such arrangement, if he intends to do so at his own or on the instruction of Railway, with prior intimation to Chief Engineer.

(f) The Contractor shall indemnify railway against any claim of subcontractor.

(g) The Contractor shall release payment to the Sub-contractor(s) promptly and shall endeavour to resolve all issues amicably and speedily with the Sub-contractor(s), so that the execution of work is not affected in any manner whatsoever.

(h) In addition to issuance of work experience certificate to Contractor, the Engineer, when, based on documents, is satisfied that subcontracted work has been carried out by subcontractor, shall issue work experience certificate to the subcontractor also for the portion of work subcontracted and successfully completed by the sub-contractor.

***Note:** Work Experience Certificate to the subcontractor shall be issued only when the contractor's work is complete and contractor is entitled for the issuance of Work Experience Certificate. However, in the same contract, when the Chief Engineer, based on documents, is satisfied that the subcontractor has successfully carried out subletted work, without issuance of work experience certificate to subcontractor at this stage, the Chief Engineer can, only once, consider the successfully completed subletted work for the fulfilment of eligibility for further subletting of work to the subcontractor in the same contract. When the contractor's work is complete and contractor is entitled for the issuance of work experience certificate, the subcontractor shall be issued one Work Experience Certificate for the total scope of work executed by the subcontractor in the contract.*

(i) The responsibility of successful completion of work by subcontractor shall lie with Contractor. Subcontracting will in no way relieve the Contractor to execute the work as per terms of the Contract.

(j) Further, in case Engineer is of the view that subcontractor's performance is not satisfactory, he may instruct the Contractor to remove the subcontractor from the work and Contractor has to comply with the above instructions with due promptness. Contractor shall intimate the actual date of discontinuation of subcontract to Engineer. No claim of Contractor whatsoever on this account shall be entertained by the Railway and this shall be deemed as 'excepted matter' (matter not arbitrable).

(k) The permitted subcontracting of work by the Contractor shall not establish any contractual relationship between the sub-contractor and the Railway and shall not relieve the Contractor of any responsibility under the Contract. (Para-7 of SGCC-2022)

**6. Assistance by Railway for the Stores to be obtained by the Contractor:** Owing to difficulty in obtaining certain materials (including Tools & Plant) in the market, the Railway may have agreed without any liability therefore to endeavour to obtain or assist the Contractor in obtaining the required quantities of such materials as may be specified in the Tender. In the event of delay or failure in obtaining the required quantities of the aforesaid material, the Contractor shall not be deemed absolved of his own responsibility and shall keep in touch with the day to day position regarding their availability and accordingly adjust progress of works including employment of labour and the Railway shall not in any way be liable for the supply of materials or for the non-supply thereof for any reasons whatsoever nor for any loss or damage arising in consequence of such delay or non-supply. (Para-8 of SGCC- 2022)

**7. Railway Passes:** No free railway passes shall be issued by the Railway to the Contractor or any of his employee/worker. (Para-9 of SGCC-2022)

For Chief Administrative Officer (Const)  
West Central Railway, Jabalpur.

**8. Representation on Works:** The Contractor shall, when he is not personally present on the site of the works place, keep a responsible agent at the works during working hours who shall on receiving reasonable notice, present himself to the Engineer and orders given by the Engineer or the Engineer's representative to the agent shall be deemed to have the same force as if they had been given to the Contractor. Before absenting himself, the Contractor shall furnish the name and address of his agent for the purpose of this clause and failure on the part of the Contractor to comply with this provision at any time will entitle the Railway to rescind the contract under Clause 62 of these Conditions. (Para-12 of SGCC-2022)

**9. Excavated Material:** The Contractor shall not sell or otherwise dispose of or remove except for the purpose of this contract, the sand, stone, clay ballast, earth, rock or other substances or materials which may be obtained from any excavation made for the purpose of the works or any building or produced upon the site at the time of delivery of the possession thereof but all the substances, materials, buildings and produce shall be the property of the Railway provided that the Contractor may, with the permission of the Engineer, use the same for the purpose of the works either free of cost or pay the cost of the same at such rates as may be determined by the Engineer. (Para-14 of SGCC-2022)

**10. Indemnity by Contractors:** The Contractor shall indemnify and save harmless the Railway from and against all actions, suit, proceedings, losses, costs, damages, charges, claims and demands of every nature and description brought or recovered against the Railways by reason of any act or omission of the Contractor, his agents or employees, in the execution of the works or in his guarding of the same. All sums payable by way of compensation under any of these conditions shall be considered as reasonable compensation to be applied to the actual loss or damage sustained, and whether or not any damage shall have been sustained. (Para-15 of SGCC-2022)

**11.(1) Security Deposit:** The Security Deposit shall be **5% of the contract value**. The Bid Security submitted by the Contractor with his tender will be retained/encashed by the Railways as part of security for the due and faithful fulfillment of the contract by the Contractor. Provided further that, if Contractor submits the Cash or Term Deposit Receipt issued from a Scheduled commercial bank of India or irrevocable Bank Guarantee Bond from a Scheduled commercial bank of India, either towards the Full Security Depositor the Part Security Deposit equal to or more than Bid Security, the Railway shall return the Bid Security, to the Contractor.

Balance of Security Deposit may be deposited by the Contractor in cash or Term Deposit Receipt issued from Scheduled commercial bank of India or irrevocable Bank Guarantee bond issued from Scheduled commercial bank of India, or may be recovered **at the rate of 6% of the bill amount** till the full Security Deposit is recovered. Provided also that in case of defaulting Contractor, the Railway may retain any amount due for payment to the Contractor on the pending "on account bills" so that the amounts so retained (including amount guaranteed through Performance Guarantee) may not exceed 10% of the total value of the contract.

The Irrevocable Bank Guarantee submitted towards Security deposit shall be initially valid up to the stipulated date of Maintenance period plus 60 days and shall be extended from time to time, depending upon extension of contract granted in terms of Clause 17A and 17B of the Standard General Conditions of Contract.

**Note:** Security Deposit deposited in cash by the Contractor or recovered from the running bills of a Contractor or submitted by contractor as Term Deposit Receipt(s) can be refunded/returned to the contractor, in lieu of irrevocable Bank Guarantee bond issued from scheduled commercial bank of India, to be submitted by him, for an amount equal to or more than the already available Security Deposit, provided however that, in a contract of value less than Rs. 50 Crore, such refund/ return of the already available Security Deposit is permitted up to two times and in a contract of value equal to or more than Rs. 50 Crore, such refund / return of the already available Security Deposit is permitted up to three times. (Para-16(1) of SGCC-2022)

**11.(2)(i) Refund of Security Deposit:** Security Deposit mentioned in sub clause (1) above shall be returned to the Contractor after the following:

- (a) Final Payment of the Contract as per clause 51.(1) of GCC and
- (b) Execution of Final Supplementary Agreement or Certification by Engineer that Railway has No Claim on Contractor and
- (c) Maintenance Certificate issued, on expiry of the maintenance period as per clause 50.(1) of GCC in case applicable. (Para-16.(2)(i) of SGCC-2022)

**11.(2)(ii) Forfeiture of Security Deposit:** Whenever the contract is rescinded as a whole under clause 62 (1) of these conditions, the Security Deposit already with railways under the contract shall be forfeited. However, in case the contract is rescinded in part or parts under clause 62 (1) of these conditions, the Security Deposit shall not be forfeited. (Para-16.(2)(ii) of SGCC-2022)

**11.(3)** No interest shall be payable upon the Bid Security and Security Deposit or amounts payable to the Contractor under the Contract, but Government Securities deposited in terms of Sub-Clause 11.(4)(b) of this clause will be payable with interest accrued thereon. (Para-16.(3) of SGCC-2022)

#### **11.(4) Performance Guarantee**

The procedure for obtaining Performance Guarantee is outlined below:

- (a) The successful bidder shall have to submit a Performance Guarantee (PG) within 21 (Twenty one) days from the date of issue of Letter of Acceptance (LOA). Extension of time for submission of PG beyond 21 (Twenty one) days and upto 60 days from the date of issue of LOA may be given by the Authority who is competent to sign the contract agreement. However, a penal interest of 12% per annum shall be charged for the delay beyond 21(Twenty one) days, i.e. from 22nd day after the date of issue of LOA. Further, if the 60th day happens to be a declared holiday in the concerned office of the Railway, submission of PG can be accepted on the next working day.

In all other cases, **if the Contractor fails to submit the requisite PG even after 60 days from the date of issue of LOA, the contract is liable to be terminated.** In case contract is terminated railway shall be entitled to forfeit Bid Security and other dues payable against that contract. In case a tenderer has not submitted Bid Security on the strength of their registration as a Startup recognized by Department of Industrial Policy and Promotion (DIPP) under Ministry of Commerce and Industry, DIPP shall be informed to this effect.

The failed Contractor shall be debarred from participating in re-tender for that work.

- (b) The successful bidder shall submit the Performance Guarantee (PG) in any of the following forms, **amounting to 5% of the original contract value:**
- (i) A deposit of Cash;
  - (ii) Irrevocable Bank Guarantee;
  - (iii) Government Securities including State Loan Bonds at 5% below the market value;
  - (iv) Pay Orders and Demand Drafts tendered by any Scheduled Commercial Bank of India;
  - (v) Guarantee Bonds executed or Deposits Receipts tendered by any Scheduled Commercial Bank of India;
  - (vi) Deposit in the Post Office Saving Bank;
  - (vii) Deposit in the National Savings Certificates;
  - (viii) Twelve years National Defence Certificates;
  - (ix) Ten years Defence Deposits;
  - (x) National Defense Bonds and
  - (xi) Unit Trust Certificates at 5% below market value or at the face value whichever is less. Also, FDR in favour of FA&CAO (free from any encumbrance) may be accepted.
- (c) The Performance Guarantee shall be submitted by the successful bidder after the Letter of Acceptance (LOA) has been issued, but before signing of the contract agreement. This P.G. shall be initially valid upto the stipulated date of completion plus 60 days beyond that. In case, the time for completion of work gets extended, the Contractor shall get the validity of P.G. extended to cover such extended time for completion of work plus 60 days.
- (d) The value of PG to be submitted by the Contractor is based on original contract value and shall not change due to subsequent variation(s) in the original contract value.
- (e) The Performance Guarantee (PG) shall be released after physical completion of the work based on 'Completion Certificate' issued by the competent authority stating that the Contractor has completed the work in all respects satisfactorily.
- (f) Whenever the contract is rescinded, the Performance Guarantee already submitted for the contract shall be encashed in addition to forfeiture of Security Deposit available with railway.
- (g) The Engineer shall not make a claim under the Performance Guarantee except for amounts to which the President of India is entitled under the contract (notwithstanding and/or without prejudice to any other provisions in the contract agreement) in the event of:
- (i) Failure by the Contractor to extend the validity of the Performance Guarantee as described herein above, in which event the Engineer may claim the full amount of the Performance Guarantee.
  - (ii) Failure by the Contractor to pay President of India any amount due, either as agreed by the Contractor or determined under any of the Clauses/Conditions of the Agreement, within 30 days of the service of notice to this effect by Engineer.
  - (iii) The Contract being determined or rescinded under clause 62 of these conditions. (Para-16.(4) of SGCC-2022)

- 12. Force Majeure Clause:** If at any time, during the continuance of this contract, the performance in whole or in part by either party of any obligation under this contract shall be prevented or delayed by reason of any war, hostility, acts of public enemy, civil commotion, sabotage, serious loss or damage by fire, explosions, epidemics/pandemics, strikes, lockouts or acts of God (hereinafter, referred to events) provided, notice of the happening of any such event is given by either party to the other within 30 days from the date of occurrence thereof, neither party shall by reason of such event, be entitled to terminate this contract nor shall either party have any claim for damages against the other in respect of such non-performance or delay in performance, and works under the contract shall be resumed as soon as practicable after such event has come to an end or ceased to exist, and the decision of the Engineer as to whether the works have been so resumed or not shall be final and conclusive, PROVIDED FURTHER that if the performance in whole or in part of any obligation under this contract is prevented or delayed by reason of any such event for a period exceeding 120 days, either party may at its option terminate the contract by giving notice to the other party. **(Para-17 of SGCC-2022)**
- 12-A Extension of Time in Contracts:** Subject to any requirement in the contract as to completion of any portion or portions of the works before completion of the whole, the Contractor shall fully and finally complete the whole of the works comprised in the contract (with such modifications as may be directed under conditions of this contract) by the date entered in the contract or extended date in terms of the following clauses:
- (i) **Extension due to Modification:** If any modifications have been ordered which in the opinion of the Engineer have materially increased the magnitude of the work, then such extension of the contracted date of completion may be granted as shall appear to the Engineer to be reasonable in the circumstances, provided moreover that the Contractor shall be responsible for requesting such extension of the date as may be considered necessary as soon as the cause thereof shall arise and in any case not less than one month before the expiry of the date fixed for completion of the works.
- (ii) **Extension for Delay not due to Railway or Contractor:** If in the opinion of the Engineer, the progress of work has any time been delayed by any act or neglect of Railway's employees or by other Contractor employed by the Railway under Sub- Clause (4) of Clause 20 of Standard General Conditions of Contract or in executing the work not forming part of the contract but on which Contractor's performance necessarily depends or by reason of proceeding taken or threatened by or dispute with adjoining or to neighbouring owners or public authority arising otherwise through the Contractor's own default etc. or by the delay authorized by the Engineer pending arbitration or in consequences of the Contractor not having received in due time necessary instructions from the Railway for which he shall have specially applied in writing to the Engineer or his authorized representative then upon happening of any such event causing delay, the Contractor shall immediately give notice thereof in writing to the Engineer within 15 days of such happening, but shall nevertheless make constantly his best endeavours to bring down or make good the delay and shall do all that may be reasonably required of him to the satisfaction of the Engineer to proceed with the works. The Contractor may also indicate the period for which the work is likely to be delayed and shall be bound to ask for necessary extension of time. The Engineer on receipt of such request from the Contractor shall consider the same and shall grant such extension of time as in his opinion is reasonable having regard to the nature and period of delay and the type and quantum of work affected thereby. No other compensation shall be payable for

works so carried forward to the extended period of time; the same rates, terms and conditions of contract being applicable as if such extended period of time was originally provided in the original contract itself.

- (iii) **Extension for Delay due to Railways:** In the event of any failure or delay by the Railway to hand over the Contractor possession of the lands necessary for the execution of the works or to give the necessary notice to commence the works or to provide the necessary drawings or instructions or any other delay caused by the Railway due to any other cause whatsoever, then such failure or delay shall in no way affect or vitiate the contract or alter the character thereof or entitle the Contractor to damages or compensation therefor, but in any such case, the Railway may grant such extension or extensions of the completion date as may be considered reasonable.

The Contractor shall indicate the period for which the work is likely to be delayed and shall seek extension of time as may be considered necessary under clause 12A(i) or/and 12A(ii) or/ and 12A(iii) above, as soon as the cause thereof shall arise and, in any case, not less than 15 days before the expiry of the date fixed for completion of the works. The Engineer shall consider the same and shall grant and communicate such extension of time as in his opinion is reasonable having regard to the nature and period of delay and the type and quantum of work affected thereby. No other compensation shall be payable for works so carried forward to the extended period of time; the same rates, terms and conditions of contract being applicable, as if such extended period of time was originally provided in the original contract itself.

The non-submission of request for extension or submission of request within less than 15 days before the expiry of the date fixed for completion of the works, shall make him ineligible for extension under these sub clauses, subject to final decision of Engineer.

**(Para-17A of SGCC-2022)**

**12-B Extension of Time with Liquidated Damages (LD) for delay due to Contractor:**

The time for the execution of the work or part of the works specified in the contract documents shall be deemed to be the essence of the contract and the works must be completed not later than the date(s) as specified in the contract. If the Contractor fails to complete the works within the time as specified in the contract for the reasons other than the reasons specified in Clause 12 and 12-A, the Railway may, if satisfied that the works can be completed by the Contractor within reasonable short time thereafter, allow the Contractor for further extension of time (Proforma at Annexure-VII of Standard General Condition of Contract) as the Engineer may decide. On such extension the Railway will be entitled without prejudice to any other right and remedy available on that behalf, to recover from the Contractor as agreed damages and not by way of penalty for each week or part of the week, a sum calculated at the rate of Liquidated Damages as decided by Engineer, between 0.05% to 0.30% of contract value of the works for each week or part of the work.

For the purpose of this Clause, the contract value of the works shall be taken as value of work as per contract agreement including any supplementary work order/contract agreement issued. ***Provided also, that the total amount of liquidated damages under this condition shall not exceed 5% of the contract value or of the total value of the item or groups of items of work for which a separate distinct completion period is specified in the contract.***

Provided further, that if the Railway is not satisfied that the works can be completed by the Contractor and in the event of failure on the part of the contractor to complete the work



within further extension of time allowed as aforesaid, the Railway shall be entitled without prejudice to any other right or remedy available in that behalf, to appropriate the contractor's Security Deposit and rescind the contract under Clause 62 of GCC, whether or not actual damage is caused by such default.

**NOTE:**

In a contract, where extension(s) of time have been allowed once under clause 12B, further request(s) for extension of time under clause 12A can also be considered under exceptional circumstances. Such extension(s) of time under clause 12A shall be without any Liquidated damages, but the Liquidated damages already recovered during extension(s) of time granted previously under clause 12B shall not be waived. **However, Price variation during such extension(s) shall be dealt as applicable for extension(s) of time under clause 12B.**

(Para-17B of SGCC-2022)

**12-C Bonus for Early Completion of Work:** In case of open tenders having value more than Rs. 20 crore and original period of completion 12 months or more, when there is no reduction in original scope of work by more than 10%, and no extension granted on either railway or Contractor's account, Contractor shall be entitled for a bonus of 1% for each 30 days early completion of work. The period of less than 30 days shall be ignored while working out bonus. The maximum bonus **shall be limited to 5% of** original contract value. The completion date shall be reckoned as the date of issuance of completion certificate by Engineer. (Para-17C of SGCC-2022)

### **EXECUTION OF WORKS**

**13.(1) Contractor's understanding:** It is understood and agreed that the Contractor has, by careful examination, satisfied himself as to the nature and location of the work, the conformation of the ground, the character, quality and quantity of the materials to be encountered, the character of equipment and facilities needed preliminary to and during the progress of the works, the general and local conditions, the labour conditions prevailing therein and all other matters which can in any way affect the works under the contract. (Para-19.(1) of SGCC-2022)

**13.(2) Commencement of Works:** The Contractor shall commence the works within 15 days after the receipt by him of an order in writing to this effect from the Engineer and shall proceed with the same with due expedition and without delay. (Para-19.(2) of SGCC-2022)

**13.(3) Accepted Programme of Work:** The Contractor who has been awarded the work shall as soon as possible but not later than 30 days after the date of receipt of the acceptance letter in respect of contracts with initial completion period of two years or less or not later than 90 days for other contracts have to submit the detailed programme of work indicating the time schedule of various items of works in the form of Bar Chart/PERT/CPM. He shall also submit the details of organisation (in terms of labour and supervisors), plant and machinery that he intends to utilize (from time to time) for execution of the work within stipulated date of completion. The programme of work amended as necessary by discussions with the Engineer, shall be treated as the agreed programme of the work for the purpose of this contract and the Contractor shall endeavor to fulfill this programme of work. The progress of work will be watched accordingly and the liquidated damages will be with reference to the overall completion date. Nothing stated herein shall preclude the Contractor in achieving earlier completion of item or whole of the works than indicated in the programme.

**In Contracts for works of New Line/Gauge Conversion/Doubling/Railway Electrification, finalized through Tenders having advertised value more than Rs.100 crores,** the Contractor shall submit a detailed time programme to the Engineer within 30 days after issue of LOA. The program shall include the physical and Financial Progress vis-à-vis program and forecast cash flow adopting Project Management Software such as Primavera/Sure Track/MS Project etc. The program must identify the milestones, interface requirements and program reporting elements. The Contractor shall supply, free of cost one set of authorized software to the Engineer and the soft copy of structured program for the project. This shall be updated every month. The Contractor shall also submit a revised programme whenever the previous programme is inconsistent with actual progress. Each programme shall include:

The order in which the Contractor intends to carry out the Works, including the anticipated timing of each stage, Contractor's Documents, procurement, manufacture of Plant, delivery to Site, construction, erection and testing, each of these stages for work by each Subcontractor, if any, the sequence and timing of inspections and tests specified in the Contract, and a supporting report which includes:

A general description of the methods which the Contractor intends to adopt, and of the major stages, in the execution of the Works, and details showing the Contractor's reasonable estimate for the number of each class of Contractor's Personnel & Equipment, required on the Site for each major stage.

Unless the Engineer, within 21 days after receiving a programme, gives notice to the Contractor stating the extent to which it does not comply with the Contract, the Contractor shall proceed in accordance with the programme, subject to his other obligations under the Contract. The Engineer shall be entitled to rely upon the programme when planning their activities.

If, at any time, the Engineer gives notice to the Contractor that a programme fails (to the extent stated) to comply with the Contract or to be consistent with actual progress and the Contractor's stated intentions, the Contractor shall submit a revised programme to the Engineer within 15 days in accordance with this Sub-Clause. (Para-19.(3) of SGCC-2022)

**13.(4) Setting out of Works:** The Contractor shall be responsible for the correct setting out of all works in relation to original points, lines and levels of reference at his cost. The Contractor shall execute the work true to alignment, grade, levels and dimensions as shown in the drawing and as directed by the Engineer's representative and check these at frequent intervals. The Contractor shall provide all facilities like labour and instruments and shall co-operate with the Engineer's representative for checking of all alignment, grades, levels and dimensions. If, at any time, during the progress of the works any error appear or arise in any part of the work, the Contractor, on being required so to do by the Engineer's representative shall, at his own cost rectify such errors, to the satisfaction of the Engineer's representative.

Such checking shall not absolve the Contractor of his own responsibility of maintaining accuracy in the work. The Contractor shall carefully protect and preserve all bench marks, sight rails, pegs and other things used in setting out the work. (Para-19.(4) of SGCC-2022)

**14.(1) Compliance to Engineer's Instructions:** The Engineer shall direct the order in which the several parts of the works shall be executed and the Contractor shall execute without delay all orders given by the Engineer from time to time; but the Contractor shall not be relieved thereby from responsibility for the due performance of the works in all respects. (Para-20.(1) of SGCC-2022)

**14.(2) Alterations to be Authorized:** No alterations in or additions to or omissions or abandonment of any part of the works shall be deemed authorised, except under instructions from the Engineer. The Contractor shall be responsible to obtain such instructions in each and every case in writing from the Engineer. (Para-20.(2) of SGCC-2022)

**14.(3) Extra Works:** Should works over and above those included in the contract require to be executed at the site, the Contractor shall have no right to be entrusted with the execution of such works which may be carried out by another Contractor or Contractors or by other means at the option of the Railway. (Para-20.(3) of SGCC-2022)

**14.(4) Separate Contracts in Connection with Works:** The Railway shall have the right to let other contracts in connection with the works. The Contractor shall afford other Contractors reasonable opportunity for the storage of their materials and the execution of their works and shall properly connect and coordinate his work with theirs. If any part of the Contractor's work depends upon proper execution or result upon the work of another Contractor(s), the Contractor shall inspect and promptly report to the Engineer any defects in such works that render it unsuitable for such proper execution and results. The Contractor's failure so-to inspect and report shall constitute an acceptance of the other Contractor's work as fit and proper for the reception of his work, except as to defects which may develop in the other Contractor's work after the execution of his work. (Para-20.(4) of SGCC-2022)

**15. Instruction of Engineer's Representative:** Any instructions or approval given by the Engineer's representative to Contractor in connection with the works shall bind the Contractor as though it had been given by the Engineer provided always as follows:

- (a) Failure of the Engineer's representative to disapprove any work or materials shall not prejudice the power of the Engineer thereafter to disapprove such work or material and to order the removal or breaking up thereof.
- (b) If the Contractor shall be dissatisfied by reason of any decision of the Engineer's representative, he shall be entitled to refer the matter to the Engineer who shall there upon confirm or vary such decision. (Para-21 of SGCC-2022)

**16.(1) Adherence to Specifications and Drawings:** The site and the detailed drawings shall be made available to the contractor commensurate with the accepted programme of work submitted under clause 13(3). The whole of the works shall be executed in perfect conformity with the specifications and drawings of the contract. If Contractor performs any works in a manner contrary to the specifications or drawings or any of them and without such reference to the Engineer, he shall bear all the costs arising or ensuing therefrom and shall be responsible for all loss to the Railway. (Para-22.(1) of SGCC-2022)

**16.(2) Drawings and Specifications of the Works:** The Contractor shall keep one copy of Drawings and Specifications at the site, in good order, and such contract documents as may be necessary, available to the Engineer or the Engineer's Representative. (Para-22.(2) of SGCC-2022)

**16.(3) Ownership of Drawings and Specifications:** All Drawings and Specifications and copies thereof furnished by the Railway to the Contractor are deemed to be the property of the Railway. They shall not be used on other works and with the exception of the signed contract set, shall be returned by the Contractor to the Railway on completion of the work or termination of the Contract. (Para-22.(3) of SGCC-2022)

**16.(4) Compliance with Contractor's Request for Details:** The Engineer shall furnish with reasonable promptness, after receipt by him of the Contractor's request, additional instructions by means of drawings or otherwise, necessary for the proper execution of the works or any part thereof. All such drawings and instructions shall be consistent with the Contract Documents and reasonably inferable therefrom. (Para-22.(4) of SGCC-2022)

**16.(5) Meaning and Intent of Specification and Drawings:** If any ambiguity arises as to the meaning and intent of any portion of the Specifications and Drawings or as to execution or quality of any work or material, or as to the measurements of the works the decision of the Engineer thereon shall be final subject to the appeal (within 7 days of such decision being intimated to the Contractor) to the Chief Engineer who shall have the power to correct any errors, omissions, or discrepancies in aforementioned items and whose decision in the matter in dispute or doubt shall be final and conclusive. (Para-22.(5) of SGCC-2022)

**17. Working during Night:** The Contractor shall not carry out any work between sun-set and sun-rise without the previous permission of the Engineer. However, if the Engineer is satisfied that the work is not likely to be completed in time except by resorting to night work, he may order the same without confirming any right on the Contractor for claiming any extra payment for the same. (Para-23 of SGCC-2022)

**18. Damage to Railway Property or Private Life and Property:** The Contractor shall be responsible for all risk to the work and for trespass and shall make good at his own expense all loss or damage whether to the works themselves or to any other property of the Railway or the lives, persons or property of others from whatsoever cause in connection with the works until they are taken over by the Railway, although all reasonable and proper precautions may have been taken by the Contractor. In case the Railway shall be called upon to make good any costs, loss or damages, or to pay any compensation, including that payable under the provisions of the Workmen's Compensation Act or any statutory amendments thereof to any person or persons sustaining damages as aforesaid by reason of any act, or any negligence or omissions on the part of the Contractor; the amount of any costs or charges including costs and charges in connection with legal proceedings, which the Railway may incur in reference thereto, shall be charged to the Contractor.

The Railway shall have the power and right to pay or to defend or compromise any claim of threatened legal proceedings or in anticipation of legal proceedings being instituted consequent on the action or default of the Contractor, to take such steps as may be considered necessary or desirable to ward off or mitigate the effect of such proceedings, charging to Contractor, as aforesaid; any sum or sums of money which may be paid and any expenses whether for reinstatement or otherwise which may be incurred and the propriety of any such payment, defence or compromise, and the incurring of any such expenses shall not be called in question by the Contractor. (Para-24 of SGCC-2022)

**19. Sheds, Storehouses and Yards:** The Contractor shall at his own expense provide himself with sheds, storehouses and yards in such situations and in such numbers as in the opinion of the Engineer is requisite for carrying on the works and the Contractor shall keep at each such sheds, storehouses and yards a sufficient quantity of materials and plant in stock as not to delay the carrying out of the works with due expedition and the Engineer and the Engineer's representative shall have free access to the said sheds, store houses and yards at any time for the purpose of inspecting the stock of materials or plant so kept in hand, and any materials or plant which the Engineer may object to shall not be brought upon or used in the works, but shall be forthwith removed from the sheds, storehouses or yards by the Contractor. The Contractor shall at his own expenses provide and maintain suitable mortar mills, soaking vats or any other equipments necessary for the execution of the works. (Para-25 of SGCC-2022)

**20. Provision of Efficient and Competent Staff at Work Sites by the Contractor:**

**20.1** The Contractor shall place and keep on the works at all times efficient and competent staff to give the necessary directions to his workmen and to see that they execute their work in sound & proper manner and shall employ only such supervisors, workmen & labourers in or about the execution of any of these works as are careful and skilled in the various trades.

**20.2** The Contractor shall at once remove from the works any agents, permitted sub-contractor, supervisor, workman or labourer who shall be objected to by the Engineer and if and whenever required by the Engineer, he shall submit a correct return showing the names of all staff and workmen employed by him.

**20.3** In the event of the Engineer being of the opinion that the Contractor is not employing on the works a sufficient number of staff and workmen as is necessary for proper completion of the works within the time prescribed, the Contractor shall forthwith on receiving intimation to this effect deploy the additional number of staff and labour as specified by the Engineer within seven days of being so required and failure on the part of the Contractor to comply with such instructions will entitle the Railway to rescind the contract under Clause 62 of Standard General Conditions of Contract. (Para-26 of SGCC-2022)

**21A. Deployment of Qualified Engineers at Work Sites by the Contractor:**

**21A.1** The Contractor shall also employ qualified Graduate Engineer(s) or equivalent or qualified Diploma Engineer(s), as prescribed in the tender document.

**21A.2** In case the Contractor fails to employ the Engineer, as aforesaid in Para 21A.1, he shall be liable to pay liquidated damages at the rates, as prescribed in the tender document.

**21A.3** No. of qualified Engineers required to be deployed by the Contractor for various activities contained in the works contract shall be specified in the tender documents as 'Special Condition of Contract.' (Para-26A of SGCC-2022)

**21B. Engagement of Minimum Technical Personnel:** The contractor shall deploy following Minimum Numbers of Qualified Graduate Engineers and qualified Diploma Holder Engineer at site for Execution of Work:

| Contract Value<br>(in crore of Rs.) | Deployment of Graduate Engineer at site |                                    | Deployment of Diploma Holder Engineer at site |                                    | Remarks                                                                                                                       |
|-------------------------------------|-----------------------------------------|------------------------------------|-----------------------------------------------|------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
|                                     | Minimum No. to be Deployed              | Minimum Experience in No. of years | Minimum No. to be Deployed                    | Minimum Experience in No. of years |                                                                                                                               |
| Upto 2                              | -                                       | -                                  | 1                                             | 3                                  | These are bare minimum numbers to be deployed at site failing which recovery shall be made as indicated under para-iii below. |
| >2 - 5                              | 1                                       | 3                                  | 1                                             | 3                                  |                                                                                                                               |
| >5 -10                              | 1                                       | 3                                  | 2                                             | 3                                  |                                                                                                                               |
| >10 - 25                            | 2                                       | 3                                  | 3                                             | 3                                  |                                                                                                                               |
| >25 - 50                            | 2                                       | 3                                  | 4                                             | 3                                  |                                                                                                                               |
| >50 - 100                           | 4                                       | 3                                  | 6                                             | 3                                  |                                                                                                                               |
| More than 100                       | 4                                       | 3                                  | 6                                             | 3                                  |                                                                                                                               |

**(i) Deployment of Qualified Engineers at Work Sites by the Contractor:**

- (a) The Contractor shall employ qualified Graduate Engineer(s) or equivalent, or qualified Diploma Engineer(s), as prescribed above. The contractor shall also employ independent technically qualified person(s) (Minimum Diploma holder) to manage the Material Testing Laboratory (ies) set up at different work sites. Proper record for availability of qualified engineers shall be maintained.
  - (b) In case the Contractor fails to employ the Engineer, as aforesaid, he shall be liable to pay liquidated damages at the rates, as prescribed in Para (iii) below. However, in case, the work at site/sites is/are stopped/restricted either as per the written directions of Engineer-in-charge due to in-sufficient fund with Railways or due to any natural calamity like COVID-19 or lockdown imposed by local authorities, the contractor may be exempted from deployment of so many no. of engineers during such period as decided by the Engineer-in-charge after submission of written request by the contractor.
  - (c) No. of qualified Engineers required to be deployed by the Contractor (over and above as aforesaid) for various activities contained in the works contract shall be specified in the tender documents as 'Special Condition of Contract' by the tender inviting authority.
- (ii)** Contractor shall submit qualification certificate and experience certificate in relevant field of construction work for Graduate Engineer / Diploma Holder Engineer to be deployed at site, for approval of Engineer-in-charge.
- (iii)** In case the Contractor fails to employ as aforesaid provision, he shall be liable to pay an amount of Rs.40,000/- and Rs.25,000/- per month or part thereof respectively for each Graduate Engineer and each Diploma Holder Engineer respectively for the default period and numbers.

- (iv) **Engagement of Minimum other Technical Staff** (with adequate Experience): In addition to deployment of above technical Engineers at site, the contractor shall deploy for the contract exclusively at least one Architect (for contracts of building construction works only), **at least one Draftsman who is skilled in Auto-CAD, one Surveyor and at least one Computer operator for the contract value more than Rs.20 (twenty) crore**, failing which recovery of Rs.20,000/- per month or part thereof shall be effected for each technical staff for the default period and numbers.

However, in case, the work at site/sites is/are stopped/restricted either as per the written directions of Engineer-in-charge due to in-sufficient fund with Railways or due to any natural calamity like COVID-19 or lockdown imposed by local authorities, the contractor may be exempted from deployment of these technical staff during such period as decided by the Engineer-in-charge after submission of written request by the contractor.

- (v) The provisions under as above should not dilute the obligations of contractor to provide sufficient number of qualified technical and other staff to comply with the approved “Method Statement” and “Quality Assurance Plan”.

**22.(1) Workmanship and Testing:** The whole of the works and/or supply of materials specified and provided in the contract or that may be necessary to be done in order to form and complete any part thereof shall be executed in the best and most substantial workman like manner with materials of the best and most approved quality of their respective kinds, agreeable to the particulars contained in or implied by the specifications and as referred to in and represented by the drawings or in such other additional particulars, instructions and drawings given during the carrying on of the works and to the entire satisfaction of the Engineer according to the instructions and directions which the Contractors may from time to time receive from the Engineer. The materials may be subjected to tests by means of such machines, instruments and appliances as the Engineer may direct and wholly at the expense of the Contractor. (Para-27.(1) of SGCC-2022)

**22.(2) Removal of Improper Work and Materials:** The Engineer or the Engineer's Representative shall be entitled to order from time to time:

(a) The removal from the site, within the time specified in the order, of any materials which in his opinion are not in accordance with the specifications or drawings.

(b) The substitution of proper and suitable materials, and

(c) the removal and proper re-execution, notwithstanding any previous tests thereof or on account payments therefor, of any work which in respect of materials or workmanship is not in his opinion in accordance with the specifications and in case of default on the part of the Contractor in carrying out such order, the Railway shall be entitled to rescind the contract under Clause 62 of GCC.

(d) The provision of Construction and Demolition Waste Management Rule 2016 issued by Ministry of Environment Forest and Climate Change dated 29.03.2016 and published in the Gazette of India, Part – II, Section -3, Sub-section (ii) are binding upon the Contractor. Contractor shall implement these provisions at worksites, for which no extra payment will be payable. (Para-27. (2) of SGCC-2022)

**23.(1) Contractor to Supply Water for Works:** Unless otherwise provided in the Contract, the Contractor shall be responsible for the arrangements to obtain supply of water necessary for the works. (Para-31.(1) of SGCC-2022)

**23.(2) Water Supply from Railway System:** The Railway may supply to the Contractor part or whole of the quantity of the water required for the execution of works from the Railway's existing water supply system at or near the site of works on specified terms and conditions and at such charges as shall be determined by the Railway and payable by the Contractor, provided that the Contractor shall arrange, at his own expense, to effect the connections and lay additional pipelines and accessories on the site and that the Contractor shall not be entitled to any compensation for interruption or failure of the water supply. (Para- 31.(2) of SGCC-2022).

**23.(3) Water Supply by Railway Transport:** In the event of the Railway arranging supply of water to the Contractor at or near the site of works by travelling water tanks or other means, the freight and other charges incurred thereby, including demurrage charges that may be levied, shall be paid by the Contractor in addition to the charges referred to in Sub-Clause

(3) of the Clause provided that the Contractor shall not be entitled to any compensation for interruption or failure of the water supply. (Para-31.(3) of SGCC-2022)

**23.(4)(a) Contractor to Arrange Supply of Electric Power for Works:** Unless otherwise provided in the contract, the Contractor shall be responsible for arrangements to obtain supply of Electric Power for the works. (Para-31.(4) (a) of SGCC-2022)

**(b) Electric Supply from the Railway System:** The Railway may supply to the Contractor part or whole of the electric power wherever available and possible, required for execution of works from the Railway's existing electric supply systems at or near the site of works on specified terms and conditions and such charges as shall be determined by the Railway and payable by the Contractor provided the cost of arranging necessary connections to the Railway's Electric Supply systems and laying of underground/overhead conductor, circuit protection, electric power meters, transmission structure, shall be borne by the Contractor and that the Contractor shall not be entitled to any compensation for interruption or failure of the Electric supply system. (Para-31.(4) (b) of SGCC-2022)

**24.(1) Tools, Plant and Materials Supplied by Railway:** The Contractor shall take all reasonable care of all tools, plant and materials or other property whether of a like description or not belonging to the Railway and committed to his charge for the purpose of the works and shall be responsible for all damage or loss caused by him, his agents, permitted sub-contractor, or his workmen or others while they are in his charge. The Contractors shall sign accountable receipts for tools, plants and materials made over to him by the Engineer and on completion of the works shall hand over the unused balance of the same to the Engineer in good order and repair, fair wear and tear excepted, and shall be responsible for any failure to account for the same or any damage done thereto. (Para-33.(1) of SGCC-2022)

**24.(2) Hire of Railway's Plant:** The Railway may hire to the Contractor such plant as concrete mixers, compressors and portable engines for use during execution of the works on such terms as may be specified in the special conditions or in a separate agreement for Hire of Plant. (Para-33.(2) of SGCC-2022)

**25.(1) Precaution During Progress of Works:** During the execution of works, unless otherwise specified, the Contractor shall at his own cost provide the materials for and execute all shoring, timbering and strutting works as is necessary for the stability and safety of all structures, excavations and works and shall ensure that no damage, injury or loss is caused or likely to be caused to any person or property. (Para-34.(1) of SGCC-2022)



**25.(2) Roads and Water Courses:** Existing roads or water courses shall not be blocked cut through, altered, diverted or obstructed in any way by the Contractor, except with the permission of the Engineer. All compensations claimed for any unauthorized closure, cutting through, alteration, diversion or obstruction to such roads or water courses by the Contractor or his agent or his staff shall be recoverable from the Contractor's bills/Security Deposit or any other dues of Contractor with the Government of India. (Para-34.(2) of SGCC-2022)

**25.(3) Provision of Access to Premises:** During progress of work in any street or thoroughfare, the Contractor shall make adequate provision for the passage of traffic, for securing safe access to all premises approached from such street or thoroughfare and for any drainage, water supply or means of lighting which may be interrupted by reasons of the execution of the works and shall react and maintain at his own cost barriers, lights and other safeguards as prescribed by the Engineer, for the regulation of the traffic, and provide watchmen necessary to prevent accidents. The works shall in such cases be executed night and day, if so ordered by the Engineer and with such vigour so that the traffic way be impeded for as short a time as possible. (Para-34.(3) of SGCC-2022)

**25.(4) Safety of Public:** The Contractor shall be responsible to take all precautions to ensure the safety of the public whether on public or railway property and shall post such look out men as may, in the opinion of the Engineer, be required to comply with regulations appertaining to the work. Contractor shall ensure placement of barricading / partitions at the place of work to ensure safety of habitants of adjacent area, failing which Engineer may advise stoppage of work as per his discretion. (Para-34.(4) of SGCC-2022)

**25.(5) Display Board:** The Contractor shall be responsible for displaying the details of works i.e. name of work, approximate cost, expected date of completion, name and address of the Contractor and address of Engineer on a proper steel Board of size not less than 1m x 1m. (Para-34.(5) of SGCC-2022)

**26. Use of Explosives:** Explosives shall not be used on the works or on the site by the Contractor without the permission of the Engineer and then also only in the manner and to the extent to which such permission is given. Where explosives are required for the works, they shall be stored in a special magazine to be provided by and at the cost of the Contractor in accordance with the Explosive Rules. The Contractor shall obtain the necessary license for the storage and the use of explosives. All operations in which or for which explosives are employed shall be at the sole risk and responsibility of the Contractor and the Contractor shall indemnify the Railway in respect thereof. (Para-35 of SGCC-2022)

**27. Rates for Items of Works:**

(i) The rates, entered in the accepted Schedule of Rates of the Contract are intended to provide for works duly and properly completed in accordance with the General and Special (if any) Conditions of the Contract and the Specifications and drawings together with such enlargements, extensions, diminutions, reductions, alterations or additions as may be ordered in terms of Clause 42 of Standard General Condition of Contract and without prejudice to the generality thereof and shall be deemed to include and cover superintendence and labour, supply, including full freight of materials, stores, patterns, profiles, moulds, fittings, centerings, scaffolding, shoring props, timber, machinery, barracks, tackle, roads, pegs, posts, tools and all apparatus and plant required on the works, except such tools, plant or materials as may be specified in the contract to be supplied to the Contractor by the Railway, the erection, maintenance and removal of all temporary works and buildings, all watching, lighting, bailing, pumping and draining, all prevention of or compensation for trespass, all

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barriers and arrangements for the safety of the public or of employees during the execution of works, all sanitary and medical arrangements for labour camps as may be prescribed by the Railway, the setting of all work and of the construction, repair and upkeep of all centre lines, bench marks and level pegs thereon, site clearance, *all fees, duties, royalties, rent and compensation to owners for surface damage* or taxes and impositions payable to local authorities in respect of land, structures and all material supplied for the work or other duties or expenses for which the Contractor may become liable or may be put to under any provision of law for the purpose of or in connection with the execution of the contract and all such other incidental charges or contingencies as may have been specially provided for in the Specifications.

However, if rates of existing GST or cess on GST for Works Contract is increased or any new tax /cess on Works Contract is imposed by Statute after the date of opening of tender but within the original date of completion/date of completion extended under clause 17 & 17A of Standard General Conditions of Contract and the Contractor there upon properly pays such taxes/cess, the Contractor shall be reimbursed the amount so paid.

Further, if rates of existing GST or cess on GST for Works Contract is decreased or any tax/cess on Works Contract is decreased / removed by Statute after the date of opening of tender, the reduction in tax amount shall be recovered from Contractor's bills/Security Deposit or any other dues of Contractor with the Government of India. (Para-37 of SGCC-2022)

**28.(1) Rates for Extra Items of Works:** Standard Schedule of Rates (SSOR) Items: Any item of work carried out by the Contractor on the instructions of the Engineer which is not included in the accepted Bill(s) of Quantities but figures in the Standard Schedule of Rates (SSOR), shall be executed at the rates set forth in the "Standard Schedule of Rates (SSOR)" modified by the tender percentage as accepted in the contract for that chapter of Standard Schedule of Rates (SSOR).

For item(s) not covered in this sub clause, the rate shall be decided as agreed upon between the Engineer and the Contractor before the execution of such items of work as per sub clause (b).

(b) Other Items: For any item of work to be carried out by the Contractor but not included in the accepted Bill(s) of Quantities and also not covered under sub clause (a) above, the Contractor shall be bound to notify the Engineer at least seven days before the necessity arises for the execution of such items of works that the accepted Bill(s) of Quantities does not include rate or rates for such extra work involved. The rates payable for such items shall be decided at the meeting to be held between the Engineer and Contractor, in as short a period as possible after the need for the special item has come to the notice. In case the Contractor fails to attend the meeting after being notified to do so or in the event of no settlement being arrived at, the Railway shall be entitled to execute the extra works by other means and the Contractor shall have no claim for loss or damage that may result from such procedure.

The assessment of rates for extra items shall be arrived at based on the prevailing rates and by taking guidance from the following documents in order of priority.

- (i) Analysis of Rates for "Unified Schedule of Rates of Indian Railways (USSOR)"
- (ii) Analysis of Rates for "Delhi Schedule of Rates issued by CPWD (DSR)"
- (iii) Market Analysis (Para-39 .(1) of SGCC-2022)

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**28.(2)** Provided that if the Contractor commences work or incurs any expenditure in regard thereto before the rates as determined and agreed upon as lastly here untofore-mentioned, then and in such a case the Contractor shall only be entitled to be paid in respect of the work carried out or expenditure incurred by him prior to the date of determination of the rates as aforesaid according to the rates as shall be fixed by the Engineer. However, if the Contractor is not satisfied with the decision of the Engineer in this respect, he may appeal to the Chief Engineer within 30 days of getting the decision of the Engineer, supported by analysis of the rates claimed. The Chief Engineer's decision after hearing both the parties in the matter would be final and binding on the Contractor and the Railway. (Para-39.(2) of SGCC-2022)

**29. Offloading of Part(s) of Work:** At the final stage of completion/ commissioning of work, in case the contractor fails to complete the final part(s) of the work and the value of such part(s) of the work is limited to 5% of the original contract value, the Engineer may allow/decide for offloading of such part(s) of works, either after the Contractor's request in writing to do so or after serving a 14 (Fourteen) days suo-moto notice (as per annexure- VIIA of GCC-2022), if the Engineer is of the opinion that :-

- (i) Such Offloading of works (up to 5% of original contract value) would enable successful completion of contract/work,
- (ii) Termination/ Part termination of the contract at this stage is not be in the interest of the Railway/work; and
- (iii) The anticipated additional cost for execution of such works through other mode would not be substantial and can be recovered from the pending dues of the contractor;

The Contractor shall be informed, in due course, by the Engineer of the mode and cost of execution of such offloaded work through other agency(ies) (as per annexure- VIIB). The extra expenditure so incurred in execution of the offloaded work, shall be recovered from subsequent Bill(s) or any other dues of the Contractor, but not exceeding the value of Performance Guarantee available in the contract. There shall be no other repercussion of such offloading on execution of the balance contract. The Contractor shall have no claim on account of above-mentioned offloading of works. (Para-40A of SGCC-2022).

**Such offloading of works (up to 5% of original contract value) would be done after taking the prior approval of PHOD** (Ref. RB letter No. 2022/CE-I/CT/GCC-2022/POLICY dated 27.04.2022).

#### **VARIATIONS IN EXTENT OF CONTRACT**

**30. Modification to Contract to be in Writing:** In the event of any of the provisions of the contract required to be modified after the contract documents have been signed, the modifications shall be made in writing and signed by the Railway and the Contractor and no work shall proceed under such modifications until this has been done. Any verbal or written arrangement abandoning, modifying, extending, reducing or supplementing the contract or any of the terms thereof shall be deemed conditional and shall not be binding on the Railway unless and until the same is incorporated in a formal instrument and signed by the Railway and the Contractor, and till then the Railway shall have the right to repudiate such arrangements. (Para-41 of SGCC-2022)

**30.(1) Powers of Modification to Contract:** The Engineer on behalf of the Railway shall be entitled by order in writing to enlarge or extend, diminish or reduce the works or make any alterations in their design, character position, site, quantities, dimensions or in the method of

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their execution or in the combination and use of materials for the execution thereof or to order any additional work to be done or any works not to be done and the Contractor will not be entitled, to any compensation for any increase/reduction in the quantities of work but will be paid only for the actual amount of work done and for approved materials supplied against a specific order. (Para-42(1) of SGCC-2022)

**30.(2)(i)** Unless otherwise specified in the special conditions of the contract, the accepted variation in quantity of each individual item of the contract would be upto 25% of the quantity originally contracted, except in case of foundation work (in which no variation limit shall apply). However, the rates for the increased quantities shall be as per sub- para (iii) below.

(ii) The Contractor shall be bound to carry out the work at the agreed rates and shall not be entitled to any claim or any compensation whatsoever upto the limit of 25% variation in quantity of individual item of works.

(iii) In case an increase in quantity of an individual item by more than 25% of the agreement quantity is considered unavoidable, then same shall be executed at following rates

- (a) Quantities operated in excess of 125% but upto 140% of the agreement quantity of the concerned item, shall be paid at 98% of the rate awarded for that item in that particular tender;
- (b) Quantities operated in excess of 140% but upto 150% of the agreement quantity of the concerned item shall be paid at 96% of the rate awarded for that item in that particular tender;
- (c) Variation in quantities of individual items beyond 150% will be avoided and would be permitted only in exceptional unavoidable circumstances and shall be paid at 96% of the rate awarded for that item in that particular tender.

(d) Variation to quantities of Minor Value Item:

The limit for varying quantities for minor value items shall be 100% (as against 25% prescribed for other items). A minor value item for this purpose is defined as an item whose original agreement value is less than 1 % of the total original agreement value.

d.(i) Quantities operated upto and including 100% of the agreement quantity of the concerned minor value item, shall be paid at the rate awarded for that item in that particular tender;

d.(ii) Quantities operated in excess of 100% but upto 200% of the agreement quantity of the concerned minor value item, shall be paid at 98% of the rate awarded for that item in that particular tender;

d.(iii) Variation in quantities of individual minor value item beyond 200% will be avoided and would be permitted only in exceptional unavoidable circumstances and shall be paid at 96% of the rate awarded for that item in that particular tender.

(iv) In case of earthwork, the variation limit of 25% shall apply to the gross quantity of earthwork items and variation in the quantities of individual classifications of soil shall not be subject to this limit.

(v) As far as Standard Schedule of Rates (SSOR) items are concerned, the variation limit of 25% would apply to the value of SOR schedule as a whole and not on individual SSOR items. However, in case of Non-Standard Schedule of Rates (Non-SSOR) items, the limit of 25% would apply on the individual items irrespective of the manner of quoting the rate (single percentage rate or individual item rate). (Para-42(2) of SGCC-2022)

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**30.(3) Valuation of Variations:** The enlargements, extensions, diminution, reduction, alterations or additions referred to in Sub-Clause (2) of this Clause shall in no degree affect the validity of the contract; but shall be performed by the Contractor as provided therein and be subject to the same conditions, stipulations and obligations as if they had been originally and expressively included and provided for in the Specifications and Drawings and the amounts to be paid therefor shall be calculated in accordance with the accepted Bills of Quantities. Any extra items/quantities of work falling outside the purview of the provisions of Sub-Clause (2) above shall be paid for at the rates determined under Clause-39 of Standard General Conditions of Contract. (Para-42(3) of SGCC-2022)

### **CLAIMS**

**31.(1) Quarterly Statement of Claims:** The Contractor shall prepare and furnish to the Engineer once in every quarter commencing from the month following the month of issue of Letter of Acceptance, an account giving full and detailed particulars of all claims for any additional expenses to which the Contractor may consider himself entitled to and of all extra or additional works ordered by the Engineer which he has executed during the preceding quarter and no claim for payment for such work will be considered which has not been included in such particulars. (Para-43(1) of SGCC-2022)

**31.(2) Signing of "No Claim" Certificate:** The Contractor shall not be entitled to make any claim whatsoever against the Railway under or by virtue of or arising out of this contract, nor shall the Railway entertain or consider any such claim, if made by the Contractor, after he shall have signed a "No Claim" Certificate in favour of the Railway in such form as shall be required by the Railway after the works are finally measured up. The Contractor shall be debarred from disputing the correctness of the items covered by "No Claim" Certificate or demanding a clearance to arbitration in respect thereof. (Para-43(2) of SGCC-2022)

### **MEASUREMENTS, CERTIFICATES AND PAYMENTS**

**32. Quantities in Bills of Quantities Annexed to Contract:** The quantities set out in the accepted Bills of Quantities with items of works quantified are the estimated quantities of the works and they shall not be taken as the actual and correct quantities of the work to be executed by the Contractor in fulfillment of his obligations under the contract. (Para-44 of SGCC-2022)

**33(i). Measurement of Works by Railway:** The Contractor shall be paid for the works at the rates in the accepted Bills of Quantities and for extra works at rates determined under Clause 39 of Standard General Conditions of Contract on the measurements taken by the Engineer or the Engineer's representative in accordance with the rules prescribed for the purpose by the Railway. The quantities for items the unit of which in the accepted Bills of Quantities is 100 or 1000 shall be calculated to the nearest whole number, any fraction below half being dropped and half and above being taken as one; for items the unit of which in the accepted Bills of Quantities is single, the quantities shall be calculated to two places of decimals. Such measurements will be taken of the work in progress from time to time and at such intervals as in the opinion of the Engineer shall be proper having regard to the progress of works. The date and time on which 'on account' or 'final' measurements are to be made shall be communicated to the Contractor who shall be present at the site and shall sign the results of the measurements (which shall also be signed by the Engineer or the Engineer's representative) recorded in the official measurements book as an acknowledgement of his acceptance of the accuracy of the measurements. Failing the Contractor's attendance, the

work may be measured up in his absence and such measurements shall, notwithstanding such absence, be binding upon the Contractor whether or not he shall have signed the measurement books provided always that any objection made by him to measurement shall be duly investigated and considered in the manner set out below:

(a) It shall be open to the Contractor to take specific objection to any recorded measurements or Classification on any ground within seven days of the date of such measurements. Any re-measurement taken by the Engineer or the Engineer's representative in the presence of the Contractor or in his absence after due notice has been given to him in consequence of objection made by the Contractor shall be final and binding on the Contractor and no claim whatsoever shall thereafter be entertained regarding the accuracy and Classification of the measurements.

(b) If an objection raised by the Contractor is found by the Engineer to be incorrect the Contractor shall be liable to pay the actual expenses incurred in measurements. (Para-45(i) of SGCC-2022)

**33(ii). Measurement of Works by Contractor's Authorized Representative (in case the contract provides for the same):**

(a) The Contractor shall be paid for the works at the rates in the accepted Bills of Quantities and for extra works at rates determined under Clause 39 of Standard General Conditions of Contract on the measurements taken by the Contractor's authorized Engineer in accordance with the rules prescribed for the purpose by the Railway.

The quantities for items the unit of which in the accepted Bills of Quantities is 100 or 1000 shall be calculated to the nearest whole number, any fraction below half being dropped and half and above being taken as one; for items the unit of which in the accepted Bills of Quantities is single, the quantities shall be calculated to two places of decimals. Such measurements will be taken of the work in progress from time to time. The date and time on which 'on account' or 'final' measurements are to be made shall be communicated to the Engineer.

The date and time of test checks shall be communicated to the Contractor who shall be present at the site and shall witness the test checks, failing the Contractor's attendance the test checks may be conducted in his absence and such test checks shall notwithstanding such absence be binding upon Contractor provided always that any objection made by Contractor to test check shall be duly investigated and considered in the manner set out below:

(i) It shall be open to the Contractor to take specific objection to test checks of any recorded measurement within 7 days of date of such test checks. Any re-test check done by the concerned Railway's authority in the presence of the Contractor or in his absence after due notice given to him in consequent of objection made by the Contractor shall be final and binding on the Contractor and no claim whatsoever shall thereafter be entertained regarding the accuracy and classification of the measurements.

(ii) If an objection raised by the Contractor is found by the Engineer to be incorrect the Contractor shall be liable to pay the actual expenses incurred in measurements.

**(b) Incorrect measurement, actions to be taken:** If in case during test check or otherwise, it is detected by the Engineer that agency has claimed any exaggerated measurement or has claimed any false measurement for the works which have not been executed; amounting to variation of 5% or more of claimed gross bill amount, action shall be taken as following:

- (i) On first occasion of noticing exaggerated/ false measurement, Engineer shall recover liquidated damages equal to 10% of claimed gross bill value.
- (ii) On any next occasion of noticing any exaggerated/false measurement, railway shall recover liquidated damages equal to 15% of claimed gross bill value. In addition, the facility of recording of measurements by Contractor as well as release of provisional payment shall be withdrawn. Once withdrawn, measurements shall be done by railway as per clause 33(i) above. (Para-45(ii) of SGCC-2022)

**34.(1) "On-Account " Payments:** The Contractor shall be entitled to be paid from time to time by way of "On-Account" payment only for such works as in the opinion of the Engineer he has executed in terms of the contract. All payments due on the Engineer's/Engineer's Representative's certificates of measurements or Engineer's certified "Contractor's authorized Engineer's measurements" shall be subject to any deductions which may be made under these presents and shall further be subject to, unless otherwise required by Clause 16 of Standard General Conditions of Contract, a retention of ten percent by way of Security Deposits, until the amount of Security Deposit by way of such retentions shall amount to 5% of the total value of the contract provided always that the Engineer may by any certificate make any correction or modification in any previous certificate which shall have been issued by him and that the Engineer may withhold any certificate, if the works or any part thereof are not being carried out to his satisfaction. (Para-46(1) of SGCC-2022)

**34.(2) Rounding off Amounts:** The total amount due on each certificate shall be rounded off to the nearest rupee, i.e. sum less than 50 paise shall be omitted and sums of 50 paise and more upto ₹1 will be reckoned as ₹ 1. (Para-46(2) of SGCC-2022)

**34.(3) On Account Payments not Prejudicial to Final Settlement:** "On-Account" payments made to the Contractor shall be without prejudice to the final making up of the accounts (except where measurements are specifically noted in the Measurement Book as "Final Measurements" and as such have been signed by the Contractor and Engineer/Engineer's Representative) and shall in no respect be considered or used as evidence of any facts stated in or to be inferred from such accounts nor of any particular quantity of work having been executed nor of the manner of its execution being satisfactory. (Para-46(3) of SGCC-2022)

**34.(4)** If payment(s) of Advances are applicable in the contract, as mentioned in the Tender Documents, Railway shall make payment(s) of Interest-bearing advances, on the request of contractor. The payment and recovery of such Advances shall be made as under:

**(a) : Mobilization Advance –**

This shall be limited to 10% of the Contract value and shall be paid in 2 stages:

**Stage 1**– 5% of Contract Value on signing of the contract agreement.

**Stage 2** – 5% on mobilization of site-establishment, setting up offices, bringing in equipment and actual commencement of work.

**The stage 1** of advance shall be payable immediately after signing of contract agreement. **The stage 2** of advance shall be payable at the time of mobilization, only after submission of an utilization certificate by the contractor that the Stage 1 advance has been properly utilized in the contract.

These Advances shall be payable against irrevocable guarantee (Bank Guarantee, FDRs) from a scheduled commercial bank of India **of at least 110%** of the value of the sanctioned advance amount (covering principal plus interest).

**(b) : Advance Against Machinery and Equipment –**

This advance shall be limited to a maximum of 10% of the contract value against new Machinery & Equipment, involving substantial outlay, brought to site and essentially required for the work. This advance shall not exceed 75% of the purchase price of such Equipment and shall be payable when Equipment is hypothecated to the President of India by a suitable bond or alternatively covered by an irrevocable Bank Guarantee from a scheduled commercial bank of India for full cost of the Plant & Equipment in a form acceptable to Railways. The Plant & Equipment shall be insured for the full value and for the entire period, they are required for the work. This Plant & Equipment shall not be removed from the site of work without prior written permission of the Engineer. No advance should be given against old Plant & Machinery.

**The advances under sub clause (a) and (b) above, are subject to the following conditions**

(i) The full amount of Advances shall be recovered from contractor dues. The recovery shall commence when the value of contract executed reaches 15% of original contract value and shall be completed when the value of work executed reaches 85% of the original contract value. The installments on each "on account bill" will be on pro-rata basis.

Interest shall be recovered on the advance outstanding for the period commencing from the date of payment of advance till date of particular on-account bill (through which recovery of principal is effected) and adjusted fully against on-account bill along with pro-rata principal recovery. In the event of any short-fall, the same shall be carried forward to the next on-account bill and shall attract interest.

(ii) The advances shall be used by the Contractor for the purpose of the Contract, and for the purpose for which they are paid. Under no circumstances, shall the advances be diverted for other purposes. Any such diversion shall be construed as a breach of the Contract and the Contractor shall be asked to return the advance at once and pay interest at 15% per annum till the advance is recovered back from him. The Contractor shall return the advance and pay the interest in one go without demur. The Contractor, if required by the Engineer shall provide the details of utilisation of Mobilisation advance.

(iii) If the Contractor is found to have contravened the provision, it will constitute a breach of contract and Railway shall be entitled to terminate the contract and forfeit his Performance Guarantee as well as Security Deposit.

(iv) In cases, where the Contract is rescinded as per clause 62 of the contract or short closed under any other condition(s) of the contract, without making full recovery of advances and accrued interest thereon, by the Railway, such balance of advances and accrued interest thereon shall immediately become due and payable by the Contractor to the Railway. The same shall be recovered from any due of Contractor with the Government of India. (Para-46(4) of SGCC-2022)

**34.(5) Manner of Payment:** Unless otherwise specified payments to the Contractor will be transferred electronically to his bank account. (Para-46(5) of SGCC-2022)

**35. Price Variation Clause (PVC):** (Para-46A of SGCC-2022)

**35.1 Applicability:** Price Variation Clause (PVC) shall be applicable **only in tender having advertised value above Rs. 2 Crores**. Provided further that, in a contract where PVC is applicable, following shall be outside the purview of price adjustments (i.e. shall be excluded from the gross value of the work for the purpose of price variation) :



- a) Materials supplied by Railway to the Contractors, either free or at fixed rate;
- b) Any extra item(s) included in subsequent variation falling outside the purview of the Bill(s) of Quantities of tender, under clause 39. (1)(b) of these Standard General Conditions, unless applicability of PVC and 'Base Month' has been specially agreed, while fixing the rates of such extra item(s).

**35.2 Base Month:** The Base Month for 'Price Variation Clause' shall be taken as the **one month prior to closing of tender including extensions, if any**, unless otherwise stated elsewhere. The quarter for applicability of PVC shall commence from the month following Base month. The Price Variation shall be based on the average Price Index of the quarter under consideration.

### 35.3 Validity:

Rates accepted by Railway Administration shall hold good till completion of work and no additional individual claim shall be admissible except:

- (a) Payment/recovery for increase/decrease in GST on works contract or imposition/removal of any tax/cess on Works Contract as per Clause 37 of Standard General Conditions of Contract,
- (b) Payment/recovery for overall market situation as per Price Variation Clause given hereunder.

**35.4** Components of various items in a contract on which variation in prices be admissible, shall be steel, cement, ferrous material, non-ferrous material, insulators, zinc and other materials, labour, plant & machinery, fuel, explosives, detonators etc. Adjustment for variation in prices of these items shall be determined in the manner prescribed.

**35.5** No price variation shall be admissible for fixed components.

**35.6** The percentages of various components in various type of works shall be as specified for all item (s)/ Bill(s) of Quantities in tender document and the same shall be fixed as per table & classifications given below:

#### (I). For Civil Engineering Works

| S<br>N | Classification                 |                 | 1A, 2 & 3A | 4A  | 5A  | 6A  | 7   | 8A  | 9A  | 1B, 3B, 4B, 5B,<br>6B 8B & 9B | 1C, 3C, 4C, 5C,<br>6C, 8C & 9C | 3D, 4D, 5D, 6D,<br>8D & 9D | 3E, 4E, 5E, 6E,<br>8E & 9E |
|--------|--------------------------------|-----------------|------------|-----|-----|-----|-----|-----|-----|-------------------------------|--------------------------------|----------------------------|----------------------------|
|        | Components                     |                 |            |     |     |     |     |     |     |                               |                                |                            |                            |
| 1      | Fixed                          | *               | 15         | 15  | 15  | 15  | 15  | 15  | 15  | 15                            | 15                             | 15                         | 15                         |
| 2      | Labour                         | L <sub>c</sub>  | 20         | 25  | 30  | 20  | 50  | 20  | 20  | 0                             | 0                              | 10                         | 25                         |
| 3      | Steel                          | S <sub>c</sub>  | 0          | 0   | 0   | 0   | 0   | 0   | 0   | 85                            | 0                              | 50                         | 0                          |
| 4      | Cement                         | C <sub>c</sub>  | 0          | 0   | 15  | 0   | 0   | 0   | 0   | 0                             | 85                             | 0                          | 0                          |
| 5      | Plant<br>Machinery<br>& Spares | PM <sub>c</sub> | 30         | 15  | 5   | 20  | 15  | 20  | 30  | 0                             | 0                              | 10                         | 30                         |
| 6      | Fuel &<br>Lubricants           | F <sub>c</sub>  | 25         | 15  | 5   | 15  | 15  | 20  | 15  | 0                             | 0                              | 10                         | 20                         |
| 7      | Other<br>materials             | M <sub>c</sub>  | 10         | 15  | 30  | 30  | 5   | 25  | 20  | 0                             | 0                              | 5                          | 10                         |
| 8      | Detonators &<br>Explosive      | E <sub>c</sub>  | 0          | 15  | 0   | 0   | 0   | 0   | 0   | 0                             | 0                              | 0                          | 0                          |
| Total  |                                |                 | 100        | 100 | 100 | 100 | 100 | 100 | 100 | 100                           | 100                            | 100                        | 100                        |

\* It shall not be considered for any price variation.

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The classification mentioned in the table above represents following type of item(s) in the work(s) –

**1 Earthwork in Formation**

- 1A All Item(s) excluding 1B or/and 1C
- 1B Item(s) for supply of Steel
- 1C Item(s) for supply of Cement

**2 Ballast Supply Works**

**3 Tunnelling Works (Without Explosives)**

- 3A All Item(s) excluding 3B or/and 3C or/and 3D or/and 3E
- 3B Item(s) for supply of Steel
- 3C Item(s) for supply of Cement or/and Grout
- 3D Item(s) for Fabrication & Erection of Structures including supply of Steel
- 3E Item(s) for Fabrication & Erection of Structures excluding supply of Steel.

**4 Tunnelling Works (With explosives)**

- 4A All Item(s) excluding 4B or/and 4C or/and 4D or/and 4E
- 4B Item(s) for supply of Steel
- 4C Item(s) for supply of Cement or/and Grout
- 4D Item(s) for Fabrication & Erection of Structures including supply of Steel
- 4E Item(s) for Fabrication & Erection of Structures excluding supply of Steel.

**5 Building Works**

- 5A All Item(s) excluding 5B or/and 5C or/and 5D or/and 5E
- 5B Item(s) for supply of Steel
- 5C Item(s) for supply of Cement
- 5D Item(s) for Fabrication & Erection of Structures including supply of Steel
- 5E Item(s) for Fabrication & Erection of Structures excluding supply of Steel.

**6 Bridges & Protection work**

- 6A All Item(s) excluding 6B or/and 6C or/and 6D or/and 6E
- 6B Item(s) for supply of Steel
- 6C Item(s) for supply of Cement
- 6D Item(s) for Fabrication, Assembly, Erection & Launching of Girders including supply of Steel
- 6E Item(s) for Fabrication, Assembly, Erection & Launching of Girders excluding supply of Steel

**7 Permanent Way linking**

**8 Platform, Passenger Amenities**

- 8A All Item(s) excluding 8B or/and 8C or/and 8D or/and 8E
- 8B Item(s) for supply of Steel item/fittings
- 8C Item(s) for supply of Cement Item
- 8D Item(s) for Fabrication & Erection of Structures including supply of Steel
- 8E Item(s) for Fabrication & Erection of Structures excluding supply of Steel

**9 Any Other Works not covered in Classification 1 to 8**

- 9A All Item(s) excluding 9B or/and 9C or/and 9D or/and 9E
- 9B Item(s) for supply of Steel
- 9C Item(s) for supply of Cement or/and Grout

9D Item(s) for Fabrication & Erection of Structures including supply of Steel

9E Item(s) for Fabrication & Erection of Structures excluding supply of Steel

**35.7 Formulae:** The Amount of variation in prices in various components (labour, material etc.) shall be worked out by the following formulae:

$$(i) \quad L = \frac{(W \text{ or } W_{SF} \text{ or } W_F \text{ or } W_{SFL} \text{ or } W_{FL}) \times (L_Q - L_B) \times L_C}{L_B \times 100}$$

$$(ii) \quad M = \frac{(W \text{ or } W_{SF} \text{ or } W_F \text{ or } W_{SFL} \text{ or } W_{FL}) \times (M_Q - M_B) \times M_C}{M_B \times 100}$$

$$(iii) \quad F = \frac{(W \text{ or } W_{SF} \text{ or } W_F \text{ or } W_{SFL} \text{ or } W_{FL}) \times (F_Q - F_B) \times F_C}{F_B \times 100}$$

$$(iv) \quad E = \frac{(W) \times (E_Q - E_B) \times E_C}{E_B \times 100}$$

$$(v) \quad PM = \frac{(W \text{ or } W_{SF} \text{ or } W_F \text{ or } W_{SFL} \text{ or } W_{FL}) \times (PM_Q - PM_B) \times PM_C}{PM_B \times 100}$$

$$(vi) \quad S = \frac{(W \text{ or } W_S \text{ or } W_{SF}) \times (S_Q - S_B) \times S_C}{S_B \times 100}$$

$$(vii) \quad C = \frac{(W \text{ or } W_C) \times (C_Q - C_B) \times C_C}{C_B \times 100}$$

**(II) For Railway Electrification Works:**

$$(i) \quad T = [0.4136 \times (C_Q - C_B) / C_B] \times 85$$

$$(ix) \quad R = [0.94 \times (R_T - R_O) / R_O + 0.06 \times (Z_T - Z_O) / Z_O] \times 85$$

$$(x) \quad N = [(P_T - P_O) / P_O] \times 85$$

$$(xi) \quad I = [(I_T - I_O) / I_O] \times 85$$

$$(xii) \quad G = [(M_Q - M_B) / M_B] \times 85$$

$$(xiii) \quad Er = [(L_Q - L_B) / L_B] \times 85$$

Where,

L Amount of price variation in Labour

M Amount of price variation in Materials F Amount of price variation in Fuel

E Amount of price variation in Explosives

PM Amount of price variation in Plant, Machinery and Spares

S Amount of price variation in Steel Supply Item

C Amount of price variation in Cement Supply Item

T Percentage variation payable on the gross value of bill of Concreting (Bill(s) of Quantities for concrete items)

|                  |                                                                                                                                                                                                                                                                                                                   |
|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| R                | Percentage variation payable on the gross value of bill of Ferrous Items (Bill(s) of Quantities for ferrous items)                                                                                                                                                                                                |
| N                | Percentage variation payable on the gross value of bill of Non-Ferrous Items(Bill(s) of Quantities for non-ferrous items)                                                                                                                                                                                         |
| I                | Percentage variation payable on the gross value of bill of Insulator (Bill(s) of Quantities for Insulator items)                                                                                                                                                                                                  |
| G                | Percentage variation payable on the gross value of bill of General Works (Bill(s) of Quantities for General items)                                                                                                                                                                                                |
| Er               | Percentage variation payable on the gross value of erection (Bill(s) of Quantities for Erection Item)                                                                                                                                                                                                             |
| L <sub>C</sub>   | % of Labour Component in the item(s)                                                                                                                                                                                                                                                                              |
| M <sub>C</sub>   | % of Material Component in the item(s)                                                                                                                                                                                                                                                                            |
| F <sub>C</sub>   | % of Fuel Component in the item(s)                                                                                                                                                                                                                                                                                |
| E <sub>C</sub>   | % of Explosive Component in the item(s)                                                                                                                                                                                                                                                                           |
| PM <sub>C</sub>  | % of Plant, Machinery and Spares Component in the item(s)                                                                                                                                                                                                                                                         |
| S <sub>C</sub>   | % of Steel Supply item Component in the item(s)                                                                                                                                                                                                                                                                   |
| C <sub>C</sub>   | % of Cement Supply item Component in the item(s)                                                                                                                                                                                                                                                                  |
| W                | Gross value of work done by Contractor as per on-account bill(s) excluding the Gross value of work under W <sub>S</sub> or/and W <sub>C</sub> or/and W <sub>SF</sub> or/and W <sub>F</sub> or/and W <sub>SFL</sub> or/and W <sub>FL</sub> and cost of materials supplied by Railway either free or at fixed rate, |
| W <sub>S</sub>   | Gross value of work done by Contractor for item(s) of supply of steel.                                                                                                                                                                                                                                            |
| W <sub>C</sub>   | Gross value of work done by Contractor for item(s) of supply of cement and /or supply of grout material.                                                                                                                                                                                                          |
| W <sub>SF</sub>  | Gross value of work done by Contractor for item(s) of Fabrication & Erection of Structures including supply of Steel.                                                                                                                                                                                             |
| W <sub>F</sub>   | Gross value of work done by Contractor for Fabrication & Erection of Structures excluding supply of Steel.                                                                                                                                                                                                        |
| W <sub>SFL</sub> | Gross value of work done by Contractor for item(s) of Fabrication, Assembly, Erection / Launching of Girders including supply of Steel.                                                                                                                                                                           |
| W <sub>FL</sub>  | Gross value of work done by Contractor for item(s) of Fabrication, Assembly, Erection / Launching of Girders excluding supply of Steel.                                                                                                                                                                           |
| L <sub>B</sub>   | Consumer Price Index for Industrial Workers - All India: Published in R.B.I. Bulletin for the base period                                                                                                                                                                                                         |
| L <sub>Q</sub>   | Consumer Price Index for Industrial Workers - All India: Published in R.B.I. Bulletin for the average price index of the 3 months of the quarter under consideration                                                                                                                                              |

|                 |                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| M <sub>B</sub>  | Wholesale Price Index: All commodities – as published in the R.B.I. Bulletin for the base period                                                                                                                                                                                                                                                                                                                               |
| M <sub>Q</sub>  | Wholesale Price Index: All commodities – as published in the R.B.I. Bulletin for the average price index of the 3 months of the quarter under consideration                                                                                                                                                                                                                                                                    |
| F <sub>B</sub>  | The average of official prices of Diesel available on the official website of ‘Petroleum Planning and Analysis cell’ under Ministry of Petroleum and Natural Gas for Delhi, Kolkata, Mumbai & Chennai, for the base period                                                                                                                                                                                                     |
| F <sub>Q</sub>  | The average of official prices of Diesel available on the official website of ‘Petroleum Planning and Analysis cell’ under Ministry of Petroleum and Natural Gas for Delhi, Kolkata, Mumbai & Chennai, for the 3 months of the quarter under consideration                                                                                                                                                                     |
| E <sub>B</sub>  | Index number of Monthly Whole Sale Price Index for the category ‘Explosive’ of (g). Manufacture of other chemical products under (J) MANUFACTURE OF CHEMICALS AND CHEMICAL PRODUCTS, published by Office of Economic Adviser, Govt. of India, Ministry of Commerce & Industry, Department of Industrial Policy & Promotion (DIPP), for the base period.                                                                        |
| E <sub>Q</sub>  | Index number of Monthly Whole Sale Price Index for the category ‘Explosive’ of (g). Manufacture of other chemical products under (J) MANUFACTURE OF CHEMICALS AND CHEMICAL PRODUCTS, published by Office of Economic Adviser, Govt. of India, Govt. of India, Ministry of Commerce & Industry, Department of Industrial Policy & Promotion (DIPP), for the average price index of 3 months of the quarter under consideration. |
| PM <sub>B</sub> | Index Number of Wholesale Prices in India by Groups and Sub Groups (Averages) for ‘Manufacture of machinery for mining, quarrying and construction’– published in RBI (Reserve Bank of India) Bulletin, for the base period.                                                                                                                                                                                                   |
| PM <sub>Q</sub> | Index Number of Wholesale Prices in India by Groups and Sub Groups (Averages) for ‘Manufacture of machinery for mining, quarrying and construction’– published in RBI (Reserve Bank of India) Bulletin, for the average price index of 3 months of the quarter under consideration.                                                                                                                                            |
| S <sub>B</sub>  | The average rate provided by the Joint Plant Committee for the relevant category of steel item as mentioned in Clause 46A.9; for the base period.                                                                                                                                                                                                                                                                              |
| S <sub>Q</sub>  | The average rate provided by the Joint Plant Committee for the relevant category of steel item as mentioned in Clause 46A.9; for the 3 months of the quarter under consideration.                                                                                                                                                                                                                                              |
| C <sub>B</sub>  | Index No. of Wholesale Price Index of sub-group Cement, Lime & Plaster as published in RBI Bulletin for the base period                                                                                                                                                                                                                                                                                                        |
| C <sub>Q</sub>  | No. of Wholesale Price Index of sub-group Cement, Lime & Plaster as published in RBI Bulletin for the average price index of the 3 months of the quarter under consideration                                                                                                                                                                                                                                                   |
| R <sub>T</sub>  | IEEMA price index for Steel Blooms (size 150mmx150mm) for the month which is two months prior to date of inspection of material.                                                                                                                                                                                                                                                                                               |
| R <sub>O</sub>  | IEEMA price index for Steel Blooms (size 150mmx150mm) for the month which is one month prior to date of opening of tender.                                                                                                                                                                                                                                                                                                     |
| P <sub>T</sub>  | IEEMA price index for Copper wire rods for the month which is two months prior to date of inspection of material.                                                                                                                                                                                                                                                                                                              |

|                |                                                                                                                                    |
|----------------|------------------------------------------------------------------------------------------------------------------------------------|
| P <sub>O</sub> | IEEMA price index for Copper wire rods for the month which is one month prior to date of opening of tender.                        |
| Z <sub>T</sub> | IEEMA price index for Zinc for the month which is two months prior to date of inspection of material                               |
| Z <sub>O</sub> | IEEMA price index for Zinc for the month which is one month prior to date of opening of tender                                     |
| I <sub>T</sub> | RBI wholesale price index for the sub-group “Insulators” for the month which is two months prior to date of inspection of material |
| I <sub>O</sub> | RBI wholesale price index for the sub-group “Insulators” for the month which is one month prior to date of opening of tender       |

**35.8** The demands for escalation of cost shall be allowed on the basis of provisional indices as mentioned above in Clause 46A.7 of Standard General Conditions of Contract. Any adjustment needed to be done based on the finally published indices shall be made as and when they become available.

**35.9: (1)** Relevant categories of steel for the purpose of operating Price Variation formula as mentioned in this Clause shall be as under:

| SL | Classification                                                 | Rates to be used for calculating S <sub>Q</sub> or S <sub>B</sub>                                                                    |
|----|----------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| 1. | Reinforcement bars and other rounds                            | Average of per tonne rates of 10mm dia TMT & 25mm dia TMT; confirming IS1786; Fe 500                                                 |
| 2. | All types and sizes of angles, channels and joists             | Average of per tonne rates of ‘Angle 75x75x6mm, Mild Steel Plate 10mm thickness and Channel 150x75mm; confirming IS2062, E250 Gr “A” |
| 3. | All types and sizes of plates                                  | Average of per tonne rates of ‘MS Plates 10mm thickness and 25mm thickness; confirming IS2062, E250 Gr “A”                           |
| 4. | Any other section of steel not covered in the above categories | Average of price for the 3 categories covered under SL 1, 2 & 3 in this table.                                                       |

**(2).** Relevant city for referring “JPC (Joint Plant Committee)” rates of steel items (S<sub>Q</sub> /S<sub>B</sub>) in different Zonal Railways shall be as under :

| SL | City    | Railway                                                                                 |
|----|---------|-----------------------------------------------------------------------------------------|
| 1. | Delhi   | Northern, North Central, North Eastern, North Western                                   |
| 2. | Kolkata | Eastern, East Central, East Coast, Northeast Frontier, South Eastern, Southeast Central |
| 3. | Mumbai  | Central, Western, West Central                                                          |
| 4. | Chennai | Southern, South Central & South Western                                                 |

### **35.10 Price Variation during Extended Period of Contract**

The price adjustment as worked out above, i.e. either increase or decrease shall be applicable upto the stipulated date of completion of work including the extended period of completion where such extension has been granted under Clause 17-A of the Standard General Conditions of Contract. However, where extension of time has been granted due to Contractor’s failure under Clause 17-B of the Standard General Conditions of Contract, price adjustment shall be done as follows:

For Chief Administrative Officer (Const)  
West Central Railway, Jabalpur.

- a. In case the indices increase above the indices applicable to the last month of original completion period or the extended period under Clause 17-A of Standard General Conditions of Contract, the price adjustment for the period of extension granted under Clause 17-B of Standard General Conditions of Contract shall be limited to the amount payable as per the Indices applicable to the last month of the original completion period or the extended period under Clause 17-A of the Standard General Conditions of Contract; as the case may be.
- b. In case the indices fall below the indices applicable to the last month of original/ extended period of completion under Clause 17-A of Standard General Conditions of Contract, as the case may be; then the lower indices shall be adopted for the price adjustment for the period of extension under Clause 17-B of the Standard General Conditions of Contract. (Para-46A of SGCC-2022)

**36. Maintenance of Works:** The Contractor shall at all times during the progress and continuance of the works and also for the period of maintenance specified in the Tender Form after the date of issue of the certificate of completion by the Engineer or any other earlier date subsequent to the completion of the works that may be fixed by the Engineer, be responsible for and effectively maintain and uphold in good substantial, sound and perfect condition all and every part of the works and shall make good from time to time and at all times as often as the Engineer shall require, any damage or defect that may during the above period arise in or be discovered or be in any way connected with the works, provided that such damage or defect is not directly caused by errors in the contract documents, act of providence or insurrection or civil riot, and the Contractor shall be liable for and shall pay and make good to the Railway or other persons legally entitled thereto whenever required by the Engineer so to do, all losses, damages, costs and expenses they or any of them may incur or be put or be liable to by reasons or in consequence of the operations of the Contractor or of his failure in any respect. (Para-47 of SGCC-2022)

**37.(1) Final Payment:** On the Engineer's certificate of completion in respect of the works, adjustment shall be made and the balance of account based on the Engineer or the Engineer's representative's certified measurements or Engineer's certified "contractor's authorized engineer's measurements" of the total quantity of work executed by the Contractor upto the date of completion and on the rates accepted in Bill(s) of Quantities and for extra works on rates determined under Clause 39 of Standard General Conditions of Contract shall be paid to the Contractor subject always to any deduction which may be made under these presents and further subject to the Contractor having signed delivered to the Engineer enclosing either a full account in detail of all claims he may have on the Railway in respect of the works or having delivered No Claim Certificate and the Engineer having after the receipt of such account given a certificate in writing that such claims are not covered under excepted matter i.e. Clauses 7(i), 8, 18, 22(5), 39.1, 39.2, 40A, 43(2), 45(i)(a), 55, 55-A(5), 57, 57A, 61(1), 61(2) and 62(1), 63(iv) and 63.2.211 of the Standard General Conditions of Contract or in any Clause (stated as excepted matter) of the Special Conditions of the Contract, that the whole of the works to be done under the provisions of the Contracts have been completed, that they have been inspected by him since their completion and found to be in good and substantial order, that all properties, works and things, removed, disturbed or injured in consequence of the works have been properly replaced and made good and all expenses and demands incurred by or made upon the Railway for or in the respect of damage or loss by from or in consequence of the works, have been satisfied agreeably and in conformity with the contract. (Para-51.(1) of SGCC-2022)

## **LABOUR**

**38. Wages to Labour:** The Contractor shall be responsible to ensure compliance with the provision of the Minimum Wages Act, 1948 (hereinafter referred to as the “said Act”) and the Rules made thereunder in respect of any employees directly or through petty Contractors or sub-contractors employed by him for the purpose of carrying out this contract.

If, in compliance with the terms of the contract, the Contractor supplied any labour to be used wholly or partly under the direct orders and control of the Railways whether in connection with any work being executed by the Contractor or otherwise for the purpose of the Railway such labour shall, for the purpose of this Clause, still be deemed to be persons employed by the Contractor.

If any moneys shall, as a result of any claim or application made under the said Act be directed to be paid by the Railway, such money shall be deemed to be moneys payable to the Railway by the Contractor and on failure by the Contractor to repay the Railway any moneys paid by it as aforesaid within seven days after the same shall have been demanded, the Railways shall be entitled to recover the same from Contractor’s bills/Security Deposit or any other dues of Contractor with the Government of India. (Para-54 of SGCC-2022)

**38-A. Apprentices Act:** The Contractor shall be responsible to ensure compliance with the provisions of the Apprentices Act, 1961 and the Rules and Orders issued thereunder from time to time in respect of apprentices directly or through petty Contractors or sub-contractors employed by him for the purpose of carrying out the Contract.

If the Contractor directly or through petty Contractors or sub-contractors fails to do so, his failure will be a breach of the contract and the Railway may, in its discretion, rescind the contract. The Contractor shall also be liable for any pecuniary liability arising on account of any violation of the provisions of the Act. (Para-54-A of SGCC-2022)

**39. Provisions of Payments of Wages Act:** The Contractor shall comply with the provisions of the Payment of Wages Act, 1936 and the rules made thereunder in respect of all employees employed by him either directly or through petty Contractors or sub-contractors in the works. If in compliance with the terms of the contract, the Contractor directly or through petty Contractors or sub-contractors shall supply any labour to be used wholly or partly under the direct orders and control of the Engineer whether in connection with the works to be executed hereunder or otherwise for the purpose of the Engineer, such labour shall nevertheless be deemed to comprise persons employed by the Contractor and any moneys which may be ordered to be paid by the Engineer shall be deemed to be moneys payable by the Engineer on behalf of the Contractor and the Engineer may on failure of the Contractor to repay such money to the Railways deduct the same from any moneys due to the Contractor in terms of the contract. The Railway shall be entitled to recover the same from Contractor’s bills/Security Deposit or any other dues of Contractor with the Government of India all moneys paid or payable by the Railway by way of compensation of aforesaid or for costs of expenses in connection with any claim thereto and the decision of the Engineer upon any question arising out of the effect or force of this Clause shall be final and binding upon the Contractor. (Para-55 of SGCC-2022)

**39-A. Provisions of Contract Labour (Regulation and Abolition) Act, 1970:**



**39-A.(1)** The Contractor shall comply with the provision of the contract labour (Regulation and Abolition) Act, 1970 and the Contract labour (Regulation and Abolition) Central Rules 1971 as modified from time to time, wherever applicable and shall also indemnify the Railway from and against any claims under the aforesaid Act and the Rules. (Para-55-A.(1) of SGCC-2022)

**39-A.(2)** The Contractor shall obtain a valid license under the aforesaid Act as modified from time to time before the commencement of the work and continue to have a valid license until the completion of the work. Any failure to fulfill the requirement shall attract the penal provision of the Act. (Para-55-A.(2) of SGCC-2022)

**39-A.(3)** The Contractor shall pay to the labour employed by him directly or through sub-contractors the wages as per provision of the aforesaid Act and the Rules wherever applicable. The Contractor shall notwithstanding the provisions of the contract to the contrary, cause to be paid the wages to labour, indirectly engaged on the works including any engaged by sub-contractors in connection with the said work, as if the labour had been immediately employed by him. (Para-55-A.(3) of SGCC-2022)

**39-A.(4)** In respect of all labour directly or indirectly employed in the work for performance of the Contractor's part of the contract, the Contractor shall comply with or cause to be complied with the provisions of the aforesaid Act and Rules wherever applicable. (Para-55-A.(4) of SGCC-2022)

**39-A.(5)** In every case in which, by virtue of the provisions of the aforesaid Act or the rules, the Railway is obliged to pay any amount of wages to a workman employed by the Contractor or his sub-contractor in execution of the work or to incur any expenditure on account of the contingent, liability of the Railway due to the Contractor's failure to fulfill his statutory obligations under the aforesaid Act or the rules, the Railway will recover from the Contractor, the amount of wages so paid or the amount of expenditure so incurred and without prejudice to the rights of the Railway under the Section 20, Sub-Section (2) and Section 2, Sub-Section (4) of the aforesaid Act, the Railway shall be at liberty to recover such amount or part thereof from Contractor's bills/Security Deposit or any other dues of Contractor with the Government of India. The Railway shall not be bound to contest any claim made against it under Sub-Section (1) of Section 20 and Sub-Section (4) of Section 21 of the aforesaid Act except on the written request of the Contractor and upon his giving to the Railway full security for all costs for which the Railway might become liable in contesting such claim. The decision of the Chief Engineer regarding the amount actually recoverable from the Contractor as stated above shall be final and binding on the Contractor. (Para-55- A.(5) of SGCC-2022)

**39-B. Provisions of Employees Provident Fund and Miscellaneous Provisions Act, 1952:** The Contractor shall comply with the provisions of Para 30 & 36-B of the Employees Provident Fund Scheme, 1952; Para 3 & 4 of Employees' Pension Scheme, 1995; and Para 7 & 8 of Employees Deposit Linked Insurance Scheme, 1976; as modified from time to time through enactment of "Employees Provident Fund & Miscellaneous Provisions Act, 1952", wherever applicable and shall also indemnify the Railway from and against any claims under the aforesaid Act and the Rules. (Para-55-B of SGCC-2022)

**39-C (i) Contractor is to abide by the provisions of various labour laws** in terms of clause 54, 55, 55A & 55B of the Standard General Condition of Contract. In order to ensure the same, an application has been developed and hosted on website

www.shramikkalyan.indianrailways.gov.in'. Contractor shall register his firm/company etc. and upload requisite details of labour and their payment in this portal. These details shall be available in public domain. The Registration / updation of Portal shall be done as under:

- (a) Contractor shall apply for onetime registration of his company/firm etc. in the Shramik kalyan portal with requisite details subsequent to issue of Letter of Acceptance. Engineer shall approve the contractor's registration in the portal within 7 days of receipt of such request.
  - (b) Contractor once approved by any Engineer, can create password with login ID (PAN No.) for subsequent use of portal for all Letter of Acceptance (LoAs) issued in his favour.
  - (c) The contractor once registered on the portal, shall provide details of his Letter of Acceptances (LoAs) / Contract Agreements on shramik kalyan portal within 15 days of issue of any LoA for approval of concerned engineer. Engineer shall update (if required) and approve the details of LoA filled by contractor within 7 days of receipt of such request.
  - (d) After approval of LoA by Engineer, contractor shall fill the salient details of contract labours engaged in the contract and ensure updating of each wage payment to them on shramik Kalyan portal on monthly basis.
  - (e) It shall be mandatory upon the contractor to ensure correct and prompt uploading of all salient details of engaged contractual labour & payments made thereof after each wage period.
- (ii) While processing payment of any 'On Account bill' or 'Final bill' or release of 'Advances' or 'Performance Guarantee / Security deposit', contractor shall submit a certificate to the Engineer or Engineer's representatives that "I have uploaded the correct details of contract labours engaged in connection with this contract and payments made to them during the wage period in Railway's Shramik Kalyan portal at 'www.shramikkalyan.indianrailways.gov.in' till Month, Year." (Para-55-C of SGCC-2022)

**39-D. Provisions of "The Building and Other Construction Workers (Regulation of Employment and Conditions of Service) Act, 1996" and "The Building and Other Construction Workers' Welfare Cess Act, 1996":**

The tenderers, for carrying out any construction work, shall get themselves registered with the Registering Officer under Section-7 of the Building and Other Construction Workers' Welfare Cess Act, 1996 and rules made thereto by the concerned State Govt., and submit certificate of Registration issued from the Registering Officer of the concerned State Govt. (Labour Dept.). The Cess shall be deducted from contractor's bills as per provisions of the Act. (Para-55-D of SGCC-2022)

**40. Reporting of Accidents:** The Contractor shall be responsible for the safety of all employees directly or through petty Contractors or sub-contractor employed by him on the works and shall report serious accidents to any of them however and wherever occurring on the works to the Engineer or the Engineers Representative and shall make every arrangement to render all possible assistance. (Para-56 of SGCC-2022)

**41. Provision of Workmen's Compensation Act:** In every case in which by virtue of the provisions of Section 12 Sub-Section (1) of the Workmen's Compensation Act 1923, Railway is obliged to pay compensation to a workman directly or through petty Contractor or sub-contractor employed by the Contractor in executing the work, Railway will recover from the Contractor the amount of the compensation so paid, and, without prejudice to the rights of Railway under Section 12 Sub-section (2) of the said Act, Railway shall be at liberty to recover such amount or any part thereof from Contractor's bills/Security Deposit or any other dues of Contractor with the Government of India. Railway shall not be bound to contest any claim made against it under Section 12 Sub-Section (1) of the said Act except on the written request of the Contractor and upon his giving to Railway full security for all costs for which Railway might become liable in consequence of contesting such claim. (Para-57 of SGCC-2022)

**41-A. Provision of Mines Act:** The Contractor shall observe and perform all the provisions of the Mines Act, 1952 or any statutory modifications or re-enactment thereof for the time being in force and any rules and regulations made thereunder in respect of all the persons directly or through the petty Contractors or sub-contractors employed by him under this contract and shall indemnify the Railway from and against any claims under the Mines Act, or the rules and regulations framed thereunder, by or on behalf of any persons employed by him or otherwise. (Para-57-A of SGCC-2022)

**42. Railway not to Provide Quarters for Contractors:** No quarters shall normally be provided by the Railway for the accommodation of the Contractor or any of his staff employed on the work. In exceptional cases where accommodation is provided to the Contractor at the Railway's discretion, recoveries shall be made at such rates as may be fixed by the Railway for the full rent of the buildings and equipment's therein as well as charges for electric current, water supply and conservancy. (Para-58 of SGCC-2022)

#### **DETERMINATION OF CONTRACT**

**43.(1) Right of Railway to Determine the Contract:** The Railway shall be entitled to determine and terminate the contract at any time should, in the Railway's opinion, the cessation of work becomes necessary owing to paucity of funds or from any other cause whatever, in which case the value of approved materials at site and of work done to date by the Contractor will be paid for in full at the rate specified in the contract. Notice in writing from the Railway of such determination and the reasons therefor shall be conclusive evidence thereof. (Para-61.(1) of SGCC-2022)

**43.(2) Payment on Determination of Contract:** Should the contract be determined under sub clause (1) of this clause and the Contractor claims payment for expenditure incurred by him in the expectation of completing the whole of the work, the Railways shall admit and consider such claims as are deemed reasonable and are supported by vouchers to the satisfaction of the Engineer. The Railway's decision on the necessity and propriety of such expenditure shall be final and conclusive. (Para-61.(2) of SGCC-2022)

**43.(3)** The Contractor shall have no claim to any payment of compensation or otherwise, howsoever on account of any profit or advantage which he might have derived from the execution of the work in full but which he did not derive in consequence of determination of contract. (Para-61.(3) of SGCC-2022)

#### **44.(1) Determination of Contract owing to Default of Contractor:**

If the Contractor should:

- (i) Becomes bankrupt or insolvent, or
- (ii) Make an arrangement for assignment in favour of his creditors, or agree to carry out the contract under a Committee of Inspection of his creditors, or
- (iii) Being a Company or Corporation, go into liquidation (other than a voluntary liquidation for the purposes of amalgamation or reconstruction), or
- (iv) Have an execution levied on his goods or property on the works, or
- (v) Assign the contract or any part thereof otherwise than as provided in Clause 7 of GCC, or
- (vi) Abandon the contract, or
- (vii) Persistently disregard the instructions of the Engineer, or contravene any provision of the contract, or
- (viii) Fail to adhere to the agreed programme of work by a margin of 10% of the stipulated period, or
- (ix) Fail to Execute the contract documents in terms of Clause 8 of the Instructions to Tenderers.
- (x) Fails to submit the documents pertaining to identity of JV and PAN in terms of Clause 17.11 of Tender Form (Second Sheet) of Annexure I available in the Instructions to Tenderers.
- (xi) Fail to remove materials from the site or to pull down and replace work after receiving from the Engineer notice to the effect that the said materials or works have been condemned or rejected under Clause 25 and 27 of GCC, or
- (xii) Fail to take steps to employ competent or additional staff and labour as required under Clause 26 of GCC, or
- (xiii) Fail to afford the Engineer or Engineer's representative proper facilities for inspecting the works or any part thereof as required under Clause 28 of GCC, or
- (xiv) Promise, offer or give any bribe, commission, gift or advantage either himself or through his partner, agent or servant to any officer or employee of the Railway or to any person on his or on their behalf in relation to the execution of this or any other contract with this Railway.
- (xv) Fail to adhere to the provisions of Clause 16 of Tender Form (Second Sheet) of Annexure I of the Instructions to Tenderers, or provision of Clause 59(9) of General Conditions of Contract.
- (xvi) Submits copy of fake documents / certificates in support of credentials, submitted by the tenderer

Then and in any of the **said Clause**, the Engineer on behalf of the Railway may serve the Contractor with a notice (Proforma at Annexure-IX of SGCC) in writing to that effect and if the Contractor does not within seven days after the delivery to him of such notice proceed to make good his default in so far as the same is capable of being made good and carry on the work or comply with such directions as aforesaid of the entire satisfaction of the Engineer, the Railway shall be entitled after giving 48 hours' notice (Proforma at Annexure-X or XII of SGCC, as the case may be) in writing under the hand of the Engineer to rescind the contract as a whole or in part or parts (as may be specified in such notice) and after expiry of 48 hours' notice, a final termination notice (Proforma at Annexure-XI or XIII of Standard General Conditions of Contract, as the case may be) should be issued.

**Note:** Engineer at his discretion may resort to the part termination of contract with notices (Proforma at Annexure- IX, XII and XIII of Standard General Conditions of Contract), only in cases where progress of work is more than or equal to 80% of the original scope of work. (Para-62.(1) of SGCC-2022)

**44.(2) Right of Railway after Rescission of Contract owing to Default of Contractor:** In the event of any or several of the courses, referred to in Sub-Clause (1) of this Clause, being adopted:

(a) The Contractor shall have no claim to compensation for any loss sustained by him by reason of his having purchased or procured any materials or entered into any commitments or made any advances on account of or with a view to the execution of the works or the performance of the contract and Contractor shall not be entitled to recover or be paid any sum for any work thereto for actually performed under the contract unless and until the Engineer shall have certified the performance of such work and the value payable in respect thereof and the Contractor shall only be entitled to be paid the value so certified.

(b) In the contract which has been rescinded as a whole, the Security Deposit already with railways under the contract shall be encashed/ forfeited and the Performance Guarantee already submitted for the contract shall be encashed. The balance work shall be got done independently without risk & cost of the failed Contractor. The failed Contractor shall be debarred from participating in the tender for executing the balance work. If the failed Contractor is a JV or a Partnership firm, then every member/partner of such a firm shall be debarred from participating in the tender for the balance work in his/her individual capacity or as a partner of any other JV /partnership firm.

Further the authorized representative of failed Contractor cannot be accepted as authorized representative in new contract.

(c) In the contract rescinded in part or parts,

(i) The full Performance Guarantee for the contract shall be recovered. No additional Performance Guarantee shall be required for balance of work being executed through the part terminated contract. The contract value of part terminated contract stands reduced to the balance value of work under the contract.

(ii) The Security Deposit of part terminated contract shall be dealt as per clause 16(2) of SGCC.

(iii) The defaulting Contractor shall not be issued any completion certificate for the contract.

(iv) The balance work shall be got done independently without risk & cost of the failed Contractor. The failed Contractor shall be debarred from participating in the tender for executing the balance work. If the failed Contractor is a JV or a Partnership firm, then every member/partner of such a firm shall be debarred from participating in the tender for the balance work in his/her individual capacity or as a partner of any other JV /partnership firm.

(v) Further the authorized representative of failed Contractor will not be accepted as authorized representative in new contract.

(d) The Engineer or the Engineer's Representative shall be entitled to take possession of any materials, tools, implements, machinery and buildings on the works or on the property on

which these are being or ought to have been executed, and to retain and employ the same in the further execution of the works or any part thereof until the completion of the works without the Contractor being entitled to any compensation for the use and employment thereof or for wear and tear or destruction thereof.

(e) The Engineer shall as soon as may be practicable after removal of the Contractor fix and determine ex-parte or by or after reference to the parties or after such investigation or enquiries as he may consider fit to make or institute and shall certify what amount (if any) had at the time of rescission of the contract been reasonably earned by or would reasonably accrue to the Contractor in respect of the work then actually done by him under the contract and what was the value of any unused, or partially used materials, any constructional plant and any temporary works upon the site. The legitimate amount due to the Contractor after making necessary deductions and certified by the Engineer should be released expeditiously. (Para-62.(2) of SGCC-2022)

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**WEST CENTRAL RAILWAY CONSTRUCTION ORGANISATION  
SPECIAL CONDITIONS OF CONTRACT**

**1 MAINTENANCE PERIOD: -**

- 1.1 The successful tenderer/contractor will have to maintain all works excepting supply of ballast **for a period of 06 (six) Calendar months or one monsoon period whichever is later**, unless otherwise specified in the Tender Conditions from the certified date of completion without any extra cost to the Railway.
- 1.2 In case of P.Way works, the contractor shall maintain the same **till six months from completion of the work or till allowing regular train service on track, whichever is earlier**, unless otherwise specified in the Tender Conditions from the certified date of completion without any extra cost to the Railway.

**2. PRICE VARIATION CLAUSE (PVC)**

**Price variation Clause (PVC) shall be applicable only in tender having advertised value above Rs. 2 Crores.**

**3. UTILIZATION OF RAILWAY STONES:**

- 3.1 Notwithstanding the inclusion of some of the rates in the schedules with contractor's stones/aggregate, the Contractor(s) shall utilize the Railway's usable surplus stones, if available for use as such or by converting them into aggregate of sizes required. The cost of boulder shall be recovered for one cubic metre net of such stones as per the prevailing rate of the area or @ Rs. 375/- per cum or as may be decided by the Railway (after deducting 30% for voids). The Contractor(s) shall collect the boulders from locations within the limit of contract section wherever they are offered to him within the section limits. The rate per cubic metre net includes royalty, collection as required. The rate also includes loading, transportation of cut spoils which has to be done by the Contractors at his/their own risk and cost and the rate is towards the cost of cut stone available on "as is and where is basis". The boulder issued to the Contractor(s) for the above purpose will be used only for the works of the agreement and shall neither be disposed off nor be used for any other agreement.

**4. REGISTRATION WITH LABOUR DEPARTMENT OF STATE GOVERNMENT:**

The tenderers for carrying out any construction work in Madhya Pradesh/ Rajasthan / Chhattisgarh/Uttar Pradesh must get themselves registered from the Registering Officer under Section-7 of the Building and Other Construction Workers' (Regulation of Employment and Conditions of Service) Cess Act 1996 and Rules made thereto. The tenderer shall be required to submit certificate of Registration issued from the Registering Officer of the Govt. of Madhya Pradesh/ Rajasthan / Chhattisgarh/Uttar Pradesh (Labour Department) as applicable. The cess shall be deducted from the contractor's bills as per provisions of the Act.

**5. USE OF PATENTED ITEMS**

- 5.1 In case, the agency supplying the patented item is not the contractor to whom the work is assigned and these items are being procured by the Contractor, the concerned contractor will obtain prior approval from the Engineer-in-charge of the work (Min. JA Grade Officer), who will ensure that all design and safety aspects are taken care

of. For any specify requirement concerning execution, warranty etc., an agreement/MOU is to be entered between the main contractor and the party supplying the patented items, clearly bringing out the responsibility of party supplying such patented items. A copy of such MOU will be furnished to the Engineer-in-charge by the main contractor to whom work has been assigned by the Railway and after the approval of Engineer- in-charge, such item can be used in work.

- 5.2. The agency supplying the patented item shall provide complete details/specifications/ drawings of the items including the manner in which it is to be used.
- 5.3. During the installation of such patented items, authorised representative of the firm supplying such patented/propriety items shall be present and after the execution of work, a certificate to be issued by the firm supplying the patented item indicating its proper installation. Such certificate will have to be kept in record by the concerned Engineer-in-charge executing the contract, before releasing payment for the work done. (Ref: Rly Board's Lr. No. 2018/CE-I/Innov/1 dtd. 18.01.2018)

#### **6. PRE-CAUTION TO BE TAKEN WHILE PLYING OF VEHICLES ADJACENT TO RUNNING LINES TO PREVENT ACCIDENT TO TRAINS.**

- 6.1 Vulnerable locations where construction work adjacent to running line can cause accident should be protected by suitable strong barrier which should be included as a paid item in contract schedule. These locations should be decided by Executive In-charge of the work at the beginning of construction and intimated to contractor in writing.
- 6.2 The barrier should be painted by retro-reflective paint at suitable interval to give warning at night.
- 6.3 No work adjacent to running track should be carried out at night without express written authority from the Engineer In-charge of the work. In fact, no contractor should do any kind of night working unless the Executive Engineer In-charge of the work gives the specified spots according to priority of work where night working has to be done. These spots should be well lit at night. In addition, the work should always; be done under supervision of Railway supervisors in addition to Contractor's supervisors. Suitable Railway personnel should be posted at site with safety equipment's like banner flags, hand signal flags, hand signal lamps and detonators to arrange protection of trains. The Railway supervisors in charge of such work should also give suitable message to adjacent stations as well as through control for issuing caution orders to the trains approaching the work site. For this purpose, he should be equipped with field telephone/walkie talkie set.
- 6.4 The Contractor shall not allow any road vehicle belonging to him or his suppliers etc., to ply in railway land next to the running line. If for execution of certain works viz, earthwork for parallel Railway Line and supply of ballast for new or existing rail line gauge conversion etc., road vehicles are necessary to be used in Railway land next to the railway line, the Contractor shall apply to the Engineer-in –Charge for permission giving the type & No. of individual vehicles, names and License particulars of the drivers, location, duration & timings for such work/movement. The Engineer in Charge or his authorised representative will personally counsel, examine & certify, the road vehicle drivers, Contractor's flagmen & supervisor and will give written permission giving names of road vehicle drivers, Contractor's flagmen and supervisor to be deployed on the work, location, period and timing of the work. This permission will be subject to the following obligatory conditions.
  - (i) The road vehicles will ply only between sunrise & sunset.



- (ii) Nominated vehicles & drivers will be utilised for work in the presence of at least one flagman & one supervisor certified for such work.
- (iii) The vehicles shall ply 6m clear of track. Any movement/work at less than 6m and up to minimum 3.5m clear of track centre, shall be done only in the presence of railway employee authorised by the Engineer-in-Charge. No part of the road vehicle will be allowed at less than 3.5m from track centre. Cost of such railway employee shall be borne by the Railway.
- (iv) The Contractor shall remain fully responsible for ensuring safety & in case of any accident, shall bear cost of all damages to this equipment and new and also damages to railway & its passengers.
- (v) The Contractor shall also be bound by the provisions of this agreement to ply the road Vehicle only with adequate margin of safety, well clear of the fixed structure profile of infringement, as stipulated in the rules made under the Indian Railway's Act and to seek and be guided by the Signals and other directions of any look-out men or other personnel retained for the purpose of ensuring safety, and to ensure extra care and vigilance while turning, reversing or moving the road Vehicles in any other manner at an inclination to the running Railway Track or the siding as the case may be. The Contractor shall employ necessary look-out; men also at his own cost, irrespective of any other arrangement that Railway may make in this regard.
- (vi) Any breach of these conditions by the Contractor and/or his agents affecting the safety of movement of Trains, Engines, or other rolling stock of the Railway shall constitute a breach of Contract by the Contractor entailing liability with termination of contract for default on the part of the Contractor.

**7.0 Mode of Payment Through Letter of Credit (LC) as option in works contract having advertised cost of Rs. 10 lakhs and above: -**

- 7.1** (i) For all the tenders having advertised cost of Rs. 10 Lakh or above, the contractor shall have the option to take payment from Railways through a letter of credit (LC) arrangement.
- (ii) This option of taking payment through LC arrangement has to be exercised in IREPS (Indian Railway Electronic Procurement System-the e-application on which tenders are called by Railways) by the tenderer at the time of bidding itself and the tenderer shall affirm having read over and agreed to the terms and conditions of the LC option.
- (iii) The option so exercised, shall be an integral part of the bidder's offer.
- (iv) The above option of taking payment through LC arrangement, once exercised by tenderer at the time of bidding, shall be final and no change shall be permitted, thereafter, during execution of contract.
- (v) In case tenderer opts for payment through LC, following shall be the procedure to deal release of payment through LC:
- (a) The LC shall be a sight LC.
  - (b) The contractor shall select his Advising/Negotiating bank for LC. The incidental cost towards issue of LC and its operation thereof shall be borne by the contractor.

- (c) SBI, New Delhi, Main Branch will be the nodal branch for issue of LC's based on online requests received from Railway Accounts Units for tenders opened in financial year 2018-19. SBI branches where the respective Railway Accounts Office has its Account (Local SBI Branch) will be the issuance/reimbursing branch for LC issued under this arrangement. The Bank shall remain same for this tender till completion of contract. The incidental cost @ 0.15% per annum of LC value, towards issue of LC and operation thereof shall be borne by the contractor and shall be recovered from his bills.
- (d) The LC shall be opened initially for duration of 180 to 365 days in consultation with contractor. The LC shall be extended time to time as per the progress of the contract, on the request of the contractor. The value of LC to be opened initially as well as extended thereafter shall be finalized by the engineer in consultation with the contractor on the basis of expected progress of work.
- (e) The LC terms and conditions shall inter-alia indemnify and save harmless the Railway from and against all losses, claims and demands of every nature and description brought or recovered against the Railways by reason of any act or omission of the contractor, his agents or employees, in relation to the letter of Credit (LC). All sums payable/borne by Railways on this account shall be considered as reasonable compensation and paid by contractor.
- (f) The LC terms and conditions shall inter-alia provide that Railways will issue a Document of authorization (**format enclosed as Annexure-P of Chapter-6**) after passing the bill for completed work to enable contractor to claim the authorized amount from their bank.
- (g) The acceptable, agreed upon document for payments to be released under the LC shall be the Document of Authorisation.
- (h) The Document of Authorisation shall be issued by Railway Accounts Office against each bill passed by Railways.
- (i) On issuance of Document of Authorisation, a copy of Document of Authorisation shall be posted on IREPS for download by the contractor. A digitally signed copy of Document of Authorisation shall also be sent by Railway Accounts Office to Railway's bank (Local SBI Branch).
- (j) The contractor shall take print out of the Document of Authorisation available on IREPS and present his claim to his bank (advising Bank) for necessary payments as per LC terms and conditions. The claim shall comprise of copy of Document of Authorisation, Bill of Exchange and Bill.
- (k) The payment against LC shall be subject to verification from Railway's Bank (Local SBI Branch).
- (l) The Contractor's bank (advising bank) shall submit the documents to the Railway's Bank (Local SBI Branch)
- (m) The railway's bank (issuing bank) shall, after verifying the claim so received w.r.t. the digitally signed Document of Authorisation received from Railway Accounts Office, release the payment to contractor's bank (advising bank) for crediting the same to contractor's account.

- (n) Any number of bills can be dealt within one LC, provided the sum total of payments to contractor is within the amount for which LC has been opened.
  - (o) The LC shall be closed after the release of final payment including PVC amount, if any, to the contractor.
  - (p) The release of performance guarantee or security deposit shall be dealt directly by railway with the contractor i.e. not through LC.
- 7.2. For opening of LC, executive department shall make a request letter to concerned Accounts Department on a format, placed as **Annexure-1 of Chapter-4**.

**8. Public Procurement (Preference to Make in India), Order-2017:-**

Whereas it is the policy of the Government of India to encourage 'Make in India' and promote manufacturing and production of goods and services in India with a view to enhancing income and employment, and Whereas procurement by the Government is substantial in amount and can contribute towards this policy objective, and

Whereas local content can be increased through partnerships, cooperation with local companies, establishing production units in India or Joint Ventures (JV) with Indian suppliers, increasing the participation of local employees in services and training them.

Now therefore the following Order is issued:

8.1. **This Order is issued pursuant to Rule 153(iii) of the General Financial Rules 2017.**

8.2. **Definitions:** For the purposes of this Order:

*'Local Content'* means the amount of value added in India which shall, unless otherwise prescribed by the Nodal Ministry, be the total value of the item procured (excluding net domestic indirect taxes) minus the value of imported content in the item (including all customs duties) as a proportion of the total value, in percent.

*'Local Supplier'* means a supplier or service provider whose product or service offered for procurement meets the minimum local content as prescribed under this Order or by the competent Ministries/Departments in pursuance of this Order.

*'L 1'* means the lowest tender or lowest bid of the lowest quotation received in a tender, bidding process or other procurement solicitation as adjudged in the evaluation process as per the tender or other procurement solicitation.

*'margin of purchase preference'* means the maximum extent to which the price quoted by a local supplier may be above the L 1 for the purpose of purchase preference.

*'Nodal Ministry'* means the Ministry or Department identified pursuant to this order in respect of a particular item of goods or services.

*'Procuring entity'* means a Ministry or department or attached or subordinate office of or autonomous body controlled by, the Government of India and includes Government companies as defined in the Companies Act.

- 8.3. **Requirement of Purchase Preference:** Subject to the provisions of this Order and to any specific instructions issued by the Nodal Ministry or in pursuance of this Order, purchase preference shall be given to local suppliers in all procurements undertaken by procuring entities in the manner specified hereunder:
- a. In procurement of goods in respect of which the Nodal Ministry has communicated that there is sufficient local capacity and local competition, and where the estimated value of procurement is Rs. 50 Lakhs or less, only local suppliers shall be eligible. If the estimated value of procurement of such goods is more than Rs. 50 Lakhs, the provisions of sub-paragraph b or c, as the case may be, shall apply.
  - b. In the procurement of goods which are not covered by paragraph 8-(3a) and which are divisible in nature, the following procedure shall be followed.
    - i. Among all qualified bids, the lowest bid will be termed as L1. If L1 is from a local supplier, the contract for full quantity will be awarded to L1.
    - ii. If L1 bid is not from a local supplier, 50% of the order quantity shall be awarded to L1. Thereafter, the lowest bidder among the local suppliers, will be invited to match the L1 price for the remaining 50% quantity subject to the local supplier's quoted price falling within the margin of purchase preference, and contract for that quantity shall be awarded to such local supplier subject to matching the L1 price. In case such lowest eligible local supplier fails to match the L1 price or accepts less than the offered quantity, the next higher local supplier within the margin of purchase preference shall be invited to match the L1 price for remaining quantity and so on, and contract shall be awarded accordingly. In case some quantity is still left uncovered on local suppliers, then such balance quantity may also be ordered on the L1 bidder.
  - c. In procurements of goods not covered by sub-paragraph 8-(3a) and which are not divisible, and in procurement of services where the bid is evaluated on price alone, the following procedure shall be followed.
    - i. Among all qualified bids, the lowest bid will be termed as L1. If L1 is from a local supplier, the contract will be awarded to L1.
    - ii. If L1 is not from a local supplier, the lowest bidder among the local suppliers, will be invited to match the L1 price subject to local supplier's quoted price falling within the margin of purchase preference, and the contract shall be awarded to such local supplier subject to matching the L1 price.
    - iii. In case such lowest eligible local supplier fails to match the L1 price, the local supplier with the next higher bid within the margin of purchase preference shall be invited to match the L1 price and so on and contract shall be awarded accordingly. In case none of the local suppliers within the margin of purchase preference matches the L1 price, then the contract may be awarded to the L1 bidder.
- 8.4. **Exemption of small purchases:** Notwithstanding anything contained in paragraph 8.3. procurements where the estimated value to be procured is less than Rs. 5 lakhs shall be exempt from this Order. However, it shall be ensured by procuring entities that procurement is not split for the purpose of avoiding the provisions of this Order.

- 8.5. **Minimum local content:** The minimum local content shall ordinarily be 50%. The Nodal Ministry may prescribe a higher or lower percentage in respect of any particular item and may also prescribe the manner of calculation of local content.
- 8.6. **Margin of Purchase Preference:** The margin of purchase preference shall be 20%.
- 8.7. **Requirement for specification in advance:** The minimum local content, the margin of purchase preference and the procedure for preference to Make in India shall be specified in the notice inviting tenders or other form of procurement solicitation and shall not be varied during a particular procurement transaction.
- 8.8. **Government E-market place:** In respect of procurement through the Government E-market place (GeM) shall, as far as possible, specifically mark the items which meet the minimum local content while registering the item for display, and shall, wherever feasible, make provision for automated comparison with purchase preference and without purchase preference and for obtaining consent of the local supplier in those cases where purchase preference is to be exercised.
- 8.9. **Verification of local content:**
- a. The local supplier at the time of tender, bidding or solicitation shall be required to provide self-certification that the item offered meets the minimum local content and shall give details of the location(s) at which the local value addition is made.
  - b. In cases of procurement for a value in excess of Rs. 10 crores, the local supplier shall be required to provide a certificate from the statutory auditor or cost auditor of the company (in the case of companies) or from a practicing cost accountant or practicing chartered accountant (in respect of suppliers other than companies) giving the percentage of local content.**
  - c. Decisions on complaints relating to implementation of this Order shall be taken by the competent authority which is empowered to look into procurement-related complaints relating to the procuring entity.
  - d. Nodal Ministries may constitute committees with internal and external experts for independent verification of self-declarations and auditor's/accountant's certificates on random basis and in the case of complaints.
  - e. Nodal Ministries and procuring entities may prescribe fees for such complaints.
  - f. False declarations will be in breach of the Code of Integrity under Rule 175(1)(i)(h) of this General Financial Rules for which a bidder or its successors can be debarred for up to two years as per Rule 151(iii) of the General Financial Rules along with such other actions as may be permissible under law.
  - g. A supplier who has been debarred by any procuring entity for violation of this Order shall not be eligible for preference under this Order for procurement by any other procuring entity for the duration of the debarment. The debarment for such other procuring entities shall take effect prospectively from the date on which it comes to the notice of other procurement entities, in the manner prescribed under paragraph 8.9h below.
  - h. The Department of Expenditure shall issue suitable instructions for the effective and smooth operation of this process, so that:

- i. The fact and duration of debarment for violation of this Order by any procuring entity are promptly brought to the notice of the Member-Convenor of the Standing Committee and the Department of Expenditure through the concerned Ministry/Department or in some other manner.
- ii. On a periodical basis such cases are consolidated and a centralized list or decentralized lists of such suppliers with the period of debarment is maintained and displayed on website(s);
- iii. In respect of procuring entities other than the one which has carried out the debarment, the debarment takes effect prospectively from the date of uploading on the website(s) in the such a manner that ongoing procurements are not disrupted.

**8.10. Specifications in Tenders and other procurement solicitations:**

- a. Every procuring entity shall ensure that the eligibility conditions in respect of previous experience fixed in any tender or solicitation do not require proof of supply in other countries or proof of exports.
- b. Procuring entities shall endeavour to see that eligibility conditions, including on matters like turnover, production capability and financial strength do not result in unreasonable exclusion of local suppliers who would otherwise be eligible, beyond what is essential for ensuring quality of creditworthiness of the supplier.
- c. Procuring entities shall, within 2 months of the issue of this Order review all existing eligibility norms and conditions with reference to sub-paragraphs 'a' and 'b' above.
- d. If a Nodal Ministry is satisfied that Indian suppliers of an item are not allowed to participate and/ or compete in procurement by any foreign government, it may, if it deems appropriate, restrict or exclude bidders from that country from eligibility for procurement of that item and/or other items relating to that Nodal Ministry. A copy of every instruction or decision taken in this regard shall be sent to the Chairman of the Standing Committee.
- e. For the purpose of sub-paragraph 8.10 d above, a supplier or bidder shall be considered to be from a country if (i) the entity is incorporated in that country, or (ii) a majority of its shareholding or effective control of the entity is exercised from that country; or (iii) more than 50% of the value of the item being supplied has been added in that country. Indian suppliers shall mean those entities which meet any of these tests with respect to India".

**9. Reverse Auction (Applicable for works tenders valued more than Rs. 50 Cr.)**

**9.1 Selection criteria for tender cases proposed through Reverse Auction (e-RA) route:-**

- (a) In the first phase, following method of purchase through Reverse Auction shall be the preferred method for tenders valued more than Rs. 50 Cr. in each case.

- (b) The process of procurement through Reverse Auction shall be followed only in case of tenders where there are at least three proven/likely competitive sources, prima facie competent for execution of work.
- (c) Financial Bids in single currency/parameter only shall be allowed.
- (d) For cases on Zonal Railways, personal approval of the PHOD/CHOD duly vetted by associate finance shall be required for any exception in tendering method for cases otherwise eligible to be processed through the method of procurement detailed herein.
- (e) Even for cases which do not satisfy the selection criteria as detailed above, Railways may follow the process of Reverse Auction, as detailed herein if they so desire, with vetting of associate finance and approval of competent authority.

## **9.2 Procedure for award of contracts through Reverse Auction: -**

- (a) The procedure discussed herein shall be fully implemented through IREPS. Any reference to Reverse Auction in these instructions shall imply e-RA.
- (b) Conduct and reporting of Reverse Auction shall be as per **Annexure-2 of Chapter-4**.
- (c) Each tender should clearly specify essential technical and commercial parameters in a transparent manner. No deviation to such essential Technical & Commercial conditions shall be permitted to the vendors in the electronic bid form.

### **9.2.1 Technical Bid and Initial Price Offer:**

- (a) e-RA shall be adopted only for those cases where evaluation is on the basis of single parameter/currency.
- (b) Bidder shall be simultaneously required to electronically submit a Technical & Commercial Bid and Initial Price Offer.  
Offer found eligible for award of contract/meeting eligibility criteria shall be categorized as Qualified for Award of Contract for the purpose of e-RA.
- (c) Offers not complying with essential technical & commercial requirements of the tender shall be declared as Ineligible for award of contract.
- (d) Technical & Commercial evaluation of bids shall be done by a Tender Committee, as per extant guidelines, delegation and the estimated value of tender. Recommendations of Tender Committee shall be considered by Tender Accepting Authority, as per existing guidelines.
- (e) Initial Price Offer of only those bidders categorized as Qualified for Award of contract shall be opened and tabulated by system separately. Extant instructions for electronic tabulation shall apply for tabulation of Initial Price Offer.

### **9.2.2 Financial Bid:**

Financial Bid shall comprise of Final Price Offer obtained through Reverse Auction. Following conditions and procedure shall be followed in selection of bidders for conduct of Reverse Auction:

(a) Selection of vendors for Reverse Auction for award of Contract:

| Number of tenderers Qualified for Award of contract | Number of tenders to be selected for Reverse Auction                                | Remarks                                                                                                                                                                                                                                                                                                                                            |
|-----------------------------------------------------|-------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| < 3                                                 | NIL*                                                                                | The bids disallowed from participating in the Reverse Auction shall be the highest bidder(s) in the tabulation of Initial Price Offer.<br>In case the highest bidders quote the same rate, the Initial Price Offer received last, as per time log of IREPS, shall be removed first, on the principle of last in first out, by IREPS system itself. |
| 3 to 6                                              | 3                                                                                   |                                                                                                                                                                                                                                                                                                                                                    |
| More than 6                                         | 50% of Vendors Qualified for award of contract (rounded off to next higher integer) |                                                                                                                                                                                                                                                                                                                                                    |

**Note:-**

- (i) \* If the number of tenderers qualified for Award of Contract is less than 3, RA shall not be done and tender may be decided on the basis of Initial Price Offer(s).
  - (ii) Make in India criteria: All bidders eligible for benefits under Public Procurement (Preference to Make in India) Order – 2017, found Qualified for Award of Contract and are within the specified range of price preference of lowest Initial Price Bid shall be permitted to participate in the Reverse Auction, irrespective of their interse ranking on the basis of Initial Price Bid. Such bidders shall be over and above the number of vendors selected for Reverse Auction, as per Para 9.2.2 (a).
  - (b) During Reverse Auction process, bidders shall not be allowed to bid a rate higher than the lowest Initial Price Offer.
- 9.2.3** Reverse Auction among bids categorized as Qualified for award of contract shall be conducted on IREPS/Suitable Platform. Bidders shall be able to see the auction screens.
- 9.2.4** After obtaining the final bids of Reverse Auction, tenders shall be finalized as per existing policy (including Make in India Order, 2017, vide **Para-8 of Chapter-4**, if applicable). All the relevant policies of Government of India at the relevant time shall be applicable.
- 9.2.5** The level of Tender Committee to consider the Final Price Offers shall be determined on the basis of lowest Initial Price Offer of bid Qualified for award of contract as opened prior to Reverse Auction. In case of level Tender Committee which evaluated technical & commercial bids as per **Para 9.2.1(d)** was higher than the level of TC competent to consider lowest Initial Price Offer of bid Qualified for award of contract, the higher level TC shall continue to finalise such tender cases.



**10. Vitiating with respect to variation in quantities: -**

The aspect of vitiating of tender with respect to variation in quantities should be checked and avoided. As a result of variations, a contract value shall be considered “vitiating” only when, the following percentage variation in contract value between tenderers are noticed to have been exceeded.

| SN | Value of contract                                                                 | Percentage difference between present Contractor and new L-1 as a result of variation (percentage shall be calculated with base as the revised contract quantities multiplied by the rates of the present contractor) |
|----|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1  | Small value contracts (Tender Value less than Rs. 50 lakh)                        | 10                                                                                                                                                                                                                    |
| 2  | Other than small value contracts (Tender Value equal to or more than Rs. 50 lakh) | 5                                                                                                                                                                                                                     |

10.1 When the percentage difference between present Contractor and new L-1 is noticed as becoming beyond the values specified above, the following action shall be taken. The Railway administration should immediately examine whether it is practicable to bring in a new agency to carry out the extra quantity of work keeping in view the progress of the work in accordance with the original contract and the nature and the lay-out of the work. If it is found that there will be no serious practical difficulty in meeting the additional quantity of work done by another agency, then fresh tenders for the extra quantity may be invited otherwise negotiating the rate with the existing contractor for arriving at a reasonable rate for the additional quantities of work, may be adopted.

10.2 The above shall be regulated as under: -

- The case shall be decided by the tender accepting authority competent for the revised quantity) and shall not be treated as a case of single tender. The provisions of Railway Board letter No. 2007/CE.I/CT/18/Pt. XII dated 31.12.2010 hereby gets suspended.
- These instructions will be similarly applicable to earning contracts with H-1, H-2 substituted for L-1, L-2 and so on.
- Executives while executing the work shall make all efforts to ensure that no vitiating takes place in normal circumstances. Vitiating should be an exception rather than a routine affair. Efforts should be made to invite bids on the basis of percentage above/below/at par.
- Vitiations should always be computed with respect to the items, rates, quantities and conditions as available at the time of Tender Opening and subsequent changes/additions by way of new items will not be counted for computing vitiating.

(Ref: RB letter No. 2017/Trans/01/Policy dated 08.02.2018)

**11. For evaluation of change in interse position/ vitiating, comparison will be made amongst the eligible tenderers quoted for the work.**

## **12. Special Conditions for Issue of Identity Cards by Contractors**

- (i) The contractor is bound to issue identity card to each and every person employed by him and deployed for execution of the contract work as per the prescribed format provided in the tender document at his cost. Failure on the part of the contractor to issue identity cards to their employees will be treated as breach of contract conditions and therefore will be dealt as per clause No-62(VII) of GCC.
- (ii) It is mandatory on the part of every employee, deployed by the contractors to keep in his possession, the identity card, issued by the contractor throughout the execution of the work. Failure to possess such identity will be treated as unauthorized presence in the Railway premises; such persons shall be liable for prosecution as per law.
- (iii) It is mandatory for the contractor to submit the list of the employees issued with the identity cards and deployed for execution of the particular; contract to the Railway's Engineer at site before commencement of the work and also for any subsequent changes made during the execution of the work.
- (iv) No claims whatsoever arising out of implementation of special conditions pertaining to issue of identity cards shall be admissible.

### **Format of Identity Card**

**Identity Card Sr. No:**

**Date of issue:**

**Contract Agreement No.:**

**Name of the contractor & Address:**

**Particulars:**

- 1. **Name of the employee** \_\_\_\_\_ :
- 2. **Date of Birth/Age** \_\_\_\_\_ :
- 3. **Personal Identification Mark** \_\_\_\_\_ :
- 4. **Permanent Address** \_\_\_\_\_ :

**(Signature of Contractor)(Signature/Thumb Impression of employee)**

(Person who signed original

Tender document or contract Agreement)

**13. Measurement and Recording of “Executed Works” by the Contractor in Railway Construction works:**

**Para-E.1316A of Engineering code (Applicable for contracts wherein the measurement of work by contractor is permitted):** Para 1316 of Engineering Code shall not be applicable for these contracts. For such contracts, contractor shall be responsible for carrying out measurements of work executed and recording of measurements for the release of on account/final payment. In such cases, the detailed procedure for recording of measurements, provisional payment, test check and final payment shall be as follows:

**Contractor's Measurement Book:**

13. 1. Railway shall arrange contractor's measurement book (CMB), each having sheet No, 1A to 4A (Form E 1313) (**Annexure-3 to 6 of Chapter-4**), followed by 100 machine number pages (Form E 1313, sheet No. 5A) (**Annexure-7 of Chapter-4**). On the top of each sheet of CMB, there shall be provision for recording the name of the work, agreement number, name of contractor and CMB number.
13. 2. CMBs shall be printed in such a way so as to keep a clear margin of 50 mm on the left side of page, Further, the left side shall have pinhole tear line at a distance of 15 mm from edge for ease of taking out sheets from these books. The binding shall be within 15mm of the margin available between edge and pinhole tear line. This shall ensure availability of minimum 35 mm clear margin to re-bind measurement books later on.

**Movement and upkeep of Contractor's Measurement Book:**

- 13.3. Dy Chief Engineer in charge of contract (Dy CE/C) shall hand over required No. of CMBs to Assistant/Executive Engineer in-charge of contract (AEN/XEN) after taking receipt of the same on sheet No. 2A (Form E.1313) (**Annexure-4 of Chapter-4**) for further issuance to contractor time to time as per progress of work.
- 13.4. CMB shall be registered with unique No. in the Register of Measurement Books (Form E.1314) maintained In the office of Dy CE/C. Separate accountal of CMBs for each agreement shall be maintained in the office of Dy CE/C and AEN/XEN.
- 13.5. In case of change of 'contractor's authorized engineer', fresh approval shall be taken from Dy. CE/C before recording of measurement.
- 13.6. While issuing the CMB to contractor, AEN/XEN shall take out sheet No. 2A to 4A (**Annexure-3 to 5 of Chapter-4**) from the CMB, take receipt of CMB from contractor on sheet No. 3A (Form E.1313) (**Annexure-5 of Chapter-4**), and keep the same in safe custody.
13. 7. Similar system as for CMB, shall be followed for issuing Field Book/Level Book (E.1317/A) to contractor for recording of levels in the field book/level book.

**Measurement**

- 13.8. The contractor's authorized engineer shall record the measurements in CMB neatly in his own handwriting, without any use of eraser/overwriting, without use of any typing fluid or any such thing. All cuttings shall be initialed. No page shall be damaged/destroyed. No page shall be kept blank in between the measurements.
- 13.9. The Contractor shall communicate the date of measurement to AEN/XEN in sufficient advance' to witness any measurement. Witnessing of measurement by railways is not compulsory except for initial levels in case of earthwork and hidden measurements. Initial levels of earthwork and hidden measurements are to be recorded in the presence of railway officials and test checked as prescribed.

- 13.10. In on account contract certificate, measurement shall be recorded for the items and quantities to be paid in the concerned on account contract certificate.
- 13.11. However, in every 4th on account contract certificate and final contract certificate, the recording of measurement for works executed shall include all the items and their quantity included in previous on account contract certificates, irrespective of whether to be paid or not in the current On Account Contract Certificate/Final Contract Certificate.
- 13.12. No payment shall be processed on Lump sum measurement taken by contractor's authorized engineer except for earthwork. For earthwork, every 4th bill shall be based on actual levels taken and detailed calculations carried out for the work done.
- 13.13. The contractor shall take out carefully from CMB the used pages of CMB with one extra blank page for processing the bill; staple them for submission to AEN/XEN along with bill, duly signing the measurements. The contractor shall keep a photocopy of the measurements with him for future reference.
- 13.14. At the time of submission of final bill, the contractor shall submit all the remaining CMBs (unused as well as partially used) with him along with bill to AEN/XEN.
- 13.15. The contractor shall submit required copies of invoice and on account contract certificate/ final contract certificate (similar to form E.1337 and E.1338) to the AEN/XEN duly marking them - original or copy. Original shall be used for release of payment whereas duplicate copies shall be used for record purpose in different offices.
- 13.16. In case contractor requires provisional payment of on-account bill; the contractor shall submit his invoice and provisional on account contract certificate for 75% of amount of work done (before deduction of taxes). The contractor shall write 'For Provisional payment' 'on top of such on-account contract certificate.
- 13.17. AEN/XEN while issuing receipt of stapled sheets of CMB to contractor shall clearly record the same in sheet 4 (E.1314) of concerned CMB, kept in the office of AEN/XEN.

#### **Release of Provisional Payment**

- 13.18. Senior Section Engineer / 'Junior Engineer with 5-year experience' (SSE/JE) and AEN/XEN shall sign & record a certificate on the original provisional 'on account contract certificate' as under:

*"Certified that the payment being made is less than the amount due for the quantities of works executed by the contractor".*

In case of payment of earthwork items in any contract, calculation of quantity of such items along with field book /level book must be enclosed. This shall be cross checked, as considered appropriate by SSE/JE & AEN/XEN, to ensure that no excess payment is being made. At this stage no test check of measurements by railway is required.

- 13.19. AEN/XEN shall keep a copy of contractor's invoice & provisional on account contract certificate in his office, and submit original invoice & original provisional on account contract certificate along with required number of duplicate copies, and used sheets of CMB to the Dy CE/C unit for passing the bill and release of payment.

For Chief Administrative Officer (Const)  
West Central Railway, Jabalpur.

- 13.20. The provisional on account contract certificate shall be passed by Dy CE/C and payment shall be released by associate based on above certification of SSE/JE and AEN/XEN. After release of payment, blank sheet of CMB (if any) shall be crossed by Dy CE/C before sending the measurement sheets back to AEN/XEN for carrying out required test checks. At this stage measurements shall not be crossed.
- 13.21. No provisional payment shall be allowed in final contract certificate, Further, once provisional payment has been released in any on-account contract certificate, the next on account contract certificate can be raised by contractor only when accounts of previous on account certificate (Provisional as well as remaining payment) has been finalized,

### **Test Check**

- 13.22. Necessary test checks shall be carried out by the SSE/JE and AEN/XEN for the works done before full payment of on-account contract certificate/final contract certificate, SSE/JE and AEN/XEN shall communicate the date of test checks to contractor in advance. The contractor can accompany during test check. The contractor shall provide support staff and all required tools& plants to facilitate test check by railway officials.
- 13.23. The stipulated test checks for AEN/XEN and SSE/JE Level is tabulated as under:

| SN  | Description of works                                                                                | Test Check in terms of % of value by |         |
|-----|-----------------------------------------------------------------------------------------------------|--------------------------------------|---------|
|     |                                                                                                     | SSE/JE                               | AEN/XEN |
| (a) | Measurement of Ballast, pitching stone, Earth work and hidden items.                                | 100%                                 | 100%    |
| (b) | Measurement of all other items.                                                                     | 100%                                 | 20%     |
| (c) | Initial and Final levels along centre line for earthwork in embankment and cutting.                 | 100%                                 | 100%    |
| (d) | Intermittent levels along centre line for earthwork in embankment and cutting.                      | 100%                                 | 20%     |
| (e) | Initial, intermittent and final levels except centre line for earth work in embankment and cutting. | 100%                                 | 20%     |

**Note:** The check regarding levels of earthwork invariably shall be carried out in cross sections having heavy cross slopes.

13. 24. Contractor's recorded measurement sheets shall be checked for any corrections/over writing during test check. All the corrections/over writing shall be initialed by. SSE/JE.
13. 25. The discrepancy noted (if any) during test check of recorded measurement shall be communicated by AEN/XEN to the contractor.
13. 26. In case of discrepancy noticed during test check, the contractor shall submit original and required copies of fresh invoice of amount corrected for discrepancy, and in case provisional payment has been released earlier, the required copies of fresh invoice of remaining amount corrected for discrepancy (if any), along with on account/final contract certificate to AEN/XEN.

### **Full payment of On Account Contract Certificate/Final Contract Certificate**

For Chief Administrative Officer (Const)  
West Central Railway, Jabalpur.

13. 27. AEN/XEN shall submit original copy of invoice and on account contract certificate of remaining amount/ Final Contract along with required number of duplicate copies and used sheets of CMBs (all used/blank CMBs in case of final contract certificate), duly signed by SSE/JE and AEN/XEN to the Dy CE/C for passing the bill and release of payment.
- 13.28. Once the payment is released, Dy CE/C shall return back the used sheets of CMB to AEN/XEN for safe custody. duly crossing of measurements by finance officer.
- 13.29. Once all used of a particular CME is received back by AEN/XEN from Dy CE/C, the AEN/XEN shall re-bind all 100 pages of CMB along with sheet No. 1A to 4A (**Annexure-3 to 6 of Chapter-4**) for submission of CMB to Dy CE/C office. Dy CE/C office shall record the receipt of same in sheet No. 2A of CMB (**Annexure-4 of Chapter-4**) and Register of Measurement Books (Form E1314).
- 13.30. The final contract certificate shall be passed by Dy CE/C only receipt of all CMBs (used/blank) from AEN/XEN.
- 13.31. The provisions of this para 1316A shall be applicable to all the departments of Indian railways and to be executed through equivalent authorities of respective departments.

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## Annexure-1 of Chapter-4

### Request Letter from Executive Branch to Accounts Office for opening of LC

Office of -----

West Central Railway

No. \_\_\_\_\_

Dated \_\_\_\_\_

The PFA/Sr. DFM/Dy. FA  
HQ/Division/Workshop/Cost

Sub: Opening of LC

Ref: Contract Agreement No.

It is requested to open a sight LC against the above referred Order/Agreement in favour of \_\_\_\_\_. The details of beneficiary are as under:

- (i) Name of Contractor
- (ii) Vendor Code
- (iii) Address
- (iv) Tender No.
- (v) Contract Agreement No.
- (vi) Description of Goods/Service
- (vii) Value of Contract
- (viii) Stages of payment
- (ix) Expected payment within 6 months (LC Amount)
- (x) Beneficiary bank details
  - (a) Bank name
  - (b) Address
  - (c) Account No.
  - (d) IFSC Code
- (xi) Validity/Period for which LC is to be opened.

It is certified that the supplier/contractor has exercised the option of taking payment due against the tender through LC arrangement in IREPS portal at the time of bidding itself and the option has been flagged in the IREPS. This has the approval of           

(Signature)

Name: \_\_\_\_\_

Designation: \_\_\_\_\_

(Official Seal)

## **Annexure-2 of Chapter-4**

### **Procedure for Conduct and Reporting of Reverse Auction**

- 1) The tendering authority shall solicit bids through an invitation to the electronic Reverse Auction to be published or communicated in accordance with the provisions similar to e- procurement.
- 2) Depending upon the nature of item/work and complexity of case on hand, following shall be indicated in the tender for e-RA itself:
  - a) Initial e-RA period: This shall be the initial time interval for e-RA. e-RA shall be open for this duration.
  - b) Auto extension period: In case any offer is received in the time period equal to auto extension period before close of initial e-RA period, the e-RA shall be extended for time equal to auto extension period from the time of last bid. There shall be no upper limit on number of auto extensions. When no offer is received in the last auto extension period, e- RA shall close.
  - c) Minimum decrement in percentage of value of the last successful bid.
- 3) Date and time for start of e-RA shall be communicated to qualified tenderers by the convener (through IREPS portal) after evaluation of the Technical Bids.
- 4) After submission of Initial Price Bid, tenderers will not be allowed to revise the taxes and other levies.
- 5) During auction period, identities of the participating tenderers will be kept hidden.
- 6) Minimum admissible bid value will be last bid value minus minimum decrement as specified by the tendering authority before starting of reverse auction. Starting point for reverse auction shall be the lowest Initial Price Bid of the tenderer eligible for award of contract.
- 7) After close of the e-RA, tabulation of last (minimum) bids received from all the tenderers will be generated and made visible to Railways and Participating tenderers.
- 8) Railway users can also view the bidding history in chronological order.
- 9) Bidders not be allowed to withdraw their last offer.
- 10) L-1 will be defined as lowest bid obtained after closure of e-RA session for tenders.

**Note:- The bidders may take help of “User Manual for Contractors for 2 stage Reverse auction” available on IREPS portal in learning center → E-tender(works)**

\*\*\*\*\*



## Annexure-3 of Chapter-4

(Sheet 1A)

|                                      |
|--------------------------------------|
| Railway.....                         |
| CMB No.....                          |
| <b>CONTRACTOR'S MEASUREMENT BOOK</b> |
| Department .....                     |
| Division/Construction Unit.....      |
| Name of Work.....                    |
| Agreement No.....                    |
| Name of Agency.....                  |
| Name to whom Issued.....             |
| Designation.....                     |
| Date of Issue.....                   |
| Date of return.....                  |

For Chief Administrative Officer (Const)  
West Central Railway, Jabalpur.

**Annexure-4 of Chapter-4**  
(Sheet 2A)

|                                                                                                                                                                               |                                         |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|
| Railway.....<br>CMB No.....<br><b>CONTRACTOR'S MEASUREMENT BOOK</b>                                                                                                           |                                         |
| Department.....                                                                                                                                                               |                                         |
| Division/Construction Unit.....                                                                                                                                               |                                         |
| Name of Work.....                                                                                                                                                             |                                         |
| Agreement No.....                                                                                                                                                             |                                         |
| Name of Agency.....                                                                                                                                                           |                                         |
| Issued to.....<br>(Name & designation)                                                                                                                                        |                                         |
| .....<br>(Station)                                                                                                                                                            | on.....<br>(date)                       |
| Received by.....<br>(Signature)                                                                                                                                               |                                         |
| .....<br>(Designation)                                                                                                                                                        | .....<br>(Station)<br>On.....<br>(date) |
| Date of first entry.....                                                                                                                                                      |                                         |
| Date of last entry.....                                                                                                                                                       |                                         |
| Date received back in Division Const. Unit                                                                                                                                    |                                         |
| Office after completion of book } .....                                                                                                                                       |                                         |
| Certified that this Measurement Book contains 100 machine numbered pages from ..... To ..... (both pages inclusive) which have been counted by me and are correct.            |                                         |
| <div style="text-align: right;">Signature.....</div> <div style="display: flex; justify-content: space-between;"> <span>Date.....</span> <span>Designation.....</span> </div> |                                         |

For Chief Administrative Officer (Const)  
West Central Railway, Jabalpur.

## Annexure-5 of Chapter-4

| Railway.....<br>CMB No.....<br><b>CONTRACTOR'S MEASUREMENT BOOK</b>                                                                                                                     |                        |                           |                |                                      |                                              |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|---------------------------|----------------|--------------------------------------|----------------------------------------------|
| Name of work.....                                                                                                                                                                       |                        |                           |                |                                      |                                              |
| Agreement No .....                                                                                                                                                                      |                        |                           |                |                                      |                                              |
| Name of Agency.....                                                                                                                                                                     |                        |                           |                |                                      |                                              |
| Issued to .....                                                                                                                                                                         |                        |                           |                |                                      |                                              |
| (Contractor's name)                                                                                                                                                                     |                        |                           |                |                                      |                                              |
| .....                                                                                                                                                                                   |                        |                           | on.....        |                                      |                                              |
| (Station)                                                                                                                                                                               |                        |                           | (date)         |                                      |                                              |
| Certified that this Measurement Book contains 100 machine numbered pages from.....<br>to.....(both pages inclusive) which have been counted by me and are correct. No<br>sheet in torn. |                        |                           |                |                                      |                                              |
| I understand that the measurement book is very important document and hence I shall ensure its<br>proper upkeep and safe custody.                                                       |                        |                           |                |                                      |                                              |
| Received by.....                                                                                                                                                                        |                        |                           |                |                                      |                                              |
| (Signature of contractor)                                                                                                                                                               |                        |                           |                |                                      |                                              |
| .....                                                                                                                                                                                   |                        | .....                     |                | .....                                |                                              |
| (Name)                                                                                                                                                                                  |                        | (Station)                 |                | (date)                               |                                              |
| Date of first entry.....                                                                                                                                                                |                        |                           |                |                                      |                                              |
| Date of last entry.....                                                                                                                                                                 |                        |                           |                |                                      |                                              |
| Certified that this Contractor's Measurement Book pages returned by contractor have been<br>counted by me and are correct. The details of pages received by me is as under:             |                        |                           |                |                                      |                                              |
| S. No.                                                                                                                                                                                  | On Account Bill<br>No. | Page No.<br>From ..... To | No of<br>Pages | Date of receipt in<br>AEN/XEN Office | Sign &<br>Designation of<br>Railway Official |
| 1                                                                                                                                                                                       |                        |                           |                |                                      |                                              |
| 2                                                                                                                                                                                       |                        |                           |                |                                      |                                              |
| 3                                                                                                                                                                                       |                        |                           |                |                                      |                                              |
| 4                                                                                                                                                                                       |                        |                           |                |                                      |                                              |
| 5                                                                                                                                                                                       |                        |                           |                |                                      |                                              |
| 6                                                                                                                                                                                       |                        |                           |                |                                      |                                              |
| 7                                                                                                                                                                                       |                        |                           |                |                                      |                                              |

(Sheet 3A)

(Sheet 4A)

CMB No.....

Name of work.....

Name of Agency.....

[illegible]

**Annexure-7 of Chapter-4**  
(Sheet 5A)

[illegible]

\*\*\*\*\*

**For Chief Administrative Officer (Const)  
West Central Railway, Jabalpur.**

## **CHAPTER-5**

### **WEST CENTRAL RAILWAY CONSTRUCTION ORGANISATION**

#### **CONTRACTOR'S FAMILARIZATION**

I/We hereby solemnly declare that I/we have visited the site of above work and have familiarized myself/ourselves of the working conditions there in all respects and in particular the following:

1. Topography of the area and existing Road network (high ways & Village, Pucca & Kacha) and availability of Service Roads,
2. Soil Conditions at the site of the work.
3. Sources and availability of construction material.
4. Rates for Construction materials.
5. Availability of local labour, both skilled and unskilled and the prevailing labour rates.
6. Availability of Water and Electricity.
7. Availability of space for putting up labour camps, offices Store Godowns, Engineering yards etc.

\*\*\*\*\*

**CHAPTER-6****WEST CENTRAL RAILWAY CONSTRUCTION ORGANISATION****ANNEXURES –A to R**

|     |                        |                                                                                                                                                                                                                                                             |
|-----|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1.  | ANNEXURE – A:          | FORMAT FOR CERTIFICATE TO BE SUBMITTED /<br>UPLOADED BY TENDERER ALONGWITH THE<br>TENDER DOCUMENTS                                                                                                                                                          |
| 2.  | ANNEXURE – A1:         | FORMAT FOR CERTIFICATE TO BE SUBMITTED /<br>UPLOADED BY ATTORNEY/AUTHORISED<br>SIGNATORY/EACH MEMBER OF PARTNERSHIP<br>FIRM/JOINT VENTURE(JV)/HINDU UNDIVIDED<br>FAMILY (HUF)/LIMITED LIABILITY PARTNERSHIP<br>(LLP) ETC. ALONGWITH THE TENDER<br>DOCUMENTS |
| 3.  | ANNEXURE – B:          | COMPLIANCE OF FINANCIAL ELIGIBILITY CRITERIA<br>BY THE TENDERER REGARDING CONTRACTUAL<br>PAYMENT RECEIVED                                                                                                                                                   |
| 4.  | ANNEXURE – C:          | HISTORY SHEET OF THE TENDERER                                                                                                                                                                                                                               |
| 5.  | ANNEXURE – D:          | DETAILS OF CONSTRUCTION MACHINERIES,<br>TOOLS & PLANTS, VEHICLES ETC                                                                                                                                                                                        |
| 6.  | ANNEXURE – E:          | DETAILS OF TECHNICAL AND OTHER<br>PERSONNELS AVAILABLE ON HAND &<br>PROPOSED TO BE ENGAGED IN WORK                                                                                                                                                          |
| 7.  | ANNEXURE – F & F1:     | DETAILS OF SIMILAR NATURE OF WORK<br>COMPLETED/SUBSTANTIALLY COMPLETED DURING<br>LAST SEVEN (07) YEARS                                                                                                                                                      |
| 8.  | ANNEXURE – G &<br>G1 : | COMPLIANCE OF ELIGIBILITY CRITERIA BY THE<br>TENDERER REGARDING BID CAPACITY                                                                                                                                                                                |
| 9.  | ANNEXURE – H:          | DECLARATION REGARDING ASSOCIATION OF<br>RAILWAY OFFICER(S) WITH TENDERER(S)                                                                                                                                                                                 |
| 10. | ANNEXURE – I:          | LIST OF COURT CASES DURING LAST 3 YEARS                                                                                                                                                                                                                     |
| 11. | ANNEXURE – J:          | LIST OF ARBITRATION CASES DURING LAST 3 YEARS                                                                                                                                                                                                               |
| 12. | ANNEXURE – K:          | BROAD PLAN OF EXECUTION OF THIS WORK WITHIN<br>THE STIPULATED COMPLETION PERIOD                                                                                                                                                                             |
| 13. | ANNEXURE – L:          | FORMAT OF MOU FOR JOINT VENTURE<br>PARTICIPATION                                                                                                                                                                                                            |
| 14. | ANNEXURE – M:          | FORMAT OF JOINT VENTURE AGREEMENT FOR JOINT<br>VENTURE PARTICIPATION                                                                                                                                                                                        |
| 15. | ANNEXURE – N:          | DETAILS OF OTHER CREDENTIALS/ FACILITIES<br>AVAILABLE WITH THE FIRM/CONTRACTOR                                                                                                                                                                              |
| 16. | ANNEXURE – O:          | MANDATE FORM OF EFT                                                                                                                                                                                                                                         |
| 17. | ANNEXURE – P:          | DOCUMENT OF AUTHORISATION                                                                                                                                                                                                                                   |
| 18. | ANNEXURE – Q:          | SPECIMEN FORMAT FOR BANK GUARANTEE FOR BID<br>SECURITY                                                                                                                                                                                                      |
| 19. | ANNEXURE – R:          | SPECIMEN FORMAT FOR BANK GUARANTEE FOR<br>PERFORMANCE GUARANTEE                                                                                                                                                                                             |

\*\*\*\*\*

For Chief Administrative Officer (Const)  
West Central Railway, Jabalpur.

## ANNEXURE-A

### FORMAT FOR CERTIFICATE TO BE SUBMITTED / UPLOADED BY TENDERER ALONG WITH THE TENDER DOCUMENTS

I..... (Name and designation) \*\* appointed as the attorney/authorized signatory of the tenderer,

M/s.....(hereinafter called the tenderer) for the purpose of the Tender documents for the work of.....as per the tender No.....of.....(Railway)\*\*, do hereby solemnly affirm and state on the behalf of the tenderer including its constituents as under:

1. I/we the tenderer (s) am/are signing this document after carefully reading the contents.
2. I/We the tenderer(s) also accept all the conditions of the tender and have signed all the pages in confirmation thereof.
3. I/we hereby declare that I/we have downloaded the tender documents from Indian Railway website [www.ireps.gov.in](http://www.ireps.gov.in) . I/we have verified the content of the document from the website and there is no addition, no deletion or no alteration to the content of the tender document. In case of any discrepancy noticed at any stage i.e. evaluation of tenders, execution of work or final payment of the contract, the master copy available with the railway Administration shall be final and binding upon me/us.
4. I/we declare and certify that I/we have not made any misleading or false representation in the forms, statements and attachments in proof of the qualification requirements.
5. **I/We also understand that my/our offer will be evaluated based on the documents/credentials submitted along with the offer and same shall be binding upon me/us.**
6. **I/We declare that the information and documents submitted along with the tender by me/us are correct and I/we are fully responsible for the correctness of the information and documents, submitted by us.**
7. I/we certify that I/we the tenderer(s) is/are not blacklisted or debarred by Railways or any other Ministry / Department of Govt. of India from participation in tender on the date of submission of bids, either in individual capacity or as a HUF/ member of the partnership firm/LLP/JV/Society/Trust.
8. I/we understand that if the contents of the certificate submitted by us are found to be forged/false at any time during process for evaluation of tenders, it shall lead to forfeiture of the tender Bid Security and may also lead to any other action provided in the contract including banning of business for a period of upto two years. Further, I/we (insert name of the tenderer) \*\*.....and all my/our constituents understand that my/our offer **shall be summarily rejected.**
9. I/we also understand that if the contents of the certificate submitted by us are found to be false/forged at any time after the award of the contract, it will lead to termination of the contract, along with forfeiture of Bid Security/Security Deposit and Performance guarantee and, may also lead to any other action provided in the contract including banning of business for a period of up to two years.

For Chief Administrative Officer (Const)  
West Central Railway, Jabalpur.



10. I/We have read the clause regarding restriction on procurement from a bidder of a country which shares a land border with India and certify that I am/We are not from such a country or, if from such a country, have been registered with the competent Authority. I/We hereby certify that I/we fulfil all the requirements in this regard and am/are eligible to be considered (evidence of valid registration by the competent authority is enclosed)

*\*\* The contents in Italics are only for guidance purpose. Details as appropriate are to be filled insuitably by tenderer.*

## ANNEXURE-A1

*(This certificate is to be given by attorney/authorized signatory/ each member of Partnership firm/ Joint Venture (JV)/ Hindu Undivided Family (HUF)/ Limited Liability Partnership (LLP) etc.)*

I/We ..... (Name), attorney/authorized signatory of the ..... (constituent firm/constituent partner) and member/partner of the ..... (tendering firm) hereby solemnly affirm and state as under:

1. I/we certify that ..... (constituent firm/constituent partner) is /are not blacklisted or debarred by Railways or any other Ministry/ Department of Govt. of India from participation in tender on the date of submission of bids, either in individual capacity or as a HUF/ member of the partnership firm/LLP/JV/Society/Trust.
2. I/we have read the clause regarding restriction on procurement from a bidder of a country which shares a land border with India and certify that I am /We are not from such a country or, if from such a country, have been registered with the competent Authority. I/We hereby certify that I/we fulfil all the requirement in this regard and am/are eligible to be considered (evidence of valid registration by the competent authority is enclosed).

SEAL AND SIGNATURE  
OF THE CONSTITUENT FIRM/CONSTITUENT PARTNER

Place:

Dated:

For Chief Administrative Officer (Const)  
West Central Railway, Jabalpur.

## ANNEXURE-B

**Compliance of Financial Eligibility Criteria by the Tenderer regarding Contractual Payments received during the qualifying period** (Reference-Para 10.2 & 17.15.2 of Tender Form (Second Sheet) of Annexure-I.

**Each Bidder or each member of a JV must fill in this form separately:**

**NAME OF BIDDER/JV PARTNER:**

| <b>Annual Contractual Turnover Data for the Previous 3/4 Years<br/>(Contractual Payment only)</b> |                            |                          |                                                  |
|---------------------------------------------------------------------------------------------------|----------------------------|--------------------------|--------------------------------------------------|
| <b>Year</b>                                                                                       | <b>Amount<br/>Currency</b> | <b>Exchange<br/>Rate</b> | <b>Indian National<br/>Rupees<br/>Equivalent</b> |
|                                                                                                   |                            |                          |                                                  |
|                                                                                                   |                            |                          |                                                  |
|                                                                                                   |                            |                          |                                                  |
|                                                                                                   |                            |                          |                                                  |
|                                                                                                   |                            |                          |                                                  |
| <b>Average Annual Contractual Turnover for last 3 years</b>                                       |                            |                          |                                                  |

1. The average annual contractual turnover shall be calculated as an average of “total contractual payments” in the previous three financial years. However, in case balance sheet of the previous year is yet to be prepared/ audited, the audited balance sheet of the fourth previous year shall be considered for calculating average annual contractual turnover.

2. The information supplied shall be substantiated by data in the audited balance sheets and profit and loss accounts for the relevant years in respect of the bidder or all members constituting the bidder.

3. Contents of this form should be certified by a Chartered Accountant duly supported by Audited Balance Sheet duly certified by the Chartered Accountant.

### **SEAL AND SIGNATURE OF THE BIDDER**

Certified that all figures and facts submitted in this form have been furnished after full consideration of all observations/notes in Auditor’s reports.

(Signature of Chartered Accountant)

Name of CA:

Registration No:

(Seal)

For Chief Administrative Officer (Const)  
West Central Railway, Jabalpur.

**ANNEXURE -C****HISTORY SHEET OF THE TENDERER**

|      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| i)   | Name of the Company                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | : |
| ii)  | Address of Registered Office                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | : |
|      | <b>Phone:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
|      | <b>Fax:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| iii) | Constitution of the Company                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | : |
| a)   | Ownership particulars whether Sole Proprietary Firm/ HUF/Partnership firm/Joint Venture (JV)/Company registered under Company Act-2013/ Limited Liability Partnership Firm registered under LLP Act-2008,/ Registered Society/ Registered Trust or any other type duly supported by the documents such as Partnership Deed, Power of Attorney, PAN Card, Memorandum of Articles/Articles of Association, Authorisation, Certificate of Incorporation, Certificate of Registration, Deem of Formation, JV related documents etc. as applicable as per tender condition. | : |
| b)   | Name and address of collaborator(s)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | : |
| c)   | Nature of participation by collaborator(s) in share holding of the Company                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | : |
| d)   | Extent and nature of proposed participation by collaborator(s) in execution of this work                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | : |
| iv)  | Number of years the firm has been in operation in India under its present                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | : |
| v)   | Any Other information                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |

Signature of tenderer

Along with Seal

For Chief Administrative Officer (Const)  
West Central Railway, Jabalpur.

## ANNEXURE – D

### DETAILS OF CONSTRUCTION MACHINERIES, TOOLS & PLANTS, VEHICLES ETC.

(AVAILABLE ON HAND AND PROPOSED TO BE UTILIZED IN WORK)

| Sl. No. | Description of equipment's | Number available (Owned/hired) | Date of purchase | Date of manufacture | Make | How driven (i.e) Petrol/ Diesel /Electrical | Condition of the equipment | Where the equipment can be inspected |
|---------|----------------------------|--------------------------------|------------------|---------------------|------|---------------------------------------------|----------------------------|--------------------------------------|
| 1       | 2                          | 3                              | 4                | 5                   | 6    | 7                                           | 8                          | 9                                    |
|         |                            |                                |                  |                     |      |                                             |                            |                                      |
|         |                            |                                |                  |                     |      |                                             |                            |                                      |
|         |                            |                                |                  |                     |      |                                             |                            |                                      |
|         |                            |                                |                  |                     |      |                                             |                            |                                      |
|         |                            |                                |                  |                     |      |                                             |                            |                                      |
|         |                            |                                |                  |                     |      |                                             |                            |                                      |
|         |                            |                                |                  |                     |      |                                             |                            |                                      |
|         |                            |                                |                  |                     |      |                                             |                            |                                      |
|         |                            |                                |                  |                     |      |                                             |                            |                                      |
|         |                            |                                |                  |                     |      |                                             |                            |                                      |
|         |                            |                                |                  |                     |      |                                             |                            |                                      |
|         |                            |                                |                  |                     |      |                                             |                            |                                      |
|         |                            |                                |                  |                     |      |                                             |                            |                                      |

Signature of tenderer

Along with Seal

For Chief Administrative Officer (Const)  
West Central Railway, Jabalpur.

## ANNEXURE – E

### DETAILS OF TECHNICAL AND OTHER PERSONNELS AVAILABLE ON HAND AND PROPOSED TO BE ENGAGED IN WORK

| Sl | N a m e | Age | Technical qualification(s) | Position with the tenderer | Commencement of present employment | Total experience | Emoluments |
|----|---------|-----|----------------------------|----------------------------|------------------------------------|------------------|------------|
| 1  | 2       | 3   | 4                          | 5                          | 6                                  | 7                | 8          |
|    |         |     |                            |                            |                                    |                  |            |
|    |         |     |                            |                            |                                    |                  |            |
|    |         |     |                            |                            |                                    |                  |            |
|    |         |     |                            |                            |                                    |                  |            |
|    |         |     |                            |                            |                                    |                  |            |
|    |         |     |                            |                            |                                    |                  |            |
|    |         |     |                            |                            |                                    |                  |            |

Signature of tenderer

Along with Seal

For Chief Administrative Officer (Const)  
West Central Railway, Jabalpur.

## ANNEXURE – F

### Details of Similar Nature of Works Completed/Substantially completed During last Seven (07) Years

| SN | Particulars                                                                                                                                                                               | Work-1 | Work-2 | Work-3 | Work-4 |
|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|--------|--------|--------|
| 1  | <b>Name of Work</b>                                                                                                                                                                       |        |        |        |        |
| 2  | Date of Award of Contract                                                                                                                                                                 |        |        |        |        |
| 3  | Contract Agreement No                                                                                                                                                                     |        |        |        |        |
| 4  | Name & Address of Client/Department                                                                                                                                                       |        |        |        |        |
| 5  | Original Value of Agreement (Rs.)                                                                                                                                                         |        |        |        |        |
| 6  | Final/Latest revised value of Contract (Rs.)                                                                                                                                              |        |        |        |        |
| 7  | Original/Extended Date of Completion                                                                                                                                                      |        |        |        |        |
| 8  | Actual Date of Completion for completed work                                                                                                                                              |        |        |        |        |
| 9  | Gross Payment Received (Including PVC) till last day of month previous to the one in which present tender notice published on IREPS website [Rs.]                                         |        |        |        |        |
| 10 | Gross Payment Received (Excluding PVC) <b>for substantially completed work</b> till last day of month previous to the one in which present tender notice published on IREPS website [Rs.] |        |        |        |        |
| 11 | Scope of Work and main Features of Contract                                                                                                                                               |        |        |        |        |
| 12 | Remarks                                                                                                                                                                                   |        |        |        |        |

**Date:** \_\_\_\_\_

**Signature of Tenderer/s with seal**

Above detail should be given only for Similar Nature of Works as defined in Tender Document and which have been Physically Completed or Substantially completed in all respects before last day of the month previous to the one in which tender is invited. Part Completed Work shall not be considered.

- **Substantially Completed Work means** an ongoing work in which payment equal to or more than 90% of the present contract value (excluding the payment made for adjustment of Price variation (PVC), if any) has been made to the contractor in that ongoing contract and no proceedings of termination of contract on Contractor's default has been initiated. The credential certificate in this regard should have been issued not prior to 60 days of date of invitation of present tender.

For Chief Administrative Officer (Const)  
West Central Railway, Jabalpur.

- **For Technical Eligibility Criteria, In case of completed work,** the value of final bill (gross amount) including the PVC amount (if paid) shall be considered as the completion cost of work. In case final bill is pending, only the total gross amount already paid including the PVC amount (if paid) shall be considered as the completion cost of work.

In case of substantially completed work, the total gross amount already paid including the PVC amount (if paid), as mentioned in the certificate, shall be considered as the cost of substantially completed work.

**Note:**

- (i) Contractual payment received by a Member in an earlier JV shall be reckoned only to the extent of the concerned member's share in that JV for the purpose of satisfying compliance of the above-mentioned financial eligibility criteria in the tender under consideration.
- (ii) Certificate from Private individual for whom such works are executed shall not be Considered for eligibility of tenderers. However, in addition to work experience certificate issued by any Govt. Organization, work experience certificate issued by Public listed company having average annual turnover of Rs 500 crore and above in last 3 financial years excluding the current financial year, listed on national Stock Exchange or Bombay Stock Exchange, incorporated/registered at least 05 years prior to the date of opening of tender, shall also be considered provided the work experience certificate has been issued by a person authorized by the Public listed company to issue such certificates.

Following documents regarding the **Public listed company** shall also be submitted along with the certificate **(Mandatory)**

- (a) Details of Average Annual turnover of the public listed company in last three financial years excluding current financial year (should be 500 Cr and above) issued by Chartered Accountant.
- (b) The copy of the documents regarding listing in the National stock exchange or Bombay stock exchange with details of status of listing as on date of opening of tender.
- (c) The copy of the document of incorporation/ registration of the Public listed company (should be at least 5 years prior to date of opening of tender).
- (d) The copy of document regarding Person Authorized by the Public listed Company to issuesuch certificate.
- (e) The relevant copy of work order, bill of quantities, bill wise details of payment received duly certified by Chartered Accountant. TDS certificates for all payments received a copy of final/last bill paid by the public listed company in support of above work experience certificate.

- (iii) **The list of works as mentioned in the above table supported with documentary proof ( Work Completion/Experience Certificate issued by the organizations for whom the work was carried out) shall only be considered for evaluating the Technical Eligibility Criteria.** In case of JV, all the members should attach copy of completion certificate issued by the organizations for whom the work was carried out. **Non-compliance with this condition is liable to result in the tender being rejected.**



## ANNEXURE -F-1

**Details of Similar Nature Works Completed or Substantially completed During last Seven (07) Years in case of tender having composite nature of work** (bifurcated as per the components of tender schedule) to be submitted along with Annexure– F

| S. No.                                                             | Description of components/ component as per tender schedule, executed in different contracts | Name of work under which components (mentioned in column-2) executed | S.N. of Annexure F, where other details of this work mentioned | Payment received under these components till ending last day of month previous to the one in which tender notice published on IREPS website | Amount received in % age under these components w.r.t estimated cost of components mentioned in tender document. |
|--------------------------------------------------------------------|----------------------------------------------------------------------------------------------|----------------------------------------------------------------------|----------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|
| 1                                                                  | 2                                                                                            | 3                                                                    | 4                                                              | 5                                                                                                                                           | 6                                                                                                                |
| <b>All components (A,B,C,D,E, ---) executed in single contract</b> |                                                                                              |                                                                      |                                                                |                                                                                                                                             |                                                                                                                  |
| 1                                                                  |                                                                                              |                                                                      |                                                                |                                                                                                                                             |                                                                                                                  |
| 2                                                                  |                                                                                              |                                                                      |                                                                |                                                                                                                                             |                                                                                                                  |
|                                                                    |                                                                                              |                                                                      |                                                                |                                                                                                                                             |                                                                                                                  |
| <b>More than onecomponent executed in single contract</b>          |                                                                                              |                                                                      |                                                                |                                                                                                                                             |                                                                                                                  |
| 1                                                                  |                                                                                              |                                                                      |                                                                |                                                                                                                                             |                                                                                                                  |
| 2                                                                  |                                                                                              |                                                                      |                                                                |                                                                                                                                             |                                                                                                                  |
|                                                                    |                                                                                              |                                                                      |                                                                |                                                                                                                                             |                                                                                                                  |
| <b>Only one component executed in single contract</b>              |                                                                                              |                                                                      |                                                                |                                                                                                                                             |                                                                                                                  |
| 1                                                                  |                                                                                              |                                                                      |                                                                |                                                                                                                                             |                                                                                                                  |
| 2                                                                  |                                                                                              |                                                                      |                                                                |                                                                                                                                             |                                                                                                                  |
|                                                                    |                                                                                              |                                                                      |                                                                |                                                                                                                                             |                                                                                                                  |

Date:

Signature of Tenderer/s  
Along with Seal

**Note: -**

- (i) Component wise details mentioned above for the similar nature of work defined for the various components in the Tender (para 10.1(b)(1-3) of Tender document Chapter-2, Annexure-1 Tender form (Second sheet)), should be supported **with documentary proof i . e** Work Completion/Experience Certificate issued by the organizations for whom the work was carried out.

For Chief Administrative Officer (Const)  
West Central Railway, Jabalpur.

- (ii) For col 5 - The total gross amount already paid for the components of the work including the PVC amount (if paid) shall be mentioned.
- (iii) Only those works will be treated as composite works which consist of more than one distinct component of work such as Civil Engg. Works, S&T work, Electrical work, OHE work etc. and there is separate schedule for each such distinct components in the tender documents.

***Format given above is for guidance only. Any certificate, containing complete/relevant required information as asked for in other format, shall be considered.***

## ANNEXURE – G

### **Compliance of Eligibility criteria by the Tenderer regarding Bid Capacity for the tenders having advertised value more than Rs. 20 Crore.**

1. Name and style of the Tenderer with address (present tenderer)
2. The available bid capacity calculated is as under:

$$\text{Available Bid Capacity} = [A \times N \times 2] - 0.33 \times N \times B$$

Where

**A** = Rs. \_\_\_\_\_ (i.e. Maximum value of construction works executed and payment received in any one financial year during the current and last three financial years immediately preceding the current financial year, up to date of opening of tender, taking into account the completed as well as works in progress.)

**N** = \_\_\_\_\_ Years (i.e. Number of years prescribed for completion of work for which bids has been invited.)

**B** = Rs. \_\_\_\_\_ (i.e. Existing commitments and balance amount of ongoing works with tenderer as per the prescribed proforma of Railway for statement of all works in progress and also the works which are awarded to tenderer but yet not started up to the date of inviting of tender)

Bid Capacity = Rs. \_\_\_\_\_

3. Details of the Chartered Accountant verifying the Bid Capacity.

- (i) Name :-
- (ii) Address:-
- (iii) Phone No:-
- (iv) Fax No:-
- (v) e-mail ID:-

4. Declaration by the tenderer: I hereby declare that the information given above are true. If any of the information given above is found to be wrong at any time, my tender will liable to be rejected.

**Signature of the Tenderer**

**Verified and signed**

**Details of Tenderer along with seal**

**Details of Chartered Accountant along with seal**

#### **Note:-**

- (a) The Tenderer(s) shall furnish the details of -
  - (i) Maximum value of construction works executed and payment received in any one of the previous three financial years or the current financial year (up to date of inviting tender) for calculating A, and
  - (ii) Existing commitments and balance amount of ongoing works with tenderer as per the prescribed proforma of Railway for statement of all works in progress and also the works which are awarded to tenderer but yet not started up to the date of inviting of tender for calculating B. In case of no works in hand, a 'NIL' statement should be furnished.

The submitted details for (i) and (ii) above should be duly verified by Chartered Accountant.

For Chief Administrative Officer (Const)  
West Central Railway, Jabalpur.

- (b) In case if a bidder is JV, the tenderer(s) must furnish the details of
- (i) Maximum value of construction works executed and payment received in any one of the previous three financial years or the current financial year (up to date of inviting tender) by each member of JV for calculating A, and
  - (ii) Existing commitments and balance amount of ongoing works with each member of JV either in individual capacity or as a member of other JV as per the prescribed proforma of Railway for statement of all works in progress and also the works which are awarded to each member of JV either in individual capacity or as a member of other JV but yet not started up to the date of inviting of tender for calculating B. In case of no works in hand, a 'NIL' statement should be furnished.

The submitted details for (i) and (ii) above should be duly verified by Chartered Accountant.

- (c) Value of a completed work/work in progress/work awarded but yet not started for a Member in an earlier JV shall be reckoned only to the extent of the concerned member's share in that JV for the purpose of satisfying his/her compliance to the above-mentioned bid capacity in the tender under consideration.
- (d) The arithmetic sum of individual "bid capacity" of all the members shall be taken as JV's "bid capacity".
- (e) The available bid capacity of tenderer shall be assessed based on the details submitted by the tenderer as per Annexure-B & Annexure-G1, Chapter-6 of Tender Document which should be duly signed by tenderer(s) and verified by Chartered Accountant. In case, the available bid capacity is lesser than estimated cost of work put to tender, his offer shall not be considered even if he has been found eligible in other eligibility criteria/tender requirement.

*For **Maximum value of construction works** executed and payment received in any one financial year i.e. **Value-"A"** considered for calculation of available Bid capacity shall be duly certified by CA supported with audited balance sheet and in case balance sheet is yet to be prepared /audited than it shall be supported by Form-16A issued by client (duly certified by CA) and/or payment received under relevant section for contractual payment of Form-26AS generated through TRACES of Income Tax Department (duly certified by CA). **Value of "A" not supported by duly certified audited balance sheet/Form-16A/Form-26AS by Chartered Accountant will not be considered for Bid capacity calculation.***

- (f) In case, the tenderer/s failed to submit the above statements along with offer, their/his offer shall be considered as incomplete and will be **rejected summarily**.

## ANNEXURE-G1

**Compliance of Eligibility criteria by the Tenderer regarding Current Commitment and Balance Amount of ongoing works for the tenders having advertised value more than Rs. 20 Crore.**

**(for calculating “B” as per the Bid Capacity formula)**

Tenderers and each partner of the JV should provide information on their current commitments and balance amount of ongoing work up to the date of inviting of present tender including works which are awarded to tenderer but not yet started up to the date of inviting of present tender and for contracts approaching completion, but for which an unqualified full completion certificate is yet to be issued.

| SN | Name of Work | LOA/Contract No. & Date | Name & Address of the Employer<br>/Tel./Fax/E-mail | Date of Award of Contract | Latest value of Contract (Rs.) | Date of Completion (Original or<br>extended date) of Work | Value of the Balance Work<br>as on Date of publishing of<br>tender notice of present<br>Tender on IREPS website | Date of publishing tender notice of<br>Present work on IREPS website | Balance Period for Completion of<br>the Work in Years = (g-i)/365 | Proportionate Amount of<br>Contract to be executed in<br>“ N” Years** (Rs.) |
|----|--------------|-------------------------|----------------------------------------------------|---------------------------|--------------------------------|-----------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|-------------------------------------------------------------------|-----------------------------------------------------------------------------|
| a  | b            | c                       | d                                                  | e                         | f                              | g                                                         | h                                                                                                               | i                                                                    | j                                                                 | k                                                                           |
|    |              |                         |                                                    |                           |                                |                                                           |                                                                                                                 |                                                                      |                                                                   |                                                                             |
|    |              |                         |                                                    |                           |                                |                                                           |                                                                                                                 |                                                                      |                                                                   |                                                                             |
|    |              |                         |                                                    |                           |                                |                                                           |                                                                                                                 |                                                                      |                                                                   |                                                                             |

“Certified that current commitments as on date of publishing of Tender notice of present tender on IREPS website for all the contracts that have been awarded or for which a Letter of Acceptance has been received or for the works in progress approaching completion, value of outstanding work and the balance completion period has been indicated in the above table correctly. It is further certified that if later on the Railway discovers that information provided in the table is incorrect then the Railway will treat our bid invalid and it will be liable for rejection.”

**Signature of the Tenderer**

**Verified and signed**

**Details of Tenderer along with seal**

**Details of Chartered Accountant along with seal**

1. The above statement should be submitted by the tenderer(s) duly signed by him and verified by Chartered Accountant. **In case, the tenderer/s failed to submit the above statements along with offer, their/his offer shall be considered as incomplete and will be rejected summarily.**
2. In case of JV, the tenderer(s) must furnish the details of each member of JV separately.
3. The tenderer(s) may submit any additional information in the above subject, if they desire so.

**For Chief Administrative Officer (Const)  
West Central Railway, Jabalpur.**

**Note:**

- (1) "N" = No. of Years prescribed for completion of work in the Present Tender.
- (2) \*\* In case, the balance period for completion of the work [assessed in column (j)] is less than the period of completion of work in the Present Tender (N), full value of balance work as contained in column (h) shall be taken in column (k).
- (3) \*\* In case, the balance period for completion of the work [assessed in column (j)] is more than the period of completion of work in the Present Tender (N), the proportionate amount to be filled in column (k) and shall be calculated as:  
= [Value of balance work as on date of publishing tender notice on IREPS website i.e. value in column (h) x N (in years)] / [Balance period for completion (in years) i.e. value in column-j]
- (4) Data should be submitted for Railway Contracts as well as all other Non-Railway contracts and all other Commitments in hand as on the date of publishing tender notice of present tender on IREPS website.
- (5) In col-f ,the Latest value of contract shall be the value of original contract or last sanctioned subsidiary /supplementary agreement/variation ( if any).

## ANNEXURE-H

### DECLARATION REGARDING ASSOCIATION OF RAILWAY OFFICER(S) WITH TENDERER(S)

- (i) Information and particulars regarding working /retired Railway Engineer (s)/ Officer(s) of Gazetted rank as proprietor of proprietary firm.

| Sr.No. | Name of retired Gazetted Officer / Engineer with designation & place of posting at the time of retirement | Date of Retirement | Particulars of permission taken for associating with the tenderer (wherever applicable) |
|--------|-----------------------------------------------------------------------------------------------------------|--------------------|-----------------------------------------------------------------------------------------|
| 1.     |                                                                                                           |                    |                                                                                         |
| 2.     |                                                                                                           |                    |                                                                                         |

- (ii) Information and particulars regarding retired Railway Engineer (s)/ Officer(s) of the Gazetted rank being one of the partner/members in the partnership Firm/ Joint venture/registered Society/ registeredfirm/ LLP etc.

| Sr. No. | Name of retired Gazetted Officer / Engineer with designation & place of posting at the time of retirement | Date of Retirement | Particulars of permission taken for associating with the tenderer (wherever applicable) |
|---------|-----------------------------------------------------------------------------------------------------------|--------------------|-----------------------------------------------------------------------------------------|
| 1.      |                                                                                                           |                    |                                                                                         |
| 2.      |                                                                                                           |                    |                                                                                         |

- (iii) Information and particulars regarding retired Railway Engineer (s)/ Officer(s) of the Gazettedrank being director in the company.

| Sr No. | Name of retired Gazetted Officer / Engineer with designation & place of posting at the time of retirement | Date of Retirement | Particulars of permission taken for associating with the tenderer (wherever applicable) |
|--------|-----------------------------------------------------------------------------------------------------------|--------------------|-----------------------------------------------------------------------------------------|
| 1.     |                                                                                                           |                    |                                                                                         |
| 2.     |                                                                                                           |                    |                                                                                         |

- (iv) Information in terms of tender document para-16(c) of Chapter-2 regarding Relative(s) employed in Gazetted capacity on the Railway:

| Sr. No. | Name of the relative who is employed in Gazetted capacity in Engineering or any other department of the Railway with Designation | Relation |
|---------|----------------------------------------------------------------------------------------------------------------------------------|----------|
| 1.      |                                                                                                                                  |          |
| 2.      |                                                                                                                                  |          |

*The information under Item(iv) is to be submitted by the tenderer, should a tenderer or Contractor being an individual, have member(s) of his family or in the case of partnership firm/ company / joint venture (JV) / registered society / registered trust etc. one or more of his partner(s)/shareholder(s) or member(s) of the family of partner(s)/shareholder(s) having share of more than 1% in the tendering entity employed in gazetted capacity in the Engineering or any other department of the railway.*

Signature of tenderer  
Along with Seal

- Note:-** 1. Details as per the above format shall be furnished by the tenderer. The format should not be left blank. In case of there being no such retired Gazetted Railway Officer/ Engineer, NIL information to be furnished in the format.
2. Document of permission from the President of India or any officer, duly authorized by him in this behalf, shall be submitted in case (i) where such Engineer or officer had not retired from government service at least 01 year prior to the date of submission of the tender and where such Engineer or officer is a proprietor, partner, member or director as the case, in proprietary firm, partnership Firm/ Joint venture/registered Society/ registered firm/ LLP, company.
- 3. If the above information is not furnished, contract is liable to be dealt in accordance with provision of clause 62 of Standard General Condition of contract.**

***Format given above is for guidance only. Any certificate, containing complete/relevant required information as asked for in other format, shall be considered.***



## ANNEXURE – I

### LIST OF COURT CASES DURING PREVIOUS THREE FINANCIAL YEARS

| Sl | Name of work | Value of work | Name of Client Deptt. | Name of the Court | Date of institution of case | Relief sought from Court | Brief reason of dispute | Final/Present position of the case |
|----|--------------|---------------|-----------------------|-------------------|-----------------------------|--------------------------|-------------------------|------------------------------------|
| 1  | 2            | 3             | 4                     | 5                 | 6                           | 7                        | 8                       | 9                                  |
|    |              |               |                       |                   |                             |                          |                         |                                    |
|    |              |               |                       |                   |                             |                          |                         |                                    |
|    |              |               |                       |                   |                             |                          |                         |                                    |
|    |              |               |                       |                   |                             |                          |                         |                                    |
|    |              |               |                       |                   |                             |                          |                         |                                    |

Signature of tenderer

Along with Seal

For Chief Administrative Officer (Const)  
West Central Railway, Jabalpur.

## ANNEXURE – J

### LIST OF ARBITRATION CASES DURING PREVIOUS THREE FINANCIAL YEARS.

| Sl | Name of work | Value of work | Name of Client Deptt. | Amount and date of claim preferred | Claim of Deptt. if any | Brief reasons of disputes | Final/ Present position of the case |
|----|--------------|---------------|-----------------------|------------------------------------|------------------------|---------------------------|-------------------------------------|
| 1  | 2            | 3             | 4                     | 5                                  | 6                      | 7                         | 8                                   |
|    |              |               |                       |                                    |                        |                           |                                     |
|    |              |               |                       |                                    |                        |                           |                                     |
|    |              |               |                       |                                    |                        |                           |                                     |
|    |              |               |                       |                                    |                        |                           |                                     |

Signature of Tenderer  
Alongwith Seal

For Chief Administrative Officer (Const)  
West Central Railway, Jabalpur.

## ANNEXURE – K

### Broad Plan of Execution of This Work Within The Stipulated Completion Period

| Sl.No | Activity | Period of completion in months |
|-------|----------|--------------------------------|
|       |          |                                |
|       |          |                                |
|       |          |                                |
|       |          |                                |
|       |          |                                |
|       |          |                                |
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|       |          |                                |
|       |          |                                |
|       |          |                                |
|       |          |                                |
|       |          |                                |

Signature of Tenderer  
Alongwith Seal

For Chief Administrative Officer (Const)  
West Central Railway, Jabalpur.

## ANNEXURE-L

### MEMORANDUM OF UNDERSTANDING (MOU) FOR JOINT VENTURE PARTICIPATION

(To be executed on non-judicial stamp paper of appropriate value in accordance with relevant stamp Act. The stamp paper is to be issued in the name of the Joint Venture Firm).

#### JOINT VENTURE PARTICIPATION

##### BETWEEN

(indicate the name of member) having its registered office at (indicate the address of the member) represented by their (indicate designation/capacity e.g Manager/General manager/Director/Managing Director/ Partner/Managing Partner/Proprietor etc.), Shri. (indicate the name) aged about .....years, S/o Shri (indicate the name of father) resident of (indicate Address) (hereinafter known as authorized signatory of the member) in the capacity of Lead Member of the first part.

##### AND

(indicate the name of member) having its registered office at (indicate the address of the member) represented by their (indicate designation/capacity e.g manager/General manager/Director/Managing Director/ Partner/Managing Partner/Proprietor etc.), Shri. (indicate the name) aged about .....years, S/o Shri (indicate the name of father) resident of (indicate Address) (hereinafter known as authorized signatory of the member) in the capacity of the constituent member of the other part.

**(In case of more than two members, include the details accordingly).**

Now, the Joint Venture Firm (JV) formed by the members i.e. (indicate name of lead member) and (indicate name of constituent member) and (indicate name of other constituent member) will be known as (indicate JV firm name and address).

The expressions (indicate name of the lead member) and (indicate name of the constituent members), shall whatever the context admits, mean and include their respective legal representatives, successors-in-interest and assigns, and shall collectively be referred to as **“Joint Venture Firm”** and individually as the **“Member”**

WHEREAS; President of India, acting through Ministry of Railways, **West Central Railway Construction Organisation** (hereinafter referred to as **“Employer”**) has invited Tender for “(indicate name of work as mentioned in Notice inviting Tender)”.

NOW, THEREFORE, THE MEMBERS AGREE AS FOLLOWS:

1. The **‘Members’** have studied the documents and have agreed to submit their Tender as Joint Venture (JV) Firm.

The following documents shall be deemed to form and be read and construed as an integral part of this MOU.

- i) Notice inviting Tender,

For Chief Administrative Officer (Const)  
West Central Railway, Jabalpur.

- ii) Tender document,
  - iii) Any Addendum/Corrigendum issued by (West Central Railway Construction organization), and
  - iv) The Tender for work submitted by Joint Venture Firm through Authorized member.
2. (a) (indicate the name of the Lead Member) shall be the **“Lead member”** of the JV Firm, for all intents and purposes having majority share (at least 51%) in Joint Venture Firm having fulfilled desired Technical & Financial Eligibility Criteria for this work.
- (b) (indicate the name of the other Constituent Members as applicable for JV upto 03 members) shall be the **“member”** of the JV Firm, for all intents and purposes having majority share (at least 20%) in Joint Venture Firm having fulfilled desired Technical & Financial Eligibility Criteria for this work.
- (c) (indicate the name of the other Constituent Members as applicable for JV more than 03 members) shall be the **“member”** of the JV Firm, for all intents and purposes having majority share (at least 10%) in Joint Venture Firm having fulfilled desired Technical & Financial Eligibility Criteria for this work.
3. The ‘Members’ resolve that the distribution of share of Financial, Technical, and other responsibilities of the constituent members of JV Firm is as under:

|                                                                      |             |
|----------------------------------------------------------------------|-------------|
| (a) <u>(indicate name &amp; Address of Lead Member)</u>              | Share ....% |
| Lead Member                                                          |             |
| (b) <u>(indicate name &amp; Address of Constituent Member)</u>       | Share ....% |
| Constituent Member                                                   |             |
| (c) <u>(indicate name &amp; Address of Other Constituent Member)</u> | Share ....% |
| Other Constituent Member                                             |             |

#### 4. **JOINT AND SEVERAL RESPONSIBILITIES**

The Members undertake that they shall be jointly and severally liable to the Employer (Railways) for execution of the work in accordance with General and Special Conditions of Contract. The JV members shall also be liable jointly and severally for the loss, damages caused to the Employer (Railways) during the course of execution of the contract or due to non-execution of the contract or part thereof.

#### 5. **ASSIGNMENT AND THIRD PARTIES**

No member of the Joint Venture firm shall have the right to assign or transfer the interest, right or liability in the contract without the written consent of the other members and that of the Employer (Railways) in respect of the said tender/contract.

6. **AUTHORIZED MEMBER**

We, authorize (*indicate lead member of JV firm nominated as authorized member*), as **Authorized member** represented by their authorized signatory Shri (*indicate the name*) to act on behalf of the Joint Venture Firm to deal with the tender, sign the agreement or enter into contract in respect of the said tender/contract, to receive payment, to witness joint measurement of work done, to sign measurement books and similar such action in respect of the said tender/ contract. However, authorized member shall not submit any such proposals, clarifications or commitments without securing clearance of other JV constituents(s).

All notices/correspondences with respect to the tender/contract would be sent by Employer (Railways) **only to the authorized signatory of Authorized member at the address of JV firm**. All such notices/correspondences sent by Employer shall be legally binding on all the members of the J.V Firm.

7. **GUARANTEES AND BONDS**

Earnest Money Deposit and all bonds/guarantees to the Employer (Railways) shall be submitted in the name JV Firm, which shall be legally binding on all the members of the J.V Firm.

8. **INDEMNITY**

Each member hereto agrees to indemnify the other members against its respective parts in case of breach/default of the respective part of the contract/tender of any liabilities sustained by the Joint Venture Firm.

9. For the execution of the respective portions of works, the members shall make their own arrangements to bring the required finance, plants and equipment, materials, manpower and other resources.

10. **DOCUMENTS & CONFIDENTIALITY.**

Each member shall maintain confidentiality regarding the information related to the tender/contract, commercial and technical information received or generated in the course of preparation and submission of the Tender.

11. **ARBITRATION**

Any dispute, controversy or claim arising out of or relating to this agreement shall be settled in the first instance amicably between the members. If an amicable settlement cannot be reached as above, it will be settled by arbitration in accordance with the Indian Arbitration and Conciliation Act 1996 or any amendments thereof. The Venue of the arbitration shall be (*indicate the name of place*). Notwithstanding settlement of any dispute among the members of the JV Firm , the JV members shall ensure satisfactory completion of awarded work as per the contract with the Employer (Railways).

12. **VALIDITY**

This MOU shall remain in force till the occurrence of the earliest of the following, unless by mutual consent, the members agree in writing to extend the validity for a further period

- a. Tender submitted by the joint venture Firm is declared unsuccessful, or
- b. Cancellation/shelving of the work by the Employer (Railways) for any reasons prior to award of work.

In case, the Tender submitted by the joint venture Firm is declared successful; the validity of this MOU shall be upto the entire period of completion (inclusive of period of extension, if any) including maintenance period.

13. This MOU is drawn in .....number of copies with equal legal strength and status. One copy is held by (indicate name of lead member) and the other by (indicate name of constituent member) & (indicate name of other constituent member) and one copy submitted with the tender to Employer (Railways).
14. This MOU shall in all respects be governed by and interpreted in accordance with Indian Laws.

15. **NOTICES/CORRESPONDANCE**

All Notices/Correspondence by Employer (Railways) in writing shall be sent by Fax confirmed, by registered post or commercial courier to the following Fax number and address. (Indicate name of authorized signatory of authorized member, name & address of JV Firm with Fax number)

16. **JV Agreement.**

We the constituent member of the JV Firm are aware that after issue of Letter of Acceptance by the Employer (Railways), an agreement among the members of the JV firm has to be executed and got registered before the Registrar of Companies under the Companies Act or with the Registrar/Sub-Registrar under the Registration Act. This agreement shall be submitted by the JV firm to the Employer (Railways) as per Standard J.V. Agreement format before signing the contract agreement for the work. In case the JV members fail to observe/comply with stipulations, the Earnest Money Deposit, deposited with the Employer (Railways) shall be forfeited and other penal actions due shall be taken against the members and the JV Firm by the Employer (Railways).

17. We, the members of JV Firm shall furnish along with the tender, requisite documents as mentioned under para 17.1, 17.2, 17.3 (as the case may be) of the Guidelines for Participation of J.V. firms in works tender.

18. **Declaration**

It is certified that we are not blacklisted or debarred by the Railways or any other Ministries/Departments of the Govt. of India from participation in tender/contracts on the date of submission of Bids either in our individual capacity or as a member of the JV Firm in which we were/are members.

IN WITNESS WHEREOF THE MEMBERS, have executed this MOU on (indicate day, month & year).

|                                                                                                               |                                                                                                                               |
|---------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| <p>(Indicate name of authorized signatory)<br/>(indicate name &amp; address of lead member)</p> <p>(Seal)</p> | <p>(Indicate name of authorized signatory)<br/><u>(Indicate name &amp; address of constituent member's)</u></p> <p>(Seal)</p> |
|---------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|

**Witness:**

1(Name & Address)

2(Name & Address)

For Chief Administrative Officer (Const)  
West Central Railway, Jabalpur.

## ANNEXURE-M

### JOINT VENTURE AGREEMENT FOR JOINT VENTURE PARTICIPATION

(The J.V. Agreement to be executed on non-judicial stamp paper of appropriate value, in accordance with relevant stamp Act. The stamp paper is to be issued in the name of the Joint Venture Firm and should not be more than six months old from the date of execution of the J.V. Agreement, on it)

### JOINT VENTURE AGREEMENT

#### BETWEEN

(indicate the name of member) having its registered office at (indicate the address of the member) represented by their (indicate designation/capacity e.g Manager/General manager/Director/Managing Director/ Partner/Managing Partner/Proprietor etc.), Shri. (indicate the name) aged about ..... years, S/o Shri (indicate the name of father) resident of (indicate Address) (hereinafter known as authorized signatory of the member) in the capacity of Lead Member of the first part.

#### And

(indicate the name of member) having its registered office at (indicate the address of the member) represented by their (indicate designation/capacity e.g manager/General manager/Director/Managing Director/ Partner/Managing Partner/Proprietor etc.), Shri. (indicate the name) aged about ..... years, S/o Shri (indicate the name of father) resident of (indicate Address) (hereinafter known as authorized signatory of the member) in the capacity of the constituent member of the other part.

**(In case of more than two members, include the details accordingly).**

Now, the Joint Venture Firm (JV) formed by the members i.e. (indicate name of lead member) and (indicate name of constituent member) and (indicate name of other constituent member) will be known as (indicate JV firm name and address).

The expressions (indicate name of the lead member) and (indicate name of the constituent members), shall whatever the context admits, mean and include their respective legal representatives, successors-in-interest and assigns, and shall collectively be referred to as **“Joint Venture Firm”** and individually as the **“Member”**

1. WHEREAS; President of India, acting through Ministry of Railways, **West Central Railway Construction Organisation** (hereinafter referred to as **“Employer”**) has invited Tender for “(indicate name of work as mentioned in Notice inviting Tender)”.

And Whereas, the above members to the Joint Venture Firm have submitted the tender in the name of the said JV Firm formed as per the MOU signed on dated... and whereas the said tender has finally been accepted by the Employer (Railways) vide Letter Of Acceptance No.....dated ....., we (indicate name of the lead member) and (indicate name of the constituent members), herewith sign the above formal JV agreement for registration of the above joint venture Firm viz (indicate JV firm name and address) and for entering into contract Agreement with the “Employer” (Railway).

2. NOW, THEREFORE, THE PARTIES AGREE AS FOLLOWS:

The following documents shall be deemed to form and be read and construed as an integral part of this J.V. Agreement.

For Chief Administrative Officer (Const)  
West Central Railway, Jabalpur.



- i) Notice inviting Tender,
  - ii) Tender document,
  - iii) Any Addendum/Corrigendum issued by (West Central Railway),
  - iv) MOU signed on ..... by us.
  - iv) Tender submitted on our behalf by the Authorized Member.
  - v) Letter Of Acceptance issued by Employer (Railways).
3. The 'Members' have studied the documents, JV guidelines and have agreed to participate in submitting the 'Tender' jointly; signed the JV MOU and submitted the tender accordingly.
4. (a) (indicate the name of the Lead Member) shall be the **"Lead member"** of the JV Firm, for all intents and purposes having majority share (at least 51%) in Joint Venture Firm having fulfilled desired Technical & Financial Eligibility Criteria for this work.
- (b) (indicate the name of the other Constituent Members as applicable for JV upto 03 members) shall be the **"member"** of the JV Firm, for all intents and purposes having majority share (at least 20%) in Joint Venture Firm having fulfilled desired Technical & Financial Eligibility Criteria for this work.
- (c) (indicate the name of the other Constituent Members as applicable for JV more than 03 members) shall be the **"member"** of the JV Firm, for all intents and purposes having majority share (at least 10%) in Joint Venture Firm having fulfilled desired Technical & Financial Eligibility Criteria for this work.
5. We, authorize (indicate lead member of JV firm nominated as authorized member), as **Authorized member** represented by their authorized signatory Shri. (indicate the name) to act on behalf of the Joint Venture Firm to deal with the tender, sign the agreement or enter into contract in respect of the said tender/contract, to receive payment, to witness joint measurement of work done, to sign measurement books and similar such action in respect of the said tender/ contract. However, authorized member shall not submit any such proposals, clarifications or commitments without securing clearance of other JV constituents(s).

All notices/correspondences with respect to the tender/contract would be sent by Employer (Railways) **only to the authorized signatory of Authorized member at the address of JV firm**. All such notices/correspondences sent by Employer shall be legally binding on all the members of the J.V Firm.

6. The 'Members' resolve that the distribution of share of Financial, Technical, and other responsibilities of the constituent members of JV Firm is as under:

|                                                                      |             |
|----------------------------------------------------------------------|-------------|
| (a) <u>(indicate name &amp; Address of Lead Member)</u>              | Share ....% |
| Lead Member                                                          |             |
| (b) <u>(indicate name &amp; Address of Constituent Member)</u>       | Share ....% |
| Constituent Member                                                   |             |
| (c) <u>(indicate name &amp; Address of Other Constituent Member)</u> | Share ....% |
| Other Constituent Member                                             |             |

For Chief Administrative Officer (Const)  
West Central Railway, Jabalpur.

7. The constitution of JV firm shall not be altered during the currency of the contract except when modification become inevitable due to succession laws etc. provided that the minimum eligibility criteria are not got vitiated. Failure to observe this stipulation shall be deemed to be breach of contract, which will entitle the Employer (Railway) to take all consequential action as per contract conditions.

**8. JOINT AND SEVERAL RESPONSIBILITY**

The Members undertake that they shall be jointly and severally liable to the Employer (Railways) for execution of the work in accordance with General and Special Conditions of Contract. The JV members shall also be liable jointly and severally for the loss, damages caused to the Employer (Railways) during the course of execution of the contract or due to non-execution of the contract or part thereof. The members solemnly affirm and declare that every possible care will be taken by them for ensuring satisfactory execution and completion of the work awarded under the contract.

**9. ASSIGNMENT AND THIRD PARTIES**

No member of the Joint Venture firm shall have the right to assign or transfer the interest, right or liability in the contract without the written consent of the other members and that of the Employer (Railways) in respect of the said tender/contract.

**10. GUARANTEES AND BONDS**

All bonds/guarantees e.g Performance Guarantee, Bank Guarantee etc. to the Employer (Railways) shall be submitted by the JV Firm as per tender conditions, only in the name of J.V Firm.

**11. INDEMNITY**

Each member hereto agrees to indemnify the other members against its respective parts in case of breach/default of the respective part of the contract/tender of any liabilities sustained by the Joint Venture Firm.

**12. USE OF MACHINERY, INSTRUMENT, LABOUR FORCE etc.**

For the execution of the respective portions of works, the members shall make full arrangements to bring the required finance, plants and equipment, materials, manpower and other resources. However, the members here to undertake that whatever the machinery, instruments, labour force, (including unskilled, skilled, inspectors, Engineer etc.) they possess at the time of entering into Joint Venture Agreement or which subsequently shall come in their possession and if such machinery, instruments, labour force is required for the speedy and efficient execution of any portion of the work, the member/members having the control over the said machinery, instruments, labour force etc. without having any regard to their share of profit and loss agreed to between the members in Joint Venture Agreement shall hand over the same which shall be placed at the disposal of the other member actually executing that portion of the work at mutually agreed terms for the purpose of execution of the contract without any hindrance and obstacle

**13. DOCUMENTS & CONFIDENTIALITY.**

Each member shall maintain confidentiality regarding the information related to the tender/contract, commercial and technical information received or generated in the course of preparation and submission of the Tender and execution of contract.

**14. ARBITRATION**

Any dispute, controversy or claim arising out of or relating to this agreement shall be settled in the first instance amicably between the members. If an amicable settlement cannot be reached as above, it will be settled by arbitration in accordance with the Indian Arbitration and Conciliation Act 1996 or any amendments thereof. The Venue of the arbitration shall be (indicate the name of place). Notwithstanding settlement of any dispute among the members of the JV Firm, the JV members shall ensure satisfactory completion of awarded work as per the contract with the Employer (Railways).

**15. DURATION OF JOINT VENTURE AGREEMENT**

It shall be valid during the entire currency of the contract including the period of extension if any and the maintenance period after the work is completed and till Security Deposit is released.

**16. NOTICES/CORRESPONDANCE**

All Notices/Correspondence by Employer (Railways) in writing shall be sent by Fax confirmed, by registered post or commercial courier to the following Fax number and address.

(Indicate name of authorized signatory of authorized member, name & address of JV Firm with Fax number)

- 17. Governing Laws** : The J.V. Agreement shall in all respect be governed by and interpreted in accordance with Indian Laws.

**18. Declaration:-**

It is certified that we are not blacklisted or debarred by the Railways or any other Ministries/Departments of the Govt. of India from participation in tender/contracts on the date of submission of Bids either in our individual capacity or as a member of the JV Firm in which we were/are members.

IN WITNESS WHEREOF THE MEMBERS, have executed this JV Agreement on (indicate day, month and year)

|                                                                                                               |                                                                                                                        |
|---------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| <p>(Indicate name of authorized signatory)<br/>(indicate name &amp; address of lead member)</p> <p>(Seal)</p> | <p>(Indicate name of authorized signatory)<br/>(Indicate name &amp; address of constituent member's)</p> <p>(Seal)</p> |
|---------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|

**Witness:**

3(Name & Address)

4(Name & Address)

Place :

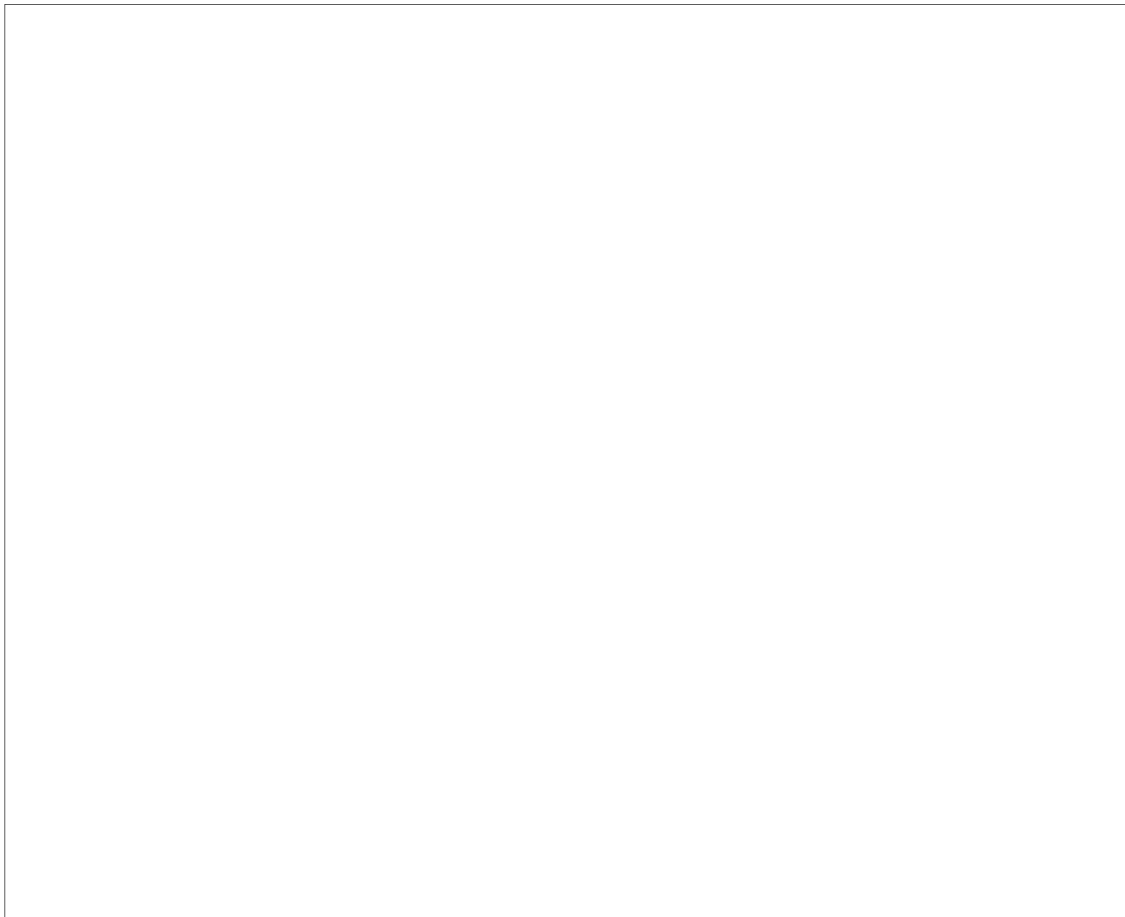
Date :

**(The J.V. Agreement should be got registered with Registrar of Company Act or with the Registrar/Sub-registrar under the registration act.)**

For Chief Administrative Officer (Const)  
West Central Railway, Jabalpur.

**ANNEXURE – N**

**OTHER CREDENTIALS/FACILITIES OF THE FIRM/CONTRACTOR**  
(WHICH ARE NOT COVERED IN ANNEXURE-A TO ANNEXURE –M)



Signature of Tenderer  
Along with Seal.

For Chief Administrative Officer (Const)  
West Central Railway, Jabalpur.

## ANNEXURE – O

### MANDATE FORM OF EFT

The Contractor/supplier having their bank accounts on any of the city centres where reserve bank of India's EFT (Electronic Fund Transfer) facility exists will receive their payments through EFT scheme in case of being awarded with the work/supply order. For that they (the agency) will have to furnish the **mandate form of EFT** (As per proforma given below)) along with the tender document.

Electronic clearing service (Credit clearing) Model Mandate form

(Investor/Customer's Option to receive payments through Credit Clearing Mechanism).

#### Name of the scheme and the periodicity of payment

No.

- 1) Contractor/supplier's Name
- 2) Particulars of their Bank Account
  - a) Name of Bank:
  - b) Name of the BranchAddress  
Telephone No.
  - c) 9 Digit Code Number of the bank and branch  
Appearing on the MICR cheque issued by the bank.
  - d) Type of the Account (S.B. Current or cash Credit)  
**With IFS Code**
  - e) Ledger and ledger folio Number.
  - f) Account number (as appearing on the cheque book)

(In lieu of the bank certificate to be obtained as under, please attach a bank cancelled cheque or photocopy of a cheque issued by your bank for verification of the above particulars)

3) Date of effect

I hereby declare that the particulars given above are correct and complete. If the transaction is delayed or not effected at all for reasons of incomplete or incorrect information, I would not hold the user institution responsible. I have read the option invitation letter and agree to discharge the responsibility expected of me as a participant under the scheme.

Signature of the Contractor/Supplier.

Date: \_\_\_\_\_

Certified that the particulars furnished above are correct as per our records.

Bank's stamp.

Date: \_\_\_\_\_

Signature of The Authorised Official of the Bank.

For Chief Administrative Officer (Const)  
West Central Railway, Jabalpur.

## ANNEXURE-P

LC/DA No(18 DIGIT IPAS GENERATED NO.)

Dated \_\_\_\_\_

### DOCUMENT OF AUTHORISATION

**Reference:** (i) Works Contract No. \_\_\_\_\_ Dated \_\_\_\_\_  
(ii) Inland Letter of Credit No. \_\_\_\_\_ Dated \_\_\_\_\_

This document is issued against Contract No. ----- (FROM IREPS) ----- dated ----- for work of -----  
(DESCRIPTION OF WORK FROM IREPS)-----

The beneficiary of aforementioned Letter of Credit M/s ..... (NAME AND VENDOR CODE)  
..... (Vendor Code ..... as per IREPS.....) is entitled to receive payment aggregating INR..... SSS ..... (FROM  
ABSTRACT OF BILL PASSED)..... out of a total LC amount of INR..... (FROM MASTER TABLE OF LC  
OPENED)..... against the first/second\* commercial Invoice No. (FROM IPAS)..... dated ..... FROM  
IPAS..... for INR (FROM IPAS)..... raised against the above contract from State Bank of India ..... (branch.....  
FROM LC MASTER TABLE).... on the strength of this Certificate.

The details of payments already made to the beneficiary under this Letter of Credit are as follows:

| Sl. No.    | Invoice No. | Invoice date | Invoice Amount (INR) | LCDA No. | LCDA date | Amount paid (INR) |
|------------|-------------|--------------|----------------------|----------|-----------|-------------------|
|            |             |              |                      |          |           |                   |
|            |             |              |                      |          |           |                   |
|            |             |              |                      |          |           |                   |
| Total Paid |             |              |                      |          |           |                   |

THIS PAYMENT \_\_\_\_\_ SSS \_\_\_\_\_

LC BALANCE AFTER THIS PAYMENT: \_\_\_\_\_

(Signature of authorised Railway authority)

Name

Designation

Official Seal

For Chief Administrative Officer (Const)  
West Central Railway, Jabalpur.

## ANNEXURE-Q

### SPECIMEN FORMAT OF BANK GUARANTEE BOND FOR SUBMISSION OF BID SECURITY

Bank Guarantee Bond from any scheduled commercial bank of India

(On non-judicial stamp paper, which should be in the name of the Executing Bank).

Name of the Bank: -----

President of India,

Acting through.....,

West Central Railway,

Beneficiary: FA & CAO (Con), West Central Railway

Bank Guarantee Bond No.:

Date:-----

In consideration of the President of India acting through ---- (***Designation & address of Contract Signing Authority***), ..... Railway, ....., .... (hereinafter called "The Railway") having invited the bid for \_\_\_\_\_ through Notice inviting tender (NIT) No..\_\_\_\_\_, We have been informed that . . . . ***[Insert name of the Bidder]***..... (hereinafter called "the Bidder") intends to submit its bid (hereinafter called "the Bid") .

WHEREAS, the Bidder is required to furnish Bid Security for the sum of ***[Insert required Value of Bid Security]***, in the form of Bank Guarantee, according to conditions of Bid.

AND

WHEREAS, .....***[Insert Name of the Bank]***, with its Branch .....***[Insert Address]*** having its Headquarters office at..... ***[Insert Address]***, hereinafter called the Bank, acting through .....***[Insert Name and Designation of the authorised persons of the Bank]***, have, at the request of the Bidder, agreed to give guarantee for Bid Security as hereinafter contained, in favour of the Railway:

1. KNOW ALL MEN that by these present that I/We the undersigned ***[Insert name(s) of authorized representatives of the Bank]***, being fully authorized to sign and incur obligations for and on behalf of the Bank, confirm that the Bank, hereby, unconditionally and irrevocably guarantee to pay to the Railway full amount in the sum of ***[Insert required Value of Bid Security]*** as above stated.
2. The Bank undertakes to immediately pay on presentation of demand by the Railway any amount up to and including aforementioned full amount without any demur, reservation or recourse. Any such demand made by the Railway on the Bank shall be final, conclusive and binding, absolute and unequivocal on the Bank notwithstanding any disputes raised/ pending before any Court, Tribunal, Arbitration or any Authority or any threatened litigation by the Bidder or Bank.
3. The Bank shall pay the amount as demanded immediately on presentation of the demand by Railway without any reference to the Bidder and without the Railway being required to show grounds or give reasons for its demand of the amount so demanded.
4. The guarantee hereinbefore shall not be affected by any change in the constitution of the Bank or in the constitution of the Bidder.
5. The Bank agrees that no change, addition, modifications to the terms of the Bid document or to any documents, which have been or may be made between the Railway and the Bidder, will in any way absolve the Bank from the liability under this guarantee; and the Bank, hereby, waives any requirement for notice of any such change, addition or modification made by Railway at any time.

For Chief Administrative Officer (Const)  
West Central Railway, Jabalpur.

6. This guarantee will remain valid and effective from.....*[insert date of issue]*till .....*[insert date, which should be minimum 90 days beyond the expiry of validity of Bid]*. Any demand in respect of this Guarantee should reach the Bank within the validity period of Bid Security.
7. The Bank Guarantee is unconditional and irrevocable.
8. The expressions Bank and Railway herein before used shall include their respective successors and assigns.
9. The Bank hereby undertakes not to revoke the guarantee during its currency, except with the previous consent in writing of the Railway. This guarantee is subject to the Uniform Rules for Demand Guarantees, ICC Publication No.758.
10. The Bank hereby confirms that it is on the SFMS (Structured Financial Messaging System) and shall invariably send the advice of this Bank Guarantee to the following bank details –

|             |                                     |
|-------------|-------------------------------------|
| IFSC CODE   | SBIN000RAIL                         |
| IFSC TYPE   | BRANCH                              |
| BANK NAME   | STATE BANK OF INDIA                 |
| BRANCH NAME | RAIL                                |
| CITY NAME   | NAVI MUMBAI                         |
| ADDRESS     | SECTOR-11, CBD BELAPUR, NAVI MUMBAI |
| DISTRICT    | NAVI MUMBAI                         |
| STATE       | MAHARASHTRA                         |
| BG ENABLED  | BG ENABLED                          |

11. The Guarantee shall be valid in addition to and without prejudice to any other security Guarantee(s) of Bidder in favour of the Railway. The Bank, under this Guarantee, shall be deemed as Principal Debtor of the Railway.

Date.....

Place.....

Bank's Seal and authorized signature(s)

[Name in Block letters] .....

[Designation with Code No.].....

[P/Attorney] No.

Witness:

1 Signature, Name & Address & Seal

2 Signature, Name& address & Seal

Bank's Seal

[P/Attorney] No.

**Note:** All italicized text is for guidance on how to prepare this bank guarantee and shall be deleted from the final document.



## ANNEXURE-R

### SPECIMEN FORMAT OF BANK GUARANTEE BOND FOR PERFORMANCE GUARANTEE

(Bank Guarantee Bond from any scheduled commercial bank of India)

(On non-judicial stamp paper, which should be in the name of the Executing Bank).

Name of the Bank: -----

President of India,

Acting through.....,

West Central Railway,

Beneficiary: FA & CAO (Con), West Central Railway

Bank Guarantee Bond No.:

Date:-----

In consideration of the President of India (herein after called the Government ) having agreed to exempt (indicate name and address of tenderer) (hereinafter called the said Contractor(s)) from the demand, under the terms and conditions of an Agreement (indicate acceptance letter no. and date) made between (indicate Dy. CE/Con or CE/Con. ,West Central Railway as applicable) and (indicate name and address of tenderer) for (indicate the name of work as mentioned in tender) (hereinafter called “the said Agreement”), of security deposit for the due fulfillment by the said Contractor(s) of the terms and conditions contained in the said Agreement, on production of a bank Guarantee for Rs.-----  
----- (Rupees\_\_\_\_\_only).

1. We, (indicate the name and address of the bank) (hereinafter referred to as “the Bank”) at the request of (indicate name and address of tenderer) (contractor’s) do hereby undertake to pay to the Government an amount not exceeding Rs. \_\_\_\_\_ against any loss or damage caused to or suffered or would be caused to or suffered by the Government by reason of any breach by the said Contractor(s) of any of the terms or conditions contained in the said Agreement.
2. We (indicate the name and address of the bank) do hereby undertake to pay the amounts due and payable under this guarantee without any demur, merely on a demand from the Government stating that the amount/claimed is due by way of loss or damage caused to or would be caused to or suffered by the Government by reason of breach by the said contractor(s) of any of the terms or conditions contained in the said Agreement or by reason of the Contractor(s) failure to perform the said Agreement. Any such demand made on the bank shall be conclusive as regards the amount due and payable by the Bank under this guarantee. However, our liability under this guarantee shall be restricted to an amount not exceeding Rs. \_\_\_\_\_.
3. We undertake to pay to the Government any money so demanded notwithstanding any dispute or disputes raised by the Contractor(s)/supplier(s) in any suite or proceeding pending before any court or Tribunal relating thereto our liability under this present being absolute and unequivocal.

The payment so made by us under this bond shall be a valid discharge of our liability for payment thereunder and the contractor(s) /supplier(s) shall have no claim against us for making such payment.

4. We, (indicate the name and address of the bank) further agree that the guarantee herein contained shall remain in full force and effect during the period that would be taken for the performance/of the said Agreement and that it shall continue to be enforceable till all the dues of the Government under or by virtue of the said Agreement have been fully paid and its claims satisfied or discharged or till (indicate Dy. CE/Con or CE/Con. ,West Central Railway as applicable) Office/ Department) Ministry of Railway certifies that the terms and conditions of the said Agreement have been fully and properly carried out by the said Contractor(s) and accordingly discharges this guarantee. Unless a demand or claim under this guarantee is made on us in writing on or before the \_\_\_\_\_ we shall be discharged from all liability under this guarantee thereafter.

For Chief Administrative Officer (Const)  
West Central Railway, Jabalpur.

5. We (indicate the name and address of the bank) further agree with the government that the Government shall have the fullest liberty without our consent and without affecting in any manner our obligations hereunder to vary any of the terms and conditions of the said Agreement or to extend time of performance by the said Contractor(s) from time to time or to postpone for any time or from time to time any of the powers exercisable by the Government against the said Contractor(s) and forbear or enforce any of the terms and conditions relating to the said agreement and we shall not be relieved from our liability by reason of any such variation, or extension being granted to the said Contractor(s) or for any forbearance, act or omission on the part of the Government or any indulgence by the Government to the said Contractor(s) or any such matter or thing whatsoever which under the law relating to sureties would, but for this provision, have effect of so relieving us.
6. This guarantee will not be discharged due to the change in the constitution of the Bank or the Contractor(s)/Supplier(s).
7. We, (indicate the name and address of the bank) lastly undertake not to revoke this guarantee during its currency except with the previous consent of the Government in writing.

Dated the \_\_\_\_\_ day of \_\_\_\_\_ 20....

For (indicate the name and address of the bank)

**CHAPTER-7**  
**ANNEXURE-1**

**WEST CENTRAL RAILWAY CONSTRUCTION ORGANISATION**  
**SPECIMEN FORMAT FOR LETTER OF ACCEPTANCE**  
**(IN CASE OF MANNUAL MODE LOAs)**

Without Prejudice  
**By Regd. Post with A.D**  
**By Special Man**

**WEST CENTRAL RAILWAY**

Office of the  
Chief Administrative Officer (Con)  
Jabalpur

No: o/o ....

Dtd:

To

-----

-----

Dear Sir(s),

**Sub :- (Name of work).**

**Ref:- E-Tender Notice No.-----**

\*\*\*\*\*

01. With references to above, your offer has been accepted by the Railway Administration at a total value of Rs. --  
----/- (Rupees ----- only) with Contract Period of -- months from the date of issue of this Acceptance Letter at the following rates shown against each item under Schedules-A,B,C as enclosed to this letter.

**Schedule-'A ' ----- % accepted rate in Above/below/at par**

**Schedule-'B ' ----- % accepted rate in Above/below/at par**

**Schedule-'C ' : ----- % accepted rate in Above/below/at par**

..... : .....

Other terms and conditions as incorporated in the tender documents issued for the work will be included in a formal agreement to be executed by you for carrying out the work.

02. An amount of Rs ----/- (Rupees----only) is to be deposited towards Security Deposit.

The EMD submitted on line vide IREPS Ref Id/Bank Trans Id No: ---- dtd. ----- for Rs----- /-(Rupees -  
---

--- only) is adjusted towards Security Deposit. Balance amount of Security Deposit i.e., Rs ----- /-(Rupees  
---

----- only) is to be deposited in cash with FA & CAO (Con) WCR Railway/Jabalpur or otherwise if you so desire amount of balance Security Deposit will be recovered at the rate of 6% from your running bills of the contract.

03. You are requested to give a performance guarantee in any acceptable form as mentioned in the tender document, for an amount of **Rs ----- ( ----- )** before signing of the agreement and it should be valid upto the stipulated date of completion plus 60 days. If the performance guarantee is in the form of B.G then it should be as per standard B.G format (enclosed). **"Bank Guarantees(BGs) to be submitted by suppliers/contractors should be sent directly to the concerned authorities by the issuing bank under Registered post. A.D.**

For Chief Administrative Officer (Const)  
West Central Railway, Jabalpur.

04. Until a formal Agreement is prepared and executed, acceptance of this Tender shall constitute a binding contract. As such you are requested to submit a Performance Guarantee (PG) within 21 (twenty one) days from the date of issue of Letter of Acceptance (LOA). Extension of time for submission of PG beyond 21 (twenty one) days and upto 60 days from the date of issue of LOA may be given by the Authority who is competent to sign the contract agreement. However, a penal interest of 12% per annum shall be charged for the delay beyond 21 (twenty one) days, i.e. from 22nd day after the date of issue of LOA. Further, if the 60th day happens to be a declared holiday in the concerned office of the Railway, submission of PG can be accepted on the next working day.

In all other cases, if the Contractor fails to submit the requisite PG even after 60 days from the date of issue of LOA, the contract is liable to be terminated. In case contract is terminated railway shall be entitled to forfeit Earnest Money Deposit and other dues payable against that contract. In case a tenderer has not submitted Earnest Money Deposit on the strength of their registration as a Startup recognized by Department of Industrial Policy and Promotion (DIPP) under Ministry of Commerce and Industry, DIPP shall be informed to this effect. The failed Contractor shall be debarred from participating in re-tender for that work.

05. Please contact Dy. CE(Con)/ .....immediately with your execution programme for further instructions to start the work on the authority of this Acceptance Letter.
06. Please acknowledge receipt of this letter of Acceptance and confirm that you are taking action as stated above.

Yours faithfully,  
Chief Engineer (Con)/-----,  
W. C. Railway/ Jabalpur  
For & on behalf of President of India.

No: CE/CON/....

Dtd.

Copy with copy of enclosures forwarded for information to: -

\*\*\*\*\*

## ANNEXURE - 2

### WEST CENTRAL RAILWAY (CONSTRUCTION ORGANISATION) CONTRACT AGREEMENT OF WORKS

|    |                                                                                                  |  |
|----|--------------------------------------------------------------------------------------------------|--|
| 1. | Contract Agreement No.                                                                           |  |
| 2. | Approximate value of contract (in figures & words)                                               |  |
| 3. | Total Security Deposit (in figures & words)                                                      |  |
| 4. | Initial SD deposited by the Contractor (details of the deposit with amount in figures and words) |  |
| 5. | Balance SD to be recovered from the contractor (in figures and words)                            |  |
| 6. | Performance guarantee & Penal interest, if any deposited.                                        |  |
| 7. | Date of completion of work                                                                       |  |
| 8. | Name of work                                                                                     |  |
| 9. | Estimate No, Allocation with Plan Head of expenditure.                                           |  |

ARTICLES OF AGREEMENT made this \_\_\_\_\_ day of \_\_\_\_\_ 20\_\_\_\_ between President of India acting through the \_\_\_\_\_ of West Central Railway Administration hereafter called the "Railway" of the one part and \_\_\_\_\_ herein after called the "Contractor" of other part.

WHEREAS the Contractor has agreed with the Railway for performance of the above works set forth in the Bills of Quantities hereto annexed upon the Standard General Conditions of Contract, updated with correction slips issued up to date of inviting tender or otherwise specified in the tender documents and the Specifications of Indian Railways Unified Standard Specifications (Works & Materials)/Specifications of CPWD DSR-2019 updated with correction slips issued up to date of inviting tender or as otherwise specified in the tender documents and the applicable Standard Schedule of Rates (SSOR) of West Central Railway updated with correction slips issued up to date of inviting tender or as otherwise specified in the tender documents and the Special Conditions and Special Specifications, if any and in conformity with the drawings here-into annexed AND WHEREAS the performance of the said works is an act in which the public are interested.

For Chief Administrative Officer (Const)  
West Central Railway, Jabalpur.

NOW THIS INDENTURE WITNESSETH that in consideration to the payments to be made by the Railways, the Contractors will duly perform the said works in the said Bills of Quantities set forth and shall execute the same with great promptness, care and accuracy in a workman like manner to the satisfaction of the Railway and will complete the same in accordance with the said specifications and said drawings and said conditions of contract on or before the..... day of..... 20.. and will maintain the said works for a period as mentioned in the tender document from the certified date of their completion and will observe, fulfill and keep all the conditions therein mentioned (which shall be deemed and taken to be part of this contract, as if the same have been fully set forth herein), AND the Railway, both hereby agree that if the Contractor shall duly perform the said works in the manner aforesaid and observe and keep the said terms and conditions, the Railway will pay or cause to be paid to the Contractor for the said works on the final completion thereof the amount due in respect thereof at the rates specified in the Bills of Quantities hereto annexed.

Contractor------(Signature)  
 Address-----  
 Date-----

Railway: Designation -----  
 (For President of India)  
 Date-----

Signature of Witnesses (to Signature of Contractor) with address:

Witnesses: .....  
 .....

-----END OF THE DOCUMENT-----



Government of India  
Ministry Of Railways  
(Railway Board)

**Indian Railways**  
**Standard**  
**General Conditions of Contract**

**GCC April 2022**

(Updated up to ACS-06 Date 21.12.2023)

New Delhi

**Engineering Department**



भारत सरकार Government of India  
रेल मंत्रालय Ministry of Railways  
(रेलवे बोर्ड Railway Board)



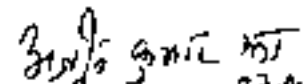
No.2022/CE-I/CT/GCC-2022/POLICY

New Delhi, Dated 27.04.2022

As per list attached

**Sub:- Indian Railways Standard General Conditions of Contract, April-2022**

1. Railway Board had issued Indian Railways Standard General Conditions of Contract, July 2020 vide letter No. 2020/CE-I/CT/3E/GCC/Policy, dated 16.07.2020.
2. On receipt of a number of suggestions from Railways, the latest version of Indian Railways Standard General Conditions of contract (GCC), issued on 16.07.2020, was examined and reviewed to expedite finalization of tenders and execution of works.
3. Board (MI, MF) has approved new Indian Railways Standard General Conditions of Contract, April 2022 (GCC, April 2022).
4. The GCC, April 2022 has been uploaded on Railway Board's website. It may be accessed through the path: [www.indianrailways.gov.in/railwayboard](http://www.indianrailways.gov.in/railwayboard) >> "About Indian Railways" >> "Railway Board Directorates" >> "Civil Engineering" >> "IR General Conditions of Contracts" >> IR General Condition of Contracts- 2022.
5. The GCC, April 2022 shall be applicable to works contracts on Indian Railways with prospective effect.
6. This issues with the concurrence of Finance Directorate of Ministry of Railways.

  
(अजीत कुमार झा) 27.4.22

कार्यपालक निदेशक/सिविल इंजी.(जी),रेलवे बोर्ड

[Phone: 030-44803: Rly: 011-23383379:MTNL]

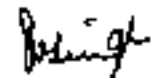
e-mail address : [edceg2022@gmail.com](mailto:edceg2022@gmail.com)

No.2022/CE-I/CT/GCC-2022/POLICY

New Delhi, Dated 27.04.2022

Copy forwarded for information to:

1. The PFAs, All Indian Railways.
2. Dy. Comptroller and Auditor General of India (Railways), Room No. 224, Rail Bhawan, New Delhi



For Member Finance



**LIST FOR DISTRIBUTION (No.2022/CE-1/CT/GCC-2022/POLICY, Dated 27.04.2022)**

1. General Managers, All Indian Railways & Production Units
2. General Manager (Com), N.F. Railway, Guwahati
3. General Manager /CORE, Allahabad
4. Chief Administrative Officers (Com), All Indian Railways (Except N.F. Railway)
5. Principal Chief Engineers, All Indian Railways
6. PFAs, All Indian Railways
7. THE Deputy Controller & Auditor General of India (Railways), Room No. 224, Rail Bhawan, New Delhi

(A)

1. CAO, COFMOW, Tilak Bridge, New Delhi
2. Principal CAO, Diesel Loco Modernization Works, Patiala (Punjab)
3. CAO (Workshop Projects), Chamber Bhawan, Judge's Court Road, Ansa Ghata, Patna-800001, Bihar

(B)

1. Director General, RDSO, Manak Nagar, Lucknow
2. Director General, NAIK, Varanasi
3. Director, IRICEN, Pune - 411 001 (Maharashtra)
4. Director, IRIFEN, PB No -233, Nasik Road-422101 (Maharashtra)
5. Director, IRISER, Taa Naka Road, Lalla Guda, Secunderabad-500017
6. Director, IRIMEE, Jamalpur Dist-Jamalpur, PIN-811214 (Bihar)
7. Director, IRITM, Sarwati Residential Estate, IRITM Campus, Manak Nagar, Lucknow

(C)

1. MD, CONCOR, Concor Bhawan, C-3, Mathura Road, Opp. Apollo Hospital, New Delhi- 110076
2. IRCON International Limited, C-4, District Centre, Saket, New Delhi- 110017
3. MD, RITES Ltd., RITES Bhawan, Plot No.1, Sector-29, Gurgaon.
4. CMD, RVNL, August Kranti Bhawan, Plot No.25, 1<sup>st</sup> Floor, Bhikaji Cama Place, New Delhi.
5. MD, MRVC Ltd., Churchgate Station Bldg, Mumbai- 400020
6. CMD, KONKAN Railway Corporation Ltd, Rail Bhawan, New Delhi-110001.
7. MD, DFCCIL, 5<sup>th</sup> Floor, Pragati Maidan Metro Station Building Complex, New Delhi-110001
8. MD, RLDA, Near Sadarjung Railway Station, Moti Bagh, Phase-I, New Delhi- 110021
9. MD, CRIS, Chanakyapur, New Delhi.
10. CMD, RailTel Corporation of India Ltd Plot No. 143, Institutional Area, Sector-44, Gurgaon - 122003
11. CME, IRCAF, 12<sup>th</sup> Floor, Core-1, Scope Minar, Dist. Centre, Laxmi Nagar, Delhi- 110092
12. Managing Director, IRFC Limited, UG Floor, East Tower, NBCC Place, Bhisham Pitamah Marg, Lodhi Road, Pragati Vihar, New Delhi.
13. CMD, IRCTC Ltd., B-148, 11th Floor, Stateman House, Barakhamba Road, New Delhi 110001
14. CMD, Braithwaite & Co. limited, 5 Hyde Road Kolkata 700043.

**Copy to:**

(A)

1. General Secretary, IRCA, DRM Office, New Delhi.
2. General Secretary, AIRF, Rail Bhawan, New Delhi
3. General Secretary, NRIK, Rail Bhawan, New Delhi
4. General Secretary, IRPOF, Rail Bhawan, New Delhi
5. General Secretary, FROA, Rail Bhawan, New Delhi
6. General Secretary, AIRPA, Rail Bhawan, New Delhi
7. General Secretary, AISC & STREA, Rail Bhawan, New Delhi
8. The Secretary, RBSS, Group (A) Officers Association, Rail Bhawan
9. The Secretary, RBSS, Group (B) Officers Association, Rail Bhawan
10. General Secretary, RBSS Association, Rail Bhawan
11. The Secretary, RBMSA, Rail Bhawan
12. The Secretary, Railway, Group (D) Employees Association, Rail Bhawan

(B)

1. Concerned PSO for kind information of Chairman cum CEO, M/Infra, M/T&RS M/O&BD, M/Finance, Railway Board
2. Chief Vigilance Officers, All Indian Railways.
3. DG(RIS), DG(RPF), AM(CE), AM(Works), AM(B), AM(Elec.), AM(RS), AM(ME), AM(Tele), AM(C&IS), AM(Sig.), AM(Pig.), PED(B&S), PED(Vigilance), PED(CE(P), PED(CE(G), EDTK(M&MC), EDCE(B&S), EDF(X)-I, EDF(X)-II, ED(Works), EDW(Pig.), ED/Project(Mon.), ED(L&A), ED(PSU), EDVE, ED(Safety), ED(Sig. Dev.), ED(Tele), EDRS(G), EDRE, EDER(G), EDFF, EDE(N), ED(Accounts), ED/T&MPP, EDMC(Chg.), EDMC(Frt.), ED/Pig., JS(conf), JS(P), JS(G), JS(D), Vigilance-III, Vig(Conf) of Railway Board.

## Indian Railways

# STANDARD GENERAL CONDITIONS OF CONTRACT I N D E X

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## PART I

### Instructions to Tenderers (ITT)

**1.0 Applicability:** These instructions and conditions of contract shall be applicable for all the tenders and contracts of railways for execution of 'Works' as defined in GFR 2017.

**1.01 Order of Precedence of Documents:** In a contract agreement, in case of any difference, contradiction, discrepancy, with regard to conditions of tender/contract, specifications, drawings, Bill(s) of Quantities etc., forming part of the tender/contract, the following shall be the order of precedence:

- i. Letter of Award/Acceptance(LOA)
- ii. Bill(s) of Quantities
- iii. Special Conditions of Contract
- iv. Technical Specifications as given in tender documents
- v. Drawings
- vi. Indian Railways Standard General Conditions of Contract updated with correction slips issued up to date of inviting tender or as otherwise specified in the tender documents.
- vii. Indian Railways Unified Standard Specification (IRUSS-2019) updated with correction slips issued up to date of inviting tender or as otherwise specified in the tender documents, if applicable in the contract.
- viii. CPWD Specifications 2019 Vol I & II updated with correction slips issued up to date of inviting tender or as otherwise specified in the tender documents, if applicable in the contract.
- ix. Indian Railways Unified Standard Specifications (Works and Material) 2010 updated with correction slips issued up to date of inviting tender or as otherwise specified in the tender documents, if applicable in the contract.
- x. IR Specifications/Guidelines updated with correction slips issued up to date of inviting tender or as otherwise specified in the tender documents.
- xi. Relevant B.I.S. Codes updated with correction slips issued up to date of inviting tender or as otherwise specified in the tender documents.

**1.1 Interpretation:** These Instructions to Tenderers shall be read in conjunction with the Standard General Conditions of Contract which are referred to herein and shall be subject to modifications additions or suppression by Special Conditions of Contract and/or Special Specifications, if any, annexed to the Tender Forms.

**1.2 Definition:** In these Instructions to Tenderers, the following terms shall have the meanings assigned hereunder except where the context otherwise requires:



(a) “Railway” shall mean the President of the Republic of India or the administrative officers of the Railway or Successor Railway authorized to deal with any matter, which these presents are concerned on his behalf.

(b) “General Manager” shall mean the Officer-in-Charge of the general superintendence and control of the Zonal Railway/Production Unit and shall also include Addl. General Manager, General Manager (Construction) and shall mean and include their successors of the Successor Railway.

(c) “Chief Engineer” shall mean the Officer-in-Charge of the Engineering Department of Railway and shall also include Chief Engineer (Construction), Chief Electrical Engineer, Chief Electrical Engineer (Construction), Chief Signal & Telecom Engineer, Chief Signal & Telecom Engineer (Construction), Chief Mechanical Engineer and shall mean and include their successors of the Successor Railway.

(d) “Divisional Railway Manager” shall mean the Officer-in-Charge of a Division of Zonal Railway and shall mean and include Divisional Railway Manager of the Successor Railway.

(e) “Engineer” shall mean the Divisional Engineer or Executive Engineer, Divisional Signal & Telecom Engineer, Divisional Electrical Engineer, Divisional Mechanical Engineer in executive charge of the works and shall include the superior officers, both Open Line and Construction Organisations, of Engineering, Signal & Telecom, Mechanical and Electrical Departments, i.e. the Senior Divisional Engineer/Deputy Chief Engineer, Senior Divisional Signal & Telecom Engineer / Dy. Chief Signal & Telecom Engineer, Senior Divisional Electrical Engineer / Deputy Chief Electrical Engineer, Senior Divisional Mechanical Engineer and shall mean & include the Engineers of the Successor Railway.

(f) “Tenderer” shall mean the person / firm / co-operative or company whether incorporated or not who tenders for the works with a view to execute the works on contract with the Railway and shall include their representatives, successors and permitted assigns.

(g) “Limited Tenders” shall mean tenders invited from all or some contractors on the approved or select list of contractors with the Railway.

(h) “Open Tenders” shall mean the tenders invited in open and public manner and with adequate notice.

(i) “Works” shall mean the works contemplated in the drawings and Bill(s) of Quantities set forth in the tender forms and required to be executed according to the specifications.

(j) “Specifications” shall mean the Specifications for Materials and Works of the Railway as specified under the authority of the Ministry of Railways or Chief Engineer or as amplified, added to or superseded by special specifications if any, appended to the Tender Forms.

(k) Standard Schedule of Rates (SSOR) shall mean the schedule of Rates adopted by the Railway, which includes-

1. “Unified Standard Schedule of Rates of the Railway (USSOR)” i.e. the Standard Schedule of Rates of the Railway issued under the authority of the Chief Engineer from time to time, updated

with correction slips issued up to date of inviting tender or as otherwise specified in the tender documents;

2. “Delhi Schedule Of Rates (DSR)” i.e. the Standard Schedule of Rates published by Director General/ Central Public Works Department, Government of India, New Delhi, as adopted and modified by the Railway under the authority of the Chief Engineer from time to time, updated with correction slips issued up to date of inviting tender or as otherwise specified in the tender documents.

(l) “Drawings” shall mean the maps, drawings, plans and tracings, or prints thereof annexed to the Tender Forms.

(m) “Contractor’s authorized Engineer” shall mean a graduate engineer or equivalent, having more than 3 years experience in the relevant field of construction work involved in the contract, duly approved by the Engineer.

(n) Date of inviting tender shall be the date of publishing tender notice on IREPS website if tender is published on website or the date of publication in newspaper in case tender is not published on website.

(o) “Bill of Quantities” shall mean Schedule of Item(s) included in the tender document along with respective quantities.

**1.3** Words importing the singular number shall also include the plural and vice versa where the context requires.

## **CREDENTIALS OF CONTRACTORS**

### **2. Application for Registration and Approved list of contractors:**

**2.1** Works of construction and of supply of material shall be entrusted for execution to contractors whose capabilities and financial status have been investigated and approved to the satisfaction of the Railway. A list of approved contractors may also be maintained by the Railway. The said list be revised periodically once in a year or so by giving wide publicity through advertisements etc. For registration, the contractor including a contractor who is already on the approved list shall apply to the concerned General Manager (Construction) / Chief Administrative Officer (Construction) / Principal Chief Engineer / Principal Chief Signal & Telecommunication Engineer / Principal Chief Mechanical Engineer / Principal Chief Electrical Engineer/ Divisional Railway Manager, furnishing particulars regarding:

(a) his position as an independent contractor specifying engineering organization available with details of partners / staff / engineers employed with qualifications and experience;

(b) his capacity to undertake and carry out works satisfactorily as vouched for by a responsible official or firm; details about the transport equipments, construction tools and plants etc. required for the work, maintained by him;

(c) his previous experience of works similar to that to be contracted for, in proof of which original certificates or testimonials may be called for and their genuineness verified, if needs be, by reference to the signatories thereof;

(d) his knowledge from actual personal investigation of the resources of the area/zone or zones in which he offers to work;

(e) his ability to supervise the work personally or by competent and duly authorized agent;

(f) his financial position;

**2.2** An applicant shall clearly state the categories of works and the area/zone/division(s)/district(s) in which he desires registration in the list of approved contractors.

**2.3** The selection of contractors for enlistment in the approved list would be done by a committee for different value of slabs as notified by Railway.

**2.4** An annual fee as prescribed by the Railway from time to time would be charged from such approved contractors to cover the cost of sending notices to them and clerkage for tenders etc. Notices shall be sent to them on registered e-mail address and registered postal address.

### **TENDERS FOR WORKS**

**3. Tender Form:** Tender Forms shall embody the contents of the contract documents either directly or by reference and shall be as per specimen form, Annexure-I. e-Tender Forms shall be issued free of cost to all tenderers.

**4. Omissions & Discrepancies:** Should a tenderer find discrepancies in or omissions from the drawings or any of the Tender Forms or should he be in doubt as to their meaning, he should at once notify the authority inviting tenders. The tender inviting authority may, if deemed necessary, clarify the same to all tenderers. It shall be understood that every endeavour has been made to avoid any error which can materially affect the basis of tender and successful tenderer shall take upon himself and provide for the risk of any error which may subsequently be discovered and shall make no subsequent claim on account thereof.

### **5. Bid Security:**

- (1) (a) The tenderer shall be required to submit the Bid Security with the tender for the due performance with the stipulation to keep the offer open till such date as specified in the tender, under the conditions of tender. The Bid Security shall be as under:

| <b>Value of the Work</b>                        | <b>Bid Security</b>                                                                                                            |
|-------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| For works estimated to cost up to ₹ 1 crore     | 2% of the estimated cost of the work                                                                                           |
| For works estimated to cost more than ₹ 1 crore | ₹ 2 lakh plus ½% (half percent) of the excess of the estimated cost of work beyond ₹ 1 crore subject to a maximum of ₹ 1 crore |

Note:

- (i) The Bid Security shall be rounded off to the nearest ₹100. This Bid Security shall be applicable for all modes of tendering.
  - (ii) Any firm recognized by Department of Industrial Policy and Promotion (DIPP) as 'Startups' shall be exempted from payment of Bid Security detailed above.
  - (iii) Labour Cooperative Societies shall submit only 50% of above Bid Security detailed above.
- (b) It shall be understood that the tender documents have been issued to the tenderer and the tenderer is permitted to tender in consideration of stipulation on his part, that after submitting his tender he will not resile from his offer or modify the terms and conditions thereof in a manner not acceptable to the Engineer. Should the tenderer fail to observe or comply with the said stipulation, the aforesaid amount shall be liable to be forfeited to the Railway.
- (c) If his tender is accepted, this Bid Security mentioned in sub para (a) above will be retained as part security for the due and faithful fulfillment of the contract in terms of Clause 16 of the Standard General Conditions of Contract. The Bid Security of other Tenderers shall, save as herein before provided, be returned to them, but the Railway shall not be responsible for any loss or depreciation that may happen thereto while in their possession, nor be liable to pay interest thereon.
- (2) The Bid Security shall be deposited either in cash through e-payment gateway or submitted as Bank Guarantee bond from a scheduled commercial bank of India or as mentioned in tender documents. The Bank Guarantee bond shall be as per **Annexure-VIA** and shall be valid for a period of 90 days beyond the bid validity period.
- (3) **In case, submission of Bid Security in the form of Bank Guarantee, following shall be ensured:**
- i. A scanned copy of the Bank Guarantee shall be uploaded on e-Procurement Portal (IREPS) while applying to the tender.
  - ii. The original Bank Guarantee should be delivered in person to the official nominated as indicated in the tender document within 5 working days of deadline of submission **before closing date for submission** of bids (i.e. excluding the last date of submission of bids).  

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  - iii. Non submission of scanned copy of Bank Guarantee with the bid on e-tendering portal (IREPS) and/or non submission of original Bank Guarantee within the specified period shall lead to summary rejection of bid.
  - iv. The Tender Security shall remain valid for a period of 90 days beyond the validity period for the Tender.

- v. The details of the BG, physically submitted should match with the details available in the scanned copy and the data entered during bid submission time, failing which the bid will be rejected
- vi. The Bank Guarantee shall be placed in an envelope, which shall be sealed. The envelope shall clearly bear the identification “**Bid for the \*\*\*\*\* Project**” and shall clearly indicate the name and address of the Bidder. In addition, the Bid Due Date should be indicated on the right hand top corner of the envelope.
- vii. The envelope shall be addressed to the officer and address as mentioned in the tender document.
- viii. If the envelope is not sealed and marked as instructed above, the Railway assumes no responsibility for the misplacement or premature opening of the contents of the Bid submitted and consequent losses, if any, suffered by the Bidder.

## **6. Care in Submission of Tenders:**

- (a) (i) Before submitting a tender, the tenderer will be deemed to have satisfied himself by actual inspection of the site and locality of the works, that all conditions liable to be encountered during the execution of the works are taken into account and that the rates he enters in the tender forms are adequate and all inclusive to accord with the provisions in Clause-37 of the Standard General Conditions of Contract for the completion of works to the entire satisfaction of the Engineer.
- (a)(ii) Tenderers will examine the various provisions of The Central Goods and Services Tax Act, 2017(CGST)/ Integrated Goods and Services Tax Act, 2017(IGST)/ Union Territory Goods and Services Tax Act, 2017(UTGST)/ respective state’s State Goods and Services Tax Act (SGST) also, as notified by Central/State Govt.& as amended from time to time and applicable taxes before bidding. Tenderers will ensure that full benefit of Input Tax Credit (ITC) likely to be availed by them is duly considered while quoting rates.
- (a)(iii) The successful tenderer who is liable to be registered under CGST/IGST/UTGST/SGST Act shall submit GSTIN along with other details required under CGST/IGST/UTGST/SGST Act to railway immediately after the award of contract, without which no payment shall be released to the Contractor. The Contractor shall be responsible for deposition of applicable GST to the concerned authority.
- (a)(iv) In case the successful tenderer is not liable to be registered under CGST/IGST/UTGST/ SGST Act, the railway shall deduct the applicable GST from his/their bills under reverse charge mechanism (RCM) and deposit the same to the concerned authority.

(a) (v) contractor shall be liable to pay/refund the amount collected as GST to the Indian Railways along with interest and penalties, if any imposed by the authorities, in case GST input tax credit of Indian Railways is denied/rejected by the tax authorities due to reasons mentioned below but not limited to:

Wrong/incorrect invoices issued by contractor;

No-filing of GST returns;

Non-payment of GST collected from Indian Railways to the authorities;

Any other non-compliance done by contractor;

**General Indemnity:** Contractor hereby agrees to indemnify and hold harmless the Indian railways from and against any and all losses, including loss on account of Input Tax credit and all losses incurred by the Indian Railways relating to or arising out of or in connection with any actual or threatened claim, legal action, proceedings, prosecution or inquiry by or against the Indian railways arising out, directly or indirectly, of failure by the contractor to comply with the provisions of GST and related laws, or based upon or arising from any failure by the contractor.

**Retention Money:** Any payment liable to be paid by Indian Railways to contractor against the goods or services or both supplied by such contractor to Indian railways shall be kept on hold in case supplier makes any non-compliance of any of the GST law provisions including non-reporting of invoices in GST returns. Such payment shall be released after proper verification of records and availability of ITC to Indian Railways as per provisions of GST law.

|                                           |
|-------------------------------------------|
| New Para as per ACS No. 6 Dt. 21.12.2023. |
|-------------------------------------------|

- (b) When work is tendered for by a firm or company, the tender shall be signed by the individual legally authorized to enter into commitments on their behalf.
- (c) The Railway will not be bound by any power of attorney granted by the tenderer or by changes in the composition of the firm made subsequent to the execution of the contract. It may, however, recognize such power of attorney and changes after obtaining proper legal advice, the cost of which will be chargeable to the Contractor.

**6.1** The tenderers shall submit a copy of certificate stating that all their statements/documents submitted along with bid are true and factual. Standard format of certificate to be submitted by the bidder is enclosed as **Annexure-V**. **In addition to Annexure-V, in case of other than Company/Proprietary firm, Annexure V-(A) shall also be submitted by the each member of a Partnership Firm/Joint venture (JV)/Hindu Undivided Family (HUF)/ Limited Liability Partnership (LLP) etc. as the case may be.** Non submission of above certificate by the bidder shall result in **summarily** rejection of his/their bid. It shall be mandatorily incumbent upon the tenderer to identify, state and submit the supporting documents duly self attested/digitally signed by which they/he is qualifying the Qualifying Criteria mentioned in the Tender Document. ACS No. 2 of 13.12.2022

## **CONSIDERATION OF TENDERS**

**7. Right of Railway to Deal with Tenders:** The Railway reserves the right of not to invite tenders for any of Railway work or works or to invite open or limited tenders and when tenders are called to accept a tender in whole or in part or reject any tender or all tenders without assigning reasons for any such action. In case if tender is accepted in part by Railway administration, Letter of Acceptance shall be issued as counter offer to the Tenderer, which shall be subject to acceptance by the Tenderer.

**7A. Two Packets System of Tendering:** With a view to assess the tenders technically without being influenced by the financial bids, 'Two Packets System of tendering' shall be adopted wherein tender documents provide for the same.

**7B. Pre Bid Conference:** Intenders having advertised value more than Rs 50 Crore or as mentioned in the tender document, Railway shall conduct Pre Bid Conference(s) with the prospective bidders.

**7C. Make in India Policy:** Provisions of Make in India Policy 2017 issued by Govt. of India, as amended from time to time, shall be followed for consideration of tenders.

**7D. Permission to Bid for a bidder from a country which shares Land boundary with India:** Any bidder from the countries sharing a land border with India will be eligible to bid in any procurement of works (including turnkey projects) only if the bidder is registered with the Competent Authority. The Competent Authority for registration will be the Registration Committee constituted by the Department for Promotion of Industry and Internal Trade (DPIIT), Government of India. For interpretation of this para, Department of Expenditure, Ministry of Finance, Government of India letter F.No.6/18/2019-PPD dated 23/07/2020 shall be referred.

**7E. Clarification of Bids:** To assist in the examination, evaluation & comparison and pre-qualification of the Tender, the Railway may, at its discretion, ask any Bidder for a clarification of its Bid. Any clarification submitted by a Bidder that is not in response to a request by the Railway shall not be entertained or considered. The Railway request for clarification and the response of the bidder in this regard shall be in writing.

However, if a Bidder does not provide clarification of its bid by the date and time communicated in the Railway request for clarification, the bid shall be evaluated as per the documents submitted along with the bid.

**8. Execution of Contract Document:** The Tenderer whose tender is accepted shall be required to appear in person at the office of General Manager/General Manager (Construction), Chief Administrative Officer (Construction), Divisional Railway Manager or concerned Engineer, as the case may be, or if tenderer is a firm or corporation, a duly authorized representative shall appear (there would be no need for appear in person if agreement is signed digitally) and execute the contract agreement within seven days of notice from Railways that the Contract Agreement is ready. Failure to do so shall constitute a breach of the agreement affected by the acceptance of the tender. The Contract Agreement shall be entered into by Railway only after submission of valid Performance Guarantee by the Contractor. In such cases the Railway may determine that such tenderer has abandoned the contract and there upon his tender and acceptance thereof shall be treated as cancelled and the Railway shall be entitled to forfeit the full amount of the Bid Security and other dues payable to the Contractor under this contract. The failed Contractor shall be debarred from participating in the re-tender for that work.

**9. Form of Contract Document:** Every contract shall be complete in respect of the document it shall so constitute. Not less than 2 copies of the contract document shall be signed by the competent authority and the Contractor and one copy given to the Contractor (there would be no need of signing two copies if agreement is signed digitally).

(a) For Zone Contracts, awarded on the basis of the percentage above or below the applicable chapter(s) of Standard Schedule of Rates (SSOR) for the whole or part of financial year, the contract agreement required to be executed by the tenderer whose tender is accepted shall be as per specimen form, Annexure-II. During the currency of the Zone Contract, work orders as per specimen form Annexures-III, for works not exceeding ₹ 5,00,000 each, shall be issued by the Divisional Railway Manager / Executive Engineer under the agreement for Zone Contract.

(b) For contracts for specific works, the contract document required to be executed by the tenderer whose tender is accepted shall be an agreement as per specimen form Annexure- IV.



**RAILWAY**  
**TENDER FORM (First Sheet)**

Tender No. \_\_\_\_\_

Name of Work \_\_\_\_\_

To

The President of India

Acting through the \_\_\_\_\_ Railway

I/We \_\_\_\_\_ have read the various conditions to tender attached hereto and agree to abide by the said conditions. I/We also agree to keep this offer open for acceptance for a period of \_\_\_\_\_ days from the date fixed for closing of the tender and in default thereof, I/We will be liable for forfeiture of my/our "Bid Security". I/We offer to do the work for \_\_\_\_\_ Railway, at the rates quoted in the attached bill(s) of quantities and hereby bind myself/ourselves to complete the work in all respects within \_\_\_\_\_ months from the date of issue of letter of acceptance of the tender.

2. I/We also hereby agree to abide by the Indian Railways Standard General Conditions of Contract, with all correction slips up-to-date and to carry out the work according to the Special Conditions of Contract and Specifications of materials and works as laid down by Railway in the annexed Special Conditions/Specifications, Standard Schedule of Rates (SSOR) with all correction slips up-to-date for the present contract.

3. A Bid Security of ₹ \_\_\_\_\_ has already been deposited online/ submitted as Bank Guarantee bond. Full value of the Bid Security shall stand forfeited without prejudice to any other right or remedies in case my/our Tender is accepted and if:

- (a) I/We do not submit the Performance Guarantee within the time specified in the Tender document;
- (b) I/We do not execute the contract documents within seven days after receipt of notice issued by the Railway that such documents are ready; and
- (c) I/We do not commence the work within fifteen days after receipt of orders to that effect.

4. (a) I/We am/are a Startup firm registered by ..... Department of Industrial Policy and Promotion (DIPP) and my registration number is ..... valid upto ..... (Copy enclosed) and hence exempted from submission of Bid Security.

5. We are a Labour Cooperative Society and our Registration No. is ..... with ..... and hence required to deposit only 50% of Bid Security.

6. Until a formal agreement is prepared and executed, acceptance of this tender shall constitute a binding contract between us subject to modifications, as may be mutually agreed to between us and indicated in the letter of acceptance of my/our offer for this work.

Signature of Witnesses:

\_\_\_\_\_

(1) \_\_\_\_\_

Signature of Tenderer(s)

(2) \_\_\_\_\_

Date \_\_\_\_\_

Address of the Tenderer(s)

\_\_\_\_\_

**TENDER FORM (Second Sheet)**

**1. Instructions to Tenderers and Conditions of Tender:** The following documents form part of Tender / Contract:

- (a) Tender Forms – First Sheet and Second Sheet
- (b) Special Conditions/Specifications (enclosed)
- (c) Bill(s) of quantities (enclosed)
- (d) Standard General Conditions of Contract and Standard Specifications (Works and Materials) of Indian Railways as amended/corrected upto latest correction slips, copies of which can be seen in the office of \_\_\_\_\_ or obtained from the office of the Chief Engineer, \_\_\_\_\_ Railway on payment of prescribed charges.
- (e) Standard Schedule of Rates (SSOR) as amended / corrected upto latest correction slips, copies of which can be seen in the office of \_\_\_\_\_ or obtained from the office of the Chief Engineer, \_\_\_\_\_ Railway on payment of prescribed charges.
- (f) All general and detailed drawings pertaining to this work which will be issued by the Engineer or his representatives (from time to time) with all changes and modifications.

**2. Drawings for the Work:** The Drawing for the work can be seen in the office of the \_\_\_\_\_ and / or Chief Engineer, \_\_\_\_\_ Railway at any time during the office hours. The drawings are only for the guidance of Tenderer(s). Detailed working drawings (if required) based generally on the drawing mentioned above, will be given by the Engineer or his representative from time to time.

**3.** The Tenderer(s) shall quote his / their rates as a percentage above or below the Standard Schedule of Rates (SSOR) of \_\_\_\_\_ Railway as applicable to \_\_\_\_\_ Division except where he/they are required to quote item rates and must tender for all the items shown in the Bill(s) of Quantities attached. The quantities shown in the attached Bill(s) of Quantities are given as a guide and are approximate only and are subject to variation according to the needs of the Railway. The Railway does not guarantee work under each item of the Bill(s) of Quantities. The tenderer(s) shall quote rates / rebates only at specified place in Tender Form supplied by Railway. Any revision of rates / rebates submitted (quoted) through a separate letter whether enclosed with the bid (Tender Form) or submitted separately or mentioned elsewhere in the document other than specified place shall be summarily ignored and will not be considered.

**4.** Tenders containing erasures and / or alterations of tender documents are liable to be rejected. Any correction made by tender(s) in his/their entries must be attested by him / them.

**5.** The works are required to be completed within a period of \_\_\_\_\_ months from the date of issue of acceptance letter.

**6. Bid Security:**

(a) Subject to exemptions provided under para 5(1) (a) of Part-1 (ITT) of this document, the tender must be accompanied by a Bid Security as mentioned in tender documents, failing which the tender shall be summarily rejected.

(b) The Tenderer(s) shall keep the offer open for a minimum period of 60 days (in case of two packet system of tendering 90days) from the date of closing of the Tender. It is understood that the tender documents have been issued to the Tenderer(s) and the Tenderer(s), is / are permitted to tender in consideration of the stipulation on his / their part that after submitting his / their tender subject to the period being extended further, if required by mutual agreement from time to time, he will not resile from his offer or modify the terms and conditions thereof in a manner not acceptable to \_\_\_\_\_ Railway. Should the tenderer fail to observe or comply with the foregoing stipulation, the amount deposited or Bank guarantee bond submitted as Bid Security for the due performance of the above stipulation, shall be forfeited to the Railway.

(c) If his tender is accepted,

- (i) the Bid Security mentioned in sub para(a) above deposited in cash through e-payment gateway will be retained as part security for the due and faithful fulfillment of the contract in terms of Clause 16 of the Standard General Conditions of Contract;
- (ii) the Bid Security mentioned in sub para(a) above submitted as Bank guarantee bond, will be encashed as part security for the due and faithful fulfillment of the contract in terms of Clause 16 of the Standard General Conditions of Contract.

The Bid Security of other Tenderers shall, save as herein before provided, be returned to them, but the Railway shall not be responsible for any loss or depreciation to the Bid Security that may happen thereto while in their possession, nor be liable to pay interest thereon.

(d) In case Contractor submits the Term Deposit Receipt/Bank Guarantee Bond towards either the Full Security Depositor the Part Security Deposit equal to or more than Bid Security, the Railway shall return the Bid Security so retained as per sub para(c) above, to the Contractor.

**7. Rights of the Railway to deal with Tender:** The authority for the acceptance of the tender will rest with the Railway. It shall not be obligatory on the said authority to accept the lowest tender or any other tender, and tenderer(s) shall neither demand any explanation for the cause of rejection of his/ their tender nor the Railway to assign reasons for declining to consider or reject any particular tender or tenders.

**8.** If the tenderer(s) deliberately gives / give wrong information in his / their tender or creates / create circumstances for the acceptance of his / their tender, the Railway reserves the right to reject such tender at any stage.

9. If any partner(s) of a partnership firm expires after the submission of its tender or after the acceptance of its tender, the Railway shall deem such tender as cancelled/contract as terminated under clause 61 of the Standard General Conditions of Contract, unless the firm retains its character as per partnership agreement. If a sole proprietor expires after the submission of tender or after the acceptance of tender, the Railway shall deem such tender as cancelled / contract as terminated under clause 61 of the Standard General Conditions of Contract.

## **10. Eligibility Criteria:**

### **10.1 Technical Eligibility Criteria:**

(a) The tenderer must have successfully completed or substantially completed any one of the following categories of work(s) during last 07 (seven) years, ending last day of month previous to the one in which tender is invited:

- (i) Three similar works each costing not less than the amount equal to 30% of advertised value of the tender, or
- (ii) Two similar works each costing not less than the amount equal to 40% of advertised value of the tender, or
- (iii) One similar work costing not less than the amount equal to 60% of advertised value of the tender.

(b) (1) In case of tenders for composite works (e.g. works involving more than one distinct component, such as Civil Engineering works, S&T works, Electrical works, OHE works etc. and in the case of major bridges – substructure, superstructure etc.), tenderer must have successfully completed or substantially completed any one of the following categories of work(s) during last 07 (seven) years, ending last day of month previous to the one in which tender is invited:

- (i) Three similar works each costing not less than the amount equal to 30% of advertised value of each component of tender, or
- (ii) Two similar works each costing not less than the amount equal to 40% of advertised value of each component of tender, or
- (iii) One similar work each costing not less than the amount equal to 60% of advertised value of each component of tender.

Note for b(1): Separate completed works of minimum required values shall also be considered for fulfillment of technical eligibility criteria for different components.

(b)(2) In such cases, what constitutes a component in a composite work shall be clearly pre-defined with estimated tender cost of it, as part of the tender documents without any ambiguity.

(b) (3) To evaluate the technical eligibility of tenderer, only components of work as stipulated in tender documents for evaluation of technical eligibility, shall be considered. The scope of work covered in other remaining components shall be either executed by tenderer himself if he has work experience as mentioned in clause 7 of the Standard General Conditions of Contractor through subcontractor fulfilling the requirements as per clause 7 of the Standard General Conditions of Contract or jointly i.e., partly himself and remaining through subcontractor, with prior approval of Chief Engineer in writing.

However, if required in tender documents by way of Special Conditions, a formal agreement duly notarised, legally enforceable in the court of law, shall be executed by the main contractor with the subcontractor for the component(s) of work proposed to be executed by the subcontractor(s), and shall be submitted along with the offer for considering subletting of that scope of work towards fulfillment of technical eligibility. Such subcontractor must fulfill technical eligibility criteria as follows:

The subcontractor shall have successfully completed at least one work similar to work proposed for subcontract, costing not less than 35% value of work to be subletted, in last 5 years, ending last day of month previous to the one in which tender is invited through a works contract.

*Note: for subletting of work costing up to Rs 50 lakh, no previous work experience of subcontractor shall be asked for by the Railway.*

In case after award of contract or during execution of work it becomes necessary for contractor to change subcontractor, the same shall be done with subcontractor(s) fulfilling the requirements as per clause 7 of the Standard General Conditions of Contract, with prior approval of Chief Engineer in writing.

Note for Item 10.1:

Work experience certificate from private individual shall not be considered. However, in addition to work experience certificates issued by any Govt. Organisation, work experience certificate issued by Public listed company having average annual turnover of Rs 500 crore and above in last 3 financial years excluding the current financial year, listed on National Stock Exchange or Bombay Stock Exchange, incorporated/registered at least 5 years prior to the date of closing of tender, shall also be considered provided the work experience certificate has been issued by a person authorized by the Public listed company to issue such certificates.

In case tenderer submits work experience certificate issued by public listed company, the tenderer shall also submit along with work experience certificate, the relevant copy of work order, bill of quantities, bill wise details of payment received duly certified by Chartered Accountant, TDS certificates for all payments received and copy of final/last bill paid by company in support of above work experience certificate.

**10.2. Financial Eligibility Criteria:** The tenderer must have minimum average annual contractual turnover of  $1.5 \frac{V}{N}$  crores; **V/N or 'V' whichever is less;** where

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V= Advertised value of the tender in crores of Rupees

N= Number of years prescribed for completion of work for which bids have been invited.

The average annual contractual turnover shall be calculated as an average of "total contractual payments" in the previous three financial years, as per the audited balance sheet. However, in case balance sheet of the previous year is yet to be prepared/ audited, the audited balance sheet of the fourth previous year shall be considered for calculating average annual contractual turnover.

The tenderers shall submit requisite information as per Annexure-VIB, along with copies of Audited Balance Sheets duly certified by the Chartered Accountant/ Certificate from Chartered Accountant duly supported by Audited Balance Sheet.

**10.3 Bid Capacity:** The tender/technical bid will be evaluated based on bid capacity formula detailed as Annexure-VI.

**10.4** No Technical and Financial credentials are required for tenders having advertised value up to Rs 50 lakh.

**10.5** Credentials if submitted in foreign currency shall be converted into Indian currency i.e., Indian Rupee as under:

The conversion rate of US Dollars into Rupees shall be the daily representative exchange rates published by the Reserve Bank of India or entity authorized by RBI to do so for the relevant date or immediately previous date for which rates have been published. Where, relevant date shall be as on the last day of month previous to the one in which tender is invited. In case of any other currency, the

same shall first be converted to US Dollars as on the last day of month previous to the one in which tender is invited, and the amount so derived in US Dollars shall be converted into Rupees at the aforesaid rate. The conversion rate of such currencies shall be the daily representative exchange rates published by the International Monetary Fund for the relevant date or immediately previous date for which rates have been published.

*[Explanation for Para 10 of the Tender Form (Second Sheet) including Para 10.1 to 10.5 - Eligibility Criteria:*

- 1. Substantially Completed Work means an ongoing work in which payment equal to or more than 90% of the present contract value (excluding the payment made for adjustment of Price variation (PVC), if any) has been made to the contractor in that ongoing contract and no proceedings of termination of contract on Contractor's default has been initiated. The credential certificate in this regard should have been issued not prior to 60 days of date of invitation of present tender.*
- 2. In case a work is started prior to 07 (seven) years, ending last day of month previous to the one in which tender is invited, but completed in last 07 (seven) years, ending last day of month previous to the one in which tender is invited, the completed work shall be considered for fulfillment of credentials.*
- 3. If a work is physically completed and completion certificate to this extent is issued by the concerned organization but final bill is pending, such work shall be considered for fulfillment of credentials*
- 4. In case of completed work, the value of final bill (gross amount) including the PVC amount (if paid) shall be considered as the completion cost of work. In case final bill is pending, only the total gross amount already paid including the PVC amount (if paid) shall be considered as the completion cost of work.*

*In case of substantially completed work, the total gross amount already paid including the PVC amount (if paid), as mentioned in the certificate, shall be considered as the cost of substantially completed work.*
- 5. If a bidder has successfully completed a work as subcontractor and the work experience certificate has been issued for such work to the subcontractor by a Govt. Organization or public listed company as defined in Note for Item 10.1 Para 10 of the Tender Form (Second Sheet), the same shall be considered for the purpose of fulfillment of credentials.*
- 6. In case a work is considered similar in nature for fulfillment of technical credentials, the overall cost including the PVC amount (if paid) of that completed work or substantially completed work, shall be considered and no separate evaluation for each component of that work shall be made to decide eligibility.*
- 7. In case of newly formed partnership firm, the credentials of individual partners from previous propriety firm(s) or dissolved previous partnership firm(s) or split previous partnership firm(s), shall be considered only to the extent of their share in previous entity on the date of dissolution / split and their share in newly formed partnership firm. For example, a partner A had 30% share in previous entity and his share in present partnership firm is 20%. In the present tender under consideration, the credentials of partner A will be considered to the extent of  $0.3 \times 0.2 \times \text{value of the work done in the previous entity}$ . For this purpose, the tenderer shall submit along with his bid all the relevant documents which include copy of previous partnership deed(s), dissolution deed(s) and proof of surrender of PAN No.(s) in case of dissolution of partnership firm(s) etc.*
- 8. In case of existing partnership firm, if any one or more partners quit the partnership firm, the credentials of remaining partnership firm shall be re-worked out i.e., the quitting partner(s) shall*

*take away his credentials to the extent of his share on the date of quitting the partnership firm (e.g. in a partnership firm of partners A, B & C having share 30%, 30% & 40% respectively and credentials of Rs 10 crore; in case partner C quits the firm, the credentials of this partnership firm shall remain as Rs 6 crore). For this purpose, the tenderer shall submit along with his bid all the relevant documents which include copy of previous partnership deed(s), dissolution deed(s) and proof of surrender of PAN No.(s) in case of dissolution of partnership firm(s) etc.*

- 9. In case of existing partnership firm if any new partner(s) joins the firm without any modification in the name and PAN/TAN no. of the firm, the credentials of partnership firm shall get enhanced to the extent of credentials of newly added partner(s) on the same principles as mentioned in item 6 above. For this purpose, the tenderer shall submit along with his bid all the relevant documents which include copy of previous partnership deeds, dissolution/splitting deeds and proof of surrender of PAN No.(s) in case of dissolution of partnership firm etc.*
- 10. Any partner in a partnership firm cannot use or claim his credentials in any other firm without leaving the partnership firm i.e., In a partnership firm of A&B partners, A or B partner cannot use credentials of partnership firm of A&B partners in any other partnership firm or propriety firm without leaving partnership firm of A&B partners.*
- 11. In case a partner in a partnership firm is replaced due to succession as per succession law, the proportion of credentials of the previous partner will be passed on to the successor.*
- 12. If the percentage share among partners of a partnership firm is changed, but the partners remain the same, the credentials of the firm before such modification in the share will continue to be considered for the firm as it is without any change in their value. Further, in case a partner of partnership firm retires without taking away any credentials from the firm, the credentials of partnership firm shall remain the same as it is without any change in their value.*
- 13. In a partnership firm "AB" of A&B partners, in case A also works as propriety firm "P" or partner in some other partnership firm "AX", credentials of A in propriety firm "P" or in other partnership firm "AX" earned after the date of becoming a partner of the firm AB shall not be added in partnership firm AB.*
- 14. In case a tenderer is LLP, the credentials of tenderer shall be worked out on above lines similar to a partnership firm.*
- 15. In case company A is merged with company B, then company B would get the credentials of company A also.]*

## **11. Tenderer Credentials:**

Documents testifying tenderer previous experience and financial status should be produced along with the tender.

Tenderer(s) who is / are not borne on the approved list of the Contractors of \_\_\_\_\_ Railway shall submit along with his / their tender:

- (i) Certificates and testimonials regarding contracting experience for the type of job for which tender is invited with list of works carried out in the past.
- (ii) Audited Balance Sheet duly certified by the Chartered Accountant regarding contractual payments received in the past.



- (iii) The list of personnel / organization on hand and proposed to be engaged for the tendered work. Similarly list of Plant & Machinery available on hand and proposed to be inducted and hired for the tendered work.
- (iv) A copy of certificate stating that they are not liable to be disqualified and all their statements/documents submitted along with bid are true and factual. Standard format of the certificate to be submitted by the bidder is enclosed as Annexure-V. **In addition to Annexure-V, in case of other than Company/Proprietary firm, Annexure V-(A) shall also be submitted by the each member of a Partnership Firm/Joint venture (JV)/ Hindu Undivided Family (HUF)/ Limited Liability Partnership (LLP) etc. as the case may be.** Non submission of a copy of certificate by the bidder shall result in summarily rejection of his/their bid. It shall be mandatorily incumbent upon the tenderer to identify, state and submit the supporting documents duly self-attested / digitally signed by which they/he are/is qualifying the Qualifying Criteria mentioned in the Tender Document. ACS No. 2 of 13.12.2022
- (v) The Railway reserves the right to verify all statements, information and documents submitted by the bidder in his tender offer, and the bidder shall, when so required by the Railway, make available all such information, evidence and documents as may be necessary for such verification. Any such verification or lack of such verification, by the Railway shall not relieve the bidder of its obligations or liabilities hereunder nor will it affect any rights of the Railway there under.
- (vi) (a) In case of any information submitted by tenderer is found to be false, forged or incorrect at any time during process for evaluation of tenders, it shall lead to forfeiture of the tender Bid Security besides banning of business for a period of upto ~~five~~ **two** years. . ACS No. 3 of 26.04.23
- (b) In case of any information submitted by tenderer is found to be false, forged or incorrect after the award of contract, the contract shall be terminated. Bid Security, Performance Guarantee and Security Deposit available with the railway shall be forfeited. In addition, other dues of the contractor, if any, under this contract shall be forfeited and agency shall be banned for doing business for a period of upto ~~five~~ **two** years. . ACS No. 3 of 26.04.23

**12.** Non-compliance with any of the conditions set forth therein above is liable to result in the tender being rejected.

**13. Execution of Contract Documents:** The successful Tenderer(s) shall be required to execute an agreement with the President of India acting through the \_\_\_\_\_, \_\_\_\_\_ Railway for carrying out the work according to the Standard General Conditions of Contract, Special Conditions / Specifications annexed to the tender and Standard Specifications (Works and Materials) of Railway as amended/corrected upto latest correction slips, mentioned in tender form (First Sheet).

**14. Documents to be Submitted Along with Tender**

- (i) The tenderer shall clearly specify whether the tender is submitted on his own (Proprietary Firm) or on behalf of a Partnership Firm / Company / Joint Venture (JV) / Registered Society / Registered Trust / Hindu Undivided Family (HUF) / Limited Liability Partnership (LLP) etc. The tenderer(s) shall enclose the attested copies of the constitution of their concern, and copy of PAN Card along with their tender. Tender Documents in such cases are to be signed by such persons as may be legally competent to sign them on behalf of the firm, company, association, trust or society, as the case may be.
- (ii) Following documents shall be submitted by the tenderer:

**(a) Sole Proprietorship Firm:**

- (i) All documents in terms of Para 10 of the Tender Form (Second Sheet) above.

**(b) HUF:**

- (i) A copy of notarized affidavit on Stamp Paper declaring that he who is submitting the tender on behalf of HUF is in the position of 'Karta' of Hindu Undivided Family (HUF) and he has the authority, power and consent given by other members to act on behalf of HUF.
- (ii) All other documents in terms of Para 10 of the Tender Form (Second Sheet) above.

**(c) Partnership Firm:**

- (i) All documents as mentioned in para 18 of the Tender Form (Second Sheet).

**(d) Joint Venture (JV):** All documents as mentioned in para 17 of the Tender Form (Second Sheet).

**(e) Company registered under Companies Act 2013:**

- (i) The copies of **MOA (Memorandum of Association) / AOA (Articles of Association)** of the company
- (ii) A copy of Certificate of Incorporation
- (iii) A copy of Authorization/Power of Attorney issued by the Company (backed by the resolution of Board of Directors) in favour of the individual to sign the tender on behalf of the company and create liability against the company.
- (iv) All other documents in terms Para 10 of the Tender Form (Second Sheet) above.

**(f) LLP (Limited Liability Partnership):**

- (i) A copy of LLP Agreement
- (ii) A copy of Certificate of Incorporation
- (iii) A copy of Power of Attorney/Authorization issued by the LLP in favour of the individual to sign the tender on behalf of the LLP and create liability against the LLP.
- (iv) An undertaking by all partners of the LLP that they are not blacklisted or debarred by Railways or any other Ministry / Department of the Govt. of India from participation in tenders / contracts as on the date of submission of bids, either in their individual capacity or in any firm/LLP or JV in which they were / are partners/members. Concealment / wrong information in regard to above shall make the contract liable for determination under Clause 62 of the Standard General Conditions of Contract.
- (v) All other documents in terms of Para 10 of the Tender Form (Second Sheet).

**(g) Registered Society & Registered Trust:**

- (i) A copy of Certificate of Registration
- (ii) A copy of Memorandum of Association of Society/Trust Deed
- (iii) A copy of Power of Attorney in favour of the individual to sign the tender documents and create liability against the Society/Trust.
- (iv) A copy of Rules & Regulations of the Society
- (v) All other documents in terms of Para 10 of the Tender Form (Second Sheet) above.

(iii) If it is NOT mentioned in the submitted tender that tender is being submitted on behalf of a Sole Proprietorship firm / Partnership firm / Joint Venture / Registered Company etc., then the tender shall be treated as having been submitted by the individual who has signed the tender.

(iv) After opening of the tender, any document pertaining to the constitution of Sole Proprietorship Firm / Partnership Firm / Registered Company/ Registered Trust / Registered Society / HUF/LLP etc. shall be neither asked nor considered, if submitted. Further, no suo moto cognizance of any document available in public domain (i.e., on internet etc.) or in Railway's record/office files etc. will be taken for consideration of the tender, if no such mention is available in tender offer submitted.

(v) A tender from JV shall be considered only where permissible as per the tender conditions.

(vi) The Railway will not be bound by any change of power of attorney or in the composition of the firm made subsequent to the submission of tender. Railway may, however, recognize such power of attorney and changes after obtaining proper legal advice, the cost of which will be chargeable to the Contractor.

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**15.** The tenderer whether sole proprietor / a company or a partnership firm / joint venture (JV) / registered society / registered trust / HUF / LLP etc if they want to act through agent or individual partner(s), should submit along with the tender, a copy of power of attorney duly stamped and authenticated by a Notary Public or by Magistrate in favour of the specific person whether he/they be partner(s) of the firm or any other person, specifically authorizing him/them to sign the tender, submit the tender and further to deal with the Tender/ Contract up to the stage of signing the agreement except in case where such specific person is authorized for above purposes through a provision made in the partnership deed / Memorandum of Understanding / Article of Association /Board resolution, failing which tender shall be summarily rejected.

A separate power of attorney duly stamped and authenticated by a Notary Public or by Magistrate in favour of the specific person whether he/they be partner(s) of the firm or any other person, shall be submitted after award of work, specifically authorizing him/them to deal with all other contractual activities subsequent to signing of agreement, if required.

*Note: A Power of Attorney executed and issued overseas, the document will also have to be legalized by the Indian Embassy and notarized in the jurisdiction where the Power of Attorney is being issued. However, the Power of Attorney provided by Bidders from countries that have signed the Hague Legislation Convention 1961 are not required to be legalized by the Indian Embassy if it carries a conforming Apostille certificate.*

## **16. Employment/Partnership etc. of Retired Railway Employees:**

(a) Should a tenderer

i) be a retired Engineer of the gazetted rank or any other gazetted officer working before his retirement, whether in the executive or administrative capacity or whether holding a pensionable post or not, in the Engineering or any other department of any of the railways owned and administered by the President of India for the time being, OR

ii) being partnership firm / joint venture (JV) / registered society / registered trust etc have as one of its partners/members a retired Engineer of the gazetted rank or any other gazetted officer working before his retirement, OR

iii) being an incorporated company have any such retired Engineer of the gazetted rank or any other gazetted officer working before his retirement as one of its directors

AND

in case where such Engineer or officer had not retired from government service at least 1 year prior to the date of submission of the tender

THEN

the tenderer will give full information as to the date of retirement of such Engineer or gazetted officer from the said service and as to whether permission for taking such contract, or if the Contractor be a partnership firm or an incorporated company, to become a partner or director as the case may be, has been obtained by the tenderer or the Engineer or officer, as the case may be from the President of India or any officer, duly authorized by him in this behalf, shall be clearly stated in writing at the time of submitting the tender.

- b) In case, upon successful award of contract, should a tenderer depute for execution of the works under or to deal matters related with this contract, any retired Engineer of gazette rank or retired gazetted officer working before his retirement in the Engineering or any other department of any of the railways owned and administered by the President of India for the time being, and now in his employment, then the tenderer will ensure that retired Engineer or retired gazetted officer had retired from government service at least 1 year prior to the date of his employment with tenderer and in case he had retired from service within a year then he possesses the requisite permission from the President of India or any officer, duly authorized by him in this behalf, to get associated with the tenderer.
- c) Should a tenderer or Contractor being an individual, have member(s) of his family or in the case of partnership firm/ company / joint venture (JV) / registered society / registered trust etc. one or more of his partner(s)/shareholder(s) or member(s) of the family of partner(s)/shareholder(s) having share of more than 1% in the tendering entity employed in gazetted capacity in the Engineering or any other department of the railway, then the tenderer at the time of submission of tender, will inform the authority inviting tenders the details of such persons.

Note: -If information as required as per 16.a), b), c) above has not been furnished, contract is liable to be dealt in accordance with provision of clause 62 of the Standard General Condition of contract.

## **JOINT VENTURE (JV) IN WORKS TENDERS**

**17. Participation of Joint Venture (JV) in Works Tender:** This para shall be applicable for works tenders wherein tender documents provide for the same.

**17.1** Separate identity/name shall be given to the Joint Venture.

**17.2** Number of members in a JV shall not be more than three, if the work involves only one department (say Civil or S&T or Electrical or Mechanical) and shall not be more than five, if the work involves more than one Department. One of the members of the JV shall be its Lead Member who shall have a majority (at least 51%) share of interest in the JV. The other members shall have a share of not less than 20% each in case of JV with upto three members and not less than 10% each in case of JV with more than three members. In case of JV with foreign member(s), the Lead Member has to be an Indian firm/company with a minimum share of 51%.

**17.3** A member of JV shall not be permitted to participate either in individual capacity or as a member of another JV in the same tender.

**17.4** The tender form shall be purchased and submitted only in the name of the JV and not in the name of any constituent member. The tender form can however be submitted by JV or any of its constituent member or any person authorized by JV through Power of Attorney to submit tender.

**17.5** Bid Security shall be submitted by JV or authorized person of JV either as :

- (i) Cash through e-payment gateway or as mentioned in tender document, or
- (ii) Bank Guarantee bond either in the name of JV, or in the name of all members of JV as per MOU irrespective of their share in the JV if the JV has not been constituted legally till the date of submission of tender.

**17.6** A copy of Memorandum of Understanding (MoU) duly executed by the JV members on a stamp paper, shall be submitted by the JV alongwith the tender. The complete details of the members of the JV, their share and responsibility in the JV etc. particularly with reference to financial, technical and other obligations shall be furnished in the MoU.

**17.7** Once the tender is submitted, the MoU shall not normally be modified / altered / terminated during the validity of the tender. In case the tenderer fails to observe/comply with this stipulation, the full Bid Security shall be liable to be forfeited.

**17.8** Approval for change of constitution of JV shall be at the sole discretion of the Railway. The constitution of the JV shall not normally be allowed to be modified after submission of the bid by the JV, except when modification becomes inevitable due to succession laws etc., provided further that there is no change in qualification of minimum eligibility criteria by JV after change of composition. However, the Lead Member shall continue to be the Lead Member of the JV. Failure to observe this requirement would render the offer invalid.

**17.9** Similarly, after the contract is awarded, the constitution of JV shall not be normally allowed to be altered during the currency of contract except when modification become inevitable due to succession laws etc. and minimum eligibility criteria should not get vitiated. Failure to observe this stipulation shall be deemed to be breach of contract with all consequential penal action as per contract conditions.

**17.10** On award of contract to a JV, a single Performance Guarantee shall be submitted by the JV as per tender conditions. All the Guarantees like Performance Guarantee, Bank Guarantee for Mobilization Advance, Machinery Advance etc. shall be accepted only in the name of the JV and no splitting of guarantees amongst the members of the JV shall be permitted.

**17.11** On issue of LOA (Letter of Acceptance), the JV entity to whom the work has been awarded, with the same shareholding pattern as was declared in the MOU/JV Agreement submitted along with the tender, shall be got registered before the Registrar of the Companies under 'The Companies Act -2013' (in case JV entity is to be registered as Company) or before the Registrar/Sub-Registrar under the 'The Indian Partnership Act, 1932' (in case JV entity is to be registered as Partnership Firm) or under 'The LLP Act 2008' (in case JV entity is to be registered as LLP). A separate PAN shall be obtained for this entity. The documents pertaining to this entity including its PAN shall be furnished to the Railways before signing the contract agreement for the work. In case the tenderer fails to observe/comply with this stipulation within 60 days of issue of LOA, contract is liable to be terminated. In case contract is terminated railway shall be entitled to forfeit the full amount of the Bid Security and other dues payable to the Contractor under this contract. The entity so registered, in the registered documents, shall have, inter-alia, following Clauses:

**17.11.1** Joint and Several Liability - Members of the entity to which the contract is awarded, shall be jointly and severally liable to the Railway for execution of the project in accordance with General and Special Conditions of Contract. The members of the entity shall also be liable jointly and severally for the loss, damages caused to the Railways during the course of execution of the contract or due to non-execution of the contract or part thereof.

**17.11.2** Duration of the Registered Entity - It shall be valid during the entire currency of the contract including the period of extension, if any and the maintenance period after the work is completed.

**17.11.3** Governing Laws - The Registered Entity shall in all respect be governed by and interpreted in accordance with Indian Laws.

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**17.12** Authorized Member - Joint Venture members in the JV MoU shall authorize Lead member on behalf of the Joint Venture to deal with the ~~tender~~ contract, sign the agreement or enter into contract in respect of the said tender, to receive payment, to witness joint measurement of work done, to sign measurement books and similar such action in respect of the said tender/contract. All notices/correspondences with respect to the contract would be sent only to this authorized member of the JV.

**17.13** No member of the Joint Venture shall have the right to assign or transfer the interest right or liability in the contract without the written consent of the other members and that of the Railway in respect of the said tender/contract.

**17.14** Documents to be enclosed by the JV along with the tender:

**17.14.1** In case one or more of the members of the JV is/are partnership firm(s), following documents shall be submitted:

- (i) A notarized copy of the Partnership Deed or a copy of the Partnership deed registered with the Registrar.
- (ii) A copy of consent of all the partners or individual authorized by partnership firm, to enter into the Joint Venture Agreement on a stamp paper,
- (iii) A notarized or registered copy of Power of Attorney in favour of the individual to sign the MOU/JV Agreement on behalf of the partnership firm and create liability against the firm.

- (iv) An undertaking by all partners of the partnership firm that they are not blacklisted or debarred by Railways or any other Ministry / Department of the Govt. of India from participation in tenders / contracts as on the date of submission of bids, either in their individual capacity or in any firm/LLP in which they were / are partners/members. Any Concealment / wrong information in regard to above shall make the bid ineligible or the contract shall be determined under Clause 62 of the Standard General Conditions of Contract.

**17.14.2** In case one or more members is/are Proprietary Firm or HUF, the following documents shall be enclosed:

- (i) A copy of notarized affidavit on Stamp Paper declaring that his Concern is a proprietary Concern and he is sole proprietor of the Concern OR he who is signing the affidavit on behalf of HUF is in the position of 'Karta' of Hindu Undivided Family (HUF) and he has the authority, power and consent given by other members to act on behalf of HUF.

**17.14.3** In case one or more members of the JV is/are companies, the following documents shall be submitted:

- (i) A copy of resolutions of the Directors of the Company, permitting the company to enter into a JV agreement,
- (ii) The copies of MOA (Memorandum of Association) / AOA (Articles of Association) of the company
- (iii) A copy of Certificate of Incorporation
- (iv) A copy of Authorization/copy of Power of Attorney issued by the Company (backed by the resolution of Board of Directors) in favour of the individual, to sign the tender, sign MOU/JV Agreement on behalf of the company and create liability against the company

**17.14.4** In case one or more members of the JV is/are LLP firm/s, the following documents shall be submitted:

- (i) A copy of LLP Agreement
- (ii) A copy of Certificate of Incorporation of LLP
- (iii) A copy of resolution passed by partners of LLP firm, permitting the Firm to enter into a JV agreement
- (iv) A copy of Authorization /copy of Power of Attorney issued by the LLP firm (backed by resolution passed by the Partners) in favour of the individual, to sign the tender and/or sign the MOU/ JV agreement on behalf of the LLP and create liability against the LLP.
- (v) An undertaking by all partners of the LLP that they are not blacklisted or debarred by Railways or any other Ministry / Department of the Govt. of India from participation in tenders / contracts as on the date of submission of bids, either in their individual capacity or in any firm/LLP or JV in which they were / are partners/members. Any Concealment / wrong information in regard to above shall make the contract liable for determination under Clause 62 of the Standard General Conditions of Contract.

**17.14.5** In case one or more members of the JV is/are Society/s or Trust/s, the following documents shall be submitted:

- (i) A copy of Certificate of Registration
- (ii) A copy of Memorandum of Association of Society/Trust Deed

- (iii) A copy of Rules & regulations of the Society
- (iv) A copy of power of Power of Attorney, in favour of the individual to sign the tender documents and create liability against the Society/Trust

**17.14.6 All other documents in terms of Para 10 of the Tender Form (Second Sheet) above.**

**17.14.7 A power of Attorney executed and issued overseas, the document will also have to be legalized by the Indian Embassy and notarized in the jurisdiction where the power of the Attorney is being issued. However, the power of Attorney provided by Bidders from countries that have signed the Hague Legislation Convention 1961 are not required to be legalized by the Indian Embassy if it carries a conforming Appostille certificate.**

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**17.15 Credentials & Qualifying Criteria:** Technical, financial eligibility and Bid capacity of the JV shall be adjudged based on satisfactory fulfillment of the following criteria:

**17.15.1 Technical Eligibility Criteria ('a' or 'b' mentioned hereunder):**

**(a) For Works without composite components**

The technical eligibility for the work as per para 10.1 above, shall be satisfied by either the 'JV in its own name & style' or 'Lead member of the JV'.

Each other (non-lead) member(s) of JV, who is/ are not satisfying the technical eligibility for the work as per para 10.1 above, shall have technical capacity of minimum 25% **10 %** of the cost of work i.e., each non-lead member of JV member must have satisfactorily completed or substantially completed during the last 07 (seven) years, ending last day of month previous to the one in which tender is invited, one similar single work for a minimum of 25% **10 %** of advertised value of the tender.

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**(b) For works with composite components**

The technical eligibility for major component of work as per para 10.1 above, shall be satisfied by either the 'JV in its own name & style' or 'Lead member of the JV' and technical eligibility for other component(s) of work as per para 10.1 above, shall be satisfied by either the 'JV in its own name & style' or 'any member of the JV'.

Each other (non-lead) member(s) of JV, who is/ are not satisfying the technical eligibility for any component of the work as per para 10.1 above, shall have technical capacity of minimum 25% **10 %** of the cost of any component of work mentioned in technical eligibility criteria. i.e., each other (non-lead) member of must have satisfactorily completed or substantially completed during the last 07 (seven) years, ending last day of month previous to the one in which tender is invited, one similar single work for a minimum of 25% **10 %** of cost of any component of work mentioned in technical eligibility criteria.

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*Note for Para 17.15.1:*

- a) *The Major component of the work for this purpose shall be the component of work having highest value. In cases where value of two or more component of work is same, any one work can be classified as Major component of work.*
- b) *Value of a completed work done by a Member in an earlier JV shall be reckoned only to the extent of the concerned member's share in that JV for the purpose of satisfying his/her compliance to the above mentioned technical eligibility criteria in the tender under consideration.*

**17.15.2 Financial Eligibility Criteria**

The JV shall satisfy the requirement of "Financial Eligibility" mentioned at para 10.2 above. The



“financial capacity” of the lead member of JV shall not be less than 51% of the financial eligibility criteria mentioned at para 10.2 above.

The arithmetic sum of individual “financial capacity” of all the members shall be taken as JV’s “financial capacity” to satisfy this requirement.

Note: Contractual payment received by a Member in an earlier JV shall be reckoned only to the extent of the concerned member’s share in that JV for the purpose of satisfying compliance of the above mentioned financial eligibility criteria in the tender under consideration.

### **17.15.3 Bid Capacity**

The JV shall satisfy the requirement of “Bid Capacity” requirement mentioned at para 10.3 above. The arithmetic sum of individual “Bid capacity” of all the members shall be taken as JV’s “Bid capacity” to satisfy this requirement.

## **18. Participation of Partnership Firms in works tenders:**

18.1 The Partnership Firms participating in the tender should be legally valid under the provisions of the Indian Partnership Act.

18.2 The partnership firm should have been in existence or should have been formed prior to submission of tender. Partnership firm should have either been registered with the Registrar or the partnership deed should have been notarized as per the Indian Partnership Act, prior to submission of tender.

18.3 Separate identity / name should be given to the partnership firm. The partnership firm should have PAN / TAN number in its own name and PAN / TAN number in the name of any of the constituent partners shall not be considered. The valid constituents of the firm shall be called partners.

18.4 Once the tender has been submitted, the constitution of the firm shall not normally be allowed to be modified / altered / terminated during the validity of the tender as well as the currency of the contract except when modification becomes inevitable due to succession laws etc., in which case prior permission should be taken from Railway and in any case the minimum eligibility criteria should not get vitiated. The re-constitution of firm in such cases should be followed by a notary certified Supplementary Deed. The approval for change of constitution of the firm, in any case, shall be at the sole discretion of the Railways and the tenderer shall have no claims what-so-ever. Any change in the constitution of Partnership firm after submission of tender shall be with the consent of all partners and with the signatures of all partners as that in the Partnership Deed. Failure to observe this requirement shall render the offer invalid and full Bid Security shall be forfeited.

If any Partner/s withdraws from the firm after submission of the tender and before the award of the contract, the offer shall be rejected and Bid Security of the tenderer will be forfeited. If any new partner joins the firm after submission of tender but prior to award of contract, his / her credentials shall not qualify for consideration towards eligibility criteria either individually or in proportion to his share in the previous firm. In case the tenderer fails to inform Railway beforehand about any such changes / modification in the constitution which is inevitable due to succession laws etc. and the contract is awarded to such firm, then it will be considered a breach of the contract

conditions liable for determination of the contract under Clause 62 of the Standard General Conditions of Contract.

18.5 A partner of the firm shall not be permitted to participate either in his individual capacity or as a partner of any other firm in the same tender.

18.6 The tender form shall be submitted only in the name of partnership firm. The Bid Security shall be submitted by partnership firm. The Bid Security submitted in the name of any individual partner or in the name of authorized partner (s) shall not be considered.

18.7 On issue of Letter of Acceptance (LOA) to the partnership firm, all the guarantees like Performance Guarantee, Guarantee for various Advances to the Contractor shall be submitted only in the name of the partnership firm and no splitting of guarantees among the partners shall be acceptable.

18.8 On issue of Letter of Acceptance (LOA), contract agreement with partnership firm shall be executed in the name of the firm only and not in the name of any individual partner.

18.9 In case the Letter of Acceptance (LOA) is issued to a partnership firm, the following undertakings shall be furnished by all the partners through a notarized affidavit, before signing of contract agreement.

(a) Joint and several liabilities:

The partners of the firm to which the Letter of Acceptance (LOA) is issued, shall be jointly and severally liable to the Railway for execution of the contract in accordance with General and Special Conditions of the Contract. The partners shall also be liable jointly and severally for the loss, damages caused to the Railway during the course of execution of the contract or due to non-execution of the contract or part thereof.

(b) Duration of the partnership deed and partnership firm agreement:

The partnership deed/partnership firm agreement shall normally not be modified/altered/terminated during the currency of contract and the maintenance period after the work is completed as contemplated in the conditions of the contract. Any change carried out by partners in the constitution of the firm without permission of Railway, shall constitute a breach of the contract, liable for determination of the contract under Clause 62 of the Standard General Conditions of Contract.

(c) Governing laws: The partnership firm agreement shall in all respect be governed by and interpreted in accordance with the Indian laws.

(d) No partner of the firm shall have the right to assign or transfer the interest right or liability in the contract without the written consent of the other partner/s and that of the Railway.

18.10 The tenderer shall clearly specify that the tender is submitted on behalf of a partnership firm. The following documents shall be submitted by the partnership firm, with the tender:

(i) A notarized copy of the Partnership Deed or a copy of the Partnership deed registered with the Registrar.

- (ii) A notarized or registered copy of Power of Attorney in favour of the individual to tender for the work, sign the agreement etc. and create liability against the firm.
- (iii) An undertaking by all partners of the partnership firm that they are not blacklisted or debarred by Railways or any other Ministry / Department of the Govt. of India from participation in tenders / contracts as on the date of submission of bids, either in their individual capacity or in any firm/LLP in which they were / are partners/members. Any Concealment / wrong information in regard to above shall make the bid ineligible or the contract shall be determined under Clause 62 of the Standard General Conditions of Contract.
- (iv) All other documents in terms of Para 10 of the Tender Form (Second Sheet) above.

#### 18.11 Evaluation of eligibility of a partnership firm:

Technical and financial eligibility of the firm shall be adjudged based on satisfactory fulfillment of the eligibility criteria laid down in Para 10 of the Tender Form (Second Sheet) above.

#### 19.0 Advances to Contractor –

If specifically provided in Tender Documents of tender having advertised value more than Rs 50 Crores, Railway shall make payment, as an Interest bearing advances, as per Contractor's request. These advances shall carry a simple interest \_\_\_\_\_ as indicated in the Tender documents. The payment and recovery of such advances shall be made as per manners prescribed in Clause 46.4 of the Standard General Conditions of Contract.

(Signature)

(Designation)

Signature of Tenderer(s) \_\_\_\_\_ Railway Date \_\_\_\_\_  
Date \_\_\_\_\_

**TENDER FORM (Third Sheet)****Name of Work:** \_\_\_\_\_**BILL OF QUANTITIES****1. Standard Schedule of Rates (SSOR) Items:**

| SL | Item No. of SSOR | Description of Item of Work | Approximate Quantity | Unit | Rates in Figures and Words to be filled by tenderer (₹) | Amount (₹) |
|----|------------------|-----------------------------|----------------------|------|---------------------------------------------------------|------------|
| 1  | 2                | 3                           | 4                    | 5    | 6                                                       | 7          |
|    |                  |                             |                      |      |                                                         |            |

**2. Non Standard Schedule of Rates (SSOR) Items:**

| SL | Item No. | Description of Item of Work | Approximate Quantity | Unit | Rates in Figures and Words to be filled by tenderer (₹) | Amount (₹) |
|----|----------|-----------------------------|----------------------|------|---------------------------------------------------------|------------|
| 1  | 2        | 3                           | 4                    | 5    | 6                                                       | 7          |
|    |          |                             |                      |      |                                                         |            |

The quantities shown in above Bill of Quantities are approximate and are as a guide to give the tenderer(s) an idea of quantum of work involved. The Railway reserves the right to increase/decrease and/or delete or include any of the quantities given above and no extra rate will be allowed on this account.

I/We undertake to do the work at \_\_\_\_\_ % above/below the Standard Schedule of Rates (SSOR) of the \_\_\_\_\_ Railway as applicable to \_\_\_\_\_ Division or at the rates quoted above for each item.

Dated \_\_\_\_\_

Signature of the Tenderer(s)

Note: Columns 1 to 5 shall be filled by the office of the Authority inviting tender. Columns 6 & 7 shall be filled by the Tenderer(s) only when percentage tenders are not invited.

**AGREEMENT FOR ZONE CONTRACT**

CONTRACT AGREEMENT No. \_\_\_\_\_ DATED \_\_\_\_\_. ARTICLES OF AGREEMENT made this \_\_\_\_\_ day of \_\_\_\_\_ between the President of India acting through the \_\_\_\_\_, \_\_\_\_\_ Railway hereinafter called the "Railway" of the one part and \_\_\_\_\_ hereinafter called the "Contractor" of the other part.

WHEREAS the Contractor has agreed with the Railway during the period of \_\_\_\_\_ months from \_\_\_\_\_ to \_\_\_\_\_ for the performance of:

(a) New Works, additions and alterations to existing structures, special repair works and supply of building materials subject to the contract value for such works not exceeding ₹ \_\_\_\_\_.

(b) All ordinary repair and maintenance works at any site between kilometer \_\_\_\_\_ and kilometre \_\_\_\_\_ as will be set forth in the work orders (which work orders shall be deemed and taken to be part of this contract) that will be issued during the said period at \_\_\_\_\_ % above/below the Standard Schedule of Rates (SSOR) of the \_\_\_\_\_ Railway, corrected up to the latest correction slips and Standard Specifications of the \_\_\_\_\_ Railway corrected upto latest correction slips and the Special Conditions and Special Specifications, if any in conformity with the drawings (if any) that will be issued with the work order, aforesaid AND WHEREAS the performance of the said work is an act in which the public are interested.

NOW THIS INDENTURE PRESENTS WITNESSETH That in consideration of the payment to be made by the Railway, the Contractor will duly perform the works set forth in the said Work Order and shall execute the same with great promptness, care and accuracy, in a workman like manner to the satisfaction of the Railway and will complete the same on or before the respective dates specified therein in accordance with the said specifications and said drawings (if any) and said conditions of contract and will observe, fulfill and keep all the conditions therein mentioned, (which shall be deemed and taken to be part of this contract as if the same had been duly set forth herein), AND the Railway both here-by agree that if the Contractor shall duly perform the said work in the manner aforesaid and observe and keep the said terms and conditions, the Railway will pay or cause to be paid to the Contractor for the said works on the completion thereof the amount due in respect thereof at the rates specified above.

Contractor \_\_\_\_\_

Designation \_\_\_\_\_

Address \_\_\_\_\_

Railway \_\_\_\_\_

(For President of India)

Witnesses (to signature of Contractor):

Signature of witnesses with address \_\_\_\_\_

Date \_\_\_\_\_

Signature of witnesses with address \_\_\_\_\_

Date \_\_\_\_\_

**ANNEXURE - III****WORK ORDER UNDER ZONE CONTRACT**

WORK ORDER NO. \_\_\_\_\_, DATED \_\_\_\_\_ UNDER CONTRACT AGREEMENT

NO. \_\_\_\_\_ DATED \_\_\_\_\_.

Name of Work \_\_\_\_\_ (SITE) \_\_\_\_\_

Schedule of Drawings \_\_\_\_\_

Authority \_\_\_\_\_ Allocation \_\_\_\_\_

The Contractor(s) \_\_\_\_\_ is / are hereby ordered to carry out the following works at \_\_\_\_\_% above/below the Standard Schedule of Rates (SSOR) of \_\_\_\_\_, updated with correction slips issued up to date of inviting tender or as otherwise specified in the tender documents under Zone Contract Agreement here-in-before referred to:

| SL                                        | Item No. | Description of Item of Work | Approximate Quantity | Unit | Rates in Figures and Words (₹) | Amount (₹) |
|-------------------------------------------|----------|-----------------------------|----------------------|------|--------------------------------|------------|
| 1                                         | 2        | 3                           | 4                    | 5    | 6                              | 7          |
| Total Approximate Value of Work = ₹ _____ |          |                             |                      |      |                                |            |

The works herein mentioned are required to be completed on or before \_\_\_\_\_ (Date). The quantities provided herein are approximate and subject to variation under Clause 42 of the Standard General Conditions of Contract updated with correction slips issued up to date of inviting tender or as otherwise specified in the tender documents.

Divisional Railway Manager/Divisional \_\_\_\_\_ Engineer

\_\_\_\_\_ Division

\_\_\_\_\_ Railway

Date \_\_\_\_\_

for President of India

I agree to complete the works herein set forth on or before the date specified under the Zone Contract Agreement herein before referred to in conformity with the drawings hereto annexed and in accordance with the General and Special (if any) Conditions of Contract updated with correction slips issued up to date of inviting tender or as otherwise specified in the tender documents and the Standard Specifications of \_\_\_\_\_ Railway updated with correction slips issued up to date of inviting tender or as otherwise specified in the tender documents .

I also agree to maintain such works for the period specified below from the date of completion:

(a) Repair and maintenance work including white/color washing: three calendar months from date of completion.

(b) All new works except earth work: Six calendar months from date of completion.

Contractor \_\_\_\_\_ (Signature)

Railway: Designation \_\_\_\_\_

Address \_\_\_\_\_

For President of India)

Date \_\_\_\_\_

Date \_\_\_\_\_

Signature of Witnesses (to Signature of Contractor) with address

1. \_\_\_\_\_  
\_\_\_\_\_  
\_\_\_\_\_
2. \_\_\_\_\_  
\_\_\_\_\_  
\_\_\_\_\_

\_\_\_\_\_ RAILWAY  
**CONTRACT AGREEMENT OF WORKS**

CONTRACT AGREEMENT NO. \_\_\_\_\_ DATED \_\_\_\_\_

ARTICLES OF AGREEMENT made this \_\_\_\_\_ day of \_\_\_\_\_ 20\_\_\_\_ between President of India acting through the Railway Administration hereafter called the "Railway" of the one part and \_\_\_\_\_ herein after called the "Contractor" of other part.

WHEREAS the Contractor has agreed with the Railway for performance of the works \_\_\_\_\_ set forth in the Bill(s) of Quantities hereto annexed upon the Standard General Conditions of Contract, updated with correction slips issued up to date of inviting tender or as otherwise specified in the tender documents and the Specifications of \_\_\_\_\_ updated with correction slips issued up to date of inviting tender or as otherwise specified in the tender documents and the applicable Standard Schedule of Rates (SSOR) of \_\_\_\_\_ updated with correction slips issued up to date of inviting tender or as otherwise specified in the tender documents and the Special Conditions and Special Specifications, if any and in conformity with the drawings here-into annexed AND WHEREAS the performance of the said works is an act in which the public are interested.

NOW THIS INDENTURE WITNESSETH that in consideration to the payments to be made by the Railways, the Contractors will duly perform the said works in the said Bill(s) of Quantities set forth and shall execute the same with great promptness, care and accuracy in a workman like manner to the satisfaction of the Railway and will complete the same in accordance with the said specifications and said drawings and said conditions of contract on or before the \_\_\_\_\_ day of \_\_\_\_\_ 20\_\_\_\_ and will maintain the said works for a period of \_\_\_\_\_ Calendar months from the certified date of their completion and will observe, fulfill and keep all the conditions therein mentioned (which shall be deemed and taken to be part of this contract, as if the same have been fully set forth herein), AND the Railway, both hereby agree that if the Contractor shall duly perform the said works in the manner aforesaid and observe and keep the said terms and conditions, the Railway will pay or cause to be paid to the Contractor for the said works on the final completion thereof the amount due in respect thereof at the rates specified in the Bill(s) of Quantities hereto annexed.

Contractor \_\_\_\_\_ (Signature)      Railway: Designation \_\_\_\_\_

Address \_\_\_\_\_ (For President of India)

Date \_\_\_\_\_      Date \_\_\_\_\_

Signature of **Witnesses** (to Signature of Contractor) with address:

**Witnesses:**

\_\_\_\_\_  
 \_\_\_\_\_



## ANNEXURE-V

Reference -Para 6.1 of ITT

### FORMAT FOR CERTIFICATE TO BE SUBMITTED / UPLOADED BY TENDERER ALONGWITH THE TENDER DOCUMENTS

I.....(Name and designation)\*\*appointed as the attorney/authorized signatory of the tenderer (including its constituents), ACS No. 2 of 13.12.2022

M/s\_\_\_\_\_ (hereinafter called the tenderer) for the purpose of the Tender documents for the work of \_\_\_\_\_ as per the tender No.\_\_\_\_\_ of \_\_\_\_\_(Railway)\*\*, do hereby solemnly affirm and state on the behalf of the tenderer including its constituents as under:

1. I/we the tenderer (s) am/are signing this document after carefully reading the contents.
2. I/We the tenderer(s) also accept all the conditions of the tender and have signed all the pages in confirmation thereof.
3. I/we hereby declare that I/we have downloaded the tender documents from Indian Railway website [www.ireps.gov.in](http://www.ireps.gov.in) . I/we have verified the content of the document from the website and there is no addition, no deletion or no alteration to the content of the tender document. In case of any discrepancy noticed at any stage i.e. evaluation of tenders, execution of work or final payment of the contract, the master copy available with the railway Administration shall be final and binding upon me/us.
4. I/we declare and certify that I/we have not made any misleading or false representation in the forms, statements and attachments in proof of the qualification requirements.
5. **I/We also understand that my/our offer will be evaluated based on the documents/credentials submitted along with the offer and same shall be binding upon me/us.**
6. **I/We declare that the information and documents submitted along with the tender by me/us are correct and I/we are fully responsible for the correctness of the information and documents, submitted by us.**
7. I/we certify that I/we the tenderer(s) is/are not blacklisted or debarred by Railways or any other Ministry / Department of Govt. of India from participation in tender on the date of submission of bids, either in individual capacity or as a HUF/ member of the partnership firm/LLP/JV/Society/Trust. ACS No. 1 of 14.07.2022
8. I/we understand that if the contents of the affidavit- **certificate** submitted by us are found to be forged/false or incorrect at any time during process for evaluation of tenders, it shall lead to forfeiture of the Bid Security **and may also lead to any other action provided in the contract including** besides banning of business for a period of upto five **two** year. Further, I/we (insert name of the tenderer) \*\*\_\_\_\_\_ and all my/our constituents understand that my/our offer shall be summarily rejected. ACS No. 4 of 7.08.2023 ACS No. 3 of 26.04.23

9. I/we also understand that if the contents of the ~~affidavit~~ **certificate** submitted by us are found to be false / forged or incorrect at any time after the award of the contract, it will lead to termination of the contract, along with forfeiture of Bid Security/Security Deposit and Performance guarantee **and may also lead to** besides any other action provided in the contract including banning of business for a period of upto five **two** year.

ACS No. 4 of 7.08.2023

ACS No. 3 of 26.04.23

10. I/We have read the clause regarding restriction on procurement from a bidder of a country which shares a land border with India and certify that I am/We are not from such a country or, if from such a country, have been registered with the competent Authority. I/We hereby certify that I/we fulfil all the requirements in this regard and am/are eligible to be considered (evidence of valid registration by the competent authority is enclosed)

SEAL AND SIGNATURE  
OF THE TENDERER

Place:

Dated:

\*\*The contents in Italics are only for guidance purpose. Details as appropriate are to be filled in suitably by tenderer.

**~~This affidavit~~ **certificate** is to be given by each member of JV, or partners of Partnership firm /LLP/etc.**

ACS No. 1 of 14.07.2022

ACS No. 2 of 13.12.2022

ANNEXURE-V (A)  
Reference -Para 6.1 of ITT

ACS No. 2 of 13.12.2022

*(This certificate is to be given by attorney/authorized signatory/each member of Partnership firm/Joint Venture (JV)/Hindu Undivided family (HUF)/Limited Liability Partnership (LLP) etc.)*

I/we .....(Name), attorney/authorized signatory of the  
.....(constituent firm/constituent partner) and member/partner of the  
.....(tendering firm) hereby solemnly affirm and state as under:

1. I/We certify that ..... (constituent firm/constituent partner) is/are not blacklisted or debarred by Railways or any other Ministry/Department of Govt. of India from participation in tender on the date of submission of bids, either in individual capacity or as a HUF/member of the partnership firm/LLP/JV/Society/Trust.
2. I/We have read the clause regarding restriction on procurement from a bidder of a country which shares a land border with India and certify that I am /We are not from such a country or, if from such a country, have been registered with the competent Authority. I/We hereby certify that I/We fulfil all the requirements in this regard and am/are eligible to be considered (evidence of valid registration by the competent authority is enclosed).

**SEAL AND SIGNATURE OF  
THE CONSTITUENT FIRM/CONSTITUENT PARTNER**

**Place:  
Dated:**

ACS No. 2 of 13.12.2022

\*\*\*\*\*

Reference -Para 10.3 & 17.15.3 of Tender Form (Second Sheet) of Annexure I of ITT

### TENDERER'S CREDENTIALS (BID CAPACITY)

\_\_\_\_\_ RAILWAY

For tenders having advertised value more than Rs 20 crore wherein eligibility criteria includes bid capacity also, the tenderer will be qualified only if its available bid capacity is equal to or more than the total bid value of the present tender. The available bid capacity shall be calculated as under:

$$\text{Available Bid Capacity} = [A \times N \times 2] - 0.33 \times N \times B$$

Where,

A = Maximum value of construction works executed and payment received in any one of the previous three financial years or the current financial year (up to date of inviting tender), taking into account the completed as well as works in progress.

N= Number of years prescribed for completion of work for which bids has been invited.

~~B = Value of existing commitments and balance amount of ongoing works with the tenderer as on date one month prior to the tender closing date to be completed in next 'N' years.~~

**B = Existing commitments and balance amount of ongoing works with tenderer as per the prescribed proforma of Railway for statement of all works in progress and also the works which are awarded to tenderer but yet not started upto the date of inviting of tender.** ACS No. 1 of 14.07.2022

Note:

- (a) The Tenderer(s) shall furnish the details of -
  - (i) Maximum value of construction works executed and payment received in any one of the previous three financial years or the current financial year (up to date of inviting tender) for calculating A, and
  - (ii) Existing commitments and balance amount of ongoing works with tenderer as per the prescribed proforma of Railway for statement of all works in progress and also the works which are awarded to tenderer but yet not started upto the date of inviting of tender for calculating B. In case of no works in hand, a 'NIL' statement should be furnished.

The submitted details for (i) and (ii) above should be duly verified by Chartered Accountant.
- (b) In case if a bidder is JV, the tenderer(s) must furnish the details of
  - (i) Maximum value of construction works executed and payment received in any one of the previous three financial years or the current financial year (up to date of inviting tender) by each member of JV for calculating A, and
  - (ii) Existing commitments and balance amount of ongoing works with each member of JV either in individual capacity or as a member of other JV as per the prescribed proforma of Railway for statement of all works in progress and also the works which are awarded to each member of JV either in individual capacity or as a member of other JV but yet not started upto the date of inviting of tender for calculating B. In case of no works in hand, a 'NIL' statement should be furnished.

The submitted details for (i) and (ii) above should be duly verified by Chartered Accountant

- (c) Value of a completed work/work in progress/work awarded but yet not started for a Member in an earlier JV shall be reckoned only to the extent of the concerned member's share in that JV for the purpose of satisfying his/her compliance to the above mentioned bid capacity in the tender under consideration.
- (d) The arithmetic sum of individual “bid capacity” of all the members shall be taken as JV’s “bid capacity”.
- (e) In case, the tenderer/s failed to submit the above statement along with offer, their/his offer shall be considered as incomplete and will be rejected **summarily**.
- (f) The available bid capacity of tenderer shall be assessed based on the details submitted by the tenderer. In case, the available bid capacity is lesser than estimated cost of work put to tender, his offer shall not be considered even if he has been found eligible in other eligibility criteria/tender requirement.

**Annexure –VIA**

Para 5 of the Instructions to Tenderers

**(Bid Security)**

Bank Guarantee Bond from any scheduled commercial bank of India

*(On non-judicial stamp paper, which should be in the name of the Executing Bank).*

**Name of the Bank: -----**

President of India,

Acting through .....,

..... Railway,

Beneficiary: ..... Railway

Date:.....

**Bank Guarantee Bond No.:**

**Date:-----**

In consideration of the President of India acting through----- (***Designation & address of Contract Signing Authority***), ..... Railway, ....., .... (hereinafter called "The Railway") having invited the bid for\_\_\_\_\_through Notice inviting tender (NIT) No.,\_\_\_\_\_, We have been informed that . . . . ***[Insert name of the Bidder]***..... (hereinafter called "the Bidder") intends to submit its bid (hereinafter called "the Bid") .

WHEREAS, the Bidder is required to furnish Bid Security for the sum of ***[Insert required Value of Bid Security]***, in the form of Bank Guarantee, according to conditions of Bid.

**AND**

WHEREAS, .....***[Insert Name of the Bank]***, with its Branch .....***[Insert Address]*** having its Headquarters office at..... ***[Insert Address]***, hereinafter called the **Bank**, acting through .....***[Insert Name and Designation of the authorised persons of the Bank]***, have, at the request of the Bidder, agreed to give guarantee for Bid Security as hereinafter contained, in favour of the Railway:

1. KNOW ALL MEN that by these present that I/We the undersigned ***[Insert name(s) of authorized representatives of the Bank]***, being fully authorized to sign and incur obligations for and on behalf of the Bank, confirm that the Bank, hereby, unconditionally and irrevocably guarantee to pay to the Railway full amount in the sum of ***[Insert required Value of Bid Security]*** as above stated.
2. The Bank undertakes to immediately pay on presentation of demand by the Railway any amount up to and including aforementioned full amount without any demur, reservation or recourse. Any such demand made by the Railway on the Bank shall be final, conclusive and binding, absolute and unequivocal on the Bank notwithstanding any disputes raised/ pending before any Court, Tribunal, Arbitration or any Authority or any threatened litigation by the Bidder or Bank.

3. The Bank shall pay the amount as demanded immediately on presentation of the demand by Railway without any reference to the Bidder and without the Railway being required to show grounds or give reasons for its demand of the amount so demanded.
4. The guarantee hereinbefore shall not be affected by any change in the constitution of the Bank or in the constitution of the Bidder.
5. The Bank agrees that no change, addition, modifications to the terms of the Bid document or to any documents, which have been or may be made between the Railway and the Bidder, will in any way absolve the Bank from the liability under this guarantee; and the Bank, hereby, waives any requirement for notice of any such change, addition or modification made by Railway at any time.
6. This guarantee will remain valid and effective from.....*[insert date of issue]*till .....*[insert date, which should be minimum 90 days beyond the expiry of validity of Bid]*. Any demand in respect of this Guarantee should reach the Bank within the validity period of Bid Security.
7. The Bank Guarantee is unconditional and irrevocable.
8. The expressions Bank and Railway herein before used shall include their respective successors and assigns.
9. The Bank hereby undertakes not to revoke the guarantee during its currency, except with the previous consent in writing of the Railway. This guarantee is subject to the Uniform Rules for Demand Guarantees, ICC Publication No.758.
10. The Bank hereby confirms that it is on the SFMS (Structured Financial Messaging System) and shall invariably send the advice of this Bank Guarantee to the following bank details –

|             |                                     |
|-------------|-------------------------------------|
| IFSC CODE   | SBIN000RAIL                         |
| IFSC TYPE   | BRANCH                              |
| BANK NAME   | STATE BANK OF INDIA                 |
| BRANCH NAME | RAIL                                |
| CITY NAME   | NAVI MUMBAI                         |
| ADDRESS     | SECTOR-11, CBD BELAPUR, NAVI MUMBAI |
| DISTRICT    | NAVI MUMBAI                         |
| STATE       | MAHARASHTRA                         |
| BG ENABLED  | YES                                 |

11. The Guarantee shall be valid in addition to and without prejudice to any other security Guarantee(s) of Bidder in favour of the Railway. The Bank, under this Guarantee, shall be deemed as Principal Debtor of the Railway.

Date ..... ..

Place.....

Bank's Seal and authorized signature(s)

*[Name in Block letters]* .....

*[Designation with Code No.]* .....

*[P/Attorney]* No.

Witness:

1 Signature, Name & Address & Seal

2 Signature, Name& address & Seal

Bank's Seal

*[P/Attorney]*No.

**Note:** All italicized text is for guidance on how to prepare this bank guarantee and shall be deleted from the final document.



Reference -Para 10.2 & 17.15.2 of Tender Form (Second Sheet) of Annexure I of ITT

**Each Bidder or each member of a JV must fill in this form separately:**

**NAME OF BIDDER/JV PARTNER:**

| <b>Annual Contractual Turnover Data for the Previous 3/4 Years<br/>(Contractual Payment only)</b> |                            |                          |                                              |
|---------------------------------------------------------------------------------------------------|----------------------------|--------------------------|----------------------------------------------|
| <b>Year</b>                                                                                       | <b>Amount<br/>Currency</b> | <b>Exchange<br/>Rate</b> | <b>Indian National Rupees<br/>Equivalent</b> |
|                                                                                                   |                            |                          |                                              |
|                                                                                                   |                            |                          |                                              |
|                                                                                                   |                            |                          |                                              |
|                                                                                                   |                            |                          |                                              |
|                                                                                                   |                            |                          |                                              |
| <b>Average Annual Contractual Turnover for last 3 years</b>                                       |                            |                          |                                              |

1. The average annual contractual turnover shall be calculated as an average of “total contractual payments” in the previous three financial years. However, in case balance sheet of the previous year is yet to be prepared/ audited, the audited balance sheet of the fourth previous year shall be considered for calculating average annual contractual turnover.
2. The information supplied shall be substantiated by data in the audited balance sheets and profit and loss accounts for the relevant years in respect of the bidder or all members constituting the bidder.
3. Contents of this form should be certified by a Chartered Accountant duly supported by Audited Balance Sheet duly certified by the Chartered Accountant.

***SEAL AND SIGNATURE OF THE BIDDER***

Certified that all figures and facts submitted in this form have been furnished after full consideration of all observations/notes in Auditor’s reports. \_\_\_\_\_

***(Signature of Chartered Accountant)***

**Name of CA:** \_\_\_\_\_

**Registration No:** \_\_\_\_\_

***(Seal)***

## Part II

### STANDARD GENERAL CONDITIONS OF CONTRACT

**1. (1) Definitions:** In these Standard General Conditions of Contract, the following terms shall have the meaning assigned hereunder except where the context otherwise requires:

- (a) "Railway" shall mean the President of the Republic of India or the Administrative Officers of the Railway or of the Successor Railway authorized to deal with any matters which these presents are concerned on his behalf.
- (b) "General Manager" shall mean the Officer in-charge of the General Superintendence and Control of the Railway and shall also include Addl. General Manager, the General Manager (Construction) and shall mean and include their successors, of the successor Railway.
- (c) "Chief Engineer" shall mean the Officer in-charge of the Engineering Department of Railway and shall also include Chief Engineer (Construction), Chief Signal & Telecommunication Engineer, Chief Signal & Telecommunication Engineer (Construction), Chief Electrical Engineer, Chief Electrical Engineer (Construction), Chief Mechanical Engineer and shall mean & include their successors, of the Successor Railway.
- (d) "Divisional Railway Manager" shall mean the Officer in-charge of a Division of the Railway and shall mean and include the Divisional Railway Manager of the Successor Railway.
- (e) "Engineer" shall mean the Divisional Engineer or the Executive Engineer, Divisional Signal & Telecom Engineer, Divisional Electrical Engineer, Divisional Mechanical Engineer in executive charge of the works and shall include the superior officers of Open Line and Construction organisations on the Railway of the Engineering, Signal & Telecom, Electrical and Mechanical Departments, i.e. the Senior Divisional Engineer/Deputy Chief Engineer, Senior Divisional Signal & Telecom Engineer / Dy.Chief Signal & Telecom Engineer, Senior Divisional Electrical Engineer / Deputy Chief Electrical Engineer, Senior Divisional Mechanical Engineer and shall mean & include the Engineers of the Successors Railway.
- (f) "Engineer's Representative" shall mean the Assistant Engineer, Assistant Signal & Telecommunication Engineer and Assistant Electrical Engineer, Assistant Mechanical Engineer in direct charge of the works and shall include any Sr. Section/Junior Engineer of Civil Engineering/Signal and Telecommunication Engineering/Mechanical Engineering/Electrical Engineering Departments appointed by the Railway and shall mean and include the Engineer's Representative of the Successor Railway.
- (g) "Contractor" shall mean the Person/Firm/LLP/Trust/Co-operative Society or Company whether incorporated or not who enters into the contract with the Railway and shall include their executors, administrators, successors and permitted assigns.
- (h) "Contract" shall mean and include the Agreement, the Work Order, the accepted Bill(s) of Quantities or Chapter(s) of Standard Schedule of Rates (SSOR) of the Railway modified by the tender percentage for items of works quantified, or not quantified, the Standard General Conditions of Contract, the Special Conditions of Contracts, if any; the Drawing, the Specifications, the Special Specifications, if any and Tender Forms, if any.

- (i) "Works" shall mean the works to be executed in accordance with the contract.
- (j) "Specifications" shall mean the Standard Specifications for Materials & Works of Railway as specified by Railway under the authority of the Chief Engineer or as amplified, added to or superseded by Special Specifications, if any.
- (k) Standard Schedule of Rates (SSOR) shall mean the schedule of Rates adopted by the Railway, which includes-
1. "Unified Standard Schedule of Rates of the Railway (USSOR)" i.e. the Standard Schedule of Rates of the Railway issued under the authority of the Chief Engineer from time to time, updated with correction slips issued up to date of inviting tender or as otherwise specified in the tender documents;
  2. "Delhi Schedule Of Rates (DSR)" i.e. the Standard Schedule of Rates published by Director General/ Central Public Works Department, Government of India, New Delhi, as adopted and modified by the Railway under the authority of the Chief Engineer from time to time, updated with correction slips issued up to date of inviting tender or as otherwise specified in the tender documents.
- (l) "Drawing" shall mean the maps, drawings, plans and tracings or prints there of annexed to the contract and shall include any modifications of such drawings and further drawings as may be issued by the Engineer from time to time.
- (m) "Constructional Plant" shall mean all appliances or things of whatsoever nature required for the execution, completion or maintenance of the works or temporary works (as hereinafter defined) but does not include materials or other things intended to form or forming part of the permanent work.
- (n) "Temporary Works" shall mean all temporary works of every kind required for the execution completion and/or maintenance of the works.
- (o) "Site" shall mean the lands and other places on, under, in or through which the works are to be carried out and any other lands or places provided by the Railway for the purpose of the contract.
- (p) "Period of Maintenance" shall mean the specified period of maintenance from the date of completion of the works, as certified by the Engineer.
- (q) 'Contractor's authorized Engineer' shall mean a graduate Engineer or equivalent, having more than 3 years experience in the relevant field of construction work involved in the contract, duly approved by Engineer.
- (r) Date of inviting tender shall be the date of publishing tender notice on IREPS website if tender is published on website or the date of publication in newspaper in case tender is not published on website.
- (s) "Bill of Quantities" shall mean Schedule of Item(s) included in the tender document along with respective quantities and rates, accepted by the Railway.

**1. (2) Singular and Plural:** Words importing the singular number shall also include the plural and vice versa where the context requires.

**1.(3) Headings and Marginal Headings:** The headings and marginal headings in these Standard General Conditions are solely for the purpose of facilitating reference and shall not be deemed to be part thereof or be taken into consideration in the interpretation or construction thereof the contract.

## **GENERAL OBLIGATIONS**

**2. (1) Execution Co-Relation and Intent of Contract Documents:** The contract documents shall be signed in triplicate by the Railway and the Contractor. The contract documents are complementary and what is called for by anyone shall be as binding as if called for by all, the intention of the documents is to include all labour and materials, equipments and transportation necessary for proper execution of work. Materials or works not covered by or properly inferable from any heading or class of the specifications shall not be supplied by the Railway to the Contractors unless distinctly specified in the contract documents. Materials or works described in words which so applied have a well-known technical or trade meaning, shall be held to refer to such recognized standards.

**2.(2)** If a work is transferred from the jurisdiction of one Railway to another Railway or to a Project authority or vice versa while contract is in subsistence, the contract shall be binding on the Contractor and the Successor Railway/Project in the same manner & take effect in all respects as if the Contractor and the Successor Railway/Project were parties thereto from the inception and the corresponding officer or the Competent Authority in the Successor Railway/Project will exercise the same powers and enjoy the same authority as conferred to the Predecessor Railway/Project under the original contract/agreement entered into.

**2. (3)** If for administrative or other reasons the contract is transferred to the Successor Railway, the contract shall, notwithstanding any things contained herein contrary there to, be binding on the Contractor and the Successor Railway in the same manner and take effect in all respects as if the Contractor and the Successor Railway had been parties thereto from the date of this contract.

**3. (1) Law Governing the Contract:** The contract shall be governed by the law for the time being in force in the Republic of India.

**3.(2) Compliance to Regulations and Bye-Laws:** The Contractor shall conform to the provision of any statute relating to the works and regulations and bye-laws of any local authority and of any water and lighting companies or undertakings, with whose system the work is proposed to be connected and shall before making any variation from the drawings or the specifications that may be necessitated by so confirming give to the Engineer notice specifying the variation proposed to be made and the reason for making the variation and shall not carry out such variation until he has received instructions from the Engineer in respect thereof. The Contractor shall be bound to give all notices required by statute, regulations or bye-laws as aforesaid and to pay all fees and taxes payable to any authority in respect thereof.

### **3.(3) Environmental and Forest clearances:**

The Railway represents and warrants that the environmental and forest clearances pertaining to the work commensurate with the progress of work/agreed programme, will be obtained by Engineer. In

the event of any delay in securing respective clearances leading to delay in execution of work, the Contractor shall be entitled to Extension of Time for the period of such delay in accordance with the provisions of Clause-17A(ii).

**4. Communications to be in Writing:** All notices, communications, reference and complaints made by the Railway or the Engineer or the Engineer's Representative or the Contractor inter-se concerning the works shall be in writing or e-mail on registered e-mail IDs i.e. the e mail id provided for correspondence in the contract agreement, otherwise email id registered with IREPS and no notice, communication, reference or complaint not in writing or through e-mail, shall be recognized.

**5. Service of Notices on Contractors:** The Contractor shall furnish to the Engineer the name, designation and address of his authorized agent and all complaints, notices, communications and references shall be deemed to have been duly given to the Contractor, if delivered to the Contractor or his authorized agent or left at or posted to the address so given and shall be deemed to have been so given in the case of posting on day on which they would have reached such address in the ordinary course of post/ e-mail or on the day on which they were so delivered or left. In the case of contract by partners, any change in the constitution of the firm shall be forthwith notified by the Contractor to the Engineer.

**6. Occupation and Use of Land:** No land belonging to or in the possession of the Railway shall be occupied by the Contractor without the permission of the Railway. The Contractor shall not use, or allow to be used the site for any purposes other than that of executing the works. Whenever non-railway bodies/persons are permitted to use railway premises with competent authority's approval, conservancy charges as applicable from time to time may be levied.

**7. Assignment or Subletting of Contract:** The Contractor shall not assign or sublet the contract or any part thereof or allow any person to become interested therein in any manner whatsoever without the special permission in writing of the Chief Engineer, save as provided below. Any breach of this condition shall entitle the Railway to rescind the contract under Clause 62 of these Conditions and also render the Contractor liable for payment to the Railway in respect of any loss or damage arising or ensuing from such cancellation; provided always that execution of the details of the work by petty Contractor under the direct and personal supervision of the Contractor or his agent shall not be deemed to be sub-letting under this clause.

In case Contractor intends to subcontract part of work, he shall submit a proposal in writing seeking permission of Chief Engineer for the same. While submitting the proposal to railway, Contractor shall ensure the following:

(a) (i) Total value of work to be assigned to sub-contractor(s) shall not be more than 50% of total contract value.

(ii) The subcontractor shall have successfully completed at least one work similar to work proposed for subcontract in last 5 years, ending date of submission of proposal by Contractor to Railway, costing not less than 35% value of work to be subletted, through a works contract. For fulfilment of above, Work Experience Certificate issued by a Govt. Department/Organisation shall be considered. Further, Work Experience Certificate issued by a Public listed company shall be considered provided the company is having average annual turnover of Rs 500 crore and above in last 3 financial years excluding the current financial year, listed on National Stock Exchange or Bombay Stock Exchange, registered at least 5 years back from the date of

submission of proposal by Contractor to Railway and work experience certificate issued by a person authorised by the Public Listed Company to issue such certificates.

*Note: for subletting of work costing up to Rs 50 lakh no previous work experience shall be asked for by the Railway.*

In case contractor submits subcontractor's work experience certificate issued by public listed company, the contractor shall also submit along with work experience certificate, the relevant copy of work order, bill of quantities, bill wise details of payment received duly certified by Chartered Accountant, TDS certificates for all payments received and copy of final/last bill paid by company in support of above work experience certificate.

- (iii) There is no banning of business with the sub-contractor in force over IR.
- (b) The Contractor shall provide to the Engineer a copy of the agreement to be entered into by Contractor with subcontractor. No subcontractor shall be permitted without a formal agreement between Contractor and subcontractor. This agreement shall clearly define the scope of work to be carried out by subcontractor and the terms of payment in clear & unambiguous manner.
- (c) On receipt of approval from Chief Engineer, Contractor shall enter into a formal agreement legally enforceable in Court of Law with subcontractor and submit a copy of the same to the Engineer.
- (d) The Contractor shall intimate to the Engineer not less than 7 days in advance, the intended date of commencement of subcontractor's work.
- (e) Once having entered into above arrangement, Contractor shall discontinue such arrangement, if he intends to do so at his own or on the instructions of Railway, with prior intimation to Chief Engineer.
- (f) The Contractor shall indemnify railway against any claim of subcontractor.
- (g) The Contractor shall release payment to the Sub-contractor(s) promptly and shall endeavour to resolve all issues amicably and speedily with the Sub-contractor(s), so that the execution of work is not affected in any manner whatsoever.
- (h) In addition to issuance of work experience certificate to Contractor, the Engineer, when, based on documents, is satisfied that subcontracted work has been carried out by subcontractor, shall issue work experience certificate to the subcontractor also for the portion of work subcontracted and successfully completed by the sub-contractor.

*Note: Work Experience Certificate to the subcontractor shall be issued only when the contractor's work is complete and contractor is entitled for the issuance of Work Experience Certificate. However, in the same contract, when the Chief Engineer, based on documents, is satisfied that the subcontractor has successfully carried out subletted work; without issuance of work experience certificate to subcontractor at this stage, the Chief Engineer can, **only once**, consider the successfully completed subletted work for the fulfilment of eligibility for further subletting of work to the subcontractor in the same contract. When the contractor's work is complete and contractor is entitled for the issuance of work experience certificate, the subcontractor shall be issued one Work Experience Certificate for the total scope of work executed by the subcontractor in the contract.*

- (i) The responsibility of successful completion of work by subcontractor shall lie with Contractor. Subcontracting will in no way relieve the Contractor to execute the work as per terms of the Contract.
- (j) Further, in case Engineer is of the view that subcontractor's performance is not satisfactory, he may instruct the Contractor to remove the subcontractor from the work and Contractor has to comply with the above instructions with due promptness. Contractor shall intimate the actual date of discontinuation of subcontract to Engineer. No claim of Contractor whatsoever on this account shall be entertained by the Railway and this shall be deemed as 'excepted matter' (matter not arbitrable).
- (k) The permitted subcontracting of work by the Contractor shall not establish any contractual relationship between the sub-contractor and the Railway and shall not relieve the Contractor of any responsibility under the Contract.

**8. Assistance by Railway for the Stores to be obtained by the Contractor:** Owing to difficulty in obtaining certain materials (including Tools & Plant) in the market, the Railway may have agreed without any liability therefore to endeavour to obtain or assist the Contractor in obtaining the required quantities of such materials as may be specified in the Tender. In the event of delay or failure in obtaining the required quantities of the aforesaid material, the Contractor shall not be deemed absolved of his own responsibility and shall keep in touch with the day to day position regarding their availability and accordingly adjust progress of works including employment of labour and the Railway shall not in any way be liable for the supply of materials or for the non-supply thereof for any reasons whatsoever nor for any loss or damage arising in consequence of such delay or non-supply.

**9. Railway Passes:** No free railway passes shall be issued by the Railway to the Contractor or any of his employee/worker.

**10. Carriage of Materials:** No forwarding orders shall be issued by the Railway for the conveyance of Contractor's materials, tools and plant by train which may be required for use in the works and the Contractor shall pay full freight charges at public tariff rates therefor.

**11. Use of Ballast Trains:** The Railway may agree to allow the Contractor use of the ballast or material trains under such conditions as shall be specially prescribed, provided that the Contractor shall pay for the use thereof charges calculated at public tariff rates on the marked carrying capacity of each vehicle subject to specified minimum charge per day or part of day and provided further that the Contractor shall indemnify the Railway against any claims or damages arising out of the use or misuse thereof and against any liabilities under the Workmen's Compensation Act, 1923 or any statutory amendments thereto.

**12. Representation on Works:** The Contractor shall, when he is not personally present on the site of the works place, keep a responsible agent at the works during working hours who shall on receiving reasonable notice, present himself to the Engineer and orders given by the Engineer or the Engineer's representative to the agent shall be deemed to have the same force as if they had been given to the Contractor. Before absenting himself, the Contractor shall furnish the name and address of his agent for the purpose of this clause and failure on the part of the Contractor to comply with this

provision at any time will entitle the Railway to rescind the contract under Clause 62 of these Conditions.

**13. Relics and Treasures:** All gold, silver, oil, other minerals of any description, all precious stones, coins, treasures relics antiquities and other similar things which shall be found in or upon the site shall be the property of the Railway and the Contractor shall duly preserve the same to the satisfaction of the Railway and shall from time to time deliver the same to such person or persons as the Railway may appoint to receive the same.

**14. Excavated Material:** The Contractor shall not sell or otherwise dispose of or remove except for the purpose of this contract, the sand, stone, clay ballast, earth, trees, rock or other substances or materials which may be obtained from any excavation made for the purpose of the works or any building or produced upon the site at the time of delivery of the possession thereof but all the substances, materials, buildings and produce shall be the property of the Railway provided that the Contractor may, with the permission of the Engineer, use the same for the purpose of the works either free of cost or pay the cost of the same at such rates as may be determined by the Engineer.

**15. Indemnity by Contractors:** The Contractor shall indemnify and save harmless the Railway from and against all actions, suit, proceedings, losses, costs, damages, charges, claims and demands of every nature and description brought or recovered against the Railways by reason of any act or omission of the Contractor, his agents or employees, in the execution of the works or in his guarding of the same. All sums payable by way of compensation under any of these conditions shall be considered as reasonable compensation to be applied to the actual loss or damage sustained, and whether or not any damage shall have been sustained.

**16.(1) Security Deposit:** The Security Deposit shall be 5% of the contract value. The Bid Security submitted by the Contractor with his tender will be retained/encashed by the Railways as part of security for the due and faithful fulfillment of the contract by the Contractor. Provided further that, if Contractor submits the Cash or Term Deposit Receipt issued from a Scheduled commercial bank of India or irrevocable Bank Guarantee Bond from a Scheduled commercial bank of India, either towards the Full Security Depositor the Part Security Deposit equal to or more than Bid Security, the Railway shall return the Bid Security, to the Contractor.

Balance of Security Deposit may be deposited by the Contractor in cash or Term Deposit Receipt issued from Scheduled commercial bank of India or irrevocable Bank Guarantee bond issued from Scheduled commercial bank of India, or may be recovered at the rate of 6% of the bill amount till the full Security Deposit is recovered. Provided also that in case of defaulting Contractor, the Railway may retain any amount due for payment to the Contractor on the pending "on account bills" so that the amounts so retained (including amount guaranteed through Performance Guarantee) may not exceed 10% of the total value of the contract.

The Irrevocable Bank Guarantee submitted towards Security deposit shall be initially valid up to the stipulated date of Maintenance period plus 60 days and shall be extended from time to time, depending upon extension of contract granted in terms of Clause 17A and 17B of the Standard General Conditions of Contract.

Note: Security Deposit deposited in cash by the Contractor or recovered from the running bills of a Contractor or submitted by contractor as Term Deposit Receipt(s) can be refunded/returned to the



contractor, in lieu of irrevocable Bank Guarantee bond issued from scheduled commercial bank of India, to be submitted by him, for an amount equal to or more than the already available Security Deposit, provided however that, in a contract of value less than Rs. 50 Crore, such refund/ return of the already available Security Deposit is permitted up to two times and in a contract of value equal to or more than Rs. 50 Crore, such refund / return of the already available Security Deposit is permitted up to three times.

**16.(2) (i) Refund of Security Deposit:** Security Deposit mentioned in sub clause (1) above shall be returned to the Contractor along with or after, the following:

- (a) Final Payment of the Contract as per clause 51.(1) **and**
- (b) Execution of Final Supplementary Agreement or Certification by Engineer that Railway has No Claim on Contractor **and**
- (c) Maintenance Certificate issued, on expiry of the maintenance period as per clause 50.(1), in case applicable.

**16. (2) (ii) Forfeiture of Security Deposit:** Whenever the contract is rescinded as a whole under clause 62 (1) of these conditions, the Security Deposit already with railways under the contract shall be forfeited. However, in case the contract is rescinded in part or parts under clause 62 (1) of these conditions, the Security Deposit shall not be forfeited.

**16.(3)** No interest shall be payable upon the Bid Security and Security Deposit or amounts payable to the Contractor under the Contract, but Government Securities deposited in terms of Sub-Clause 16.(4)(b) of this clause will be payable with interest accrued thereon.

**16.(4) Performance Guarantee**

The procedure for obtaining Performance Guarantee is outlined below:

(a) The successful bidder shall have to submit a Performance Guarantee (PG) within 21 (Twenty one) days from the date of issue of Letter of Acceptance (LOA). Extension of time for submission of PG beyond 21 (Twenty one) days and upto 60 days from the date of issue of LOA may be given by the Authority who is competent to sign the contract agreement. However, a penal interest of 12% per annum shall be charged for the delay beyond 21(Twenty one) days, i.e. from 22<sup>nd</sup> day after the date of issue of LOA. Further, if the 60<sup>th</sup> day happens to be a declared holiday in the concerned office of the Railway, submission of PG can be accepted on the next working day.

In all other cases, if the Contractor fails to submit the requisite PG even after 60 days from the date of issue of LOA, the contract is liable to be terminated. In case contract is terminated railway shall be entitled to forfeit Bid Security and other dues payable to the contractor against that particular contract, subject to maximum of PG amount. In case a tenderer has not submitted Bid Security on the strength of their registration as a Startup recognized by Department of Industrial Policy and Promotion (DIPP) under Ministry of Commerce and Industry, DIPP shall be informed to this effect.

The failed Contractor shall be debarred from participating in re-tender for that work.

- (b) The successful bidder shall submit the Performance Guarantee (PG) in any of the following forms, amounting to 5% of the original contract value:-
  - (i) A deposit of Cash;
  - (ii) Irrevocable Bank Guarantee;

- (iii) Government Securities including State Loan Bonds at 5% below the market value;
  - (iv) Pay Orders and Demand Drafts tendered by any Scheduled Commercial Bank of India;
  - (v) Guarantee Bonds executed or Deposits Receipts tendered by any Scheduled Commercial Bank of India;
  - (vi) Deposit in the Post Office Saving Bank;
  - (vii) Deposit in the National Savings Certificates;
  - (viii) Twelve years National Defence Certificates;
  - (ix) Ten years Defence Deposits;
  - (x) National Defence Bonds and
  - (xi) Unit Trust Certificates at 5% below market value or at the face value whichever is less. Also, FDR in favour of FA&CAO (free from any encumbrance) may be accepted.
- (c) The Performance Guarantee shall be submitted by the successful bidder after the Letter of Acceptance (LOA) has been issued, but before signing of the contract agreement. This P.G. shall be initially valid upto the stipulated date of completion plus 60 days beyond that. In case, the time for completion of work gets extended, the Contractor shall get the validity of P.G. extended to cover such extended time for completion of work plus 60 days.
- (d) The value of PG to be submitted by the Contractor is based on original contract value and shall not change due to subsequent variation(s) in the original contract value.
- (e) The Performance Guarantee (PG) shall be released after physical completion of the work based on 'Completion Certificate' issued by the competent authority stating that the Contractor has completed the work in all respects satisfactorily.
- (f) Whenever the contract is rescinded, the Performance Guarantee already submitted for the contract shall be encashed.
- (g) The Engineer shall not make a claim under the Performance Guarantee except for amounts to which the President of India is entitled under the contract (not withstanding and/or without prejudice to any other provisions in the contract agreement) in the event of:
- (i) Failure by the Contractor to extend the validity of the Performance Guarantee as described herein above, in which event the Engineer may claim the full amount of the Performance Guarantee.
  - (ii) Failure by the Contractor to pay President of India any amount due, either as agreed by the Contractor or determined under any of the Clauses/Conditions of the Agreement, within 30 days of the service of notice to this effect by Engineer.
  - (iii) The Contract being determined or rescinded under clause 62 of these conditions.

**17. Force Majeure Clause:** If at any time, during the continuance of this contract, the performance in whole or in part by either party of any obligation under this contract shall be prevented or delayed by reason of any war, hostility, acts of public enemy, civil commotion, sabotage, serious loss or damage by fire, explosions, epidemics/pandemics, strikes, lockouts or acts of God (hereinafter, referred to events) provided, notice of the happening of any such event is given by either party to the other within 30 days from the date of occurrence thereof, neither party shall by reason of such event, be entitled to terminate this contract nor shall either party have any claim for damages against the other in respect of such non-performance or delay in performance, and works under the contract shall be resumed as soon as practicable after such event has come to an end or ceased to exist, and the decision of the Engineer as to whether the works have been so resumed or not

shall be final and conclusive, PROVIDED FURTHER that if the performance in whole or in part of any obligation under this contract is prevented or delayed by reason of any such event for a period exceeding 120 days, either party may at its option terminate the contract by giving notice to the other party.

**17A Extension of Time in Contracts:** Subject to any requirement in the contract as to completion of any portion or portions of the works before completion of the whole, the Contractor shall fully and finally complete the whole of the works comprised in the contract (with such modifications as may be directed under conditions of this contract) by the date entered in the contract or extended date in terms of the following clauses:

- (i) **Extension due to Modification:** If any modifications have been ordered which in the opinion of the Engineer have materially increased the magnitude of the work, then such extension of the contracted date of completion may be granted as shall appear to the Engineer to be reasonable in the circumstances, provided moreover that the Contractor shall be responsible for requesting such extension of the date as may be considered necessary as soon as the cause thereof shall arise.
- (ii) **Extension for Delay not due to Railway or Contractor:** If in the opinion of the Engineer, the progress of work has any time been delayed by any act or neglect of Railway's employees or by other Contractor employed by the Railway under Sub-Clause (4) of Clause 20 of these Conditions or in executing the work not forming part of the contract but on which Contractor's performance necessarily depends or by reason of proceeding taken or threatened by or dispute with adjoining or to neighbouring owners or public authority arising otherwise through the Contractor's own default etc. or by the delay authorized by the Engineer pending arbitration or in consequences of the Contractor not having received in due time necessary instructions from the Railway for which he shall have specially applied in writing to the Engineer or his authorized representative then upon happening of any such event causing delay, the Contractor shall immediately give notice thereof in writing to the Engineer within 15 days of such happening, but shall nevertheless make constantly his best endeavours to bring down or make good the delay and shall do all that may be reasonably required of him to the satisfaction of the Engineer to proceed with the works. The Contractor may also indicate the period for which the work is likely to be delayed and shall be bound to ask for necessary extension of time.
- (iii) **Extension for Delay due to Railways:** In the event of any failure or delay by the Railway to hand over the Contractor possession of the lands necessary for the execution of the works or to give the necessary notice to commence the works or to provide the necessary drawings or instructions or any other delay caused by the Railway due to any other cause whatsoever, then such failure or delay shall in no way affect or vitiate the contract or alter the character thereof or entitle the Contractor to damages or compensation therefor, but in any such case, the Railway may grant such extension or extensions of the completion date as may be considered reasonable.

The Contractor shall indicate the period for which the work is likely to be delayed and shall seek extension of time as may be considered necessary under clause 17A(i) or/and 17A(ii) or/ and 17A(iii) above, as soon as the cause thereof shall arise and, in any case, not less than ~~one month~~ **15 days** before the expiry of the date fixed for completion of the works. The Engineer shall consider the same and shall grant and communicate such extension of time as in his opinion is reasonable having regard to the nature and period of delay and the type and quantum of work affected thereby. No other compensation shall be payable for works so carried forward to the extended period of time; the same rates, terms and conditions of contract being applicable, as if such extended period of time was originally provided in the original contract itself.

**ACS No. 1 of 14.07.2022**

The non-submission of request for extension or submission of request within less than ~~one month~~ **15 days** before the expiry of the date fixed for completion of the works, shall make him ineligible for extension under these sub clauses, subject to final decision of Engineer. **ACS No. 1 of 14.07.2022**

**17B Extension of Time with Liquidated Damages (LD) for delay due to Contractor:** The time for the execution of the work or part of the works specified in the contract documents shall be deemed to be the essence of the contract and the works must be completed not later than the date(s) as specified in the contract. If the Contractor fails to complete the works within the time as specified in the contract for the reasons other than the reasons specified in Clause 17 and 17A, the Railway may, if satisfied that the works can be completed by the Contractor within reasonable short time thereafter, allow the Contractor for further extension of time (Proforma at Annexure-VII) as the Engineer may decide. On such extension the Railway will be entitled without prejudice to any other right and remedy available on that behalf, to recover from the Contractor as agreed damages and not by way of penalty for each week or part of the week, a sum calculated at the ~~following rates of the contract value of the works.~~ **rate of Liquidated Damages as decided by Engineer between 0.05 % to 0.30 % of the contract value of the works for each week or part of the week.** **ACS No. 1 of 14.07.2022**

For the purpose of this Clause, the contract value of the works shall be taken as value of work as per contract agreement including any supplementary work order/contract agreement issued. Provided also, that the total amount of liquidated damages under this condition shall not exceed 5% of the contract value or of the total value of the item or groups of items of work for which a separate distinct completion period is specified in the contract.

| S.No. | <del>Duration of extension of time under Clause 17B</del>                                                                                                       | Rate of Liquidated Damages                                                                                    |
|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| (i)   | <del>Up to Twenty Five percent of original period of completion including period of extension of time granted under Section 17A(i)</del>                        | <del>As decided by Engineer, between 0.01% to 0.05% of contract value for each week or part of the week</del> |
| (ii)  | <del>Above Twenty Five percent but upto Fifty percent of original period of completion including period of extension of time granted under Section 17A(i)</del> | 0.10 % of contract value for each week or part of the week                                                    |
| (iii) | <del>Above Fifty percent of original period of completion including period of extension of time granted under Section 17A(i)</del>                              | 0.30 % of contract value for each week or part of the week                                                    |

Provided further, that if the Railway is not satisfied that the works can be completed by the Contractor and in the event of failure on the part of the contractor to complete the work within further extension of time allowed as aforesaid, the Railway shall be entitled without prejudice to any other right or remedy available in that behalf, to appropriate the contractor's Security Deposit and rescind the contract under Clause 62 of these Conditions, whether or not actual damage is caused by such default.

NOTE:

In a contract, where extension(s) of time have been allowed once under clause 17B, further request(s)

for extension of time under clause 17A can also be considered under exceptional circumstances. Such extension(s) of time under clause 17A shall be without any Liquidated damages, but the Liquidated damages already recovered during extension(s) of time granted previously under clause 17B shall not be waived. However, Price variation during such extension(s) shall be dealt as applicable for extension(s) of time under clause 17B.

**17C Bonus for Early Completion of Work:** In open tenders having advertised value more than Rs.50 crore and original period of completion 12 months or more, when there is no reduction in original scope of work by more than 10%, and no extension granted on either railway or Contractor's account, Contractor shall be entitled for a bonus of 1% for each 30 days early completion of work. The period of less than 30 days shall be ignored while working out bonus. The maximum bonus shall be limited to 5% of original contract value. The completion date shall be reckoned as the date of issuance of completion certificate by Engineer.

**18.(1) Illegal Gratification:** Any bribe, commission, gift or advantage given, promised or offered by or on behalf of the Contractor or his partner or agent or servant or anyone on his behalf, to any officer or employee of the Railway or to any person on his behalf in relation to obtaining or execution of this or any other contract with the Railway shall, in addition to any criminal liability which he may incur, subject Contractor to the rescission of the contract and all other contracts with the Railway and to the payment of any loss or damage resulting from such decision and the Railway shall be entitled to deduct the amounts so payable from the Contractor's bills/Security Deposit or any other dues of Contractor with the Government of India.

**18.(2)** The Contractor shall not lend or borrow from or have or enter into any monetary dealings or transactions either directly or indirectly with any employee of the Railway and if he shall do so, the Railway shall be entitled forthwith to rescind the contract and all other contracts with the Railway. Any question or dispute as to the commission of any such offence or compensation payable to the Railway under this Clause shall be settled by the General Manager of the Railway, in such a manner as he shall consider fit & sufficient and his decision shall be final & conclusive. In the event of rescission of the contract under this Clause, the Contractor will not be paid any compensation whatsoever except payments for the work done upto the date of rescission.

## EXECUTION OF WORKS

**19.(1) Contractor's understanding:** It is understood and agreed that the Contractor has, by careful examination, satisfied himself as to the nature and location of the work, the conformation of the ground, the character, quality and quantity of the materials to be encountered, the character of equipment and facilities needed preliminary to and during the progress of the works, the general and local conditions, the labour conditions prevailing therein and all other matters which can in any way affect the works under the contract.

**19.(2) Commencement of Works:** The Contractor shall commence the works within 15 days after the receipt by him of an order in writing to this effect from the Engineer and shall proceed with the same with due expedition and without delay

**19.(3) Accepted Programme of Work:** The Contractor who has been awarded the work shall as soon as possible but not later than 30 days after the date of receipt of the acceptance letter in respect

of contracts with initial completion period of two years or less or not later than 90 days for other contracts have to submit the detailed programme of work indicating the time schedule of various items of works in the form of Bar Chart/PERT/CPM. He shall also submit the details of organisation (in terms of labour and supervisors), plant and machinery that he intends to utilize (from time to time) for execution of the work within stipulated date of completion. The programme of work amended as necessary by discussions with the Engineer, shall be treated as the agreed programme of the work for the purpose of this contract and the Contractor shall endeavor to fulfill this programme of work. The progress of work will be watched accordingly and the liquidated damages will be with reference to the overall completion date. Nothing stated herein shall preclude the Contractor in achieving earlier completion of item or whole of the works than indicated in the programme.

In Contracts for works of New Line/Gauge Conversion/Doubling/Railway Electrification, finalized through Tenders having advertised value more than ~~Rs.50 crores~~ **Rs.100 crores**, the Contractor shall submit a detailed time programme to the Engineer within 30 days after issue of LOA. The program shall include the physical and Financial Progress vis-à-vis program and forecast cash flow adopting Project Management Software such as **Primavera/Sure Track/MS Project etc.** The program must identify the milestones, interface requirements and program reporting elements. The Contractor shall supply, free of cost one set of authorized software to the Engineer and the soft copy of structured program for the project. This shall be updated every month. The Contractor shall also submit a revised programme whenever the previous programme is inconsistent with actual progress. Each programme shall include:

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The order in which the Contractor intends to carry out the Works, including the anticipated timing of each stage, Contractor's Documents, procurement, manufacture of Plant, delivery to Site, construction, erection and testing, each of these stages for work by each Subcontractor, if any, the sequence and timing of inspections and tests specified in the Contract, and a supporting report which includes:

a general description of the methods which the Contractor intends to adopt, and of the major stages, in the execution of the Works, and

details showing the Contractor's reasonable estimate for the number of each class of Contractor's Personnel & Equipment, required on the Site for each major stage.

Unless the Engineer, within 21 days after receiving a programme, gives notice to the Contractor stating the extent to which it does not comply with the Contract, the Contractor shall proceed in accordance with the programme, subject to his other obligations under the Contract. The Engineer shall be entitled to rely upon the programme when planning their activities.

If, at any time, the Engineer gives notice to the Contractor that a programme fails (to the extent stated) to comply with the Contract or to be consistent with actual progress and the Contractor's stated intentions, the Contractor shall submit a revised programme to the Engineer within 15 days in accordance with this Sub-Clause.

**19.(4) Setting out of Works:** The Contractor shall be responsible for the correct setting out of all works in relation to original points, lines and levels of reference at his cost. The Contractor shall execute the work true to alignment, grade, levels and dimensions as shown in the drawing and as

directed by the Engineer's representative and check these at frequent intervals. The Contractor shall provide all facilities like labour and instruments and shall co-operate with the Engineer's representative for checking of all alignment, grades, levels and dimensions. If, at any time, during the progress of the works any error appear or arise in any part of the work, the Contractor, on being required so to do by the Engineer's representative shall, at his own cost rectify such errors, to the satisfaction of the Engineer's representative.

Such checking shall not absolve the Contractor of his own responsibility of maintaining accuracy in the work. The Contractor shall carefully protect and preserve all bench marks, sight rails, pegs and other things used in setting out the work.

**20.(1) Compliance to Engineer's Instructions:** The Engineer shall direct the sequence in which the several parts of the works shall be executed and the Contractor shall execute without delay all orders given by the Engineer from time to time; but the Contractor shall not be relieved thereby from responsibility for the due performance of the works in all respects.

**20.(2) Alterations to be Authorized:** No alterations in or additions to or omissions or abandonment of any part of the works shall be deemed authorised, except under written instructions from the Engineer.

**20.(3) Extra Works:** Should works over and above those included in the contract require to be executed at the site, the Contractor shall have no right to be entrusted with the execution of such works which may be carried out by another Contractor or Contractors or by other means at the option of the Railway.

**20.(4) Separate Contracts in Connection with Works:** The Railway shall have the right to let other contracts in connection with the works. The Contractor shall afford other Contractors reasonable opportunity for the storage of their materials and the execution of their works and shall properly connect and coordinate his work with theirs. If any part of the Contractor's work depends upon proper execution or result upon the work of another Contractor(s), the Contractor shall inspect and promptly report to the Engineer any defects in such works that render it unsuitable for such proper execution and results. The Contractor's failure so-to inspect and report shall constitute an acceptance of the other Contractor's work as fit and proper for the reception of his work, except as to defects which may develop in the other Contractor's work after the execution of his work.

**21. Instruction of Engineer's Representative:** Any instructions or approval given by the Engineer's representative to Contractor in connection with the works shall bind the Contractor as though it had been given by the Engineer provided always as follows:

- (a) Failure of the Engineer's representative to disapprove any work or materials shall not prejudice the power of the Engineer thereafter to disapprove such work or material and to order the removal or breaking up thereof.
- (b) If the Contractor shall be dissatisfied by reason of any decision of the Engineer's representative, he shall be entitled to refer the matter to the Engineer who shall there upon confirm or vary such decision.

**22.(1) Adherence to Specifications and Drawings:** The site and the detailed drawings shall be made available to the contractor commensurate with the accepted programme of work submitted under clause 19(3). The whole of the works shall be executed in perfect conformity with the specifications and drawings of the contract. If Contractor performs any works in a manner contrary to the specifications or drawings or any of them and without such reference to the Engineer, he shall bear all the costs arising or ensuing therefrom and shall be responsible for all loss to the Railway.

**22.(2) Drawings and Specifications of the Works:** The Contractor shall keep one copy of Drawings and Specifications at the site, in good order, and such contract documents as may be necessary, available to the Engineer or the Engineer's Representative.

**22.(3) Ownership of Drawings and Specifications:** All Drawings and Specifications and copies thereof furnished by the Railway to the Contractor are deemed to be the property of the Railway. They shall not be used on other works and with the exception of the signed contract set, shall be returned by the Contractor to the Railway on completion of the work or termination of the Contract.

**22.(4) Compliance with Contractor's Request for Details:** The Engineer shall furnish with reasonable promptness, after receipt by him of the Contractor's request, additional instructions by means of drawings or otherwise, necessary for the proper execution of the works or any part thereof. All such drawings and instructions shall be consistent with the Contract Documents and reasonably inferable there from.

**22.(5) Meaning and Intent of Specification and Drawings:** If any ambiguity arises as to the meaning and intent of any portion of the Specifications and Drawings or as to execution or quality of any work or material, or as to the measurements of the works the decision of the Engineer thereon shall be final subject to the appeal (within 7 days of such decision being intimated to the Contractor) to the Chief Engineer who shall have the power to correct any errors, omissions, or discrepancies in aforementioned items and whose decision in the matter in dispute or doubt shall be final and conclusive.

**23. Working during Night:** The Contractor shall not carry out any work between sun-set and sun-rise without the previous permission of the Engineer. However, if the Engineer is satisfied that the work is not likely to be completed in time except by resorting to night work, he may order the same without confirming any right on the Contractor for claiming any extra payment for the same.

**24. Damage to Railway Property or Private Life and Property:** The Contractor shall be responsible for all risk to the work and for trespass and shall make good at his own expense all loss or damage whether to the works themselves or to any other property of the Railway or the lives, persons or property of others from whatsoever cause in connection with the works until they are taken over by the Railway, although all reasonable and proper precautions may have been taken by the Contractor. In case the Railway shall be called upon to make good any costs, loss or damages, or to pay any compensation, including that payable under the provisions of the Workmen's Compensation Act or any statutory amendments thereof to any person or persons sustaining damages as aforesaid by reason of any act, or any negligence or omissions on the part of the Contractor; the amount of any costs or charges including costs and charges in connection with legal proceedings, which the Railway may incur in reference thereto, shall be charged to the Contractor. The Railway



shall have the power and right to pay or to defend or compromise any claim of threatened legal proceedings or in anticipation of legal proceedings being instituted consequent on the action or default of the Contractor, to take such steps as may be considered necessary or desirable to ward off or mitigate the effect of such proceedings, charging to Contractor, as aforesaid; any sum or sums of money which may be paid and any expenses whether for reinstatement or otherwise which may be incurred and the propriety of any such payment, defence or compromise, and the incurring of any such expenses shall not be called in question by the Contractor.

**25. Sheds, Storehouses and Yards:** The Contractor shall at his own expense provide himself with sheds, storehouses and yards in such situations and in such numbers as in the opinion of the Engineer is requisite for carrying on the works and the Contractor shall keep at each such sheds, storehouses and yards a sufficient quantity of materials and plant in stock as not to delay the carrying out of the works with due expedition and the Engineer and the Engineer's representative shall have free access to the said sheds, store houses and yards at any time for the purpose of inspecting the stock of materials or plant so kept in hand, and any materials or plant which the Engineer may object to shall not be brought upon or used in the works, but shall be forthwith removed from the sheds, storehouses or yards by the Contractor. The Contractor shall at his own expenses provide and maintain suitable mortar mills, soaking vats or any other equipments necessary for the execution of the works.

**26. Provision of Efficient and Competent Staff at Work Sites by the Contractor:**

**26.1** The Contractor shall place and keep on the works at all times efficient and competent staff to give the necessary directions to his workmen and to see that they execute their work in sound & proper manner and shall employ only such supervisors, workmen & labourers in or about the execution of any of these works as are careful and skilled in the various trades.

**26.2** The Contractor shall at once remove from the works any agents, permitted sub-contractor, supervisor, workman or labourer who shall be objected to by the Engineer and if and whenever required by the Engineer, he shall submit a correct return showing the names of all staff and workmen employed by him.

**26.3** In the event of the Engineer being of the opinion that the Contractor is not employing on the works a sufficient number of staff and workmen as is necessary for proper completion of the works within the time prescribed, the Contractor shall forthwith on receiving intimation to this effect deploy the additional number of staff and labour as specified by the Engineer within seven days of being so required and failure on the part of the Contractor to comply with such instructions will entitle the Railway to rescind the contract under Clause 62 of these conditions.

**26A. Deployment of Qualified Engineers at Work Sites by the Contractor:**

**26A.1** The Contractor shall also employ qualified Graduate Engineer(s) or equivalent, or qualified Diploma Engineer(s), as prescribed in the tender documents.

**26A.2** In case the Contractor fails to employ the Engineer, as aforesaid in Para 26A.1, he shall be liable to pay liquidated damages at the rates, as prescribed in the tender documents.

**26A.3** No. of qualified Engineers required to be deployed by the Contractor for various activities contained in the works contract shall be specified in the tender documents as ‘Special Condition of Contract’.

**27.(1) Workmanship and Testing:** The whole of the works and/or supply of materials specified and provided in the contract or that may be necessary to be done in order to form and complete any part thereof shall be executed in the best and most substantial workman like manner with materials of the best and most approved quality of their respective kinds, agreeable to the particulars contained in or implied by the specifications and as referred to in and represented by the drawings or in such other additional particulars, instructions and drawings given during the carrying on of the works and to the entire satisfaction of the Engineer according to the instructions and directions which the Contractors may from time to time receive from the Engineer. The materials may be subjected to tests by means of such machines, instruments and appliances as the Engineer may direct and wholly at the expense of the Contractor.

**27.(2) Removal of Improper Work and Materials:** The Engineer or the Engineer's Representative shall be entitled to order from time to time:

(a) The removal from the site, within the time specified in the order, of any materials which in his opinion are not in accordance with the specifications or drawings.

(b) The substitution of proper and suitable materials, and

(c) the removal and proper re-execution, notwithstanding any previous tests thereof or on account payments therefor, of any work which in respect of materials or workmanship is not in his opinion in accordance with the specifications and in case of default on the part of the Contractor in carrying out such order, the Railway shall be entitled to rescind the contract under Clause 62 of these conditions.

(d) The provision of Construction and Demolition Waste Management Rule 2016 issued by Ministry of Environment Forest and Climate Change dated 29.03.2016 and published in the Gazette of India, Part – II, Section -3, Sub-section (ii) are binding upon the Contractor. Contractor shall implement these provisions at worksites, for which no extra payment will be payable.

**28. Facilities for Inspection:** The Contractor shall afford the Engineer and the Engineer's Representative every facility for entering in and upon every portion of the work at all hours for the purpose of inspection or otherwise and shall provide all labour, materials, planks, ladders, pumps, appliances and things of every kind required for the purpose and the Engineer and the Engineer's Representative shall at all times have free access to every part of the works and to all places at which materials for the works are stored or being prepared.

**29. Examination of Work before Covering Up:** The Contractor shall give 7 days' notice to the Engineer or the Engineer's Representative whenever any work or materials are intended to be covered up in the earth, in bodies or walls or otherwise to be placed beyond the reach of measurements in order that the work may be inspected or that correct dimensions may be taken before being so covered, placed beyond the reach of measurement in default whereof, the same shall at the option of

the Engineer or the Engineer's Representative be uncovered and measured at the Contractor's expense or no allowance shall be made for such work or materials.

**30. Temporary Works:** All temporary works necessary for the proper execution of the works shall be provided and maintained by the Contractor and subject to the consent of the Engineer shall be removed by him at his expenses when they are no longer required and in such manner as the Engineer shall direct. In the event of failure on the part of the Contractor to remove the temporary works, the Engineer will cause them to be removed and cost as increased by supervision and other incidental charges shall be recovered from the Contractor. If temporary huts are provided by the Contractor on the Railway land for labour engaged by him for the execution of works, the Contractor shall arrange for handing over vacant possession of the said land after the work is completed; if the Contractor's labour refuse to vacate, and have to be evicted by the Railway, necessary expenses incurred by the Railway in connection therewith shall be borne by the Contractor.

**31.(1) Contractor to Supply Water for Works:** Unless otherwise provided in the Contract, the Contractor shall be responsible for the arrangements to obtain supply of water necessary for the works.

**31.(2) Water Supply from Railway System:** The Railway may supply to the Contractor part or whole of the quantity of the water required for the execution of works from the Railway's existing water supply system at or near the site of works on specified terms and conditions and at such charges as shall be determined by the Railway and payable by the Contractor, provided that the Contractor shall arrange, at his own expense, to effect the connections and lay additional pipelines and accessories on the site and that the Contractor shall not be entitled to any compensation for interruption or failure of the water supply.

**31.(3) Water Supply by Railway Transport:** In the event of the Railway arranging supply of water to the Contractor at or near the site of works by travelling water tanks or other means, the freight and other charges incurred thereby, including demurrage charges that may be levied, shall be paid by the Contractor in addition to the charges referred to in Sub-Clause (2) of the Clause provided that the Contractor shall not be entitled to any compensation for interruption or failure of the water supply.

**31.(4)(a) Contractor to Arrange Supply of Electric Power for Works:** Unless otherwise provided in the contract, the Contractor shall be responsible for arrangements to obtain supply of Electric Power for the works.

**(b) Electric Supply from the Railway System:** The Railway may supply to the Contractor part or whole of the electric power wherever available and possible, required for execution of works from the Railway's existing electric supply systems at or near the site of works on specified terms and conditions and such charges as shall be determined by the Railway and payable by the Contractor provided the cost of arranging necessary connections to the Railway's Electric Supply systems and laying of underground/overhead conductor, circuit protection, electric power meters, transmission structure, shall be borne by the Contractor and that the Contractor shall not be entitled to any compensation for interruption or failure of the Electric supply system.

**32. Property in Materials and Plant:** The materials and plant brought by the Contractor upon the site or on the land occupied by the Contractor in connection with the works and intended to be

used for the execution thereof shall immediately be deemed to be the property of the Railway. Such of them as during the progress of the works are rejected by the Engineer under Clause 25 of these conditions or are declared by him not to be needed for the execution of the works or such as on the grant of the certificate of completion remain unused shall immediately on such rejection, declaration or grant cease to be deemed the property of the Railway and the Contractor may then (but not before) remove them from the site or the said land. This clause shall not in any way diminish the liability of the Contractor nor shall the Railway be in any way answerable for any loss or damage which may happen to or in respect of any such materials or plant either by the same being lost, stolen, injured or destroyed by fire, tempest or otherwise.

**33.(1) Tools, Plant and Materials Supplied by Railway:** The Contractor shall take all reasonable care of all tools, plant and materials or other property whether of a like description or not belonging to the Railway and committed to his charge for the purpose of the works and shall be responsible for all damage or loss caused by him, his agents, permitted sub-contractor, or his workmen or others while they are in his charge. The Contractors shall sign accountable receipts for tools, plants and materials made over to him by the Engineer and on completion of the works shall hand over the unused balance of the same to the Engineer in good order and repair, fair wear and tear excepted, and shall be responsible for any failure to account for the same or any damage done thereto.

**33.(2) Hire of Railway's Plant:** The Railway may hire to the Contractor such plant as concrete mixers, compressors and portable engines for use during execution of the works on such terms as may be specified in the special conditions or in a separate agreement for Hire of Plant.

**34.(1) Precaution During Progress of Works:** During the execution of works, unless otherwise specified, the Contractor shall at his own cost provide the materials for and execute all shoring, timbering and strutting works as is necessary for the stability and safety of all structures, excavations and works and shall ensure that no damage, injury or loss is caused or likely to be caused to any person or property.

**34.(2) Roads and Water Courses:** Existing roads or water courses shall not be blocked cut through, altered, diverted or obstructed in any way by the Contractor, except with the permission of the Engineer. All compensations claimed for any unauthorized closure, cutting through, alteration, diversion or obstruction to such roads or water courses by the Contractor or his agent or his staff shall be recoverable from the Contractor's bills/Security Deposit or any other dues of Contractor with the Government of India.

**34.(3) Provision of Access to Premises:** During progress of work in any street or thoroughfare, the Contractor shall make adequate provision for the passage of traffic, for securing safe access to all premises approached from such street or thoroughfare and for any drainage, water supply or means of lighting which may be interrupted by reasons of the execution of the works and shall react and maintain at his own cost barriers, lights and other safeguards as prescribed by the Engineer, for the regulation of the traffic, and provide watchmen necessary to prevent accidents. The works shall in such cases be executed night and day, if so ordered by the Engineer and with such vigour so that the traffic way be impeded for as short a time as possible.

**34.(4) Safety of Public:** The Contractor shall be responsible to take all precautions to ensure the safety of the public whether on public or railway property and shall post such look out men as may,

in the opinion of the Engineer, be required to comply with regulations appertaining to the work. Contractor shall ensure placement of barricading / partitions at the place of work to ensure safety of habitants of adjacent area, failing which Engineer may advise stoppage of work as per his discretion.

**34.(5) Display Board:** The Contractor shall be responsible for displaying the details of works i.e. name of work, approximate cost, expected date of completion, name and address of the Contractor and address of Engineer on a proper steel Board of size not less than 1m x 1m.

**35. Use of Explosives:** Explosives shall not be used on the works or on the site by the Contractor without the permission of the Engineer and then also only in the manner and to the extent to which such permission is given. Where explosives are required for the works, they shall be stored in a special magazine to be provided by and at the cost of the Contractor in accordance with the Explosive Rules. The Contractor shall obtain the necessary license for the storage and the use of explosives. All operations in which or for which explosives are employed shall be at the sole risk and responsibility of the Contractor and the Contractor shall indemnify the Railway in respect thereof.

**36.(1) Suspension of Works:** The Contractor shall on the order of the Engineer, suspend the progress of the works or any part thereof for such time or times and in such manner as the Engineer may consider necessary and shall during such suspension properly protect and secure the work so far as is necessary in the opinion of the Engineer. If such suspension is:

- (a) Provided for in the contract, or
- (b) Necessary for the proper execution of the works or by the reason of weather conditions or by some default on the part of the Contractor, and or
- (c) Necessary for the safety of the works or any part thereof, or
- (d) Necessary for the safety of adjoining public or other property or safety of the public or workmen or those who have to be at the site, or
- (e) Necessary to avoid disruption of traffic and utilities, as also to permit fast repair and restoration of any damaged utilities, or
- (f) Due to instruction of The National Green Tribunal or any other statutory authority due to high level of pollution in the city of worksite.

**36.(2)** The Contractor shall not be entitled to the extra costs, if any, incurred by him during the period of suspension of the works, but in the event of any suspension ordered by the Engineer for reasons other than aforementioned and when each such period of suspension exceeds 14 days, the Contractor shall be entitled to such extension of time for completion of the works as the Engineer may consider proper having regard to the period or periods of such suspensions and to such compensations as the Engineer may consider reasonable in respect of salaries or wages paid by the Contractor to his employees during the periods of such suspension.

**36.(3) Suspension Lasting More than 3 Months:** If the progress of the works or any part thereof is suspended on the order of the Engineer for more than three months at a time, the Contractor may serve a written notice on the Engineer requiring permission within 15 days from the receipt thereof to proceed with the works or that part thereof in regard to which progress is suspended and if such permission is not granted within that time the Contractor by further written notice so served may, but is not bound to, elect to treat the suspension where it affects part only of the works as an omission of such part or where it affects the whole of the works, as an abandonment of the contract by the Railway.

**37. Rates for Items of Works:**

(i) The rates, entered in the accepted Bill(s) of Quantities of the Contract are intended to provide for works duly and properly completed in accordance with the General and Special (if any) Conditions of the Contract and the Specifications and drawings together with such enlargements, extensions, diminutions, reductions, alterations or additions as may be ordered in terms of Clause 42 of these conditions and without prejudice to the generality thereof and shall be deemed to include and cover superintendence and labour, supply, including full freight of materials, stores, patterns, profiles, moulds, fittings, centerings, scaffolding, shoring props, timber, machinery, barracks, tackle, roads, pegs, posts, tools and all apparatus and plant required on the works, except such tools, plant or materials as may be specified in the contract to be supplied to the Contractor by the Railway, the erection, maintenance and removal of all temporary works and buildings, all watching, lighting, bailing, pumping and draining, all prevention of or compensation for trespass, all barriers and arrangements for the safety of the public or of employees during the execution of works, all sanitary and medical arrangements for labour camps as may be prescribed by the Railway, the setting of all work and of the construction, repair and upkeep of all center lines, bench marks and level pegs thereon, site clearance, all fees duties, royalties, rent and compensation to owners for surface damage or taxes and impositions payable to local authorities in respect of land, structures and all material supplied for the work or other duties or expenses for which the Contractor may become liable or may be put to under any provision of law for the purpose of or in connection with the execution of the contract and all such other incidental charges or contingencies as may have been specially provided for in the Specifications.

However, if rates of existing GST or cess on GST for Works Contract is increased or any new tax /cess on Works Contract is imposed by Statute after the date of opening of tender but within the original date of completion/date of completion extended under clause 17 & 17A and the Contractor thereupon properly pays such taxes/cess, the Contractor shall be reimbursed the amount so paid.

Further, if rates of existing GST or cess on GST for Works Contract is decreased or any tax/cess on Works Contract is decreased / removed by Statute after the date of opening of tender, the reduction in tax amount shall be recovered from Contractor's bills/Security Deposit or any other dues of Contractor with the Government of India.

**38. Demurrage and Wharfage Dues:** Demurrage charges calculated in accordance with the scale in force for the time being on the Railway and incurred by the Contractor failing to load or unload any goods or materials within the time allowed by the Railway for loading as also wharfage charges, of materials not removed in time, as also charges due on consignments booked by or to him shall be paid by the Contractor, failing which such charges shall be debited to the Contractor's

account in the hands of the Railway and shall be deducted from any sums which may become due to him in terms of the contracts.

**39.(1) Rates for Extra Item(s) of Works:**

(a) Standard Schedule of Rates (SSOR) Items: Any item of work carried out by the Contractor on the instructions of the Engineer which is not included in the accepted Bill(s) of Quantities but figures in the Standard Schedule of Rates (SSOR), shall be executed at the rates set forth in the "Standard Schedule of Rates (SSOR)" modified by the tender percentage as accepted in the contract for that chapter of Standard Schedule of Rates (SSOR). ~~However, the cumulative value of all such extra item(s) together (modified by the respective tender percentage) shall not exceed 10 % of the original contract value.~~ **ACS**

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For item(s) not covered in this sub clause, the rate shall be decided as agreed upon between the Engineer and the Contractor before the execution of such items of work as per sub clause (b).

(b) Other Items: For any item of work to be carried out by the Contractor but not included in the accepted Bill(s) of Quantities and also not covered under sub clause (a) above, the Contractor shall be bound to notify the Engineer at least seven days before the necessity arises for the execution of such items of works that the accepted Bill(s) of Quantities does not include rate or rates for such extra work involved. The rates payable for such items shall be decided at the meeting to be held between the Engineer and Contractor, in as short a period as possible after the need for the special item has come to the notice. In case the Contractor fails to attend the meeting after being notified to do so or in the event of no settlement being arrived at, the Railway shall be entitled to execute the extra works by other means and the Contractor shall have no claim for loss or damage that may result from such procedure.

The assessment of rates for extra item(s) shall be arrived at based on the prevailing market rates of labour, machinery & materials and by taking guidance from the following documents in order of priority:

- i. Analysis of Rates for "Unified Standard Schedule of Rates of Indian Railways (USSOR)"
- ii. Analysis of Rates for "Delhi Schedule of Rates issued by CPWD (DSR)"
- iii. Market Analysis

**39.(2)** Provided that if the Contractor commences work or incurs any expenditure in regard thereto before the rates as determined and agreed upon as lastly hereuntofore-mentioned, then and in such a case the Contractor shall only be entitled to be paid in respect of the work carried out or expenditure incurred by him prior to the date of determination of the rates as aforesaid according to the rates as shall be fixed by the Engineer. However, if the Contractor is not satisfied with the decision of the Engineer in this respect, he may appeal to the Chief Engineer within 30 days of getting the decision of the Engineer, supported by analysis of the rates claimed. The Chief Engineer's decision after hearing both the parties in the matter would be final and binding on the Contractor and the Railway.

**40. (1) Handing over of Works:** The Contractor shall be bound to hand over the works executed under the contract to the Railway complete in all respects to the satisfaction of the Engineer. The Engineer shall determine the date on which the work is considered to have been completed, in support of which his certificate shall be regarded as sufficient evidence for all purposes. The Engineer shall determine from time to time, the date on which any particular section of the work shall

have been completed, and the Contractor shall be bound to observe any such determination of the Engineer.

**40.(2) Clearance of Site on Completion:** On completion of the works, the Contractor shall clear away and remove from the site all constructional plant, surplus materials, rubbish and temporary works of every kind and leave the whole of the site and works clean and in a workman like condition to the satisfaction of the Engineer. No final payment in settlement of the accounts for the works shall be paid, held to be due or shall be made to the, Contractor till, in addition to any other condition necessary for final payment, site clearance shall have been affected by him, and such clearance may be made by the Engineer at the expense of the Contractor in the event of his failure to comply with this provision within 7 days after receiving notice to that effect. Should it become necessary for the Engineer to have the site cleared at the expenses of the Contractor, the Railway shall not be held liable for any loss or damage to such of the Contractor's property as may be on the site and due to such removal there from which removal may be affected by means of public sales of such materials and property or in such a way as deemed fit and convenient to the Engineer.

**40A Offloading of Part(s) of Work:** At the final stage of completion/ commissioning of work, in case the contractor fails to complete the final part(s) of the work and the value of such part(s) of the work is limited to 5% of the original contract value, the Engineer may allow/decide for offloading of such part(s) of works, either after the Contractor's request in writing to do so or after serving a 14 (Fourteen) days suo-moto notice (as per annexure- VIIA), if the Engineer is of the opinion that :-

- (i) Such Offloading of works (up to 5% of original contract value) would enable successful completion of contract/work,
- (ii) Termination/ Part termination of the contract at this stage is not be in the interest of the Railway/work;, and
- (iii) The anticipated additional cost for execution of such works through other mode would not be substantial and can be recovered from the pending dues of the contractor;

The Contractor shall be informed, in due course, by the Engineer of the mode and cost of execution of such offloaded work through other agency(ies) (as per annexure- VIIB). The extra expenditure so incurred in execution of the offloaded work, shall be recovered from subsequent Bill(s) or any other dues of the Contractor, but not exceeding the value of Performance Guarantee available in the contract. There shall be no other repercussion of such offloading on execution of the balance contract. The Contractor shall have no claim on account of above mentioned offloading of works.

## **VARIATIONS IN EXTENT OF CONTRACT**

**41. Modification to Contract to be in Writing:** In the event of any of the provisions of the contract required to be modified after the contract documents have been signed, the modifications shall be made in writing and signed by the Railway and the Contractor and no work shall proceed under such modifications until this has been done. Any verbal or written arrangement abandoning, modifying, extending, reducing or supplementing the contract or any of the terms thereof shall be deemed conditional and shall not be binding on the Railway unless and until the same is incorporated in a formal instrument and signed by the Railway and the Contractor, and till then the Railway shall have the right to repudiate such arrangements.



**42.(1) Powers of Modification to Contract:** The Engineer on behalf of the Railway shall be entitled by order in writing to enlarge or extend, diminish or reduce the works or make any alterations in their design, character position, site, quantities, dimensions or in the method of their execution or in the combination and use of materials for the execution thereof or to order any additional work to be done or any works not to be done and the Contractor will not be entitled, to any compensation for any increase/reduction in the quantities of work but will be paid only for the actual amount of work done and for approved materials supplied against a specific order.

42.(2) (i) Unless otherwise specified in the special conditions of the contract, the accepted variation in quantity of each individual item of the contract would be upto 25% of the quantity originally contracted, except in case of foundation work (in which no variation limit shall apply). However, the rates for the increased quantities shall be as per sub- para (iii) below.

(ii) The Contractor shall be bound to carry out the work at the agreed rates and shall not be entitled to any claim or any compensation whatsoever upto the limit of 25% variation in quantity of individual item of works.

(iii) In case an increase in quantity of an individual item by more than 25% of the agreement quantity is considered unavoidable, then same shall be executed at following rates

- a. Quantities operated in excess of 125% but upto 140% of the agreement quantity of the concerned item, shall be paid at 98% of the rate awarded for that item in that particular tender;
- b. Quantities operated in excess of 140% but upto 150% of the agreement quantity of the concerned item shall be paid at 96% of the rate awarded for that item in that particular tender;
- c. Variation in quantities of individual items beyond 150% will be avoided and would be permitted only in exceptional unavoidable circumstances and shall be paid at 96% of the rate awarded for that item in that particular tender.
- d. Variation to quantities of Minor Value Item:  
The limit for varying quantities for minor value items shall be 100% (as against 25% prescribed for other items). A minor value item for this purpose is defined as an item whose original agreement value is less than 1 % of the total original contract value.
  - d.(i) Quantities operated upto and including 100% of the agreement quantity of the concerned minor value item, shall be paid at the rate awarded for that item in that particular tender;
  - d.(ii) Quantities operated in excess of 100% but upto 200% of the agreement quantity of the concerned minor value item, shall be paid at 98% of the rate awarded for that item in that particular tender;
  - d.(iii) Variation in quantities of individual minor value item beyond 200% will be avoided and would be permitted only in exceptional unavoidable circumstances and shall be paid at 96% of the rate awarded for that item in that particular tender.

(iv) In case of earthwork items, the variation limit of 25% shall apply to the gross quantity of earthwork items and variation in the quantities of individual classifications of soil shall not be subject to this limit.

(v) As far as Standard Schedule of Rates (SSOR) items are concerned, the variation limit of 25% would apply to the value of SSOR schedule(s) as a whole and not on individual SSOR items. However, in case of Non Standard Schedule of Rates (SSOR) items, the limit of 25% would apply on

the individual items irrespective of the manner of quoting the rate (single percentage rate or individual item rate).

**42.(3) Valuation of Variations:** The enlargements, extensions, diminution, reduction, alterations or additions referred to in Sub-Clause (2) of this Clause shall in no degree affect the validity of the contract; but shall be performed by the Contractor as provided therein and be subject to the same conditions, stipulations and obligations as if they had been originally and expressively included and provided for in the Specifications and Drawings and the amounts to be paid therefor shall be calculated in accordance with the accepted Bill(s) of Quantities. Any extra item(s)/quantities of work falling outside the purview of the provisions of Sub-Clause (2) above shall be paid for at the rates determined under Clause-39 of these Conditions.

## CLAIMS

**43.(1) Quarterly Statement of Claims:** The Contractor shall prepare and furnish to the Engineer once in every quarter commencing from the month following the month of issue of Letter of Acceptance, an account giving full and detailed particulars of all claims for any additional expenses to which the Contractor may consider himself entitled to and of all extra or additional works ordered by the Engineer which he has executed during the preceding quarter and no claim for payment for such work will be considered which has not been included in such particulars.

**43.(2) Signing of "No Claim" Certificate :** The Contractor shall not be entitled to make any claim whatsoever against the Railway under or by virtue of or arising out of this contract, nor shall the Railway entertain or consider any such claim, if made by the Contractor, after he shall have signed a "No Claim" Certificate in favour of the Railway in such form as shall be required by the Railway after the works are finally measured up. The Contractor shall be debarred from disputing the correctness of the items covered by "No Claim" Certificate or demanding a clearance to arbitration in respect thereof.

## MEASUREMENTS, CERTIFICATES AND PAYMENTS

**44. Quantities in Bill(s) of Quantities Annexed to Contract:** The quantities set out in the accepted Bill(s) of Quantities with items of works quantified are the estimated quantities of the works and they shall not be taken as the actual and correct quantities of the work to be executed by the Contractor in fulfillment of his obligations under the contract.

**45(i). Measurement of Works by Railway:** The Contractor shall be paid for the works at the rates in the accepted Bill(s) of Quantities and for extra works at rates determined under Clause 39 of these Conditions on the measurements taken by the Engineer or the Engineer's representative in accordance with the rules prescribed for the purpose by the Railway. The quantities for items the unit of which in the accepted Bill(s) of Quantities is 100 or 1000 shall be calculated to the nearest whole number, any fraction below half being dropped and half and above being taken as one; for items the unit of which in the accepted Bill(s) of Quantities is single, the quantities shall be calculated to two places of decimals. Such measurements will be taken of the work in progress from time to time and at such intervals as in the opinion of the Engineer shall be proper having regard to the progress of works. The date and time on which 'on account' or 'final' measurements are to be made shall be communicated

to the Contractor who shall be present at the site and shall sign the results of the measurements (which shall also be signed by the Engineer or the Engineer's representative) recorded in the official measurements book as an acknowledgement of his acceptance of the accuracy of the measurements. Failing the Contractor's attendance, the work may be measured up in his absence and such measurements shall, notwithstanding such absence, be binding upon the Contractor whether or not he shall have signed the measurement books provided always that any objection made by him to measurement shall be duly investigated and considered in the manner set out below:

(a) It shall be open to the Contractor to take specific objection to any recorded measurements or Classification on any ground within seven days of the date of such measurements. Any re-measurement taken by the Engineer or the Engineer's representative in the presence of the Contractor or in his absence after due notice has been given to him in consequence of objection made by the Contractor shall be final and binding on the Contractor and no claim whatsoever shall thereafter be entertained regarding the accuracy and Classification of the measurements.

(b) If an objection raised by the Contractor is found by the Engineer to be incorrect the Contractor shall be liable to pay the actual expenses incurred in measurements.

**45(ii). Measurement of Works by Contractor's Authorized Representative (in case the contract provides for the same):**

(a) The Contractor shall be paid for the works at the rates in the accepted Bill(s) of Quantities and for extra works at rates determined under Clause 39 of these Conditions on the measurements taken by the Contractor's authorized Engineer in accordance with the rules prescribed for the purpose by the Railway. The quantities for items the unit of which in the accepted Bill(s) of Quantities is 100 or 1000 shall be calculated to the nearest whole number, any fraction below half being dropped and half and above being taken as one; for items the unit of which in the accepted Bill(s) of Quantities is single, the quantities shall be calculated to two places of decimals. Such measurements will be taken of the work in progress from time to time. The date and time on which 'on account' or 'final' measurements are to be made shall be communicated to the Engineer.

The date and time of test checks shall be communicated to the Contractor who shall be present at the site and shall witness the test checks, failing the Contractor's attendance the test checks may be conducted in his absence and such test checks shall notwithstanding such absence be binding upon Contractor provided always that any objection made by Contractor to test check shall be duly investigated and considered in the manner set out below:

- (i) It shall be open to the Contractor to take specific objection to test checks of any recorded measurement within 7 days of date of such test checks. Any re-test check done by the concerned Railway's authority in the presence of the Contractor or in his absence after due notice given to him in consequent of objection made by the Contractor shall be final and binding on the Contractor and no claim whatsoever shall thereafter be entertained regarding the accuracy and classification of the measurements.
- (ii) If an objection raised by the Contractor is found by the Engineer to be incorrect the Contractor shall be liable to pay the actual expenses incurred in measurements.

**(b) Incorrect measurement, actions to be taken:** If in case during test check or otherwise, it is detected by the Engineer that agency has claimed any exaggerated measurement or has claimed any false measurement for the works which have not been executed; amounting to variation of 5% or more of claimed gross bill amount, action shall be taken as following:

- (i) On first occasion of noticing exaggerated/ false measurement, Engineer shall recover liquidated damages equal to 10% of claimed gross bill value.
- (ii) On any next occasion of noticing any exaggerated/false measurement, railway shall recover liquidated damages equal to 15% of claimed gross bill value. In addition, the facility of recording of measurements by Contractor as well as release of provisional payment shall be withdrawn. Once withdrawn, measurements shall be done by railway as per clause 45(i) above.

**46.(1) "On-Account " Payments:** The Contractor shall be entitled to be paid from time to time by way of "On-Account" payment only for such works as in the opinion of the Engineer he has executed in terms of the contract. All payments due on the Engineer's/Engineer's Representative's certificates of measurements or Engineer's certified "Contractor's authorized Engineer's measurements" shall be subject to any deductions which may be made under these presents and shall further be subject to, unless otherwise required by Clause 16 of these Conditions, a retention of six percent by way of Security Deposits, until the amount of Security Deposit by way of such retentions shall amount to 5% of the total value of the contract provided always that the Engineer may by any certificate make any correction or modification in any previous certificate which shall have been issued by him and that the Engineer may withhold any certificate, if the works or any part thereof are not being carried out to his satisfaction.

**46.(2) Rounding off Amounts:** The total amount due on each certificate shall be rounded off to the nearest rupee, i.e. sum less than 50 paise shall be omitted and sums of 50 paise and more upto ₹1 will be reckoned as ₹ 1.

**46.(3) On Account Payments not Prejudicial to Final Settlement:** "On-Account" payments made to the Contractor shall be without prejudice to the final making up of the accounts (except where measurements are specifically noted in the Measurement Book as "Final Measurements" and as such have been signed by the Contractor and Engineer/Engineer's Representative) and shall in no respect be considered or used as evidence of any facts stated in or to be inferred from such accounts nor of any particular quantity of work having been executed nor of the manner of its execution being satisfactory.

**46.(4)** If payment(s) of Advances are applicable in the contract, as mentioned in the Tender Documents, Railway shall make payment(s) of Interest bearing advances, on the request of contractor. The payment and recovery of such Advances shall be made as under:

**(a): Mobilisation Advance –**

This shall be limited to 10% of the Contract value and shall be paid in 2 stages :

Stage 1– 5% of Contract Value on signing of the contract agreement.

Stage 2 – 5% on mobilization of site-establishment, setting up offices, bringing in equipment and actual commencement of work.

The stage 1 of advance shall be payable immediately after signing of contract agreement.

The stage 2 of advance shall be payable at the time of mobilisation, only after submission of an utilization certificate by the contractor that the Stage 1 advance has been properly utilized in the contract.

These Advances shall be payable against irrevocable guarantee (Bank Guarantee, FDRs) from a scheduled commercial bank of India of at least 110% of the value of the sanctioned advance amount (covering principal plus interest).

**(b): Advance Against Machinery and Equipment –**

This advance shall be limited to a maximum of 10% of the contract value against new Machinery & Equipment, involving substantial outlay, brought to site and essentially required for the work. This advance shall not exceed 75% of the purchase price of such Equipment and shall be payable when Equipment is hypothecated to the President of India by a suitable bond or alternatively covered by an irrevocable Bank Guarantee from a scheduled commercial bank of India for full cost of the Plant & Equipment in a form acceptable to Railways. The Plant & Equipment shall be insured for the full value and for the entire period, they are required for the work. This Plant & Equipment shall not be removed from the site of work without prior written permission of the Engineer. No advance should be given against old Plant & Machinery.

**The advances under sub clause (a) and (b) above, are subject to the following conditions -**

(i) The full amount of Advances shall be recovered from contractor dues. The recovery shall commence when the value of contract executed reaches 15% of original contract value and shall be completed when the value of work executed reaches 85% of the original contract value. The installments on each "on account bill" will be on pro-rata basis.

Interest shall be recovered on the advance outstanding for the period commencing from the date of payment of advance till date of particular on-account bill (through which recovery of principal is effected) and adjusted fully against on-account bill along with pro-rata principal recovery. In the event of any short-fall, the same shall be carried forward to the next on-account bill and shall attract interest.

(ii) The advances shall be used by the Contractor for the purpose of the Contract, and for the purpose for which they are paid. Under no circumstances, shall the advances be diverted for other purposes. Any such diversion shall be construed as a breach of the Contract and the Contractor shall be asked to return the advance at once and pay interest at 15% per annum till the advance is recovered back from him. The Contractor shall return the advance and pay the interest in one go without demur. The Contractor, if required by the Engineer shall provide the details of utilisation of Mobilisation advance.

(iii) If the Contractor is found to have contravened the provision, it will constitute a breach of contract and Railway shall be entitled to terminate the contract and forfeit his Performance Guarantee as well as Security Deposit.

(iv) In cases, where the Contract is rescinded as per clause 62 of the contract or short closed under any other condition(s) of the contract, without making full recovery of advances and accrued interest thereon, by the Railway, such balance of advances and accrued interest thereon shall immediately become due and payable by the Contractor to the Railway. The same shall be recovered from any due of Contractor with the Government of India.

**46.(5) Manner of Payment:** Unless otherwise specified payments to the Contractor will be transferred electronically to his bank account.

**46A. Price Variation Clause (PVC):**

**46A.1 Applicability:** Price Variation Clause (PVC) shall be applicable only in tender having advertised value above **Rs. 2 Crores** ~~and having completion period above 12 months~~. Provided further that, in a contract where PVC is applicable, following shall be outside the purview of price adjustments (i.e. shall be excluded from the gross value of the work for the purpose of price variation) :

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- a) Materials supplied by Railway to the Contractors, either free or at fixed rate;
- b) Any extra item(s) included in subsequent variation falling outside the purview of the Bill(s) of Quantities of tender, under clause 39. (1)(b) of these Standard General Conditions, unless applicability of PVC and 'Base Month' has been specially agreed, while fixing the rates of such extra item(s).

**46A.2 Base Month:** The Base Month for 'Price Variation Clause' shall be taken as the one month prior to closing of tender, unless otherwise stated elsewhere. The quarter for applicability of PVC shall commence from the month following the Base month. The Price Variation shall be based on the average Price Index of the quarter under consideration.

**46A.3 Validity:**

Rates accepted by Railway Administration shall hold good till completion of work and no additional individual claim shall be admissible except:

- (a) Payment/recovery for increase/decrease in GST on works contract or imposition/removal of any tax/cess on Works Contract as per Clause 37,
- (b) Payment/recovery for overall market situation as per Price Variation Clause given hereunder.

**46A.4** Components of various items in a contract on which variation in prices be admissible, shall be steel, cement, ferrous material, non-ferrous material, insulators, zinc and other materials, labour, plant & machinery, fuel, explosives, detonators etc. Adjustment for variation in prices of these items shall be determined in the manner prescribed.

**46A.5** No price variation shall be admissible for fixed components.

**46A.6** The percentages of various components in various type of works shall be as specified for all item (s)/ Bill(s) of Quantities in tender document and the same shall be fixed as per table & classifications given below:

**(I). For Civil Engineering Works**

| S<br>N | Classification                 |                 | 1A, 2 & 3A | 4A       | 5A       | 6A       | 7        | 8A       | 9A  | 1B, 3B, 4B, 5B,<br>6B 8B & 9B | 1C, 3C, 4C, 5C,<br>6C, 8C & 9C | 3D, 4D, 5D, 6D,<br>8D & 9D | 3E, 4E, 5E, 6E,<br>8E & 9E |
|--------|--------------------------------|-----------------|------------|----------|----------|----------|----------|----------|-----|-------------------------------|--------------------------------|----------------------------|----------------------------|
|        | Components                     |                 |            |          |          |          |          |          |     |                               |                                |                            |                            |
| 1      | Fixed                          | *               | 15         | 15       | 15       | 15       | 15       | 15       | 15  | 15                            | 15                             | 15                         | 15                         |
| 2      | Labour                         | L <sub>c</sub>  | 15<br>20   | 20<br>25 | 25<br>30 | 15<br>20 | 15<br>50 | 15<br>20 | 20  | 5<br>0                        | 5<br>0                         | 10                         | 25                         |
| 3      | Steel                          | S <sub>c</sub>  | 0          | 0        | 20<br>0  | 0        | 0        | 0        | 0   | 75<br>85                      | 0                              | 50                         | 0                          |
| 4      | Cement                         | C <sub>c</sub>  | 0          | 0        | 15       | 0        | 0        | 0        | 0   | 0                             | 75<br>85                       | 0                          | 0                          |
| 5      | Plant<br>Machinery<br>& Spares | PM <sub>c</sub> | 35<br>30   | 15       | 0<br>5   | 20       | 20<br>15 | 20       | 30  | 0                             | 0                              | 10                         | 30                         |
| 6      | Fuel &<br>Lubricants           | F <sub>c</sub>  | 25         | 15       | 10<br>5  | 35<br>15 | 35<br>15 | 35<br>20 | 15  | 5<br>0                        | 5<br>0                         | 10                         | 20                         |
| 7      | Other<br>materials             | M <sub>c</sub>  | 10         | 15       | 15<br>30 | 15<br>30 | 15<br>5  | 15<br>25 | 20  | 0                             | 0                              | 5                          | 10                         |
| 8      | Detonators<br>&<br>Explosive   | E <sub>c</sub>  | 0          | 20<br>15 | 0        | 0        | 0        | 0        | 0   | 0                             | 0                              | 0                          | 0                          |
| Total  |                                |                 | 100        | 100      | 100      | 100      | 100      | 100      | 100 | 100                           | 100                            | 100                        | 100                        |

\* It shall not be considered for any price variation.

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The classification mentioned in the table above represents following type of item(s) in the work(s) –

**1 Earthwork in Formation**

1A All Item(s) excluding 1B or/and 1C

1B Item(s) for supply of Steel

1C Item(s) for supply of Cement

**2 Ballast Supply Works**

**3 Tunnelling Works (Without Explosives)**

3A All Item(s) excluding 3B or/and 3C or/and 3D or/and 3E

3B Item(s) for supply of Steel

3C Item(s) for supply of Cement or/and Grout

3D Item(s) for Fabrication & Erection of Structures including supply of Steel

3E Item(s) for Fabrication & Erection of Structures excluding supply of Steel.

**4 Tunnelling Works (With explosives)**

4A All Item(s) excluding 4B or/and 4C or/and 4D or/and 4E

4B Item(s) for supply of Steel

4C Item(s) for supply of Cement or/and Grout

4D Item(s) for Fabrication & Erection of Structures including supply of Steel

4E Item(s) for Fabrication & Erection of Structures excluding supply of Steel.

## 5 Building Works

- 5A All Item(s) excluding 5B or/and 5C or/and 5D or/and 5E
- 5B Item(s) for supply of Steel
- 5C Item(s) for supply of Cement
- 5D Item(s) for Fabrication & Erection of Structures including supply of Steel
- 5E Item(s) for Fabrication & Erection of Structures excluding supply of Steel.

## 6. Bridges & Protection work

- 6A All Item(s) excluding 6B or/and 6C or/and 6D or/and 6E
- 6B Item(s) for supply of Steel
- 6C Item(s) for supply of Cement
- 6D Item(s) for Fabrication, Assembly, Erection & Launching of Girders including supply of Steel
- 6E Item(s) for Fabrication, Assembly, Erection & Launching of Girders excluding supply of Steel

### Permanent Way linking

#### Platform, Passenger Amenities

- 8A All Item(s) excluding 8B or/and 8C or/and 8D or/and 8E
- 8B Item(s) for supply of Steel item/fittings
- 8C Item(s) for supply of Cement Item
- 8D Item(s) for Fabrication & Erection of Structures including supply of Steel
- 8E Item(s) for Fabrication & Erection of Structures excluding supply of Steel

### Any Other Works not covered in Classification 1 to 8

- 9A All Item(s) excluding 9B or/and 9C or/and 9D or/and 9E
- 9B Item(s) for supply of Steel
- 9C Item(s) for supply of Cement or/and Grout
- 9D Item(s) for Fabrication & Erection of Structures including supply of Steel
- 9E Item(s) for Fabrication & Erection of Structures excluding supply of Steel

**46A.7 Formulae:** The Amount of variation in prices in various components (labour, material etc.) shall be worked out by the following formulae:

- (i) 
$$L = \frac{(W \text{ or } W_S \text{ or } W_C \text{ or } W_{SF} \text{ or } W_F \text{ or } W_{SFL} \text{ or } W_{FL}) \times (L_Q - L_B) \times L_C}{L_B \times 100}$$
$$L = \frac{(W \text{ or } W_{SF} \text{ or } W_F \text{ or } W_{SFL} \text{ or } W_{FL}) \times (L_Q - L_B) \times L_C}{L_B \times 100}$$
- (ii) 
$$M = \frac{(W \text{ or } W_{SF} \text{ or } W_F \text{ or } W_{SFL} \text{ or } W_{FL}) \times (M_Q - M_B) \times M_C}{M_B \times 100}$$
- (iii) 
$$F = \frac{(W \text{ or } W_S \text{ or } W_C \text{ or } W_{SF} \text{ or } W_F \text{ or } W_{SFL} \text{ or } W_{FL}) \times (F_Q - F_B) \times F_C}{F_B \times 100}$$
$$F = \frac{(W \text{ or } W_{SF} \text{ or } W_F \text{ or } W_{SFL} \text{ or } W_{FL}) \times (F_Q - F_B) \times F_C}{F_B \times 100}$$
- (iv) 
$$E = \frac{(W \text{ or } W_S \text{ or } W_C \text{ or } W_{SF} \text{ or } W_F \text{ or } W_{SFL} \text{ or } W_{FL}) \times (E_Q - E_B) \times E_C}{E_B \times 100}$$
$$E = \frac{(W) \times (E_Q - E_B) \times E_C}{E_B \times 100}$$
- (v) 
$$PM = \frac{(W \text{ or } W_S \text{ or } W_C \text{ or } W_{SF} \text{ or } W_F \text{ or } W_{SFL} \text{ or } W_{FL}) \times (PM_Q - PM_B) \times PM_C}{PM_B \times 100}$$
$$PM = \frac{(W \text{ or } W_{SF} \text{ or } W_F \text{ or } W_{SFL} \text{ or } W_{FL}) \times (PM_Q - PM_B) \times PM_C}{PM_B \times 100}$$
- (vi) 
$$S = \frac{(W \text{ or } W_S \text{ or } W_{SF}) \times (S_Q - S_B) \times S_C}{S_B \times 100}$$
- (vii) 
$$C = \frac{(W \text{ or } W_C) \times (C_Q - C_B) \times C_C}{C_B \times 100}$$



**(II) For Railway Electrification Works:**

- (viii)  $T = [0.4136x(C_Q - C_B) / C_B] \times 85$
- (ix)  $R = [0.94x(R_T - R_O) / R_O + 0.06x(Z_T - Z_O) / Z_O] \times 85$
- (x)  $N = [(P_T - P_O) / P_O] \times 85$
- (xi)  $I = [(I_T - I_O) / I_O] \times 85$
- (xii)  $G = [(M_Q - M_B) / M_B] \times 85$
- (xiii)  $Er = [(L_Q - L_B) / L_B] \times 85$

*Where,*

|                 |                                                                                                                                                                                                                                                                                                                   |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| L               | Amount of price variation in Labour                                                                                                                                                                                                                                                                               |
| M               | Amount of price variation in Materials                                                                                                                                                                                                                                                                            |
| F               | Amount of price variation in Fuel                                                                                                                                                                                                                                                                                 |
| E               | Amount of price variation in Explosives                                                                                                                                                                                                                                                                           |
| PM              | Amount of price variation in Plant, Machinery and Spares                                                                                                                                                                                                                                                          |
| S               | Amount of price variation in Steel Supply Item                                                                                                                                                                                                                                                                    |
| C               | Amount of price variation in Cement Supply Item                                                                                                                                                                                                                                                                   |
| T               | Percentage variation payable on the gross value of bill of Concreting (Bill(s) of Quantities for concrete items)                                                                                                                                                                                                  |
| R               | Percentage variation payable on the gross value of bill of Ferrous Items (Bill(s) of Quantities for ferrous items)                                                                                                                                                                                                |
| N               | Percentage variation payable on the gross value of bill of Non-Ferrous Items (Bill(s) of Quantities for non-ferrous items)                                                                                                                                                                                        |
| I               | Percentage variation payable on the gross value of bill of Insulator (Bill(s) of Quantities for Insulator items)                                                                                                                                                                                                  |
| G               | Percentage variation payable on the gross value of bill of General Works (Bill(s) of Quantities for General items)                                                                                                                                                                                                |
| Er              | Percentage variation payable on the gross value of erection (Bill(s) of Quantities for Erection Item)                                                                                                                                                                                                             |
| L <sub>C</sub>  | % of Labour Component in the item(s)                                                                                                                                                                                                                                                                              |
| M <sub>C</sub>  | % of Material Component in the item(s)                                                                                                                                                                                                                                                                            |
| F <sub>C</sub>  | % of Fuel Component in the item(s)                                                                                                                                                                                                                                                                                |
| E <sub>C</sub>  | % of Explosive Component in the item(s)                                                                                                                                                                                                                                                                           |
| PM <sub>C</sub> | % of Plant, Machinery and Spares Component in the item(s)                                                                                                                                                                                                                                                         |
| S <sub>C</sub>  | % of Steel Supply item Component in the item(s)                                                                                                                                                                                                                                                                   |
| C <sub>C</sub>  | % of Cement Supply item Component in the item(s)                                                                                                                                                                                                                                                                  |
| W               | Gross value of work done by Contractor as per on-account bill(s) excluding the Gross value of work under W <sub>S</sub> or/and W <sub>C</sub> or/and W <sub>SF</sub> or/and W <sub>F</sub> or/and W <sub>SFL</sub> or/and W <sub>FL</sub> and cost of materials supplied by Railway either free or at fixed rate, |

|           |                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $W_S$     | Gross value of work done by Contractor for item(s) of supply of steel.                                                                                                                                                                                                                                                                                                                                                         |
| $W_C$     | Gross value of work done by Contractor for item(s) of supply of cement and /or supply of grout material.                                                                                                                                                                                                                                                                                                                       |
| $W_{SF}$  | Gross value of work done by Contractor for item(s) of Fabrication & Erection of Structures including supply of Steel.                                                                                                                                                                                                                                                                                                          |
| $W_F$     | Gross value of work done by Contractor for Fabrication & Erection of Structures excluding supply of Steel.                                                                                                                                                                                                                                                                                                                     |
| $W_{SFL}$ | Gross value of work done by Contractor for item(s) of Fabrication, Assembly, Erection / Launching of Girders including supply of Steel.                                                                                                                                                                                                                                                                                        |
| $W_{FL}$  | Gross value of work done by Contractor for item(s) of Fabrication, Assembly, Erection / Launching of Girders excluding supply of Steel.                                                                                                                                                                                                                                                                                        |
| $L_B$     | Consumer Price Index for Industrial Workers - All India: Published in R.B.I. Bulletin for the base period                                                                                                                                                                                                                                                                                                                      |
| $L_Q$     | Consumer Price Index for Industrial Workers - All India: Published in R.B.I. Bulletin for the average price index of the 3 months of the quarter under consideration                                                                                                                                                                                                                                                           |
| $M_B$     | Wholesale Price Index: All commodities – as published in the R.B.I. Bulletin for the base period                                                                                                                                                                                                                                                                                                                               |
| $M_Q$     | Wholesale Price Index: All commodities – as published in the R.B.I. Bulletin for the average price index of the 3 months of the quarter under consideration                                                                                                                                                                                                                                                                    |
| $F_B$     | The average of official prices of Diesel available on the official website of ‘Petroleum Planning and Analysis cell’ under Ministry of Petroleum and Natural Gas for Delhi, Kolkata, Mumbai & Chennai, for the base period                                                                                                                                                                                                     |
| $F_Q$     | The average of official prices of Diesel available on the official website of ‘Petroleum Planning and Analysis cell’ under Ministry of Petroleum and Natural Gas for Delhi, Kolkata, Mumbai & Chennai, for the 3 months of the quarter under consideration                                                                                                                                                                     |
| $E_B$     | Index number of Monthly Whole Sale Price Index for the category ‘Explosive’ of (g). Manufacture of other chemical products under (J) MANUFACTURE OF CHEMICALS AND CHEMICAL PRODUCTS, published by Office of Economic Adviser, Govt. of India, Ministry of Commerce & Industry, Department of Industrial Policy & Promotion (DIPP), for the base period.                                                                        |
| $E_Q$     | Index number of Monthly Whole Sale Price Index for the category ‘Explosive’ of (g). Manufacture of other chemical products under (J) MANUFACTURE OF CHEMICALS AND CHEMICAL PRODUCTS, published by Office of Economic Adviser, Govt. of India, Govt. of India, Ministry of Commerce & Industry, Department of Industrial Policy & Promotion (DIPP), for the average price index of 3 months of the quarter under consideration. |
| $PM_B$    | Index Number of Wholesale Prices in India by Groups and Sub Groups (Averages) for ‘Manufacture of machinery for mining, quarrying and construction’– published in RBI (Reserve Bank of India) Bulletin, for the base period.                                                                                                                                                                                                   |
| $PM_Q$    | Index Number of Wholesale Prices in India by Groups and Sub Groups (Averages) for ‘Manufacture of machinery for mining, quarrying and construction’– published in RBI (Reserve Bank of India) Bulletin, for the average price index of 3 months of                                                                                                                                                                             |

|                |                                                                                                                                                                                   |
|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                | the quarter under consideration.                                                                                                                                                  |
| S <sub>B</sub> | The average rate provided by the Joint Plant Committee for the relevant category of steel item as mentioned in Clause 46A.9; for the base period.                                 |
| S <sub>Q</sub> | The average rate provided by the Joint Plant Committee for the relevant category of steel item as mentioned in Clause 46A.9; for the 3 months of the quarter under consideration. |
| C <sub>B</sub> | Index No. of Wholesale Price Index of sub-group Cement, Lime & Plaster as published in RBI Bulletin for the base period                                                           |
| C <sub>Q</sub> | No. of Wholesale Price Index of sub-group Cement, Lime & Plaster as published in RBI Bulletin for the average price index of the 3 months of the quarter under consideration      |
| R <sub>T</sub> | IEEMA price index for Steel Blooms (size 150mmx150mm) for the month which is two months prior to date of inspection of material.                                                  |
| R <sub>O</sub> | IEEMA price index for Steel Blooms (size 150mmx150mm) for the month which is one month prior to date of opening of tender.                                                        |
| P <sub>T</sub> | IEEMA price index for Copper wire rods for the month which is two months prior to date of inspection of material.                                                                 |
| P <sub>O</sub> | IEEMA price index for Copper wire rods for the month which is one month prior to date of opening of tender.                                                                       |
| Z <sub>T</sub> | IEEMA price index for Zinc for the month which is two months prior to date of inspection of material                                                                              |
| Z <sub>O</sub> | IEEMA price index for Zinc for the month which is one month prior to date of opening of tender                                                                                    |
| I <sub>T</sub> | RBI wholesale price index for the sub-group “Insulators” for the month which is two months prior to date of inspection of material                                                |
| I <sub>O</sub> | RBI wholesale price index for the sub-group “Insulators” for the month which is one month prior to date of opening of tender                                                      |

### **(III) SIGNALING & TELECOMMUNICATION WORKS:**

- (a) The following expressions and meanings are assigned to the value of the work done for signalling and telecommunication works:

SIGWK = Value of signalling works for a stage payment of the item signalling works;

INVSIG = Value of inventory for signalling works for a stage payment of the item inventory for signalling works;

INTGTESTSIG = Value of integrated testing and commission for signalling works of the Railway Project;

COMWK= Value of telecommunication works for a stage payment of the item telecommunication works;

INVCOM = Value of inventory for telecommunication works for a stage payment of the item inventory for telecommunication works; and

INTGTESTCOM = Value of integrated testing and commission for telecommunication

works of the Railway Project.

- (b) Price adjustment for changes in cost of signalling works and telecommunication works shall be paid in accordance with the following formula:
- (i) 
$$\text{VSIGWK} = 0.85 \text{ SIGWK} \times [\text{PELEX} \times (\text{ELEX}_i - \text{ELEX}_o) / \text{ELEX}_o + \text{POFC} \times (\text{OFC}_i - \text{OFC}_o) / \text{OFC}_o + \text{PLB} \times (\text{LBI} - \text{LBo}) / \text{LBo} + \text{POTH} \times (\text{OTHi} - \text{OTHo}) / \text{OTHo} + \text{S30C} \times (\text{P30C}_i - \text{P30C}_o) / \text{P30C}_o + \text{S24C} \times (\text{P24C}_i - \text{P24C}_o) / \text{P24C}_o + \text{S19C} \times (\text{P19C}_i - \text{P19C}_o) / \text{P19C}_o + \text{S12C} \times (\text{P12C}_i - \text{P12C}_o) / \text{P120C}_o + \text{S9C} \times (\text{P9C}_i - \text{P9C}_o) / \text{P9C}_o + \text{S6C} \times (\text{P6C}_i - \text{P6C}_o) / \text{P6C}_o + \text{S4C} \times (\text{P4C}_i - \text{P4C}_o) / \text{P4C}_o + \text{S2C} \times (\text{P2C}_i - \text{P2C}_o) / \text{P2C}_o + \text{S12C2.5} \times (\text{P12C2.5}_i - \text{P12C2.5}_o) / \text{P12C2.5}_o + \text{S2C2.5} \times (\text{P2C2.5}_i - \text{P2C2.5}_o) / \text{P2C2.5}_o + \text{S2C25} \times (\text{P2C25}_i - \text{P2C25}_o) / \text{P2C25}_o + \text{QC} \times (\text{PQC}_i - \text{PQC}_o) / \text{PQC}_o];$$
  - (ii) 
$$\text{VINVSIG} = 0.85 \text{ SIGWK} \times [\text{PELEX} \times (\text{ELEX}_i - \text{ELEX}_o) / \text{ELEX}_o + \text{POTH} \times (\text{OTHi} - \text{OTHo}) / \text{OTHo}];$$
  - (iii) 
$$\text{VINTGTESTSIG} = 0.85 \text{ INTGTESTSIG} \times [\text{PLB} \times (\text{LBI} - \text{LBo}) / \text{LBo} + \text{POTH} \times (\text{OTHi} - \text{OTHo}) / \text{OTHo}];$$
  - (iv) 
$$\text{VCOMWK} = 0.85 \text{ COMWK} \times [\text{PELEX} \times (\text{ELEX}_i - \text{ELEX}_o) / \text{ELEX}_o + \text{POFC} \times (\text{OFC}_i - \text{OFC}_o) / \text{OFC}_o + \text{PLB} \times (\text{LBI} - \text{LBo}) / \text{LBo} + \text{POTH} \times (\text{OTHi} - \text{OTHo}) / \text{OTHo} + \text{S30C} \times (\text{P30C}_i - \text{P30C}_o) / \text{P30C}_o + \text{S24C} \times (\text{P24C}_i - \text{P24C}_o) / \text{P24C}_o + \text{S19C} \times (\text{P19C}_i - \text{P19C}_o) / \text{P19C}_o + \text{S12C} \times (\text{P12C}_i - \text{P12C}_o) / \text{P120C}_o + \text{S9C} \times (\text{P9C}_i - \text{P9C}_o) / \text{P9C}_o + \text{S6C} \times (\text{P6C}_i - \text{P6C}_o) / \text{P6C}_o + \text{S4C} \times (\text{P4C}_i - \text{P4C}_o) / \text{P4C}_o + \text{S2C} \times (\text{P2C}_i - \text{P2C}_o) / \text{P2C}_o + \text{S12C2.5} \times (\text{P12C2.5}_i - \text{P12C2.5}_o) / \text{P12C2.5}_o + \text{S2C2.5} \times (\text{P2C2.5}_i - \text{P2C2.5}_o) / \text{P2C2.5}_o + \text{S2C25} \times (\text{P2C25}_i - \text{P2C25}_o) / \text{P2C25}_o + \text{QC} \times (\text{PQC}_i - \text{PQC}_o) / \text{PQC}_o + \text{PCEQP} \times (\text{CEQP}_i - \text{CEQP}_o) / \text{CEQP}_o];$$
  - (v) 
$$\text{VINVCOM} = 0.85 \text{ SIGWK} \times [\text{PELEX} \times (\text{ELEX}_i - \text{ELEX}_o) / \text{ELEX}_o + \text{PCEQP} \times (\text{CEQP}_i - \text{CEQP}_o) / \text{CEQP}_o + \text{POTH} \times (\text{OTHi} - \text{OTHo}) / \text{OTHo}];$$
 and
  - (vi) 
$$\text{VINTGTESTCOM} = 0.85 \text{ INTGTESTCOM} \times [\text{PLB} \times (\text{LBI} - \text{LBo}) / \text{LBo} + \text{POTH} \times (\text{OTHi} - \text{OTHo}) / \text{OTHo}].$$

Where

VSIGWK = Increase or decrease in the cost of signalling works during the period under consideration due to changes in the rates for relevant components as specified in sub-paragraph (h);

VINVSIG = Increase or decrease in the cost of inventory for signalling during the period under consideration due to changes in the rates for relevant components as specified in sub-paragraph (h);

VINTGTESTSIG = Increase or decrease in the cost of integrated testing and commissioning of signalling works of the Railway Project during the period under consideration due to changes in the rates for relevant components as specified in sub-paragraph (h);

VCOMWK = Increase or decrease in the cost of communication works during the period under consideration due to changes in the rates for relevant components as specified in sub-paragraph (h);

VINVCOM = Increase or decrease in the cost of inventory for telecommunications works

during the period under consideration due to changes in the rates for relevant components as specified in sub-paragraph (h);

VINTGTESTCOM = Increase or decrease in the cost of integrated testing and commissioning of telecommunication works of the Railway Project during the period under consideration due to changes in the rates for relevant components as specified in sub-paragraph (h);

PCEQP, PELEX, PIC, PLB, POFC, and POTH are the percentages of communication equipment, electronics, PVC insulated cables, labour, optical fibre cables, and other materials respectively;

CEQP<sub>0</sub> = The wholesale price index as published by the Ministry of Commerce and Industry, Government of India (hereinafter called “WPI”) for communication equipment for the month of the Base Month;

CEQP<sub>i</sub> = The WPI for communication equipment for the average price index of the 3 months of the quarter under consideration;

ELEX<sub>0</sub> = The WPI for electronics for the month of the Base Month;

ELEX<sub>i</sub> = The WPI for electronics for the average price index of the 3 months of the quarter under consideration;

P30C<sub>i</sub> = Price payable per Km as adjusted in accordance with price variation Clause for size 30C x 1.5 sq mm signalling cable

P30C<sub>0</sub> = Price per Km of cable as per purchase order/ Contract agreement.

S30C = Percentage of size 30C x 1.5 sq mm signalling cable shall govern the price adjustment of the contract price for signalling and telecommunication works.

P24C<sub>i</sub> = Price payable per Km as adjusted in accordance with price variation Clause for size 24C x 1.5 sq mm signalling cable

P24C<sub>0</sub> = Price per Km of cable as per purchase order/ Contract agreement.

S24C = Percentage of size 24C x 1.5 sq mm signalling cable shall govern the price adjustment of the contract price for signalling and telecommunication works.

P19C<sub>i</sub> = Price payable per Km as adjusted in accordance with price variation Clause for size 19C x 1.5 sq mm signalling cable

P19C<sub>0</sub> = Price per Km of cable as per purchase order/ Contract agreement.

S19C = Percentage of size 19C x 1.5 sq mm signalling cable shall govern the price adjustment of the contract price for signalling and telecommunication works.

P12C<sub>i</sub> = Price payable per Km as adjusted in accordance with price variation Clause for size 12C x 1.5 sq mm signalling cable

P12C<sub>0</sub> = Price per Km of cable as per purchase order/ Contract agreement.

S12C = Percentage of size 12C x 1.5 sq mm signalling cable shall govern the price adjustment of the contract price for signalling and telecommunication works.

P9C<sub>i</sub> = Price payable per Km as adjusted in accordance with price variation Clause for size 9C x 1.5 sq mm signalling cable

$P9C_o$  = Price per Km of cable as per purchase order/ Contract agreement.

$S9C$  = Percentage of size 9C x 1.5 sq mm signalling cable shall govern the price adjustment of the contract price for signalling and telecommunication works.

$P6C_i$  = Price payable per Km as adjusted in accordance with price variation Clause for size 6C x 1.5 sq mm signalling cable

$P6C_o$  = Price per Km of cable as per purchase order/ Contract agreement.

$S6C$  = Percentage of size 6C x 1.5 sq mm signalling cable shall govern the price adjustment of the contract price for signalling and telecommunication works.

$P4C_i$  = Price payable per Km as adjusted in accordance with price variation Clause for size 4C x 1.5 sq mm signalling cable

$P4C_o$  = Price per Km of cable as per purchase order/ Contract agreement.

$S4C$  = Percentage of size 4C x 1.5 sq mm signalling cable shall govern the price adjustment of the contract price for signalling and telecommunication works.

$P2C_i$  = Price payable per Km as adjusted in accordance with price variation Clause for size 2C x 1.5 sq mm signalling cable

$P2C_o$  = Price per Km of cable as per purchase order/ Contract agreement.

$S2C$  = Percentage of size 2C x 1.5 sq mm signalling cable shall govern the price adjustment of the contract price for signalling and telecommunication works.

$P12C2.5_i$  = Price payable per Km as adjusted in accordance with price variation Clause for size 12C x 2.5 sq mm signalling cable

$P12C2.5_o$  = Price per Km of cable as per purchase order/ Contract agreement.

$S12C2.5$  = Percentage of size 12C x 2.5 sq mm signalling cable shall govern the price adjustment of the contract price for signalling and telecommunication works.

$P2C2.5_i$  = Price payable per Km as adjusted in accordance with price variation Clause for size 2C x 2.5 sq mm signalling cable

$P2C2.5_o$  = Price per Km of cable as per purchase order/ Contract agreement.

$S2C2.5$  = Percentage of size 2C x 2.5 sq mm signalling cable shall govern the price adjustment of the contract price for signalling and telecommunication works.

$P2C25_i$  = Price payable per Km as adjusted in accordance with price variation Clause for size 2C x 25 sq mm signalling cable

$P2C25_o$  = Price per Km of cable as per purchase order/ Contract agreement.

$S2C25$  = Percentage of size 2C x 25 sq mm signalling cable shall govern the price adjustment of the contract price for signalling and telecommunication works.

$PQC_i$  = Price payable per Km as adjusted in accordance with price variation Clause for size 0.9mm dia, 6 Quad cable.

$PQC_o$  = Price per Km of cable as per purchase order/ Contract agreement.

$QC$  = Percentage of size 0.9mm dia, 6 Quad cable shall govern the price.

LBo = The consumer price index for industrial workers – All India, published by Labour Bureau, Ministry of Labour, Government of India, (hereinafter called “CPI”) for the month of the Base Month;

LBi = The CPI for industrial workers – All India for the average price index of the 3 months of the quarter under consideration;

OFCo = The WPI for fibre cables for the month of the Base Month;

OFCi = The WPI for fibre cables for the average price index of the 3 months of the quarter under consideration;

OTHo = The WPI for all commodities for the month of the Base Month; and

OTHi = The WPI for all commodities for the average price index of the 3 months of the quarter under consideration.

- (c) The following percentages shall govern the price adjustment of the Contract Price for signalling and telecommunication works:

| Works<br><br>Component                  | Signalling       |                      |                                      | Telecommunication       |                    |                                      |
|-----------------------------------------|------------------|----------------------|--------------------------------------|-------------------------|--------------------|--------------------------------------|
|                                         | Signalling Works | Signalling inventory | Integrated testing and Commissioning | Telecommunication Works | Telecomm inventory | Integrated testing and Commissioning |
| Electronics (PELEX)                     | ***%             | ***%                 | —                                    | ***%                    | ***%               | —                                    |
| Communication Equipment (PCEQP)         | —                | —                    | —                                    | ***%                    | ***%               | —                                    |
| Optical Fibre Cable (POFC)              | ***%             | —                    | —                                    | ***%                    | —                  | —                                    |
| 30C x 1.5 sq mm signalling cable(S30C)  | ***%             | —                    | —                                    | ***%                    | —                  | —                                    |
| 24C x 1.5 sq mm signalling cable (S24C) | ***%             | —                    | —                                    | ***%                    | —                  | —                                    |
| 19Cx 1.5 sq mm signalling cable (S19C)  | ***%             | —                    | —                                    | ***%                    | —                  | —                                    |
| 12C x 1.5 sq mm signalling cable (S12C) | ***%             | —                    | —                                    | ***%                    | —                  | —                                    |
| 9C x 1.5 sq mm signalling cable (S9C)   | ***%             | —                    | —                                    | ***%                    | —                  | —                                    |
| 6C x 1.5 sq mm signalling cable (S6C)   | ***%             | —                    | —                                    | ***%                    | —                  | —                                    |
| 4C x 1.5 sq mm signalling cable (S4C)   | ***%             | —                    | —                                    | ***%                    | —                  | —                                    |

|                                            |      |      |      |      |      |      |
|--------------------------------------------|------|------|------|------|------|------|
| 2C x 1.5 sq mm signalling cable (S2C)      | ***% | —    | —    | ***% | —    | —    |
| 12C x 2.5 sq mm signalling cable (S12C2.5) | ***% | —    | —    | ***% | —    | —    |
| 2C x 2.5 sq mm signalling cable (S2C2.5)   | ***% | —    | —    | ***% | —    | —    |
| 2C x 25 sq mm signalling cable (S2C25)     | ***% | —    | —    | ***% | —    | —    |
| 0.9 mm dia, 6Quad cable (QC)               | ***% | —    | —    | ***% | —    | —    |
| Labour (PLB)                               | ***% | —    | ***% | ***% | ***% | ***% |
| Other materials                            | ***% | ***% | ***% | ***% | ***% | ***% |
| Total                                      | 100% | 100% | 100% | 100% | 100% | 100% |

*(Note- the percentages may be finalized by tendering authority depending on BOQ)*

### **FORMULAE FOR SIGNALING & TELECOM CABLE**

The price payable for signalling cables is variable as per Price Variation Formula given below:

**For Signalling Copper Cables:**

$$P_i = P_o + CuF (Cu - Cu_o) + CCFcu(CC - CCo) + FeF (Fe - Fe_o)$$

**For Telecom Copper Cables For Jelly Filled, 0.9 mm dia, 6 quad cable**

$$P_i = P_o + CuF (Cu - Cu_o) + AlFcu(Al - Alo) + CCFcu (CC - CCo) + FeF (Fe - Fe_o)$$

**For Aluminium Power Cables:**

$$P_i = P_o + AlF (Al - Alo) + CCFAI(CC - CCo) + FeF (Fe - Fe_o)$$

Where,

$P_i$  = Price payable per KM as adjusted in accordance with Price variation clause.

$P_o$  = Price per KM of cable as per Purchase order.

$CuF$  = Variation factor for Copper

$Cu_o$  = Price of copper Rod in Rs. Per MT

$CCFCu$  = Variation factor for PVC Compound for Copper Signalling & Telecom cable

$CCo$  = Price of PVC Compound in Rs. Per MT

$AlF$  = Variation factor for Aluminium

$Alo$  = Price of EC grade LME Aluminium rods (Properzi rods) in Rs. Per MT.

$CCFAI$  = Variation factor for PVC Compound for Aluminium power cable

$FeF$  = Variation factor for Steel

$Fe_o$  = Price of Steel for Armour (Flat strip 4 mm. x 0.8mm/ Round 1.4mm dia) in Rs. Per MT



**(Prices per MT for Cu, CCo, Fe, Al as applicable on the 1<sup>st</sup> working day of the month, one month prior to the deadline for submission of bids. The above prices and indices are as published by IEEMA vide circular reference no. IEEMA (PVC) /CABLE --/--/-- one month prior to the deadline for submission of bids.)**

Cu= Price of Copper Rod in Rs. Per MT.

C<sub>C</sub>= Price of PVC Compound in Rs. Per MT.

Fe= Price of Steel for Armouring (Flat strip 4mm x 0.8 mm/ Round 1.4mm dia) in Rs. Per MT.

Al = Price of EC grade LME Aluminium rods (Properzi rods) in Rs. Per MT.

**(Prices per MT for Cu, CC, Fe, Al as prevailing on 1<sup>st</sup> working day of the calendar month covering the date One month prior to the date of inspection call letter will be applicable for the calculation of updated price. The above prices and indices are as published by IEEMA vide circular reference no. IEEMA (PVC) /CABLE --/--/-- one month prior to the date of inspection.)**

The value of variation factors for copper, steel and PVC Compound are different for different sizes of signalling cables. Accordingly, the PVC formula for some of the types of signalling cable is as given under:-

Underground Railway Signalling Cable unscreened and armoured copper conductor

- (i) Size 30 C x 1.5 sq.mm.

$$P30C_i = P30C_o + 0.391(Cu - C_{uo}) + 0.557(CC - C_{Co}) + 0.425(Fe - Fe_o)$$

For armouring, price of steel flat strip of size 4mmx0.8mm is to be taken into consideration.

- (ii) Size 24C x 1.5 sq.mm

$$P24C_i = P24C_o + 0.313(Cu - C_{uo}) + 0.481(CC - C_{Co}) + 0.398(Fe - Fe_o)$$

For armouring, value of steel flat strip of size 4mmx0.8mm is to be taken into consideration.

- (iii) Size 19C x 1.5 sq.mm

$$P19C_i = P19C_o + 0.248(Cu - C_{uo}) + 0.395(CC - C_{Co}) + 0.343(Fe - Fe_o)$$

For armouring, value of steel flat strip of size 4mmx0.8mm is to be taken into consideration.

- (iv) Size 12C x 1.5 sq.mm

$$P12C_i = P12C_o + 0.157(Cu - C_{uo}) + 0.277(CC - C_{Cu}) + 0.289(Fe - Fe_o)$$

For armouring, value of steel wire size 1.4mm dia is to be taken into consideration.

- (v) Size 9C x 1.5 sq.mm

$$P9C_i = P9C_o + 0.117(Cu - C_{uo}) + 0.241(CC - C_{Cu}) + 0.383(Fe - Fe_o)$$

For armouring, value of steel wire size 1.4mm dia is to be taken into consideration.

- (vi) Size 6Cx 1.5 sq.mm

$$P6C_i = P6C_o + 0.078(Cu - C_{uo}) + 0.199(CC - CC_u) + 0.329(Fe - Fe_o)$$

For armouring, value of steel wire size 1.4mm dia is to be taken into consideration.

(vii) Size 4C x 1.5 sq.mm

$$P4C_i = P4C_o + 0.052(Cu - C_{uo}) + 0.152(CC - CC_o) + 0.277(Fe - Fe_o)$$

For armouring, value of steel wire size 1.4mm dia is to be taken into consideration.

(viii) Size 2C x 4 sq.mm(multistrand)

$$P2C_i = P2C_o + 0.073(Cu - C_{uo}) + 0.156(CC - CC_o) + 0.3(Fe - Fe_o)$$

For armouring, value of steel wire size 1.4mm dia is to be taken into consideration.

(ix) Size 12C x 2.5 sq.mm

$$P12C_{2.5} = P12C_{2.5o} + 0.282(Cu - C_{uo}) + 0.371(CC - CC_o) + 0.342(Fe - Fe_o)$$

For armouring, value of steel flat strip of size 4mmx0.8mm is to be taken into consideration.

(x) Size 2C x 2.5 sq.mm

$$P2C_{2.5} = P2C_{2.5o} + 0.047(Cu - C_{uo}) + 0.139(CC - CC_o) + 0.277(Fe - Fe_o)$$

For armouring, value of steel wire size 1.4mm dia is to be taken into consideration.

(xi) Size 2C x 25 sq.mm PVC insulated, armoured, Aluminium power cable

$$P2C_{25} = P2C_{25o} + 0.146(Al - Al_o) + 0.303(CC - CC_o) + 0.306(Fe - Fe_o)$$

For armouring, value of steel flat strip of size 4mmx0.8mm is to be taken into consideration.

(xii) For Jelly filled, 0.9mm dia, 6 quad cable

$$PQC_i = PQC_o + 0.135(Al - Al_o) + 0.139(Cu - C_{uo}) + 0.515(CC - CC_o) + 0.693(Fe - Fe_o)$$

For PVC Compound Grade CW-22, is to be taken into consideration.

**46A.8** The demands for escalation of cost shall be allowed on the basis of provisional indices as mentioned above in Clause 46A.7. Any adjustment needed to be done based on the finally published indices shall be made as and when they become available.

**46A.9: (1)** Relevant categories of steel for the purpose of operating Price Variation formula as mentioned in this Clause shall be as under:

| SL | Classification                                     | Rates to be used for calculating $S_Q$ or $S_B$                                                                                      |
|----|----------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| 1. | Reinforcement bars and other rounds                | Average of per tonne rates of 10mm dia TMT & 25mm dia TMT; confirming IS1786; Fe 500                                                 |
| 2. | All types and sizes of angles, channels and joists | Average of per tonne rates of 'Angle 75x75x6mm, Mild Steel Plate 10mm thickness and Channel 150x75mm; confirming IS2062, E250 Gr "A" |
| 3. | All types and sizes of plates                      | Average of per tonne rates of 'MS Plates 10mm thickness and 25mm thickness; confirming IS2062, E250 Gr "A"                           |

|    |                                                                |                                                                                |
|----|----------------------------------------------------------------|--------------------------------------------------------------------------------|
| 4. | Any other section of steel not covered in the above categories | Average of price for the 3 categories covered under SL 1, 2 & 3 in this table. |
|----|----------------------------------------------------------------|--------------------------------------------------------------------------------|

(2). Relevant city for referring “JPC (Joint Plant Committee)” rates of steel items (SQ /SB) in different Zonal Railways shall be as under :

| SL | City    | Railway                                                                                 |
|----|---------|-----------------------------------------------------------------------------------------|
| 1. | Delhi   | Northern , North Central, North Eastern, North Western                                  |
| 2. | Kolkata | Eastern, East Central, East Coast, Northeast Frontier, South Eastern, Southeast Central |
| 3. | Mumbai  | Central, Western, West Central                                                          |
| 4. | Chennai | Southern, South Central&South Western                                                   |

#### 46A.10 Price Variation during Extended Period of Contract

The price adjustment as worked out above, i.e. either increase or decrease shall be applicable upto the stipulated date of completion of work including the extended period of completion where such extension has been granted under Clause 17A of the Standard General Conditions of Contract. However, where extension of time has been granted due to Contractor’s failure under Clause 17B of the Standard General Conditions of Contract, price adjustment shall be done as follows:

- a. In case the indices increase above the indices applicable to the last month of original completion period or the extended period under Clause 17A, the price adjustment for the period of extension granted under Clause 17B shall be limited to the amount payable as per the Indices applicable to the last month of the original completion period or the extended period under Clause 17A of the Standard General Conditions of Contract; as the case may be.
- b. In case the indices fall below the indices applicable to the last month of original/ extended period of completion under Clause 17A, as the case may be; then the lower indices shall be adopted for the price adjustment for the period of extension under Clause 17B of the Standard General Conditions of Contract.

**47. Maintenance of Works:** The Contractor shall at all times during the progress and continuance of the works and also for the period of maintenance specified in the Tender Form after the date of issue of the certificate of completion by the Engineer or any other earlier date subsequent to the completion of the works that may be fixed by the Engineer, be responsible for and effectively maintain and uphold in good substantial, sound and perfect condition all and every part of the works and shall make good from time to time and at all times as often as the Engineer shall require, any damage or defect that may during the above period arise in or be discovered or be in any way connected with the works, provided that such damage or defect is not directly caused by errors in the contract documents, act of providence or insurrection or civil riot, and the Contractor shall be liable for and shall pay and make good to the Railway or other persons legally entitled thereto whenever required by the Engineer so to do, all losses, damages, costs and expenses they or any of them may incur or be put or be liable to by reasons or in consequence of the operations of the Contractor or of his failure in any respect.

**48.(1) Certificate of Completion of Works:** As soon as in the opinion of the Engineer, the work has been completed and has satisfactorily passed any final test or tests that may be prescribed, the

Engineer shall issue a certificate of completion duly indicating the date of completion in respect of the work and the period of maintenance of the work shall commence from the date of completion mentioned in such certificate. The certificate, inter alia, should mention that the work has been completed in all respects and that all the contractual obligations have been fulfilled by the Contractor and that there is no due from the Contractor to Railways against the contract concerned.

The Engineer may also issue such a certificate indicating date of completion with respect to any part of the work (before the completion of the whole of work), which has been both completed to the satisfaction of the Engineer and occupied or used by the Railway. When any such certificate is given in respect of part of a work, such part shall be considered as completed and the period of maintenance of such part shall commence from the date of completion mentioned in the completion certificate issued for that part of the work.

**48.(2) Contractor not Absolved by Completion Certificate:** The Certificate of Completion in respect of the works referred to in Sub-Clause (1) of this Clause shall not absolve the Contractor from his liability to make good any defects imperfections, shrinkages or faults which may appear during the period of maintenance specified in the tender arising in the opinion of the Engineer from materials or workmanship not in accordance with the drawings or specifications or instruction of the Engineer, which defects, imperfections, shrinkages or faults shall upon the direction in writing of the Engineer be amended and made good by the Contractor at his own cost; and in case of default on the part of Contractor, the Engineer may employ labour and materials or appoint another Contractor to amend and make good such defects, imperfections, shrinkages and faults and all expenses consequent thereon and incidental thereto shall be borne by the Contractor and shall be recoverable from any moneys due to him under the contract.

**48(3) Final Supplementary Agreement:** After the work is completed or otherwise concluded by the parties with mutual consent, and taken over by the Railway as per terms and conditions of the contract agreement, and there is unequivocal no claim on either side under the Contract other than as mentioned in item 4 of Annexure XIV, the parties shall execute the Final Supplementary Agreement as per Annexure XIV.

**49. Approval only by Maintenance Certificate:** No certificate other than Maintenance Certificate, if applicable, referred to in Clause 50 of the Conditions shall be deemed to constitute approval of any work or other matter in respect of which it is issued or shall be taken as an admission of the due performance of the contract or any part thereof.

**50.(1) Maintenance Certificate:** The Contract shall not be considered as completed until a Maintenance Certificate, if applicable, shall have been signed by the Engineer stating that the works have been completed and maintained to his satisfaction. The Maintenance Certificate shall be given by the Engineer upon the expiration of the period of maintenance or as soon thereafter as any works ordered during such period pursuant to Sub Clause (2) to Clause 48 of these Conditions shall have been completed to the satisfaction of the Engineer, and full effect shall be given to this Clause notwithstanding the taking possession of or using the works or any part thereof by the Railway.

The Competent Authority to issue above Maintenance Certificate shall normally be the authority who is competent to sign the contract. If this Competent Authority is of the rank lower than JA Grade, then a JA Grade Officer (concerned with the work) should issue the Certificate. The Certificate, inter

alia, should mention that the work has been completed in all respects and that all the contractual obligations have been fulfilled by the Contractor and that there is no due from the Contractor to Railways against the contract concerned

**50.(2) Cessation of Railway's Liability:** The Railway shall not be liable to the Contractor for any matter arising out of or in connection with the contract for execution of the works unless the Contractor has made a claim in writing in respect thereof before the issue of the Maintenance Certificate under this clause.

**50.(3) Unfulfilled Obligations:** Notwithstanding the issue of the Maintenance Certificate the Contractor and (subject to Sub-Clause (2) of this Clause) the Railway shall remain liable for the fulfillment of any obligation incurred under the provision of the contract prior to the issue of the Maintenance Certificate which remains unperformed at the time such Certificate is issued and for the purposes of determining the nature and extent of any such obligations, the contract shall be deemed to remain in force between the parties thereto.

**51.(1) Final Payment:** On the Engineer's certificate of completion in respect of the works, adjustment shall be made and the balance of account based on the Engineer or the Engineer's representative's certified measurements or Engineer's certified "contractor's authorized engineer's measurements" of the total quantity of work executed by the Contractor upto the date of completion and on the rates accepted in Bill(s) of Quantities and for extra works on rates determined under Clause 39 of these Conditions shall be paid to the Contractor subject always to any deduction which may be made under these presents and further subject to the Contractor having signed delivered to the Engineer enclosing either a full account in detail of all claims he may have on the Railway in respect of the works or having delivered No Claim Certificate and the Engineer having after the receipt of such account given a certificate in writing that such claims are not covered under excepted matter i.e. Clauses 7(j), 8, 18, 22(5), 39.1, 39.2, 40A, 43(2), 45(i)(a), 55, 55-A(5), 57, 57A, 61(1), 61(2) and 62(1), 63(iv) and 63.2.11 of the Standard General Conditions of Contract or in any Clause (stated as excepted matter) of the Special Conditions of the Contract, that the whole of the works to be done under the provisions of the Contracts have been completed, that they have been inspected by him since their completion and found to be in good and substantial order, that all properties, works and things, removed, disturbed or injured in consequence of the works have been properly replaced and made good and all expenses and demands incurred by or made upon the Railway for or in the respect of damage or loss by from or in consequence of the works, have been satisfied agreeably and in conformity with the contract.

**51.(2) Post Payment Audit:** It is an agreed term of contract that the Railway reserves to itself the right to carry out a post-payment audit and/ or technical examination of the works and the Final Bill including all supporting vouchers, abstracts etc. and to make a claim on the Contractor for the refund of any excess amount paid to him till the release of security deposit or settlement of claims, whichever is later, if as a result of such examination any over-payment to him is discovered to have been made in respect of any works done or alleged to have been done by him under the contract.

**51-A. Production of Vouchers etc. by the Contractor:**

- (i) For a contract of more than one crore of rupees, the Contractor shall, whenever required, produce or cause to be produced for examination by the Engineer any quotation, invoice, cost

or other account, book of accounts, voucher, receipt, letter, memorandum, paper of writing or any copy of or extract from any such document and also furnish information and returns verified in such manner as may be required in any way relating to the execution of this contract or relevant for verifying or ascertaining cost of execution of this contract (the decision of the Engineer on the question of relevancy of any documents, information or return being final and binding in the parties). The Contractor shall similarly produce vouchers etc., if required to prove to the Engineer, that materials supplied by him, are in accordance with the specifications laid down in the contract.

- (ii) If any portion of the work in a contract of value more than one crore of rupees be carried out by a sub-contractor or any subsidiary or allied firm or company (as per Clause 7 of the Standard General Conditions of Contract), the Engineer shall have power to secure the books of such sub-contract or any subsidiary or allied firm or company, through the Contractor, and such books shall be open to his inspection.
- (iii) The obligations imposed by Sub Clause (i) & (ii) above is without prejudice to the obligations of the Contractor under any statute rules or orders binding on the Contractor.

**52. Withholding and Lien in Respect of Sums Claimed:** Whenever any claim or claims for payment of a sum of money arises out of or under the contract against the Contractor, the Railway shall be entitled to withhold and also have a lien to retain such sum or sums in whole or in part from the security, if any, deposited by the Contractor and for the purpose aforesaid, the Railway shall be entitled to withhold the said cash Security Deposit or the Security if any, furnished as the case may be and also have a lien over the same pending finalization or adjudication of any such claim. In the event of the security being insufficient to cover the claimed amount or amounts or if no security has been taken from the Contractor, the Railway shall be entitled to withhold and have a lien to the extent of the such claimed amount or amounts referred to supra, from any sum or sums found payable or which at any time thereafter may become payable to the Contractor under the same contract or any other contract with this or any other Railway or any Department of the Central Government pending finalization or adjudication of any such claim.

It is an agreed term of the contract that the sum of money or moneys so withheld or retained under the lien referred to above, by the Railway will be kept withheld or retained as such by the Railways till the claim arising out of or under the contract is determined by the arbitrator (if the contract governed by the Arbitration Clause) or by the competent court as the case may be and that the Contractor will have no claim for interest or damages whatsoever on any account in respect of such withholding or retention under the lien referred to supra and duly notified as such to the Contractor. For the purpose of this clause, where the Contractor is a partnership firm or a company, the Railway shall be entitled to withhold and also have a lien to retain towards such claimed amount or amounts in whole or in part from any sum found payable to any partner / company, as the case may be whether in his individual capacity or otherwise.

**52-A Lien in Respect of Claims in other Contracts:**

- (i) Any sum of money due and payable to the Contractor (including the Security Deposit returnable to him) under the contract may be withheld or retained by way of lien by the Railway, against any claim of this or any other Railway or any other Department of the Central

Government in respect of payment of a sum of money arising out of or under any other contract made by the Contractor with this or any other Department of the Central Government.

- (ii) However, recovery of claims of Railway in regard to terminated contracts may be made from the Final Bill, Security Deposits and Performance Guarantees of other contract or contracts, executed by the Contractor. The Performance Guarantees submitted by the Contractor against other contracts, if required, may be withheld and encashed. In addition, 10% of each subsequent 'on-account bill' may be withheld, if required, for recovery of Railway's dues against the terminated contract.
- (iii) It is an agreed term of the contract that the sum of money so withheld or retained under this Clause by the Railway will be kept withheld or retained as such by the Railway till the claim arising out of or under any other contract is either mutually settled or determined by arbitration, if the other contract is governed by Arbitration Clause or by the competent court as the case may be and Contractor shall have no claim for interest or damages whatsoever on this account or on any other ground in respect of any sum of money withheld or retained under this Clause and duly notified as such to the Contractor.

**53. Signature on Receipts for Amounts:** Every receipt for money which may become payable or for any security which may become transferable to the Contractors under these presents, shall, if signed in the partnership name by anyone of the partners of a Contractor's firm be a good and sufficient discharge to the Railway in respect of the moneys or security purported to be acknowledged thereby and in the event of death of any of the Contractor, partners during the pendency of the contract, it is hereby expressly agreed that every receipt by anyone of the surviving Contractor partners shall if so signed as aforesaid be good and sufficient discharge as aforesaid provided that nothing in this Clause contained shall be deemed to prejudice or effect any claim which the Railway may hereafter have against the legal representative of any Contractor partner so dying for or in respect to any breach of any of the conditions of the contract, provided also that nothing in this clause contained shall be deemed to prejudice or effect the respective rights or obligations of the Contractor partners and of the legal representatives of any deceased Contractor partners interse.

## **LABOUR**

**54. Wages to Labour:** The Contractor shall be responsible to ensure compliance with the provision of the Minimum Wages Act, 1948 (hereinafter referred to as the "said Act") and the Rules made thereunder in respect of any employees directly or through petty Contractors or sub-contractors employed by him for the purpose of carrying out this contract.

If, in compliance with the terms of the contract, the Contractor supplied any labour to be used wholly or partly under the direct orders and control of the Railways whether in connection with any work being executed by the Contractor or otherwise for the purpose of the Railway such labour shall, for the purpose of this Clause, still be deemed to be persons employed by the Contractor.

If any moneys shall, as a result of any claim or application made under the said Act be directed to be paid by the Railway, such money shall be deemed to be moneys payable to the Railway by the Contractor and on failure by the Contractor to repay the Railway any moneys paid by it as aforesaid within seven days after the same shall have been demanded, the Railways shall be entitled to recover the same from Contractor's bills/Security Deposit or any other dues of Contractor with the Government of India.

**54-A. Apprentices Act:** The Contractor shall be responsible to ensure compliance with the provisions of the Apprentices Act, 1961 and the Rules and Orders issued thereunder from time to time in respect of apprentices directly or through petty Contractors or sub-contractors employed by him for the purpose of carrying out the Contract.

If the Contractor directly or through petty Contractors or sub-contractors fails to do so, his failure will be a breach of the contract and the Railway may, in its discretion, rescind the contract. The Contractor shall also be liable for any pecuniary liability arising on account of any violation of the provisions of the Act.

**55. Provisions of Payments of Wages Act:** The Contractor shall comply with the provisions of the Payment of Wages Act, 1936 and the rules made thereunder in respect of all employees employed by him either directly or through petty Contractors or sub-contractors in the works. If in compliance with the terms of the contract, the Contractor directly or through petty Contractors or sub-contractors shall supply any labour to be used wholly or partly under the direct orders and control of the Engineer whether in connection with the works to be executed hereunder or otherwise for the purpose of the Engineer, such labour shall nevertheless be deemed to comprise persons employed by the Contractor and any moneys which may be ordered to be paid by the Engineer shall be deemed to be moneys payable by the Engineer on behalf of the Contractor and the Engineer may on failure of the Contractor to repay such money to the Railways deduct the same from any moneys due to the Contractor in terms of the contract. The Railway shall be entitled to recover the same from Contractor's bills/Security Deposit or any other dues of Contractor with the Government of India all moneys paid or payable by the Railway by way of compensation of aforesaid or for costs of expenses in connection with any claim thereto and the decision of the Engineer upon any question arising out of the effect or force of this Clause shall be final and binding upon the Contractor.

**55-A. Provisions of Contract Labour (Regulation and Abolition) Act, 1970:**

**55-A.(1)** The Contractor shall comply with the provision of the contract labour (Regulation and Abolition) Act, 1970 and the Contract labour (Regulation and Abolition) Central Rules 1971 as modified from time to time, wherever applicable and shall also indemnify the Railway from and against any claims under the aforesaid Act and the Rules.

**55-A.(2)** The Contractor shall obtain a valid license under the aforesaid Act as modified from time to time before the commencement of the work and continue to have a valid license until the completion of the work. Any failure to fulfill the requirement shall attract the penal provision of the Act.

**55-A.(3)** The Contractor shall pay to the labour employed by him directly or through sub-contractors the wages as per provision of the aforesaid Act and the Rules wherever applicable. The



Contractor shall notwithstanding the provisions of the contract to the contrary, cause to be paid the wages to labour, indirectly engaged on the works including any engaged by sub-contractors in connection with the said work, as if the labour had been immediately employed by him.

**55-A.(4)** In respect of all labour directly or indirectly employed in the work for performance of the Contractor's part of the contract, the Contractor shall comply with or cause to be complied with the provisions of the aforesaid Act and Rules wherever applicable.

**55-A.(5)** In every case in which, by virtue of the provisions of the aforesaid Act or the rules, the Railway is obliged to pay any amount of wages to a workman employed by the Contractor or his sub-contractor in execution of the work or to incur any expenditure on account of the contingent, liability of the Railway due to the Contractor's failure to fulfill his statutory obligations under the aforesaid Act or the rules, the Railway will recover from the Contractor, the amount of wages so paid or the amount of expenditure so incurred and without prejudice to the rights of the Railway under the Section 20, Sub-Section (2) and Section 2, Sub-Section (4) of the aforesaid Act, the Railway shall be at liberty to recover such amount or part thereof from Contractor's bills/Security Deposit or any other dues of Contractor with the Government of India. The Railway shall not be bound to contest any claim made against it under Sub-Section (1) of Section 20 and Sub-Section (4) of Section 21 of the aforesaid Act except on the written request of the Contractor and upon his giving to the Railway full security for all costs for which the Railway might become liable in contesting such claim. The decision of the Chief Engineer regarding the amount actually recoverable from the Contractor as stated above shall be final and binding on the Contractor.

**55-B.Provisions of Employees Provident Fund and Miscellaneous Provisions Act, 1952:**The Contractor shall comply with the provisions of Para 30 & 36-B of the Employees Provident Fund Scheme, 1952; Para 3 & 4 of Employees' Pension Scheme, 1995; and Para 7 & 8 of Employees Deposit Linked Insurance Scheme, 1976; as modified from time to time through enactment of "Employees Provident Fund & Miscellaneous Provisions Act, 1952", wherever applicable and shall also indemnify the Railway from and against any claims under the aforesaid Act and the Rules.

**55-C** (i) Contractor is to abide by the provisions of various labour laws in terms of above clause 54, 55, 55-A and 55-B of the Standard General Conditions of Contract. In order to ensure the same, an application has been developed and hosted on website 'www.shramikkalyan.indianrailways.gov.in'. Contractor shall register his firm/company etc. and upload requisite details of labour and their payment in this portal. These details shall be available in public domain. The registration/ updation in Portal shall be done as under:

- (a) Contractor shall apply for onetime registration of his company/firm etc. in the Shramikkalyan portal with requisite details subsequent to issue of Letter of Acceptance. Engineer shall approve the contractor's registration in the portal within 7 days of receipt of such request.
- (b) Contractor once approved by any Engineer, can create password with login ID (PAN No.) for subsequent use of portal for all Letter of Acceptances (LoAs) issued in his favour.
- (c) The contractor once registered on the portal, shall provide details of his Letter of Acceptances (LoAs) / Contract Agreements on shramikkalyan portal within 15 days of issue

of any LoA for approval of concerned Engineer. Engineer shall update (if required) and approve the details of LoA filled by contractor within 7 days of receipt of such request.

- (d) After approval of LoA by Engineer, contractor shall fill the salient details of contract labours engaged in the contract and ensure updating of each wage payment to them on shramikkalyan portal on monthly basis.
- (e) It shall be mandatory upon the contractor to ensure correct and prompt uploading of all salient details of engaged contractual labour & payments made thereof after each wage period.

(ii) While processing payment of any 'On Account Bill' or 'Final Bill' or release of 'Advances' or 'Performance Guarantee / Security Deposit', contractor shall submit a certificate to the Engineer or Engineer's representatives that "I have uploaded the correct details of contract labours engaged in connection with this contract and payments made to them during the wage period in Railway's Shramikkalyan portal at 'www.shramikkalyan.indianrailways.gov.in' till \_\_\_\_Month, \_\_\_\_Year."

**55-D. Provisions of "The Building and Other Construction Workers (Regulation of Employment and Conditions of Service) Act, 1996" and "The Building and Other Construction Workers' Welfare Cess Act, 1996":**

The tenderers, for carrying out any construction work, shall get themselves registered with the Registering Officer under Section-7 of the Building and Other Construction Workers Act, 1996 and Rules made thereto by the concerned State Govt., and submit Certificate of Registration issued by Registering Officer of the concerned State Govt. (Labour Dept.). The Cess shall be deducted from contractor's bills as per provisions of the Act.

**56. Reporting of Accidents:** The Contractor shall be responsible for the safety of all employees directly or through petty Contractors or sub-contractor employed by him on the works and shall report serious accidents to any of them however and wherever occurring on the works to the Engineer or the Engineers Representative and shall make every arrangement to render all possible assistance.

**57. Provision of Workmen's Compensation Act:** In every case in which by virtue of the provisions of Section 12 Sub-Section (1) of the Workmen's Compensation Act 1923, Railway is obliged to pay compensation to a workman directly or through petty Contractor or sub-contractor employed by the Contractor in executing the work, Railway will recover from the Contractor the amount of the compensation so paid, and, without prejudice to the rights of Railway under Section 12 Sub-section (2) of the said Act, Railway shall be at liberty to recover such amount or any part thereof from Contractor's bills/Security Deposit or any other dues of Contractor with the Government of India. Railway shall not be bound to contest any claim made against it under Section 12 Sub-Section (1) of the said Act except on the written request of the Contractor and upon his giving to Railway full security for all costs for which Railway might become liable in consequence of contesting such claim.

**57-A. Provision of Mines Act:** The Contractor shall observe and perform all the provisions of the Mines Act, 1952 or any statutory modifications or re-enactment thereof for the time being in force

and any rules and regulations made thereunder in respect of all the persons directly or through the petty Contractors or sub-contractors employed by him under this contract and shall indemnify the Railway from and against any claims under the Mines Act, or the rules and regulations framed thereunder, by or on behalf of any persons employed by him or otherwise.

**58. Railway not to Provide Quarters for Contractors:** No quarters shall normally be provided by the Railway for the accommodation of the Contractor or any of his staff employed on the work. In exceptional cases where accommodation is provided to the Contractor at the Railway's discretion, recoveries shall be made at such rates as may be fixed by the Railway for the full rent of the buildings and equipments therein as well as charges for electric current, water supply and conservancy.

**59.(1) Labour Camps:** The Contractor shall at his own expense make adequate arrangements for the housing, supply of drinking water and provision of latrines and urinals for his staff and workmen, directly or through the petty Contractors or sub-contractors and for temporary creche (Bal-Mandir) where 50 or more women are employed at a time. Suitable sites on Railway land, if available, may be allotted to the Contractor for the erection of labour camps, either free of charge or on such terms and conditions that may be prescribed by the Railway. All camp sites shall be maintained in clean and sanitary conditions by the Contractor at his own cost.

**59.(2) Compliance to Rules for Employment of Labour:** The Contractor(s) shall conform to all laws, bye-laws rules and regulations for the time being in force pertaining to the employment of local or imported labour and shall take all necessary precautions to ensure and preserve the health and safety of all staff employed directly or through petty contractors or sub-contractors on the works.

**59.(3) Preservation of Peace:** The Contractor shall take requisite precautions and use his best endeavours to

(i) Prevent any riotous or unlawful behaviour by or amongst his workmen and other employed directly or through the petty Contractors or sub-contractors on the works and for the preservation of peace and protection of the inhabitants and

(ii) Security of property in the neighbourhood of the works. In the event of the Railway requiring the maintenance of a Special Police Force at or in the vicinity of the site during the tenure of works, the expenses thereof shall be borne by the Contractor and if paid by the Railway shall be recoverable from the Contractor.

**59.(4) Sanitary Arrangements:** The Contractor shall obey all sanitary rules and carry out all sanitary measures that may from time to time be prescribed by the Railway Medical Authority and permit inspection of all sanitary arrangements at all times by the Engineer, the Engineer's Representative or the Medical Staff of the Railway. Should the Contractor fail to make the adequate sanitary arrangements, these will be provided by the Railway and the cost thereof recovered from the Contractor.

**59.(5) Outbreak of Infectious Disease:** The Contractor shall remove from his camp such labour and their families as refuse protective inoculation and vaccination when called upon to do so by the Engineer or the Engineer's Representative on the advice of the Railway Medical Authority. Should cholera, plague, or other infectious disease break out, the Contractor shall burn the huts, beddings,

clothes and other belongings of or used by the infected parties and promptly erect new huts on healthy sites as required by the Engineer, failing which within the time specified in the Engineer's requisition, the work may be done by the Railway and the cost thereof recovered from the Contractor.

**59.(6) Treatment of Contractor's Staff in Railway Hospitals:** The Contractor and his staff, other than labourers and their families requiring medical aid from the railway hospital and dispensaries will be treated as private patients and charged accordingly. The Contractors' labourers and their Families will be granted free treatment in railway hospitals and dispensaries where no other hospitals or dispensaries are available provided the Contractor pays the cost of medicines, dressing and diet money according to the normal scale and additional charges for special examinations such as pathological and bacteriological examination, X-Ray, etc. and for surgical operation.

**59. (7) Medical Facilities at Site:** The Contractor shall provide medical facilities at the site as may be prescribed by the Engineer on the advice of the Railway Medical Authority in relation to the strength of the Contractor's resident staff and workmen.

**59. (8) Use of Intoxicants:** The sale of ardent spirits or other intoxicating beverages upon the work or in any of the buildings, encampments or tenements owned, occupied by or within the control of the Contractor or any of his employees shall be forbidden and the Contractor shall exercise his influence and authority to the utmost extent to secure strict compliance with this condition.

**59.(9) Restrictions on the Employment of Retired Engineers of Railway Services Within One Year of their Retirement:** The Contractor shall not, if he is a retired Government Engineer of Gazetted rank, himself engage in or employ or associate a retired Government Engineer of Gazetted rank, who has not completed one year from the date of retirement, in connection with this contract in any manner whatsoever without obtaining prior permission of the President and if the Contractor is found to have contravened this provision it will constitute a breach of contract and administration will be entitled to terminate the contract and forfeit his Performance Guarantee as well as Security Deposit.

**60.(1) Non-Employment of Labourers below the age of 15:** The Contractor shall not employ children below the age of 15 as labourers directly or through petty Contractors or sub-contractors for the execution of work.

**60.(2) Medical Certificate of Fitness for Labour:** It is agreed that the Contractor shall not employ a person above 15 and below 19 years of age for the purpose of execution of work under the contract unless a medical certificate of fitness in the prescribed form (Proforma at Annexure-VIII) granted to him by a certifying surgeon certifying that he is fit to work as an adult, is obtained and kept in the custody of the Contractor or a person nominated by him in this behalf and the person carries with him, while at work; a token giving a reference to such certificate. It is further agreed that the responsibility for having the adolescent examined medically at the time of appointment or periodically till he attains the age of 19 years shall devolve entirely on the Contractor and all the expenses to be incurred on this account shall be borne by him and no fee shall be charged from the adolescent or his parent for such medical examination.

**60.(3) Period of Validity of Medical Fitness Certificate:** A certificate of fitness granted or renewed for the above said purposes shall be valid only for a period of one year at a time. The

certifying surgeon shall revoke a certificate granted or renewed if in his opinion the holder of it, is no longer fit for work in the capacity stated therein. Where a certifying surgeon refuses to grant or renew a certificate or revoke a certificate, he shall, if so required by the person concerned, state his reasons in writing for doing so.

**60.(4) Medical Re-Examination of Labourer:** Where any official appointed in this behalf by the Ministry of Labour is of the opinion that any person employed in connection with the execution of any work under this contract in the age group 15 to 19 years is without a certificate of fitness or is having a certificate of fitness but no longer fit to work in the capacity stated in the certificate, he may serve on the Contractor, or on the person nominated by him in this regard, a notice requiring that such persons shall be examined by a certifying surgeon and such person shall not if the concerned official so directs, be employed or permitted to do any work under this contract unless he has been medically examined and certified that he is fit to work in the capacity stated in the certificate.

#### **EXPLANATIONS:**

- (1) Only Qualified Medical Practitioners can be appointed as "Certifying Surgeons" and the term "Qualified Medical Practitioners" means a person holding a qualification granted by an authority specified in the Schedule to the Indian Medical Degrees Act, 1916 (VII to 1916) or in the Schedule to the Indian Medical Council Act, 1933 (XXVII) of 1933.
- (2) The Certifying surgeon may be a medical officer in the service of State or Municipal Corporation.

#### **DETERMINATION OF CONTRACT**

**61.(1) Right of Railway to Determine the Contract:** The Railway shall be entitled to determine and terminate the contract at any time should, in the Railway's opinion, the cessation of work becomes necessary owing to paucity of funds or from any other cause whatever, in which case the value of approved materials at site and of work done to date by the Contractor will be paid for in full at the rate specified in the contract. Notice in writing from the Railway of such determination and the reasons therefor shall be conclusive evidence thereof.

**61.(2) Payment on Determination of Contract:** Should the contract be determined under sub clause (1) of this clause and the Contractor claims payment for expenditure incurred by him in the expectation of completing the whole of the work, the Railways shall admit and consider such claims as are deemed reasonable and are supported by vouchers to the satisfaction of the Engineer. The Railway's decision on the necessity and propriety of such expenditure shall be final and conclusive.

**61.(3)** The Contractor shall have no claim to any payment of compensation or otherwise, howsoever on account of any profit or advantage which he might have derived from the execution of the work in full but which he did not derive in consequence of determination of contract.

#### **62.(1) Determination of Contract owing to Default of Contractor:**

If the Contractor should:

- (i) Becomes bankrupt or insolvent, or

- (ii) Make an arrangement for assignment in favour of his creditors, or agree to carry out the contract under a Committee of Inspection of his creditors, or
- (iii) Being a Company or Corporation, go into liquidation (other than a voluntary liquidation for the purposes of amalgamation or reconstruction), or
- (iv) Have an execution levied on his goods or property on the works, or
- (v) Assign the contract or any part thereof otherwise than as provided in Clause 7 of these Conditions, or
- (vi) Abandon the contract, or
- (vii) Persistently disregard the instructions of the Engineer, or contravene any provision of the contract, or
- (viii) Fail to adhere to the agreed programme of work by a margin of 10% of the stipulated period, or
- (ix) Fail to Execute the contract documents in terms of Para 8 of the Instructions to Tenderers.
- (x) Fail to submit the documents pertaining to identity of JV and PAN in terms of Para 17.11 of Tender Form (Second Sheet) of Annexure I available in the Instructions to Tenderers.
- (xi) Fail to remove materials from the site or to pull down and replace work after receiving from the Engineer notice to the effect that the said materials or works have been condemned or rejected under Clause 25 and 27 of these Conditions, or
- (xii) Fail to take steps to employ competent or additional staff and labour as required under Clause 26 of these Conditions, or
- (xiii) Fail to afford the Engineer or Engineer's representative proper facilities for inspecting the works or any part thereof as required under Clause 28 of these Conditions, or
- (xiv) Promise, offer or give any bribe, commission, gift or advantage either himself or through his partner, agent or servant to any officer or employee of the Railway or to any person on his or on their behalf in relation to the execution of this or any other contract with this Railway.
- (xv) Fail to adhere to the provisions of Para 16 of Tender Form (Second Sheet) of Annexure I of the Instructions to Tenderers, or provision Clause 59(9) of these Conditions.
- (xvi) Submits copy of fake documents / certificates in support of credentials, submitted by the tenderer

Then and in any of the **said Clause**, the Engineer on behalf of the Railway may serve the Contractor with a notice (Proforma at Annexure-IX) in writing to that effect and if the Contractor does not within seven days after the delivery to him of such notice proceed to make good his default in so far as the same is capable of being made good and carry on the work or comply with such directions as aforesaid of the entire satisfaction of the Engineer, the Railway shall be entitled after giving 48 hours' notice (Proforma at Annexure-X or XII, as the case may be) in writing under the hand of the Engineer to rescind the contract as a whole or in part or parts (as may be specified in such notice) and after expiry of 48 hours' notice, a final termination notice (Proforma at Annexure-XI or XIII, as the case may be) should be issued.

Note: Engineer at his discretion may resort to the part termination of contract with notices (Proforma at Annexure- IX, XII and XIII), only in cases where progress of work is more than or equal to 80% of the original scope of work.

**62.(2) Right of Railway after Rescission of Contract owing to Default of Contractor:** In the event of any or several of the courses, referred to in Sub-Clause (1) of this Clause, being adopted:

(a) The Contractor shall have no claim to compensation for any loss sustained by him by reason of his having purchased or procured any materials or entered into any commitments or made any advances on account of or with a view to the execution of the works or the performance of the contract and Contractor shall not be entitled to recover or be paid any sum for any work thereto for actually performed under the contract unless and until the Engineer shall have certified the performance of such work and the value payable in respect thereof and the Contractor shall only be entitled to be paid the value so certified.

(b) In the contract which has been rescinded as a whole, the Security Deposit already with railways under the contract shall be encashed/ forfeited and the Performance Guarantee already submitted for the contract shall be encashed. The balance work shall be got done independently without risk & cost of the failed Contractor. The failed Contractor shall be debarred from participating in the tender for executing the balance work. If the failed Contractor is a JV or a Partnership firm, then every member/partner of such a firm shall be debarred from participating in the tender for the balance work in his/her individual capacity or as a partner of any other JV /partnership firm.

Further the authorized representative of failed Contractor cannot be accepted as authorized representative in new contract.

(c) In the contract rescinded in part or parts,

(i) The full Performance Guarantee available for the contract shall be recovered. No additional Performance Guarantee shall be required for balance of work being executed through the part terminated contract. The contract value of part terminated contract stands reduced to the balance value of work under the contract.

(ii) The Security Deposit of part terminated contract shall be dealt as per clause 16(2) of these Conditions.

(iii) The defaulting Contractor shall not be issued any completion certificate for the contract.

(iv) The balance work shall be got done independently without risk & cost of the failed Contractor. The failed Contractor shall be debarred from participating in the tender for executing the balance work. If the failed Contractor is a JV or a Partnership firm, then every member/partner of such a firm shall be debarred from participating in the tender for the balance work in his/her individual capacity or as a partner of any other JV /partnership firm.

(v) Further the authorized representative of failed Contractor will not be accepted as authorized representative in new contract.

(d) The Engineer or the Engineer's Representative shall be entitled to take possession of any materials, tools, implements, machinery and buildings on the works or on the property on which these are being or ought to have been executed, and to retain and employ the same in the further execution of the works or any part thereof until the completion of the works without the Contractor being entitled to any compensation for the use and employment thereof or for wear and tear or destruction thereof.

(e) The Engineer shall as soon as may be practicable after removal of the Contractor fix and determine ex-parte or by or after reference to the parties or after such investigation or enquiries as he may consider fit to make or institute and shall certify what amount (if any) had at the time of rescission of the contract been reasonably earned by or would reasonably accrue to the Contractor in respect of the work then actually done by him under the contract and what was the value of any unused, or partially used materials, any constructional plant and any temporary works upon the site. The legitimate amount due to the Contractor after making necessary deductions and certified by the Engineer should be released expeditiously.

## **SETTLEMENT OF DISPUTES – INDIAN RAILWAY ARBITRATION AND CONCILIATION RULES**

### **63. Conciliation of Disputes:**

- (i) This clause is applicable in the tender having advertised value less than or equal to Rs 50 Crore.
- (ii) All disputes and differences of any kind whatsoever arising out of or in connection with the contract, whether during the progress of the work or after its completion and whether before or after the determination of the contract, shall be referred by the Contractor to the "Chief Engineer" or "Divisional Railway Manager" through "Notice of Dispute" provided that no such notice shall be served later than 30 days after the date of issue of Completion Certificate by the Engineer. Chief Engineer or Divisional Railway Manager shall, within 30 days after receipt of the Contractor's "Notice of Dispute", notify the name of conciliator(s) to the Contractor.
- (iii) The Conciliator(s) shall assist the parties to reach an amicable settlement in an independent and impartial manner within the terms of contract.
- (iv) If the parties reach agreement on a settlement of the dispute, they shall draw up and sign a written settlement agreement duly signed by Engineer In-charge, Contractor and conciliator(s). When the parties sign the settlement agreement, it shall be final and binding on the parties.
- (v) The parties shall not initiate, during the conciliation proceedings, any arbitral or judicial proceedings in respect of a dispute that is the subject matter of the conciliation proceedings.
- (vi) The conciliation proceedings shall be terminated as per Section 76 of 'The Arbitration and Conciliation Act, 1996.

**63.1 Matters Finally Determined by the Railway:** All disputes and differences of any kind whatsoever arising out of or in connection with the contract, whether during the progress of the work or after its completion and whether before or after the determination of the contract, shall be referred by the Contractor to the GM and the GM shall, within 120 days after receipt of the Contractor's representation, make and notify decisions on all matters referred to by the Contractor in writing provided that matters for which provision has been made in Clauses 7(j), 8, 18, 22(5), 39.1, 39.2, 40A, 43(2), 45(i)(a), 55, 55-A(5), 57, 57A, 61(1), 61(2), 62(1), 63(iv) and 63.2.11 of the Standard General Conditions of Contract or in any Clause (stated as excepted matter) of the Special Conditions of the Contract, shall be deemed as 'excepted matters' (matters not arbitrable) and



decisions of the Railway authority, thereon shall be final and binding on the Contractor; provided further that 'excepted matters' shall stand specifically excluded from the purview of the Dispute Adjudication Board (DAB) and Arbitration.

**63.2 Dispute Adjudication Board (DAB):** This clause is applicable in the tender having advertised value more than Rs 50 Crore.

**63.2.1** Any dispute/s if not settled with the Engineer, shall be referred to DAB.

The DAB shall consist of a panel of three Retired Railway Officers, retired not below senior administrative grade (SAG). The DAB shall be formed within 90 days of signing of Contract Agreement. For this purpose, a panel of DAB members shall be maintained in the General Manager's office. The complete panel, which shall not be less than five members, shall be sent by Chief Engineer to the Contractor to nominate one member of the DAB from the panel as Contractor's nominee within two weeks of receipt of the panel. On receipt of Contractor's nominee, the Chief Engineer shall nominate one member from the same panel as Railway nominee for the DAB. Both above nominees shall jointly select presiding member of the DAB from the same panel.

**63.2.2** The appointment of DAB shall be effectuated by way of a tri-partite agreement among the Railway, Contractor and the respective DAB members. The terms of the remuneration of each member shall be as fixed by Ministry of Railways from time to time. Each party shall be responsible for paying one-half of this remuneration.

**63.2.3** If one or more of the members appointed refuses to act as DAB member, or is unable or unwilling to perform his functions as DAB member for any reason whatsoever or dies or in the opinion of the Chief Engineer fails to act without undue delay, the parties shall terminate the mandate of such DAB member and thereupon new DAB member shall be appointed in the same manner, as the outgoing DAB member had been appointed.

**63.2.4** The appointment of any member may be terminated by mutual agreement of both Parties, but not by the Railway or the Contractor acting alone. Unless otherwise agreed by both the Parties, the appointment of the DAB (including each member) shall expire upon expiry of this Contract Agreement.

**63.2.5** Before start of DAB proceedings, each DAB member shall give the following certificate to the Railway and the Contractor:

*"I have no any past or present relationship in relation to the subject matter in dispute, whether financial, business, professional or other kind. Further, I have no any past or present relationship with or interest in any of the parties whether financial, business, professional or other kind, which is likely to give rise to justifiable doubts as to my independence or impartiality."*

**63.2.6** DAB proceedings shall be conducted as decided by the DAB. The DAB shall give its decision within 90 days of a Dispute referred to it by any of the Parties, duly recording the reasons before arriving at the decision. The DAB shall decide the issue within terms and conditions of the contract. This time limit shall be extendable subject to the Parties mutual agreement.

**63.2.7** The DAB decision shall not be binding on both the Parties. In case any party is not satisfied by the decision of DAB, then the aggrieved party may approach Arbitral Tribunal for arbitration proceedings.

**63.2.8** No dispute shall be referred to Arbitral Tribunal unless the same has been referred to DAB for adjudication. However, in case DAB is not formed due to any reason, the disputes can be directly referred to Arbitral Tribunal to adjudicate the dispute.

**63.2.9** In the specific cases of any misconduct by any of the members of the DAB, the parties shall have the right to specifically bring it to the notice of the DAB such conduct, through a statement filed with necessary documents in proof of such misconduct and the DAB, after taking NOTICE of such conduct initiate the replacement of the member concerned, in the same manner the member to be replaced was appointed.

**63.2.10** Once the decision is given by DAB, DAB cannot review the decision at its own or on the request of one party, unless both parties agree for review of decision by DAB.

**63.2.11** In case DAB decision is not challenged by either party within 180 days of receipt of decision of DAB, the decision shall be considered as final and parties would be barred for referring the same to Arbitral Tribunal for adjudication.

**63.2.12** The obligation of the Railway and the Contactor shall not be altered by reasons of issue being or under reference to DAB.

**63.2.13** The DAB shall conduct the proceedings at any convenient venue which shall be decided by DAB in consultations with parties.

**63.2.14** It is a term of this contract that the Parties shall not approach any Court of Law for settlement of such disputes or differences unless an attempt has first been made by the parties to settle such disputes or differences through DAB and Arbitral Tribunal.

#### **64.(1) : Demand for Arbitration:**

**64.(1)(i):** In the event of any dispute or difference between the parties hereto as to the construction or operation of this contract, or the respective rights and liabilities of the parties on any matter in question, dispute or difference on any account or as to the withholding by the Railway of any certificate to which the Contractor may claim to be entitled to, or if the Railway fails to make a decision within 120 days, then and in any such case, but except in any of the “excepted matters” referred to in Clause 63.1 of these Conditions, the Contractor, after 120 days but within 180 days of his presenting his final claim on disputed matters shall demand in writing that the dispute or difference be referred to arbitration.

**64.(1)(ii)(a):** The demand for arbitration shall specify the matters which are in question, or subject of the dispute or difference as also the amount of claim item-wise. Only such dispute or difference, in respect of which the demand has been made, together with counter claims or set off, given by the Railway, shall be referred to arbitration and other matters shall not be included in the reference.

**64.(1)(ii)(b):** The parties may waive off the applicability of Sub-Section 12(5) of Arbitration and Conciliation (Amendment) Act 2015, if they agree for such waiver in writing, after dispute having arisen between them, in the format given under Annexure XV of these conditions.

**64.(1)(iii)(a):** The Arbitration proceedings shall be assumed to have commenced from the day, a written and valid demand for arbitration is received by the Railway.

**64.(1)(iii)(b):** The claimant shall submit his claims stating the facts supporting the claims alongwith all the relevant documents and the relief or remedy sought against each claim within a period of 30 days from the date of appointment of the Arbitral Tribunal.

**64.(1)(iii)(c):** The Railway shall submit its defence statement and counter claim(s), if any, within a period of 60 days of receipt of copy of claims from Tribunal, unless otherwise extension has been granted by Tribunal.

**64.(1)(iii)(d):** Place of Arbitration: The place of arbitration would be within the geographical limits of the Division of the Railway where the cause of action arose or the Headquarters of the concerned Railway or any other place with the written consent of both the parties.

**64.(1)(iv):** No new claim shall be added during proceedings by either party. However, a party may amend or supplement the original claim or defense thereof during the course of arbitration proceedings subject to acceptance by Tribunal having due regard to the delay in making it.

**64.(1)(v):** If the Contractor(s) does/do not prefer his/their specific and final claims in writing, within a period of 90 days of receiving the intimation from the Railways that the final bill is ready for payment, he/they will be deemed to have waived his/their claim(s) and the Railway shall be discharged and released of all liabilities under the contract in respect of these claims.

**64.(2): Obligation During Pendency of Arbitration:** Work under the contract shall, unless otherwise directed by the Engineer, continue during the arbitration proceedings, and no payment due or payable by the Railway shall be withheld on account of such proceedings, provided, however, it shall be open for Arbitral Tribunal to consider and decide whether or not such work should continue during arbitration proceedings.

**64.(3) : Appointment of Arbitrator:**

**64.(3)(a) : Appointment of Arbitrator where applicability of section 12 (5) of Arbitration and Conciliation Act has been waived off:**

**64.(3)(a)(i):** In cases where the total value of all claims in question added together does not exceed ₹ 1,00,00,000/- (Rupees One Crore), the Arbitral Tribunal shall consist of a Sole Arbitrator who shall be a Gazetted Officer of Railway not below Junior Administrative Grade, nominated by the General Manager. The sole arbitrator shall be appointed within 60 days from the day when a written and valid demand for arbitration is received by General Manager.

**64.(3)(a)(ii):** In cases not covered by the Clause 64(3)(a)(i), the Arbitral Tribunal shall consist of a panel of three Gazetted Railway Officers not below Junior Administrative Grade or 2 Railway Gazetted Officers not below Junior Administrative Grade and a retired Railway Officer, retired not below the rank of Senior Administrative Grade Officer, as the arbitrators. For this purpose, the

Railway will send a panel of at least four (4) names of Gazetted Railway Officers of one or more departments of the Railway which may also include the name(s) of retired Railway Officer(s) empanelled to work as Railway Arbitrator to the Contractor within 60 days from the day when a written and valid demand for arbitration is received by the General Manager.

Contractor will be asked to suggest to General Manager at least 2 names out of the panel for appointment as Contractor's nominee within 30 days from the date of dispatch of the request by Railway. The General Manager shall appoint at least one out of them as the Contractor's nominee and will, also simultaneously appoint the balance number of arbitrators either from the panel or from outside the panel, duly indicating the 'presiding arbitrator' from amongst the 3 arbitrators so appointed. General Manager shall complete this exercise of appointing the Arbitral Tribunal within 30 days from the receipt of the names of Contractor's nominees. While nominating the arbitrators, it will be necessary to ensure that one of them is from the Accounts Department. An officer of Selection Grade of the Accounts Department shall be considered of equal status to the officers in Senior Administrative Grade of other departments of the Railway for the purpose of appointment of arbitrator.

**64.3.(a).iii:** The serving railway officer working in arbitral tribunal in the ongoing arbitration cases as per clause 64.(3)(a)(i) and clause 64.(3)(a)(ii) above, can continue as arbitrator in the tribunal even after his retirement.

**64.(3)(b): Appointment of Arbitrator where applicability of Section 12 (5) of Arbitration and Conciliation Act has not been waived off:**

(i) In cases where the total value of all claims in question added together does not exceed ₹ 50,00,000/- (Rupees Fifty Lakh), the Arbitral Tribunal shall consist of a Retired Railway Officer, retired not below the rank of Senior Administrative Grade Officer, as the arbitrator. For this purpose, the Railway will send a panel of at least four (4) names of retired Railway Officer(s) empanelled to work as Railway Arbitrator duly indicating their retirement dates to the Contractor within 60 days from the day when a written and valid demand for arbitration is received by the General Manager.

Contractor will be asked to suggest to General Manager at least 2 names out of the panel for appointment as arbitrator within 30 days from the date of dispatch of the request by Railway. The General Manager shall appoint at least one out of them as the arbitrator within 30 days from the receipt of the names of Contractor's nominees.

(ii) In cases where the total value of all claims in question added together exceed ₹ 50,00,000/- (Rupees Fifty Lakh), the Arbitral Tribunal shall consist of a Panel of three (3) retired Railway Officer, retired not below the rank of Senior Administrative Grade Officer, as the arbitrators. For this purpose, the Railway will send a panel of at least four (4) names of retired Railway Officer(s) empanelled to work as Railway Arbitrator duly indicating their retirement date to the Contractor within 60 days from the day when a written and valid demand for arbitration is received by the General Manager.

Contractor will be asked to suggest to General Manager at least 2 names out of the panel for appointment as Contractor's nominee within 30 days from the date of dispatch of the request by Railway. The General Manager shall appoint at least one out of them as the Contractor's nominee and will, also simultaneously appoint the balance number of arbitrators either from the panel or from

outside the panel, duly indicating the 'Presiding Arbitrator' from amongst the 3 arbitrators so appointed. General Manager shall complete this exercise of appointing the Arbitral Tribunal within 30 days from the receipt of the names of Contractor's nominees. While nominating the arbitrators, it will be necessary to ensure that one of them has served in the Accounts Department.

**64.(3)(c)(i):** If one or more of the arbitrators appointed as above refuses to act as arbitrator, withdraws from his office as arbitrator, or vacates his/their office/offices or is/are unable or unwilling to perform his functions as arbitrator for any reason whatsoever or dies or in the opinion of the General Manager fails to act without undue delay, the General Manager shall appoint new arbitrator/arbitrators to act in his/their place in the same manner in which the earlier arbitrator/arbitrators had been appointed. Such re-constituted Tribunal may, at its discretion, proceed with the reference from the stage at which it was left by the previous arbitrator (s).

**64.(3) (c) (ii):** (a) The Arbitral Tribunal shall have power to call for such evidence by way of affidavits or otherwise as the Arbitral Tribunal shall think proper, and it shall be the duty of the parties hereto to do or cause to be done all such things as may be necessary to enable the Arbitral Tribunal to make the award without any delay. The proceedings shall normally be conducted on the basis of documents and written statements.

(b) Before proceeding into the merits of any dispute, the Arbitral Tribunal shall first decide and pass its orders over any plea submitted/objections raised by any party, if any, regarding appointment of Arbitral Tribunal, validity of arbitration agreement, jurisdiction and scope of the Tribunal to deal with the dispute (s) submitted to arbitration, applicability of time 'limitation' to any dispute, any violation of agreed procedure regarding conduct of the arbitral proceedings or plea for interim measures of protection and record its orders in day to day proceedings. A copy of the proceedings duly signed by all the members of tribunal should be provided to both the parties.

**64.3(c)(iii):** (i) Qualification of Arbitrator (s):

(a) Serving Gazetted Railway Officers of not below JA Grade level.

(b) Retired Railway Officers not below SA Grade level, one year after his date of retirement.

(c) Age of arbitrator at the time of appointment shall be below 70 years.

(ii) An arbitrator may be appointed notwithstanding the total number of arbitration cases in which he has been appointed in the past.

(iii) While appointing arbitrator(s) under Sub-Clause 64.(3)(a)(i), 64.(3)(a)(ii), 64.(3)(b)(i) & 64.(3)(b)(ii) above, due care shall be taken that he/they is/are not the one/those who had an opportunity to deal with the matters to which the contract relates or who in the course of his/their duties as Railway servant(s) expressed views on all or any of the matters under dispute or differences. A certification to this effect as per annexure- XVI shall be taken from Arbitrators also. The proceedings of the Arbitral tribunal or the award made by such Tribunal will, however, not be invalid merely for the reason that one or more arbitrator had, in the course of his service, opportunity to deal with the matters to which the contract relates or who in the course of his/their duties expressed views on all or any of the matters under dispute.

**64.(3)(d)(i):** The arbitral award shall state item wise, the sum and reasons upon which it is based. The analysis and reasons shall be detailed enough so that the award could be inferred therefrom.

**64.(3)(d)(ii):** A party may apply for corrections of any computational errors, any typographical or clerical errors or any other error of similar nature occurring in the award of a Tribunal and interpretation of a specific point of award to Tribunal within 60 days of receipt of the award.

**64.(3)(d)(iii):** A party may apply to Tribunal within 60 days of receipt of award to make an additional award as to claims presented in the arbitral proceedings but omitted from the arbitral award.

**64.(4):** In case of the Tribunal, comprising of three members, any ruling on award shall be made by a majority of members of Tribunal. In the absence of such a majority, the views of the Presiding Arbitrator shall prevail.

**64.(5):** Where the arbitral award is for the payment of money, no interest shall be payable on whole or any part of the money for any period till the date on which the award is made.

**64. (6):** The cost of arbitration shall be borne by the respective parties. The cost shall inter-alia include fee of the arbitrator(s), as per the rates fixed by Railway Board from time to time and the fee shall be borne equally by both the parties, provided parties sign an agreement in the format given at Annexure XV to these conditions after/ while referring these disputes to Arbitration. Further, the fee payable to the arbitrator(s) would be governed by the instructions issued on the subject by Railway Board from time to time irrespective of the fact whether the arbitrator(s) is/are appointed by the Railway Administration or by the court of law unless specifically directed by Hon'ble court otherwise on the matter.

**64.(7)** Subject to the provisions of the aforesaid Arbitration and Conciliation Act 1996 and the rules thereunder and relevant para of the Standard General Conditions of Contract and any statutory modifications thereof shall apply to the appointment of arbitrators and arbitration proceedings under this Clause.

**64.(8)** In case arbitration award is challenged by a party in the Court of Law, 75% of award amount, pending adjudication by Court of Law, shall be made by party to other party. In case payment is to be made by Railway to Contractor, the terms & conditions as incorporated in the Ministry of Railways letter No. 2016/CE(I)/CT/ARB/3(NITI Aayog)/Pt. dated 08<sup>th</sup> Mar,2017 as amended from time to time, shall be followed. In case Contractor has to pay to the Railway, then 75% of the award amount shall be deducted by the Railway from the Contractor's bills, Performance Guarantee/ Security Deposit or any other dues of Contractor with the Government of India.

## PART-II ANNEXURES

### ANNEXURE – VII

Reference Para 17B

Registered Acknowledgement Due

#### PROFORMA FOR TIME EXTENSION

No. \_\_\_\_\_

Dated: \_\_\_\_\_

Sub: (i) \_\_\_\_\_ *(name of work)*.

(ii) Acceptance letter no. \_\_\_\_\_

(iii) Understanding/Agreement no. \_\_\_\_\_

Ref: \_\_\_\_\_ *(Quote specific application of Contractor for extension to the date received)* \_\_\_\_\_

Dear Sir,

1. The stipulated date for completion of the work mentioned above is \_\_\_\_\_. From the progress made so far and the present rate of progress, it is unlikely that the work will be completed by the above date (or 'However, the work was not completed on this date').

2. Expecting that you may be able to complete the work if some more time is given, the competent authority, although not bound to do so, hereby extends the time for completion from \_\_\_\_\_ to \_\_\_\_\_.

3. Please note that an amount equal to the liquidated damages for delay in the completion of the work after the expiry of \_\_\_\_\_ *(give here the stipulated date for completion with/without any liquidated damage fixed earlier)* will be recovered from you as mentioned in Clause 17B of the Standard General Conditions of Contract for the extended period, notwithstanding the grant of this extension. You may proceed with the work accordingly.

4. The above extension of the completion date will also be subject to the further condition that no increase in rates on any account will be payable to you.

5. Please intimate within a week of the receipt of this letter your acceptance of the extension of the conditions stated above.

6. Please note that in the event of your declining to accept the extension on the above said conditions or in the event of your failure after accepting or acting upto this extension to complete the work by \_\_\_\_\_ *(here mention the extended date)*, further action will be taken in terms of Clause 62 of the Standard General Conditions of Contract.

Yours faithfully

For and on behalf of the President of India

**PROFORMA OF 14 DAYS NOTICE FOR OFFLOADING OF PART OF CONTRACT  
WORK**

\_\_\_\_\_ **RAILWAY**  
(Without Prejudice)

To

M/s \_\_\_\_\_

Dear Sir,

Contract Agreement No. \_\_\_\_\_

In connection with \_\_\_\_\_

In spite of repeated instructions to you by the subordinate offices as well as by this office through various letters of even no. \_\_\_\_\_, dated \_\_\_\_\_; you have failed to show adequate progress of work so as to complete the contract within the original / extended date of completion of contract and part(s) of contract work are yet to be started/ still lagging behind the agreed program of work, listed as under:

*(Details of part(s) of work which is delayed and can be executed independently, to be mentioned).*

2. Your attention is invited to this office/Chief Engineer's office letter no. \_\_\_\_\_, dated \_\_\_\_\_ in reference to your representation, dated \_\_\_\_\_.

3. As you have failed to abide by the instructions issued to commence the work /to show adequate progress of work, you are hereby given 14 days' notice in accordance with Clause 40A of the Standard General Conditions of Contract to deploy adequate resources i.e. *(the details of resource requirement, to be mentioned)* and commence / to make good the progress for part(s) of works detailed above, failing which action as provided in Clause 40A of the Standard General Conditions of Contract shall be commenced after expiry of 14 days' notice period viz. to offload few/ all part(s) of work mentioned above to any of the existing or new contractor without your participation and at your Risk & Cost, not exceeding the value of Performance Guarantee of this contract, which may please be noted.

Kindly acknowledge receipt.

Yours faithfully  
For and on behalf of the President of India



**ANNEXURE – VIIB**

(Reference Clause 40(A))

Registered Acknowledgement Due

**NOTICE FOR PART OF CONTRACT WORK OFFLOADED**

\_\_\_\_\_ **RAILWAY**  
(Without Prejudice)

To

M/s \_\_\_\_\_

Dear Sir,

Contract Agreement No. \_\_\_\_\_

In connection with \_\_\_\_\_

1. Fourteen days' notice under Clause 40A of the Standard General Conditions of Contract was given to you under this office letter of even no., dated \_\_\_\_\_; but you have taken no/inadequate action to deploy adequate resources to commence the part(s) of work/show adequate progress of the part(s) of work, mentioned therein.

As you have failed to abide by the instructions issued to commence the part(s) of work/show adequate progress of the part(s) of work even at the lapse of 14 days' notice period under Clause 40A of the Standard General Conditions of Contract, few part(s) of the work under the contract have been offloaded and being executed by other mode(s) at the cost detailed below:

*Or,*

1. Please refer your request letter no..... dated ....., wherein it was requested under clause 40 A of the Standard General Conditions of Contract to offload part(s) of works at your risk & cost. The details of part(s) of the work under the contract which have been offloaded and being executed by other mode(s) at the cost detailed below:

*(List of Part(s) of work offloaded, Details of mode of execution of such offloaded work alongwith approximate cost thereof to be mentioned)*

2. The final measurement of work(s) already executed for above part(s) of work recorded as per clause 45 (A) or/and 45 (B) of the Standard General Conditions of Contract is enclosed herewith.

3. The Bill(s) of Quantities for Part(s) of work offloaded is enclosed herewith.

4. The additional cost in execution of offloaded work through mode(s) mentioned in para (1) above is determined as Rs. \_\_\_\_\_, over& above the cost of execution under this contract (including the

PVC amount payable as per contract, as on the date of issue of this notice). This additional cost shall be recovered from your next on account bill(s) or any other dues payable to you under contract.

5. The Contract value gets reduced to Rs.....:

6. You are requested to continue with the balance work in the contract subsequent to offloading of above part(s) of work.

Kindly acknowledge receipt.

Yours faithfully

For and on behalf of the President of India

**ANNEXURE – VIII**

Reference Para 60.(2)

**CERTIFICATE OF FITNESS**

1. (a) Serial Number \_\_\_\_\_  
(b) Date \_\_\_\_\_
2. Name of person examined \_\_\_\_\_
3. Father's Name: son/daughter of \_\_\_\_\_  
Residing at \_\_\_\_\_
4. Sex \_\_\_\_\_
5. Residence: \_\_\_\_\_
6. Physical fitness \_\_\_\_\_
7. Identification marks \_\_\_\_\_
8. Date of birth, if available, and/or certified age \_\_\_\_\_  
I certify that I have personally examined (name) \_\_\_\_\_ who is desirous of being employed in a factory or on a work requiring manual labour and that his/her age as nearly as can be ascertained from my examination, is \_\_\_\_\_ years.  
  
I certify that he/she is fit for employment in a factory or on a work requiring manual labour as an adult/child.
9. Reasons for :  
(a) Refusal to grant certificate, or \_\_\_\_\_  
(b) Revoking the certificate \_\_\_\_\_

Signature or left hand

Thumb impression of the person examined.

Signature of Certifying Surgeon

**Note:** In case of physical disability, the exact details and cause of the physical disability should be clearly stated.

**ANNEXURE – IX**

(Reference Clause 62. (1)

Registered Acknowledgement Due

**PROFORMA OF 7 DAYS NOTICE FOR WORKS AS A WHOLE/ IN PARTS**

**(DETAILS OF PART OF WORK TO BE MENTIONED)**

\_\_\_\_\_ **RAILWAY**

(Without Prejudice)

To

M/s \_\_\_\_\_  
\_\_\_\_\_

Dear Sir,

Contract Agreement No. \_\_\_\_\_

In connection with \_\_\_\_\_

In spite of repeated instructions to you by the subordinate offices as well as by this office through various letters of even no. \_\_\_\_\_, dated \_\_\_\_\_; you have failed to start work/show adequate progress and/or submit detailed programme for completing the work/ part of work (details of part of work to be mentioned).

2. Your attention is invited to this office/Chief Engineer's office letter no. \_\_\_\_\_, dated \_\_\_\_\_ in reference to your representation, dated \_\_\_\_\_.

3. As you have failed to abide by the instructions issued to commence the work /to show adequate progress of work you are hereby given 7 days' notice in accordance with Clause 62 of the Standard General Conditions of Contract to commence works / to make good the progress, failing which further action as provided in Clause 62 of the Standard General Conditions of Contract viz. to terminate your Contract and complete the balance work without your participation will be taken.

Kindly acknowledge receipt.

Yours faithfully

For and on behalf of the President of India

**ANNEXURE – X**

Reference Para 62(1)

Registered Acknowledgement Due

**PROFORMA OF 48 HRS. NOTICE FOR WHOLE WORK**

\_\_\_\_\_ **RAILWAY**

(Without Prejudice)

To

M/s \_\_\_\_\_

\_\_\_\_\_

Dear Sir,

Contract Agreement No. \_\_\_\_\_

In connection with \_\_\_\_\_

Seven days' notice under Clause 62 of the Standard General Conditions of Contract was given to you under this office letter of even no., dated \_\_\_\_\_; but you have taken no action to commence the work/show adequate progress of the work.

2. You are hereby given 48 hours' notice in terms of Clause 62 of the Standard General Conditions of Contract to commence works / to make good the progress of works, failing which and on expiry of this period your above contract will be rescinded and the work under this contract will be carried out independently without your participation and your Security Deposit shall be forfeited and Performance Guarantee shall also be encashed and any other consequences which may please be noted.

Kindly acknowledge receipt.

Yours faithfully

For and on behalf of the President of India

**ANNEXURE – XI**

Reference Para 62.(1)

Registered Acknowledgement Due

**PROFORMA OF TERMINATION NOTICE**

\_\_\_\_\_ **RAILWAY**

(Without Prejudice)

No. \_\_\_\_\_

Dated \_\_\_\_\_

To

M/s \_\_\_\_\_

Dear Sir,

Contract Agreement No. \_\_\_\_\_

In connection with \_\_\_\_\_

Forty eight hours (48 hrs.) notice was given to you under this office letter of even no., dated \_\_\_\_\_; but you have taken no action to commence the work/show adequate progress of the work.

Since the period of 48 hours' notice has already expired, the above contract stands rescinded in terms of Clause 62 of the Standard General Conditions of Contract and the balance work under this contract will be carried out independently without your participation. Your participation as well as participation of every member/partner in any manner as an individual or a partnership firm/JV is hereby debarred from participation in the tender for executing the balance work and your Security Deposit shall be forfeited and Performance Guarantee shall also be encashed.

Kindly acknowledge receipt.

Yours faithfully

For and on behalf of the President of India

**ANNEXURE – XII**

Reference Para 62.(1)

Registered Acknowledgement Due

**PROFORMA OF 48 HRS. NOTICE FOR PART OF THE WORK.....**

**(DETAILS OF PART OF WORK TO BE MENTIONED)**

\_\_\_\_\_ **RAILWAY**

(Without Prejudice)

To

M/s \_\_\_\_\_

\_\_\_\_\_

Dear Sir,

Contract Agreement No. \_\_\_\_\_

In connection with \_\_\_\_\_

1. Seven days' notice under Clause 62 of the Standard General Conditions of Contract was given to you under this office letter of even no., dated \_\_\_\_\_; but you have taken no action to commence the work/show adequate progress of the part of work.....(details of part to be mentioned).

2. You are hereby given 48 hours' notice in terms of Clause 62 of the Standard General Conditions of Contract to commence works / to make good the progress of works, failing which and on expiry of this period your above part of work.....(Details of part to be mentioned) in contract will be rescinded and the work will be carried out independently without your participation.

3. Your full Performance Guarantee for the contract shall be forfeited and you shall not be issued any completion certificate for the contract. However, no additional Performance Guarantee shall be required for balance of work being executed through the part terminated contract.

4. The contract value of part terminated contract shall stands reduced to .....

Kindly acknowledge receipt.

Yours faithfully

For and on behalf of the President of India

**ANNEXURE – XIII**

Reference Para 62.(1)

Registered Acknowledgement Due

**PROFORMA OF TERMINATION NOTICE FOR PART OF THE WORK.....  
(DETAILS OF PART OF WORK TO BE MENTIONED)**

**RAILWAY**

(Without Prejudice)

No. \_\_\_\_\_

Dated \_\_\_\_\_

To

M/s \_\_\_\_\_

Dear Sir,

Contract Agreement No. \_\_\_\_\_

In connection with \_\_\_\_\_

1. Forty eight hours (48 hrs.) notice was given to you under this office letter of even no., dated \_\_\_\_\_; but you have taken no action to commence the work/show adequate progress of the part of work.....(details of part to be mentioned).
2. Your above part of work in contract .....(details of part to be mentioned) stands rescinded in terms of Clause 62 of the Standard General Conditions of Contract and the same will be carried out independently without your participation. Your participation as well as participation of every member/partner in any manner as an individual or a partnership firm/JV is hereby debarred from participation in the tender for executing the balance work
3. Your full Performance Guarantee for the contract shall be forfeited and you shall not be issued any completion certificate for the contract. However, no additional Performance Guarantee shall be required for balance of work being executed through the part terminated contract.
4. The contract value of part terminated contract stands reduced to .....

Kindly acknowledge receipt.

Yours faithfully

For and on behalf of the President of India



## ANNEXURE – XIV

Reference Para 48.(3)

**FINAL SUPPLEMENTARY AGREEMENT**

1. Articles of agreement made this day \_\_\_\_\_ in the year \_\_\_\_\_ between the President of India, acting through the \_\_\_\_\_ Railway Administration having his office at \_\_\_\_\_ herein after called the Railway of the one part and \_\_\_\_\_ of the second part.
2. Whereas the party hereto of the second part executed an agreement with the party hereto of the first part being agreement Number \_\_\_\_\_ dated \_\_\_\_\_ for the performance \_\_\_\_\_ herein after called the 'Principal Agreement'.
3. And whereas it was agreed by and between the parties hereto that the works would be completed by the party hereto of the second part on \_\_\_\_\_ date last extended and whereas the party hereto of the second part has executed the work to the entire satisfaction of the party hereto of the first part.
4. And whereas the party hereto of the first part already made payment to the party hereto of the second part diverse sums from time to time aggregating to ₹ \_\_\_\_\_ including the Final Bill bearing voucher No. \_\_\_\_\_ dated \_\_\_\_\_ of value \_\_\_\_\_ duly adjusted as per price variation clause, if applicable (the receipt of which is hereby acknowledged by the party hereto of the second part in full and final settlement of all his /its claims under the principal agreement.

And whereas the party hereto of the second part have received sum of ₹ \_\_\_\_\_ through the Final Bill bearing voucher No. \_\_\_\_\_ dated \_\_\_\_\_ duly adjusted as per price variation clause (PVC), if applicable (the receipt of which is hereby acknowledged by the party thereto of the second part) from the party hereto of the first part in full and final settlement of all his/its disputed claims under principal agreement.

Now, it is hereby agreed by and between the parties in the consideration of sums already paid by the party hereto of the first part to the party hereto of the second part against all outstanding dues and claims for all works done under the aforesaid principal agreement excluding the Security Deposit, the party hereto of the second part have no further dues of claims against the party hereto of the first part under the said Principal Agreement. It is further agreed by and between the parties that the party hereto of the second part has accepted the said sums mentioned above in full and final satisfaction of all its dues and claims under the said Principal Agreement.

(Applicable in case Final Supplementary Agreement is signed after release of Final Payment)

**Or**

*And whereas the party hereto of the first part already made payment to the party hereto of the second part diverse sums from time to time aggregating to ₹ \_\_\_\_\_ through various On Account Bills (the receipt of which is hereby acknowledged by the party hereto of the second part).*

*And whereas the party hereto of the second part have received sum of ₹ \_\_\_\_\_ through various On Account Bills (the receipt of which is hereby acknowledged by the party thereto of the second part) from the party hereto of the first part and party hereto of the second part have accepted final measurements recorded on Page No..... to Page No.... of Measurement Book No.....and corresponding Final Bill duly adjusted as per price variation clause (PVC), if applicable, for full and final settlement of all his/its disputed claims under principal agreement.*

*Now, it is hereby agreed by and between the parties in the consideration of sums already paid through various On Account Bills and sums to be paid through Final Bill duly adjusted as per price variation clause (PVC), if applicable, based on accepted final measurements including the Security Deposit by the party hereto of the first part to the party hereto of the second part against all outstanding dues and claims for all works done under the aforesaid principal agreement, the party hereto of the second part have no further dues of claims against the party hereto of the first part under the said Principal Agreement.*

*(Applicable in case Final Supplementary Agreement is signed before release of Final Payment)*

5. It is further agreed and understood by and between the parties that the arbitration clause contained in the said principal agreement shall cease to have any effect and/or shall be deemed to be non-existent for all purposes.

Signature of the Contractor/s

for and on behalf of the President of India

Witnesses

ADDRESS: \_\_\_\_\_

**ANNEXURE-XV**

Reference Para 64.3 & 64.6

**Agreement towards Waiver under Section 12(5) and Section 31A (5) of Arbitration and Conciliation (Amendment) Act**

I/we..... (Name of agency/Contractor) with reference to agreement no..... raise disputes as to the construction and operation of this contract, or the respective rights and liabilities, withholding of certificate and demand arbitration in respect of following claims :

Brief of claim:

- (i) Claim 1- Detailed at Annexure-
- (ii) Claim 2 –
- (iii) Claim 3 –

I/we..... (post of Engineer) with reference to agreement no..... hereby raise disputes as to the construction and operation of this contract, or the respective rights and liabilities, withholding of certificate and demand arbitration in respect of following claims:

I/we.....do/do not agree to waive off applicability of section 12(5) of Arbitration and Conciliation (Amendment) Act.

Signature of Claimant\_\_\_\_\_ Signature of Respondent \_\_\_\_\_

**Agreement under Section 31(5)**

I/we..... (Name of claimant) with reference to agreement no..... hereby waive off the applicability of sub section 31-A (2) to 31-A (4) of the Arbitration and Conciliation (Amendment) Act. We further agree that the cost of arbitration will be shared by the parties as per Clause 64(6) of the Standard General Conditions of Contract.

.

Signature of Claimant\_\_\_\_\_ Signature of Respondent \_\_\_\_\_

\*Strike out whichever not applicable.

**Certification by Arbitrators appointed under Clause 63 & 64 of Indian Railways General Conditions of Contract**

1. Name:
2. Contact Details:
3. Prior experience (Including Experience with Arbitrations):
4. **I do not have more than ten on-going Arbitration cases with me.**
5. I hereby certify that I have retired from Railways w.e.f. \_\_\_\_\_ and empanelled as Railway Arbitrator as per 'The Arbitration and Conciliation Act- 1996'.
6. I have no any past or present relationship in relation to the subject matter in dispute, whether financial, business, professional or other kind.  
Or  
I have past or present relationship in relation to the subject matter in dispute, whether financial, business, professional or other kind. The list of such interests is as under:
7. I have no any past or present relationship with or interest in any of the parties whether financial, business, professional or other kind, which is likely to give rise to justifiable doubts as to my independence or impartiality in terms of The Arbitration and Conciliation Act-1996.  
Or  
I have past or present relationship with or interest in any of the parties whether financial, business, professional or other kind, which is likely to give rise to justifiable doubts as to my independence or impartiality in terms of The Arbitration and Conciliation Act-1996. The details of such relationship or interests are as under:
8. There are no concurrent Circumstances which are likely to affect my ability to devote sufficient time to the arbitration and in particular to finish the entire arbitration within twelve months.

Or

There are Circumstances which are likely to affect my ability to devote sufficient time to the arbitration and in particular to finish the entire arbitration within twelve months. The list of such circumstances is as under:

**LOCATION OF SITE & SCOPE OF WORK:**

**Name of work:** - "Execution of balance works in newly constructed staff quarters at Rewa in connection with the Rewa-Sidhi new BG rail line project".

\*\*\*\*\*

**LOCATION OF SITE & SCOPE OF WORK:**

The work is required in connection with Rewa-Sidhi new BG rail line project. The work proposed under this tender at Rewa in connection with Rewa-Sidhi new BG rail line project. The site is connected by road network at all locations and tenderer, if required, may have to make approach roads at his own cost. Tenderer may please go through the Safety instructions in tender document. Tenderer will be bound by Safety instructions of Construction Engineers and work will have to be carried without any extra cost.

**The scope of work under this tender is construction of following items-**

- (a) Execution of balance work in newly constructed staff quarters at Rewa
- (b) There are 23 nos of Type-II quarters which are partially completed, 01 nos Type-III quarters in which work is done up to lintel level, 01 nos Type-IV quarters in which work done up to plinth level.
- (c) 01 nos Type-V quarter to be newly constructed under the scope of instant tender.

**Maintenance period:** The successful tenderer/contractor will have to maintain all works excepting supply of ballast for a period of 06 (six) Calendar months or one monsoon period whichever is later, unless otherwise specified in the Tender Conditions from the certified date of completion without any extra cost to the Railway as per item 1.1 of Chapter-4 of the standard tender document.

Note: The scope of works as specified above in this tender is indicative in nature and may change in during actual execution of work as per requirements of site conditions. The above information is for guidance of the tenderer; however, the tenderer is required to inspect the site before tendering and get acquainted with the existing site conditions.

Signature of Tenderer

for Chief Administrative Officer  
(Const.) West Central Railway,

**Chapter -1**  
**ADDITIONAL SPECIAL CONDITIONS OF CONTRACT**

1. Unless otherwise specified in contract agreements, quality and workmanship of all work shall conform to the relevant specifications mentioned in "Indian Railway Unified Standard Specifications for Works & Materials" and any other standard specification referred therein. In case certain specification is not covered in the Indian Railway Unified Standard specification then relevant I. S. Codes shall be applicable. In case of absence of relevant I. S. Codes, worldwide reputed standards like ASTM or British Standards as decided by Engineer-in-charge shall be applicable.
2. Work shall be carried out strictly in accordance with the approved drawings and no deviation from drawings shall be permitted unless specifically approved / ordered by Engineer. Decision of Railway in this regard would be final and binding on the contractor. The contractor shall arrange to prepare site specific sketches / drawings as per direction of Engineer – in – charge. No extra payment shall be made on this account except otherwise specified in tender schedule.
3. The intending tenderer are required to survey the availability of the construction material in the project area including its availability during the entire construction period and its transportation to the site of work. No claim on account of non-availability of survey & other materials nor its transportation to site of work, shall be entertained or paid by Railways.
4. Rates include, unless otherwise specified, all freight charges and lead and lift to the site of work, for all materials, equipment, tools and plants, etc. that may be used by the Contractor on the work.

Rates include, unless otherwise specified, all work with Contractor's materials, labour, tools, and plants that may be required to complete the works under the scope of individual Standard Schedule item, including scaffolding, shuttering, form work, etc. up to all heights and down to all depths, unless specified otherwise.

- 4.(a) The contractor shall ensure that the materials such as bitumen, tar, paint, water - proofing compound, distemper and the like are packed in original sealed containers / packing, bearing manufacturer's marking and brands, etc., (except where the quantity required is a fraction of the smallest packing). Materials not complying with the required specifications shall be rejected.
- 4.(b) The Contractor shall carry out such instructions as are given to him in writing by the Engineer to ensure that the full quantity of materials as specified, go into the work.
5. It should be specifically noted that some of the detailed drawings may not have been finalized by the Railway and will, therefore, be supplied to the contractors on specific requisition or demand from contractors as and when they are finalized. No compensation, whatsoever, on this account shall be payable by the Railway.
6. Codes, Manuals & By-laws: - The contractor shall arrange all relevant codes (IS, IRC etc.), manuals, by- laws which are required for execution of work or as considered to be required by engineer, at site at their own cost and no extra will be paid on this account. In case of failure Railway may procure / purchase the same and recover charges from contractor/s bills or any other dues payable.
7. The contractor will have to arrange for sufficient plant, machineries, and labour at all times during the execution of work and ensure strictly that there is no delay on this account. Any decision by Railway's representative as to minimum number of shuttering sets, plants, machinery in this regard shall be final and binding on the contractors.

Signature of Tenderer

For Chief Administrative Officer  
(Const.) West Central Railway,

8. **Site Office for Engineer and other Supervisory staff:** -Contractor shall arrange office at site or any other location as per instruction of Engineer in charge or his representative during the execution of the work with required furniture items like tables, chairs, cupboards, lockers etc. Work includes providing electric supply & all electrical items like lights, fans and complete wiring, telephone etc. complete. The Contractor shall provide these office accommodations within 1 month from the date of the commencement of work or 1 month from the date of providing the land. No payment will however be made for site office construction.

Contractor shall arrange the computational facility for computation (1 Computer + 1 Printer multifunctional & accessories and internet facility). This facility is required mainly for (i) Preparation and printing various reports and to update the site records as per requirement. (ii) To prepare the required charts like progress report, bar chart, pert chart etc. and this will be utilized at site for better and good presentation at site as per requirement with computer operator having knowledge of MS Office + AutoCAD. The above computer operator shall also attend the Railway Office to carry out various drawings, charts, reports incidental to the work on behalf of contractor.

Contractor shall arrange the good conditioned vehicle to the engineer In-charge or his representative during the execution/inspection of the work. No payment will however be made for vehicle. This facility is incidental to the work. Hence contractor is required to provide this facility till completion of work including extended period. No additional payment shall be made for the same being incidental to the work.

9. Contractor shall depute two skilled and two semi – skilled staff as soon as acceptance letter is received by contractor to assist the Railway supervisor / site Engineer for carrying out necessary day to day works like layout marking, assisting in measurement, handling of various survey instruments, records, register etc. They shall be under exclusive and total control of Railway Supervisors / Site Engineers and no payment for this will be made.
10. Contractor has to provide all necessary signages, boards etc. and required works for road traffic diversion if required, being incidental to work & no extra payment for the same will be made. Contractor has to take all necessary precaution for the safety of the road traffic.
11. The tenderer should visit the site and get himself acquainted thoroughly with the site conditions before tendering especially the constraints, encumbrances etc. & quote their rates accordingly. Tenderers are advised to examine approachability, availability of roads, water etc. Also, tenderer should study the location, site conditions as well local conditions including labour etc. which are likely to be encountered where work is to be carried out. The site is approachable by main roads and village roads. However, the contractor/s may have to make service/approach roads wherever required even through the private land, at their own cost and consequences. The element of making the service road with prior permission of land owner is deemed to have been considered while quoting the rates & no extra compensation or claim whatsoever will be payable to the contractor on this account.
12. The Railway administration reserves the rights to operate the items in schedule fully or partly, up to any extent or delete any item of this tender schedule as required as per site conditions or any technical or other consideration & the contractor will have no right to claim extra cost or compensation on this account.
13. No separate item is provided for jungle clearance, cutting of bushes/plants/trees, removal of obstruction for working and marking of center line and layout for work, the cost of this work is deemed to be included in and considered, while quoting the percentage rate for this tender. Contractor will also have to clear bushes, grass etc. free of cost.
14. No claim whatsoever will be entertained by the Railway on account of any delay or hold up of the works arising out of, by or in consequence of delay in approval of drawings, changes, modification, alterations,

Signature of Tenderer

For Chief Administrative Officer  
(Const.) West Central Railway,

- additions, omission & the site layout plans of details drawings and design and or late supply of such material as are required to be arranged by the Railway or due to any other factors on Railway Accounts.
- 15 No claim for idle labour and or idle machinery etc. on any account whatsoever will be entertained. Similarly, no claim shall also be entertained for business loss or any such loss even if preferred or raised by the contractors
- 16 It is primary responsibility of the contractor to obtain all necessary permissions, permits from all concern authorities. However, Railway may provide all necessary help for obtaining the permission. Contractor is not entitled for any compensations such as idling of machinery, staff etc. on account of any delay in work for getting these permissions.
- 17 It shall be noted that while carrying out the work, other agencies such as Electrical, S&T and Civil will also be working in the area and contractor will have to execute the work in total co-ordination with other agencies etc. The contractors should ensure that no obstruction is caused to the working of other contractors in the vicinity and nuisance to the residents of the adjacent area. In the section, some works may be in progress, Contractor has to co-ordinate with those agencies for executing the work smoothly, without disturbing the work of other agencies.
- 18 It may also be borne in mind that High Tension / Low tension Electrical wires are running across at the site. The contractors will have to take precaution while doing work of construction of bridges etc
- 19 Also, other underground electrical cables & pipe line of farmers / Govt. are running across the alignment. The contractors are advised to detect them and informed to Railway in advance well before starting the excavation for foundation/cutting for formation.
- 20 The rate quoted by the tenderer should be inclusive of all taxes, levies, royalty charges leviable by State / Central Govt. or by local authorities. The rate shall be inclusive of all sales / turn over tax, Service Tax, VAT including all other cess, etc., on contract in the State of Madhya Pradesh. No claims for increase in the rates / taxes / cess shall be entertained or paid even if preferred so by the contractor.
- 21 At the close of daily work special care is to be taken to clear sites from loose materials which can be misused for sabotage.
- 22 Any items, if not required as per site conditions can be deleted or new items required to be executed will be operated according to the modification of drawing or as per the instructions of Engineer-in-charge, governed by the provision of General Condition of Contract. No claim whatsoever in this regard shall be entertained or paid even if so, preferred by the contractor/s.
- 23 Unless specified otherwise cement shall be supplied/arranged by contractor as per specification & relevant clauses in Tender / Agreement attached and the rates quoted by the contractor for various items should be inclusive of cost of their own cement.
- 24 Reinforcement steels of various diameters shall be supplied by the contractor as per specification and relevant clauses in Tender/Agt. attached, payment for which will be made under relevant item of schedule.
- 25 The contractor shall submit a programme for completion of the work in form of Bar / Pert Chart before the commencement of work.

| Sr.No | Description of material | Approved Manufacturers                                                                                                                             |
|-------|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| 1.    | Cement                  | Ultratech, ACC, GRASIM, Ambuja, JK, RAJSHREE& Birla or approved equivalent decided by Engineer- in-charge                                          |
| 2.    | Reinforcing steel bars  | TMT reinforcement steel bars (500D/550D) to be procured <b>SAIL /TISCO / RINL / Jindal Panther /JSW Neo</b> and as approved by Engineer-in-charge. |

Signature of Tenderer

For Chief Administrative Officer  
(Const.)West Central Railway,



|    |                               |                                                                           |
|----|-------------------------------|---------------------------------------------------------------------------|
| 3. | Structural steel              | TISCO, SAIL, RINL, IISCO, JINDAL                                          |
| 4. | Admixture & De-shuttering oil | FOSROC, SIKKA or approved equivalent decided by Engineer -in -charge      |
| 5. | Non shrink grout              | FOSROC, SIKKA, MBT or approved equivalent decided by Engineer -in -charge |

**26 LIST OF APPROVED MANUFACTURERS & SUPPLIERS**

*Note: - Contractor shall ensure that Material source and material are approved by Engineer-in-charge before commencement of work. Also, contractor/s should ensure & submit separate Design mix with the change in source of cement.*

- 27 It is pertinent to mention that the work is at different locations. Considering that the entire work is to be completed in within the specified completion period, contractor is required to plan simultaneous execution of various works by mobilizing required resources like shuttering materials, skilled labours etc which will be critical in ensuring completion of work within specified completion period. Contractor shall therefore make note of this & accordingly make timely arrangement for required material for enabling smooth and speedy execution of work.
- 28 The payment of deck slab, parapet wall, wearing coat and crash barrier of ROB shall be done in USSOR Item No - 022050 / 022054.
- 29 All excavated materials (Earth & Boulders) transported and poured from Rewa to Satna doubling project noextra will be paid for the same.
- 30 *The schedule rate of Earthwork item is inclusive of all taxes including royalty charges. For execution of work contractor at his own expenses shall obtain all required permits or parwana necessary for carrying out the work. Railway Administration shall not under any circumstances be liable to obtain any permit or parwana whatsoever for the contractor.*

*No recovery of royalty charges from on account bills of contractor will be made by Railway Administration. However, it is the sole responsibility of contractor to obtain the royalty clearance certificate from the concerned revenue Authority/Collector and submit the same @ every 2nd on account bill.*

*In case of failure in submission of royalty clearance certificate @ every 2nd on account bill, an amount equal to the amount of unpaid royalty charges as intimated by the Revenue Authority/Collector or as calculated on the basis of relevant rates for payment of royalty charges as applicable to the area, will be deducted from the contractor's bill.*

*On completion of work, contractor will be required to obtain a final royalty clearance certificate from the concerned State Revenue Authorities/Collector and produce the same to Railway before the release of final bill. If contractor fails to produce the royalty clearance certificate for royalty charges, final bill of the contractor will not be delayed on this account and final bill will be passed after retaining an amount equal to the amount of unpaid royalty charges as intimated by the Revenue Authority/Collector or as calculated on the basis of relevant rates for payment of royalty charges as applicable to the area.*

*Amount so retained by the Railway Administration, shall be released only after receipt of authentic final royalty clearance certificate from the concerned Revenue Authority/Collector. No claim regarding interest charges for delay in payment of retained amount on royalty account shall be entertained.*

*Any increase in rate of royalty after opening of tender (bid closing date) shall be on Railway account. Similarly, decrease in rate of royalty shall be deducted from contractor's bill. In case of change of royalty charges, contractor should get the measurement recorded in M.B. within 7 days of effective date*

Signature of Tenderer

For Chief Administrative Officer  
(Const.) West Central Railway,

T. No. **DYCEC1REWA-07-2024**

**6**

*of revision of royalty charges, to segregate the work done before change and after change. In the event of failure to do so, Railways decision shall be final and conclusive.*

Signature of Tenderer

For Chief Administrative Officer  
(Const.)West Central Railway,

**Chapter-2****SPECIAL CONDITIONS OF CONTRACT: TECHNICAL SPECIFICATIONS****1) GENERAL**

The Special Conditions are applicable to all items including USSOR & N.S. items. These will be over and above the Indian Railways Unified Standard Specifications (IRUSS) (Works and Materials) 2010.

**CODAL PROVISIONS:** All the work is to be carried out by and large as per the latest amended versions of the following relevant IRS and IS codes. In case of any discrepancy or disagreement between different specifications to be followed for any item of work, the following references shall be made in order of precedence as they appear below:

Provisions in these documents  
IRS codes of Practice/Standard  
specifications. IS codes  
Provision of any other relevant code.

However, the above precedence can be changed by the Engineer in charge and his decision with respect to the applicability/order of priority shall be final and binding. Contractor will be responsible to arrange adequate copies of each of the relevant codes/specifications/manuals at site of work at his own cost

- 1.1. **Indian Railways Unified Standard Specifications (IRUSS) (Works and Materials) 2010:-** For Materials & works shall be adopted. Some additional specifications are added herewith. Notwithstanding any provisions made in the IRUSS 2010, the provisions made in the technical specifications shall be binding. In the absence of any specification for any work on material the relevant Indian standard specification would be applicable and where no Indian standard specification exists, relevant international specification or the specification given by Railway would be followed. Decision of Railway in this regard would be final and binding on the contractor.
- 1.2. **Field Laboratory:** - The work covers the provision and maintenance of an adequately equipped field laboratory as required for site control on the quality of materials and the works. The lab should be manned, managed & maintained by suitable, qualified personnel from the contractor. The following minimum items/equipment shall be provided in the field laboratory.

**Description**

The Contractor shall arrange to provide fully furnished and adequately equipped field laboratory. The field laboratory shall preferably be located adjacent to the site office of the Engineer and provided with amenities like water supply, electric supply etc. as for the site office of the Engineer in Clause 1.2.

The laboratory should have space for the installation of equipment, laboratory tables and cupboards, working space for carrying out various laboratory tests, besides a wash basin, toilet facility and a curing tank for the curing of samples, around 4m x 2m x 1m in size and a fume chamber. Wooden/concrete working table with a working platform area of about 1m x 10m shall be provided against the walls, also providing wooden cupboards above and below the working tables to store accessories such as sample moulds etc. At least 2 racks of slotted angles and M.S. sheets and at least 4 stools for laboratory test operators shall also be provided

Signature of Tenderer

For Chief Administrative Officer  
(Const.) West Central Railway,

**1.2.3 Laboratory Equipment**

The following items of laboratory equipment shall be provided in the field laboratory.

**General**

- |       |                                                                                                                                                                                                                                                  |                                 |
|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|
| i)    | Oven - Electrically operated, thermostatically controlled, range upto 200°C sensitivity 1°C                                                                                                                                                      | 1 No.                           |
| ii)   | Platform balance 300 kg capacity                                                                                                                                                                                                                 | 1 No.                           |
| iii)  | Balance 20 kg capacity self-indicating type                                                                                                                                                                                                      | 1 No.                           |
| iv)   | Electronic Balance 5 kg capacity accuracy 0.5 gm                                                                                                                                                                                                 | 2 Nos.                          |
| v)    | Water-bath electrically operated and thermostatically controlled with adjustable shovels, sensitivity 1°C                                                                                                                                        | 1 No.                           |
| vi)   | Thermometers :<br>Mercury-in-glass thermometer range 0° to 250°C                                                                                                                                                                                 | 4 Nos.                          |
| vii)  | Kerosene or gas stove or electric hot plate                                                                                                                                                                                                      | 1 No.                           |
| viii) | Glasswares, spatulas, wire gauzes, steel scales, measuring tape, casseroles, karahis, enameled trays of assorted sizes, pestle-mortar, porcelain dishes, gunny bags, plastic bags, chemicals, digging tools like pickaxes, shovels etc.          | As required                     |
| ix)   | Set of IS sieves with lid and pan:<br>450 mm diameter:<br>63 mm, 53 mm, 37.5 mm, 26.5 mm, 13.2 mm, 9.5 mm, 6.7 mm and 4.75 mm size<br>200 mm diameter:<br>2.36 mm, 2.0 mm, 1.18 mm, 600 micron, 425 micron, 300 micron, 150 micron and 75 micron | 1 set<br><br><br><br><br>2 sets |
| x)    | Water testing kit for chloride, sulphate and pH value                                                                                                                                                                                            | 1 set at each site              |
| xi)   | Digital Moisture Meter                                                                                                                                                                                                                           | 1 No                            |
| xii)  | First aid box                                                                                                                                                                                                                                    | 1 set                           |

**For cement and cement concrete**

- |      |                                                                                                                    |                      |
|------|--------------------------------------------------------------------------------------------------------------------|----------------------|
| i)   | Vicat apparatus for testing setting times                                                                          | 1 No.                |
| ii)  | Slump testing apparatus                                                                                            | 4 Set                |
| iii) | Compression and Flexural strength testing machine of 200 tonnes capacity with additional dial for flexural testing | 1 No.                |
| iv)  | Needle Vibrator (Electrical driven)                                                                                | 2 Nos.<br>(Minimum)  |
| v)   | Air meter apparatus                                                                                                | 1 No.                |
| vi)  | Permeability testing apparatus                                                                                     | 1 No.                |
| vii) | Cube mould of size 15 cm x 15 cm x 15 cm                                                                           | 36 Nos.<br>(Minimum) |
| iii) | Accelerated cured concrete test apparatus (IS: 9013-1978)                                                          | 1 No.                |

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**For soil and aggregates**

|      |                                                                                            |             |
|------|--------------------------------------------------------------------------------------------|-------------|
| i)   | Modified Procter Test Equipment                                                            | 1 Set       |
| ii)  | Dry Bulk Density Test apparatus (sand pouring cylinder, tray, can etc.) complete.          | 2 Sets      |
| iii) | Set for sand replacement method for measuring the density of the compacted soil            | 2 Sets      |
| iv)  | Oven – electrically operated, thermostatically controlled range upto 200°C sensitivity 1°C | 1 No.       |
| i)   | Rapid moisture meter                                                                       | 1 No.       |
| ii)  | Mechanical sieve shaker with complete set of sieves                                        | 1 No.       |
| iii) | Flakiness & Elongation gauges                                                              | 2 Nos. each |
| iv)  | CBR testing equipment.                                                                     | 1 Set       |
| v)   | Relative density apparatus                                                                 | 1 Set.      |

**Note:** The items and their numbers listed above are tentative and shall be finally decided by the Engineer in charge as per the actual requirement of the same on the works.

**1.4 Testing to be done by contractor at his own cost:** - The tenderers should note that the rates of the items are inclusive of all the testing charges required for on-site or laboratory testing of all the materials for passing of the materials as specified in this tender document or required as per IRS / IS Codes or relevant Codes. All outside lab testing shall be from labs approved by NABL. However, the periodic and timely testing of materials will be the responsibility of the contractor/s only.

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**Chapter-3****SPECIFICATIONS FOR EARTH WORK AND BLANKETING**

1. Formation works: -
  - (i) Earthwork in Embankments with Railway's earth.
  - (ii) Earthwork in Embankments with Contractor's earth.
  - (iii) Earthwork in Cuttings in Soil.
  - (iv) Earthwork in Cuttings in Rock not requiring blasting.
  - (v) Earthwork in Cuttings in Rock requiring blasting.
  - (vi) Supplying and Spreading Blanketing material.
2. The above items of work envisage earthwork in Railway Formations and other works incidental thereto including earthwork in level crossing, road approaches, trolley refuges, man refuges, side drains catch water drains, nallah diversions, road diversions and all other earthwork items incidental to the Railway Formation works as directed by the Engineer whose decision in this regard being final and conclusive.
3. Without in anyway restricting or modifying the provisions of the General Conditions of Contract in this regard, the rates for the above works shall be inclusive of all costs of the following elements.
  - (i) Site clearance & dagh-belling as required; setting out profiles, maintenance and provisions of level pegs, center line and other pegs, bench marks, providing necessary dressing and drainage of borrow pits in the Railway land as required, etc. complete with all Contractor's labor, materials, tools, plants and machinery including fuel, crew etc.
  - (ii) Excavations in borrow pits, cuttings in all types of soils, rocks etc. as applicable and in all conditions. Whether dry, moist, wet or slushy inclusive of pumping out of water as may be required for the execution of work etc. complete with all contractor's Labour, materials, explosives, other consumable materials, tools, plants and machinery, fuel crew etc.
  - (iii) All leads; all crossings, of Railway lines, road and nallahs as required; all lifts & all descents etc. required for forming the embankments or disposing the excavated materials from the cuttings by stacking or in any other manner, in embankment, spoil dumps, stacks or any other location as directed by the Engineer whose decision in this regard being final and conclusive; providing earth bunds for stagnating water on embankments and dressing up of all slopes, surfaces etc. complete with all contractor Labour, material, tools, plants and machinery including fuel, crew etc.
  - (iv) Spreading the earth or the excavated materials from cuttings or blanketing material in layers of specified thickness adding water as required, compaction with contractor's approved type of compacting / rolling equipment etc. complete with all contractor's Labour, materials, tools, plants and machinery, fuel, crew etc.
  - (v) For earthwork in embankments with Contractor's earth and supplying and spreading of blanketing material in addition to the above, the rates will be inclusive of the cost of acquisition of the earth or blanketing material, by purchase of land or payment of royalty charges or in any other manner, transportation and all other incidental works, complete, for bringing the earth or blanketing material from outside the Railway land.
4. The payment for the earthwork and supplying and spreading of blanketing material will be made on the basis of cross sections for the compacted work as accepted by the Engineer.
5. The final profile of the top is to be achieved by deploying motor graders, use of motor grader is compulsory & must be ensured for achieving better top finishing quality.

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6. The rate for earth work in bank shall also include excavation in all condition of soil whether dry or wet, loading into and unloading from contractor's own transport, lead and lifts as mentioned in item, filling bank in layers, dressing all slopes and surfaces to final dimensions.
7. The rates quoted by the tenderer/s shall include clearing the site from all bushes, roots, grass etc. demarcating by furrows the toes/edges of slopes of banks/cutting and cost of setting out by stacks bamboos, strings pegs etc. and benching of slopes. After site clearance, all pockets and depressions left in the soil shall be made good and completed.
8. Rate for earth work in bank /cutting shall also include the cost of maintenance of the bank/cutting by the contractor to the correct profile including repairs of all rain cut and removal of slips and making good the profile etc. until final measurement has been taken by the Engineer-in-charge. Completed bank/cutting shall be taken over after final measurements are recorded.
9. No extra payment shall be made for any interruption of shrinkage, settlement, slips, damages etc. that take place due to rains during the course of work and/or for wet excavation. Final measurement will be taken and finished bank/cutting to required slope taken over after the work is completed to the satisfaction of the Engineer.
10. The affected trees which are coming in the way of work will be marked by the Engineer's representative and only those trees which have been thus marked and listed will be removed by the contractor/s. No claim will be admissible from the contractor/s in regard to tree branches, bushes woods etc. stolen from the site after their cutting will be admissible.
11. The work shall be carried out as per the longitudinal section plan and cross section profile furnished by the Engineer in charge. Changes in 'L' section and cross section profiles and Railway yard plans can be made during the course of work, if necessary. Railway reserves the right to make changes in alignment and formation level as considered necessary to suit the site conditions and no compensation in any form will be admissible on this account. The contractors will have to execute the work as per final plans at the rate/rates quoted by him/them.
12. The earthwork gross quantities in embankment will be reduced by 10% towards shrinkage allowance as per para 1.1.6.10 of IRUSS 2010. The shrinkage allowance will, however, be reduced to 5% if embankment is compacted by heavy machineries as specified in para 1.9 of Indian Railways Unified Standard Specification of Works & Materials, the shall also be applied in case NS items operated for earth work in embankment.
13. Stretches of bank or cutting when completed will not be less than 1 km in length before they are offered for final measurement and while taking over by the Engineer's representative unless otherwise directed by Engineer in charge.
14. Some of the working area may be full of waste, garbage and night soil and it shall be the responsibility of the contractor to dispose of this as well as other organic soil, peat, muck or black cotton soil etc. such type of soil shall not be allowed to use on banks. The unusable soil shall be disposed off as per instruction of engineer in charge.
15. **LEVELLING INTERVALS MEASUREMENTS ETC.**
  - (i) Normally longitudinal levels shall be taken at interval of 20m. Where the ground is uneven or there is sudden rise or fall in levels, levels may be taken at closer intervals. For cross sections, levels shall be taken at interval of not more than 5 m. Levels for cross sections may be taken at closer intervals where warranted.

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- (ii) Before the commencement of the work, the cross-sectional levels of ground will be taken in the presence of contractor/s and will be plotted in due course by the Engineer's representative. The contractor/s will be required to sign both the level books and the profile sheets and start work only after this.

**16. CODES OF SPECIFICATIONS:**

- (i) The execution of earthwork including completion of the same shall conform to the latest version of codes of practice mentioned below with the amendment incorporated upto the date of opening of this tender.

- IS-2720 Part II 1973 method of tests for soil determination of water contents (II Edition).
- IS-2720 Part III section I of 1980 method of tests for soil determination of specific gravity.
- IS-2720 Part IV 1983 method of test for soil grain size analysis.
- IS-2720 Part V 1985 determination of liquid and plastic limit.
- IS-2720 Part VIII 1983 determination of moisture contents dry density using heavy compaction.
- IS-2720 Part XIV 1983 determination of density index (Relative density) of a cohesionless soil).
- IS-2720 Part XVII 1992 method of test for soil determination of field moisture equivalents.
- IS-2720 Part XXIX 1975 determination of dry density of soil in place by the core cutter methods.
- IS-1498-1970 classification and identification of soils for General Engineering purpose.
- IS-10379-1982 code of practice for field control of moisture and compaction of soils forembankment and sub-grade.
- Indian Railways Unified Standard Specifications (IRUSS) (Works and Materials) 2010.
- Comprehensive Guidelines and Specifications for Railway Formation issued by RDSO(RDSO/2020/GE: IRS-0004, Sept-2020)

XIII IS 2720 Part XVII 1987 – Determination of CBR Value in lab.

XIV IS 2720 Part XXXI (Part 31) 1969 – Determination of CBR value in field.

- (ii) A copy of all above codes will be kept at site before starting of earth work by contractor. Nothing extra will be paid on this account.
- (iii) In case there is any variation in the different codal practices, the decision of Dy Chief Engineer (C) in regard to the practice to be followed shall be final and binding on the contractor/s.

**17. COMPACTION OF SANDY OR SILTY SOILS WITH MODERATE COHESION.**

- (i) With the moderate cohesion, the compaction in layers by rollers is most effective. Vibratory rollers have been found more effective than the static rollers and greater thickness of layers can be allowed.

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- (ii) The water content at which a soil is compacted has an effect on all the physical properties of compacted soil. Therefore, design parameters should be carefully selected to cater for the most unfavorable conditions.
- (iii) Water content and densities should be specified as obtained in the field trials as per IS 10379-1982  
. Laboratory tests for heavy compaction as per IS 2720 (Pt.VIII) –1983 should be carried out for obtaining these. Densities attained in field trials normally should be around maximum dry densities as obtained from these tests and should form the basis for specifications and control. The moisture content controls may not be specified and 98% of such densities as achieved in field trials are only specified.
- (iv) For compaction controls Guide lines for Earthwork in Railway Project Published by RDSO-Lucknow & RDSO SPECIFICATION No. RDSO/2020/GE: IRS-0004 issued in September, 2020 should be used for field trials.
- (v) Any relaxation in the above-mentioned specifications, if necessary due to special circumstances, should have the approval of the Chief Engineer in charge of the project.

**18. COMPACTION OF CLAYS.**

- (i) Main objective of compacting predominantly clayey soils is to achieve a uniform mass of soil with no voids between the chunks of clays which are placed during the entire work. If moisture content is too high, rollers tend to sink into the soil. Maximum dry densities and optimum moisture contents should be found from laboratory tests for heavy compaction as specified in IS:2720 (Pt.VIII)1983. The laboratory results may only be used for arriving at those practically achievable densities and optimum moisture contents as obtained from the field trials as per IS 10379-1982.
- (ii) Sheep-foot rollers are most effective in breaking the clods and filling large spaces. The layer thickness should be equal to the depth of the feet of roller plus 50 mm. For specifications of sheep- foot rollers IS 4616-1968 may be referred.

**19. COMPACTION OF COHESIONLESS SOILS**

- (i) The effective method of compacting cohesion less soil is the use of vibratory roller and soil has water content near OMC. However, in spite of compaction, railway embankment may show small settlement during the initial stages of traffic after the line is opened. Moreover, introduction of new type of stock with different vibratory characteristics and axle loads etc. may also result in small settlements due to embankment soil undergoing further compaction.
- (ii) Poorly graded sands and gravels with uniformity co-efficient of less than 2.0 should not be used in earthwork for the banks to safeguard against liquefaction under moving loads or especially due to an earthquake tremor.
- (iii) IS Code No. 2720 (Pt.XIV 1983) should be followed for compaction in cohesion less soils. Minimum 70% Relative Density must be achieved during compaction which shall be done in layers as specified in RDSO guidelines.

**20. GENERAL POINTS**

1. Before the work is commenced, field compaction trials should be conducted as per RDSO/2020/GE: IRS- 0004 for deciding upon the most desirable moisture content and economical thickness of layers and the number of passes required to achieve the specified density. Heavy compaction tests should be first carried

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out on the soil from the borrow pit areas to ascertain the optimum moisture content and the maximum dry density. A cast area should then be selected and the fill material spread over this area in strips varying the depth of the loose materials and compacting with suitable type of roller at or near the OMC. The dry density of the compacted soil should be determined after different number of passes. Based on these trials the optimum depth of loose layers and the number of passes required should be decided such trials may have to be repeated at different locations wherever there is a change in the type of soil.

2. Adequate drainage must be ensured for the works in service conditions. The formation bed should have a cross slope of 1 in 30 from the centre towards the side drains on either side.
3. The side slopes of the embankments and cuttings may have to be protected by special measures if required such as turfing, GJT etc. to prevent erosion and payment will be done in relevant items.
4. It will be necessary to keep the borrow pits sufficiently away from the toe of the embankments to prevent base failures due to lateral escapement of the soil. The distance of borrow pit from the bank will have to be decided in each case on its merits.
5. In the case of embankment in highly cohesive or clayey soils, special treatment may be necessary to ensure a stable formation. Such measures will have to be determined after thorough investigation and study of the soil properties.
6. Special investigation such as slope stability analysis will also be necessary in regard to high fill (more than 6m) construction on swampy ground or marshy lands and deep cuttings.

## **21. SPECIFICATIONS FOR MECHANICAL COMPACTION OF EARTH WORK.**

1. Guidelines for execution of earthwork in embankments and cuttings on new constructions, doublings and conversion projects have been separately formulated. These envisage mechanical compaction of earthwork. The following specifications for such compaction work are recommended for adoption. These may require some modifications to suit local conditions and may also have to be amended on experience at a future date.
2. SUB GRADE COMPACTION: - After site clearance all pockets and depressions left in the soil, if any, should be made good and compacted suitably.
3. METHOD OF COMPACTION: - Earth work should be done in layers not exceeding 300 mm and 600 mm thick in the loose state with the static and vibratory rollers respectively. The thickness of the loose layer should be decided after preliminary trial with the equipment proposed to be used as per RDSO/2020/GE: IRS-0004. The layer should be compacted preferably at or near the water content thus determined with suitable rollers to achieve the desired density of 98% of MDD as obtained from heavy compaction as per IS 2720 (Pt. VIII) – 1983. It would be necessary to; use right type of compaction equipment for proper and speedy compaction for quality control of earthwork, however, only practically achievable values of densities would be indicated as obtained from the field trials.
  - (i) Where compaction is done by sheep foot rollers, the depth of each layer of loose soil should not exceed by more than 50 mm the length of their feet.
  - (ii) The suitability of various rolling equipment for different types of soils is given in Annexure – F of RDSO guide line RDSO/2020/GE: IRS-0004 and also reproduced below:

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| Material<br>(1)             | Major<br>divisions<br>(2)            | Sub groups<br>(3)                                                                                                                                                                                                                                                                                                                                                                                               | Suitable type<br>of<br>compaction<br>plant<br>(4)                                                                                                                                                                                                                                                | Maximum<br>number of<br>passes for<br>satisfactory<br>compaction<br>(5) | Maximum<br>thickness of<br>compacted<br>layer (mm)<br>(6) | Remarks<br>(7)                                                                                                                                                                                                                                                                      |
|-----------------------------|--------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|-----------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Rock-like<br>materials      | Natural<br>rocks                     | All rock fill (except<br>chalk)                                                                                                                                                                                                                                                                                                                                                                                 | Heavy<br>vibratory<br>roller not less<br>than 180 kg<br>per 100 mm<br>of roll<br><br>Grid roller not<br>less 180 kg<br>per 100 mm<br>of roll<br>Self- propelled<br>tamping<br>rollers                                                                                                            | 4 to 12                                                                 | 500 to 1500<br>depending<br>on plant<br>used              | If well<br>graded or<br>easily<br>broken down<br>then this can<br>be classified<br>as a coarse-<br>grained soil<br>for the<br>purpose of<br>compaction.<br>The<br>maximum<br>diameter of<br>the rock<br>fragment<br>should not<br>exceed two<br>third of the<br>layer<br>thickness. |
| Coarse-<br>grained<br>soils | Gravel<br>sand,<br>gravelly<br>soils | Well graded gravel<br>and gravel/sand<br>mixture: little or<br>no fines<br><br>Well graded<br>gravel/ sand<br>mixtures with<br>excellent clay<br>binder<br><br>Uniform gravel:<br>little or no fines<br><br>Poorly graded<br>gravel and<br>gravel/sand<br>mixtures: little or<br>no fines<br><br>Gravel with excess<br>fines, silty gravel,<br>clayey<br>gravel, poorly<br>graded gravel/<br>sand/clay mixtures | Grid roller<br>over 540 kg<br>per 100mm of<br>roll<br><br>Pneumatic<br>tired over<br>2000 kg per<br>wheel<br>Vibratory plate<br>compactor<br>over 1100<br>kg/sq.m. of<br>base plate<br><br>Smooth wheel<br>roller<br>Vibratory<br>roller<br><br>Vibro-rammer<br>Self-propelled<br>tamping roller | 3 to 12<br>depending<br>on type of<br>plant                             | 75 to 275<br>depending<br>on type of<br>plant             |                                                                                                                                                                                                                                                                                     |

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|             |                                |                                                                                                                                                                                                      |                                                                                                                                                                                                                               |                                    |                                       |                                                                                                                                                                 |
|-------------|--------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
|             | Sand and sandy soils           | Well graded sands and gravelly sands; little or no fines<br><br>Well graded sands with excellent clay binder                                                                                         |                                                                                                                                                                                                                               |                                    |                                       |                                                                                                                                                                 |
| <b>1</b>    | <b>2</b>                       | <b>3</b>                                                                                                                                                                                             | <b>4</b>                                                                                                                                                                                                                      | <b>5</b>                           | <b>6</b>                              | <b>7</b>                                                                                                                                                        |
|             | Uniform sands and gravels      | Uniform gravels; little or no fines<br>Uniform sands; little or no fines<br>Poorly graded sands; little or no fines<br>Sands with fines, silty sands, clayey sands, poorly graded sand/clay mixtures | Smooth wheeled roller below 500kg per 100mm of roll<br><br>Grid roller below 540kg per 100mm of rolling<br><br>Pneumatic tired roller below 1500kg per wheel<br>Vibratory roller<br>Vibrating plate compactor<br>Vibro-tamper | 3 to 16 depending on type of plant | 75 to 300 depending on type of plant  |                                                                                                                                                                 |
| Fines soils | Soils having low plasticity    | Silts (inorganic) and very fine sands, rock flour, silty or clayey fine sands with slight plasticity<br>Clayey silts (inorganic)<br>Organic silts of low plasticity                                  | Sheep foot roller<br>Smooth wheeled roller<br>Pneumatic tired roller<br>Vibratory roller over 70 kg per 100 mm of roll<br>Vibratory plate compactor over 1400 kg/sq.m of base plate<br>Vibro-tamper<br>Power rammer           | 4 to 8 depending on type of plant  | 100 to 450 depending on type of plant | If water content is low, it may be preferable to use vibratory roller. Sheep foot rollers are best suited to soils at water contents below their plastic limit. |
|             | Soils having medium plasticity | Silty and sandy clays (inorganic) of medium plasticity<br>Clays (inorganic) of medium plasticity                                                                                                     |                                                                                                                                                                                                                               |                                    |                                       |                                                                                                                                                                 |

However, decision of Engineer in charge regarding the maximum thickness of compacted layer to be used at individual sites will be final and work will be done accordingly.

## 22. RECOMMENDED TESTS FOR QUALITY CONTROL

- (i) Coarse grained soil which contains fines passing 75 microns up to 50% should be compacted to get

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the density index (Relative density) a minimum of 70% as obtained in accordance with IS:2720 (Pt. XIV)-

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1983.

- (ii) All other types of soil when compacted should normally have at least 98% of the maximum dry density as determined by using heavy compaction in accordance with IS: 2720 (Pt. VIII) – 1983. However, the actual densities obtained during field trials should form the basis to eliminate the difficulties in the field due to any reason of weather conditions or non-availability of appropriate compaction equipment.

**23. MOISTURE CONTENT**

- (i) The objective should be compact near OMC or above to achieve the density as specified in most efficient manner.

**24. PRECAUTION**

- i. Earth layer should be compacted to the desired density over its entire width commencing from the sides, before the next layer is commenced.
- ii. There should be a minimum overlap of 150 mm between each run; of the rollers.
- iii. Care should be taken during rolling to suitably slope the surface of the bank to facilitate the shedding and to minimize absorption of rain water, special attention being given to the prevention of ponding.
- iv. The top of the formation should be finished to a slope of 1 in 30 away from the centre.
- v. Extra wide bank by 50cm on either side shall be rolled and then dressed to size for avoiding any loose earth at the shoulders.

**b. QUALITY CONTROL OF WORK**

- i. Adequate arrangements for control of compaction must be ensured during the construction so that the required degree of density is obtained in each layer of earthwork.
- ii. The density of each layer of compacted soil should be ascertained by taking soil samples, collected mostly on either side of the centre line at intervals of 10 m or so with a few taken at random near the two sides using sand replacement as per IS: 2720 (Pt. XXVIII)-1974 or core cutter method as per IS: 2770 (Pt. XXIX) 1975. The number of tests will of course, depend on the width of the formation.
- iii. The above tests for quality control may not be necessary while mechanically compacting banks formed with cohesion less soil except for the top one metre.

**c. EARTHWORK WITH CONTRACTOR'S OWN EARTH**

**Specification of earth:** Soil to be used in embankment shall conform to specifications laid down in RDSO/2020/GE:IRS-0004.

Following soil should be avoided:

- Peat and organic soils
- chalk etc. which are likely to disintegrate.
- Poorly graded sands with  $C_u < 2$ .

**d. Selection of Borrow Area: -**

- (i) Borrow area should be selected sufficiently away from the alignment, as far as possible at the extreme of Railway land but normally not less than 3 m plus height of the embankment to prevent base failure due to lateral escapement of the soil.
- (ii) Borrow area should be selected for soil suitable to be used in construction.

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- (iii) Till such time the embankment is completed and final measurements have been recorded and handed over; if subsidence taken places due to elements such as winds, floods, cyclone etc. all such rain cuts, subsidence etc. have to be made good by the contractor/s for which no payment what so ever shall be payable.
- (iv) The cross section of ground will be recorded by the engineer or engineer's representative in the prescribed level books only and not in any book or register. The contractor/s shall not start the work unless he/they is/are satisfied that the proper ground levels have been recorded in proper level books by the engineer's representative and the same are signed and accepted by him, failing which the level recorded in level books by the engineer's representative will be binding on the contractor/s.
- (v) No dispute of any sort subsequently will be accepted as it will also be the responsibility of the contractor/s not to start the work unless ground levels are recorded in proper level books provided for the purpose and the same are signed by him/them.
- (vi) In case of banks no payment over the required theoretical profile will made. In case earthwork is done above the required formation level, the same shall have to be brought to correct level banks which are smaller than the required profile, the same will have to be made up before final measurements are recorded.
- (vii) The area of the cross section covered by standard theoretical profile shall only be considered for payment. The extra 50 cm wide earth work (berm) on either side shall not be payable. The entire 50 cm berm on either side will not be allowed to be recovered.
- (viii) The earth work shall be done without causing any disturbances to the rail or road traffic and tenderer/s is/ are not entitled for any extra payment for obstructions such as rail or road traffic and also for crossing track etc. The rate quoted by the tenderer/s shall be deemed to include all cost of work, including excavation lead, lift, spreading, watering, mechanical compaction etc. required for the satisfactory completion of works.
- (ix) If due to any reasons, the work is left incomplete in any stretch it would be dressed by the contractor/s so that measurements can be taken. If the contractor/s fails to do so it will be done by the Railway at the cost of contractor/s.

**25. Specification for Mechanically Produced Blanketing Material for Railway Formations including Guidelines of Laying as per RDSO/2020/GE: IRS-0004**

These are as under:

1. Scope: This specification and guidelines are applicable for supply and laying of mechanically produced blanket material for use on railway formations along with its manufacture, transportation, compaction, testing, acceptance and measurement.
2. Definitions:
  - Formation:** An earthen structure of specified dimensions, both for cuttings and for embankment.
  - Blanket:** Blanket is a layer of specified coarse, granular material of designed thickness provided over full width of formation between sub grade and ballast.
  - Mechanical Production:** Production of blanket material from naturally available raw material at a plant.
  - Blending:** Mechanical mixing of more than one natural or mechanically produced granular soil material in tentatively predefined ratio in order to get final product of desired properties

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## 26. RECOMMENDED SPECIFICATIONS OF BLANKET MATERIAL (Mandatory)

Specifications of the material for blanket layer over prepared sub-grade should be such that it is well-graded sandy gravel layer of adequate hardness. Particles size gradation curve should be more or less within Enveloping Curves of blanket material as shown in Fig. 9 below & Grading Percentages within the range given in Table-5 below and should also have following criteria satisfied:

- (i)  $C_u > 7$  and  $C_c$  between 1 and 3.
- (ii) Fines (passing 75 microns) : 3% to 10%.
- (iii) Los Angeles Abrasion value < 35%.
- (iv) Minimum required Soaked CBR value 25 of the blanket material compacted at 100% of MDD.
- (v) In exceptional cases on technical and economic considerations, LAA value may be relaxed upto 40% by Pr CE on Open Line & CAO/C in construction projects.
- (vi) Filter Criteria should be satisfied with prepared sub grade/sub grade layer just below blanket layer, as given below :
  - Criteria-1:  $D_{15}(\text{blanket}) < 5 \times D_{85}(\text{sub-grade})$  Criteria-2:  $D_{15}(\text{blanket}) > 4 \times D_{15}(\text{sub-grade})$
  - Criteria-3:  $D_{50}(\text{blanket}) < 25 \times D_{50}(\text{sub-grade})$
  - (vii) This item is only for mechanically crushed material. The mechanically crushed material may be directly produced in gradation as per RDSO guidelines issued in Sept.-2020.
  - (viii) If graded material is not produced by crusher then different sizes crushed material will be mechanically blended so that gradation % comes as per RDSO specifications.
  - (ix) No natural material will be used or blended for the blanketing material.

27. Filter Criteria is optional, at present. This can be adopted with the experience gained of its compliance for different types of soils with blanket.

**Table-5 : Grading Percentage of Blanket Material**

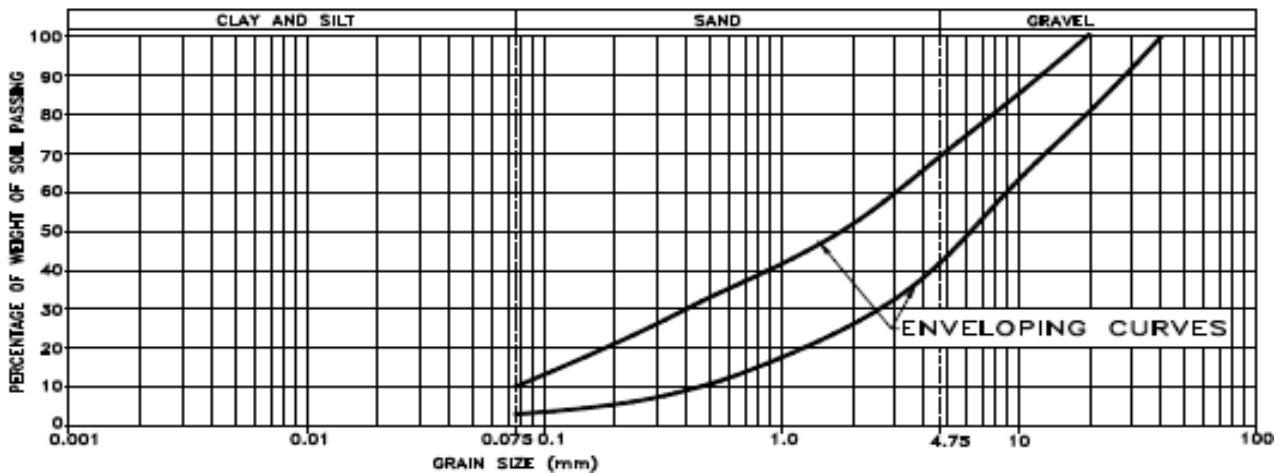
| SL | IS Sieve Size | Percent Passing (by weight) |
|----|---------------|-----------------------------|
| 1. | 40 mm         | 100                         |
| 2. | 20 mm         | 80 - 100                    |
| 3. | 10 mm         | 63 - 85                     |
| 4. | 4.75 mm       | 42 - 68                     |
| 5. | 2 mm          | 27 - 52                     |
| 6. | 600 micron    | 13 - 35                     |
| 7. | 425 micron    | 10 - 32                     |
| 8. | 212 micron    | 6 - 22                      |
| 9. | 75 micron     | 3 - 10                      |

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**Fig. 9 : Enveloping Curves for Blanket Material:**



## 28. QUALIFYING AND QUALITY ASSURANCE TESTS (Mandatory)

Qualifying tests as part of pre-selection of good earth for track sub grade, embankment fill is required to be carried out. Also, quality of compaction is required to be done to ensure good quality construction.

**i) Selection of soil:** For selection of soil to be used as embankment fill **CBR test** is required to be conducted on material. CBR test is conducted on ground soil, embankment fill, prepared sub-grade & blanket material to ensure the minimum specified CBR value of these materials to be used in construction.

This test is carried out on soil sample in laboratory as per procedure given in IS:2720 (Part 16)-1987 & in field as per IS:2720 (Part 31)– 1969.

**ii) Quality Assurance Test on Compacted Layer:** Quality Assurance Tests are required to be conducted on part completion stages of formation, prior to clearing for further earthwork, track linking work:

**Heavy Proctor test** is required to be conducted to determine the Maximum Dry Density of soil as per IS: 2720 (part 8). In-situ density is measured in the field by Sand Replacement Method (IS: 2720 – part 28) or Core Cutter Method

(IS: 2720 – part 29) to calculate the degree of compaction. this shall be determined in laboratory as per BIS procedure with the specified frequency of earthwork quantity, as envisaged in RDSO/2020/GE:IRS- 0004.

**Second Step Plate Load Test (Optional)** is required to be conducted in-situ for measurement of Deformation Modulus EV2 of compacted layers of embankment, blanket, prepared subgrade etc. The test procedure has been detailed in German Code DIN:18134- 2001, 'Determining Deformation & Strength Characteristics Of Soil By Plate Loading Test' and in RDSO/2020/GE:IRS-0004. The minimum value of EV2 should be ensured at different levels as specified. Brief procedure of these above tests has been given in Annexure-2.

### i. Frequency of Quality Assurance Tests

a) CBR test for selection of formation materials and other tests required for ensuring conformation of the materials (blanket, sub grade) as per specification e.g., size gradation, Cu, Cc, Los Angeles Tests, OMC/MDD etc.

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shall be conducted at following frequency:

- i) Embankment Fill : one set of tests for every 5000 cum
- ii) Prepared sub grade : one set of tests for every 2000 cum
- iii) Blanket material : one set of tests for every 5000 cum

b) In-situ Degree of Compaction (or In-situ dry density measurement) test shall be conducted on each compacted layers in random pattern at following frequency for the different layers:

- i) Embankment Fill : one density measurement at every 500 sqm surface area of each compacted layers
- ii) Blanket and Prepared Sub grade : one density measurement at every 200sqm surface area of each compacted layers.

c) Second step Plate Load Test. This should be done for EV2 measurement at top of each formation layers eg. at sub-soil, compacted sub grade, prepared sub grade, blanket etc. at the frequency of one test per km length of section.

**Table 3.3: Specification and Thickness of Formation Layers for 25T axle load:  
Single layer system**

| Layers                         | Specification                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Thickness                                                                                             |
|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| <b>Blanket</b>                 | <ul style="list-style-type: none"><li>i) <math>C_u &gt; 7</math> and <math>C_c</math> between 1 and 3.</li><li>ii) Fines (passing 75 microns) : 3% to 10%</li><li>iii) Minimum soaked CBR value <math>\geq 25</math>, (Soil compacted at 100% of MDD * in Lab)</li><li>iv) Los Angeles Abrasion value <math>&lt; 40\%</math></li><li>v) Field Compaction : Min. 100% of MDD * in field trial</li><li>vi) Minimum <math>E_{v2}</math> ** = 100 MPa</li><li>vii) Size gradation – within specified range (as table-3.7) or should lie more or less within enveloping curves (as Fig-3.8)</li><li>viii) Filter criteria (***) <b>Optional</b> should be satisfied with sub-grade layer as given below:<br/>Criteria-1: <math>D_{15}</math> (blanket) <math>&lt; 5 \times D_{85}</math> (sub-grade)<br/>Criteria-2: <math>D_{15}</math> (blanket) <math>&gt; 4</math> to <math>5 \times D_{15}</math> (sub-grade)<br/>Criteria-3: <math>D_{50}</math> (blanket) <math>&lt; 25 \times D_{50}</math> (sub-grade)</li></ul> | <b>30 cm</b> over SQ3 sub-grade<br><b>40 cm</b> over SQ2 sub-grade<br><b>55 cm</b> over SQ1 sub-grade |
| <b>Sub-grade</b><br>Top Layer: | <p>SQ1/SQ2/SQ3 soil</p> <p>SQ1 soils (To be used only with dispensation of PCE/ CAO)</p> <ul style="list-style-type: none"><li>i) For SQ2/SQ3 soil, CBR <math>\geq 6</math> ( soil compacted at 98% of MDD *)</li><li>ii) For SQ1 soil, CBR <math>\geq 4</math> soil compacted at 98% of MDD *</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 100 cm                                                                                                |

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|                             |                                                                                                                                                                                                                                                                                                             |                          |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|
| Lower layer (fill)          | iii) Field Compaction : Min. 98% of MDD *<br>iv) Minimum $E_{v2}$ = 45 MPa (for SQ1)<br>60 MPa (for SQ2/SQ3)<br>SQ1/SQ2/SQ3 soil (+)<br>i) $CBR \geq 3$ (soil compacted at 97% of MDD *)<br>ii) Field Compaction : Min. 97% of MDD *                                                                        | As per Embankment height |
| Ground Soil/Sub-soil Strata | i) Undrained Cohesion of soil ( $c_u$ ) $\geq 25$ KPa (only for soils having particles finer than 75 micron exceeding 12%)<br>ii) $E_{v2}$ (determined from PLT) $\geq 20$ MPa<br>iii) N (determined from SPT) $\geq 5$<br>Ground Improvement is required, if any of the above parameters not complied with | --                       |

\* MDD mentioned in the above table

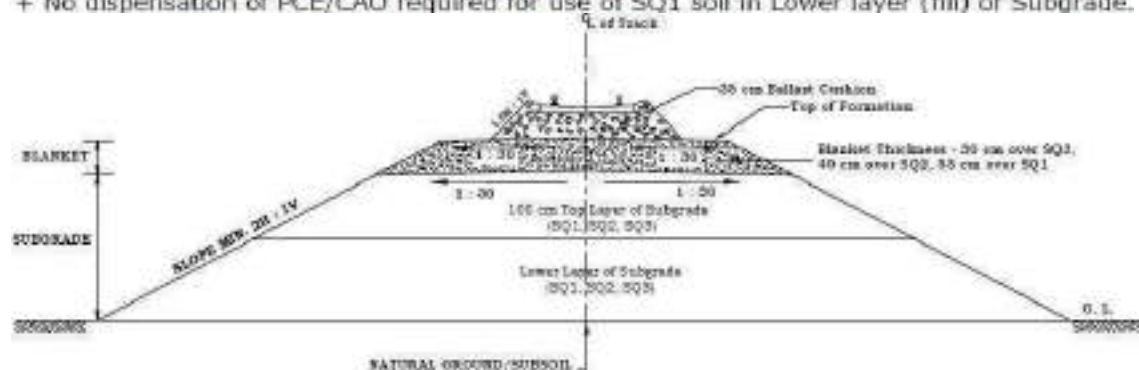
(a) For determination of CBR - MDD achieved in Lab

(b) For field compaction - MDD achieved in field compaction trials which should not be less than 98% of MDD in lab.

\*\*  $E_{v2}$  is Modulus of deformation (**Para 2.0 of Appendix-H**)

\*\*\* With the application of Non-woven Geotextile as a separator layer below the blanket, filter criteria will not be required or mandatory.

+ No dispensation of PCE/CAO required for use of SQ1 soil in Lower layer (fill) of Subgrade.



**Fig-3.4: Track Formation for 25 T Axle Load (Single layer system)**

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| Layers                             | Specification                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Thickness                                                                                |
|------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| <b>Blanket</b>                     | i) $C_u > 7$ and $C_c$ between 1 and 3.<br>ii) Fines (passing 75 microns) : 3% to 10%<br>iii) Los Angeles Abrasion value < 40%<br>iv) Minimum soaked CBR value $\geq 25$ , (soil compacted at 100% of MDD * in lab)<br>v) Field compaction: 100% of MDD* in field trial<br>vi) Minimum $E_{v2}$ ** = 100 MPa<br>vii) Size gradation – within specified range (as table-3.7) or should lie more or less within enveloping curves (as fig.-3.8)<br>viii) Filter criteria (***)Optional should be satisfied with prepared sub-grade layer as given below:<br><br>Criteria-1: $D_{15} \text{ (blanket)} < 5 \times D_{95} \text{ (prepared sub-grade)}$<br>Criteria-2: $D_{15} \text{ (blanket)} > 4 \text{ to } 5 \times D_{15} \text{ (prepared sub-grade)}$<br>Criteria-3: $D_{90} \text{ (blanket)} < 25 \times D_{90} \text{ (prepared sub-grade)}$ | <b>30 cm over SQ3 Prepared Sub-grade</b><br><br><b>40 cm over SQ2 Prepared Sub-grade</b> |
| <b>Prepared Subgrade</b>           | SQ2/SQ3<br>i) CBR $\geq 8$ (soil compacted upto 98% of MDD *)<br>ii) Plasticity Index $\leq 12$<br>iii) Field Compaction :Min. 98% of MDD *<br>iv) Minimum $E_{v2}$ = 60 MPa                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>50 cm over SQ1 fill</b><br><br><b>35 cm over SQ2 fill</b>                             |
| <b>Subgrade Top Layer</b>          | SQ1/SQ2/SQ3<br>(SQ1 soils (To be used only with dispensation of PCE/ CAO)<br>i) CBR $\geq 5$ ( soil compacted at 97% of MDD *)<br>for SQ2/SQ3 soils<br>ii) For SQ1 soil, CBR $\geq 4$ ( soil compacted at 97% of MDD *)<br>iii) Field Compaction : Min. 97% of MDD *<br>iv) Minimum $E_{v2}$ = 30 MPa (for SQ1)<br>45 MPa (for SQ2/SQ3)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <b>50 cm</b>                                                                             |
| <b>Lower layer (fill)</b>          | SQ1/SQ2/SQ3 soil (+)<br>i) CBR $\geq 3$ (soil compacted at 97% of MDD *)<br>ii) Field Compaction :Min. 97% of MDD *                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>As per Embankment height</b>                                                          |
| <b>Ground Soil/Sub-soil Strata</b> | i) Undrained Cohesion of soil ( $c_u$ ) $\geq 25$ KPa (only for soils having particles finer than 75 micron exceeding 12%)<br>ii) $E_{v2}$ (determined from PLT) $\geq 20$ MPa<br>iii) N (determined from SPT) $\geq 5$<br>Ground Improvement is required, if any of the above parameters not complied with                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | --                                                                                       |

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(Const.) West Central Railway,

\* MDD mentioned in above table

- For determination of CBR - MDD achieved in Lab,
- For field compaction - MDD achieved in field compaction trials which should not be less than 98% of MDD in lab.

\*\* $E_{v2}$  is Modulus of deformation (Para 2.0 of Appendix-H)

\*\*\* With the application of Non-woven Geotextile as a separator layer below the blanket, filter criteria will not be required or mandatory.

+ No dispensation of PCE/CAO required for use of SQ1 soil in Lower layer (fill) of Subgrade.



Fig-3.5: Track Formation for Two layer system (for 25 T Axle load)

## ii. Method Statement for Mechanical Production of Blanketing Material:

### 29. Crushing Methodology:

In the event of non-availability of natural source of blanket material, depending on the proximity of project site from the parent rock/boulder sources, it may be decided to crush the rock/boulder in order to produce crushed blanket material. Salient features of this methodology are:

- Crushed blanket material may be produced as sole product or in conjunction with ballast or any other nominal size.
- Trials and permutations of feed speed, crushing cycle, and sieve combinations may be required to arrive at the required particle size gradation.
- It is possible to achieve near total produce of desired gradation through cycle management. Alternately, it may be possible to get by-products of other sizes in the desired proportion and blanket material as main produce or vice-versa.
- Optimization of production rates and costs can be achieved by controlling the output at each sieve stage.
- It is ideal to mix the required quantity of water for OMC (accounting for loss/gain of moisture due to weather conditions) at the crusher plant and transport the material in wet condition.
- In all such cases, where large scale crushing is taking place for production of ballast, trials may be conducted to arrive at ideal particle size gradation of the by-product also.

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- No natural material shall be blended with crushed graded blanket material. All different sizes graded material shall only be mechanically crushed material used for blanket.

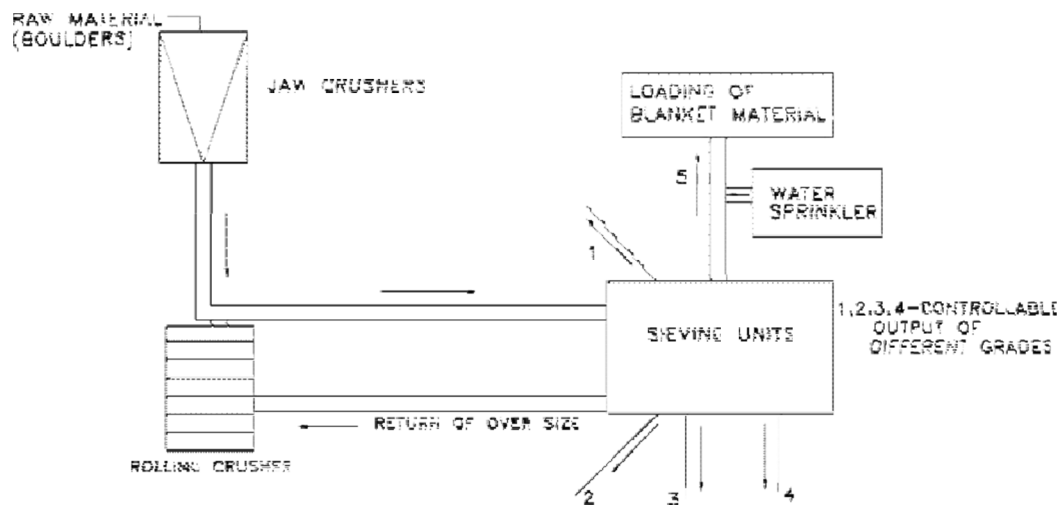
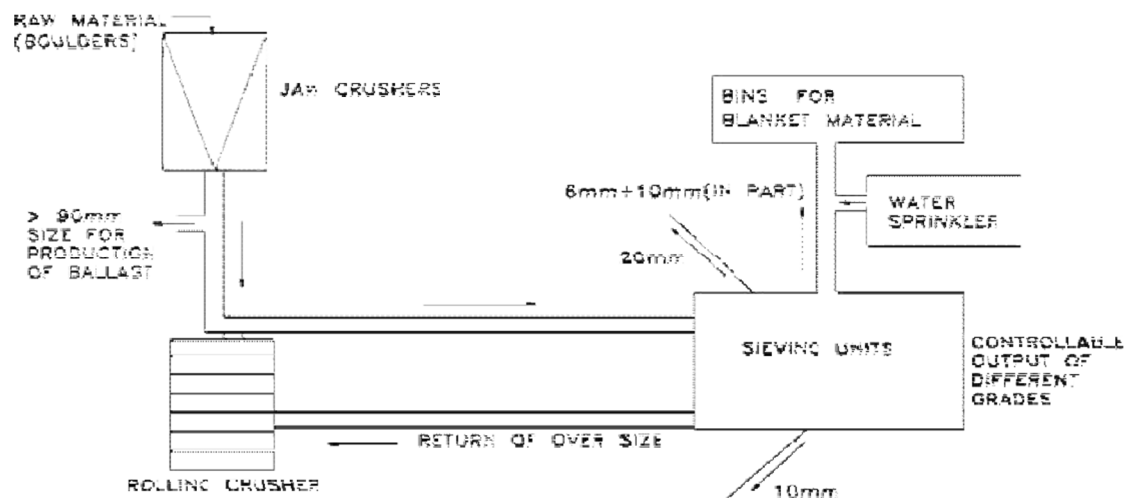


Fig.2 SCHEMATIC DIAGRAM OF MANUFACTURE OF BLANKET MATERIAL BY CRUSHERS

The schematic diagram of for production of blanket at this crusher is given below:



SCHEMATIC DIAGRAM OF MANUFACTURE OF BLANKET MATERIAL BY CRUSHERS

### 30. Transportation:

The blanket material should be transported wet after mixing water in order to achieve OMC, in tippers for direct unloading on formation.

### 31. Laying, spreading and Compaction:

- The blanket material must be spread with a tractor mounted grader or a paver-finisher in layers of uniform thickness, before allowing compaction.
- Compaction to specified levels of RD or percentage of MDD will be carried out through a number of passes of vibratory rollers of 100-120 kN static weight or equivalent capacity. A combination of vibrating rolling initially and static finishing rolling may be established through trials. Speed of roller shall not exceed 5 km/hr.

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3. Proper control of moisture is required to optimize the compaction effort. Optimum moisture content may be established through Modified Proctor Apparatus (IS:2720, part 8) and moisture may be added by sprinkling at the plant or at site as per the requirement.
4. Rolling is to be carried out in layers of not more than 300 mm each, following the same camber profile as provided in the sub grade layer and to be maintained upto the top layer.
5. No provision for uncompacted portion may be made on the edges of embankment. The sides may be hand rammed with a suitable rammer.
6. The contractor will be required to arrange the nuclear testing gauge of approved make to measure the in-situ moisture content and field density and nothing extra will be paid for the same.

**32. Quality Control Checks on Finished blanket work:**

1. Degree of compaction of each layer of compacted blanket should be ascertained by measurement of dry density/Relative Density of soil at locations selected in specified pattern. The method of sampling, frequency of tests, method of tests to be conducted and acceptance criteria to be adopted are as under.
2. Method of Sampling:
  - (i) The sampling adopted has to be such that effectiveness of proper compaction having been done for the entire area under consideration can be judged. For this, the Engineer in-charge should lay down the method adopted in detail depending on site conditions and accordingly records of checks done are properly maintained. However, in absence of such procedure laid down, following method should be adopted:
  - (ii) Suggested method of sampling: For each layer, a minimum of one sample at a predetermined interval (in compliance with the requirement stated in next para) along the centerline of the alignment would be taken. The checking points may be staggered to the extent possible.
  - (iii) Frequency of Tests: Density check would be done for every layer of blanket material as per following minimum frequency:
  - (iv) At least one density check for every 30 m length for blanket layers and top one metre of prepared Subgrade/subgrade along the alignment in a staggered pattern of each compacted layer.
  - (v) At least one density check for layers other than as specified in (i) above, every 500 m<sup>2</sup> or 75 m c/c whichever occurs earlier along the alignment in a staggered pattern of each compacted layer.
  - (vi) In case of important bridge approaches (100 m length on either side), at least one density check for every 25 m length shall be adopted.

**33. Method of in-situ dry density measurements:**

Any of the following methods could be adopted as per the requirements at site.

| Method of measurement      | of | Procedure of test             | Parameters to be measured                     | Remarks                                 |
|----------------------------|----|-------------------------------|-----------------------------------------------|-----------------------------------------|
| i) Sand Replacement Method |    | As per IS-2720 (Part 28) 1974 | a) In situ Dry Density<br>b) Moisture content | May be adopted for all type of material |

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For Chief Administrative Officer  
(Const.) West Central Railway,

| Method of measurement                                                           | Procedure of test             | Parameters to be measured                                                           | Remarks                                                                                                                                                                      |
|---------------------------------------------------------------------------------|-------------------------------|-------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ii) Core Cutter Method                                                          | As per IS-2720 (Part 29) 1975 | -do-                                                                                | In some of the coarse-grained soils (with little fines) taking core cutter samples is difficult. In such cases, sand replacement method may be used for density measurement. |
| iii) Nuclear Moisture Density Gauge                                             | As issued by RDSO             | a) Bulk density<br>b) Moisture content<br>c) Dry density<br>d) Degree of compaction | May be used in consultation with RDSO                                                                                                                                        |
| iv) Compactor meters fitted on roller (On roller continuous compaction control) | As issued by RDSO             | As issued by RDSO                                                                   | May be used in consultation with RDSO.                                                                                                                                       |

i. Acceptance Criteria:

- The blanket material, which contains fines passing 75 micron IS Sieve, upto 5 percent should have the Density Index (Relative Density) a minimum of 70% as obtained in accordance with IS: 2720 (Part 14) – 1983.
- For other materials, field dry density should not be less than maximum attainable dry density obtained in field compaction trial. However, in field compaction trial, the maximum attainable dry density should not be less than 98% of MDD values as obtained by Heavy Compaction Test (IS: 2720 (part 8) – 1983) in the laboratory. In case, there are difficulties in achieving 98% of the MDD values as obtained by Laboratory test, in the field trials, the same may be relaxed upto 95% of MDD with the specific approval of Chief Engineer/construction, recording reasons of such relaxation.

ii. Measurement:

- Measurement of blanket material should be done on the basis of finished cross section after the material and workmanship have been accepted as per the above criteria. No deduction is to be made towards voids/shrinkage.
- In very rare cases, where it is not possible to take blanket material on finished sub grade, measurement may be done on the basis of stack measurement with the permission of Chief Engineer in-charge. It may be necessary to frame different schedule items for different methods of measurement. There should be no occasion to change the method of measurement unless specifically provided for in the tender documents duly approved by competent authority.
- It is advisable to tally the quantity and quality measured at site with the 'Incoming Materials Register' maintained at plant as a means of double check.

iii. Side slopes:

- Embankment: Side slopes of blanket layer are to be finished to the same slope as designed for the subgrade (formation layer below the blanket), which may be 2H:1V (minimum) in most cases. However, slope stability analysis and other suitable investigation should be carried out

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in case of high fill (more than 6m) construction or on swampy ground or marshy lands

2. Cuttings: Where blanket is provided in cuttings replacing the parent material, depth of cutting and side drains is to be so adjusted that the filled material forms an embankment with side slopes of 2:1. Pitching of blanket slope may be planned upto 1.0 m height to check erosion due to flowing water. In very deep cuttings, where it is not possible to increase the width of cutting due to the costs involved, it may be cheaper to entrench the blanket material. In such cases, proper drainage is to be planned below the sub-grade level to drain out water from the bottom of blanket layer.
3. For the purpose of determining the suitability of the soil, the blanketing material & the measures to be adopted for the field quality control, the contractor shall extract adequate number of samples of the soil/blanketing material from the nominated borrow areas and cuttings as directed by the Engineer from time to time. The samples shall be extracted in manner specified in the relevant Indian Standard Specifications (ISS) or the Indian Standard Code of practice (ISC).
4. The soil/blanketing material samples mentioned herein above shall be tested as prescribed in the relevant IS Code or IRS to determine.
  - a) The suitability of the soil and the blanketing material for the work.
  - b) The natural dry density of soil in the top 300 mm of the formation in the cuttings.
  - c) For the soil to be used in the embankments and as obtained in the top surface of the cuttings finished to final level, the Maximum Dry Density (MDD) & the Optimum Moisture Content (OMC) shall be obtained from Heavy Compaction in accordance with IS-2720 (Part VIII)
  - d) For the blanketing material, the MDD & OMC as obtained from Heavy Compaction in accordance with IS-2720 (Part VIII).
  - e) Suitable type of Rolling/Compaction equipment.
  - f) The measures to be adopted for Field quality control of compaction.
  - g) Any other parameter of the material characteristics as considered necessary by the Engineer for the proper execution of the work.
5. The decision of the Engineer on the number of samples, their locations, the frequency of the extractions, soil/blanketing material characterization / field control measures to be determined from the testing of the samples, the acceptance and application of the test results to decide if the soil/blanketing material is suitable for the work and determine the quality control measures for the execution of the work etc. shall be final and conclusive.
6. The expenditure on extraction and testing of soil/blanketing material samples (when done through an outside agency, other than Railway, as decided by the Engineer whose decision in this regard being final and conclusive) as mentioned herein above shall be borne by the Contractor and is deemed to have been included in the contracted rates for the work.
7. Before commencing the work and during the progress of the work as and when considered necessary by the Engineer, the method for field control of moisture and compaction will be determined by the Engineer adopting any of the methods specified in IS-10379. All the tests required to be performed for this purpose to determine the field quality control system shall be carried out by the contractor as specified herein above with contractor's equipment's.
8. For earthwork in embankments, after site clearance all pockets and depressions left in the soil, if any shall be made good and compacted.
9. The blanketing work shall cover the entire width of the embankment and/or cutting from shoulder to shoulder. If the material is erodible, it shall be confined in a trench with berms of width as specified by the Engineer. In such cases adequate sand drains as directed by the Engineer shall be provided across the cess.
10. The earthwork in embankments / spreading of the blanketing materials shall be done in layers not exceeding 600 mm thick where vibratory rollers are used, or 300 mm thick in other cases

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or the optimum thickness as determined after carrying out the specified field test as applicable, the thickness of the layers being reckoned in the loose state. Where the compaction is done with sheep-foot rollers, the depth of each layer of loose soil/blanketing material shall not exceed by more than 10 mm the length of their feet. The type of the rolling/compaction equipment to be used in the work, the number of passes required of the rolling/compacting equipment etc. shall be as directed by the Engineer whose decision in this regard shall be final and conclusive.

11. The earthwork in embankments, the formation level surface of cutting and the blanketing work shall be compacted as directed by the Engineer whose decision in this regard shall be final and conclusive. The compaction should preferably be done at the OMC. Depending on site conditions, the Engineer, at his sole discretion may permit the variation in the moisture content at the compaction stage of say (-) 2% with respect to the OMC subject to the acceptance criteria for the field compacted density. The compaction at moisture content higher than the OMC may be permitted by the Engineer at his sole discretion subject to the moisture content being less than the plastic limit for the type of material being used in the embankment/blanketing layers.
12. Where the soil/blanketing material is dry, water shall be sprinkled either in the borrow pit or over the spread layers, as convenient, in order to attain a workable moisture content before rolling/compaction is commenced. Where the natural moisture of the soil/blanketing material in any layer is higher than OMC, soil/blanketing material may be left for drying for a suitable period till the moisture content is reduced to the OMC level or the rolling/compaction commenced under specific permission of the Engineer as provided herein above.
13. The acceptance criteria for the compaction work shall be as under:-
  - a) Coarse grained or cohesion less soil/blanketing material which may contain fines passing 75 microns IS sieve up to 5% for plastic fines and up to 12% for non-plastic fines shall be compacted so as to obtain a minimum Density Index (Relative density of 70% as obtained in accordance with the I.S. 2720 (Part XIV).
  - b) All other types of soils used in the embankments and all types of blanketing materials other than coarse grained blanketing material as specified herein above shall be compacted so as to obtain 98% of the MDD as determined using heavy compaction in accordance with I.S. 2720 (Part VIII) 1983, the actual percentage of the MDD to be obtained within the above range shall be determined by the Engineer based on the field trials carried out in the manner specified herein above.

**V. Mechanical Compaction of side slopes**

Compaction of side slopes will be carried out in accordance with Appendix E of RDSO/2020/GE: IRS-0004.

**Vi. Earth work in cutting:**

For earth work in cutting RDSO Guidelines for cutting in Railway formation No. GE: G-2 shall be followed

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For Chief Administrative Officer  
(Const.) West Central Railway,

**OPEN FOUNDATION**

4.1 **EXCAVATION**

:

After the site has been cleared, the limits of excavation shall be set out true to lines, curves and slopes.

- 4.2 Excavation shall be taken to width at the lowest step of the foundation and the sides shall be left plumb where the nature of soil allows it. Where the nature of soil or the depth of the trench and season of the year do not permit vertical sides, the Contractor at his own expense shall put up necessary shoring, strutting and planking or cut slopes to a safer angle or both with due regard to the safety of personnel and works and to the satisfaction of the Engineer.
- 4.3 The depth to which the excavation is to be carried out shall be as shown on the drawings, unless the type of material encountered is such as to require changes, in which case the depth shall be as ordered by the Engineer.
- 4.4 Where blasting is to be resorted to, the same shall be carried out to prevent any damage. The work has to be carried out with supervision by qualified personnel.
- 4.5 Dewatering and Protection: Open foundations shall be laid dry. If water is met within excavation, the Contractor shall take adequate measures such as bailing, pumping constructing diversion channels, drainage channels, bunds, depression of water level by well point system and other necessary works to keep the foundation trenches dry and to protect the green concrete against damage by erosion or sudden rising of water level.
- 4.6 Pumping from the interior of any foundation enclosure shall be done in such a manner as to preclude the possibility of the movement of water through any fresh concrete. No pumping shall be permitted during the placing of concrete or for any period of at least 24 hours thereafter, unless it is done from a suitable sump separated from the concrete work by a watertight wall or other similar means.
- 4.7 At the discretion of the Contractor, cement grouting or other approved methods may be used to prevent or reduce seepage and to protect the excavation area.
- 4.8 The Contractor shall take all precautions in diverting channels and, in discharging the drained water as not to cause damage to the works or any other property. No separate payment shall be made on this account.
- 4.9 **Preparation of Foundation:** - The bottom of the foundation shall be leveled both longitudinally and transversely or stepped as directed by the Engineer. As rock or other hard strata is being to be encountered, it shall be freed of all soft and loose material, cleaned and cut to a firm surface either level and stepped as directed by the Engineer.
- 4.10 **Backfilling:** Backfilling shall be done with approved material after concrete is fully set and carried out in such a way as not to cause undue thrust on any part of the structure. All space between foundation concrete and the sides of excavation shall be refilled to the original surface in layers not exceeding 150 mm compacted thickness. The compaction shall be done with the help of suitable equipment such as mechanical tamper, rammer, plate vibrator etc., after necessary watering, so as to achieve a density not less than the field density before excavation.
- 4.11 **Disposal of Excavated Materials:** All the excavated materials shall be the property of the Railways. The good/usable material obtained from the excavation of foundation shall be used for back filling of foundation, and the existing pits in the right-of-way and filling in that embankment, including loading of surplus good earth into truck, transporting, unloading, spreading, watering, ramming, leveling, etc. with all lifts and lead as per specifications of Earthwork and as directed by the Engineer. Unsuitable and surplus material not intended for use shall also, if necessary, be transported with all lifts and lead, disposed at specified location as directed by the Engineer and no extra payment shall be admissible.
- 4.12 **Measurement for Payment:** Excavation for structures shall be measured in cum. for all class of material encountered, limited to the dimensions shown on the drawings or as directed by the Engineer. Excavation over increased width, cutting of slopes, shoring, shuttering and planking shall be deemed as convenience for the Contractor in executing the work and shall not be measured and paid. Rate of excavation is inclusive of dewatering & refilling. No separate payment will be made for dewatering and refilling.

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**Chapter- 5**  
**Pile Foundation**

- 5.1 **General:** The construction of pile foundation shall conform to the requirements laid down in IS : 2911 Part – I concrete piles – Section 1 to 3 bored cast in situ piles – 1979 and as amended up to date accepting the criteria specified hereunder. Load test on piles shall be carried out as per IS : 2911 (partIV ) 1985.
- 5.2 The pile shall be cast in situ RCC piles. The rate for pile foundation includes building up of earth filled island platform, drilling/boring for all piles through all type of soils including soft and hard rocks, boulder and casting of piles. The payment will be made from cut off level to foundation level.
- i) **Boring of Piles:** Only hydraulic rotary rigs shall be used for boring of RCC caste in situ piles Bored cast-in-situ piles; manner of concreting only by tremie and choice of boring tools in orderto permit a satisfactory installation of a pile. The pile shall be installed as accurately as possibleas per the designs and drawings.
    - (a) Diameters of the piles shall be the concrete shaft diameters and shall not be less than the diameters specified in the drawing.
    - (b) These shall be formed by boring to the founding strata specified on the drawings or as directed on the site. The sides of the boring shall be prevented from collapsing by temporary mild steel liner.
    - (c) Piles shall be constructed in a sequence approved by the Engineer. During boring, the contractor shall, where required by the Engineer, take soil, rock or ground water samples and transport them to an approved testing laboratory or carry out soil tests as directed.
    - (d) The method adopted shall be chosen giving due considerations to the subsoil data, ground water conditions and to the other relevant.
    - (e) The bottom of the steel liner shall be sufficiently in advance of the boring tool so as to prevent settlement of outside soil and formation of cavities.
    - (f) Mild steel liners shall continue up to soft rock or hard rock whichever is struck earlier.
    - (g) Boring shall be done using hydraulic rotary drilling rigs suiting to different kinds of strata encountered.
    - (h) The size of the cutting tool shall not be less than the diameter of the pile minus 75mm.
    - (i) The quantum of liners required up to depth of cut off level shall be measured as per drawing.
    - (j) Pumping from a borehole shall not be permitted unless a liner has been driven into a stable stratum which prevents flow of external ground water from other strata in significant quantities.
    - (k) In case of end bearing piles founded on rock the piles shall be embedded by minimum one pile diameter to three times the pile diameter depending on the condition of the rock or as specifiedin the drawing. Rock samples of the founding strata shall be preserved at site.
  - ii) **Inspection:** Each pile boring shall be inspected prior to the placing of concrete in it. Where the diameter of a dry boring is 750 mm or greater, equipment approved by the Engineer shall be provided to enable the contractor and the Engineer to descent into the boring for the purpose ofinspection.
  - iii) **Cleanliness of Pile Bases:** On completion of boring, disturbed or remolded soil shall be removedfrom the base of the piles flushing with tremie pipe clean water.
  - iv) **Concreting**
    - a) Concreting of boreholes shall start as soon as possible after the boring of piles. Bore hole

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which is not cased, if left un concreted for more than two hours, it shall be cleaned thoroughly before placing concrete. Concreting under water shall be done in one operation. The method of placing and the workability of the concrete shall be such that continuous monolithic concrete shaft of the full cross section is formed.

- b) The concrete shall be placed without such interruption as would allow the previously placed batch to have hardened. The engineer shall approve the method of placing.
- c) The contractor shall take all precautions in the design of mix and placing of the concrete to avoid arching of the concrete in the casting. No spoil, liquid or other foreign matter shall be allowed to contaminate the concrete.
- d) Placing concrete in dry holes (borings) Approved measures shall be taken to avoid segregation and bleeding and that the concrete at the piles is not deficient in grout.
- e) Concreting of full pile has to be carried out uninterrupted and in one operation. If the concreting is interrupted in between for any reasons, sufficient portion from top surface of the partly concreted pile containing laitance, weak concrete etc. shall be removed so as to obtain a sound concrete surface, which should be cleaned by sand blasting. The engineer shall start further concreting only after inspections and approval of cleaned concrete surface. Precautions as detailed in clause 7.5 IS 2911 (pt. I/Sec.2)-1979 shall be taken in tremie concreting.
- f) Whenever practicable the concrete should be placed in clean dry hole. Concreting of bored piles in water bearing ground shall be done with utmost care/especially when the hole cannot be bailed of water or to be made dry before tremie concreting. Under such circumstances plug of concrete should be deposited under water in base of the hole by tremie pile, as soon as the concrete has hardened sufficiently the hole should be pumped free of water and the concreting of the shaft shall be done in dry by tremie method up to surface. In exceptional circumstances where concreting in the dry is not possible dewatering should be done to the extent possible before resorting to concreting using a tremie pipe.
- g) Prolonged delays in the commencement after the completion of the boring shall not be permitted. The time interval between the completion of boring and placing of concrete shall not exceed 6 hours.
- h) The concrete shall have a minimum slump of 150mm in case of concreting in water free bore. Suitable precautions shall be taken for prevention of segregation. Internal vibrators shall not be used unless the contractor is satisfied that segregation will not result because of vibration and unless the method of use has been approved by the Engineer.
- i) The concrete for piles under water or in drilling mud shall be placed with a Tremie pipe. The Tremie pipe shall not be less than 200 mm diameter for 20mm aggregate. The joint between the hopper and Tremie pipe as well as the joints in the Tremie pipe shall be watertight and the Tremie pipes shall be thoroughly cleaned after each use. The concrete shall have a minimum slump of 150mm.
- j) The contractor shall ensure that heavily contaminated drilling mud has not accumulated at the base of boring since this could impair free flow of concrete from the Tremie pipe.
- k) If the specific gravity of the drilling mud at the base of the bore exceeds 1.2, the placing of concrete shall not proceed.
- l) The first charge of concrete shall be placed in the hopper over a sliding plate of the bottom of the hopper.
- m) The Tremie pipe shall at all times penetrate the previously placed concrete with an adequate margin against accidental withdrawal. The Tremie pipe shall not be withdrawn until the completion of concreting. At all times a sufficient quantity of concrete shall be maintained within the pipe to ensure that the pressure from it exceeds that from the seepage water.

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- n) Spot measurements shall be taken at suitable intervals to check that the Tremie pipe has an adequate penetration into previous concrete.
- o) Concreting of the pile shall be in one single and continuous operation. In case of long piles of large diameter, large size mixers or more mixers shall be used.
- p) The top of concrete in a pile shall be brought above the cut off level since the top concrete is loose and is weak because of contamination with water/drilling mud. This ensures good concrete at the cut off level.
- v. **Disposal of Mud:** The borehole should be maintained full with the drilling fluid, where used throughout the concreting operation. Mud displaced from the borehole by the concrete shall be channeled away or pumped into suitable receptacles for disposal to waste or re use.
- vi. **Control of Alignment:** The pile shall be positioned as accurately as possible to the vertical. Any pile deviating from its proper alignment to such an extent that the resulting eccentricity cannot be taken care of by strengthening the foundation, pile cap or pile ties shall at the discretion of the engineer in charge, be replaced or supplemented by an additional pile, the cost of which shall be borne by the contractor. As a general guide, the permissible positional deviation for piles shall not be greater than 7.5 cm at the level of bottom of pile cap and shall not exceed 1.5 percent (about 1) from the specified inclination.
- vii. **Cut off Level:** The concrete shall be brought out sufficiently above the cut off level (minimum 600 mm) as directed by the engineer to permit all laitance and weak concrete to be removed. Reinforcement from piles shall project out sufficiently to achieve adequate anchorage in to the pile cap. Cut off level of piles shall be indicated in working drawings or as indicated by Engineer In Charge. The top of concrete in pile shall be brought above the cut off level to remove all laitance and weak concrete and to ensure good concrete at cut off level. As per the guideline, for cut off level up to 1.5m below working level, the concrete shall be cast 300mm above COL. For each additional 0.3 m increase in depth of COL an additional coverage of 50mm shall be required. In the circumstances where COL is below ground water level, the need to maintain a pressure should be observed and accordingly length of extra concrete above COL shall be determined by the contractor and approval of Engineer In Charge obtained before concreting. Nothing extra shall be paid for pile chipping above the COL. It is deemed to be included in the quoted rates.
- viii. **Setting Out:** The contractor shall check the casing position for each pile.
- ix. **Defective Piles:** In case defective piles are formed, they shall be removed or left in place which ever convenient without affecting the performance of adjacent pile or the cap as a whole. Additional piles shall be provided to replace them as suggested by the engineer at site at the cost of the contractor.
  - a Depth of foundation will be reckoned with respect to average founding level of piles.
  - b Ring beam having dimensions as shown in the approved drawing shall be provided over the pile. The reinforcement from the pile shall be well anchored into the ring beam.
- x) **Piling Programme:** The contractor shall inform the engineer each day of the programme of piling for the following day and shall give adequate notice of his intention to work outside normal hours at weekends.
- xi) **Records:** The contractor shall keep records in a tabulated form as given in below and shall submit to two signed copies of these records to the engineer on the next working day after the pile was installed.
- xii) **Damage to Piles:** The contractor shall ensure that the damage does not occur to completed piles. The contractor shall submit to the engineer his proposed sequence and timing for driving or boring piles having regard to the avoidance of damage to adjacent piles.

For each pile boring bore log will be plotted in pile register showing the variations in soil strata. The

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soil samples can be kept in plastic bottle indicating pile No. & level of boring.

Details of pile

Pile reference No. \_\_\_\_\_

1. Nominal cross sectional dimensions or diameter.
2. Standing ground water level.
3. Date and time of boring
4. Date of concreting
5. Length of pile
6. Soil samples taken and in situ test carried out
7. Length and details of reinforcement
8. Concrete Mix
9. Volume of concrete supplied to pile where this can be measured in practice.
10. Length of socketing in rocks.
11. Bottom level (founding level ) of pile, Cut off level of pile.

**5.3 Payment of the pile:**

Length of the pile will be paid from bottom of founding level to the bottom of pile cap as per the approved drawing. Nothing extra will be paid for empty boring. Rate is inclusive of dismantling/chipping of pile head up to cut off level, and no extra payment will be made.

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## Chapter-6 BEARINGS

### 6.1 Elastomeric Bearing

6.2 Elastomeric bearings shall be used as per approved dimensions and materials as specified in technical specification.

6.3 In this bearing system individual girders are supported on elastomeric bearings.

6.4 The elastomeric bearings shall have the following specification:

#### 6.5 Raw material

Chloroprene (CR) only shall be used in the manufacture of bearing. Grades of raw elastomer of proven use in elastomeric bearings, with low crystallization rates and adequate shelf life (e.g. Neoprene WRT, Bayprene 110, Skyprene B-5 and Denka S-40V) shall be used. No reclaimed rubber or vulcanised wastes or natural rubber shall be used. The raw elastomer content of the compound shall not be lower than 60 percent by its weight. The ash content shall not exceed 5 per cent (as per tests conducted in accordance with ASTM D-297, sub-section 10). EPDM and other similar candidate elastomer, for bridge bearing use, shall not be permitted.

Properties: The elastomer shall conform to the properties specified in Table 1.

Table 1. Properties of Elastomer

| Properties                         | Unit     | Test Method, IS specification reference | Value of the characteristic specified                           |
|------------------------------------|----------|-----------------------------------------|-----------------------------------------------------------------|
| 1. Physical Properties             |          |                                         |                                                                 |
| 1.1 Hardness                       | IRHD     | IS: 3400 (Part II)                      | 60 ± 5                                                          |
| 1.2 Minimum Tensile Strength       | Mpa      | IS: 3400 (Part I)                       | 17                                                              |
| 1.3 Minimum Elongation at break    | Per cent | IS: 3400 (Part I)                       | 400                                                             |
| 2. Maximum Compression Set         | Per cent | IS: 3400 (Part X) duration              | temperature                                                     |
|                                    |          | (h) +0 to 24.2                          | (deg. C) 100 ± 1                                                |
| 3. Accelerated Ageing              | CR       | IS: 3400 duration                       | temperature                                                     |
|                                    |          | (h) 70                                  | (deg. C) 100 ± 1                                                |
| 3.1 Max change in Tensile Strength | IRHD     |                                         | + 15                                                            |
| 3.2 Max change in Tensile Strength | Per cent |                                         | For Chief Administrative Officer (Const.) West Central Railway, |

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|     |               |          |      |
|-----|---------------|----------|------|
| 3.3 | Max change in | Per cent | - 40 |
|-----|---------------|----------|------|

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Elongation

Shear modulus of the elastomer bearing shall neither be less than 0.80 Mpa nor greater than 1.0 Mpa.

- (i) The adhesion strength of elastomer to steel plates determined according to IS: 3400 (Part XIV) method shall not be less than 7 kN/m.
- (ii) For elastomeric bearings (CR) used in adverse climatic conditions the following ozone resistance test shall be satisfied :
- (iii) The ozone resistance of elastomer shall be proved satisfactory when assessed by test according to IS: 3400 (Part XX). The strain, temperature, duration and ozone concentration of the test shall be 20 per cent,  $40 \pm 1$  degree Celsius, 96 h and 50 pphm by volume respectively.
- (iv) No cracking detected by visual observation at the end of the test shall be considered satisfactory. No specific tests for assessment of low temperature resistance may be deemed necessary.
- (v) Laminates of mild steel conforming to IS: 226 shall only be permitted to be used. Use of any other material like fibre glass or similar fabric as laminates shall not be permitted.
- (vi) The manufacturers of elastomeric bearings shall satisfy the Engineer that they have in-house facilities for testing the elastomer for carrying out the following tests in accordance with the relevant provisions of ASTM D-297:-

|    |                            |                                                            |
|----|----------------------------|------------------------------------------------------------|
| a) | Identification of polymers | To confirm the usage of Chloroprene (Appendix X-2 of ASTM) |
| b) | Ash content test           | To determine the percentage (sub-section 34 of ASTM)       |
| c) | Specific gravity test      | (sub-section 15 of ASTM)                                   |
| d) | Polymer content test       | (sub-section 10 of ASTM)                                   |

- (vii) The Engineer shall invariably get the test (a) performed within his presence or in the presence of his authorised representative to satisfy the requirement. In case of any disputes regarding interpretation of results the Engineer may carry out test as per ASTM S-3452-78 (Chromatography test) at the manufacturer's cost in a recognised test laboratory.
- (viii) The elastomer specimen to conduct the test shall be obtained from the bearings selected at random for destructive test. Remaining part of the test bearing shall be preserved by the Engineer for any test to be done in future, if required.

**6.6 Fabrication**

- (i) Bearing shall be cast as a single unit in a mould and vulcanised under heat and pressure. (ii) Casting of elements in separate units and subsequent bonding shall not be permitted, nor shall cutting from large size cast be permitted.
- (iii) Bearings of similar size to be used in particular bridge project shall be produced by identical process and in one lot as far as practicable. Phased production may only be resorted to when the total number of bearings is large enough.
- (iv) The mould used shall have standard surface finish adequate to produce bearings free from any surface blemishes.
- (v) Steel plates for laminates shall be sand blasted, clean of all mill scales and shall be free from all contaminants prior to bonding by vulcanisation. Rusted plates with pitting shall not be used. All edges of plates shall be rounded.

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- (vi) Spacers used in mould to ensure cover and location of laminates shall be maximum size and number practicable. Any hole at surface or in edge cover shall be filled in subsequently.
- (vii) Care shall be taken to ensure uniform vulcanizing conditions and homogeneity of elastomer through the surface and body of bearings.
- (viii) The bearings shall be fabricated with the tolerances specified in Table 2.

Table 2.

|     | <u>Items</u>                                                          | <u>Tolerances</u>           |
|-----|-----------------------------------------------------------------------|-----------------------------|
| 1.  | Overall plan dimensions                                               | -0, + 6 mm                  |
| 2.  | Total bearing thickness                                               | -0, + 5 %                   |
| 3.  | Parallelism                                                           |                             |
| a)  | Of top surface of bearing with respect to the bottom surface as datum | 1 in 200                    |
| b)  | Of one side surface with respect to the other as datum                | 1 in 100                    |
| 4a. | Thickness of individual internal layer of elastomer                   | ± 20 percent (max. of 2 mm) |
| b)  | Thickness of individual outer layer                                   | -0, + 1 mm                  |
| 5a. | Plan dimensions of laminates                                          | - 3 mm, + 0                 |
| b)  | Thickness of laminates                                                | ± 10 per cent               |
| c)  | Parallelism of laminate with respect to bearing base as datum         | 1 in 100                    |

- (ix) The vulcanising equipment/press shall be such that between the platens of press the pressure and temperature are uniform and capable of being maintained at constant values as required for effecting a uniform vulcanisation of the bearing.
- (x) The moulding dies utilised for manufacturing the bearings shall be so set inside the platen of the press, that the pressure developed during vulcanisation of the product is evenly distributed and the thickness maintained at all places are within acceptable tolerance limits taking into consideration the shrinkage allowance of vulcanizate.
- (xi) The raw compound which has been introduced inside the metal dies for vulcanisation shall be accurately weighed each time and it must be ensured that sufficient quantity has been put inside the die for proper flow of material at every place so that a homogeneous and compact bearing is produced without any sign of sponginess or deficiency of material at any place.
- (xii) Before any vulcanizate of any batch of production is used for producing vulcanised bearings, test pieces in the form of standard slab and buttons shall be prepared in accordance with prescribed standards and salient properties tested and recorded regularly against each batch of production to monitor the quality of the products.

**6.7 Acceptance Specifications**

- (i) The manufacturer shall have all the test facilities required for the process and acceptance control tests installed at his plant to the complete satisfaction of the Engineer. The test facilities and their operation shall be open to inspection by the Engineer on demand.
- (ii) All acceptance and process control tests shall be conducted at the manufacturer's plant. Cost of all materials, equipment and labour shall be borne by the manufacturer unless otherwise specified or specially agreed to between the manufacturer and Engineer-in- Charge.
- (iii) Any acceptance testing delayed beyond 180 days of production shall require special approval of the Engineer and modified acceptance specification, if deemed necessary by him.

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- (iv) All acceptance testing shall be conducted by the Inspector with aid of the personnel having adequate expertise and experience in rubber testing provided by the manufacturer, working under the supervision of the Inspector and to his complete satisfaction.
- (v) Lot by lot inspection and acceptance shall be made.

**6.8 Acceptance lot**

- (i) A lot under acceptance shall comprise all bearings, including the pair of extra test bearings where applicable of equal or near equal size produced under identical conditions of manufacture to be supplied for a particular project.
- (ii) The size and composition of acceptance lot shall be got approved by the Engineer.
- (iii) For the purpose of grading levels of acceptance, testing lots shall be classified as follows:
  - (a) A lot size of 100 or larger number of bearings shall be defined as an extra large lot.
  - (b) A lot size of 20 to 100 bearings shall be defined as a large lot.
  - (c) A lot size of less than 20 bearing shall be defined as a small lot.

**6.9 Levels of acceptance inspection**

- 1. The level of acceptance testing shall be based on number of bearing manufactured in one lot. This shall involve manufacture of extra bearing/s for each lot to be used as test bearing and eventually consumed in destructive testing. The cost of extra bearings shall be borne by the contractor.

**6.10 Testing**

- 1. Acceptance testing shall comprise of general inspection, test on specially moulded test pieces and test on complete bearings or sections for measurement of various quality characteristics detailed below:

**6.11 Acceptance testing level**

**1 General Inspection**

- 1. All bearings of the lot shall be visually inspected for any defects in surface finish, shape or any other disassemble superficial defects.
- 2. All bearings of the lot shall be checked for tolerances specified in Table 2.
- 3. All bearings of the lot shall be subjected axial load to correspond to  $\sigma_m$  (i.e. average compressive stress) = 15 MPa applied in steps and held constant while visual examination is made to check for discernible defects like:
  - a) Misalignment of reinforcing plates.
  - b) Poor bond at laminate/steel interface.
  - c) Variation in thickness of elastomer layers.
  - d) Any surface defects.
  - e) Low stiffness
- 4. Deflection under loads between  $\sigma_m = 5$  MPa and  $\sigma_m = 15$  MPa shall be measured and recorded for all bearings with sufficient accuracy (+ 5 per cent), Variation in stiffness of any individual bearing from the mean of the measured values for all such bearings of the lot shall not be larger than 20 per cent (of the mean value).
- 5. Test on specially moulded test pieces
  - (a) Test pieces shall be moulded by the manufacturer with identical compound and under identical vulcanising conditions as used in the manufacture of the bearings of the acceptance, lot. The process shall be open to inspection by the Engineer.
  - (b) Test pieces offered for inspection shall be identified by suitable markings and duly certified by the manufacturer.

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- (c) The quality characteristics to be tested are listed below. The specification references in parenthesis shall define the corresponding specification for test piece, test method and criterion for acceptance.

Composition (see note 1 below)  
Hardness (Table-1, 1.1)  
Tensile strength (Table-1, 1.2)  
Elongation at Break (Table-1, 1.3)  
Compression Set (Table- 1, 2)  
Accelerated Ageing (Table-1, 3)  
Adhesion strength(Clause 10.2.2.1)  
Ozone Resistance (see Note 2 below)

**Note 1.**

For acceptance testing the properties enumerated in Clause 10.2.2.1 and specific gravity of elastomer of test pieces from test bearing shall be compared with those for corresponding specially moulded test pieces furnished by the manufacturer. The following variations shall be deemed maximum acceptable:

Specific Gravity  $\pm 2$   
Ash Content  $\pm 0.5$  per cent  
Hardness (Table-1, 1.1)  
Tensile strength (Table-1, 1.2)  
Elongation at Break (Table-1, 1.3)  
Compression Set (Table-1, 2)  
Accelerated Ageing (Table-1,3)  
Adhesion strength (Clause 10.2.2.1)

**Note 2.**

Ozone resistance test can be waived by the Engineer for bearings of CR when satisfactory results of ozone resistance tests on similar grade of elastomer may be available from process control records or development test data furnished by the manufacturer.

Where such process control data are not available or the frequency of testing not deemed adequate, ozone resistance test shall be mandatory for acceptance of bearings of CR.

Process and acceptance control tests for ozone resistance by an independent testing agency shall be acceptable.

**6.12 Tests on Complete Bearings or Sections: -**

1. One bearing shall be selected at random from the lot as test bearings. In place of complete bearing, manufacturer can be allowed to fabricate smaller size test piece in similar conditions for destructive and adhesion testing. Decision of Engineer in charge will be final and binding. This bearing/s shall be excluded from the lot accepted.
2. The following tests shall be conducted on test bearings:
  - (a) Determination of vertical stress/strain curve
  - (b) Determination of modulus G.
  - (c) Determination of ultimate strength in compression.
  - (d) Determination of adhesion strength between elastomer & plate.
  - (e) The test specifications and acceptance criteria shall conform to those given in UIC 772 R.

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**6.13** Inspection certificate A lot under inspection shall be accepted by the Inspector and so certified, when no defect is found with respect to any of the quality characteristics tested on samples drawn from the lot according to specifications laid down in Clause 10.2.2.6 covering general inspection tests on specially moulded test pieces and on complete bearings.

- (a) In case of any bearing with defect, the lot shall be rejected by the Inspector and so certified.
- (b) In case any bearing is found to be defective with respect to any quality characteristic, discerned by general inspection test specified in Clause.
- (c) 2.6 shall nevertheless be completed. If the said lot, rejected by general inspection, satisfies the acceptance criteria in respect of these other tests, the lot and individual bearings found defective shall be clearly identified in the inspection certificate.
- (d) The manufacturer shall obtain from the inspector, authorised by the Engineer-in-Charge, immediately on completion of his inspection, an inspection certificate which shall include the details of a lot or lots accepted/rejected by him and records of all test measurements.

**6.14** Quality control certificate

1. The manufacturer shall certify for each lot of bearing under acceptance:
2. That an adequate system of continuous quality control was operated in his plant.
3. That the entire process remained in control during the production of the lot of bearings under acceptance as verified from the quality control records/charts, which shall be open to inspection of Engineer/Inspector on demand.
4. A certified copy of results of process control testing done on samples of elastomer used in the production of the lot shall be appended and shall include at least the following information:
5. Composition of compound-raw elastomer and ash content, the grade of raw elastomer used (including name, source, age on shelf), test results of hardness, tensile strength, elongation at break, compression set, accelerated ageing, etc.

**6.15** Acceptance

The manufacturer shall furnish the following to Engineer for the acceptance judgment:

1. Quality control certificate as laid down in Clause 10.2.2.8.
2. Inspection certificate as laid down in Clause 10.2.2.7.
3. The manufacturer shall furnish any supplementary information on the system of quality control and/or process and acceptance control testing as may be deemed necessary by the Engineer.
4. In case of any evidence of process or acceptance control testing being deemed unsatisfactory by him, Engineer at his sole discretion may call for a special acceptance of the lot according to specifications laid down by him, without any prejudice to his right to reject the lot. The entire cost of such supplementary inspection shall be borne by the manufacturer.
5. The Engineer shall be the sole authority for acceptance of a lot on scrutiny of the certificates along with any supplementary evidence and complete satisfaction therewith.
6. In case of rejection of a lot, the Engineer shall reserve the right to call for special acceptance inspection for the succeeding lots offered for inspection, according to the specifications laid down by him. The entire cost of such tightened inspection shall be borne by the manufacturer.

**6.16** Certification and Marking

1. Bearing shall be transported to bridge site after final acceptance by Engineer and shall be accompanied by an authenticated copy of the certificate to that effect.

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2. An information card giving the following details for the bearings, duly certified by the manufacturer shall also be appended:  
Name of manufacturer.  
Date of manufacture.  
Elastomer grade used.  
Bearing dimensions. Production batch no. Acceptance lot no.  
Date of testing.  
Specific bridge location, if any.  
Explanation of markings used on the bearing.
3. All bearings shall have suitable index markings identifying the information. The markings shall be made in indelible ink or flexible paint and if practicable should be visible after installation. The top of the bearing and direction of installation shall be indicated.

**6.17 Storage and Handling**

1. Each elastomeric bearing shall be clearly labelled or marked. The bearing shall be wrapped in a cover. They shall be packed in timber crates with suitable arrangement to prevent movement and to protect corners and edges.
2. Care shall be taken to avoid mechanical damage, contamination with oil, grease and dirt, undue exposure to sunlight and weather to the bearings during transport and handling prior to and during installation.

**6.18 Installation**

1. Installation of multiple bearings one behind the other on a single line of support shall be of identical dimensions.
2. Bearings must be placed between true horizontal surfaces (maximum tolerance 0.2 per cent perpendicular to the load) and at true plan position of their control lines marked on receiving surfaces (maximum tolerance + 1 mm in height).
3. Concrete surfaces shall be free from local irregularities (maximum tolerance + 1 mm in height).
4. For cast-in place concrete construction of superstructure, where bearings are installed prior to its concreting, the forms around the bearings shall be soft enough for easy removal. Forms shall also fit the bearings snugly and prevent any leakage of mortar grout. Any mortar contaminating the bearings during concreting shall be completely removed before setting.
5. For precast concrete superstructure elements, fixing of bearing to them may be done by application of epoxy resin adhesive to interface, after specified surface preparation. The specifications for adhesive material, workmanship and control shall be approved by the Engineer. Care shall be taken to guard against faulty application and consequent behaviour of the adhesive layer as lubricant. The bonding by the adhesive shall be deemed effective only as a device for installation and shall not be deemed to secure bearings against displacement for the purpose of design.

**6.19 Inspection Of Bearings**: - The bearings should be manufactured from RDSO approved firms, and inspected by Railway's representative.

- (a) Measurement for payment:- Bearings shall be measured in volume of finished size of bearing as per approved drawing. Rate shall be inclusive of cost of all material, labour, tools, equipments, testing (Raw material as well as complete finished product), transportation and installation of bearings with all lead & lift.

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**Chapter-7**  
**STRUCTURAL STEEL**

1.0 Description

- a) This section is for general structural steel work required for any of the schedules. It shall include supply, fabrication, transportation, erection and painting structural steel, rivet steel, cast steel, steel forgings, cast iron and other incidental metal construction of the kind, size and quantity in conformity with drawings, specifications and as desired by Engineer.
- b) All steel works whether in fabrication, transportation or erection shall be done in accordance with relevant IS and IRS specifications and codes, subject to any further provisions contained in these specifications. In case of any variance, these specifications shall prevail.

**2.0 General:**

- a. General requirements relating to supply of material shall conform to specification of IS: 1387, for the purpose of which the supplier shall be the contractor and the purchaser shall be the Engineer.
- b. Finished rolled material shall be free from cracks, flaws, injurious seams, laps, blisters, ragged and imperfect edges and other defects. It shall have a smooth, uniform finish, and shall be straightened in the mill before shipment. They shall also be free from loose mill scale, rust, pitting or other defects affecting strength and durability.
- c. The acceptance of any material on inspection at the mill, i.e., rolling mills, foundry or fabricating plant where material for the work is manufactured, shall not be a bar to its subsequent rejection, if found defective.
- d. All steel materials, plates, bars and structural shall have straight edges, flat surfaces and be free from twist. If necessary, they shall be cold straightened or flattened by pressure before being worked or assembled unless they are required to be of curvilinear form. Even formation of curve should be by cold working. Pressure applied should be such as not to injure the material. Adjacent surfaces or edges shall be in close contact or at uniform distance throughout.
- e. Unless specified otherwise, high tensile steel rivet conforming to IS: 1149 shall be used for members of high tensile steel conforming to IS: 961 and shall not be used for mild steel members.
- f. The use of cast steel shall be limited to bearings and other similar parts. Generally all steel pins (including knuckle pins) and rollers shall be of forged steel.
- g. Cast iron shall not be used in any portion of a steel structure, except where it is subject to direct compression.

**3.0 Materials:**

All materials shall conform to relevant Clause of tender documents. Special requirements are given below:-

- a) Mild steel for bolts and nuts shall conform to IS: 2062 but have a minimum tensile strength of 44 Kg/Sqm. and minimum percentage elongation of 14. High tensile steel for bolts and nuts shall conform to IS: 961 but with a minimum tensile strength of 58 Kg/Sq.m. High strength friction

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gripbolts shall be permitted for use only on satisfactory evidence of performance to the requirements of BS. ASTM, DIN or other equivalent specifications as specified by Engineer or included inspecial provisions.

- b) For cast steel the yield stress shall be determined and shall not be less than 50 percent of minimum tensile strength.
- c) Plain washers shall be of steel. Tapered or other specially shaped washers shall be of steel or malleable cast iron.
- d) Parallel barrel drifts shall have a tensile strength of not less than 55 Kg/sq. mm with elongation of not less than 20 per cent measured on a gauge length of 4 So (So= cross sectional area).
- e) Copper alloys, mechanite or special steels used for bearings or similar other parts shall conform to requirements specified by Engineer or included in special provision.
- f) Mild Steel electrodes shall conform to IS: 814 and those of high tensile steel shall conform to IS: 1442.
- g) All paints and enamels used shall conform to requirements specified on drawings or other special provisions laid down by Engineer. Unless otherwise specified, paints shall conform to relevant IS specification.

**4.0 Fabrication**

**1. General**

- a) All work shall be in accordance with drawings and as specified in these specifications, care being taken that all parts of an assembly fit accurately together. These shall form the provisions of relevant IS codes supplemented and prevailed by IRS specifications B-1-62, B-3-61, Parts 2 and 4 and IRS steel Bridge Code.
- b) Normally for repetitive works like bridge girders, in industrial structures and sheds when different components are built in units for assembling at site, similar units will be required to be made interchangeable.
- c) Unless specially required under the contract, corresponding parts need not be interchangeable but the parts shall be match marked as required under relevant Sub-Clause.
- d) Templates, jigs and other appliances used for ensuring accuracy of work shall be of mild steel; where specially required, these shall be bushed with hard steel. All measurements shall be made by means of steel tape or other device properly calibrated. Where bridge materials have been used as templates for drilling, these shall be inspected and passed by Engineer before they are used in the finished structure.
- e) All structural steel members and parts shall have straight edges and blunt surfaces. If necessary, they shall be straightened or flattened by pressure unless they are required to be of curvilinear forms. They shall also be free from twist. Pressure applied for straightening or flattening shallbe such as would not injure the materials. Adjacent surfaces or edges shall be in close contact or at uniform distance or edges shall be in close contact or at uniform distance throughout.

**5. Preparation of Edges and Ends**

- 1. All structural steel parts, where required shall be sheared, cropped, sawn or flame cut and groundaccurately to required dimensions and shape.
- 2. In the case of high tensile steel at least 6 mm of material from the flame-cut edge shall beremoved by machining.

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3. Longitudinal edges of all plates and cover plates in plate girders and built-up members shall be machined except in the following areas:
  - a. Rolled edges of single universal plates or flats may not be machined.
  - b. Covers to single flange plates may be left unmachined.
  - c. Machine flame cutting instead of machining is acceptable for edges of single plates in compression and for edges of single plate, 25 mm or less thick, in tension.
  - d. Edges of single shaped plates over 25 mm thick not capable of being machined by ordinary method may be machine flame cut and the end surface ground.
  - e. Edges of universal plates or flats of same nominal width used in tiers may be left unmachined, if so authorized by Engineer.
4. All edges of splice and gusset plates 12 mm thick and over shall be machined and those less than 12 mm thick may be sheared and ground.
5. The ends of plates and sections forming main components of plate girders or of built-up members shall be machined, machine flame cut, sawn, or hand flame cut and ground.
6. In joints and splices of compression members, in girder flanges and in tension members where so specified on drawings, the abutting surfaces shall be faced and brought to an even bearing. A tolerance of 0.5 mm may be permitted locally. Where close fitting is not specified, any clearance shall not exceed 3 mm.
7. Where ends of stiffeners are required to be fitted, they shall be machined, machine flame cut, sawn, sheared and ground, or hand flame cut and ground. The ends of lacing bar shall be rounded unless otherwise required. Other edges and ends of mild steel parts may be sheared and any burrs at edges shall be removed.

#### Preparation of Holes

### **7. Drilling and Sub-punching**

- a) All holes for rivets shall be drilled or drilled in smaller diameter and reamed. However, if preferred, the holes may be sub-punched to a diameter of 6 mm less than the finished size and then reamed to the proper size. Sub-punching is not permitted for truss members of open web girders.
- b) Where several plates or sections form a compound member, they shall, where practicable, be firmly connected together by clamps or tacking bolts and holes drilled through the group in one operation or alternatively, in the case of repetition work plates and sections may be drilled separately using jigs and templates. The jigs and templates shall be checked at least once after 25 operations. All burrs shall be removed.
- c) Forming holes by flame cutting is severely forbidden. Holes finally formed for rivets and bolts shall be 1.5 mm larger in diameter than that of bar or bolt. Care shall be taken not to exceed this limit. In case of holes for turned bolts they shall be drilled to tolerances as per IS or IRS specification B-1 clause 29.5.
- d) In the case of repetition of spans, the erection of every span shall not be insisted upon, except where close tolerance or turned bolts are used, provided that acceptable methods are adopted to ensure strict interchangeability. In such cases, one span in ten or any number less than ten of each type shall be

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erected from pieces selected at random by Engineer and should there be any failure of piece to fit, all similar spans shall be erected complete. In the event of the spans being proved completely interchangeable, all corresponding parts shall carry the same mark so that sorting of materials at site is facilitated.

**8. Block Drilling**

Where the number of plates to be riveted exceeds three or the total thickness is 90 mm or more, the rivet holes, unless they have been drilled through steel bushed jigs, shall be drilled out in place 3 mm all round after assembling. In such cases, the work shall be thoroughly bolted together.

**9. Size of Holes**

The sizes of holes in millimetre are given in the following table. (Table 800.1). (TABLE 800.1) DIAMETER OF HOLES FOR RIVETS

| <u>Nominal dia. of Rivets</u><br><u>(mm)</u> | <u>Dia. of Holes</u><br><u>(mm)</u> |
|----------------------------------------------|-------------------------------------|
| 12                                           | 13.5                                |
| 14                                           | 15.5                                |
| 16                                           | 17.5                                |
| 18                                           | 19.5                                |
| 20                                           | 21.5                                |
| 22                                           | 23.5                                |
| 24                                           | 25.5                                |
| 27                                           | 29.0                                |
| 30                                           | 32.0                                |
| 33                                           | 35.0                                |

**1. Close tolerance bolts and barrel bolts**

- (a) The diameter of holes shall be equal to nominal diameter of the bolt shank or barrel minus 0.15 mm to 0.0 mm.
- (b) The holes not drilled through all thicknesses at one operation shall be drilled to smaller size and reamed after assembly.

**2. Rivet and Riveting**

- a) The diameter of rivets shown on drawings shall be the size before heating. Each rivet shall be of sufficient length so as to fill in the hole thoroughly and to form a head of standard dimensions as given in Indian Standard Handbook on Steel Sections, Part I. It shall be free from burrs on the underside of the head. Table 800.1 gives the sizes of holes which are generally found sufficient.
- b) When countersunk heads are required, the heads shall fill the countersunk. The included angle of the head shall be as follows:
  - a. for plates over 14 mm thick: 90 degrees
  - b. For plate's up to and including 14 mm thick: 120 degrees
- c) The tolerance on diameter of rivets shall be in accordance with IS: 1148 and IS: 1149 for mild steel rivets and high tensile steel rivets respectively and unless otherwise specified the tolerance shall be minus tolerance.

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- d) Rivets shall be heated uniformly to a "light cherry red" and shall be red hot from head to the point when inserted and shall be upset in its entire length so as to fill the hole as completely as possible when hot. In no case shall tip of the rivet be hotter than the head. Rivets after being heated and before being inserted in the hole shall be made free from scale by striking the hot rivet on a hard surface. Rivets less than 10 mm in diameter may be driven cold.
- e) The rivets shall be machine driven, preferably by direct acting riveters. The driving pressure shall be maintained on the rivets for a short time after the upsetting is completed. High tensile steel rivets shall be heated up to 1100 degree Celsius. Any rivet whose point is heated more than prescribed shall not be driven. Hand riveting shall not be resorted to except with prior approval of Engineer.
- f) Where flush surface is required, any projecting metal shall be chipped or ground off.
- g) Before riveting is commenced, all work shall be properly and tightly bolted together. The bolts should be frequently tightened as the riveting proceeds.
- h) Drifts shall be used only for drawing the work into position and shall not be used to such an extent as to distort the holes. Drifts of a larger size than the nominal diameter of the hole shall not be used. The riveting shall be done by hydraulic or pneumatic machine unless otherwise specified by Engineer.
- i) Driven rivets, when struck sharply on the head with a quarter-pound rivet testing hammer, shall be free from movement and vibration.
- j) All loose or burnt rivets and rivets with cracked or badly formed defective heads or with heads which are unduly eccentric with the shanks, shall be removed and replaced. In removing rivets, the head shall be sheared off and the rivet punched out so as not to injure adjacent metal and, if necessary, they shall be drilled out. Recouping and caulking shall not be permitted.

**10. Bolts, Nuts and Washers**

1. Black Bolts (black all over)

Black bolts are forged bolts in which the shanks, heads and nuts do not receive any further treatment except cutting of screw threads. They shall be true to shape and size and shall have standard dimensions as shown on drawings.

2. Close Tolerance Bolts

Close tolerance bolts shall be faced under the head and turned the shank.

3. Turned Barrel Bolts

The diameter of the screwed portion of turned barrel bolts shall be 1.5 mm smaller than the diameter of the barrel unless otherwise specified by Engineer. The diameter of the bolts as given on drawing shall be the nominal diameter of the barrel. The length of the barrel shall be such that it bears fully on all the parts connected. The threaded portion of each bolt shall project through the nut by at least one thread. Faces of heads and nuts bearing on steel work shall be machined.

4. High Strength Structural Bolts

HSFG bolts are high strength structural bolts which have been tightened such as to induce predefined tension in the bolt shank. Due to the tension in the bolt, the interface between the plies (steel members in a joint) cannot move relative to each other because of the frictional resistance.

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**5. Washers**

- a) In all cases where the full bearing area of the bolt is to be developed, the bolt shall be provided with a steel washer of sufficient thickness under the nut to avoid any threaded portion of the bolt being within the thickness of the part bolted together and to prevent the nut when screwed up, from bearing on the shank of the bolt.
- b) For close tolerance of turned barrel bolts, steel washers whose faces give a true bearing shall be provided under the nut. The washer shall have a hole diameter not less than 1.5 mm larger than the barrel and a thickness of not less than 6 mm so that the nut, when screwed up, will not bear on the shoulder of the bolt.
- c) Taper washers with a correct angle of taper shall be provided under all heads and nuts bearing on bevelled surfaces.
- d) Spring washers may be used under nuts to prevent slackening of the nuts when excessive vibrations occur.
- e) Where the heads or nuts bear on timber, square washers having a length of each side not less than three times the diameter of the bolts or round washers having a diameter of 3.5 times the diameter of bolts and with a thickness not less than one quarter of diameter shall be provided.
- f) The Direct Tension Indicators (DTI washer) are special type of washers with indentations which get pressed when tension is applied. The pressing of indentations to required level indicates that the required tension has been applied in the bolts. HSFG bolts shall be provided with DTI washers.

**6. Studs**

Ordinary studs may be used for holding parts together, the holes in one of the parts being tapped to take the thread of the stud. Counter-sunk studs may be used for making connections where the surfaces are required to be clear of all obstruction, such as protruding heads of bolts or rivets, studs may also be welded on steel work in positions as required.

**7. Service Bolts**

- a) Service bolts shall have the same clearance as black bolts and where it is required that there should be no movement prior to final riveting, sufficient drifts or close tolerance bolts shall be used to assemble and locate the work.
- b) But no drifting to match up unfair holes shall be allowed. Any apparent error in fabrication which prevents assembly or fitting up of parts by the use of drifts shall be investigated immediately. If some ream ring is required and uses of special rivets are called for, same will not be undertaken without prior approval of Engineer. No extra payment shall be admissible for such additional work unless fabrication is done by a different agency and is established to be defective.

**8. Drifts**

The barrel shall be drawn or machined to the required diameter for a length of not less than one diameter over the combined thickness of the metal through which the drifts have to pass. The diameter of the parallel barrel shall be equal to the nominal diameter of the hole subject

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to a tolerance of +0 mm and -0.125 mm. Both ends of the drift for a length of equal to 1.5 times the diameter of the parallel portion of the bar shall be turned down with a taper to a diameter at the end equal to one-half of that of parallel portion.

**9. Use of Drifts and Bolts**

- a) In cases where the joint seems to withstand stresses arising from special methods of erection before being riveted up, parallel barrelled drifts and bolts shall be used to withstand the full stresses/forces arising thereof. In such case number of drifts used will be subject to a maximum of 40%.
- b) In the event of anticipated emergencies such as staying being in danger of being affected by floods or storms before riveting can be completed, the joints to be riveted shall be made secure by filling 40% holes by drifts and 40% with service bolts fully tightened up. Otherwise, in normal course filling not less than 50% of holes with service bolts shall make joints.

**10. Pins and Pin**

**Holes**

- a) Pins shall be parallel throughout and shall have a smooth surface free from flaws. They shall be of sufficient length to ensure that all parts connected thereby shall have a full bearing on them. Where the ends are threaded, they shall be turned to a smaller diameter at the ends for the thread and shall be provided with a pilot nut, where necessary, to protect the thread when being drawn to place.
- b) Pins more than 175 mm in length or diameter shall be forged and annealed.

**2. Pin Holes**

- a) Pin holes shall be bored true to gauge, smooth, straight, at right angles to the axis of the member and parallel to each other, unless otherwise required. Tolerance in the length of tension members from outside to outside of pin-holes and of compression members from inside to inside of pin holes shall be one millimetre. In built-up members, boring shall be done after the members have been riveted or welded.
- b) The specified diameter of the pin hole shall be its minimum diameter. The resulting clearance between the pin and the hole shall not be less than 0.5 mm and not more than 1.0mm.

**3. Shop Erection and Match-Marking**

- a) Engineer and inspecting officials shall be given free access to the templating fabrication shop and for frequent inspection of different parts under fabrication.
- b) Before being dispatched, steel work shall be temporarily erected in fabrication shop for inspection by Engineer or other inspecting official either wholly or in such portion as Engineer may require so that he may be satisfied both in respect of alignment camber and fit of all connections. For this purpose, sufficient number of parallel drifts and service bolts tightly screwed up shall be employed. All parts shall fit accurately and be in accordance with drawings and specifications.
- c) After the work has been passed by Engineer and before being dismantled, each part shall be carefully marked for re-erection with distinguishing marks and stamped with durable markings. Drawings showing these markings correctly shall be supplied to Engineer.
- d) Unloading, handling and storage of steel work as per these specifications shall be the responsibility of the contractor. The cost of repairs or of rejected material, its removal and the cost of transporting replacement material to the site shall be borne by the Contractor.

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- e) Where close tolerance or turned barrel bolts are used for those cases where interchangeability is not insisted upon, each span shall be erected and members of each span marked distinctly.

**11. Welding**

- (a) All welding shall be done with the prior approval of Engineer and workmanship shall conform to provisions of IS: 823 or other relevant Indian Standards as appropriate.
- (b) All the welds shall be done by submerged arc welding process either fully automatic or semi-automatic. Manual metal arc welding may be done only with prior permission of Engineer. In case of bridge girders this would be permissible only in case
- (c) of very short runs or of minor importance or where access or locations of welds do not permit automatic or semi-automatic welding.
- (d) Except for special types of edge or end preparation, like double "U" or single "J" or double "J" (which would be prepared by machining or grouting) the fusion edges of all plates to be jointed shall be prepared by using mechanically controlled automatic flame cutting equipment and then ground to a smooth surface.
- (e) When material thickness is 20 mm or more, special precautions like preheating shall be taken, as laid down in IS: 823.
- (f) The general welding procedures including particulars of preparation of fusion faces for metal arc welding shall be carried out in accordance with IS: 9595. Welding shall not be done when the air temperature is less than 10 degree Celsius. Welding shall not be done when the surfaces are moist, during periods of strong winds unless the work and the welding operations are adequately protected.
- (g) Welding shall be permitted to be performed for the project either in the shop or in the field, only by operators who have passed qualification tests to the satisfaction of Engineer.

**12. Tolerances:** Tolerances work shall be specified on the drawings and shall be subject to the approval of Engineer.

**13. Erection**

**1. General**

The contractor shall erect structural steel, remove temporary construction, and undertake all work required to complete the construction of works included in the contract in accordance with approved drawings and specifications to the entire satisfaction of Engineer.

**2. Organization and Equipment**

- a. The contractor shall submit erection plans prepared by the fabricator, showing the method and procedure of erection, compatible with the details of fabrication.
- b. Unless otherwise provided in the contract, the contractor shall supply and erect all necessary false work and staging and shall supply all labour, tools, tackles, erection plant and other materials necessary to carry out the work complete in all respects.
- c. The contractor shall supply all rivets, bolts, nuts, washers, etc., required to complete erection at site with an allowance for wastage, etc., of 12.5 per cent of the net number of field rivets, bolts, washers required, or a minimum of five number of each item whichever is higher.
- d. Service bolts and nuts, ordinary platens, washers, and drifts for use in erection work shall be supplied at 60 percent (45 percent bolts and 15 percent drifts) of the number of field rivets per span in

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each size (this includes wastage). A reduction in quantities of service bolts, etc., may, however, be specified by Engineer if more than one span of each type is ordered.

- e. Deflection and vibratory tests shall be conducted in respect of supporting structures, launching truss as

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also the structure under erection and unusual observations reviewed; looseness of fittings are to be noted.

- f. For welded structures, welder's qualifications and skill are to be checked as per standard norms. Non-destructive tests of joints as per designer's directives are to be carried out.
- g. Precision non-destructive testing instruments available in the market should be used for noting various important parameters of the structures frequently and systematic record is to be kept.
- h. Safety requirements should conform to IS:7205; IS: 7273 and IS: 7269 as applicable and should be a consideration of safety, economy and rapidity.
- i. Erection work should start with complete resources mobilized as per latest approved drawings and after a thorough survey of foundations and other related structural work. In case of work of greater magnitude, maximum mechanization is to be adopted.
- j. The structure should be divided into erectable modules as per the scheme. This should be pre-assembled in a suitable yard/platform and its matching with members of the adjacent module checked by trial assembly before erection.
- k. The structure shall be set out to the required lines and levels. The stocks and masses are to be carefully preserved. The steelwork should be erected, adjusted and completed in the required position to the specified line and levels with sufficient drifts and bolts. Packing materials are to be available to maintain this condition. Organized "Quality Surveillance" checks need to be exercised frequently.
- l. Before starting work, the contractor shall obtain necessary approval of Engineer as to the method of erection, the number and character of tools and plants, type and quantity of labour to be maintained, and safety precautions taken. Approval of Engineer shall not relieve the contractor of his responsibility for safety of his method or equipment or from carrying out the work fully in accordance with drawings and specifications.
- m. During the progress of the work, the contractor shall have a competent engineer or foreman in charge of work, who shall be adequately experienced in steel erection and acceptable to Engineer.

#### **14. Handling and Storing Materials**

- (i) Materials to be stored shall be placed on skids above ground. It shall be kept clean and properly drained. Long members shall be supported on skids placed near enough together to prevent injury from deflection. The contractor shall be held responsible for loss or damage to any material paid for by the Department while in his care or for any damage to such material resulting from his work.
- (ii) IS: 7293 and IS: 7969 dealing with handling of materials and equipment for safe working should be followed.

#### **15. Formwork and Staging**

Formwork, staging and supports shall be properly designed, substantially built and maintained for all anticipated loads. The contractor, if required, shall submit plans and designs for approval to Engineer. Approval of plans, however, shall not relieve the contractor of his responsibility

#### **16. Straightening Bent Material**

The straightening of plates, angles and other shapes which are slightly bent in transit or handling shall be done by methods not likely to produce fracture or any injury to the material. The metal shall not be heated unless permitted by Engineer for special cases when heating shall not be done to a temperature higher than that producing a dark 'cherry red' colour followed by as slow cooling as possible. Following the straightening of a bend or buckle, the surface shall be carefully investigated for evidence of fracture. Sharp kinks and bends and sign of any fracture may be cause for rejection of material.

#### **17. Assembling Steel**

- a. The parts shall be accurately assembled as shown on drawings and match marks shall be followed.

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The material shall be carefully handled so that no parts will be bent, broken or otherwise damaged.

- b. Hammering, which will injure or distort members, shall not be done. Bearing surface or surfaces to be in permanent contact shall be cleaned, before the members are assembled.
- c. All fasteners shall have a washer under nut or bolt head, whichever is turned in tightening.
- d. Any connection to be riveted or bolted shall be secured in close contact with service bolts or with a sufficient number of permanent bolts before the rivets are driven or before the connections are finally bolted. Joints shall normally be made by filling not less than 50 percent of holes with service bolts and barrel drifts in the ratio 4:1. The service bolts are to be fully tightened up as soon as the joint is assembled. Connections to be made by close tolerance or barrel bolts shall be completed as soon as practicable after assembly.
- e. Any connections, to be site welded, shall be securely held in position by approved methods to ensure accurate alignment, camber and elevation before welding is commenced.
- f. Field riveting, welding and bolted and pin connections shall conform to the requirements of appropriate Sub-Clause.
- g. The correction of minor misfits involving harmless amounts of reaming, cutting and chipping will be considered a legitimate part of erection. However, any error in shop fabrication or deformation resulting from handling and transportation which prevents proper assembling and fitting up of parts by moderate use of drifts or by a moderate amount of reaming and slight chipping or cutting shall be reported immediately to Engineer and his approval of the method of correction obtained. The correction shall be made in the presence of Engineer.

**18. Field Inspection**

- a. All materials, equipment and work of erection shall be subject to inspection of Engineer who shall be provided with all facilities including labour and tools required at all reasonable times. Any work found defective is liable to be rejected and no material or work shall be painted until inspected and approved.
- b. Structural steel and components viz., rivets, bolts, nuts, washers, welding consumables, etc., shall be tested for mechanical and chemical properties to the requirements of appropriate IS codes.
- c. Rivets and riveted connection shall be inspected and tested for compliance of Codal requirements. Welding procedure, welded connection and testing shall be in compliance with Codal requirements. All facilities necessary for stage inspection during welding and on completion shall be provided to Engineer or inspecting official.
- d. Adequate means of identification either by identification mark or other record shall be provided to enable each weld to be traced to the welder(s) by whom it was carried out. The fabricated member/component made out of rolled and built-up section shall be checked for compliance of the tolerances specified in these specifications or as stipulated in relevant IS Codes and the checking for deviations shall be made over the full length.
- e. During inspection, the component/member shall not have any load or external restraint.

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**19. Painting**

**(a) General**

- (i) Unless otherwise specified, all metal work shall be given approved shop coats as well as field coats of painting. The item of work shall include preparation of metal surfaces, application of protective covering, drying of paint coats and supply of all tools, scaffolding, labour and materials necessary:
- (ii) General steel work shall be protected against corrosion by :
- (iii) Minimum of three coats of paint, or
- (iv) A metal coating followed by two coats of paint.
- (v) Unless otherwise specified, all painting and protective coating work shall generally be done in accordance with IS: 1477 (Part 1) and IRS specification B1-79.

**b. Surface Preparation**

- a. Steel surface to be painted either at fabrication shop or at site of work shall be prepared in a thorough manner with a view to ensuring complete removal of mill scale by one of the following processes as agreed to between fabricator and Engineer:
  - i. Grit and sand blasting;
  - ii. Pickling which should be restricted to single plates, bars and sections;
  - iii. Flame cleaning;
  - iv. Scraping and wire brushing.
- b. Primary coat shall be applied as soon as practicable after cleaning and in case of flame cleaning, primary coat shall be applied while the metal is still warm.
- c. All slag from welds shall be removed before painting. Care shall be taken to brush the surface clean prior to painting. Surfaces shall be maintained dry and free from dirt and oil. Work out of doors in frosty or humid weather shall be avoided.
- d. Coatings
  - a. Prime coat to be used shall conform to specification of primers approved by Engineer. Metal coatings shall be regarded as priming coats.
  - b. All coats shall be compatible with each other. When metal coatings are used, the undercoat shall be compatible with the metal concerned. The undercoat and the finishing coat shall preferably be from the same manufacturer. Successive coats of paints shall be of different shades or colours and each coat shall be allowed to dry thoroughly before the next coat is applied. Particular care shall be taken with the priming and painting of edges, corners, welds and rivets.

**e. Painting in Shop**

- a. All fabricated steel shall be painted in shops, after inspection and acceptance, with at least one priming coat, unless the exposed surfaces are subsequently to be cleaned at site or metal coated.
- b. Shop contact surfaces, if specifically required to be painted, shall be brought together while the paint is still wet.

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- c. Surface not in contact but inaccessible after shop assembly shall receive the fully specified protectivetreatment before assembly.

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d. Where surfaces are to be welded, the steel shall not be painted or metal coated within a suitable distance from any edges to be welded, if the specified paint or metal coating would be harmful to welders or is expected to impair the quality of site welds.

e. Exposed machined surfaces shall be adequately

protected. f. Painting at Site:

a. The methods of application of all paint coatings shall be in accordance with the manufacturer's written recommendation and shall be as approved by the Engineer. Spray painting may be permitted provided it will not cause inconvenience to the public and is appropriate to the type of structure being coated. Areas hard to gain access to for painting and areas shaded for spray application shall be coated first by brushing.

b. Oil based red lead primers must be applied by brush only, taking care to work into all corners and crevices.

c. The primer, intermediate and finishing coats shall all be applied so as to provide smooth coatings of uniform thickness. Wrinkled or blistered coatings or coatings with pinholes, sags, lumps or other blemishes shall not be accepted. Where the Engineer so directs, the coating shall be removed by abrasive blast cleaning and replaced at the contractor's expense.

d. Surfaces, which will be inaccessible after site assembly, shall receive the full-specified protective treatment before assembly.

e. Surfaces, which will be in contact after site assembly, shall receive a coat of paint (in addition to any shop priming) and shall be brought together while the paint is still wet.

f. Damaged or deteriorated paint surfaces shall first be made good with the same type of coat as the shop coat.

g. Where steel has received a metal coating in the shop, this coating shall be completed at site so as to be continuous over any welds, bolts and site rivets.

h. Specified protective treatment shall be completed after erection

## **20. Tests And Standards of Acceptance**

a. The materials shall be tested in accordance with relevant IS and/or IRS specifications/codes and necessary test certificates shall be furnished. Additional tests, if required by Engineer, shall be got carried out by the Contractor at his own cost.

b. The fabrication, furnishing, erecting, painting of structural steel work shall be in accordance with these specification and shall be got checked and approved by the Engineer.

## **21. Measurements For Payment**

i. Measurements of this item shall be in tones based on net weight of metal in the fabricated structure computed on the basis of nominal weight of materials.

ii. The weight of rolled and cast steel and cast iron shall be determined from the dimensions shown on drawings on the following basis:

Rolled or cast steel:

7.85 gm/cu. cm Cast

iron:

7.21

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gm/cu.cm

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- iii. Weight of structural sections shall be nominal weight.
- iv. The weight of castings shall be computed from dimensions shown on drawings with an addition of 5(five) percent for fillets and over-runs.
- v. The weight of weld fillets and the weight of protective coatings shall not be included.
- vi. The weight of rivet heads shall be computed by taking the weight of 100 snap head as given in Table 800.2.

**WEIGHT OF RIVET HEADS**

| Dia. of rivet<br>as<br>manufactured (mm) | Weight of<br>100<br>snapheads<br>(kg) |
|------------------------------------------|---------------------------------------|
| 12                                       | 1.3                                   |
| 14                                       | 2.1                                   |
| 16                                       | 3.4                                   |
| 18                                       | 4.45                                  |
| 20                                       | 6.1                                   |
| 22                                       | 8.1                                   |
| 24                                       | 10.5                                  |
| 27                                       | 15.0                                  |
| 30                                       | 20.5                                  |
| 33                                       | 27.2                                  |

- vii. When specially agreed upon, allowance for snap heads may be taken as a flat percentage of the total weight unless otherwise specified in tender documents. This percentage may be taken as 3 per cent or modified by mutual agreement.
- viii. The Contractor shall supply detailed calculation sheets for the weight of metal in the fabricated structures. No additions shall be made for the weight of protective coatings or weld fillets.
- ix. Where computed weight forms the basis for payment, the weight shall be calculated for exact cut sizes of members used in the structure, deductions being made for all cuts, except for rivet holes. Additions shall be made for rivet heads as mentioned above.
- x. When especially agreed upon, the basis for payment may be the fabricated structure weight complete, according to specifications included in special provisions of the Contract.

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For Chief Administrative Officer  
(Const.) West Central Railway,

**Chapter-8**  
**STRUCTURAL CONCRETE**

**1. MATERIALS**

**2. Cement:**

- (i) Cement older than 3 months from the date of manufacture shall not be used.
- (ii) The cement used shall be any of the following, with prior approval of the engineer in charge.
- (iii) Ordinary Portland Cement (53 grade) conforming to IS 12269- 1987
- (iv) Ordinary Portland Cement (43 grade) conforming to IS:8112-1989
- (v) Ordinary Portland Cement (33 grade) conforming to IS:269-1989
- (vi) Portland slag cement confirming to IS:455-1989
- (vii) Portlandpozzolana cement confirming to IS:1489 (Part-I) -1991 fly ash

basedNote:-

- 1) All updated specifications upto date of opening of tender are applicable.
- 2) Only readymade Portland Pozzolona Cement [confirming to IS:1489 (Part-I) -1991 fly ash based] shall be allowed to use. **Mixing of fly ash and OPC cement at site/ RMC plant to produce PortlandPozzolona Cement shall not be allowed.**
- 3) In case of blended cement i.e. Portland pozzalana cement and Portland slag cement rate of development of strength is slow as compared to ordinary Portland cement hence period of removal of formwork and period of curing should be suitable increased.
- 4) Only Ordinary Portland Cement (53 grade) conforming to IS 12269- 1987 shall be allowed for Prestressed Concrete work (PSC).
- (viii) Cement manufactured by any of the following brands/ companies shall only be allowed to used.
  - 1) Ambuja
  - 2) ULTRATECH
  - 3) ACC
  - 4) Birla Uttam
  - 5) Grasim
  - 6) JK

OR approved equivalent after due testing.

**Testing frequency for cement:**

| Sr. | Type of test   | Test frequency                                                                                                                                                                                            | Remark.                                                                                                                                                        |
|-----|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1   | Physical test  | One sample for every 50 MT of cement consumed or part thereof or each batch of cement in case of time lapse of 15 days in supply of consecutive batches and manufacturers test certificate for each lot.  | Test to be done.<br>a Initial and final setting time.<br>b Fineness.<br>c Compressive strength.<br>d Soundness<br>e Specific Gravity.                          |
| 2   | Chemical test. | One sample for every 100 MT of cement consumed or part thereof or each batch of cement in case of time lapse of 15 days in supply of consecutive batches and manufacturers test certificate for each lot. | Lime saturation factor.<br>Alumina Iron ratio<br>Insoluble residue<br>Magnesium oxide.<br>Sulphuric Anhydride<br>Loss of Ignition<br>Chlorides<br>C3A content. |

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|  |  |  |                      |
|--|--|--|----------------------|
|  |  |  | Specific<br>Surface. |
|--|--|--|----------------------|

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For Chief Administrative Officer  
(Const.)West Central Railway,

**3. Fine Aggregates (Sand)**

- a) Creek sand shall not be used in the construction of any concrete/Masonry work .
- b) Sand, if found too coarse, shall be suitably blended with finer sand obtained from approved resources to obtain the desired grading. The provision of two types of sand, their stacking separately and their mixing in the specified proportions shall be at the contractor's cost.
- c) The river sand shall not contain silt more than a total of 3% by weight. Chloride content in sand shall not be greater than 0.04% by weight.
- d) Artificial or crushed sand may be allowed to be used subject to passing from 75 micron sieve shall not be more than as provided in the IS code.
- e) The grading of the sand shall conform to IS: 383.
- f) Test frequency for FINE AGGREGATE [Confirming to IS: 383-1970, IS-2386-1963 )

| Sr. No   | Characteristic to be schecked                                   | Acceptance Criteria      |                                                                                                                                                                                                                                                                                                                                                    | Method of testing         | Test Frequency                                              |       |          |    |     |      |        |      |        |      |       |     |       |     |      |     |      |                     |                                                                                      |
|----------|-----------------------------------------------------------------|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|-------------------------------------------------------------|-------|----------|----|-----|------|--------|------|--------|------|-------|-----|-------|-----|------|-----|------|---------------------|--------------------------------------------------------------------------------------|
|          | <b><u>PHYSICAL TESTS</u></b>                                    | Code                     | Value                                                                                                                                                                                                                                                                                                                                              |                           |                                                             |       |          |    |     |      |        |      |        |      |       |     |       |     |      |     |      |                     |                                                                                      |
| 1        | Determination of particle size – Grading zones (Sieve Analysis) | Table 4 of IS:383 – 1973 | <table><tr><td colspan="2">Zone- II</td></tr><tr><td>Sieve</td><td>Passing%</td></tr><tr><td>10</td><td>100</td></tr><tr><td>4.75</td><td>90-100</td></tr><tr><td>2.36</td><td>75-100</td></tr><tr><td>1.18</td><td>55-90</td></tr><tr><td>600</td><td>35-59</td></tr><tr><td>300</td><td>8-30</td></tr><tr><td>150</td><td>0-10</td></tr></table> | Zone- II                  |                                                             | Sieve | Passing% | 10 | 100 | 4.75 | 90-100 | 2.36 | 75-100 | 1.18 | 55-90 | 600 | 35-59 | 300 | 8-30 | 150 | 0-10 | Part –I of IS :2386 | One sample per Lot of every 50 M <sup>3</sup> or part from each source for each site |
| Zone- II |                                                                 |                          |                                                                                                                                                                                                                                                                                                                                                    |                           |                                                             |       |          |    |     |      |        |      |        |      |       |     |       |     |      |     |      |                     |                                                                                      |
| Sieve    | Passing%                                                        |                          |                                                                                                                                                                                                                                                                                                                                                    |                           |                                                             |       |          |    |     |      |        |      |        |      |       |     |       |     |      |     |      |                     |                                                                                      |
| 10       | 100                                                             |                          |                                                                                                                                                                                                                                                                                                                                                    |                           |                                                             |       |          |    |     |      |        |      |        |      |       |     |       |     |      |     |      |                     |                                                                                      |
| 4.75     | 90-100                                                          |                          |                                                                                                                                                                                                                                                                                                                                                    |                           |                                                             |       |          |    |     |      |        |      |        |      |       |     |       |     |      |     |      |                     |                                                                                      |
| 2.36     | 75-100                                                          |                          |                                                                                                                                                                                                                                                                                                                                                    |                           |                                                             |       |          |    |     |      |        |      |        |      |       |     |       |     |      |     |      |                     |                                                                                      |
| 1.18     | 55-90                                                           |                          |                                                                                                                                                                                                                                                                                                                                                    |                           |                                                             |       |          |    |     |      |        |      |        |      |       |     |       |     |      |     |      |                     |                                                                                      |
| 600      | 35-59                                                           |                          |                                                                                                                                                                                                                                                                                                                                                    |                           |                                                             |       |          |    |     |      |        |      |        |      |       |     |       |     |      |     |      |                     |                                                                                      |
| 300      | 8-30                                                            |                          |                                                                                                                                                                                                                                                                                                                                                    |                           |                                                             |       |          |    |     |      |        |      |        |      |       |     |       |     |      |     |      |                     |                                                                                      |
| 150      | 0-10                                                            |                          |                                                                                                                                                                                                                                                                                                                                                    |                           |                                                             |       |          |    |     |      |        |      |        |      |       |     |       |     |      |     |      |                     |                                                                                      |
| 2        | Silt Content                                                    |                          | Not more than 3% by wt.& shale, clay, & other structurally weak particles- 2% by weight.                                                                                                                                                                                                                                                           |                           |                                                             |       |          |    |     |      |        |      |        |      |       |     |       |     |      |     |      |                     |                                                                                      |
| 3        | Specific Gravity                                                |                          | 2.6 to 2.85                                                                                                                                                                                                                                                                                                                                        |                           | One sampl randomly 500 m3 of concrete or change of source.  |       |          |    |     |      |        |      |        |      |       |     |       |     |      |     |      |                     |                                                                                      |
| 4        | Absorption Value                                                |                          | Not more than 3%                                                                                                                                                                                                                                                                                                                                   | IS:2386 –1963             | One sample randomly 500 m3 of concrete or change of source. |       |          |    |     |      |        |      |        |      |       |     |       |     |      |     |      |                     |                                                                                      |
| 5        | Soundness of Aggregate                                          | Cl. 3.6 of IS ; 383      | Not more than 10% with Na2SO4& 15%with MgSO4                                                                                                                                                                                                                                                                                                       | Part IV of IS: 2386 –1963 |                                                             |       |          |    |     |      |        |      |        |      |       |     |       |     |      |     |      |                     |                                                                                      |
| 6        | Presence of deleterious material (total)                        |                          | 5% Max(Total)                                                                                                                                                                                                                                                                                                                                      |                           |                                                             |       |          |    |     |      |        |      |        |      |       |     |       |     |      |     |      |                     |                                                                                      |
| 7        | Coal and Lignites                                               | Table (1) of IS ; 383    | 1.0% Max.                                                                                                                                                                                                                                                                                                                                          | Part –II of IS :2386      |                                                             |       |          |    |     |      |        |      |        |      |       |     |       |     |      |     |      |                     |                                                                                      |
| 8        | Clay lumps                                                      |                          | 1.0% Max.                                                                                                                                                                                                                                                                                                                                          |                           |                                                             |       |          |    |     |      |        |      |        |      |       |     |       |     |      |     |      |                     |                                                                                      |
| 9        | Material finer than 75micron                                    |                          | 3.0% Max.                                                                                                                                                                                                                                                                                                                                          |                           |                                                             |       |          |    |     |      |        |      |        |      |       |     |       |     |      |     |      |                     |                                                                                      |
| 10       | Soft fragments                                                  |                          | -Nil-                                                                                                                                                                                                                                                                                                                                              |                           |                                                             |       |          |    |     |      |        |      |        |      |       |     |       |     |      |     |      |                     |                                                                                      |
| 11       | Shale                                                           |                          | 1.0% Max.                                                                                                                                                                                                                                                                                                                                          |                           |                                                             |       |          |    |     |      |        |      |        |      |       |     |       |     |      |     |      |                     |                                                                                      |

Signature of Tenderer

For Chief Administrative Officer  
(Const.)West Central Railway,

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|    |              |                     |            |  |
|----|--------------|---------------------|------------|--|
|    | CHEMICAL     |                     |            |  |
| 12 | Cl – content | Cl.-4.2.2<br>of CBC | 0.04% Max. |  |

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For Chief Administrative Officer  
(Const.)West Central Railway,

|    |              |                   |           |  |  |
|----|--------------|-------------------|-----------|--|--|
| 13 | SO3- Content | Cl. -4.2.2 of CBC | 0.4% Max. |  |  |
|----|--------------|-------------------|-----------|--|--|

#### 4. Coarse Aggregates

Coarse aggregates for the works shall be crushed stone aggregates conforming to IS:383 obtained from approved sources. Coarse aggregate containing flat or flaky pieces or mica shall be rejected.

#### Test frequency for COARSE AGGREGATE

[Confirming to IS: 383-1970, IS-2386-1963 & IS: 456-2000]

#### Physical Tests

| Sr. No | Characteristics tobe checked                    | Acceptance Criteria  |                                                                              |            |        | Method oftesting          | Test Frequenc y                                                                              |
|--------|-------------------------------------------------|----------------------|------------------------------------------------------------------------------|------------|--------|---------------------------|----------------------------------------------------------------------------------------------|
|        |                                                 | Code                 | Value                                                                        |            |        |                           |                                                                                              |
| 1      | Determination of particle size (Sieve Analysis) | Table (2) of IS :383 | Sieve size                                                                   | 40 mm      | 20mm   | IS :2386 (part I) – 1963  | One sample ever y 50m <sup>3</sup> or part thereof f                                         |
|        |                                                 |                      | 63m m                                                                        | 100 85–100 | - 100  |                           |                                                                                              |
|        |                                                 |                      | 40m m                                                                        | 0–20       | 85-100 |                           |                                                                                              |
|        |                                                 |                      | 23m m                                                                        | 0-5        | 0-20   |                           |                                                                                              |
|        |                                                 |                      | 10 mm                                                                        |            | 0-5    |                           |                                                                                              |
|        |                                                 |                      | 4.75m m                                                                      |            |        |                           |                                                                                              |
| 2      | Moisture Content                                |                      | As per site condition Max. 4 %                                               |            |        |                           |                                                                                              |
| 3      | Flakiness Index                                 | Client               | Not more than 30%                                                            |            |        |                           |                                                                                              |
| 4      | Elongation index                                | --do --              | Not more than 30%                                                            |            |        |                           |                                                                                              |
| 5      | Specific gravity                                |                      | 2.60 to 2.85                                                                 |            |        |                           | One sample per Lot of every 500 M <sup>3</sup> of concrete or part thereof from each source. |
| 6      | Absorption value                                |                      | Not more than 3%                                                             |            |        |                           |                                                                                              |
| 7      | Crushing Value                                  | Cl.3.3 of IS: 383    | Not more than 30% for concrete for wearing surfaces & 45% for other concrete |            |        | IS :2386 (part IV) – 1963 |                                                                                              |
| 8      | Impact Value                                    | Cl.3.4 of IS: 383    | Not more than 30% for concrete for wearing surface & 45% for other concrete  |            |        | --DO --                   |                                                                                              |
| 9      | Abrasion Value                                  | Cl.3.5 of IS: 383    | Not more than 30% for concrete for wearing surface & 50% for other concrete  |            |        | --DO --                   |                                                                                              |

#### 5. Water :

Creek water shall not be used. Potable water of suitable quality should be used.

a) The permissible limits for solids shall be as follows:

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|                                     | Permissible limits (Max.) |
|-------------------------------------|---------------------------|
| Organic                             | 200 mg/lit.               |
| Inorganic                           | 3000                      |
| mg/lit.Sulphates (SO <sub>4</sub> ) | 400                       |
| mg/lit. Chlorides (Cl)              | 300                       |
| mg/lit. Suspended matter            | 2000                      |
| mg/lit.Acidic material              | 2                         |
| mg/lit.                             |                           |

- b) The PH value shall not be less than 6.

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c) Test frequency for water.

[Confirming to Cl.5.4 of IS: 456 – 2000 & IR Bridge Code]

| Sr. No | Characteristics to be checked | Acceptance Criteria      |                                                             | Method of testing        | Test Frequency             |
|--------|-------------------------------|--------------------------|-------------------------------------------------------------|--------------------------|----------------------------|
|        |                               | Code                     | Value                                                       |                          |                            |
| 1      | PH Value                      | Cl.- of 5.4.2 IS : 456   | Not less than 6                                             | IS: 3025                 | 100 m3 of concrete at site |
| 2      | Chloride content              |                          | Not more than 300 ppm (0.03 %) for RCC and 200 ppm for PSC. | Part -32                 |                            |
| 3      | Sulphate content              | Table (1) Cl. 5.4        | Not more than 400 ppm                                       | Part -24                 |                            |
| 4      | Organic impurities            | Table (1) Cl. 5.4 IS:456 | 200 mg/ lt.                                                 | Part –18                 | Every 3 month              |
| 5      | Inorganic impurities          |                          | 3000 mg /lit                                                | IS : 3025                |                            |
| 6      | Suspended matter              |                          | 2000 mg/lt                                                  | Part –17 of IS; 3025     |                            |
| 7      | Alkalinity                    |                          | 10mg/lit                                                    | Part 22 & 23 of IS :3025 |                            |
| 8      | Acidity                       |                          | 2 mg / lit                                                  |                          |                            |

**6. Concrete Admixtures:**

Concrete admixtures are proprietary items of manufacture and shall be obtained only from established manufacturers with proven track record, quality assurance and full fledged laboratory facilities. The brand and type of admixture should be tested at site for various properties imparted to the concrete and to be used only if found suitable with prior approval of the engineer in charge.

a) Test frequency for admixture: - [Confirming to IS: 9103 & IRC CBC 1997]

| Sr. No | Characteristic to be checked | Acceptance Criteria Value           | Method of testing    | Test Frequency                     |
|--------|------------------------------|-------------------------------------|----------------------|------------------------------------|
| 1      | Chloride content             | NIL                                 | IS:6925              | Each batch Up to 4000Kg            |
| 2      | Air content                  | < Air content of controlled mix +1% | IS:1199              |                                    |
| 3      | Uniformity test              | As per table 1 as given below.      | Annexure E OF 9103   | Each batch                         |
| 4      | Bleeding                     | kk < 5%                             | TABLE 1 A OF IS:9103 | Regular at site with various trial |

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For Chief Administrative Officer  
(Const.) West Central Railway,

**Table 1****Uniformity test & Requirements of Admixtures**

| <b>Sr no</b> | <b>Properties to be checked</b> | <b>Values</b>                                                          | <b>Test method</b>    |
|--------------|---------------------------------|------------------------------------------------------------------------|-----------------------|
| 1            | Dry moisture content            | Within 3 % of MTC                                                      | Annexure E of IS-9103 |
| 2            | Ash content                     | Within 1 % of MTC                                                      | Annexure E of IS-9103 |
| 3            | Relative density                | Within 0.02 % of MTC                                                   | Annexure E of IS-9103 |
| 4            | Chloride ion content            | Within 10% of the value N OR within 0.2 % of MTC, whichever is greater | Annexure E of IS-9103 |
| 5            | Ph value                        | 7 – 8                                                                  | Annexure E of IS-9103 |
| 6            | SO <sub>3</sub>                 | NIL                                                                    | CBC:4.4.2             |

\* MTC= Manufacturers Test Certificate.

**7. Storage of Materials****a. General:**

All materials should be stored at proper places so as to prevent their deterioration or intrusion by foreign matter and to ensure their satisfactory quality and fitness for the work. All such materials even though stored in approved godowns/places, must be subjected to acceptance test prior to their immediate use.

**b. Aggregates :**

Aggregate stockpiles should be made on ground that is denuded of vegetation, is hard and well drained. If necessary, the ground shall be covered with 50 mm plank. Aggregates placed directly on the ground shall not be removed from the stockpile within 15 cm of the ground until the final cleaning up of the work, and then only the clean aggregate will be permitted to be used. Aggregate shall be stored in such a way as to prevent segregation of sizes and avoid contamination with fines and other undesirable material.

**c. Cement:**

Cement shall be transported, handled and stored on the site in a godown in such a manner as to avoid deterioration or contamination. Cement shall be stored above ground level in perfectly dry and water-tight sheds and shall be stacked not more than eight bags high.

d. The Contractor shall prepare and maintain proper records on site in respect of delivery, handling, storage and use of cement and these records shall be available for inspection by the Engineer at all times.

**8. Design Mix :**

For all items of concrete only design mix shall be used. Prior to the start of construction, the contractor shall design the mix, the proportions of materials, including chemical admixtures to be used and submit to the Engineer for approval. Water-reducing admixtures (including plasticisers or super-plasticisers, retarders) or mineral admixture such as fly ash, GGBS or silica fumes may be used at the Contractor's option, subject to the approval of the Engineer, attaining the required strength and other properties of the concrete. Following are the requirements for Designed Mixes.

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- a) The mix design shall be done considering the cement content limits as prescribed for various grade of concrete **(CVC circular No.34/10/10 dated 07.10.2010 circulated vide PCE/WCR letter No.W HQ/PCE/Instruction dated 23.06.2015).**

| Grade of concrete | Cement content in kg. per cum of concrete |
|-------------------|-------------------------------------------|
| M 15              | 250 to 300                                |
| M 20              | 280 to 330                                |
| M 25              | 310 to 360                                |
| M 30              | 340 to 390                                |
| M 35              | 360 to 420                                |
| M 40              | 380 to 450                                |
| M 45 to M 55      | 400 to 450                                |

- b) If any change are required in cement content as specified above due to change in ingredient (Course aggregate, fine aggregate, cement, water etc.) of design mix, special approval of Chief Engineer (Constn.) to be taken before use in field.
- c) The contractor may be allowed to use mineral additive like silica fumes etc. to achieve the required strengths at his own cost.
- d) Whenever there is significant change in the quality of any of the ingredients for concrete, the Engineer may at his discretion order to carry out of fresh trial mixes. All costs for trial mixes & tests shall be to the contractors account and held to be included in the contract rates.
- e) When the proportions of a concrete mix have been approved by the Engineer, the contractor shall not vary the quality or source of the materials or the mix without the written approval of the Engineer.

#### **9. Special Measures**

- a) Mixing Concrete: Concrete shall be mixed in automatic batching and mixing plant. Hand mixing shall not be permitted. The mixers or the plant shall invariably be fitted with water measuring (metering) devices. The Engineer shall approve the concrete mixer or the plant.
- b) Mixing shall be continued till materials are uniformly distributed and a uniform colour of the entire mass is obtained, and each individual particle of the coarse aggregate shows complete coating of mortar containing its proportionate amount of cement.
- c) The method of transporting and placing concrete shall be approved by the Engineer. Concrete shall be transported and placed as near as practicable to its final position, so that no contamination, segregation or loss of its constituent materials takes place.
- d) When concrete is conveyed by chute, the plant shall be of such size and design as to ensure practically continuous flow. Slope of the chute shall be so adjusted that the concrete flows without any segregation of its ingredients. The delivery end of the chute shall be as close as possible to the point of deposit. Concrete shall not be moved by vibrators. The chute shall be thoroughly flushed with water before and after each working period and the water used for this purpose shall be discharged outside the formwork.
- e) All formwork and reinforcement contained in it shall be cleaned and made free from standing water, dust, snow or ice immediately before placing of concrete.
- f) No concrete shall be placed in any part of the structure until the approval of the Engineer has been obtained. If concreting is not started within 24 hours of the approval being given, approval shall

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to be obtained again from the Engineer. Concreting then shall proceed continuously over the area between the construction joints

- g) Except where otherwise agreed to by the Engineer, concrete shall be deposited in horizontal layers to a compacted depth of not more than 300 mm
- h) Concrete for RCC and PSC, when deposited shall have a temperature of not less than 5 degrees Celsius, and not more than 30 degree Celsius. Use of ice may be required to lower the temperature of the concrete at the discretion of the engineer in charge.
- i) Concrete shall be thoroughly compacted by vibration using electrically driven needle vibrators or other suitable means to produce a dense homogeneous void-free mass having the required surface finish. When immersion type vibrator needles are used, contact of needle with reinforcement and all inserts like ducts etc. shall be avoided. Additional vibrators in serviceable condition shall be kept at site so that they can be used in the event of breakdowns.
- j) The permanent structure shall not be allowed to come in contact with sea water for at least 72 hours.

**10. Construction Joints :-**

Construction joints shall be avoided as far as possible. Sequencing of concrete placement should be organized in such a way that cold joints are totally eliminated. The sequence of concreting shall be submitted for approval of Railway prior to concreting of the structural element. Construction joints shall be located so as not to impair the strength of the concrete. Concrete placed to form the face of a construction joint shall have all laitance removed. One of the methods to remove laitance is applying retarders by brush on freshly laid concrete and wash it by water jet after initial setting of concrete. The laitance at top does not get set and easily removed by washing. Before concreting is resumed, all loose matter on the existing concrete surface shall be removed.

**11. Concreting Under Water:**

All concreting shall preferably be done under dry conditions & concreting under water conditions shall be avoided as far as possible. If concreting under water is not avoidable then following precaution shall be taken.

When it is necessary to deposit concrete under water, the methods, equipment, materials and proportions of mix to be used shall be got approved from the Engineer before any work is started and in such cases concrete shall contain 10 per cent more cement than that required for the same mix placed in the dry.

All under-water concreting shall be carried out by tremie method only, using tremie of appropriate diameter. The tremie concreting when started should continue without interruption for the full height of the member being concreted. The concrete production and placement equipment should be sufficient to enable the underwater concrete to be completed uninterrupted within the stipulated time

**12. Protection and Curing:**

Concreting operations shall not commence until adequate arrangements for curing have been made by the Contractor. Curing and protection of concrete shall start immediately after compaction of the concrete to protect it from:

Premature drying out particularly by solar radiation and wind  
Avoiding plastic shrinkage cracks  
High internal thermal gradients  
Leaching out by rain and flowing water

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- a) Water Curing: Only potable water shall be used. Sea/creek water shall not be used for curing. Sea/creek water shall not come into contact with concrete members unless it has attained adequate strength. Exposed surface of concrete shall be kept continuously in a damp or wet condition by pounding or by covering with a layer of sacks, canvas, Hessian or similar materials and shall be kept constantly wet for a period of not less than 21 days from the date of placing of concrete. In case of the Portlandpozzolana cement and Portland slag cement curing period shall be 28 days.
- b) Curing compound : Curing compounds shall normally not be used. If proposed to be used, they can be permitted at selected locations, at the directive of the Engineer.

**13. Finishing :-**

- a) Immediately after removal of forms, exposed bars or bolts, if any, shall be cut inside the concrete member to a depth of at least 50 mm below the surface of the concrete and the resulting holes filled with polymer cement grout.
- b) All construction and expansion joints in the completed work shall be left carefully tooled and free from any mortar and concrete. Expansion joint filler shall be left exposed for its full length with clean and true edges.
- c) The finished surfaces of concrete after removal of formwork should be such that no touching up is required. All kind of fins / protrusion caused by formwork joints, if any, shall be ground using electric surface grinder.
- d) Immediately on removal of forms, the concrete work shall be examined by the Engineer before any defects are made good.
- e) The work that has sagged or contains honeycombing to an extent detrimental to structural safety or architectural appearance shall be rejected.
- f) Surface defect of a minor nature may be accepted. On acceptance of such work by the Engineer, the same shall be rectified as directed by Engineer.

**14. Equipment :-**

All equipment used for concreting shall have the approval of engineer in charge.

**15. Measurement for Payment :**

Structural concrete shall be measured in cubic meters. In reinforced or pre-stressed concrete, the volume occupied by reinforcement or pre-stressing cables and sheathing shall not be deducted. Payment shall be made based on the dimensions as per the drawing if the structure is within prescribed tolerance.

The item for pre-stressed concrete girders/slabs is for providing in-situ girders/slabs. Subject to approval of methodology by engineer, the contractor will be allowed to cast and stress the PSC girder / slabs in casting yard and launch the same using approved methodology. However, payment will be made as per schedule item and nothing extra will be paid for casting and launching the girders / slabs separately.

**16. TESTS AND STANDARDS OF ACCEPTANCE**

Concrete shall conform to the surface finish and tolerance as prescribed in these specifications

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for respective components.

a) Open Foundation

- i) Variation in dimensions. : + 50 mm , - 10 mm
- ii) Misplacement from specified position in plan. : 15 mm
- iii) Surface irregularities measured with 3m straight edge. : 3 mm

b) Substructure /Superstructure

- i) Variation in cross-sectional dimensions. : + 10 mm, - 5 mm
- ii) Misplacement from specified position in plan. : 10 mm
- iii) Variations from plumb over full height. :  $\pm 10$  mm

**17. Sampling and testing :**

a) Compressive strength

- (i) Green concrete, for making 3 test cubes, shall be taken from a batch of concrete at point of delivery into construction, according to procedure laid down in IS:1199. Frequency shall be as per IS:456 –

| Quantity of Concrete<br>work, m <sup>3</sup> | No. of samples For 28 days                                                                        | No. of samples For 7 Days                                                                            |
|----------------------------------------------|---------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| 1-5                                          | 1 Set                                                                                             | 1 Set                                                                                                |
| 6-15                                         | 2 Sets                                                                                            | 2 Sets                                                                                               |
| 16-30                                        | 3 Sets                                                                                            | 2 Sets                                                                                               |
| 31-50                                        | 4 Sets                                                                                            | 3 Sets                                                                                               |
| 51 and above                                 | 4 Sets plus one Set additional<br>sample for each additional 50m <sup>3</sup><br>or part thereof. | 3 Sets plus one Set<br>additional sample for each<br>additional 50m <sup>3</sup><br>or part thereof. |

At least one sample shall be taken from each shift of work. Each set consists of 3 cubes.

- (ii) 150 mm cubes shall be made, cured and tested at the age of 28 days for compressive strength in accordance with IS: 456-2000.
- (iii) Three test specimens shall be made from each sample for testing at 28 day. Additional cubes shall be made for various purposes such as to determine the strength of concrete at 7 days for any other purpose as given in above table.
- (iv) The test strength of the sample shall be the average of the strength of 3 cubes.

b) Permeability test to be done as follows :-

1. Prepare the cylindrical test specimen 150 mm dia and 160 mm high.
2. After 28 days of curing, the test specimen is fitted in a machine such that the specimen can be

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placed in water under pressure upto 7 bars.

3. At first a pressure of one bar is applied for 48 hours, followed by 3 bars for 24 hours and 7 bars for next 24 hours.
4. After the passage of the above period, the specimen is taken out and split in the middle by compression applied on two round bars on opposite sides above and below.
5. The water penetration in the broken core is to be measured with a scale and the depth of

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penetration assessed in mm (max. permissible limit 25 mm).

6. Frequency of testing for permeability: 2 sets of 3 sample each per 100 m<sup>3</sup> of RCC/PSC concrete.

- Test for Total chlorides and Sulphates in concrete.

| Sr No. | Characteristic to be checked              | Acceptance Criteria ( Value)                                  | Test Frequency                        |
|--------|-------------------------------------------|---------------------------------------------------------------|---------------------------------------|
|        |                                           |                                                               |                                       |
| 1      | Total Chlorides in concrete as cl         | 0.06% by wt. Of Cement for PSC 0.15% by wt. of Cement for RCC | Every 500 m <sup>3</sup> of concrete. |
| 2      | Total Sulphate content as SO <sub>3</sub> | 4% by the mass of cement                                      |                                       |

**18. Acceptance criteria:** The concrete shall be deemed to comply the strength requirement if the test results on the concrete cubes satisfy the acceptance criteria given in IS-456 (Clause 16) i.e. any particular grade of concrete will be considered satisfactory if the 28 days' compressive strengths of individual sets (each set consists of 3 cubes) and of individual cubes satisfy the following requirements:

- a) The mean strength determined from any group of four consecutive test results is more or equal to the characteristic strength (fck) plus 0.825 times the established standard deviation (rounded off to nearest 0.5 N/mm<sup>2</sup>), ( $s \geq fck + 0.825 \sigma$  N/mm<sup>2</sup>).

OR

The mean strength determined from any group of four consecutive test results is more or equal to characteristic strength (fck) plus 4 N/mm<sup>2</sup> ( $s \geq fck + 4$  N/mm<sup>2</sup>).

- b) The strength from any individual test result does not fall below the characteristic strength (fck) minus 4 N/mm<sup>2</sup>.

**19.** Whenever a mix is redesigned due to a change in the quality of aggregate or cement or for any other reason, it shall be considered a new mix and initially subject to the acceptability criteria above.

**20.** If the concrete produced at site does not satisfy the above strength requirements, the Engineer will reserve the right to require the contractor to improve the methods of batching, the quality of the ingredients and redesign the mix with increased cement content, if necessary. The contractor shall not be entitled to claim any extra cost for the extra cement used for the modifications stipulated by the Engineer for fulfilling the strength requirements specified.

**21.** It is the complete responsibility of the contractor to design the concrete mixes by approved standard methods and to produce the reinforced concrete conforming to the specification and the strength requirements approved by the Engineer.

**22.** As frequently as the Engineer may require, testing shall be carried out in the field for:

1. Moisture content and absorption and density of sand and aggregate.
2. Silt content of sand.
3. Grading of sand and aggregates.
4. Slump test of concrete.
5. Concrete cube test
6. Permeability test for concrete
7. Density & PH value of plasticizer

**23.** Before concreting commences on any section of the works the contractor shall obtain approval of the

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Engineer or his representative as regards the formworks and reinforcements conforming to the drawings. He shall also indicate to the Engineer in writing and obtain his approval for positions of construction joints.

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24. **Cracks:** If cracks develop in concrete construction, which in the opinion of the Engineer may be detrimental to the strength of the construction, the contractor shall dismantle the construction, carry away the debris, replace the construction and carry out all consequential work thereto.

If any cracks develop in the concrete construction, which in the opinion of the Engineer, are not detrimental to the stability of the construction, the Contractor at his own expense shall grout the cracks with polymer cement grout of approved quality and also at his own expense and risk shall make good to the satisfaction of the Engineer.

The Engineer's decision as to the extent of the liability of the Contractor in the above matter shall be final and binding.

25. **Finishing of concrete:** Should any concrete be found honeycombed or in any way defective which may be suspected to affect the performance of the structure, shall be rejected outright. However, some surface defects like minor honey combing etc. not affecting the structural properties shall on the instruction of the Engineer be repaired as per the approved procedure. Immediately on removal of forms, the RCC work shall be examined by the Engineer, before any defects are made good.

1) The work that has sagged or contains honey combing to an extent detrimental to structural safety or architectural concept shall be rejected as given in Para 4.4.10.5 based on visual inspection test.

2) Surface defects of a minor nature may be accepted. On acceptance of such a work by the Engineer, the same shall be rectified as follows: -

i) Surface defects which require repair when forms are removed, usually consist of bulges due to movement of forms, ridges at form joints, honey-combed areas, damage resulting from the stripping of forms, and bolt holes.

ii) Bulges and ridges are removed by careful chipping or tooling and the surface is then rubbed with a grinding stone.

iii) Honey-combed and other defective areas must be chipped out, the edges being cut as straight as possible and perpendicularly to the surface, or preferably slightly undercut to provide a key at the edge of the patch-

a) Shallow patches are first treated with a coat of thin grout composed of one part of cement and one part of fine sand and then filled with mortar similar to that used in the concrete. The mortar is placed in layers not more than 10 mm thick and each layer is given a scratch finish to secure bond with the succeeding layer. The last layer is finished to match the surrounding concrete by floating, rubbing or tooling on formed surfaces by pressing the form material against the patch while the mortar is still plastic.

b) Large and deep patches require filling up with concrete held in place by forms. Such patches are reinforced and carefully dowelled to the hardened concrete.

3) Holes left by bolts are filled with mortar carefully packed into places in small amounts. The mortar is mixed as dry as possible, with just enough water so that it will be tightly compacted when forced into place.

4) Tiered holes extending right through the concrete may be filled with mortar with a pressure gun similar to the gun used for greasing motor cars.

5) Normally, Patches appear darker than the surrounding concrete, possibly owing to the presence on their surface of less cement laitance. Where uniform surface colour is important this defect shall be remedied by adding 10 to 20 percent of white Portland cement to the patching mortar, the exact quantity

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6) The same amount of care to cure the material in the patches should be taken as with the whole structure. Curing must be started as soon as possible, after the patch is finished to prevent early drying. Damp hessian may be used but in some locations it may be difficult to hold it in place. A membrane curing compound in these cases will be most convenient.

26. Wherever IRS/BIS or other codes have been revised and their provisions are to be adopted superseding above provisions, the Engineer can decide so, the decision of Engineer in charge in this regard will be binding on the contractor.

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**Chapter-9**  
**FORMWORK**

1. Description: Formwork shall include all temporary or permanent forms required for forming the concrete of the shape, dimensions and surface finish as shown on the drawing or as directed by the Engineer, together with all props, staging, centering, scaffolding and temporary construction required for their support.
2. Materials: Materials and components used for formwork shall be examined for damage or excessive deterioration before use and shall be used only if found suitable after necessary repairs. Forms shall be constructed with metal. The metal used for forms shall be of such thickness that the forms remain true to shape. The use of approved internal steel ties or steel or plastic spacers shall be permitted. All shuttering plates, props, H frames, screw jacks and all necessary fittings required for formwork shall be of steel. Wooden planks and props shall not be allowed to be used for formwork and staging work.
3. Design of Formwork : The Contractor shall furnish the drawing of complete formwork (i.e. the forms as well as their supports) for approval of the Engineer before any erection is taken up. Notwithstanding any approval or review of drawing and design by the Engineer, the Contractor shall be entirely responsible for the adequacy and safety for formwork. The foundation of all supports shall be designed to suit the bearing capacity of soil to support the designed loads without settlement.
4. The formwork shall be robust and strong and the joints shall be leak-proof. Tunnel formwork or monolithic formwork will be preferred for RCC boxes.
5. The number of joints in the formwork shall be kept to a minimum by using large size panels. The design shall provide for proper “soldiers” to facilitate alignment. All joints shall be leak proof and must be properly sealed. Use of PVC JOINT sealing tapes, foam rubber or PVC T-section is essential to prevent leakage of grout.
6. As far as practicable, clamps shall be used to hold the forms together.
7. Use of ties shall be restricted, as far as practicable. Wherever ties are used they shall be used with HDPE sheathing so that the ties can easily be removed. No parts prone to corrosion shall be left projecting or near the surface.
8. Unless otherwise specified, or directed, chamfers or fillets of sizes 25 mm x 25 mm shall be provided at all angles of the formwork to avoid sharp corners. The chamfers, bevelled edges and moulding shall be made in the formwork itself. Opening for fixtures and other fittings shall be provided in the shuttering as directed by the Engineer.
9. Shuttering for walls, sloping members and thin sections of considerable height shall be provided with temporary opening to permit inspection and cleaning out before placing of concrete.
10. The formwork shall be so made as to produce a finished concrete true to shape, line and levels and dimensions as shown on the drawings, subject to the tolerances specified in respective sections of these specifications, or as directed by the Engineer.
11. Where metal forms are used, all bolts and rivets shall be countersunk and well ground to provide a smooth, plane surface.
12. Forms shall be made sufficiently rigid by the use of ties and bracings to prevent any displacement or sagging between supports. They shall be strong enough to withstand all pressure, ramming and vibration during and after placing the concrete. Screw jacks or hard wood wedges, where required, shall be provided to make up any settlement in the formwork either before or during the placing of concrete.
13. The formwork shall be coated with an approved release agent that will effectively prevent sticking and will not stain the concrete surface.

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14. The formwork shall be constructed with pre-camber to the soffit to allow for deflection of the formwork.

15. **Formed Surface and Finish** :- The formwork shall be lined with material approved by the Engineer

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so as to provide a smooth finish of uniform texture and appearance. This material shall leave no stain on the concrete and so fixed to its backing as not to impart any blemishes. It shall be of the same type and obtained from only one source throughout for the construction of any one structure. The contractor shall make good any imperfections in the resulting finish as required by the Engineer. Internal ties and embedded metal parts shall be carefully detailed and their use shall be subject to the approval of the Engineer.

**16. Precautions**

- A. Special measures in the design of formwork shall be taken to ensure that it does not hinder the shrinkage of concrete. The soffit of the formwork shall be so designed as to ensure that the formwork does not restrain the shortening and/or hogging of beams during pre-stressing. The forms may be removed at the earliest opportunity subject to the minimum time for removal of forms with props retained in position.
- B. Provision shall be made for safe access on, to and about the formwork at the levels as required.
- C. Close watch shall be maintained to check for settlement of formwork during concreting. Any settlement of formwork during concreting shall be promptly rectified

**17. Preparation of Formwork before Concreting**

The inside surfaces of forms shall, except in the case of permanent form work or where otherwise agreed to by the Engineer be coated with a release agent supplied by approved manufacturer or of an approved material to prevent adhesion of concrete to the formwork.

**18. Before re-use of forms, the following actions shall be taken :**

- i) The contact surfaces of the forms shall be cleaned carefully and dried before applying a release agent.
- ii) It should be ensured that the release agent is appropriate to the surface to be coated. The same type and make of release agent shall be used throughout on similar formwork materials and different types should not be mixed.
- iii) The form surfaces shall be evenly and thinly coated with release agent. The vertical surface shall be treated before horizontal surface and any excess wiped out.
- iv) The release agent shall not come in contact with reinforcement or the hardened concrete.
- v) All forms shall be thoroughly cleaned immediately before concreting.
- vi) The Contractor shall give the Engineer due notice before placing any concrete in the forms to permit him to inspect and approve the formwork, but such inspection shall not relieve the contractor of his responsibility for safety of formwork, men, machinery, materials and finish or tolerances of concrete.

**vii) Removal of Formwork**

1) The scheme for removal of form work (i.e. de-shuttering and de-centering) shall be planned in advance and furnished to the Engineer for scrutiny and approval. The form work shall be so removed as not to cause any damage to concrete. Centering shall be gradually and uniformly lowered in such a manner as to permit the concrete to take stresses due to its own weight uniformly and gradually and any shock or vibration to structure is avoided.

2) Where not specifically approved, the time of removal of form work (when ordinary Portland Cement is used without any admixtures at an ambient temperatures exceeding 10 degrees Celsius) shall be as under:

|    |                                                                           |                                                  |
|----|---------------------------------------------------------------------------|--------------------------------------------------|
| a) | Walls, piers, abutments, columns and vertical faces of structural members | 12 to 48 hours as may be decided by the Engineer |
| b) | Soffits of Slabs (with props left under)                                  | 3 days                                           |
| c) | Props (left under slabs)                                                  | 14 days                                          |
| d) | Soffit of Girders (with props left under)                                 | 7 days                                           |
| e) | Props (left under girders)                                                | 21 days                                          |

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- 3) Where there are re-entrant angles in the concrete sections, the form work should be removed at these sections as soon as possible after the concrete has set, in order to avoid cracking due to shrinkage of concrete.
- 4) When form work is dismantled, its individual components shall be examined for damage and damaged

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pieces shall be removed for rectification. Such examination and rectification shall always be carried out before being used again. Before re-use all components shall be cleaned of deposits of soil, concrete or other unwanted materials. Threaded parts shall be oiled after cleaning.

All bent steel props shall be straightened before re-use. The maximum deviation from straightness is 1/600 of the length. The maximum permissible axial loads in used props shall be suitably reduced depending upon their condition. The condition of the timber components, plywood and steel shuttering plates shall be examined closely for distortion and defects before re-use

5) Measurements for Payment:

The rate for concrete otherwise specified in Plain Concrete or Reinforced Concrete does not include cement, formwork & temporary work & they shall be paid in respectively in appropriate item.

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**Chapter- 10**  
**REINFORCEMENT**

**1. Steel:** Steel used in the works shall be Thermo Mechanically Treated high yield strength bars of FE-500 D / FE 550 D conforming to IS: 1786-1985 (latest revision). TMT reinforcement steel bars to be procured from approved list of RDSO vendors and as approved by Engineer -in -charge

**2. Inspection & Testing**

Every bar shall be inspected before assembling on the works and any defective, brittle, excessively rusted or burnt bars shall be removed. Cracked ends of bars shall be cut out. Physical and Chemical tests. (IS – 2062, IS – 432) Test Frequency:-

- (i) Upto 10 mm. dia. : one sample for 25 MT or part thereof for each dia. of bar.
- (ii) Above 10 mm. dia. : one sample for 40MT or part thereof for each dia. of bar.

**3. Physical requirement & testing**

Test Frequency: -

- (i) Upto 10 mm. dia. : one sample for 25 MT or part thereof for each dia. of bar.
- (ii) Above 10 mm. dia. : one sample for 40MT or part thereof for each dia. of bar.

**4. Physical tests:** Following lab tests shall be carried out.

- (iii) Tensile test. This shall be done as per IS:1608.
- (iv) Bend Test : This shall be done as per IS:1599.
- (v) Rebend Test : This shall be done as per IS : 1786.
- (vi) Elongation percent minimum.
- (vii) 0.2% percent proof stress / yield stress.

**5. Chemical Tests:**

The manufacturer's test certificate shall be produced during procuring of material. If required same shall be tested in laboratory. The cost of such testing shall be borne by the contractor.

**6. Stacking and storage :**

- (i) Steel for reinforcement shall be stored in such a way as to prevent distortion and corrosion. Care shall be taken to protect the reinforcement from exposure to saline atmosphere during storage, fabrication and use. It may be achieved by treating the surface of reinforcement with cement wash or by other suitable methods. Bars of different classifications, sizes and lengths shall be stored separately to facilitate issue in such sizes and lengths to cause minimum wastage in cutting from standard lengths.
- (ii) In case of long storage reinforcement bars shall be stacked above ground level by at least 150 mm upon platforms, skids, or other suitable supports to avoid distortion of sections. A coat of cement wash shall be given for such long storage to prevent scaling and rusting.
- (iii) Lapping & Welding: Reinforcement includes authorized spacer bars and overlaps. Wastage and unauthorized overlaps shall not be paid for. The rate of reinforcement shall include cost of labour and materials required for all operation including cleaning of bars, straightening, cutting, hooking, bending, binding placing in position etc as required or directed including tack welding on crossing of bars in lieu of binding with wires as well as welding of wires in lieu of overlaps.
- (iv) Welding of Bars: Wherever facility for electric welding is available, welding of bars shall be done in lieu of overlap. The location and type of welding shall be got approved by the Engineer. Welding shall be as per IS: 2751 and IS: 9417.
- (v) Spacing, Assembly & Supporting: General Requirements: Steel conforming to para 4.1.2 for

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reinforcement shall be clear and free from loose mill scales, dust, loose rust, coats of paints, oil or other coatings, which may destroy or reduce bond. It shall be stored in such a way as to avoid distortion and to prevent deterioration and corrosion. Prior to assembly of reinforcement on no account and oily substance shall be used for removing the rust.

- (vi) Assembly of reinforcement: Bars shall be bent correctly and accurately to the size and shape as shown in the detailed drawing or as directed by Engineer. Preferably bars of full length shall be used. Necessary cutting straightening is also including in the rates and no extra payment will be paid for the same. Over lapping of bars, where necessary, shall be done as directed by the Engineer. The overlapping bars shall not touch each other and these shall be kept apart with concrete between them by 25 mm or 1 ¼ times the maximum size of the coarse aggregate whichever is greater. But where this is not possible, the overlapping bars shall be bound together at intervals not exceeding twice the dia of such bars with two strands annealed steel wire of 0.90 mm to 1.6 mm twisted tight. The overlaps / splices shall be staggered as per directions of the Engineer. But in no case the over lapping shall be provided in more than 50% of cross sectional area at one section.
- (vii) Supporting: The contractor shall provide approved types of supports as specified on the drawings for maintaining the top bars of the slab in position during concreting. All cover blocks shall be of concrete (not sand cement mortar) and of the same strength as that of the surrounding concrete and properly compacted and vibrated on a vibrating table. They shall be cured for a minimum period of 21 days before they are used the work. PVC or similar material cover block can be permitted with specific approval of engineer in charge.
- (viii) Cleaning: Bars must be cleaned before concreting commences of all scale, rust or partially set concrete which may have been deposited there during placing of previous lift of concrete.

**7. Measurement for Payment**

- a. Reinforcement including authorized spacer bars and laps shall be measured in length of different diameters, as actually (not more than as specified in the drawings) used in the work nearest to a centimeter and their weight calculated on the basis of standard weight table. Wastage and unauthorized overlaps shall not be paid for. Annealed steel wire required for binding or tack welding shall not be measured, its cost being included in the rate of reinforcement.
- b. Wherever tack welding is used in lieu of binding, such welds shall not be measured. Chairs, separators etc. shall be provided as directed by the Engineer and measured separately and paid for.

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**Chapter-11**  
**PRESTRESSING**

**1. Materials**

- a. Prestressing Steel- Low Relaxation strands conforming to IS:14268-1995 would be used. The prestressing steel should be subjected to an acceptance test prior to their actual use, in accordance with BS: 4447.
- b. Prestressing Accessories-Prestressing accessories i.e. anchorages and grips should be subjected to an acceptance test, prior to their actual use on the works, in accordance with BS:4447.
- c. Sheathing-The sheathing ducts shall be of the spiral corrugated type. Unless otherwise specified, the material shall be either galvanized Cold Rolled Cold Annealed (CRCA) Mild Steel conforming to IS:513 intended for mechanical treatment and surface refining but not for quench hardening or tempering or HDPE sheathing of approved manufacturer.
- d. In view of the aggressive environment, galvanised mild steel strips or HDPE sheathing conforming to IRC: 18-2000 D.3.6.2 shall be used.
- e. The thickness of sheathing shall be as shown on the drawing, but shall not be less than 0.5 mm with sheathing ducts having external diameter of 90 mm or as shown in drawing.
- f. The galvanised sheathing shall conform to the requirement as per tests specified in IRC 18.
- g. The sheathing ducts shall be manufactured at the project site utilising appropriate machines. The manufacturing shed should have concrete flooring. Sheathing will be manufactured just prior to its use. Storage at site shall be kept absolutely minimum. Where sheathing duct joints are unavoidable, such joints shall be made slurry tight by the use of corrugated threaded sleeve couplers which may be tightly screwed onto the outer side of the sheathing ducts.
- h. The length of the coupler should not be less than 150 mm but should be increased up to 200 mm wherever practicable. The joints between the ends of the coupler and the duct shall be sealed with heat shrink adhesive sealing tape to prevent penetration of cement slurry during concreting. The couplers of adjacent ducts should be staggered wherever practicable. As far as possible, couplers should not be located in curved zones.
- i. Cables shall be threaded after concreting. A temporary tendon shall be inserted in the sheathing or the sheathing shall be stiffened by other suitable method during concreting to prevent its collapse or ovalisation. As an alternative HDPE tube of slightly less diameter may also be inserted inside the sheathing.
- j. Anchorage  
Anchorage shall be procured from authorized/approved manufacturers only viz. Freyssinet system, Dynamic, CCL system etc.. Anchorages shall conform to BS:4447. Test certificates from a laboratory fully equipped to carry out the tests shall be furnished to the Engineer. Such test certificates shall not be more than 12 months old at the time of making the proposal for adoption of a particular system for the project. However, the anchorages will be required to be tested in an approved laboratory.  
No damaged anchorages shall be used. Steel parts shall be protected from corrosion at all times. Threaded parts shall be protected by greased wrappings and tapped holes shall be protected by suitable plugs until used. The anchorage components shall be kept free from mortar and loose rust and any other deleterious coating.
- k. Storage of Materials

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(i) HTS strands should be coated with water soluble oil and wrapped in HDPE wrappers.

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Prestressing strands should be stored in a godown where humidity is controlled up to 60 percent with the help of electric heaters and a number of perforated pouches of dehumidifying powder may be suspended in the storage godown. Dry & wet bulb thermometers (Hygrometers) may be installed inside the godown to check the humidity from time to time. Double track doors are to be provided in the storing place.

(ii) Sheathing coils, properly wrapped in waterproof wrapper, is to be stored in a humidity controlled godown.

(iii) Workmanship

(iv) Cleaning

Tendons shall be free of loose rust, oil, grease, tar, paint, mud or any other deleterious substance. Cleaning of the steel may be carried out by immersion in suitable solvent solution and compressed air.

(v) Straightening-High tensile strand shall be supplied in coils of sufficiently large diameter such that tendons shall retain their physical properties and shall be straight as it unwinds from the coil. Tendons of any type that are damaged, kinked or bent shall not be used.

(vi)The packing of prestressing strand shall be removed only just prior to making of cable for placement. Suitable stands shall be provided to facilitate uncoiling of strands without damage to steel. Care shall be taken to avoid the possibility of strand coming into contact with the ground.

(vii) Positioning- Prestressing tendons shall be accurately located and maintained in position, both vertically and horizontally, as per drawings.

(viii) Tendon shall be so arranged that they have a smooth profile without sudden bends or kink

(ix) Installation of Strands

(x) For making the cable at site firstly the strands are cut to length, then made into a cable on a working bench using a device called a former. The cable is tied with wires placed about every 80.0cm. In most of the cases the ties are removed as the cable is introduced in the duct. Since the duct diameter is sufficiently large the ties may be left in place. The cable is threaded into the duct either by pulling it from the far end by suitable means, or by pushing it from the near end. Individual strands can also be threaded using a cable threading machine.

(xi) Cable Supporters-Sheathing shall be placed in correct position and profile by providing suitable ladders and spacers. Such ladders may be provided at intervals of approximately 0.7 m. Sheathing shall be tied rigidly with such ladders/spacer bars so that they do not get disturbed during concreting.

(xii) The method of supporting and fixing shall be such that profile of cables is not disturbed during vibrations, by pressure of wet concrete, by workmen or by construction traffic.

(xiii) Profiling- The profiling of cable should be such as to give a distribution of prestressing in each section which corresponds to the stresses under the applied loads. The profiling of cable is done in accordance with the ordinates of x-axis and y-

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axis as given in the working drawings. Sufficient number of cable ordinates shall be given to fix exact location of cable. For profiling, the cables are attached to horizontal supporter bars fixed to the stirrups of the beams. The stirrups must be fairly strong and rigid to support the weight of the cable. "U" shaped saddles of 8 mm mild steel welded to one branch of the stirrup are often used to fix the cable. The supporters shall be spaced 70.0 cms. apart. The first support should be 30.00 cm. from the anchorage so as to ensure that the cable is coaxial with the anchorage and firmly held during concreting.

(xiv) Precaution During Profiling

(j) The following precautions should be taken during profiling:-

- a) Tendons shall be handled with care to avoid damage or contamination, to either the tendon or the sheathing. Any tendons damaged or contaminated shall be cleaned or replaced.
- b) Between the points given, the cable must retain a regular curvature without undulations or sharp changes in the direction which will cause increase in the friction loss at the time of stressing.
- c) The cable axis must coincide with the anchorage axis for a length of, at least 40 cm behind the anchorage.
- d) Strand should be uncoiled in such a manner that it is not twisted.
- e) All the strands of the cable should be numerically identified with paint, at both ends, before threading. This will avoid any internal twist of the strands.
- f) When the strands are pulled into the tendon ducts together in bundle, special care should be taken that it does not damage the duct or the strands.
- g) Strands should be cut evenly by abrasive cutter to desired length. Flame cutting shall not be permitted.
- h) Identification label from strand coil should be kept in safe case. Strand data from identification label and the tendon numbers in which the strands are used should be entered in construction note.
- i) Checking of Cable After Cable Placing-Successful and quick stressing and grouting operations largely depend upon the care taken with cable and reinforcement placing. After completion of profiling and placing the following must be checked: -

a) Cable Profile

The cable must follow a regular profile, without wobbles. This condition is important to avoid abnormally high friction losses when stressing. The cable ordinates must be checked. The cable must be straight and coaxial with the anchorage for a distance of at least of 70.0cm, from the end of the girder. It must be firmly fixed so as to prevent sagging of cable under the weight of concrete.

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b) Water Tightness of the Cable

The sheath should be inspected when placed. Damages caused by handling and placing must be repaired with adhesive tape.

c) Strength of the Supports

The cable must be firmly held vertically and horizontally and must remain in position during concrete placing and vibration. Welded saddles are the most reliable cable supports.

d) Anchorage Zone

The sheath connection with a precast end block must be water tight. The joint between Anchorage and form work must also be water tight.

e) Temporary Tendon- A temporary cable comprising of 19T13 strands shall be inserted in the ducts to prevent opalization of the ducts. HDPE tube may also be used.

f) Cutting- Cutting and trimming of wires or strands shall be done by suitable mechanical cutters only. The ends of prestressing steel, keeping 20 mm projection beyond the wedge shall be cut after the grout has set.

g) Protection of Prestressing Steel-Prestressing steel shall be continuously protected against corrosion, until grouted. The corrosion protector shall have no deleterious effect on the steel or concrete or on the bond strength of steel to concrete.

h) Concreting-

After completion of the profiling, the concreting of the member is done. Special care is required during this operation. The following precautions should be taken during concreting :

- a) The sheathing should be protected from getting punctured by any sharp object. In case of any damage to the sheath it should be repaired immediately with adhesive tape.
- b) Care should be taken that profiling is not disturbed.
- c) The coupler joints should be checked at regular intervals to ensure that joints are sealed properly and mortar is not entering the duct.
- d) The temporary cable/HDPE tube should be hammered from both the ends to prevent the setting of such mortar that may have entered the duct from any loose joint or hole.
- e) The concrete should be of sufficient workability.
- f) When the cable supports are not welded, the concrete must not be dumped on the cables in a manner which would tear away the support.

(i) Stressing Data & Record to be Maintained

- 1. Stressing sequence
- 2. Tendon identification
- 3. Tendon length (from anchorage end to anchorage end)
- 4. Design elongation of tendon ( $l_z$ )
- 5. Elastic shortening of concrete = ( $l_b$ )
- 6. Wedge Set at anchorage ( $l_s$ )
- 7. Elongation of the prestressing steel within the stressing jack (from bearing plate to master grip) ( $l_e$ )
- 8. Total calculated elongation  $l_o = l_z + l_b + l_e + l_s$
- 9. Specified stressing force
- 10. Specified cross section area of prestressing steel
- 11. Specified Modulus of prestressing steel
- 12. Actual cross section area of prestressing steel installed in the tendon
- 13. Actual Modulus of Elasticity

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14. Elongation that may be obtained should be estimated on the basis of modulus of elasticity and

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cross-sectional area of prestressing steel used.

(j) Stressing

(k) Stressing Schedule- The following information is of prime requirement before stressing operation. (l) The elongation and jacking force for each cable: The initial prestressing force required and the stress strain diagram of the proposed steel must be used for these calculations.

- (i) The minimum concrete strength before stressing.
- (ii) The stressing sequence of the cable.
- (iii) Special requirements concerning removal of scaffolding corresponding to the stressing phase.
- (iv) Prestart Checks- Stressing can begin when the following conditions are met :
- (v) Friction loss of the jacks have been determined.
- (vi) The concrete strength specified has been reached.
- (vii) The cable ends are accessible.
- (viii) The stressing equipment's are in working order and the gauges are calibrated.
- (ix) The jack is placed in position.

(x) Stressing Procedure

a) Preparation of the Cable and the Anchorage

The strand surface must be cleaned of all dirt by emery paper so that adequate friction is generated between strand and grips. Traces of mortar must be removed with a steel brush. The internal surface of the guide must be cleaned.

b) Placing of Bearing Plate

The strands may be threaded through the bearing plate and the grips placed in position.

c) Placing the Jack-The grooves of the jack may be greased for easy de-wedging of jacks using graphite grease or if not available, using paraffin wax. The inside of the barrels of anchors must be greased.

d) The jack may be placed on the anchorage with each strand in a groove of the jack. The jack may be pushed to bring it as close as possible to the concrete face. The wedges are then placed using sharp hammer blows making sure that the strands lie correctly aligned in the grooves of the jack. For jacks with individual grips, the strands may be attached firstly by threading of the barrels and then placing the grips and driving them home using a piece of tube supplied with the jack. The grip serrations may be checked and cleaned when necessary.

e) Stressing of Cables-Tensioning force shall be applied in gradual and steady steps and carried out in such a manner that the applied tensions and elongations can be measured at all times. The sequence of stressing, applied tensions and elongations shall be in accordance with the approved drawing or as directed by the Engineer.

f) It shall be ensured that in no case, the load is applied to the concrete before it attains the strength specified on the drawing or as stipulated by the prestressing system supplier, whichever is more.

g) After prestressing steel has been anchored, the force exerted by the tensioning equipment shall be decreased gradually and steadily so as to avoid shock to the prestressing steel or anchorage.

h) The tensioning force applied to any tendon shall be determined by direct reading of the pressure gauges or dynamo-meters and by comparison of the measured elongation with the calculated elongation. The calculated elongation shall be invariably adjusted with respect to the modulus of elasticity of steel for the particular lot as given by the manufacturer.

i) The operation must be carried out as follows:

a. All the leads are plugged.

b. One end of the second lead, should be attached to the blocking chamber of the jack so as to discharge the fluid in the blocking chamber into the reservoir of the pump.

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c. After stressing to 50.0 Kg/cm<sup>2</sup> approximately, release the jack holding arrangement.

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- d. The pump may be operated until the cable has reached the required stress and elongation. The pressure drop should not be allowed while taking readings. Elongation reading may be taken when the pump is stopped.
- e. The tension in the stressing chamber may be held by closing the valve on the lead.
- f. If the calculated elongation  $A_0$  is reached before the calculated gauge pressure  $P_0$ , stressing is continued until  $P_0$ , provided the elongation does not exceed  $1.05 A_0$ .
- g. In case this elongation  $1.05 A_0$  is reached before  $P_0$ , stressing may be stopped.
- h. If at pressure  $P_0$  the elongation  $A_0$  has not been reached, tensioning may be continued until elongation  $A_0$  has been reached provided the pressure does not exceed  $1.05 P_0$ .
- i. If the elongation at  $1.05 P_0$  is less than  $0.95 A_0$  the following measures must be taken in succession to define the cause of this lack of elongation.
- j. The pressure gauge may be recalibrated.
- k. The correct functioning of the pump and leads may be checked.
- l. The cable is detensioned and slid in its duct to check that it is not blocked by mortar which may have entered through holes in the sheath.
- m. The jack's internal friction should be determined by testing.
- n. Re-establish the modulus of elasticity of steel.
- o. A complete record of prestressing operations along with elongation and jack pressure data shall be maintained. A format shall be submitted well in advance for approval.
- p. Grouting-  
The following conditions must be present in order to ensure a good grouting operation:
  - a. The grout must completely fill the duct, without air pockets or bleeding pockets.
  - b. The grout must not contain any component which could attack the steel.
  - c. The duct must in no way have obstacles to the flow of grout and must be as uniform as possible without sudden changes of cross section.
  - d. The area of the free space in the sheath must be at least equal to the area of the prestressing steel.
  - e. The grouting equipment must be sufficiently powerful to ensure the passage of the grout from one end to the other end of the duct, in spite of the head loss.
  - f. Properties of Grout- The water/cement ratio should be as low as possible, consistent with workability, but not exceeding 0.40. The grout should be homogeneous and sufficiently fluid, having maximum bleeding not more than 2%. Only Portland cement not more than 2 months old shall be used. Admixture if used shall be chloride free. The quantum of admixture to be used shall be as per manufacturer's recommendation and field trials.
  - g. The compressive strength of 100 mm cubes of grout shall not be less than 17 MPa at 7 days. The properties and suitability of the grout shall be established by a mock-up trial.
  - h. Batching & Mixing of Grout- All materials shall be batched by weight except the mixing water and liquid admixture which may be batched by weight or volume. The accuracy of batching shall be:
    - + 2% for cement and admixture
    - + 1% for mixing water
  - i. The mixing water shall include the water content of liquid admixture. Depending upon the environmental or material influence e.g. temperature, configuration of the tendon and properties of the materials used, the w/c ratio shall be kept as low as possible giving regard to the required

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properties ofthe grout.

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- j. The maximum water cement ratio (w/c) shall be 0.40. The grout shall be mixed in a mixing tank filled with motorised high speed colloidal mixer. The grout will then be transferred to an agitation tank where the mixed grout will be continuously agitated with a slow speed agitator. The outlet of the agitation tank will be connected to the grout pump. Unless manufacturers specify otherwise, water shall be added to the mixer first, followed by the dry materials which may be added as a whole or in part in sequence until the total quantities are added. The minimum mixing time determined from grouting trials shall be adhered to and shall generally not exceed 3 minutes.
  - k. The temperature of freshly mixed grout shall be 5 C minimum and 25 C maximum. Grouting procedure shall be as per Appendix-5 of IRC:18.
- (a) Anchorage
- (b) All bearing surfaces of the anchorages shall be cleaned prior to concreting and tensioning.
  - (c) Anchor cones, blocks and plates shall be securely positioned and maintained during concreting such that the centre line of the duct passes axially through the anchorage assembly.
  - (d) The cast iron guide shall be recessed from the concrete surface by a minimum cover of 150 mm.
  - (e) After the prestressing operations are completed and prestressing strands are cut, the surface shall be painted with two coats of epoxy of suitable formulation having a dry film thickness of 80 microns per coat and entire recess shall be filled with concrete or non-shrink/pre-packaged mortar or dense cementitious polymer modified mortar. Plastic cap or strands to be fixed with epoxy glue.
  - (f) Tensioning Equipment- All tensioning equipment shall be procured from authorised manufacturers only and be approved by the Engineer prior to use. Where hydraulic jacks are used, they shall be power-driven unless otherwise approved by the Engineer. The tensioning equipment shall satisfy the following requirement :
    - i) The means of attachments of the prestressing steel to the jack or any other tensioning apparatus shall be safe and secure.
    - ii) A single multipull stressing jack shall be used which is capable of tensioning simultaneously all the strands of the tendon. Suitable facilities for handling and attaching the multipull jack to the tendons shall be provided. The jacks shall have hydraulic blocking system.
    - iii) The tensioning equipment shall be such that it can apply controlled total force gradually on the concrete without inducing dangerous secondary stress in steel, anchorage or concrete; and
    - iv) Means shall be provided for direct measurement of the force by use of dynamo-meters or pressure gauges fitted in the hydraulic system itself to determine the pressure in the jacks. Facilities shall also be provided for the linear measurement of the extension of prestressing steel to the nearest mm and of any slip of the gripping devices at transfer.
    - v) Stressing jack shall be tested for friction loss prior to use. Friction loss should not exceed 3 %. all dynamo meters and pressure gauges including a master gauge shall be calibrated by an approved laboratory immediately prior to use and then at intervals not exceeding 3 months and the true force determined from the calibration curve.

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- vi) Pressure gauges shall be concentric scale type gauges accurate to within two per cent of their full capacity. The minimum nominal size of gauge shall be 100 mm. The gauge shall be so selected that when the tendon is stressed to 75 per cent of its breaking load, the gauge is reading between 50 per cent and 80 per cent of its full capacity. Suitable safety devices shall be fitted to protect pressure gauges against sudden release of pressure.
- vii) Provision shall be made for the attachment of the master gauge to be used as a check whenever requested for by the Engineer.
- viii) Grouting Equipment- All grouting equipment's shall be procured from authorized manufacturers only and be approved by the Engineer prior to use.
- ix) Two separate tanks will be fabricated - one for mixing and one for agitation. Grout will be mixed in the mixing tank with a high-speed mixer capable of producing a colloidal mix. Mixing time will be generally between 2 to 3 minutes. Tests will be done at site to arrive at the optimum mixing time for a particular equipment. After mixing, grout will be transferred to agitation tank. The grout will be kept under continuous agitation by a slow speed agitator so that there is no settlement of cement particles in the period between mixing and pumping.
- x) Grout pump shall be positive displacement type and should be capable of pumping the grout in a continuous operation and not by way of pulse. The pump must be fitted with a pressure gauge. The pump must have a relief arrangement for bypass of the grout in case of pressure built up. Grout shall be pumped at pressure around 0.3 Mpa which should be gradually increased to 0.5 Mpa. The speed of grouting should not be more than 10 metres per minute.
- xi) A direct feed high pressure water pump shall be available for emergency. The grouting equipment should contain a screen having a mesh size of IS:106. This should be fitted at the exit of the mixing tank.
- xii) Safety Precautions During Tensioning
  - (a) Care shall be taken during tensioning to ensure the safety of all persons in the vicinity.
  - (b) Jacks shall be secured in such a manner that they will be held in position, should they lose their grip on the tendons.
  - (c) No person shall be allowed to stand behind the jacks or close to the line of the tendons while tensioning is in progress.
  - (d) The operations of the jacks and the measurement of the elongation and associated operations shall be carried out in such a manner and from such a position that the safety of all concerned is ensured.
  - (e) A safety barrier shall be provided at both ends to protect against accidental flying of the wedges. During actual tensioning operation, warning sign shall be displayed at both ends of the tendon.
  - (f) After prestressing, concrete shall neither be drilled nor any portion cut nor chipped away nor disturbed, without express approval of the Engineer. Neither welding shall be permitted on/near tendons, nor any heat shall be applied to tendons. Any tendon which has been affected by welding, weld spatter or heat shall be rejected.
- (xiii) Transportation and Storage of Precast Units-
  - (a) Precast girders shall be transported in an upright position. Points of support and the direction of

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reactions with respect to the girder shall approximately be the same during transportation, and storage as and when the girder is placed in final position. Precast girders shall not be handled before complete hardening of the grout.

- (b) When members are to be stacked, they shall be firmly supported at such bearing positions as will ensure that the stresses induced in them are always less than the permissible design stresses. Further, inclined side supports shall be provided at the ends and along the length of a precast girder to prevent lateral movements or instability.
- (c) Care shall be taken during storage, hoisting and handling of the precast units to prevent their cracking or damaging. Units damaged by improper storing or handling shall be replaced by the Contractor at his expense.

i. Tolerances

Permissible tolerances for positional deviation of prestressing tendons shall be limited to the following :

|    |                                                  |   |      |
|----|--------------------------------------------------|---|------|
| a) | Variation from the specified horizontal profile. | : | 5 mm |
| b) | Variation from the specified vertical profile.   | : | 5 mm |
| c) | Variation from the specified position in member. | : | 5 mm |

ii. Measurement for Payment

Prestressed concrete shall be measured in cubic metres. The volume occupied by mild steel reinforcement/HYSD bars, high tensile steel, sheathing and anchorages shall not be deducted. High tensile (prestressing) steel shall be paid for separately and its length shall be measured as actually incorporated in the finished work. From the length so measured its weight shall be calculated in tonnes on theoretical basis and paid for.

Anchorage devices, additional length of cables for attaching jack, ducts or sheathing, grout, non-prestressed steel reinforcement fixed to the anchorage devices, making of recesses and filling the same, protection by painting with epoxy and furnishing samples for testing shall all be deemed to be included in the item of high tensile steel and shall not be measured and paid separately.

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**Chapter-12**  
**MISCELLANEOUS ITEMS**

**Coal Tar epoxy**

1. General :- All RCC substructure in contact with earth shall be given approved 2 coats of coal tar epoxy compound painting. The work shall include preparation of surfaces, application of protective covering and drying of the paint coatings.
2. Surface Preparation :- Surface to be painted shall be cleaned with wire brush & water to remove earth, dust etc..
3. Quality of Paint :- The paints which have been tested for the following qualities as per specifications given in the relevant IS codes should only be used.
  - Weight test (weight per 10 liters of paint thoroughly mixed)
  - Drying time
  - Flexibility and Adhesion
  - Consistency
  - Dry thickness and rate of consumption
4. Type of Paint :- Primer coat shall consist of 1 coats of epoxy zinc phosphate primer of approved make approx. 60 micron thick. Finishing coat shall consist of 2 coats of coal tar epoxy Hi Build paint of approved make. Total thickness of paint including primer shall not be less than 250 micron.
  - i. Method of Application: - Paint shall be applied either by brush, roller or spray. All painting will be done in accordance with IS: 1477 (Part-1). The primer, intermediate and finishing coats shall all be applied so as to provide smooth coatings of uniform thickness. Wrinkled or blistered coatings or coatings with pinholes, sags, lumps or other blemishes shall not be accepted.
  - ii. Tests :- The materials shall be tested in accordance with relevant IS Specifications and necessary test certificates shall be furnished.

| Sr. No | Properties               | Acceptability criteria       | Frequency of test                                                  |
|--------|--------------------------|------------------------------|--------------------------------------------------------------------|
|        |                          | Value                        |                                                                    |
| 1      | Weight test              | Manufacturer's specification | Each batch shall be Compared with Manufacturer's Test Certificate. |
| 2      | Drying time              |                              |                                                                    |
| 3      | Flexibility and adhesion |                              |                                                                    |
| 4      | Consistency              |                              |                                                                    |
| 5      | Dry thickness.           |                              |                                                                    |
| 6      | Rate of consumption      |                              |                                                                    |

- iii. Measurement for Payment: - Protective coating of coal tar epoxy shall be measured in square meters of surface area painted.
5. Pitching
- i. Description: - This work shall consist of covering the slopes of embankment or banks of creek with stone/boulders, over a layer of granular material called filter.
  - ii. Pitching: - The pitching shall be provided as per West Central Railway specification 2002, & as indicated in the drawings. The stone shall be sound, hard, durable & fairly regular in shape. Quarry stone should be used. Round boulders shall not be allowed. The stones subject to marked deterioration by water or weather shall not be accepted. No stone, weighing less than 30 kg shall, be used. The sizes of spalls shall be a minimum of 25 mm and shall be suitable to fill the voids in the pitching.
  - iii. Filter media: - The material for the filter shall consist of sand, gravel, stone or coarse sand. To prevent escape of the earth through the voids of the stone pitching as well as to allow free movement of water without creating any uplift head on the pitching, one or more layers of graded materials, commonly known as a filter medium, shall be provided underneath the pitching.

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- iv. The gradation of the filter material shall satisfy the following requirements:-Provision of a suitably designed filter is necessary under the slope pitching to prevent the escape of underlying earth material through the voids of stone pitching when subjected to the attack of flowing water and wave action, etc. In order to achieve this requirement, the filter may be provided in one or more layers satisfying the following criteria :

$$\frac{D_{15}(\text{Filter})}{D_{85}(\text{Base})} \leq 5 \quad \text{and} \quad 4 < \frac{D_{15}(\text{Filter})}{D_{15}(\text{Base})} \leq 20 \quad \text{and} \quad \frac{D_{50}(\text{Filter})}{D_{50}(\text{Base})} \leq 25$$

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1. In the foregoing, D 15 means the size of that sieve which allows 15 per cent by weight of the filtermaterial to pass through it and similar is the meaning of D 50 and D 85.
2. If more than one filter layer is required, the same requirement as above shall be followed for each layer. The finer filter shall be considered as base material for selection of coarser filter.
3. The filter shall be compacted to a firm condition. The thickness of filter is generally of the order of 20mm to 300 mm. Where filter is provided in two layers, thickness of each layer shall be 150 mm..

v. Construction Operations :-

- a) Before laying the pitching, the sides of banks shall be trimmed to the required slope and profiles put up by means of line and pegs at intervals of 3 metres to ensure regular straight work and a uniform slope throughout. Depressions shall be filled & thoroughly compacted.
  - b) The filter granular material shall be laid over the prepared base and suitably compacted to the thickness specified on the drawings.
  - c) The lowest course of pitching shall be started from the toe wall and built up in courses upwards. The toe wall shall be in dry rubble masonry (uncoursed).
  - d) The stone pitching shall commence in a trench below the toe of the slope. Stone shall be placed by derrick or by hand to the required length, thickness and depth conforming to the drawings. Stonesshall be set normal to the slope, and placed so that the largest dimension is perpendicular to the face of the slope, unless such dimension is greater than the specified thickness of pitching.
  - e) The largest stones shall be placed in the bottom courses and for use as headers for subsequent courses.
  - f) In hand placed pitching, the stone of flat stratified nature should be placed with the principal bedding plane normal to the slope. The pattern of laying shall be such that the joints are broken and voids are minimum by packing with spalls, wherever necessary, and the top surface is as smooth as possible.
  - g) When full depth of pitching can be formed with a single stone, the stones shall be laid breaking joints and all interstices between adjacent stones shall be filled in with spalls of the proper size and wedged in with hammers to ensure tight packing.
  - h) When two or more layers of stones must be laid to obtain the design thickness of pitching, dry masonry shall be used and stones shall be well bonded. To ensure regular and orderly disposition of the full intended quantity of stone as shown, template cross walls in dry masonry shall be built about a metre wide and to the full height of the specified thickness at suitable intervals and all along the length and width of the pitching. Within these walls the stones shall be hand packed as specified.
  - i) Measurement for Payment :- The filter & boulder pitching shall be measured in cubic meters.
6. Expansion joint
- i. Expansion joint shall be provided as per RDSO's drawing No.RDSO/B-1596.
  - ii. The expansion joint shall be measured in running metres.
7. Integrated 4 coat system - Protective coating for Concrete surface.

All concrete surface of superstructure and part of substructure above road level shall be

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provided with Integrated 4 coat system developed by CECRI Karaikudi.

- i. Material :- Material for epoxy painting shall be procured from the approved licensee of Central Electrochemical Research Institute, Karaikudi and shall be got approved from the Engineer. The following tests shall be carried out on epoxy material prior to use.
  1. Salt spray Test as per ASTM B 117
  2. Scratch hardness Test as per ASTM D2197-68
  3. Adhesion Test as per ASTM D3359-83
  4. Water absorption Test as per ASTM D-570
  5. UV test as per ASTM D3361-81
  6. Pot life Test.
  7. Chemical Resistance Test for Chloride, Sulphates, Dilute Acids (HCl 5% & H<sub>2</sub>SO<sub>4</sub> 5%), Alkalis (NaOH) etc.. Each batch of material shall be got tested.
- ii. Surface Preparation: - The surface of concrete to be coated shall be free from dirt, dust, residue of grease, demolding oil, curing compound and other deleterious material. The surface shall be cleaned with wire brushes. Use of cup brushes attached to the electric grinder shall be preferable. The cracks, depressions in the concrete surface shall be sealed with sealant EPCO-1010. Before application of epoxy paint, it shall be ensured that the concrete surface shall be cured and free of moisture and dust.
- iii. Properties of Materials of various coats

|    |               | <b>First coat primer</b>                                       | <b>Second Coat</b>                                                         | <b>Third Coat</b>                                                 | <b>Fourth &amp; Final Coat</b> |
|----|---------------|----------------------------------------------------------------|----------------------------------------------------------------------------|-------------------------------------------------------------------|--------------------------------|
|    |               | EPOXY – Polymide Primer For Coatings Over Concrete Surfaces.   | EPOXY – Micaceous Iron Oxide Undercoat For Coatings Over Concrete Surfaces | EPOXY – Polymide Finish Paint For Coatings Over Concrete Surfaces | Aliphatic Acrylic Polyurethane |
| 1. | Pigment       |                                                                |                                                                            |                                                                   |                                |
|    | Main pigment  | Red iron oxide 82.5% by Wt of total pigment.                   | Micaceous iron oxide – 50.6% by Wt. Of total pigment.                      | TiO <sub>2</sub> 68.4% Wt. Of total pigment.                      |                                |
| 2. | Other pigment | Talc.                                                          | Silicates & clay materials – 49.4%.                                        | Talc & Silica.                                                    |                                |
| 3. | Binder        | Epoxy Resin ( Bis-Phenol “A” type) Epoxide equivalent 475-520) | Epoxy Resin (Bis-Phenol “A” type) Epoxide equivalent 475-520).             | Epoxy Resin (Bis-Phenol “A” type) Epoxide equivalent 475-520)     | Acrylic Resin                  |
| 4. | Curing Agent  | Symbol 125 polyamide (versal)                                  | Symbol 125 polyamide                                                       | Symbol 125 polyamide                                              | Aliphatic Isocyanate           |

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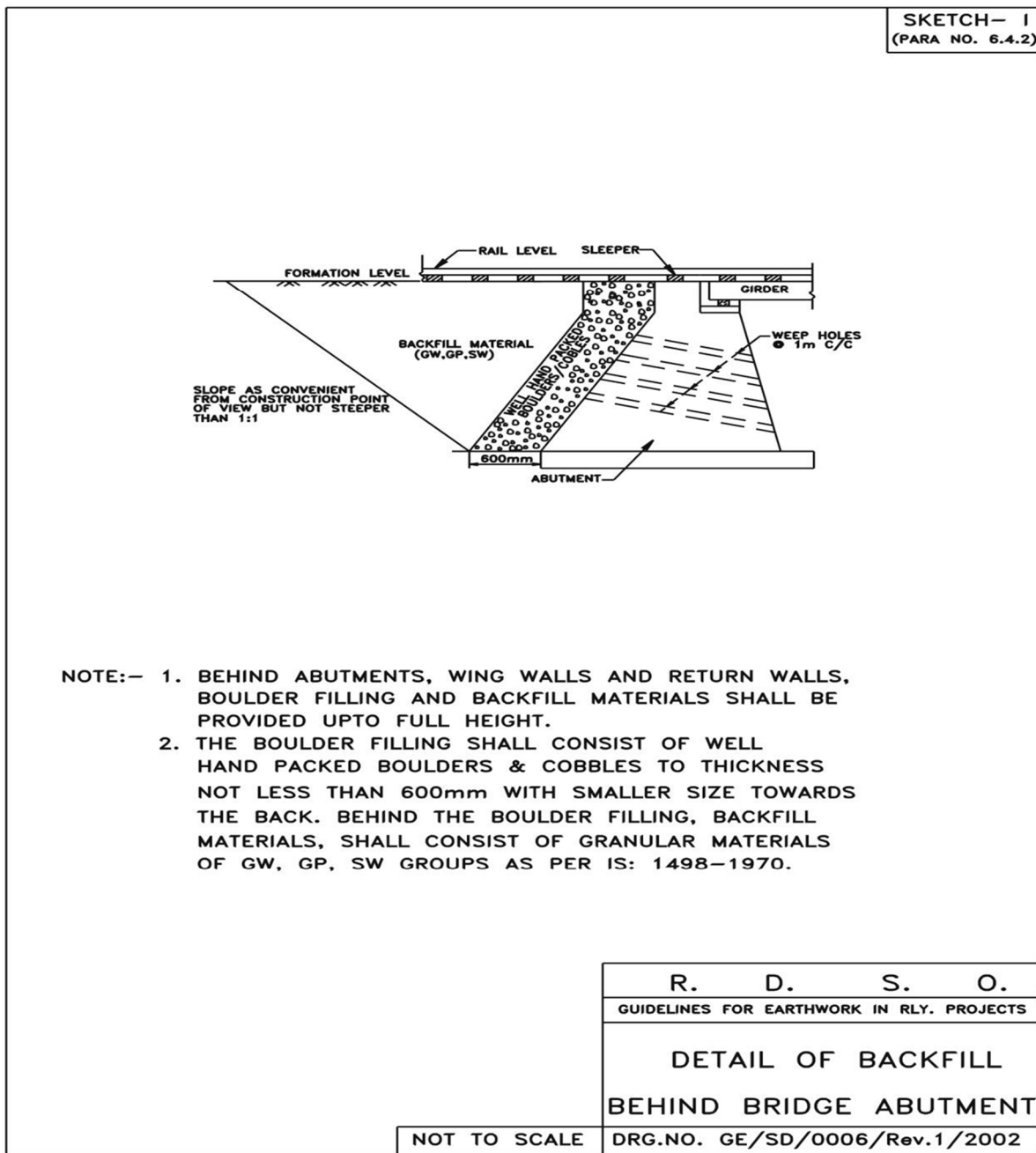
|     |                                | mid type)                     | (versamide type).                    | (versamide type)                    | te                                |
|-----|--------------------------------|-------------------------------|--------------------------------------|-------------------------------------|-----------------------------------|
| 5.  | Solvents                       | 1:1 Xylene – MIBK mixture.    | 1:1 Xylene O MIBK mixture.           | 1:1 Xylene MIBK mixture.            | -                                 |
| 6.  | PVC                            | 33%                           | 36%.                                 | 18%.                                | -                                 |
| 7.  | Volume solid.                  | 55%                           | 60%                                  | 80%                                 | 50%                               |
| 9.  | Specific gravity               | 1.5                           | 1.5                                  | 1.35                                | -                                 |
| 10. | Mixing ratio. (Base :catalyst) | 3.1 by volume                 | 3.1 by volume (Base : curing agent). | 3.1 by volume (Base : curing agent) | 4:1 (Base : Curing Agent)         |
| 11. | Mode of Application            | Spray / Brush.                | Brush/ Air – less spray.             | Brush / Air – less spray.           | Brush / Roller/ Air – less spray. |
| 12. | Coverage                       | 5.5 – 6m <sup>2</sup> /litre. | 4-5m <sup>2</sup> / litre.           | 5-6 m <sup>2</sup> / litre.         | 11.4 – 13.3 Sq. m per liter       |
| 13. | Dry Film Thickness (Brush)     | 55 ± 5 micron                 | 110-120 micron                       | 110 ± 5 micron.                     | 30±5 micron per coat              |
| 14. | Re-cotability                  | After 24 hrs.                 | 24 hrs.                              | After 24 hrs.                       | After 24 Hrs.                     |

Notwithstanding the provisions above any modifications in composition or application procedure, stipulated by CECRI, Karaikudi who would be associated with Railways, in advisory capacity shall be adopted during execution.

- iv. Measurement for Payment :- Protective coating of **Integrated 4 coat system protective coating for Concrete surface** shall be measured in square meters of surface area painted.

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Inspection Registers and Records

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**1. Site Order Book**

| S. No. | Date | Instructions issued on the Inspection of work with Signature and designation | Contractor/ Contractor's representative acknowledgement with signature, Name & Date | Compliance re Contractor/ Contractor's representative signature, Name & Date |
|--------|------|------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|------------------------------------------------------------------------------|
| 1.     | 2.   | 3.                                                                           | 4.                                                                                  | 5.                                                                           |
|        |      |                                                                              |                                                                                     |                                                                              |
|        |      |                                                                              |                                                                                     |                                                                              |

**2. Hindrance Register**

| S. No. | Nature of Hindrance | Date of Occurrence | Date of Clearance | Period | Over Taping period if any | Weight age of Hindrance | Net effective days of Hindrance | Remarks & references |
|--------|---------------------|--------------------|-------------------|--------|---------------------------|-------------------------|---------------------------------|----------------------|
| 1.     | 2                   | 3                  | 4                 | 5      | 6                         | 7                       | 8                               | 9                    |
|        |                     |                    |                   |        |                           |                         |                                 |                      |
|        |                     |                    |                   |        |                           |                         |                                 |                      |

**3. Drawing Register**

| S. No. | Drg. No and revision no.if any | Date of Receipt | Details of Drg. | Date of issue to Contractor | Acknowledgement of contractor |
|--------|--------------------------------|-----------------|-----------------|-----------------------------|-------------------------------|
| 1.     | 2                              | 3               | 4               | 5                           | 6                             |
|        |                                |                 |                 |                             |                               |
|        |                                |                 |                 |                             |                               |

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**4. Cement Register**

| Sr .<br>N<br>o. | Date<br>of<br>Rece<br>ipt | Sour<br>ce of<br>Rece<br>ipt | Bil<br>l/<br>C<br>h.<br>No. | Manufac<br>turer<br>Test<br>Certificat<br>e<br>referenc<br>e | Qty<br>Rec<br>d.<br>(bag<br>s) | Progressi<br>ve Total<br>of<br>Receipt<br>(Bags) | Date<br>of<br>Issu<br>e | Qty<br>Issu<br>ed<br>(bag<br>s) | Qty<br>Return<br>ed at<br>the<br>end of<br>the<br>Day<br>(bags) | Net<br>Qty<br>Issu<br>ed<br>(bag<br>s) | Progr<br>essi<br>ve<br>Total<br>of<br>Issue<br>(Bags<br>) |
|-----------------|---------------------------|------------------------------|-----------------------------|--------------------------------------------------------------|--------------------------------|--------------------------------------------------|-------------------------|---------------------------------|-----------------------------------------------------------------|----------------------------------------|-----------------------------------------------------------|
| 1               | 2                         | 3                            | 4                           | 5                                                            | 6                              | 7                                                | 8                       | 9                               | 10                                                              | 11                                     | 12                                                        |
|                 |                           |                              |                             |                                                              |                                |                                                  |                         |                                 |                                                                 |                                        |                                                           |
|                 |                           |                              |                             |                                                              |                                |                                                  |                         |                                 |                                                                 |                                        |                                                           |

**5. Steel Register**

| S. N | Date<br>of<br>Receip<br>t | Sourc<br>e of<br>Recei<br>pt &<br>Bill/<br>Ch.<br>No. | Qty<br>Recd.<br>(MT) | Cum<br>Qty<br>Recd.<br>(MT) | Date<br>of<br>Issue | Qty<br>Issued<br>(MT) | Cum<br>Qty<br>Issued<br>(MT) | Balanc<br>e at the<br>end of<br>the<br>Day<br>(MT) | Items of<br>work for<br>which<br>Consumed | Sign<br>Ofsi<br>Eng<br>With<br>date |
|------|---------------------------|-------------------------------------------------------|----------------------|-----------------------------|---------------------|-----------------------|------------------------------|----------------------------------------------------|-------------------------------------------|-------------------------------------|
| 1    | 2                         | 3                                                     | 4                    | 5                           | 6                   | 7                     | 8                            | 9                                                  | 10                                        | 11                                  |
|      |                           |                                                       |                      |                             |                     |                       |                              |                                                    |                                           |                                     |
|      |                           |                                                       |                      |                             |                     |                       |                              |                                                    |                                           |                                     |

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#### 6. Sieve Analysis of Stone Aggregate Nominal Size

| S. N | Date | Weight of sample in Gms | Size of sieves | Weight retained on each sieve | % of weight retained | Cumulative % of weight Retained | % of weight passing | Specified % of weight passing | S |
|------|------|-------------------------|----------------|-------------------------------|----------------------|---------------------------------|---------------------|-------------------------------|---|
| 1    | 2    | 3                       | 4              | 5                             | 6                    | 7                               | 8                   | 9                             | 1 |
|      |      |                         |                |                               |                      |                                 |                     |                               |   |
|      |      |                         |                |                               |                      |                                 |                     |                               |   |

#### 7. Silt Contents of Fine Sand/ Coarse Sand

| S. N | Date | Source of material | Height of Silt after Setting (V-1) | Height of Sand after Setting (V-2) | % of Silt Content V1/V2x100 | Acceptability as per specification | Sign. Of site Engg with date | Sign. Of contractor with date |
|------|------|--------------------|------------------------------------|------------------------------------|-----------------------------|------------------------------------|------------------------------|-------------------------------|
| 1    | 2    | 3                  | 4                                  | 5                                  | 6                           | 7                                  | 8                            | 9                             |
|      |      |                    |                                    |                                    |                             |                                    |                              |                               |
|      |      |                    |                                    |                                    |                             |                                    |                              |                               |

#### 8. Slump Test

| S. N | Date of Testing | Item of work and location | Vibrators used Yes/No | Quantity of water added per bag of cement (Ltr.) | Height of specimen after removal of mould in (mm) | Slump (mm) | Acceptability of result of action taken | Sign. Of site Engineer with date |
|------|-----------------|---------------------------|-----------------------|--------------------------------------------------|---------------------------------------------------|------------|-----------------------------------------|----------------------------------|
| 1    | 2               | 3                         | 4                     | 5                                                | 6                                                 | 7          | 8                                       | 9                                |
|      |                 |                           |                       |                                                  |                                                   |            |                                         |                                  |
|      |                 |                           |                       |                                                  |                                                   |            |                                         |                                  |

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**9. Cube Test**

| S<br>No. | Date<br>of<br>colle<br>ction | Gra<br>de<br>of<br>Mix. | Mark<br>of<br>speci<br>men | 7 Days Test Result |               |                                                     |                                                            | 28 Days Test Result |               |                                                     |                                                                    | R<br>S<br>ie<br>st<br>th |
|----------|------------------------------|-------------------------|----------------------------|--------------------|---------------|-----------------------------------------------------|------------------------------------------------------------|---------------------|---------------|-----------------------------------------------------|--------------------------------------------------------------------|--------------------------|
|          |                              |                         |                            | Date of<br>testing | Load<br>in KN | Compre<br>ssi<br>ve<br>strengt<br>h<br>(KN/M<br>M2) | Average<br>Compr<br>essive<br>strengt<br>h<br>(KN/m<br>m2) | Date of<br>testing  | Load<br>in KN | Compre<br>ssi<br>ve<br>strengt<br>h<br>(KN/M<br>M2) | Averag<br>e<br>Compr<br>essive<br>strengt<br>h<br>(KN/m<br>m<br>2) |                          |
| <b>1</b> | <b>2</b>                     | <b>3</b>                | <b>4</b>                   | <b>5</b>           | <b>6</b>      | <b>7</b>                                            | <b>8</b>                                                   | <b>9</b>            | <b>10</b>     | <b>11</b>                                           | <b>12</b>                                                          | <b>13</b>                |
|          |                              |                         |                            |                    |               |                                                     |                                                            |                     |               |                                                     |                                                                    |                          |
|          |                              |                         |                            |                    |               |                                                     |                                                            |                     |               |                                                     |                                                                    |                          |

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**10. Density Test by Core Cutter  
Method MDD as per lab test  
WS.....**

| S. N | Location (C.H.)/Area Represented by the test | Core Cutter Nos. | Weight of core cutter + weight of soil (in gram) (W1) | Weight of Empty core cutter (in gram) (W2) | Weight of Wet soil (in gram) W=W <sub>1</sub> -W <sub>2</sub> | Volume of Core Cutter (in CC) V | Bulk Density (gram/cc W <sub>4</sub> =W <sub>3</sub> =W/V | Moisture Content of compaction layers (M) | Dry Density gram/cc W <sub>4</sub> =W <sub>3</sub> /(1+M) | Density of compaction W <sub>4</sub> |
|------|----------------------------------------------|------------------|-------------------------------------------------------|--------------------------------------------|---------------------------------------------------------------|---------------------------------|-----------------------------------------------------------|-------------------------------------------|-----------------------------------------------------------|--------------------------------------|
| 1    | 2                                            | 3                | 4                                                     | 5                                          | 6                                                             | 7                               | 8                                                         | 9                                         | 10                                                        | 11                                   |
|      |                                              |                  |                                                       |                                            |                                                               |                                 |                                                           |                                           |                                                           |                                      |
|      |                                              |                  |                                                       |                                            |                                                               |                                 |                                                           |                                           |                                                           |                                      |

**11. Test for Thickness and Density of the Compacted Layer (By Sand Replacement Method) for Asphalt Concrete/ Bitumen CC/ Pavement**

Last Test Density in gms/ CC

| S N O. | Date of test | Qty represented by the test | Location of holes | Thickness of layer | Weight of materials removed from the carpet Hole | Initial weight of sand taken in cylinder | Weight of sand filling in cone of cylinder | Weight of sand remaining in cylinder | Predetermined bulk density of sand | Density = $\frac{A.d}{W - (W_1 + W_2)}$ |
|--------|--------------|-----------------------------|-------------------|--------------------|--------------------------------------------------|------------------------------------------|--------------------------------------------|--------------------------------------|------------------------------------|-----------------------------------------|
|        |              |                             |                   |                    |                                                  |                                          |                                            |                                      |                                    |                                         |
|        |              |                             |                   |                    |                                                  |                                          |                                            |                                      |                                    |                                         |
|        |              |                             |                   |                    |                                                  |                                          |                                            |                                      |                                    |                                         |

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|  |  |  |  |                        |                      |      |  |  |  |  |  |
|--|--|--|--|------------------------|----------------------|------|--|--|--|--|--|
|  |  |  |  | Individ<br>ual<br>(mm) | Av<br>g.<br>(m<br>m) | A gm |  |  |  |  |  |
|  |  |  |  |                        |                      |      |  |  |  |  |  |
|  |  |  |  |                        |                      |      |  |  |  |  |  |

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West Central Railway,

**Chapter-13**

**SPECIFICATION FOR NS ITEMS**

**1. NS- Earth work in Filling in Embankment in the state of Madhya Pradesh:-**

Earthwork under item NS-1 shall be executed in accordance with the specifications issued vide following guidelines/ reports / directives:

- a. RDSO Guidelines No. GE. G-1 for Earthwork in Railway Projects.
  - b. RDSO Guidelines and Specifications for Design of Formation for Heavy Axle Load Report No. RDSO/2007/GE:0014.
  - c. RDSO Guidelines for design/ construction of formation on Indian Railway Tracks issued vide letter No. RS/G/108/ Heavy Axle Load, dated 19.10.2015.
  - d. RDSO Guidelines for design/ construction of formation on Indian Railway Tracks issued vide letter No. RS/G/108/ Heavy Axle Load, dated 26.10.2016.
  - e. Any subsequent amendments made in above guidelines/ reports/ directives specifications.  
**The test should be done as per RDSO Guideline Specification No.- RDSO/2018/GE: IRS-0004 (D) Part-IV in July 2019 and amended / updated from time to time.**
1. Rates quoted for NS-1 is deemed to be inclusive of taxes, duties, octroi or any form of levies etc and excluding Royalty.
    - a. It is the responsibility of Contractor to make due Payments towards Royalty on Earthwork in filling, if applicable, directly to State Governments. Railways shall not be responsible for non-payment of Royalty by Contractor to the State Govt.
    - b. Re- imbursement of Royalty paid by Contractor to State Government shall be made on actual basis separately on submission of proof of payment to State Govt. by Contractor and Confirmation of Payment received by State Govt.
  2. At present, Rate of Royalty are as per rates circulated by the Government of Madhya Pradesh vide resolution dated 06 September, 2014 which is @ Rs. 50/- per m<sup>3</sup>.

Or

At present, the rate of royalty are as per rates circulated by the Government of Rajasthan vide resolution dated 28th February, 2017 which is @ Rs. 3/- per m<sup>3</sup>.
  3. Claim towards such Royalty should be unfailingly accompanied by proof of payment and rate thereof. In the absence of these, royalty payment will not be released.
  4. Amount reimbursed against Royalty to the Contractor shall not be considered for variation purpose. It will be charged directly to the estimate of the work.
    - a. In case of all other taxes, duties, octroi, (GST if applicable) or any form of levies, etc. (except Royalty), no additional amount will be paid or claim entertained on this account by the Railway. Contractor shall not have any claim whatsoever as a result of the increase in the rates for such taxes, duties or any other form of levies etc.
  5. As per specification, deduction of earthwork qty for shrinkage will be done from payment (a) In case of mechanical compaction 5% of measured earthwork qty. (b) In case of manual compaction 10% of measured earthwork qty.

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For Chief Administrative Officer  
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**1. NS (Supplying & fabrication protective screen made of MS angles) –**

Rate is inclusive of fabrication of MS angles/pipe frame with mesh, G.I. plain sheet including all nut, bolts/clamps etc. complete & fixing in peripherals with suitable clamps, filling gaps making good in cement mortar complete with primer & painting. All fabrication work shall confirm to relevant IS standards. The work shall be carried out to the complete satisfaction of Engineer-In-Charge.

**2. NS- Preparation and submission of structural design and detailed drawing**

Design and Drawing shall be submitted within 45 days of issue of acceptance letter, duly proof checked. The tenderer shall specially note that while every effort shall be made to approve the design and drawing expeditiously, no claim shall be entertained on account of delay in approval of design and drawings for whatsoever reason.

The main items of the work involved in the design and drawings are as under:

- a) Boring for collection of samples for safe bearing capacity.
  - b) Foundations which include open foundation or Piles, pile cap as per the soil met with.
  - c) Provision of piers, pier cap, capping beam, and pedestal.
  - d) Provision of bearings.
  - e) Bow string type of open web steel Girder is not within the scope of work as RDSO approved drawing will be used. However, launching scheme of Bow string type of open web steel Girder is within the scope of Design and Drawing and is to be got proof checked and approved by the Railway.
  - f) PSC solid slab Deck slab crash Barrier, RCC waling etc complete
- 3.** The contractor is expected to submit his own design and drawings based on West Central Railway's GAD, which shows only general details of ROB as proposed by railway. This GAD indicates the arrangements of the proposed bridge structures with main dimensions and clearances required. However, contractor shall make out their design and drawing strictly adhering to the leading dimensions as indicated in railway's drawing. Design and drawing shall be submitted duly proof checked.
- 4.** The designer may visit the site and acquaint himself with the condition under which the ROB to be designed.

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**5. Loading and Design Criteria:**

All the members of the bridge shall be designed for the following loadings as specified in Standard Code of Practice for Road Bridges (IRC Codes), amended from time to time.

- i) Live Loads
  - a) One lane of IRC Class 'AA' or
  - b) Two lanes of IRC Class 'A' or
  - c) One lane of IRC Class '70 R' whichever produces maximum stresses.
- ii) Footpath loading, wind, temperature, braking force etc. shall be as per IRC codes of practice as amended from time to time.
- iii) Contractor shall submit structural design & drawings of all components of Road over Bridge.  
Contractor's design should fulfill the following requirements: -
  - a) It shall ensure soundness of design.
  - b) It shall be economical and ensure speedy construction.
  - c) It should match the local surround aesthetically.
- iv) The design of the ROB including bearing shall be done strictly in accordance with the provisions contained in the latest version of following relevant IRC codes: -
  - a) All relevant standards and specifications of MOST IRC codes including SP-33.
  - b) Those standards which are not covered by above to be referred from relevant IS codes.
  - c) Indian Railway Schedule of Dimensions for BG.
  - d) The standard specification and code of practice for Road Bridges IRC-6 (Loading and forces) IRC- 24 (Steel Road Bridges) IRC -21 (Cement Concrete) IRC-22 (Composite construction) IRC 83(iii) POT PTFE Bearing) and IS 3935 (Shear Connector)
  - e) The substructure and super structure has to be designed for Zone V by analyzing earth Quake Forces as per standard codal provisions.
  - f) All structural steel Shall be of E350/ E250 Grade B confirming to IS:2062 (weld able quality)
  - v) Designs to be suitable to National Highway Authority of India's latest specifications.
  - vi) Indian Railway Schedule of Dimensions for BG.
  - vii) For any items not covered by any of the code of practice standard specifications shall be adopted and mentioned.

**NOTE:- In all the above cases, railway's decision shall be final and binding on the contractor.**

**6. Structural details of various components: -**

**1. Foundation:**

Type and depth of foundation and safe bearing capacity of soil near the site of foundation as shown in the railway's GAD is tentative. The type and depth of foundation will be ascertained by the designer based on the trial bore details and Plate Load test etc.

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**2. Sub-structure:**

The component of sub-structure comprising Raft Piles, Pile Cap (Depending upon the type of foundation), Columns, Pier, Abutments Capping beams, Bearings, etc. should be designed as shown in the GAD. The abutments pier and pier cap shall be designed to have future inspections of bearings. Pedestals and pier caps shall be designed to accommodate POT-PTFE bearing.

**3. Super-structure:**

Contractor should design the superstructure as PSC solid slab along with Deck slab, wearing coat, Crash barrier, Parapet wall etc to suit the width of the ROB as per the GAD. Launching scheme of Bow string type of open web steel Girder shall be designed to suit the relevant drawings for superstructure and to suit the site conditions.

- a. Safety measures and precautions to be incorporated in design and drawing: All safety measures and precautions required for the safe execution and for safety of the bridge from wind, seismic and other considerations should be incorporated in the design and drawing.

**The contractor should submit a list indicating the deviation from IRC codes/specification if any and list out the specifications which are being adopted from codes other than IRC/IRS along with the Design.**

- 4.0 Submission of design and drawings including design for temporary works required for execution of works and launching of girders etc:
- 4.1 Launching scheme of Bow string open web steel Girder shall be designed to suit the relevant Structural drawings of superstructure and to suit the site conditions. Preliminary general arrangement drawings are available which are based on site details collected. Contractor shall proceed further for preparation of detailed drawings based on the data available. The general arrangement drawing may be modified by the contractor/s as and when required by the railway, for which no extra payment shall be admissible.
- 4.2 The contractor shall submit the detailed design calculations for approval to the railway. The design shall be submitted in sufficient details giving references to the prescribed literature books /codes referred to in the design. After incorporating the changes which are found necessary in the opinion of the railway, the contractor/s shall submit two copies of revised design calculations for the ROB which again will be subjected to technical scrutiny, if found necessary, in the opinion of the railway.
- 4.3 Contractor shall submit structural design & drawings of all components. Contractor shall submit drawings, designs and specifications, including temporary arrangements, design of deck slab shuttering, launching scheme of steel girders, load testing arrangements, bar bending schedule, etc., duly proof checked by any IIT/NIT, for approval of the railway administration.
- 4.4 Design is to be done with due consideration of RDSO guidelines. Provision of inspection ladder and inspection platform shall be designed as per railway's requirement.

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- 4.5** The design calculations to be submitted to the railway for approval shall be in any of the following formats: -
- i) Design done manually supported by the step- by- step detailed structural calculations.
  - ii) Alternatively, design may be submitted in the form of computer input file/data along one sample manual calculations accompanied by a source code computer programme on compact disk along with the listing with description of the variables used in the analysis and design of the structure.
- 4.6** Design shall be submitted by the contractor/s in SI units and shall be submitted with full explanatory notes wherever required. If during the scrutiny to the detailed design calculations and drawings, any changes therein which are necessary in the opinion of the railway/proof checker shall be incorporated.
- 4.7** The contractor shall modify the design and drawings as considered necessary as result of check made by railways. All necessary clarifications must be furnished by the contractor in writing or in personal discussions as considered necessary by railways. For this no extra payment, no travelling expenses etc. will be paid.
- 4.8** A complete set of final design for the ROB on bond paper, neatly typed in double space in bind volumes shall be submitted by the contractor/s for Railway's record. The design report shall contain the index, the design methodology detail design calculations (each calculation being neatly explained with reference to clauses of various codes and references). Each page of design **should be signed by Designer, Contractor & Proof checking authority.**
- 4.9)** Approval to the drawing and design calculation by the Railway shall in no way relieve the responsibility of contractor for the soundness, correctness and safety of the structure designed by him/them.

**5.0 Preparation of Drawings**

a. The drawing shall be considered as final only after the approval of competent authority of the Railways. Even after final approval, Railway reserves the right to ask for modifications, alterations in the drawings without any extra cost.

All the drawings prepared at various stages shall be on computerized prints like AUTO CAD. The contractor/s shall submit six sets of working arrangement drawings and six sets of detailed drawing including revisions, if any, complete in all respect, for being used as contract drawing for construction after approval by the railways.

**b. Submission of Drawings**

The drawings shall be submitted in original in reproducible tracing film of minimum thickness 75 microns double matt type of approved quality available in the market in A-1 size 594 x 841 mm indicating the details of proposed structures in black, colour and with all notes of general and technical nature of the drawing.

The drawings and design so submitted shall be treated as the property of West Central Railway, Jabalpur. The railway shall reserve the rights to use these drawings and design at location other than for which the drawing and design have been submitted by the contractor. No extra payment shall be made on this account.

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**c. Clarification**

Notwithstanding the approval in the design/drawing by the railways if any doubt arises during the execution of work, the contractor shall always issue clarifications as and when required by the railways.

In addition to the above contractor shall submit the following details:-

- i. The contractor shall submit detailed technical specifications for the various items involved.
- ii. Any changes required in the design at the execution stage shall be carried out by the designer within the rates quoted by him for the works. The rates quoted should cover the cost of site visit as required during the design and construction stages.

iii. **Designer should also sign all the drawings and design along with contractor.**

**SCOPE AND DETAILS OF WORK** for Preparation of detailed structural design & drawings and getting them proof checked from any IIT/NIT, including making all corrections as suggested by proof checker / Railways.

**6. Mode of Payment:**

The mode of payment for this item shall be as under:

|   |                                                                                  |                            |
|---|----------------------------------------------------------------------------------|----------------------------|
| 1 | After submission of detail design and drawing duly proof checked from NIT / IIT. | 40% of the accepted rates. |
| 2 | After approval by Railway.                                                       | 60% of the accepted rates. |

**NS- Casting, Supplying, Transporting & Launching precast reinforced concrete fascia panel of thickness 180mm in M-35 design mix:-**

- 1 This item involves the work of providing and laying in position cement concrete of M-35 grade for **RCC fascia panel of thickness 180mm in M-35 design mix** as per approved drawing.
- 2 The items includes cost of casting, supplying, transporting & launching in position precast RCC fascia panel of thickness 180mm including all labour, machineries, materials, shuttering, cantering vibrating, curing etc **except cement & reinforcement steel which will be paid separately under relevant USSOR-2019 item** as per consumption required according to mix design duly approved by Railway for M-35 grade of concrete.
- 3 The cement used shall be generally 43/53 grade ordinary Portland cement conforming to IS: 8112 and 12269.
- 4 Only machine mixed and automatic weigh batched concrete shall be used during

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entire work. No hand Mix concrete shall be allowed. The concrete shall be placed with the help of transit mixer or concrete pump.

- 5 Measurement: Dimensions of length, breadth and thickness shall be measured correct to nearest Cm.  
Areas shall be worked out to nearest 0.01 sqm. and the cubic contents of consolidated concrete shall be worked out to nearest 0.01 cum. Any work done in excess over the specified dimensions or sections shown in the drawing shall be ignored.
- 6 Concrete shall be of M-35 grade as per mix design only.
- 7 Date of casting and name of casting depot shall be engraved segment on each unit.
- 8 The depot where casting of RCC facia panel of thickness 180mm in M-35 design mix is proposed shall be inspected by railway officials in order to ensure quality of raw material and quality control in concreting. All arrangements for casting will be done in presence of concerned SSE (W) / XEN / AXEN. Casting of RCC facia panel of thickness 180mm in M-35 shall be done in presence of Engineer's representative who will put a passing mark on each piece. Transportation of RCC facia panel of thickness 180mm in M-35 will be allowed only after inspection by the Engineer-in-charge.
- 9 After casting and inspection of the precast RCC facia panel of thickness 180mm in M-35, the contractor will have to transport them by his own mechanical means viz. Tractor or trucks/Trailers/crane etc. to both bridge sites in Jirapur- Khilchipur Road and Chhatta-Biaora Road at ROB location. All loading and unloading of casted segment of RCC facia panel of thickness 180mm in M-35 at the ROB site will be done by means of crane or any other mechanical means to avoid any damage to the precast RCC facia panel of thickness 180mm in M-35 .
- 10 Damage to RCC facia panel of thickness 180mm in M-35 during transportation and stacking etc. will be rectified with the epoxy mortar and re-offer for inspection in case of minor damage. In case of major damage, a fresh RCC ballast retainers, slabs will be cast free of cost and penalty as deemed fit shall be imposed on both cases.
- 11 **Mode of Payment:**

The mode of payment for this item shall be as under:

|   |                                                                                                                       |                            |
|---|-----------------------------------------------------------------------------------------------------------------------|----------------------------|
| 1 | Casting RCC <b>facia panel of thickness 180mm in M-35</b> as directed by Engineer in charge                           | 50% of the accepted rates. |
| 2 | Transporting & Unloading RCC <b>facia panel of thickness 180mm in M-35</b> ROB site as directed by Engineer in charge | 30% of the accepted rates. |
| 3 | Launching in position RCC ballast retainers, slabs including finishing and cleaning at site as directed               | 20% of the accepted rates. |

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|  |                       |  |
|--|-----------------------|--|
|  | by Engineer in charge |  |
|--|-----------------------|--|

**NS Casting supplying & transporting RCC components segments in M-35/40 Grademix design:**

- (i) These items involves the work of casting supplying & transporting RCC component in M- 35/40 grade mix design as per approved drawing.
- (ii) The items includes cost of all labour, machineries & tools, materials, shuttering, centering vibrating, curing etc except cement & reinforcement steel which will be paid separately under relevant USSOR-2010 item as per consumption required according to mix design duly approved by Railway.
- (iii) The cement used shall be generally 43/53 grade ordinary Portland cement conforming to IS: 8112 and 12269.
- (iv) Measurement: Dimensions of length, breadth and thickness shall be measured correct to nearest Cm. Areas shall be worked out to nearest 0.01 Sqm. & the cubic contents of consolidated concrete shall be worked out to nearest 0.01 cum. Any work done in excess over the specified dimensions or sections shown in the drawing shall be ignored.
- (v) Date of casting and name of casting depot shall be engraved on each segment unit.
- (vi) The depot where casting of RCC components is proposed shall be inspected by railway officials in order to ensure quality control of concrete & raw material. All arrangements for casting and casting of RCC components shall be done in presence of Engineer or his representative. Transportation of RCC components to the work site will be allowed only after inspection by the Engineer or his representative.
- (vii) After casting and inspection of the RCC components, the contractor shall transport them by his own mechanical means viz. Tractor or trucks/Trailers/crane etc. to the work site. All loading and unloading of casted RCC components at the concrete depot and at the work site will be by means of crane or any other mechanical means to avoid any damage to the precast RCC components.
- (viii) Damage to RCC components during transportation and stacking etc. will be rectified with the epoxy mortar and re-offered for inspection in case of minor damage. In case of major damage, a fresh RCC component will be casted free of cost.
- (ix) The contractor is advised to visit the site of work for which casting of RCC components has to be done. Preparation of service roads etc. for transportation will be the responsibility of the contractor.
- (x) The precast RCC components shall be placed near the site in such a manner that no component should infringe the road way as well as SOD of railways.
- (xi) Contractor will have to provide sufficient labour, tools & plants, machinery for launching of precast RCC components in position, for which no extra payment will be made.
- (xii) **Mode of payment:**  
The mode of payment for this item shall be as under:-
  - (i) After casting RCC components. = 40%
  - (ii) After Transporting & Unloading RCC components at site = 30%
  - After launching in position RCC components including finishing and cleaning at site. =30%.

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However, before releasing such payment the contractor has to execute an Indemnity Bond in favour of Railway's against losses / damages etc. The Indemnity Bond shall be legally scrutinized by Railway before releasing such payment.

**NS: Placement of precast RCC components:**

- (i) This item consist of Placement of Precast reinforced Cement Concrete component with the help of suitable capacity road crane at desired location with all contractor's labour, material, tools and machinery, crane etc., preparation of surface as directed by Engineer representative including filling of joints by epoxy mortar including supply of epoxy and sand, with all lead, lift etc. complete.
- (ii) The density of precast RCC component will be taken as 2500 Kg/cum for calculation purpose to work out the weight of RCC component.

**NS:- Providing, placing and compacting to desired density approved backfill material**

The fill material in the reinforced soil zone shall have drained or effective angle of friction not less than 30° measured in accordance with IS: 2720 (Part 13), by conducting a drained direct shear test using large shear box may be conducted (IS:2720: Part 39:Section 1). Thus gradation of fill soil shall be as per following limits.

| Sieve Size | Percentage Passing |
|------------|--------------------|
| 75 mm      | 100%               |
| 425 micron | 0-60%              |
| 75 micron  | less than 15       |
| PI<6       |                    |

Materials with more than 15 percent passing 75 micron sieve, but less than 10 percent of particles smaller than 15 microns are acceptable provided PI is less than 6 and angle of friction is not less than 30°.

Fly ash may be used as fill material in reinforced soil walls provided its angle of internal friction is not less than 30° and PI is less than 6. Gradation requirements need not be completely satisfied. Reference may be made to IRC Guide lines on Use of fly ash in Road Embankments (IRC:SP-58). Fly ash also satisfy requirements concerning PH and environmental conditions of the fill vis-a-vis the reinforcement type as specified in Clause 3014.1

The fill material used in the reinforced soil zone be free from organic or other deleterious materials and shall not react adversely (chemically, electrically or biologically) with the reinforcement material and/ or facing material.

Properties of fill soil in the reinforced zone, unreinforced zone (or retained / back fill) soil and the foundation soil shall be determined accurately during the construction phase, as per quality assurance plans and directions of Engineer so as to ensure that these are the same as those considered in the design phase.

The fill soil in the unreinforced zone shall conform to the requirements specified in the design.

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**NS- Providing and erecting a “Thrie” Metal Beam Crash barrier**

**1 Materials**

- 1.1** Metal beam rail shall be corrugated sheet steel beams of the class, type, section and thickness indicated on the drawings. Railing posts shall be made of steel of the section and thickness indicated on the drawings. Railing posts shall be made of steel of the section, weight and length as shown on the drawings. All complete steel rail elements, terminal sections, posts, bolts, nuts, hardware and other steel fittings shall be galvanized. All elements of the railing shall be free from abrasions, rough or sharp edges and shall not be kinked, twisted or bent.
- 1.2** The “W” beam type safety barrier shall consist of a steel post and a 3mm thick “W” beam rail element. The steel post and the blocking out spacer shall both be channel section of 75 mmx150 mm & size 5 mm thick. The rail shall be 70cm above the ground level and posts shall be spaced 2 m center to center. Double “W” beam barrier shall be as indicated in IRC:5- 1998. The thrie beam safety barrier shall have posts and spacers and fasteners for steel barriers shall be galvanized by hot dip process. (zinc coated , 0.55 kg per square meter; minimum single spot ) unless otherwise specified . The galvanizing on all other steel parts shall conform to the relevant IS Specifications. All fittings (bolts, nuts, washers) shall conform to the IS: 1367 and IS: 1364. All galvanizing shall be done after fabrication.
- 1.3** Concrete for bedding and anchor assembly shall conform to section 1700 of these Specifications.

**1.3.1 Construction Operations**

- 1.3.2** The line and grade of railing shall be true to that shown on the plans. The railing shall be carefully adjusted prior to fixing in place, to ensure proper matching at abutting joints and correct alignment and camber throughout their length. Holes for field connections shall be drilled with the railing in place in the structure at proper grade and alignment.
- 1.3.3** Unless otherwise specified on the drawing, railing steel posts shall be given one shop coat of paint (primer) and three coats of paint on structural steel after erection, if the sections are not galvanized. Any part of assembly below ground shall be painted with three coats of red lead paint.
- 1.3.4** Splices and end connections shall be of the type and designs specified or shown on the plans and shall be of such strength as to develop full design strength of the rail elements.

**1.3.5 Installation of Posts.**

- 1.3.6** Holes shall be dug or drilled to the depth indicated on the plans or posts may be driven by approved methods and equipment, provided these are erected in proper position and are free from distortion and burring or any other damage.
- 1.3.7** All post holes that are dug or drilled shall be of such size as will permit proper setting of the

posts and allow sufficient room for back filling and tapping.

- 1.3.8** Holes shall be backfilled with selected earth or stable materials in layers not exceeding 100 mm thickness and each layer shall be thoroughly tamped and rammed. When backfilling and tamping are completed, the posts or anchors shall be held securely in place.
- 1.3.9** Post holes that are drilled in rock and holes for anchor posts shall be backfilled with concrete.
- 1.3.10** Posts for metal beam guard rail on bridges shall be bolted to the structure as detailed on the plans. The anchor bolts shall be set to proper location and elevation with templates and carefully checked.

#### **1.4 Erection**

- 1. All guard rail anchors shall be set and attachments made and placed as indicated on the plans or as directed by the Engineer.
- 2. All bolts or clips used for fastening the guardrail of fittings to the posts shall be drawn up tightly. Each bolt shall have sufficient length to extend at least 6 mm through and beyond the full nut, except where such extensions might interfere with or endanger traffic in which case the bolts shall be cut off flush with the nut.
- 3. All railings shall be erected, drawn and adjusted so that the longitudinal tension will be uniform throughout the entire length of the rail.
- 4. End Treatment for Steel Barrier- End treatments shall form an integral part of safety barriers which should not spear, vault or roll a vehicle for head on or angled impacts. The two end treatments recommended for steel barriers are "Turned down guardrail" and "Anchored in back slope", as shown on the drawings or as directed by the Engineer.

#### **1.5 Tolerance**

The posts shall be vertical with a tolerance not exceeding 6 mm in a length of 3 m. The railing barrier shall be erected true to line and grade.

#### **1.6 Measurements for Payment**

- 1.6.1 Metal beam railing barriers will be measured by linear metre of completed. Length as per plans and accepted in place. Terminals / anchors of various types shall be paid for by numbers.
- 1.6.2 Furnishing and placing anchor bolts and / or devices for guard rail posts on bridges shall be considered incidental to the construction and the costs thereof shall be included in the price for other items of construction.
- 1.6.3 No measurement for payment will be made for excavation of backfilling performed in connection with this construction.

#### **1.7 Rate**

The contractor unit rate shall include full compensation for furnishing of labour, materials, tools, equipment and incidental costs necessary for doing all the work involved in constructing the metal beam railing barrier complete in place in all respects as per these Specifications.

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**NS Submission of completion drawings of ROB/as per Railway's Completion** drawing is to be prepared and submitted as directed by Engineer in-charge. It shall be the responsibility of the contractor to submit the completion drawing on Auto CAD with one tracing film in A-0 size, 4 sets of drawings prepared on plotter and one soft copy in CD. No extra payment will be made for this. In case of non-fulfillment of this condition, Rs. 20,000/- will be deducted from contractor's bill.

**Last Page of Tender  
Document(Part-I)**

Chapter-13

**ADDITIONAL SPECIAL CONDITION AND SPECIFICATION FOR**  
**TUNNELING WORK**

**1.0 GENERAL INFORMATION:**

The proposed work is mainly comprising of tunneling related works in tunnel no 1 at km 23600 to 26968 m including cut and cover in soil/rock including approach earthwork, portal walls, support, lining, shotcreting, minor bridges, drains and other miscellaneous works between Churhat and Sidhi station in connection of Rewa- Sidhi- Singrauli New Line.

The tenderer/s should inspect and study the site conditions of the section, availability of materials, approach roads etc. before submission of the tender.

The Tender Schedule for this work consists of four parts as follows: -

|  |                                                                                                                              |
|--|------------------------------------------------------------------------------------------------------------------------------|
|  | Items of works that are covered by the West Central Railway USSOR-2021                                                       |
|  | Items of works that are not covered by the West Central Railway<br>USSOR-2021 vide<br>Schedule "D" - NS items – tunnel work. |

**2.0 RATES FOR ITEMS OF WORK**

The rates quoted by the tenderer/s shall be deemed to include work necessary for setting out of works in the different phases as ordered by the engineer's representative. No. extra payment shall be made for such works or for phase work carried out which are necessary for satisfactory execution of works.

The rates quoted by the tenderer/s shall include clearing the site of all bushes, honey bees, roots, grass etc, demarcating by furrows the toes / edges of slopes of bank/cutting and cost of setting out by stakes, bamboos, strings, pegs etc. After site clearance, all pockets and depressions in the soil shall be made good and completed.

Tenderer/s will quote in their offer after full inspection of site and site conditions made known to him/them.

The affected trees and branches which are coming in the way of work will be marked by the Engineer's representative and only these trees and branches which have been thus marked and listed will be removed by the contractor. No extra payment shall be made for cutting of tree.

Cess and water charges will be recoverable as per extant rules.

In case of any dispute about quality of work, decision of Engineer in Charge at site will be final and no claim on this account will be entertained.

The depth of foundation will be decided by Engineer in charge considering the strata met with.



## **GENERAL INFORMATION**

### **3.1 APPROACHES**

The proposed tunnel approach is in general sloppy terrain with partial Inhabitation and fields. There does not exists proper approach roads and the agency who has to take up the work needs to provide proper way for approaches of work site either by leasing/purchasing from surrounding landowners. No extra payment will be given for this and the quoted rates are deemed to be included this expenditure.

### **3.2 SALIENT FEATURES OF TUNNEL**

A tentative salient feature of the proposed tunnel is shown in Annexure-I. This is only tentative for guidance and actual execution may change according to the site condition/Design of Tunnel.

### **3.3 GEOLOGICAL CONDITIONS**

The terrain of the proposed site is of sloppy in nature partly wooded with bushes and trees. The surface soil is basically of hard in nature with rock outcrop. The hard rock strata in general expected from a depth of 8m to 10m from the surface, except a few locations, where loose boulders and soft rock exists. However, firms may please note that the experience in the domain of tunneling, during actual excavation, the strata likely to be met may be substantially different from that reported upon and the one indicated through geological survey. Railway will therefore not entertain any claims on account of such variations. A detailed geological report and sub soil condition along with soil strata will be issued to the eligible tenderer at the time of issue of LOA for taking up the design details.

### **3.4 CLIMATIC CONDITION**

The above tunnel location falls in Sidhi- District of MP state. The annual rainfall is in the range of 1000-1100mm.

### **3.5 SITE OFFICE FOR ENGINEERS & OTHER SUPERVISORY STAFF**

The contractor shall arrange to provide fully furnished office accommodation of approximately 75 Sqm size (minimum two air cooled cabin) at his own cost with three executive office table, three executive chairs, six office chairs, two steel almirah, sufficient number of display board etc. within 15 days from the commencement of work as per requirement/ directions of the Engineer in charge, including electric & water facility, security and maintenance of the same and shall be removed after completion of the work. All dismantled items of the site office and all furniture items shall be property of the contractor at the completion of work. In case contractor does not provide the office accommodation with above details recovery of Rs. 50,000/- per month will be made from the contractor's dues.

### **4.0 DECLARATION**

The information given in clauses above is for general guidance only. The tenderers are advised to visit the site themselves and make themselves aware and conversant of all the data relating to site including availability and rates of all labour and materials, service road, land for campsites and arrangements for water, climatic conditions, labour laws, power situation, water availability transport problems etc.

## **5.0 DRAWING FOR CONSTRUCTION OF TUNNEL BY UNDERGROUND EXCAVATIONS & CUT AND COVER METHODS**

A typical sketch showing the maximum moving dimensions and finished dimensions and profiles for tunnels along with permanent support with trolley refuges are given in Annexure – I. Trolley and man refuges shall be as per requirements of Indian Railway permanent Way manual and as per approved Railway drawings. The cross- section of the tunnels and cut & cover in any portion shall be as per the clearance laid down in the “Schedule of Dimensions” for single line broad gauge electrified section. The drawings attached are for guidance only.

The Railway Administration reserves the right to modify the plans and drawings as referred to in the special data and specifications without assigning any reasons as and when considered necessary by the Railway. The rates quoted and accepted against the Schedules shall hold good irrespective of any changes, modifications, alterations, additions & in the locations of structures and detailed drawings, specifications and/or manner of executing the work.

No claim whatsoever will be entertained by the Railway on account of any delay or hold up of the works arising out of delay in approval of drawings, changes, modifications, alteration, additions & omissions & the site layout plans or detailed drawings.

#### **5.1 LIST OF CODES TO BE FOLLOWED: -**

Construction shall be as per various codes of practice of BIS, IRS and IRC some of them are mentioned below for guidance. The following list is for guidance and is by no means exhaustive and provisions of relevant codes even not mentioned below shall be binding on the contractor. In case of contradictions between provisions of various codes or between codes and clauses mentioned in this document, the decision of the Engineer in charge shall be binding and final on the contractor.

- i) IS 5878 (various parts) Codes of Practices relating to Tunnelling and Underground Excavations.
- ii) IS 4081-1967 Safety Code of Blasting and related Drilling operations.
- iii) IS 4576-1978 Safety Code of practice for Tunneling.
- iv) IS 9103 Admixtures for Concrete.
- v) IS 9012 Recommended practice for Shotcreting.
- vi) IS 3764 Safety Code for excavation work.
- vii) IS 7293 Safety Code for working with construction machinery.
- viii) IS 2062 Structural Steel Codes.
- ix) IS 823 Manual Metal Arc Welding Codes.
- x) IS 456 Code of Practice for Plain and Reinforced Concrete 2000
- xi) IS 1893 Criteria for Earthquake Resistant Design of Structures.
- xii) IS 800 Code of Practice for use of Structural Steel.
- xiii) IS 2720 Code of Practice for Soil Investigation (All parts)
- xiv) IS 1343 Code of Practice for Pre-stressed Concrete.
- xv) **IMPORTANT CODES**
  - a) IS 13365 part-1-1998-Rock mass rating (RMR for prediction of engineering properties)
  - b) IS 13365 part-2-1992 (reaffirmed 2001)- Rock mass quantity for prediction of support pressure in underground openings.
  - c) IS 15026-2002-Tunneling methods in rock masses-guideline.
  - d) IS 11315 part-2-1987 (reaffirmed 2001)
  - e) IS 11315 part-3-1987 (reaffirmed 2005) is 11315 part-3-1987 (reaffirmed 2005)
  - f) IS 11315 part-11-1985 (reaffirmed 2001) is 8764-1998 (reaffirmed 2008)

- g) IS 9143 1979 (reaffirmed 2001) shall be referred (reaffirmed 2008)
- h) Any other relevant code.
- xvi) IRS Indian Railway Concrete Bridge code.
- xvii) IRS Indian Railway Standard for Sub-structure Code.
- xviii) Indian Congress specifications for Road & Bridges.
- xix) Indian Explosive Act 1984.
- xx) Indian Explosive Rules 1940/1983
- xxi) Indian Railway schedule of dimensions (BG)
- xxii) Indian Railway ways & works Manual.
- xxiii) IRS Bridge rules.
- xxiv) IS 816-1969 Code of Practice for use of metal arc welding for general construction in mild steel (first revision) (Amendment 2)
- xxv) IS 1200 part XXV-1971 Method of measurement of building and civil engineering works part 25 Tunneling (second Revision) (Amendment 4)
- xxvi) 10262 code of practice for concrete mix design. Drilling of boreholes is required to be carried out in accordance with specifications of relevant codes of Bureau of Indian Standard as given below:
  - xxvii) IS: 4078 Code of practice for indexing and storage of drill cores.
  - xxviii) IS: 2131 Method for standard penetration test
  - xxix) IS: 1892 Code of practice for sub-surface investigation for foundations.
  - xxx) IS: 6926 Diamond core drilling site investigation for river valley projects.
  - xxxi) IS: 5313 Guide for core drilling observation.
  - xxxii) IS: 4464 Code of practice for presentation of drilling information and core description in foundation investigation.
  - xxxiii) IS: 5529 (Part-I & Part-II) Code of practice for in-situ permeability tests in overburden and rock respectively.

**NOTE:**(1) The latest edition of all the codes of practice along with their upto date correction slips shall be applicable. (As revised upto the date of opening/negotiation of the present tender).

(2) In absence of relevant provision of specification in above mentioned IRS/IRC/BIS Codes the reference shall be made to best available Engineering Practice/International codes as per direction of Engineer in Charge.

(3) The decision of Chief Engineer/Construction/Jabalpur regarding the interpretation of specifications shall be final and binding on the contractor.

(4) All relevant codes shall be made available by the contractor at site at his own cost.

## **6.0 SITE INSPECTION REGISTER: -**

A site inspection register shall be maintained by the Engineer or his authorized representative in which the contractor will be bound to sign, day to day entries made by the Engineer in Charge or his representative. The Contractor is required to take note of the instructions given to them through the Site Inspection Register and comply with the same within a reasonable time. The Contractor shall also arrange to receive all the letters etc. issued to him at the site of works.

## **6.1 SAFETY: -**

Safety to the existing Formation/ Tunnel, Track and its approaches etc. The proposed work has to be executed in the near vicinity of existing broad-gauge line. New Tunnel, side drains, Gabion walls etc have to be constructed in such a manner that the existing Formation/ Tunnel, its approach etc., do not get affected in any way. If any disturbance to the existing Formation/ track / Tunnel and its approach, takes place, same will be to the account of the contractor and decision of the Engineer in charge will be final in such cases. The Contractor at his own cost shall provide upon the works to the satisfaction of the Engineer-in-Charge and at such places as he may nominate, proper and sufficient lifesaving, firefighting and first aid appliances which shall at all times be available for use.

All the safety precautions for open and underground excavation as per relevant ISI codes as well as 'Safety Manual' published by National Hydroelectric Power Corporation, Delhi shall be observed by the contractor / tenderer.

Besides above the contractor/tenderer shall be bound to carry out the other instructions of Engineer-in-charge which the Engineer-in-charge may deem necessary for safety arrangements.

The contractor's responsibility for safe working shall not be conditioned by any instructions as may be given to him by Engineer-in-charge and the contractor/ tenderer shall on his own ascertain and adopt all necessary precautions.

The cost of all safety arrangements including cost of all labour and material, tools and plants and shuttering/ scaffolding/ shoring wherever required is deemed to have been included in various rates quoted in schedule of items, rates and quantities and nothing extra is payable under this contract.

**6.2 High tension lines and Sources Electro Magnetic Radiation:** Work in vicinity of High-Tension Lines and Sources Electro Magnetic Radiation: Special precaution are necessary, especially so when blasting is to be carried out, in the vicinity of high tension electric lines and sources of electro-magnetic radiation. All rules and procedures laid down by Electricity Authority in this regard shall be strictly followed.

The contractor should strive and maintain near zero safety hazard record while excavation of tunnel by blasting/mechanical means, providing temporary support systems, followed strictly the provisions of Safety Manual and arranging personal protective equipment. The agency should also coordinate with '108' emergency service for any eventuality.

### **6.3 Ventilation:**

All underground excavation work shall be properly and adequately ventilated by system of pipes and fans to the satisfaction of the Engineer-in-charge. Positive artificial means of ventilation shall be deployed and shall be in operation at all times when people are working under ground. The ventilation plans and ducts employed shall be of size and capacity as to be capable of delivering to or exhausting from the vicinity of the working face at least 4.5 cum of air per minute per person and in addition an extra 2.5 cum per minute per person of the internal combustion engines if permitted to be used by the Engineer-in-charge. After blasting the ventilation plant shall be set to exhaust the explosive gases for time which will depend on the number and size of the blasts. The workman shall not be allowed to enter and start the

work unless and until all objectionable gassier removed to the extent considered satisfactory by the Engineer-in-charge. Adequate ventilation will be provided at all underground works areas, where necessary by mechanical means sufficient to keep the working cavities safe for workers.

#### **6.4 LIGHTING OF THE TUNNEL**

Adequate lighting shall be provided for works carried out in night shifts and where day light is not adequate. Adequate lighting shall also be provided at the faces and any other point where works are in progress and also at equipment installations. Any obstructions, such as jumbo frame work etc. inside the tunnel shall be well lighted to avoid accident when hauling units are moving. Lighting at the working face shall be adequate, however, no single light shall be so powerful as to cause temporary blinding effect when looked at. All the excavation work shall be adequately illuminated with electric lights to the satisfaction of the Engineer-in-charge. Point of intensive works like the face, shall be properly and carefully illuminated to ensure safety during working. Special light shall be provided as necessary for checking alignment. Proper illumination shall be provided also during non-working hours as may be required at each location. The cost of providing installing operating and maintaining all the lighting to required standards shall be considered as included in the unit rates for various items of schedule of items, rates and quantities.

#### **7.0 COMMUNICATION FACILITIES: -**

The contractor shall install and maintain Modern Radio communication facilities in good working order at all times during the period of the work at or near the working face of the tunnel shafted., and other points so as to afford prompt communications over such telephone. The contractor shall also install, maintain and operate such other communication and signal facilities as may be necessary for safe and efficient execution of the work. All such facilities shall be subject to the approval of the Engineer-in-charge. Authorized employees of Government shall have free use of such communication and signal facilities.

The entire cost of providing and maintaining such a communication and signal system shall be deemed to have been included in the unit rates tendered for underground works in schedule of item rates and quantities.

#### **8.0 CONSTRUCTION OF TUNNEL BY CUT AND COVER & EARTHWORK IN EXCAVATION IN CUTTING: -**

Construction of Tunnel by the Cut and Cover Method will be executed at the locations mostly in the approaches where adequate cover is not available

##### **8.1 SCOPE: -**

Excavation in all types of soil and rocks including rock requiring blasting to proper line, level, grade and dimensions as required and maintaining the same till taken over by Railways.

**8.2** Back filling with selected earth over the RCC D-shaped box structures of Cut and Covers portion.

**8.3** Works required ensuring safety, Temporary works, dewatering, setting of works etc.

**8.4** The contractor should commence work systematically from one or more points in

consultation with the Engineer-in-charge and maintain continuous progress completing the cuttings in continuous length in all respects including levelling and dressing.

**8.5** The contractor at his own cost shall provide, fix and be responsible for the maintenance of all stakes, templates, profiles and mark points, buries, monuments, center line pillars, reference pillars etc. and shall take all necessary precautions to prevent their being removed, altered or disturbed and will be responsible for the consequences of such removal, alterations or disturbance and their efficient reinstatement. Contractor will also preserve the samples of strata wherever asked for in such a manner so as to enable assessment of the type of strata encountered. This work will be done within the quoted rates and nothing extra will be paid on this account.

**8.6** Generally, only L-section of the alignment will be supplied to the contractors. The profile of cutting with side slopes and side drains shall be as per approved plans. The formation level shall be reckoned to be the level of the centre of the cutting. Suitable berms where the depth of cutting is more shall have to be provided as per the direction of the Engineer.

**8.7** The widths, side slopes of cutting may be changed by the Engineer, if so found necessary for the stability of the cutting. The contractor shall have no claim whatsoever on account of change in side slopes or formation widths. The decision of the Engineer for determining the slopes initially or subsequently and for modifying the slopes and formation widths shall be final and binding and no claim whatsoever shall be entertained by Railways.

**8.8** The dumping site for disposing off cut spoils including rocks shall be so chosen by contractor so that it is away from Railway Land enough to avoid any falling back of soil to Railway land. Such dumping site shall be taken on lease / purchase by contractor at his own cost and nothing shall be payable extra to him on this account.

**8.9 BACK FILLING, SIDE FILLING & CONSOLIDATION: -**

The contractor shall programme his works in such a way that the excavation material approved for back filling goes immediately for the purpose. If this is not found feasible for any reason and when so decided by the Engineer, the contractor shall arrange for suitable back filling material at his own cost and for which nothing extra will be paid for the same. Work of back filling should be commenced only when the Engineer-in-charge permits the same. The space between RCC section of Cut and Cover and trench shall be filled with approved excavated material / granular material in layers of 300mm thickness and shall be fully consolidated manually. Top of RCC Cut and Cover shall be filled in layers as approved by Engineer-in-charge with selected excavated material, approved up to the desired height as directed by the Engineer. Payment will be made under relevant Item of schedule of items, rates and quantities. The contractor shall take special and particular care to consolidate the ground, where utilities are to be transferred so that no settlement occurs to these utilities.

**8.10 RATES: -**

The tendered rate against Item of Schedule of Rates and quantities for open excavation in cutting will be applicable to all classes of soils and rocks under each category and include all operations, incidental charges & royalty etc. The tendered rates payable for various items of open excavation for leading and disposal of surplus earth shall be as per Schedule of Rates and includes the cost of all labours, materials, equipment all lead, lift, and ascent, descent, nallah crossing including that for temporary works, safety arrangements, temporary stabilization of slopes outside the Railway Land for safe execution of the work, setting out, clearing of site, benching in ground, maintenance of slopes and formation till taken over by

Railway.

#### **8.11 DEPLOYMENT OF TECHNICAL STAFF: -**

8.11.1 The contractor shall employ proper managerial and technical personnel during the execution of this work and the personnel deployed shall have adequate experience and thorough knowledge of the works executed including the specifications and proceedings involved. The list of managerial and technical personnel proposed to be engaged by the contractor shall be submitted to the Dy. Chief Engineer in charge of the project along with the programme chart and approval of Engineer to be obtained for engaging them for work.

8.11.2 Tunnelling is a special nature of work and the minimum requirement of technical personnel To be deployment for this contract is enclosed in Contract agreement.

8.11.3 The above specified technical personnel is minimum requirement only and contractor has To deploy additional technical person of all category like Engineer, supervisors, surveyors, technicians, khalasi for achieving the required progress within the Currency period/approved program by Railways.

#### **9.0 UNDERGROUND EXCAVATION AND SUPPORTS:**

Tunnelling by Underground excavation will be done where adequate cushion / cover is available. The decision of Site in Charge in this regard shall be final and binding.

##### **9.1 SCOPE: -**

Underground excavation shall generally include but not limited to following items/operations as per approved drawings and or as directed by the Engineer-in-charge:

The underground excavations will also include the following categories:

- i. Subsequent enlargement of fully excavated section.
- ii. Excavation by controlled blasting, line drilling, pre-splitting, multiple drift etc.
- iii. Excavation by heading and benching etc.

Drilling and Controlled Blasting (if required) in all types of strata actually met with. Drilling by drilling jumbos where directed by Engineer-in-charge to reduce the cycle time.

Where blasting is prohibited by Engineer-in-charge, drilling and excavating by mechanical excavators.

Removal, conveyance and disposal of the excavated materials as and where directed by the Engineer-in-charge.

Lighting and ventilation arrangements required for construction arrangement and including their safe operations.

Providing communication facilities required for successful completion of the tunnel work.

Providing necessary arrangements and facilities for inspection and measurements at any time and when required by the Engineer-in-charge.



Providing all tools, plants, labour and material necessary and ancillary works thereto for satisfactory completion of the work.

Construction and maintenance of all haul roads, ramps etc. for mucking. Any

steel structure required for drilling and other operation etc.

Any other connected work in connection with above. No extra payment shall be given to temporary spreading finer materials (obtained by excavation) in the invert floor to form approximate level surface for track and for use as working floors. The nature and character of soil / rock likely to be met within the underground excavations is indicated in the para 4.4 above. The Railway does not, however, take any responsibility for any variations that may be detected in actual excavations.

## **9.2 SETTING OUT: -**

Setting Out of the tunnel shall be done strictly as per IS: 5878 (Part I) –1971 with up-to-date amendments. Suitable permanent bench marks shall be established near the portals and adits of tunnels and shafts and cut and cover etc. as the work progresses, center line marks shall be marked on pegs inserted into the roof of tunnels, cut and cover and other hollow cuts at convenient intervals to the satisfaction of the Engineer-in-charge for checking alignment, grades levels etc. and also the dimensions of the “minimum excavation lines”. The contractor shall at all times remain responsible for the sufficiency and accuracy of all such bench marks, center line pegs and reference points. The cost of these shall be deemed to have been included in the rates tendered for work.

## **9.3 ACCURACY OF ALIGNMENT, GRADES, AND LEVELS ETC.**

The Contractor shall establish bench marks and fixed reference points by the co-ordinates with the value of the levels as shown in working plans provided by the Railways or at least two reference points on the alignment at either portal and a benchmark in working area of the proposed reach work. The contractors shall make arrangements to protect and safe guard such points/benchmarks and preserve same up to completion of work. The contractor shall provide free of cost all facilities like labour and latest instruments etc. and all facilities to the Engineer-in-charge to check the alignment, grades levels etc. whenever and anytime they are asked for. Such checking by the Engineer-in-charge shall not absolve the contractor from his responsibility of maintaining the accuracy of the work. Any discrepancy or error detected during the course of excavations and/or at the end of the work when the faces meet shall be set right by the contractor at his own cost in a manner satisfactory to the Engineer-in-charge. The department will have the right to demand at the site from the contractor at his cost, all instruments considered necessary for proper control of line and level like electronic total station, automatic levels, laser discs, theodolites etc.

The contractor shall provide suitable rig and make all necessary arrangements to the satisfaction of the Engineer-in-charge for moving the rig along the alignment and grade of the tunnel, shaft etc. to check the dimensions of minimum excavation lines, lining etc. as and when required by the Engineer-in-charge.

**9.4 DRILLING AND BLASTING: -** Contractor shall comply with all statutory rules and regulations in regard to procuring, transporting, storing and using explosive that may be required for this work. Utmost care will be taken to ensure safety of all labour/staff or general



public while handling the explosives.

The drilling and blasting shall be generally as per provision of I.S: 5878 (Part –II) 1979.

## **9.5 SUBMISSION OF DESIGN & DRAWINGS**

**9.5.1** The tunnel shall be executed as per Railway's approved design and drawings. The contractor has to take Guidance & for design pattern of holes required for Tunnel Blasting/Excavation verified from firm such as "CENTRAL INSTITUTE OF MINING & FUELRESEARCH, REGIONAL CENTRE (CIMFR), NAGPUR' for safety to avoid over cut and to achieve required progress," (i.e. Design of Tunnel including support system for stability based on Geotechnical data, preparation of report indicating locations of permanent lining, rock bolting and shotcreting etc., from firm such as CIMFR, Regional centre, Nagpur.)

**9.5.2** The contractor shall submit proposed method of drilling and blasting for underground excavation. This shall consist of the scheme of excavation, the pattern and type of holes to be drilled, quantity of explosives to be used and fixing method etc. No underground excavation shall commence without prior approval of the Engineer.

**9.5.3** Approval by Engineer of any schemes etc., above does not absolve the contractor of his responsibility in any way.

**9.5.4** Controlled Drilling and Blasting: Wherever in the opinion of the Engineer it is considered necessary to adopt controlled blasting the same shall be resorted to and no extra payment shall be made.

**9.5.5** The Contractor shall make the arrangement for supply, transport, storage and safe custody of the explosives at his own cost. Railway will only give assistance by issuing the necessary certificates/letters as required by the concerned government department for giving license for the use of explosives and its supply.

## **9.6 VENTILATION, LIGHTING, MUCKING AND DEWATERING**

Contractor shall make arrangement for Ventilation, Lighting, Mucking and Dewatering as laid down in IS: 5878 (Part II/Sec2) –1971. Adequate Safety requirements, in regard to ventilation, lighting, mucking and dewatering of tunnel, shall be taken as specified in IS: 4756 – 1968. The capacity of the ventilation plant should be adequate enough to allow starting of mucking operation, commensurate with the schedule progress of work. The cost of furnishing, installing, maintaining and operation of the ventilation, lighting, mucking and dewatering system & arrangement for measuring the concentration of the poisonous gases shall be deemed to have been included in the unit rates tendered in the Schedule of Rates for the underground excavation.

## **9.7 DRAINAGE AND PUMPING**

**9.7.1** All water from the underground excavation shall be satisfactorily drained away and pumped out if necessary. In case, any underground springs are met with, the water from them shall be soled away as not to damage or endanger any work. The pumps to be used shall be operated either by compressed air or electric motor and not by any type of internal combustion engine unless specifically permitted. Drainage of all tunnels, shafts etc, excavated will be by gravity preferably. The cost of drainage by gravity during construction will be considered included in the rate accepted for the various items in the Schedule of Rates for the

underground excavation. If a channel has to be excavated outside the lines of excavations shown, the contractor may excavate such channel with the approval of the Engineer-in-charge. This channel has to be filled back by the contractor with concrete as specified for the back filling of tunnels. The excavation and back filling of such channel carried out by the contractor shall be at his cost. When, however, the water has to be pumped out, the contractor shall install the necessary pumps and pipes as may be approved by the Engineer-in-charge in writing the cost for which will be paid separately through the items provided in schedule of works. The channels or the pumping unit shall be maintained satisfactorily till 21 days after concrete back fill is laid.

The underground and the water emanating from the construction sites required to be pumped out may be contaminated with dust, cement, slurry, dirt, sand etc. The pumping arrangements to be provided shall be suitable for such drainage work. It shall be seen that at no location there is pooling of water, which has the potential to obstruct the work. The contractor must also take all precautions essential to prevent water from the open area flowing into the tunnels so as to keep the work sites including the face of the tunnel free from water during all seasons of the year.

## **9.8 EXCAVATION LINES AND RATES**

**9.8.1** The tunnels, shafts, etc, shall be excavated to the lines and grades shown in the drawings, or as directed by the Engineer-in-charge. The 'Excavation Lines' indicated on the drawings are, minimum excavation lines, within which no unexcavated material of any kind shall be allowed to remain, The 'Payment Lines' are the outside limits up to which the contractor will be paid regardless of the fact whether the limits of actual excavation fall inside or outside these lines except in the case of natural cleavages etc. in rock unless specifically provided herein below. Unless specifically mentioned otherwise the 'Payment Lines' will be taken normally or radially to the center line as the case may be 150mm beyond the 'minimum' excavation lines.

**9.8.2** Any enlargement beyond the 'Payment Line' considered necessary by the contractor for the convenience of his work may be done by him with the prior approval of the Engineer-in-charge. This shall be filled back with concrete of class similar to the backfill concrete. All works carried out for such enlargement including the backfill will be purely at the contractor's cost.

**9.8.3** **OVERBREAK:** - Any excavation carried out or which takes place inadvertently beyond the 'Payment Line' shall be deemed to be an 'OVERBREAK'. In case cavity and chimney formation occur during the pendency of the contract, they shall be deemed to be covered in this definition of the 'OVERBREAK'.

**9.8.4** "GEOLOGICAL OVERBREAK" consists of that portion of the over break outside a line parallel to and 300mm beyond the "Payment Line", the occurrence of which, as per the decision of Engineer-in-charge, is an unavoidable result of adverse geological conditions due to concealed joints, faults and other structural defects in rock and not due to lack of reasonable care and skill in excavation on the part of the contractor. This definition of 'GEOLOGICAL OVERBREAK' shall be applicable for the term "GEOLOGICAL OVERBREAK" used elsewhere in this document.

**9.8.5** **REPORTING OF GEOLOGICAL OVER BREAK:** - An occurrence report of 'OVERBREAK' extending outside a line parallel to and 300mm beyond the "Payment Line" shall be prepared and forwarded by contractor immediately after its occurrence to Engineer-in-charge so that Engineer-in-charge may evaluate the cause of such OVERBREAK and decide if such OVERBREAK shall be defined as "GEOLOGICAL OVERBREAK". Decision

of Engineer-in-charge shall be find in this regard.

#### **10. MEASUREMENT & PAYMENT**

Except as otherwise provided herein, no OVERBREAK beyond "Payment Line" shall be measured for payment. Measurement for payment of "GEOLOGICAL OVERBREAK" will be made only for the volume of rock removed between the payment line and the actual excavated rock profile but limited to average parallel line at 300mm beyond the 'Payment Line'. Nothing extra shall be paid beyond average parallel line at 300mm from payment line.

##### **10.1 PAYMENT OF OVERBREAK/ GEOLOGICAL OVERBREAK**

Payment as under will be made to contractor and beyond 10% no payment will be made. Up to 5% - @ 50% of the accepted rate for tunnel excavation.

Beyond 5% to 10% - @ 25% of the accepted rate for tunnel excavation  
Beyond 10% - No payment.

These percentages will be reckoned on the total quantity of excavation executed for entire length of the particular tunnel.

Over-break beyond 10% of pay line profile quantity will not be paid for and this element shall be taken care in the rate quoted for the relevant item in the Schedule.

Over-breaks that are adjudged by the engineer in charge as due to bad workmanship will automatically result in all consequent works such as providing temporary supports, backfilling of M-20 concrete, plugging etc. as are considered necessary being done by the contractor at his own cost.

When the excavation of tunnel is in progress, if directed by Engineer-in-charge, contractor shall backfill portions of "OVER BREAK" and "GEOLOGICAL OVER BREAK" as defined in para 12.8 and 12.9 by concrete of suitable grade & quality as approved by Engineer-in-charge. Measurement and payment of such a backfill concrete shall be regulated in accordance with relevant provisions made under item of concrete.

Cavity and chimney formation, if occurred during execution of work, shall be dealt as per provisions applicable to "OVER BREAK" and "GEOLOGICAL OVER BREAK" depending upon which portion of cavity and chimney formation lies in areas of "OVERBREAK" and which portion lies in areas of "GEOLOGICAL OVER BREAK". The contractor shall use every precaution to avoid excavation beyond the payment lines marked on the drawings. All drilling and blasting shall be carefully and skillfully performed so that the material beyond the required lines is not shattered. No blasting that might damage the rock shall be permitted and any damage done in the rock by blasting shall be repaired by and at the expense of the contractor and in a manner satisfactory to the Engineer-in-charge.

##### **10.2 PAYMENT OF UNDER GROUND EXCAVATION FOR TUNNELING (SOIL /ROCK)**

Recovery for unused cut spoil: - Railway will recover @ Rs. 25 per cum for the unused surplus cut spoils (excluding hard rock) coming out from items (1) & (2) of Schedule 'B' to be dumped outside railway boundary / area which will be property of contractor. Since the exact recovery on this account can be finalized only after completion of work, stage

payment is envisaged for running on account bills.

12.13.2

- (a) For underground tunnel excavation initially a payment of 80 % of quoted rate will be paid on executed quantity.
- (b) Further 10% payment for actual quantity of work done when excavation is done for catch side drains, side slopes completed and surface dressed as per instructions of engineer in charge.
- (c) Balance 10% payment for actual quantity of work done shall be made when excavation is completed in all respects and side drains with masonry/concrete is built.

**10.3 SUBSEQUENT INCREASE OR REDUCTION IN DIMENSIONS: -**

If during the progress of excavation of any particular element of the work it is required to increase or reduce the cross – sectional dimensions of the excavations from those shown in the typical cross- sections on the drawings, the contractor shall on being so directed by the Engineer-in-charge, excavate the further portion to such increased or reduced dimensions at the same unit rate as tendered for the particular work. The Engineer-in-charge will establish fresh, minimum, excavation line and the 'Payment Line' shall be 150mm beyond such fresh 'minimum' excavation line. The excavation already done shall also be enlarged to the increased dimensions if so ordered by the Engineer-in-charge. The quantity measured under this item will be between the original Payment Line and the new Payment Line, which will be fixed by Engineer-in-charge. The quantity paid for GEOLOGICAL OVER BREAKS due to natural cleavages, will, however, be excluded. This will not, however, apply to the excavation of soft layers of rock already excavated and intended to be plugged back with concrete. The payment covered under this clause shall be governed by relevant item of Schedule.

All excavation shall be so done that the rock around the periphery of the excavated final section shall not be shattered, loosened or otherwise weakened. Large over-breaks will be avoided by suitable means such as light charges, ream holes or similar measures. The cost of this measure shall be considered as included in the accepted rate of excavation. In a special location, where specifically indicated or ordered by the Engineer-in-charge in writing, the use of explosives shall be discontinued and the rock excavated by line drilling, broaching, barring, chiselling, wedging, as may be approved and directed by the Engineer-in-charge. The payment of such excavation will be made under relevant item of Tunnel excavation in schedule and no separate rate is admissible for purpose of calculating the quantity to be paid, thus the depth of excavation shall be taken as 150mm (One hundred fifty millimetres) from minimum excavation line.

**11. SCALING OF ROCK**

After face is blasted, gases removed and the face ventilated, all loose, or loosened rock frothed top and side shall be removed by approved methods to avoid danger to workmen and equipment. Later, if any loose or loosened rock is noticed anywhere in the excavation, it shall be removed expeditiously. As the excavation advances trimming of rock surface remaining within the excavations lines shall be undertaken simultaneously. In no case shall trimming be postponed and left to be tackled last. This work if left to be done later takes lot of

time and may involve avoidable delay resulting in postponement of completion dates. The cost of such scaling and removal of loose or loosened rock, trimming etc., shall be considered as included in the unit rate for excavation.

**12. ADITS AND SHAFTS: -**

If for adhering to the prescribed time schedule, additional faces are to be opened and adits are found necessary, the contractor may arrange to execute the work accordingly after obtaining the written approval of Engineer. In such cases the basic rate for excavation for adit as also the tunnel excavation carried out through the adit will be the same. If provision of a vertical shaft is found necessary, the basic rate admissible for tunnel excavation will be enhanced as under:

- i. For tunnel excavation involved in sinking the vertical or near vertical shaft. 50% Extra over basic rate of excavation in tunnel
- ii. For excavation of main tunnel through the shaft 0% (i.e., no enhancement)

**b. TEMPORARY SUPPORTS: -**

By temporary supports it is meant support whether of timber or steel required to support the underground excavated section during excavation but which can be removed before or during the concrete lining. Contractor shall erect temporary supports, when so desired by the Engineer-in-charge. It shall be the contractor's responsibility to bring in writing to the notice of the Engineer-in-charge location where the contractor may consider the provision of temporary supports necessary. Consequence, if any for his failure to do so in good time, shall be the sole responsibility of the contractor. The methods, the extent of supporting, sizes and spacing of supports etc., shall be approved by Railway.

Temporary supports may be of steel or timber and shall be removed before placing the concrete lining. No payment will be made for erection of temporary supports. If, however, steel supports are used, cost of steel and its fabrication will be paid separately. The methodology and type of material to be used for temporary supports has to be approved by Engineer before actual erection. If steel supports are used the contractor has to take all measures for repeated use in tunnel excavation cycle. Once the supports are used extensively and become scrap, it is the property of the contractor.

**12.1 IMMEDIATE SHOTCRETING AND SHOTCRETING WITH STEEL FIBRES (SFRS)**

**DESCRIPTION: -**

Work consists of the production and application of Shot Crete with or without steel fibre to the specified thickness at the locations required, using wet Shot Crete machine with required materials, equipment, tools and labour necessary to perform the preparation, application and the cleanup pertaining thereto.

**DEFINITION: -**

Shot Crete, for this work, shall be defined as spraying premixed Portland cement concrete as per approved design with admixtures including Silica fumes if required and additives (i.e., water reducing agents, thixotropic agent, accelerators etc.) from a spray nozzle by means of compressed air using wet Shot Crete machine with Robot arm. When steel fibers and micro silica are added to shotcrete, it shall form SFRS. Shot Crete/SFRS shall be applied in accordance with relevant ACI/ASTM/EFNARC specifications/guidelines.

Rebound is defined as the constituents of Shot Crete /SFRS that rebound from surfaceduring the application of Shotcrete /SFRS.

### **MATERIALS:**

Ingredients of Shot Crete / SFRS include (i) Cement, (ii) Aggregate, (iii) Admixtures, (iv) Silica Fume and (v) Steel Fibres. Specifications of ingredients are given below:

- (i) Cement: Cement (OPC Gr. 43/53) or as directed by Engineer-in-charge shall be used.
- (ii) Aggregate: Aggregate shall conform to the requirements of relevant IS code or as directed by Engineer-in-charge. The aggregate size shall not exceed 10 mm. The grain size distribution of aggregate shall be as under: -

#### **Percentage passing through sieve**

| <b>Sieve size(mm)</b> | <b>Minimum %</b> | <b>Maximum %</b> |
|-----------------------|------------------|------------------|
| 10                    | 100              | 100              |
| 8                     | 90               | 100              |
| 4                     | 73               | 100              |
| 2                     | 55               | 90               |
| 1                     | 37               | 75               |
| 0.5                   | 22               | 50               |
| 0.25                  | 11               | 25               |
| 0.12                  | 4                | 10               |
| 5                     |                  | 2                |

The aggregate shall be well graded and no fraction shall constitute more than 30% of the total mass.

- (iii) Admixtures: - Only proven admixtures which meet the requirement of the specification on field trials and satisfactory performance on other projects already constructed in India and elsewhere shall be used to ensure a homogeneous mix and better dispersions of various constituents of Shot Crete / SFRS. A two-component system to control the consistency shall be used to ensure good workability and low rebound. The admixture system shall produce slump killing effect and thixotropic consistency thus adhering in thick layers without trickling and subsequently reducing the fibre rebound. The contractor shall submit along

with the design a certificate for satisfactory performance of admixtures and chemical compatibility between two components from user department and manufacturer. Admixture shall consist of following: -

- a) Water reducing/thixotropic agents: This shall be added at the batching plant/concrete mixture to keep the Shotcrete /SFRS mix workable during transportation and to ensure pump ability attain acceptable low W/C ratio. The chloride contents shall not exceed 0.1% by weight.
- b) Accelerators: Accelerator shall meet the following requirements: -

- i) Only liquid accelerator shall be used.
- ii) Water glass (sodium silicate) shall not be used unless in a modified polymer based form approved by the Engineer-in-charge.
- iii) Only the minimum quantity of the accelerator necessary shall be permitted in normal shotcreting operations subject to maximum dosage of 5% by weight of cementations material.
- iv) Accelerator should have pH value less than 12.
- v) Accelerator shall be selected such that at the chosen dosage for use in the works, the decrease in strength at 28 days compared with the base concrete without any accelerators shall not exceed 20%. There shall be no further reduction in strength of the shotcrete mix between 28and 90 days.
- vi) Accelerator shall have the property to provoke sharp slump loss and shotcrete should have thixotropic consistency as it hits the substrate to reduce the fibre rebound.
- (vii) Micro silica (Silica Fumes): - Micro silica produced / manufactured by only approved supplier as approved by Engineer-in-charge shall be used for the work. It should be as per specifications of the supplier confirming to relevant ASTM/Canadian Standards. The doze of silica fumes shall be5% to 7% of cement weight subject to site conditions. Silica fumes shall have bulk density between 500 Kg/Cum to 700 Kg/Cum. A Few important requirements of Canadian code are reproduced below for guidance: -

**A. CHEMICAL REQUIREMENT**

| <b>SN.</b> | <b>Description/ Property Specification</b>                       | <b>Frequency of Test</b> |
|------------|------------------------------------------------------------------|--------------------------|
| 1.         | Content of SiO2- More than 85%<br>directedby engineer in charge) | Lot or 100 MT (or as     |
| 2.         | Content of SiO3- Less than 1%<br>directedby engineer in charge)  | Lot or 100 MT (or as     |
| 3.         | Loss on Ignition- Less than 6%<br>directedby engineer in charge) | Lot or 100 MT (or as     |

**B. PHYSICAL REQUIREMENT**

|    |                                                                                                                           |                                                         |
|----|---------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------|
| 1. | Accelerated Pozzolonic Activity index with<br>directedPortland Cement at 7 days (Min % age of<br>control) - More than 85% | Lot or 100 MT (or as<br>by engineer in charge)          |
| 2. | Increase in drying shrinkage of Mortar<br>ars at 28 days (Max%age joints) - 5%                                            | Lot or 100 MT (or as directed<br>by engineer in charge) |



3. Reactivity with cement reduction of mortar expansion at 14 days (Min % age)-80%

Lot or 100 MT (or as directed by engineer in charge)

### C. METHOD OF TESTS

| <b><u>S. N.</u></b>                                                                               | <b><u>Chemical Tests</u></b>      | <b><u>Reference Standards</u></b> |
|---------------------------------------------------------------------------------------------------|-----------------------------------|-----------------------------------|
| 1.                                                                                                | Silica (SiO <sub>2</sub> )        | ASTMC 114                         |
| 2.                                                                                                | Sulphur trioxide                  | CSCAN 3A5                         |
| 3.                                                                                                | Moisture content                  | ASTMC 311                         |
| 4.                                                                                                | Loss in ignition.                 | ASTMC 311                         |
| 5.                                                                                                | Pozzolonic Activity Index 7 days  | ASTMC 311 (Section 29 – 32)       |
| 6.                                                                                                | Pozzolonic Activity Index 28 days | ASTMC 311 (Section 29 – 32)       |
| 7.                                                                                                | Slag Activity Index               | ASTMC 311 Clause 6                |
| 8.                                                                                                | Soundness                         | ASTMC 311 (Section 25)            |
| 9.                                                                                                | Drying shrinkage                  | ASTMC 311 (Section 22-24)         |
| (except that for testing silicafume<br>50 gm. of silica fume and 1325gm. of sand in test mixture) |                                   | use 500g. of Portland cement      |
| 10.                                                                                               | Reactivity with cement alkalis    | ASTMC 441                         |
| 11.                                                                                               | Fineness                          | CSN CAN 3-A5                      |
| 12.                                                                                               | Relative density                  | ASTMC 311 (Section 20)            |
| 13.                                                                                               | Uniformity of air content         | ASTMC 311 (Section 27 & 28)       |

The contractor shall submit test certificate from a reputed laboratory / supplier certificate for all ingredients brought and intended to be used in Shot Crete / SFRS by him. Engineer-in-charge may however get the samples of these materials tested independently at contractor's own cost. The cost of such test shall be deemed to be included in the quoted rates.

#### iv) STEEL FIBRES:

The steel fibres to be used in SFRS shall be of high tensile steel, either cold rolled or cold drawn wires. Fiber shall be dry and free from oil, grease and chlorides. The fiber shall be in accordance with ASTM A 820-85. However, some of the control parameters are given below for guidance:

- a. Length of steel fibers: - 25mm – 40mm.
- b. Geometric shape: - The steel fibers shall have suitably deformed profile to ensure proper matrix of SFRS mix end to develop better and anchorage with no fibre balling.
- c. Aspect ratio: - 45-75
- d. (Length/diameter)
- e. Fiber tensile strength: - Not less than 1000 Mpa.
- f. Quantity of steel fibres: - Approximately 50 Kg / Cum in the sprayed
- g. (In situ fibre contents) concrete (subject to design requirements)
- h. The mixing procedure adopted by the contractor should be such that there is no fibre balling.

#### i. STRENGTH CRITERIA: -

The final product shall meet the strength criteria as per relevant EFNARC / ACI standards.



The frequency of testing shall be as specified in the codes or as directed by Engineer-in-charge. The strength requirements for SFRS/Plain Shot Crete are given below:

**A. Steel Fibre Reinforced Shot Crete;**

The SFRS shall meet the following strength requirements:

| <b><u>SN Description Days after which testing done (MPA)</u></b> |    | <b><u>Required Strength</u></b> |
|------------------------------------------------------------------|----|---------------------------------|
| 1. Compressive Strength on Cylindrical Cores                     | 3  | 10                              |
| 2. Compressive Strength on Cylindrical Cores                     | 28 | 20                              |
| 3. Flexural Strength                                             | 28 | 3.8                             |
| 4. Toughness                                                     | 28 | 2.4                             |

**B. Plain Wet Concrete**

The mix shall be designed to give the following strength requirements.

**SN Description Days after which testing done Required Strength (MPA)**

|                                              |    |    |
|----------------------------------------------|----|----|
| 1. Compressive Strength on Cylindrical Cores | 3  | 10 |
| 2. Compressive Strength on Cylindrical Cores | 28 | 20 |
| 3. Flexural Strength                         | 28 | 3  |

**MIX DESIGN: -**

The contractor shall submit mix design to meet the strength as specified for prior approval of Engineer-in-charge prior to commencement of Shot Crete / SFRS. Mix design should give following details: -

|                                       |     |
|---------------------------------------|-----|
| i. Ordinary Portland cement-          | Kg. |
| ii. Micro silica -                    | Kg. |
| iii. Natural fine aggregate-          | Kg. |
| iv. Crushed coarse aggregate -        | Kg. |
| v. Crushed fine aggregate-            | Kg. |
| vi. Steel fibres -                    | Kg. |
| vii. Water reducer/thixotropic agent- | Kg. |
| viii. Accelerator-                    | Kg. |
| ix. W/C ratio                         |     |
| x. Slump                              |     |
| xi. Density (Wet)                     |     |

- xii. Compressive strength on 3 days/7 days/28 days on cylindrical cores.
- xiii. Equivalent cubes strength 28 days
- xiv. Flexural strength 28 days
- xv. Toughness 28 days

### **13. TEST PROCEDURE:**

Tests of SFRS/Shotcrete shall be done as per relevant ASTM / JSCE / EFNARC standard or as directed by Engineer-in-charge. All equipment for testing shall be arranged by the contractor at his own cost. Some of the tests of the Shotcrete / SFRS are given below for guidance:

#### **13.1 COMPRESSIVE STRENGTH:**

The specimen cylindrical cores measuring minimum 75 mm dia or as directed by Engineer-in-charge or concrete cubes 100 mm size, shall be drilled / sawn / extracted out of prepared test panels. Panels for testing shall be at least 600 X 600 mm in size and shall be of thickness as per design. The panel shall be prepared by Shotcreting into vertical moulds, which shall be constructed of steel or other non-water absorbent material and shall have sidesplayed outmoulds at 15 degree to prevent the entrapment of rebound.

**13.2 FLEXURAL STRENGTH AND TOUGHNESS PROPERTIES:** - Flexural strengths and toughness properties shall be measured in accordance with Japanese standard JSCE/ASTM standards. The beam shall be cut out from splayed panels. The beam shall be stored in water for a minimum of 3 days after sawing and immediately before testing and kept moist during testing.

#### **13.3 PLACING EQUIPMENT: -**

Contractor should have wet Shotcrete machine with Robot arm, which can do Shotcreting efficiently without any wastage of working hours. The machine should be able to place the Shotcrete & SFRS as per design mix. The air supply system shall be capable of supplying air at the pressure and volumes needed for the efficient operation of the machine. No air supply system shall be used that delivers air contaminated by oil or that is incapable of maintaining constant pressure. In addition, a separate air hose and blowpipe shall be available to remove dust & rebound during Shotcrete application.

#### **13.4 METHOD OF APPLICATION: -**

Only wet mix method shall be used for the purpose spraying Shotcrete/SFRS. Water cement ratio shall be 0.45 (maximum) by weight of cementations material. The workability shall be measured by slump test and should be between 10 cm to 15 cm. The slump shall also be checked after interval of 15 minutes, which should be within prescribed limits. The Shotcrete older than 2 hours and having slump not within prescribed limits shall not be used unless treated with hydration control additives. The required workability shall be ensured before transporting the mix for execution at batching plants / transit mixer itself. Surface shall be suitably prepared by scaling followed by cleaning with compressed air and water under pressure through nozzle. All surfaces shall be wet and clean at the time of applying SFRS / Shotcrete. Before filling the mix in the hopper of the Shotcrete machine, slurry shall be mixed and put into the concrete hose. The application procedure of SFRS/Shotcrete shall be developed in the field to give minimum rebound, minimum shrinkage to the satisfaction of Engineer-in-charge. The nozzle shall be held at a predetermined distance approximately one metre but in no case greater than 1.5 metre from the surface to be covered, and positioned so that the stream of flowing material shall be applied as early as possible at right angles to the surface to be covered. Nozzle shall be held in steady motion so that the Shotcrete is applied uniformly to build up the required thickness of layer. Acceptable Shotcrete

shall consist of dense and uniform concrete without rebound, inclusions, segregation or discernible weakness of bond between layers. The nozzle men shall apply shotcrete with a uniform consistency in order to minimize binding cohesion and density, minimize rebound and segregation and prevent sagging of applied Shot Crete. Shot Crete shall always be started from the bottom to avoid rebound material getting locked in. The first 50 mm thick layer of SFRS shall be applied uniformly and before applying SFRS, local depressions shall be filled with plain Shot Crete or as directed by Engineer-in-charge. Before laying the second layer of SFRS, the first layer shall be allowed to take its initial set and have loose materials removed. Care shall be taken to ensure that no air pockets formed behind the SFRS during application of SFRS. All applied SFRS / Shotcrete shall be kept wet for at least 7 days to ensure proper curing of the Shot Crete. Sufficient air shall be added to the nozzle to get good compaction and higher compressive strength. All necessary precautions shall be taken to ensure that there is no damage to the instrument installed in rock and the drainage holes do not get plugged.

### **13.5 PROFICIENCY OF WORKMEN:**

Nozzle man shall have previous experience in the application of shotcrete on atleast two projects of comparable nature and shall work under immediate supervision of a foreman or instructor with at least five years of experience. Proficiency of workmen will be checked by CE/CN/ERS 192 Contractor Engineer-in-charge before allowing them to work.

### **13.6 CONTROL OF WATER:**

Water flow and seepage shall be controlled in such a manner that detrimental effects are completely and permanently eliminated. The contractor may drain such water by pipes chases or other appropriate methods approved by the Engineer-in-charge.

### **13.7 SAFETY MEASURES:**

Alkali hydroxides and other chemicals contained in Shot Crete admixtures are moderately toxic and can cause skin and respiratory irritation unless adequate safety measures are undertaken. In applying Shot Crete containing toxic admixtures, the nozzle men should have air to breathe, which shall be free of toxic or objectionable material. Gloves and necessary protective clothing shall be worn to protect against dermatitis.

### **13.8 MEASUREMENT AND PAYMENT:**

- i. Measurement for payment of Shot Crete/SFRS shall be made of volume in cubic metres of Shot Crete / SFRS placed computed by multiplying the payment area by the specified (i.e. the design) thickness. The payment area shall be defined as under: -
- ii. When excavated profile does not extend to portion of 'GEOLOGICAL OVERBREAK' as defined in para 12.9, payment area shall be projection of irregular area on to the Payment Line as defined in para 12.7.1 i.e. Payment Line perimeter multiplied by length along the tunnel.
- iii. When excavated profile extends to portion of 'GEOLOGICAL OVERBREAK' as defined in para 12.9, an imaginary line parallel to 'Payment Line' shall be drawn in such a way that it depicts average line of the actual excavation. In case perimeter of such an imaginary line is less than or equal to the perimeter of a line parallel to and 300mm beyond 'Payment Line' the payment area shall be calculated by considering perimeter of Payment Line only as described in para (ii) above. However, in case perimeter of such an imaginary

line is more than the perimeter of a line parallel to and 300mm beyond 'Payment Line', the Payment area shall be calculated by multiplying the perimeter of such an imaginary line with length along the tunnel.

**iv.** Where Shot Crete is used only for filling of local depressions before applying SFRS, measurement and payment of Shot Crete shall be made on the basis of actual volume of concrete deposited to obtain the required thickness of Shot Crete.

**v.** Shot Crete placed by the contractor in excess of thickness shown on the Drawings or as specified by Engineer-in-charge shall not be measured for payment.

**vi.** Rates quoted for Shotcrete / SFRS should include cost of all materials (excluding cost of cement), providing and incorporating admixtures or other additives in Shot Crete / SFRS etc complete.

**vii.** Wire mesh as per specifications shall be arranged by contractor at his own cost himself and is provided for in schedule of items, rates and quantities.

**viii.** No separate payment shall be made for Shot Crete /SFRS consumed in rebound and / or wasted Shotcrete /SFRS. The cost of various ingredients of rebound / wastage is deemed to be included in the unit price of Shotcrete / SFRS.

**13.9 SUPPORTING BOLTS:** - Where, in the opinion of the Engineer-in-charge blocks of rock are likely to get loose and fall down, the contractor shall arrange and install supporting bolts. Such supporting bolts are expected to be needed mostly in the roofs only. It shall however be the sole responsibility of the contractor to bring in writing to the notice of the Engineer-in-charge, the location where the contractor may consider it necessary to provide supporting bolts. The final decision about installation of supporting bolts at any location and the total quantity thereof shall however, vest with the Engineer-in-charge. Specification for rock bolts are detailed below: -

#### **13.9.1 ROCK BOLTING**

Where in the opinion of the Engineer-in-charge the rock is highly jointed and blocks of rock are likely to get loose and fall down, the contractor shall install supporting rock bolts. Such rock bolts are expected to be needed mostly in the roof & side walls. The location and number of such bolts may be determined after discussion and joint inspection with the Engineer-in-charge. It shall, however, be the sole responsibility of the contractor to bring in writing to the notice of the Engineer-in-charge the location where contractor may consider it necessary to provide supporting bolts. Geological conditions and fractured nature of rock shall be the main determining factor for establishing necessary lengths of bolts, the pattern, spacing and the area to be bolted. Systematic pattern of bolting is shown on the drawing. However, the pattern of bolting is subject to change at the discretion of Engineer in charge.

#### **13.10 INSTALLATION OF ROCK BOLTS (25 mm HYSD bars): -**

The contractor shall drill 38mm to 45mm diameter holes in roof or the sides of the excavation allocations and to depths as directed or approved by the Engineer-in-charge. These would vary in different situations and locations. The rock bolts shall be fabricated as per approved drawings. The contractor shall carry out drilling and fixing including grouting of rock bolts with approved grout mix using cement, sand and admixtures for quick setting of grout. The admixtures shall be conforming to relevant codes and as approved by Engineer-in-Charge. Admixtures shall be provided by the contractor at his own cost, which shall be deemed to be included in the quoted rates. Prior to installation of rock bolts, holes shall be flushed out until the return water is clear. The contractor then shall pour / inject

sufficient cement grout into the hole such that when the bolt/ dowel is fully inserted into the hole, it will displace the grout, leaving the annulus completely filled with grout. When necessary, due to inclination of hole, contractor shall place the grout in the hole using perforated metal sleeves acceptable to Engineer – in - Charge. The grout mix to be used shall have following properties:

- i. Maximum water cement ratio to be limited to 0.4 to ensure barrier against corrosion and other aggressive agents and stress. The grout shall be workable 1:1 cement / sand mix.
- ii. Thixotropic consistency to make grouts suitable for vertical holes.
- iii. Minimum strength development of 12 MPa in 24 hours. 28 days strength of grout mix shall not be less than 30 MPa.
- iv. It should have enough workability for easy pumping and placement.

The bolt / dowel shall be protected against disturbance for a minimum time of 48 hours after installation or more as required by Engineer-in-charge. After the grout has gained adequate strength, washer plates shall tighten against the rock face, so as to induce positive compression in rock mass around the bolts. Alternatively the contractor may be permitted to use commercial epoxy grouts/Cement capsules manufactured by reputed manufacturers of construction chemicals, if it substantially reduces the setting time for the grout and expedites work no extra rate will be paid for this. The contractor shall maintain on site a reserve supply of rock bolts/dowels complete with accessories.

All bolts within 10m of a blasting operation shall be checked within 6 hours of blasting and retightened.

#### **13.11 PULL OUT TEST ON INSTALLED ROCK BOLTS**

The contractor shall perform pull out tests on rock bolts selected by the Engineer-in-charge. The number of such pull out tests shall not exceed 1 in every 100 bolts. The bolts shall resist a pull out tension satisfying design requirements. If any rock bolt fails due to improper workmanship or material defect, the Engineer-in-charge may order a test for all adjacent rock bolts, and all rock bolts so failing shall be replaced and retested at the contractor's expense. The design requirement for 25mm dia rock bolts shall be 9 tonnes pull or higher at failure unless otherwise specified. The contractor shall supply testing apparatus and necessary operator and labourers to operate it. The testing apparatus shall be equipped with a gauge which shall give the applied load at all times during the testing operation. All testing shall be done in the presence of Engineer-in-charge or his representative. The cost of the testing shall be included in the quoted rates. The Engineer-in-charge reserves the right to ask for pull out tests in any location where he deems it necessary.

#### **13.12 MEASUREMENTS AND PAYMENT**

Payment for furnishing and installing rock bolts shall be made as per linear length of the steel rod installed inside the drilled holes and not per meter of drilling and the rate tendered shall include the cost of fabrication of rock bolt, drilling of hole, grouting, materials etc., tightening the bolts to the required torque and also the temporary timber supports and all other works in connection therewith. Any bolt furnished and installed of a length less than 3.0m deep at the direction of the Engineer - in - Charge shall be paid for as actual length inside the rock. Rock Bolts which because of blockage, caving or loose cannot serve their intended purpose shall not be paid for.

#### **13.13 PERMANENT SUPPORTS**

Permanent tunnel supports are those supports erected during the excavation of the tunnel

and shaft which will be left permanently in the tunnel and shafts forming part of the finished tunnel. At the discretion of the railway the method of providing permanent steel supports in the tunnel may be altered or deleted.

### **13.14 FABRICATION OF PERMANENT TUNNEL SUPPORTS**

1. The main members in the permanent steel supports for tunnel shall normally be sawn to exact lengths and holes drilled to template. The butt and footplates may be sheared to sizes and shall be carefully cut and holes accurately drilled, so that when the members are in position the holes shall be truly concentric.
2. Rolled steel joists shall be cold bent to radius indicated in the drawing and the butt and footplates shall then be welded in. When finished, the support shall be true to profile and free from all kinds of twists and open joints and the material shall not be defective or strained in any manner.
3. No bolt hole shall be more than 1.5 mm larger than the diameter of the bolt. Sizes of the bolts used shall be in accordance with drawing.
4. Fabrication of the steel support shall be done in accordance with the drawings. Workmanship and finish shall be comparable to the best general practice in modern steel fabrication shop. All ring beam segment shall be fabricated in such a manner that required number of segment selected at random might be easily assembled and bolted together to form a complete support of specified dimensions. Finished diameters of the holes in the beams, butt plates splice plate or wall plates joints etc. shall not be less than that shown in the drawing which is more by 1.5mm than normal diameter of the bolt. The holes will be drilled and reamed with the parts to be bolted in assembled position.
5. All welding work carried out in the fabrication of steel supports shall conform to IS: 823-1964 (Code of Procedure for Manual Metal Arc Welding of Mild Steel) as amended from time to time.
6. Welding shall be by the electric arc welding process using a method, which excludes the atmosphere from the molten metal. Welding shall conform to the applicable provision of IS: 816-1956 "Code of practice for use of Metal Arc Welding for general construction in Mild Steel". Inspection of Weld shall be done as per IS: 822 – 1970.
7. Surface to be welded shall be free from loose scales, slag rust, grease, paint and any other foreign material except that mild scale, which withstands, a vigorous wire brushing, may remain. Joint surface shall be free from fines and tears, preparation of edge by gas cutting shall whenever practicable be done with a mechanically guided torch.
8. All welding shall be structurally safe. Only qualified welders shall be allowed to weld. For this purpose, the Engineer-in-charge or his authorized agent shall periodically test the welders. A welder shall be allowed to do fillet weld only on such locations for which he is qualified. For the above test IS 817-1967 "Code of practice for Training and Testing of Metal Arch. Welder" shall be applicable.

**13.15** The dimensions and number of the foot blocks, lagging, spreaders, wedges and blocking, where not shown on the drawings shall, in all cases, be as necessary to serve their functions and for safety. The material used for foot block, lagging, blocking and spreader shall be of Contractor's steel, payment for which will be made in the relevant item of Schedule of

Rates. All equipment including welding material shall be arranged and provided by the contractor at his own cost and nothing extra shall be paid.

**13.16** The contractor shall assemble, complete set of steel supports in a horizontal position before dispatch from the work, any of the support as may be required by the Railway for the purpose of checking.

**13.17** No painting of any sort is to be carried out on the steel support.

**14. ERECTION OF SUPPORTS:**

The section and spacing to which supports shall generally be erected is shown in approved drawing. Actual section and spacing at which arch ribs and posts are to be erected will be decided by the Engineer as the work is in progress to suit ground conditions. The decision of the Engineer-in-charge will be final and binding on the contractor.

14.1 When the break out is affected in two portions as aforesaid the wall plate on both sides shall first be erected correct to line and level and secured in position on timber packing. The length of wall plates will depend upon the "SAFE BRIDGE ACTION PERIOD OF THE ROCK" i.e. the time that can be allowed to elapse after blasting and before the steel support are erected, the cycle of operations and the advance to be obtained in one cycle. Wall plates laid on either side shall be connected loose to the previously laid wall plates by means of connection plates and bolts. Arch rib shall then be guyed up and lined and leveled & connected to the previously erected arch rib by means of the tie rods and nuts at both ends.

14.2 In case of "OVER BREAK" that are judged as due to defective workmanship of the contractor, no payment will be made for any new steel work erected by the contractor for rectification of the damaged supports and no compensation will be paid on this account. Any extra steel required for rectification / replacement of damaged permanent supports shall be arranged by the Contractor at his own cost.

14.3 Any protective measure required for preventing / damage to support already carried out should be done by the contractor at his cost or the work has to be carried with controlled blasting in order to avoid such damages. The contractor will not be entitled for any compensation on this account.

14.4 Repair / Replacement of the support if necessary due to negligence on the part of the contractor will be done entirely at the cost of the contractor. Nothing contained in this para shall prevent the contractor from erecting, at his own expenses permanent supports heavier than approved by the Engineer-in-charge, nor shall it be construed to relieve the contractor from the sole responsibility for the safety of the excavation or the liability for injuries of deaths of persons or damage to property.

14.5 The materials, bolts, nuts and washer will conform to the relevant IS specification for the intended use and shall be arranged and provided by contractor at his own cost. The sequence and method of excavation shall be selected so as to enable the installation, of the necessary supports, before the BRIDGE ACTION PERIOD for the excavated profile has expired. The quantity for this substantially depends upon the strata encountered during the excavation. The contractor will not be entitled for any claim for such variations. The tentative spacing



and number of support are shown in the approved drawing. The detailed plan will be supplied to the contractor from time to time for different types of rock encountered and contractor will be required to provide the support as per these drawings.

14.6 The steel support shall be installed with proper workmanship, true to the line and grade and as directed by the Engineer-in-charge and shall be maintained by the contractor in proper condition and alignment until the concrete lining is placed against them.

14.7 The contractor shall correct improper installation of supports within 48 hours after the defects are brought to his attention.

14.8 In supported sections of the tunnel, the contractor shall securely brace the support with spreader, blocking and wedges as per approved drawing or as directed by Engineer at site. Where it is necessary to place structural struts across the invert, the struts shall be placed as shown on the approved drawings or as directed.

14.9 Due to insufficient or improper support in certain rock condition the load and side pressures may increase due to gradual subsidence of rock and if support initially provided are found inadequate substantially due to such increase in load it shall be the responsibility of the contractor to take appropriate and prompt measures to strengthen the supports and package material (cement concrete block, stone boulder etc.) behind the support and grout the backpacking material later after the lining has been placed or any other measures as approved by the Engineer-in-charge. The additional steel provided by way of strengthening shall be measured for payment as a steel support against relevant item of schedule rates and quantities.

**The back packing materials shall form part of the first stage concrete of lining and therefore, will not be paid for separately.**

#### **15.0 STAGE PAYMENT**

The stage payment for structural steel works of permanent steel support and temporary steel support will be as follows: -

- a) 75% of the payment on supply of structural steel to the site of work or approved workshop against irrevocable Bank guarantee or Indemnity bond (as the case may be) and on necessary production of records.
- b) Balance 25% payment after fabrication and erection of steel components satisfactory inside tunnel in all respects.

The contractor shall employ proper managerial and technical personnel during the execution of this work and the personnel deployed shall have adequate experience and thorough knowledge of the works executed including the specifications and proceedings involved. The list of managerial and technical personnel proposed to be engaged by the contractor shall be submitted to the Dy. Chief Engineer in -charge of the project along with the program- chart and approval of Engineer to be obtained for engaging them for work.



## **16.0 MAINTENANCE**

During the maintenance period, the Contractor shall bear the responsibility and be liable for maintenance as envisaged in the clause of general conditions of the contract.

**16.1 Handing over of Tunnel and approaches:** - The tunnel and approaches shall be handed over to the Railway on the date of completion of the agreement (including extension granted to if any) duly finishing to the profile including rectification of the tunnel surfaces, portals tunnel approaches, slopes, rain cuts and top surfaces to the satisfaction of Engineer in charge. No additional payment will be made for such rectification of slopes and surfaces.

## **16.2 CEMENT CONCRETE**

For all concrete work, the technical specification for concrete (Conc-09) available in this tender is applicable. In addition, the following requirements and condition needs to be satisfied during concreting work for Part-II items.

## **16.3 SCOPE OF WORK: -**

The work covered by this section consists of furnishing all materials equipment and labour for the manufacture, transport, placing finishing and curing of concrete in the portal structures, lagging and back fill concrete behind R.C.C. lagging and for construction of mass CC / RCC work in CUT and COVER construction tunnels and shafts. The item of concrete is split up into several items according to the class of concrete to be used and its location and will be measured and paid for accordingly.

## **16.4 BATCHING AND MIXING EQUIPMENT**

All aggregates shall be batched by weight. The contractor shall have to install preferably batching plant (RMC plant) of the requisite capacity to maintain the required progress on different items of work. The batching plant shall be capable of determining accurately by direct weighing the prescribed amounts of the various ingredients including water, cement, admixtures etc. and each individual size aggregate entering the concrete and combining them to give them a uniform mix within the prescribed time and discharging the mix without segregation. The equipment and its operation shall at all the times are subject to the approval of the Engineer-in-charge. **Alternatively, the contractor can arrange RMC from outside agencies duly taking prior approval from Engineer in charge.**

## **16.5 MEASUREMENTS AND PAYMENT**

Measurement and payment shall be made on the basis of the actual volume of the concrete placed but limited to the quantity upto the payment line indicated in drawings. No deduction shall be made for the space occupied by reinforcement and other metal work, electric conduit lines etc. The quantities of all holes and passages and embedded parts greater than 0.05 Sqm in cross-section shall however be deducted from the total quantity to arrive at the concrete work for payment. The Over Break / Cavity due to contractor's fault formed beyond the Pay Line shall be filled up by the contractor with back fill concrete and no payment for such extra concrete will be made. In case of **"Geological Over-break"** the concreting will be paid at the tendered rate. No payment will, however, be made for embedding fixtures or providing other installations, electric conduits etc. All labour materials, plants, etc., involved in providing cement slurry and mortar on rock surface and construction joints shall be deemed to be included in the unit rate to be paid for concrete.

The rate quoted for the concrete items specified in Part-II exclusively for tunnel constructions and in cut and cover portion is inclusive of all labour, materials, mixing, transportation, centering, shuttering, leveling, compaction, curing etc. (except cost of cement, which will be paid separately). It may be noted that nothing extra will be paid for using RMC plant and transit mixers and the percentage rate quoted by the tenderer is deemed to include all these costs.

**Employment of Key Personnel****DETAILS OF KEY TECHNICAL PERSONAL REQUIRED FOR TUNNEL PROJECT:-**

The Contractor shall employ the following minimum key personnel during the execution of the contracted work.

| <b>Estimated Value of Tender</b> | <b>Position</b>           | <b>Min. No.</b> | <b>Personal Qualification Experience</b>                                                                                                                          |
|----------------------------------|---------------------------|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>1.</b>                        | <b>Project Manager</b>    | <b>1</b>        | Graduate in Civil Engineering/ Project Management Minimum 10 years experience in civil Engineering Work with at least 2 years in Tunneling work.                  |
| <b>2.</b>                        | <b>Tunnel Engineer</b>    | <b>0</b>        | Graduate in Civil Engineering Minimum 3 years experience exclusively in Tunneling projects, including Survey using modern equipments.                             |
| <b>3.</b>                        | <b>Safety Officer</b>     | <b>0</b>        | Graduate in Mining (Safety) Minimum 3 years Experience in Mining industry / Tunneling work.                                                                       |
| <b>4.</b>                        | <b>Geologist</b>          | <b>0</b>        | MSc./PhD. in applied Geology. Minimum 5 years experience in subsoil investigation & Tunneling work/Mining (Service required during all stages of Tunneling work). |
| <b>5.</b>                        | <b>Foreman/Supervisor</b> | <b>1</b>        | Diploma in civil Engineering/Graduate in Civil Engineering Minimum 8 years experience in civil Engineering Work / 3 years in Tunneling work.                      |

**The following recovery for not providing the above personnel at site will be recovered from the contractor.**

1. For SL. No. 1 :- @ Rs.50,000/- per month/Person
2. For SL. No. 2 :- @ Rs.30,000/- per month/Person
3. For SL. No. 3 :- @ Rs.25,000/- per month/Person
4. For SL. No. 4 :- @ Rs.25,000/- per month/Person
5. For SL. No. 5 :- @ Rs.15,000/- per month/Person

The Biodata (Name, Qualification, Years of experience etc.) of the key personnel as mentioned above, who are proposed to be deployed, shall be furnished in the tender offer. It will be essential to retain the same key personnel for continuity of the project. However in case a situation arises to replace the key personnel for whatever reason, the same will be done in consultation with the Railways.

The decision of the engineer as to the period for which the required key personnel were not employed by the Contractor and as to the reasonableness of the amount to be deducted on this account shall be final and binding on the Contractor.

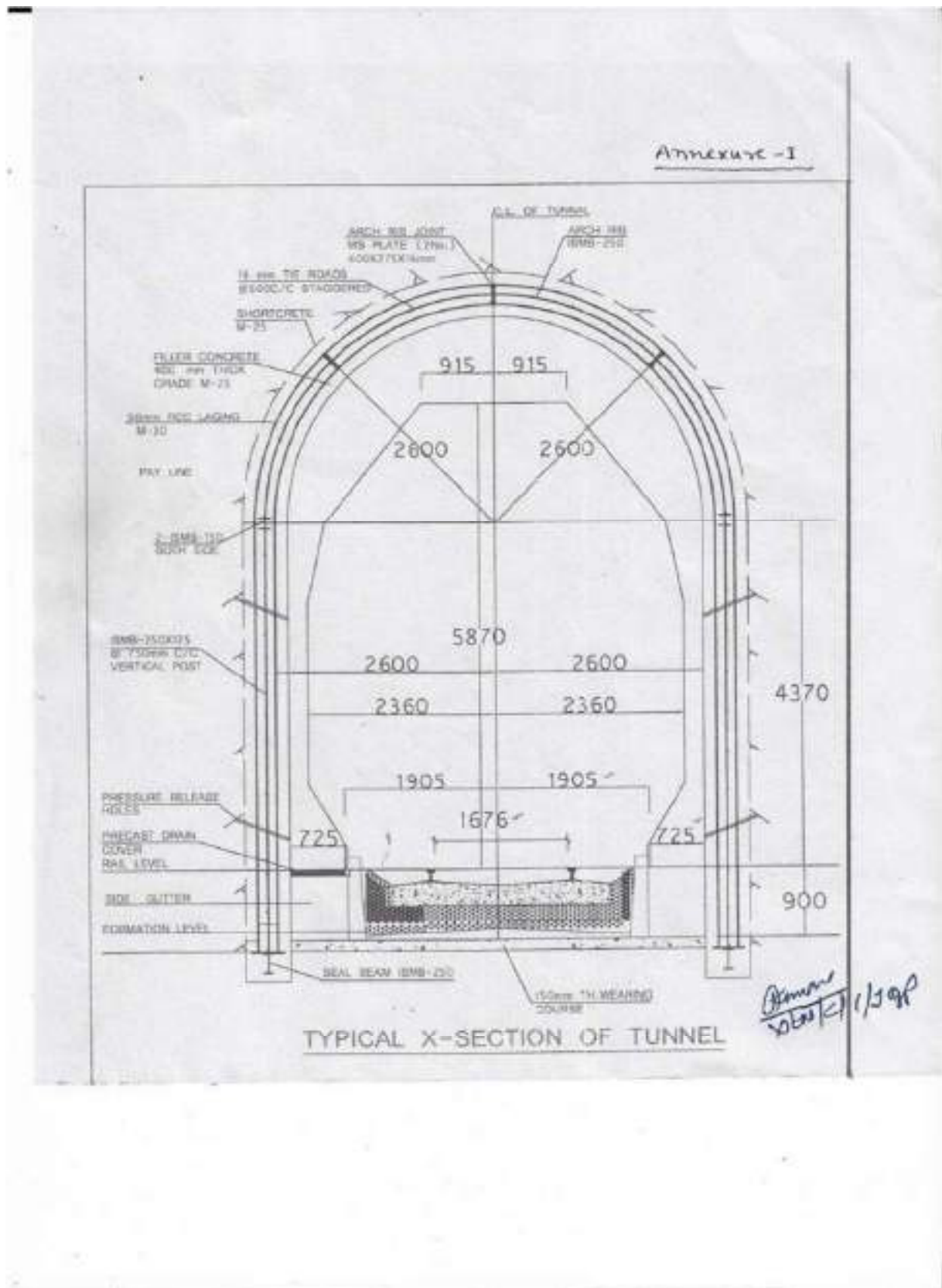
**16.6 DEPLOYMENT OF MACHINERY**

The agency is required to deploy the following minimum scale of vital machinery with requisite resources right from beginning and simultaneously for main Tunnel work execution.

| <b>S.N.</b> | <b>Type of Machinery</b>                                                                                                                                      | <b>Nos</b> | <b>Remarks</b>                                                                                                 |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|----------------------------------------------------------------------------------------------------------------|
| (1)         | Equipment's set for drilling holes for blasting tunnel excavation (As per approved methodology & pattern).                                                    | 0 Sets     | The average monthly progress for tunnel excavation is to be achieved at the rate of 100 meters/ Calendar month |
| (2)         | Dozer                                                                                                                                                         | 0 Set      | Matching to site locality & capacity to clear off all outgoing muck/debris.                                    |
| (3)         | Rock drilling equipment's with compressor and accessories for rock bolting work and drilling cavity holes Progress pressure relief holes for various diameter | 1 Full Set | Capacity & accessories to match the of tunneling work                                                          |
| (4)         | RMC Plant with all connected accessories (Preferable).                                                                                                        | 1 Full Set | Capacity not less than 25cum per hour is desired                                                               |
| (5)         | Concrete pneumatic placer /pump of farthest locations suitable capacity.                                                                                      | 2 sets     | Suitable to operate for the while tunneling work in progress.                                                  |
| (6)         | Transit mixers for RMC transportation.                                                                                                                        | One pair   | Minimum capacity of 6cum per vehicle.                                                                          |
| (7)         | Excavation.                                                                                                                                                   | 0 Nos      | Tata Hitachi of (PC 200)/ equivalent.                                                                          |
| (8)         | Automatic Dumpers.                                                                                                                                            | 0 Nos      | Minimum capacity of 8cum per load.                                                                             |
| (9)         | Complete set of shotcreting machine.                                                                                                                          | 1 Full Set | Minimum capacity of 24cum per hour make Schwing Stetter or equivalent.                                         |
| (10)        | Hydra Cranes                                                                                                                                                  | 0 No       | ---                                                                                                            |

Note:-1) The above machinery is the minimum specified vital items exclusively required for the completion of Tunnel work only (part-II). Contractor has to deploy all additional necessary machine to achieve the progress as approved by Railways.

2) In case tenderer fails to deploy the above minimum plants/machinery and resulting in slow progress, action as per clause 62 of GCC may be taken by Railways and penalty as decided by Engineer in charge will be imposed. Decisions of Engineer in charge in this regard will be final & binding on contractor.



## **Chapter-14**

### **SPECIAL CONDITION & SPECIFICATION OF CONTRACT FOR BUILDING WORK**

#### **1. SCOPE OF WORK:**

- 1.1 Any other work incidental to the construction of buildings.

#### **2. STANDARD SPECIFICATIONS AND CODE OF PRACTICE:**

- 2.1 The following standard specifications, architectural and structural drawings to be supplied by Railway, Indian Railway Standard & Indian Standard Rules & codes of practices as revised from time to time available from the Manager, Government of India, Publication Branch, Patiala House, New Delhi/ Director, ISI / Manak Bhawan, Bahadur Saha Zafar Nagar, New Delhi respectively must be followed with regard to design, material and workmanship.

- (i) I.S. Code of practice for plain and reinforced concrete (I.S.456).
- (ii) I.S. Code of practice for use of structural steel (I.S.800)
- (iii) I.S. 1742 Code of practice for building drainage.
- (iv) IRS code of practice for Electric Arc welding of mild steel structures.
- (v) IS specification for fine and coarse aggregate from natural sources from concrete (IS-515).
- (vi) Standing specification contained in Indian Railway Unified Standard Specification (Work and Materials)-2010 (Volume-I and Volume-II) with corrections slips issued up to date of opening of tender and Eastern Railway Engineering Department Unified Standard Schedule of rates (Work and Materials)-2011 as amended by up-to-date correction slip

#### **3. PLAIN/REINFORCED CEMENT CONCRETE WORKS :**

- 3.1 The ISI code of practice for the structural use of reinforced concrete in building shall from part of these additional specification and this code and the standard practice in reinforced concrete construction that has been evolved by the adoption of this code shall be followed.
- 3.2 Fine and coarse aggregate: Fine and coarse aggregate for all type of concrete works shall conform to IRUSS-2010 (Vol - I & Vol - II).
- 3.3 In addition to the routine test / special test on material will be carried out whenever required by the Engineer. The cost of the special test will be borne by the Railways, the results are as per standard laid down, failing which the cost of these tests will be borne by the contractor. Necessary facility in the form of moulds, cones, scales, materials, labour for casting, curing, specimens and such other facilities as per prerequisite required to any standard concrete test will be provided by the contractor free of cost. Cement, however, will be supplied free by the Railway for the purpose.

#### **4 FORM WORKS:**

- 4.1 GENERAL:-Shuttering shall be either of wooden planks of 35mm minimum thickness with steel sheet lining or shuttering lining or of steel plates stiffened by steel angles, it should be ensured that the shuttering should be leak proof and there should not be any leakage of cement slurry during casting of the concrete. The shuttering shall be supported on wooden battens and beams and prop of vertical bellies properly coarse branched together, so as to make the formwork rigid. In place of bellies props, brick pillar of adequate section, built in mud mortar may be used.
- 4.2 The form work shall conform to the shape, lines and dimension as showing in the plain. It shall be sufficiently rigid and strong to maintain correct shape of the members during deposition of concrete and shall be able to resist forces caused by vibration of concrete and incidental loads, associated with men working over it. The battering shall have smooth and even surface and its joints shall not permit leakage of cement grout.

- 4.3 If at any stage of working during or after placing of concrete in the structure, the formwork bulges out beyond the required shape of the structure, the concrete shall be removed and work redone with fresh concrete and adequately rigid formwork at contractor's cost. Details of shuttering and centering shall be subject to the approval of the Engineer-in-charge the completed formwork shall be inspected by Engineer-in-charge, before the reinforced cement bars are placed in positions.
- 4.4 **CAMBER**: The shuttering on beams and slabs shall have camber of 4mm per meter (1 in 250) or as directed by Engineer-in-charge, so as to offset the subsequent deflection. The cantilevers, the camber at free end shall be 1/50th of the projected length or as directed by the Engineer-in-charge.
- 4.5 Provision for holes shall be made in the shuttering for inserting fan hole clamps and provision of conduits etc. for concealed wiring and providing architectural finishing grooves if any at the junction of slabs with beams or walls or columns wherever required for architectural consideration, concealed sanitary and water supply pipes and fittings etc. as are required to be built in connection with the provision of various services in the buildings for service or architectural reasons. It may also be necessary to make holes in the shuttering of RCC columns for projecting bars. The tendered rates shall include the cost providing the above mentioned services, grooves etc. and no extra payment shall be made to the contractor for making these provisions nor any deduction shall be made on account of any saving in RCC due to these.

The formwork for the RCC chajjas will be so made that the drip coarse band can be casted along with the chajjas. No extra payment shall be made for this drip coarse band.

## **5 REINFORCEMENTS:**

- 5.1 Reinforcement may be either with M.S. Round or Tor steel as decided by the Engineer as per approved drawing. No extra payment shall be made in case tor steel is used in lieu of mild steel.
- 5.2 It shall be the responsibility of the contractor to clean the reinforcement bars with dry gunny bags, if they are coated with rust of impurities and nothing extra shall be paid for the same.
- 5.3 The rate for reinforcement cement concrete including strengthening and uncoiling of rolls of reinforcement. No extra payment for strengthening and/or uncoiling of reinforcement shall be payable by the Railway.

## **6. CONCRETING:**

- 6.1 The concrete shall be mixed properly in approved type mechanical mixer as per IRUSS-2010 (Vol - I & Vol - II). The proper consistency shall be determined by the Engineer-in-charge by a slump test, which shall be carried out. Cost of moulds, labour, tools and plants etc. for slumps tests of concrete shall be borne by the contractor except cement which shall be supplied free by the Railway or will be paid under relevant item.
- 6.2 The concrete shall be compacted immediately after placing by means of mechanical vibrator of suitable design for continuous operation.

## **7 MEASUREMENTS:**

- 7.1 All work will be paid for at the tendered rates on the basis of actual measurements at site. No account will be taken for heights and thickness over those shown in the plans, unless they are authorized by Engineer-in-charge, in writing.
- 7.2 Measurement shall be made according to IRUSS-2010 (Vol - I & Vol - II).

## **8 RATES:**

- 8.1 For all items of reinforcement concrete, the tendered rates shall include supplying and removal of scaffolding, supply of formwork, shuttering and centering etc. of approved design, their erection, dismantling, clearing and oiling, etc. cutting, hooking, bonding, binding, bailing and straightening of steel section, binding and placing in position of reinforcement etc. complete manufacturing of the reinforcement in required shape as per drawings, screening or washing the aggregate, mechanically mixing and placing the same in position & use of equipment including mechanical mixers, vibrator etc. all

watering during the work and curing for the prescribed period after-works & finished exposed surface.

**9 FOUNDATIONS:**

**9.1 TIMELY NOTICE FOR INSPECTION OF FOUNDATIONS OF WORKS TO BE COVERED UP:**

The contractor shall give notice to the Engineer when and as soon as the excavation of any portion of the site for obtaining a foundation or bottom, whether above or below water, has reached the depth and width shown in the drawings. The contractor shall also give further notice to the Engineer whenever any bottom foundation is ready for inspection and whenever it is necessary to cover up any work in respect of which the Engineer desires previous inspection, so that the Engineer may inspect the same before it is covered up. Bottom of foundation should be maintained by the contractor till execution of the work without any extra cost as directed by the Engineer-in-charge.

**10 BRICK WORK:**

10.1 All brickwork shall be done in well-burnt bricks as per IRUSS-2010 (Vol - I & Vol - II) in cement mortar in proportion as may be specified in the drawings or as instructed by the Engineer-in-charge.

10.2 All pipes, clamps or other fittings as may be required shall be fixed in position as the work proceeds. Chases will have to be cut in the brick walls before housing the fittings and the contractor's rate for brickwork shall be inclusive of the cost of cutting chases. Any change specially made after the masonry work has been completed will be at the cost of the Railway Administration.

10.3 No claim for any additional labour involved in doing the masonry work around the boards, telephone boxes etc. shall be entertained.

**11 ROUGH CAST/SAND FACED CEMENT PLASTER 20mm THICK.**

11.1 All brick masonry shall be thoroughly wetted and joints raked out to a depth of at least 20mm well washed with clean fresh water to ensure a clean depth of 13mm free from any mortar, and must be kept watered for a week before the plaster is applied if the masonry is old, otherwise the watering should be done for 2 days.

11.2 Samples of rough cast/sand faced shall be got approved from Engineer-in-charge before commencement of work and work shall be done strictly according to IRUSS-2010 (Vol - I & Vol - II).

**12 WATER SUPPLY AND SANITARY INSTALLATIONS:**

12.1 For execution of sanitary installations and water supply works, the contractor shall arrange a Licensed plumber and employ especially skilled artisans for these works. The work shall be executed as per IRUSS-2010 (Vol - I & Vol - II).

12.2 The work of providing GI and/or HCI pipes, as required, shall proceed along with the construction of building to avoid demolition or breaking up of masonry at a later stage.

12.3 Samples of sanitary installations and fittings such as W.C. Pan, wash basins, sinks etc. shall be submitted to the Engineer-in-charge for approval before supplying and fixing & shall be provided strictly in accordance with the approved samples.

12.4 G.I. pipes and fittings shall conform to BIS specifications and samples be got approved from the Engineer-in-charge before using the materials in the work.

12.5 RCC pipes for sewer should conform to NP-2 class of pipe specification as per BIS. These should be tested and certified by approved testing agencies/laboratories and the contractor, if asked, for approval of materials by the Engineer should produce certificate to this effect. Nothing extra will be paid for testing and certification by testing agency laboratory.

**13 FLOORING:**

13.1 Flooring shall be laid using approved quality sand and coarse aggregates and as per IRUSS-2010 (Vol - I



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& Vol - II). Floor shall be laid in panels and if dividing strips shall be provided at the discretion of Engineer-in-charge. No extra payment will be made for the same.

**14 ROOFING:**

14.1 The roof shall be laid either with RCC or as per approved plan.

14.2 Openings for fan clamps and other fittings, connection with services shall be provided in Shuttering as directed for which nothing shall be paid.

**15. TESTING OF BUILDING MATERIALS:**

15.1 Regular testing of building materials such as bricks, sand, aggregates, tiles, water proofing compounds, doors and windows etc. should be done.

15.2 Day to day quality control, sample testing facilities must be available at work sites.

15.3 Test cubes for concrete should be made and tested as per IS specifications.

15.4 Concrete mix as specified in the tender documents should be followed at work site. In case of design mix, IS specifications for designing, producing, using, testing and accepting/rejecting must be followed.

15.5 Cement should be used by weight only in case of design mix concrete.

15.6 In case of cement, steel, HTS wires, besides obtaining test certificates from the contractors, regular independent tests to check the quality as per IS specifications should be done.

Witness:

1.

2.

Signature of Tenderer

Date.....